SUBTASK 32-33-31-210-001

(5) Examine the hydraulic connections for leaks.

SUBTASK 32-33-31-480-003

(6) Extend the nose landing gear and install the downlock pin (TASK 32-00-01-480-801).

SUBTASK 32-33-31-860-003

(7) Remove the power from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-33-31-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(8) Lower the airplane and remove the jacks (TASK 07-10-01-580-802).

H. Nose Gear Lock Valve Manifold Installation Test (Airplane not on Jacks)

NOTE: This test gives does a check of the manifold assembly [1] with the airplane on the ground, if jacks are not available.

SUBTASK 32-33-31-860-004

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCK PINS ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-31-700-001

(2) Use the override trigger to move the landing gear lever from the DN position to the UP position for three times.

NOTE: Hold the lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-33-31-700-002

(3) Move the landing gear lever to the DN position.

(a) Make sure that the nose lock actuator tries to extend.

NOTE: The hydraulic line to the actuator should pulse.

SUBTASK 32-33-31-700-003

(4) Move the landing gear lever to the UP position.

(a) Make sure that the nose lock actuator tries to retract.

NOTE: The hydraulic line to the actuator should pulse.

SUBTASK 32-33-31-700-004

(5) Move the landing gear lever to the DN position.

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SUBTASK 32-33-31-210-002

(6) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-33-31-860-005

(7) Remove the power from hydraulic system A (TASK 29-11-00-860-805).

SUBTASK 32-33-31-210-003

(8) Make sure that there are no leaks in the hydraulic connections.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-31-200-001

(1) Service the reservoirs if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR TRANSFER CYLINDER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Gear Transfer Cylinder Removal

(2) Nose Gear Transfer Cylinder Installation.

TASK 32-33-41-000-801

2. Nose Gear Transfer Cylinder Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-41-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-41-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-33-41-580-001

(3) Install chocks around the tires of the nose landing gear (PAGEBLOCK 10-11-05/201).

E. Nose Gear Transfer Cylinder Removal

SUBTASK 32-33-41-020-001

(1) Disconnect the hydraulic lines from the transfer cylinder [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-41-480-002

(2) Install caps/plugs, STD-14686, on the hydraulic lines and in the ports on the transfer cylinder [1].

SUBTASK 32-33-41-020-002

(3) Remove the screws [3] and washers [4] from the clamps [2].

SUBTASK 32-33-41-020-003

(4) Remove the clamps [2].

SUBTASK 32-33-41-020-004

(5) Remove the transfer cylinder [1] from the airplane.

SUBTASK 32-33-41-020-005

(6) Do these steps if the transfer cylinder [1] does not have unions [6] installed:

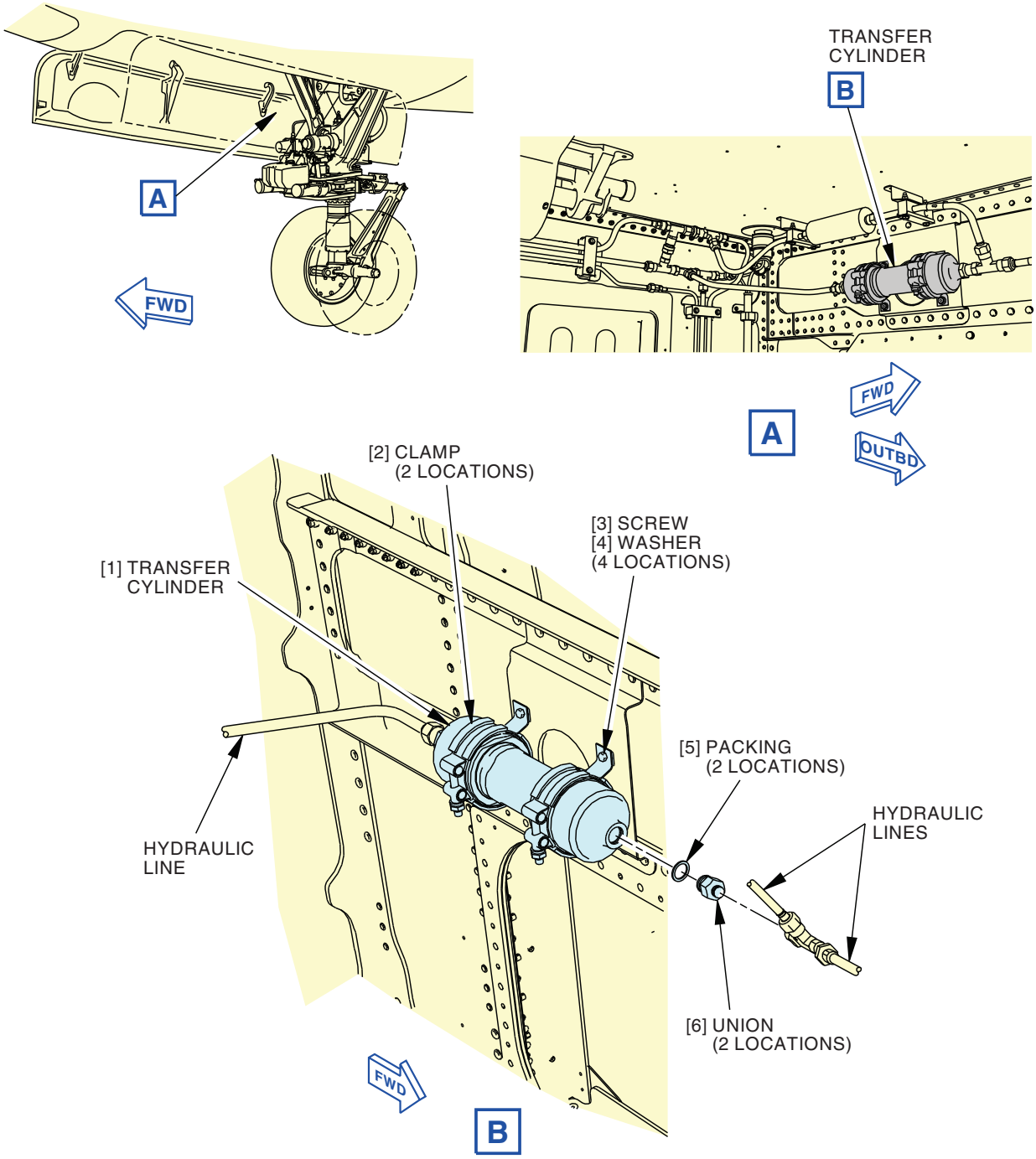
(a) Remove the unions [6] from the transfer cylinder [1].

(b) Remove and discard the packings [5].

(c) Install caps/plugs, STD-14686, in the ports of the transfer cylinder [1].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Landing Gear Transfer Cylinder Installation
Figure 401/32-33-41-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-41-400-801

3. Nose Gear Transfer Cylinder Installation

(Figure 401)

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transfer cylinder	32-33-41-01-015	PTF ALL
5	Packing	32-33-41-01-010	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Nose Gear Transfer Cylinder Installation

SUBTASK 32-33-41-420-001

(1) Do these steps if the transfer cylinder [1] does not have the unions [6] installed:

(a) Lubricate these items with hydraulic fluid, D00153:

1) The packings [5].

2) The union [6].

(b) Remove the caps/plugs, STD-14686, from the ports of the transfer cylinder [1].

(c) Install the packings [5] and the unions [6] in the transfer cylinder [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-41-610-001

(2) Fill the transfer cylinder [1] from the two ends with hydraulic fluid, D00153.

(a) Put a cap on the union [6].

SUBTASK 32-33-41-420-002

(3) Put the transfer cylinder [1] in the clamps [2].

SUBTASK 32-33-41-420-003

(4) Adjust the position of the transfer cylinder [1] to align it with the hydraulic lines.

SUBTASK 32-33-41-080-001

(5) Remove the caps from the unions [6] and caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-33-41-410-001

(6) Connect the hydraulic lines to the transfer cylinder [1].

(a) Tighten the B-nuts on the hydraulic lines to 170 ±9 in-lb (19 ±1 N·m).

SUBTASK 32-33-41-420-004

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(7) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and shank of the screws [3] prior to installation.

SUBTASK 32-33-41-420-005

(8) Install the screws [3] and washers [4] in the clamps [2].

SUBTASK 32-33-41-420-006

(9) Tighten the hydraulic lines.

G. Nose Gear Transfer Cylinder Installation Test

SUBTASK 32-33-41-580-002

WARNING

MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE NOSE LANDING GEAR. FAST MOVEMENT OF THE NOSE LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-33-41-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-41-080-002

(3) Remove the downlock pin for the nose landing gear (TASK 32-00-01-080-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-41-870-001

WARNING MAKE SURE THAT YOU INSTALL THE GROUND LOCK ASSEMBLIES IN THE LANDING GEAR TO PREVENT THE ACCIDENTAL OPERATION. IF THE LANDING GEARS RETRACT, IT CAN CAUSE INJURY TO PERSONS OR DAMAGE TO THE STRUCTURE.

(4) Extend and retract the nose landing gear several times.

NOTE: This will bleed the air from the transfer cylinder [1] and the hydraulic lines.

(a) Make sure that the nose landing gear operates correctly.

SUBTASK 32-33-41-210-001

(5) Examine the hydraulic connections for leaks.

SUBTASK 32-33-41-480-003

(6) Extend the nose landing gear and install the downlock pin (TASK 32-00-01-480-801).

SUBTASK 32-33-41-860-003

(7) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-41-580-003

(8) Do this task: Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802.

SUBTASK 32-33-41-840-001

(9) Examine the hydraulic reservoirs for the correct servicing.

(a) Do a servicing if it is necessary (TASK 12-12-00-610-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR LOCK MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Gear Lock Mechanism Removal

(2) Nose Gear Lock Mechanism Installation.

B. The Nose Gear Lock Mechanism has these parts:

(1) Forward Lock Link Assembly

(2) Spring Assembly

(3) Aft Lock Link Assembly.

TASK 32-33-51-000-801

2. Nose Gear Lock Mechanism Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-51-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-51-580-001

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE AREA AROUND THE NOSE GEAR. THE MOVEMENT OF THE NOSE GEAR CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Do this task: Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

SUBTASK 32-33-51-860-001

(3) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Nose Landing Gear Lock Mechanism Removal

SUBTASK 32-33-51-020-001

(1) Do these steps to disconnect the spring assembly [2] from the aft lock link assembly [3] and the wheel well fitting:

(a) Remove the pins [8], nuts [4], and washers [5] from the shaft [6] where the spring assembly [2] is attached to the aft lock link assembly [3] and the wheel well fitting (View B).

(b) Use the expander set, SPL-1866, to extend the spring assembly [2].

(c) Remove the pins [8], nuts [9] and washers [5] from the bolts [10] that hold the spring assembly [2] to the forward lock link assembly [1] (View D).

(d) Remove the spring assembly [2], bolts [10], washers [11], and sleeves [12] from the forward lock link assembly [1].

(e) Remove the spring assembly [2] from the aft lock link assembly [3].

(f) Install the bolts [10], washers [11], spring assembly [2], and sleeves [12] to attach the spring assembly [2] to the forward lock link assembly [1].

(g) Install the washers [5] and nuts [9], but do not tighten the nuts [9].

SUBTASK 32-33-51-020-002

(2) Remove the pin [18], nut [19], washer [20], washer [21], washers [23], and pin [22] to disconnect the rod end of the lock actuator from the aft lock link assembly [3] (View C).

SUBTASK 32-33-51-020-003

(3) Remove the pin [30], nut [29], and bolt [27] from the nut [28], on the pin [25] that connects the forward lock link assembly [1] to the drag strut (View F).

SUBTASK 32-33-51-020-004

(4) Remove the nut [28], washer [26], and pin [25] to disconnect the forward lock link assembly [1] from the drag strut.

SUBTASK 32-33-51-020-005

(5) Do the steps that follow to remove the lock sensor [14] from the aft lock link assembly [3] (View E):

(a) Remove the screws [17] and washers [24] from the clamps that attach the wire bundle to the aft lock link assembly [3].

(b) Remove the nuts [13], washers [15], and screws [16] from the lock sensor [14].

(c) Remove the lock sensor [14] from the aft lock link assembly [3].

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SUBTASK 32-33-51-020-006

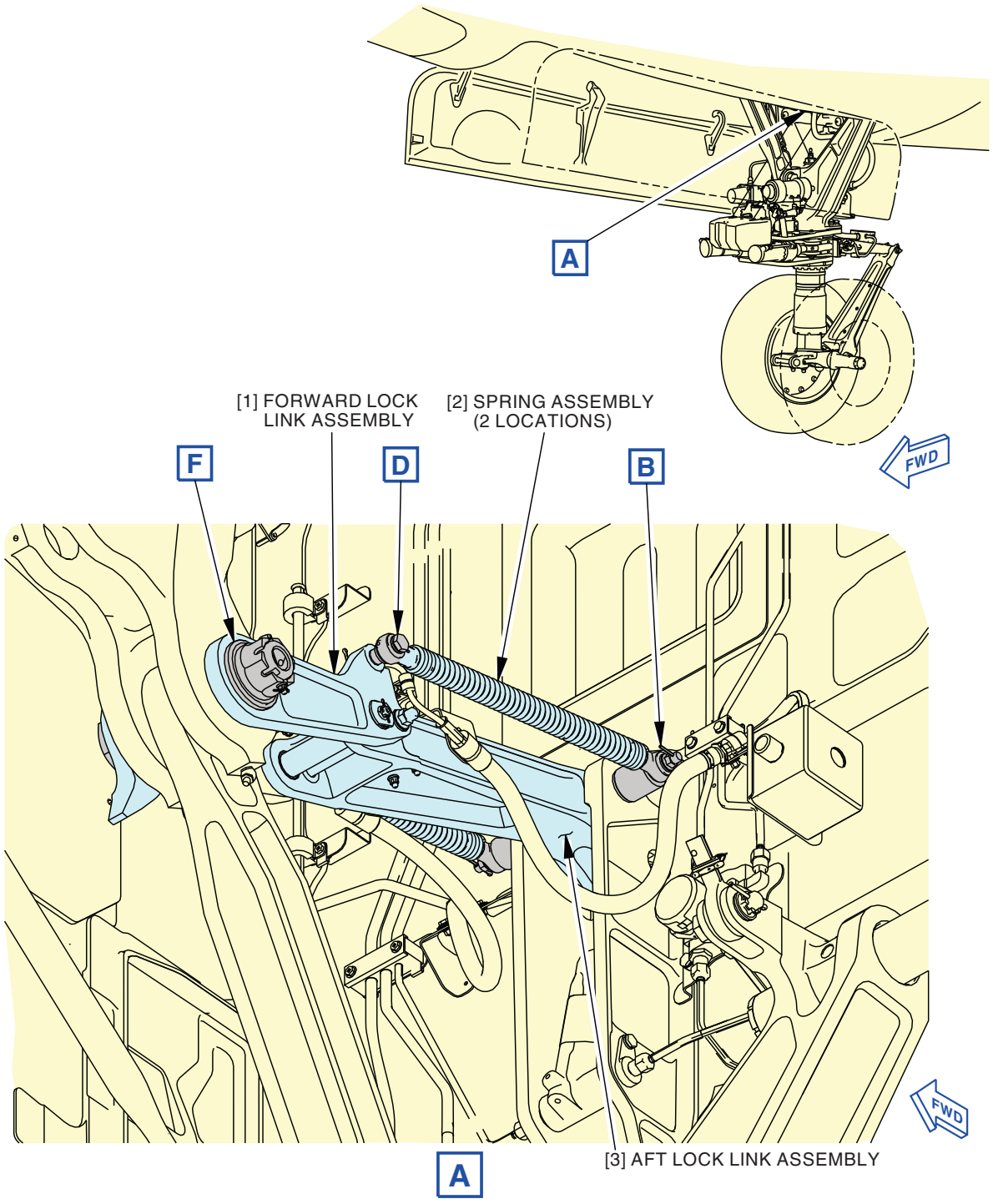
(6) Remove the bushing [7] and shaft [6] to disconnect the aft lock link assembly [3] from the wheel well fitting (View C).

SUBTASK 32-33-51-020-007

(7) Remove the lock mechanism as a complete assembly.

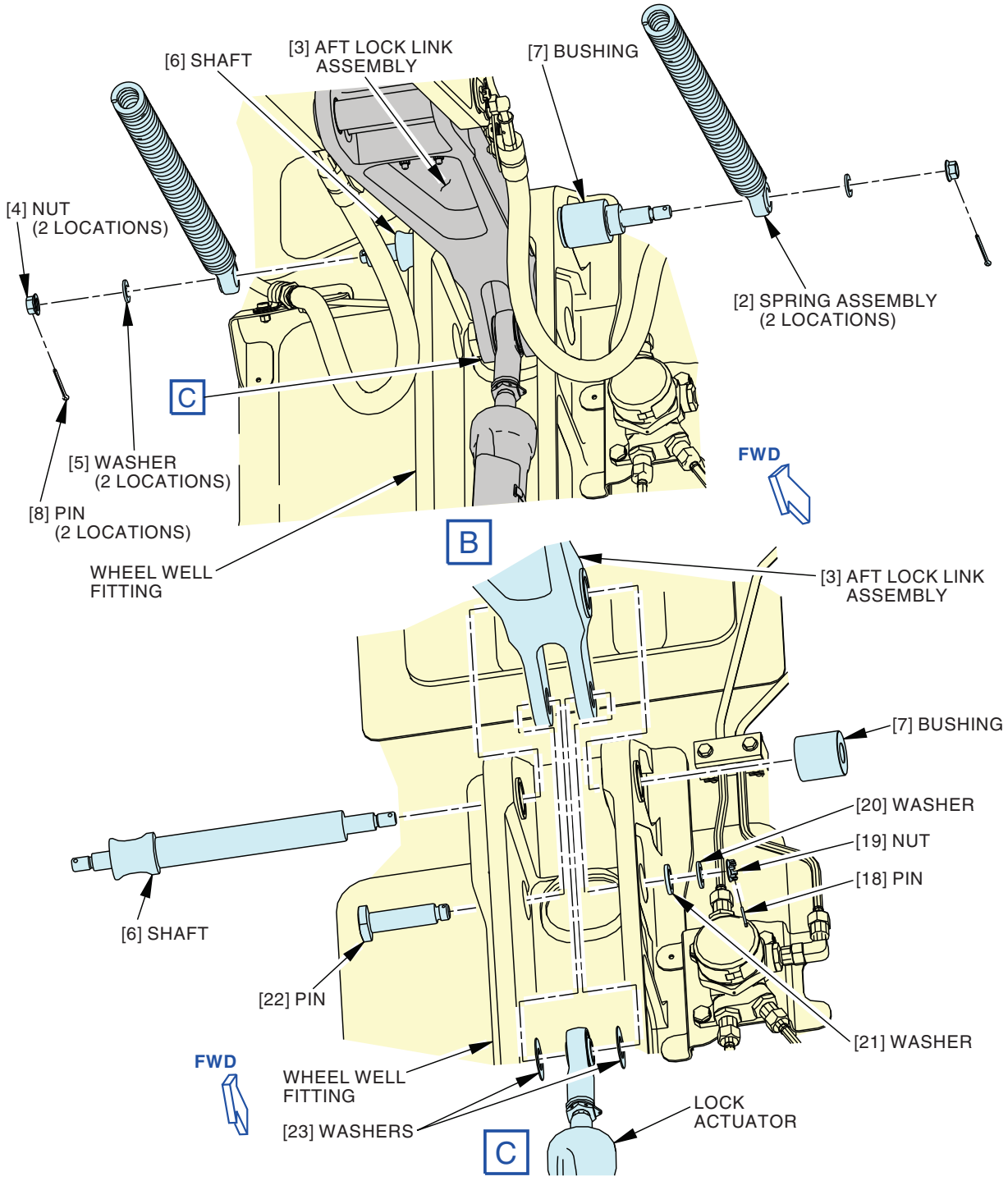
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



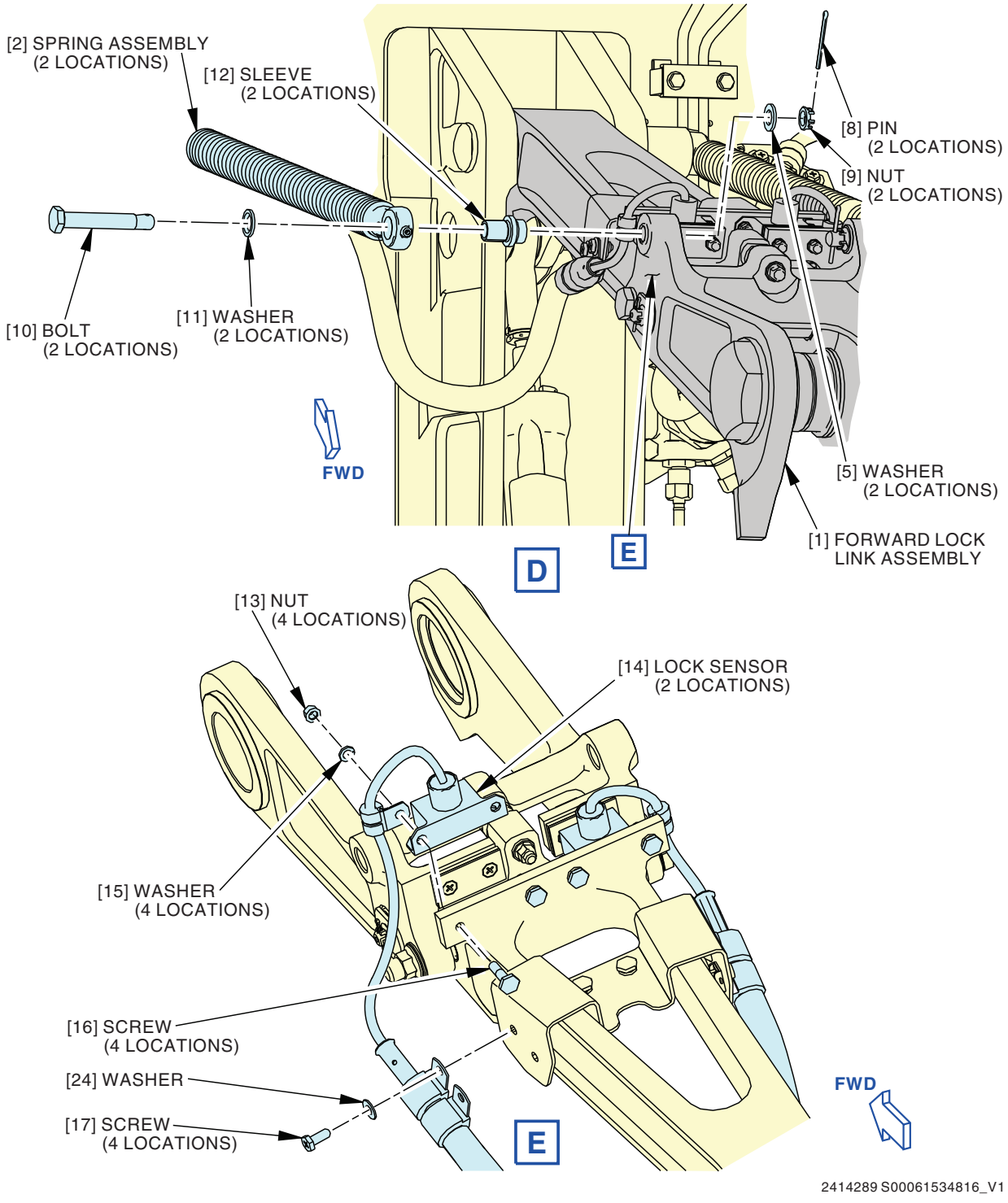
Nose Gear Lock Mechanism Installation
Figure 401/32-33-51-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



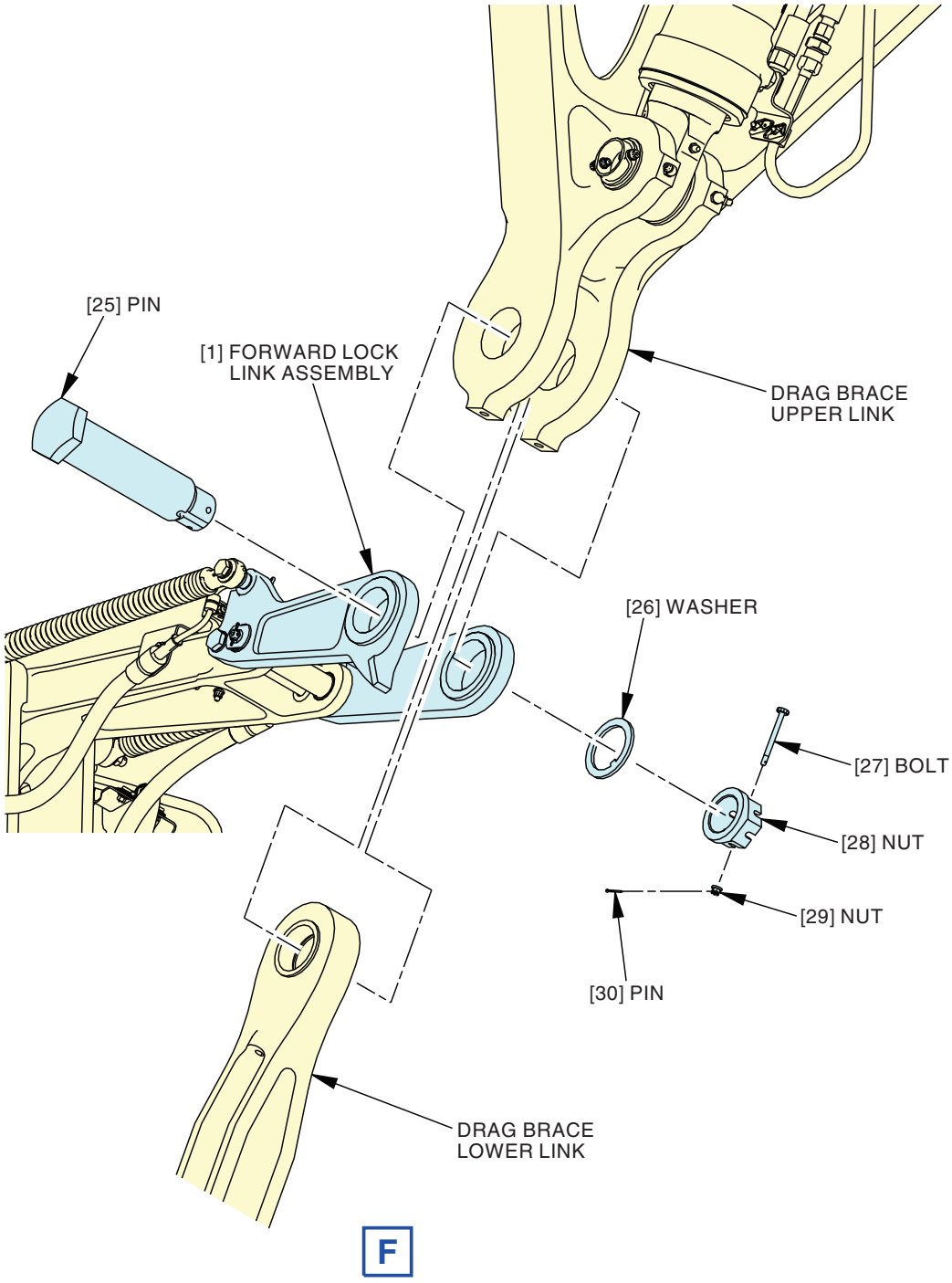
Nose Gear Lock Mechanism Installation
Figure 401/32-33-51-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Lock Mechanism Installation
Figure 401/32-33-51-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Lock Mechanism Installation
Figure 401/32-33-51-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-51-400-801

3. Nose Gear Lock Mechanism Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-61-41-400-801	Nose Landing Gear Lock Sensor Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Forward lock link assembly	32-33-51-01-120	PTF ALL
2	Spring assembly	32-33-51-01-030	PTF ALL
3	Aft lock link assembly	32-33-51-01-190	PTF ALL
8	Pin	32-33-51-01-005	PTF ALL
18	Pin	32-33-51-01-045	PTF ALL
30	Pin	32-33-51-01-045	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

F. Nose Gear Lock Mechanism Installation

SUBTASK 32-33-51-420-006

(1) If you replace one of the lock links, you must install or remove the shims on the completed lock mechanism assembly to adjust the overcenter dimension.

NOTE: The overcenter dimension is set on the assembly.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-51-420-001

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Do the steps that follow to connect the aft lock link assembly [3] to the fitting on the aft wheel well wall (Figure 401, Views B, C):

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread reliefs of the shaft [6] and the nuts [4].

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the faces of the washers [5].

(c) Apply a thin layer of corrosion inhibiting material, G50136, to the new pins [8].

(d) Move the aft lock link assembly [3] into the fitting.

(e) Install the shaft [6] and bushing [7] to connect the aft lock link assembly [3] to the fitting.

SUBTASK 32-33-51-420-002

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Do the steps that follow to connect the forward lock link assembly [1] to the drag strut (Figure 401, View F):

(a) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread reliefs of the pin [25] and nut [28], and to the faces of the washer [26].

CAUTION REMOVE UNWANTED CORROSION-INHIBITING COMPOUND FROM SURFACES WHICH WILL BE LUBRICATED. IF YOU APPLY CORROSION-INHIBITING COMPOUND TO JOINTS THAT TURN, FAILURE OF THE LANDING GEAR TO EXTEND OR RETRACT CAN OCCUR.

(b) Apply grease, D00633, to the chrome plate on the pin [25].

(c) Install the pin [25], washer [26], and nut [28].

I (d) Tighten the nut [28] to 95 in-lb (10.7 N·m) - 115 in-lb (13.0 N·m).

1) If it is necessary, loosen the nut [28] to align the nearest castellation with the hole in the pin [25].

(e) Apply a thick layer of corrosion inhibiting material, G50136, to the shank and threads of the bolt [27].

(f) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the nut [29], and the new pin [30].

(g) Install thebolt [27] and nut [29].

(h) Torque the nut [29] to 18 in-lb (2.0 N·m) - 25 in-lb (2.8 N·m) above the run-on torque.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If it is necessary, loosen the nut [29] to align the nearest castellation with the hole in the bolt [27].

(i) Install the new pin [30].

SUBTASK 32-33-51-420-003

(4) Do the steps that follow to connect the aft lock link assembly [3] of the lock mechanism to the rod end of the lock actuator (Figure 401, View C):

(a) Apply grease, D00633, to the bore of the rod end and aft lock link assembly [3], and the pin [22].

(b) Install the pin [22], washer [20], washer [21], washers [23], and nut [19].

(c) Torque the nut [19] to 18 ft-lb (24.4 N·m) - 24 ft-lb (32.5 N·m).

1) If it is necessary, loosen the nut [19] to align the nearest castellation with the hole on the pin [22].

(d) Install the new pin [18].

SUBTASK 32-33-51-420-004

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Do these steps to connect the spring assembly [2] to the forward lock link assembly [1] and the wheel well fitting (Figure 401, Views B, D):

(a) Remove the washers [5] and nuts [9] that attach the spring assembly [2] to the forward lock link assembly [1].

(b) Remove the spring assembly [2], bolts [10], washers [11], and sleeves [12] from the forward lock link assembly [1].

(c) Apply a thick layer of corrosion inhibiting material, G50136, to the shank and threads of the bolts [10].

(d) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the nuts [9], the faces of the washers [5] and sleeves [12], and the new pins [8].

(e) Apply grease, D00633, to the chrome plated surfaces of the sleeves [12] and the bolts [10].

(f) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread reliefs of the nuts [4], the faces of the washers [5], and the pins [8].

(g) Apply grease, D00633, to the chrome plated surfaces of the shaft [6].

(h) Use the expander set, SPL-1866, to extend the spring assembly [2].

(i) Connect the spring assembly [2] to the forward lock link assembly [1] (Figure 401, View D):

1) Put the spring assembly [2] in its location at the forward lock link assembly [1].

2) Install the bolts [10], sleeves [12], and the washers [11] to connect the spring assembly [2] to the forward lock link assembly [1].

3) Install the washers [5] and nuts [9] on the bolts [10].

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4) Torque the nuts [9] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) above the run-on torque.
   a) Install the new pins [8].

(j) Connect the spring assembly [2] to the shaft [6] in the wheel well fitting (Figure 401, View B):
1) Install the shaft [6] in the wheel well fitting.
2) Put the spring assembly [2] in its location on the shaft [6].
3) Install the washers [5] and nuts [4] on the shaft [6].
4) Torque the nuts [4] to 20 in-lb (2.3 N·m) - 24 in-lb (2.7 N·m) above the run-on torque
   a) Install the new pins [8].

(k) Remove the expander set, SPL-1866, from spring assembly [2].

SUBTASK 32-33-51-420-005

(6) Do the steps that follow to install the lock sensors [14] on the aft lock link assembly [3] (Figure 401, View E):
(a) Put the lock sensors [14] on the aft lock link assembly [3].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread reliefs of the screws [16], and the faces of the washers [15].
(c) Install the screws [16], washers [15], and nuts [13] on the lock sensor [14].
(d) Install the screws [17] to attach clamps for the wire bundles to the aft lock link assembly [3].

SUBTASK 32-33-51-720-001

(7) Do a post installation check of the nose gear lock sensors [14] (TASK 32-61-41-400-801).

G. Nose Gear Lock Mechanism Installation Test

SUBTASK 32-33-51-480-005

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-33-51-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

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SUBTASK 32-33-51-580-002

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE AREA AROUND THE NOSE GEAR. THE MOVEMENT OF THE NOSE GEAR CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Extend and retract the nose landing gear several times.

NOTE: Examine the nose gear and lock mechanism for the correct operation.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-51-480-003

(1) Extend the nose landing gear and install the ground lock assembly (TASK 32-00-01-480-801).

SUBTASK 32-33-51-860-003

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-51-580-003

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(3) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR LOCK SPRING - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Nose Gear Lock Spring Removal

(2) Nose Gear Lock Spring Installation.

TASK 32-33-52-000-801

2. Nose Gear Lock Spring Removal

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-52-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-52-860-001

(2) For the hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

E. Nose Gear Lock Spring Removal

SUBTASK 32-33-52-020-001

(1) Remove the pin [1], nut [7], and washer [3] from the shaft where the lock spring [6] is attached to the aft lock link and the wheel well fitting.

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SUBTASK 32-33-52-480-002

(2) Use the expander set, SPL-1866, to extend the lock spring [6].

SUBTASK 32-33-52-020-002

(3) Remove the pin [1], nut [2], and washer [3] from the bolt [4] that holds the lock spring [6] to the forward lock link.

SUBTASK 32-33-52-020-003

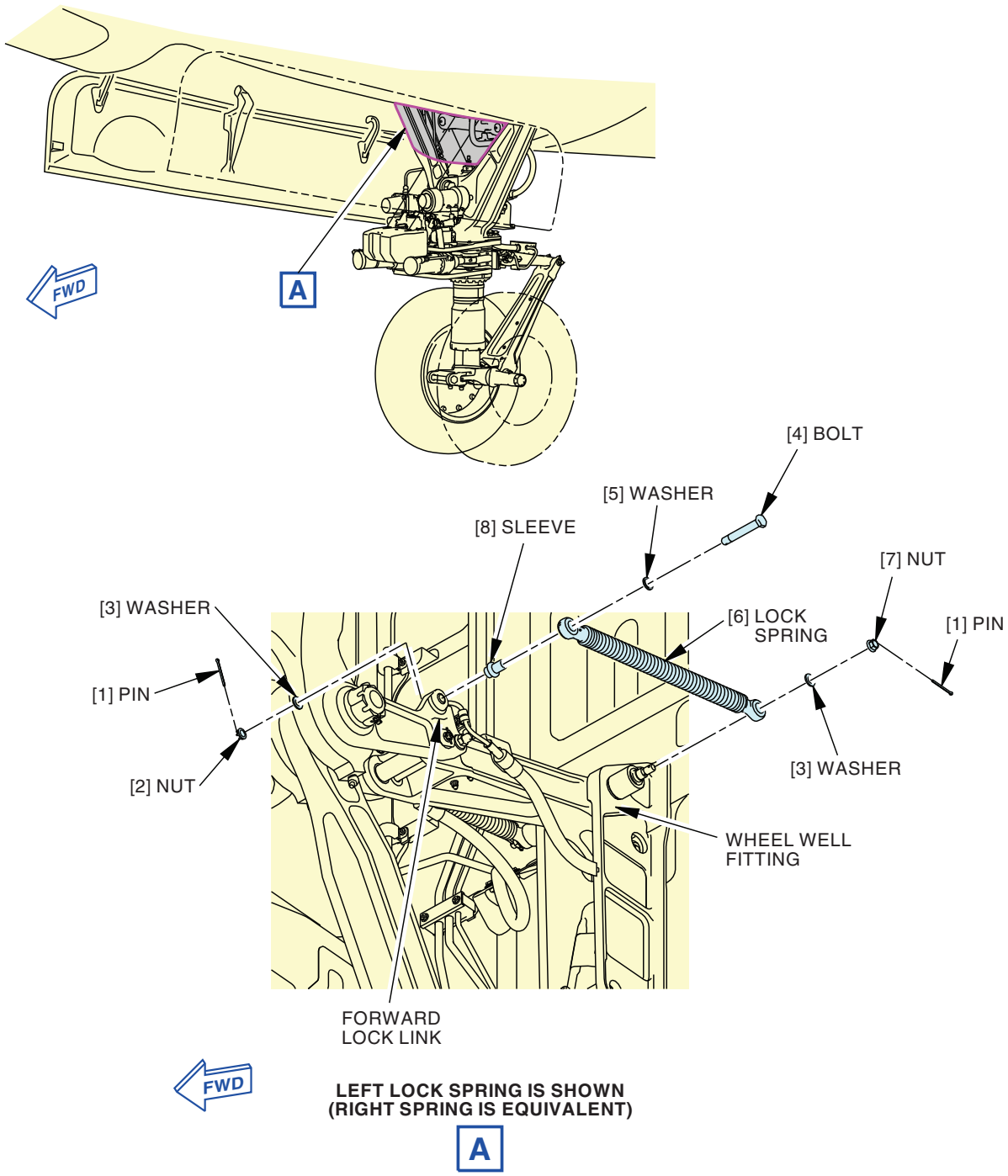
(4) Remove the lock spring [6], bolt [4], washer [5], and sleeve [8].

SUBTASK 32-33-52-020-004

(5) Remove the lock spring [6].

——— END OF TASK ———

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Nose Gear Lock Spring Installation
Figure 401/32-33-52-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-52-400-801

3. Nose Gear Lock Spring Installation

A. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1866	Expander Set - Spring, Main Landing Gear and Nose Landing Gear Part #: C32014-29 Supplier: 81205 Opt Part #: C32014-1 Supplier: 81205 Opt Part #: C32014-20 Supplier: 81205

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pin	32-33-51-01-005	PTF ALL
6	Lock spring	32-33-51-01-030	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Nose Gear Lock Spring Installation Procedure

SUBTASK 32-33-52-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thick layer of corrosion inhibiting material, G50136, to the shank and threads of the bolt [4].

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SUBTASK 32-33-52-600-005

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply a thin layer of corrosion inhibiting material, G50136, to the threads of the nut [2], faces of the washer [5] and sleeve [8], and the new pin [1].

SUBTASK 32-33-52-600-002

(3) Apply grease, D00633, to the chrome plated surfaces of the sleeve [8] and bolt [4].

SUBTASK 32-33-52-600-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(4) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread reliefs of the nut [7], the faces of the washer [3], and the new pin [1].

SUBTASK 32-33-52-600-004

(5) Apply grease, D00633, to the chrome plated surface of the shaft on the wheel well fitting.

SUBTASK 32-33-52-480-003

(6) Use the expander set, SPL-1866, to extend the lock spring [6].

SUBTASK 32-33-52-420-001

(7) Do the steps that follow to connect the lock spring [6] to the shaft on the wheel well fitting.

(a) Put the lock spring [6] in its location on the shaft on the wheel well fitting.

(b) Install the washer [3] and nut [7] to attach the lock spring [6] to the wheel well fitting and aft lock link.

1) Tighten the nut [7] to 22 ±2 in-lb (2.5 ±0.2 N·m) above the run-on torque.

(c) Install the pin [1].

SUBTASK 32-33-52-420-002

(8) Do the steps that follow to connect the lock spring [6] to the forward lock link:

(a) Put the lock spring [6] in its location on the forward lock link.

(b) Install bolt [4], washer [5], and the sleeve [8] to connect the lock spring [6] to the forward lock link.

(c) Install the washer [3] and nut [2] on the bolt [4].

1) Tighten the nut [2] to 22 ±2 in-lb (2.5 ±0.2 N·m) above the run-on torque.

(d) Install the pin [1].

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SUBTASK 32-33-52-080-001

(9) Remove the expander set, SPL-1866, from the lock spring [6].

——— END OF TASK ———

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NOSE GEAR WHEEL WELL DOOR LINKAGE - REMOVAL/INSTALLATION

1. General

A. This procedure contains the following tasks:

(1) A removal Nose Gear Wheel Well Door Linkage

(2) An installation Nose Gear Wheel Well Door Linkage.

B. This linkage, called the door mechanism assembly, has these parts:

(1) Upper Pushrod Assembly

(2) Lower Pushrod Assembly

(3) Bellcrank Assembly.

TASK 32-33-61-000-801

2. Nose Gear Wheel Well Door Linkage - Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Prepare for the Removal

SUBTASK 32-33-61-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-61-580-001

(2) Install chocks around the tires of the nose landing gear (PAGEBLOCK 10-11-05/201).

D. Nose Gear Wheel Well Door Linkage Removal

SUBTASK 32-33-61-020-001

(1) Remove the pin [8], nut [7], washer [9], and bolt [5] to disconnect the universal [10] on the lower pushrod assembly [3] from the door hinge (View B, Figure 401).

SUBTASK 32-33-61-020-002

(2) Remove the pin [16], nut [14], washer [15], bolt [11], washer [12], and bushing [13] to disconnect the lower pushrod assembly [3] from the bellcrank assembly [4] (View C, Figure 401).

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SUBTASK 32-33-61-020-003

(3) Remove the pin [16], nut [20], washer [21], bolt [19], washer [18], and bushing [17] to disconnect the upper pushrod assembly [2] from the trunnion clevis (View D, Figure 401).

SUBTASK 32-33-61-020-004

(4) Remove the pin [16], nut [20], washer [21], bolt [19], washer [18], and bushing [17] to disconnect the upper pushrod assembly [2] from the bellcrank assembly [4] (View E, Figure 401).

SUBTASK 32-33-61-020-005

(5) Do these steps to remove the bellcrank assembly [4] (View G, Figure 401):

(a) Remove the lockwire from the pin [24] at the inboard end of the door operator bolt [23] and remove the pin [24].

(b) Remove the nut [25] from the door operator bolt [23].

(c) Remove the bellcrank assembly [4] from the door operator bolt [23].

SUBTASK 32-33-61-020-006

(6) Do these steps to remove the door operator bolt [23]:

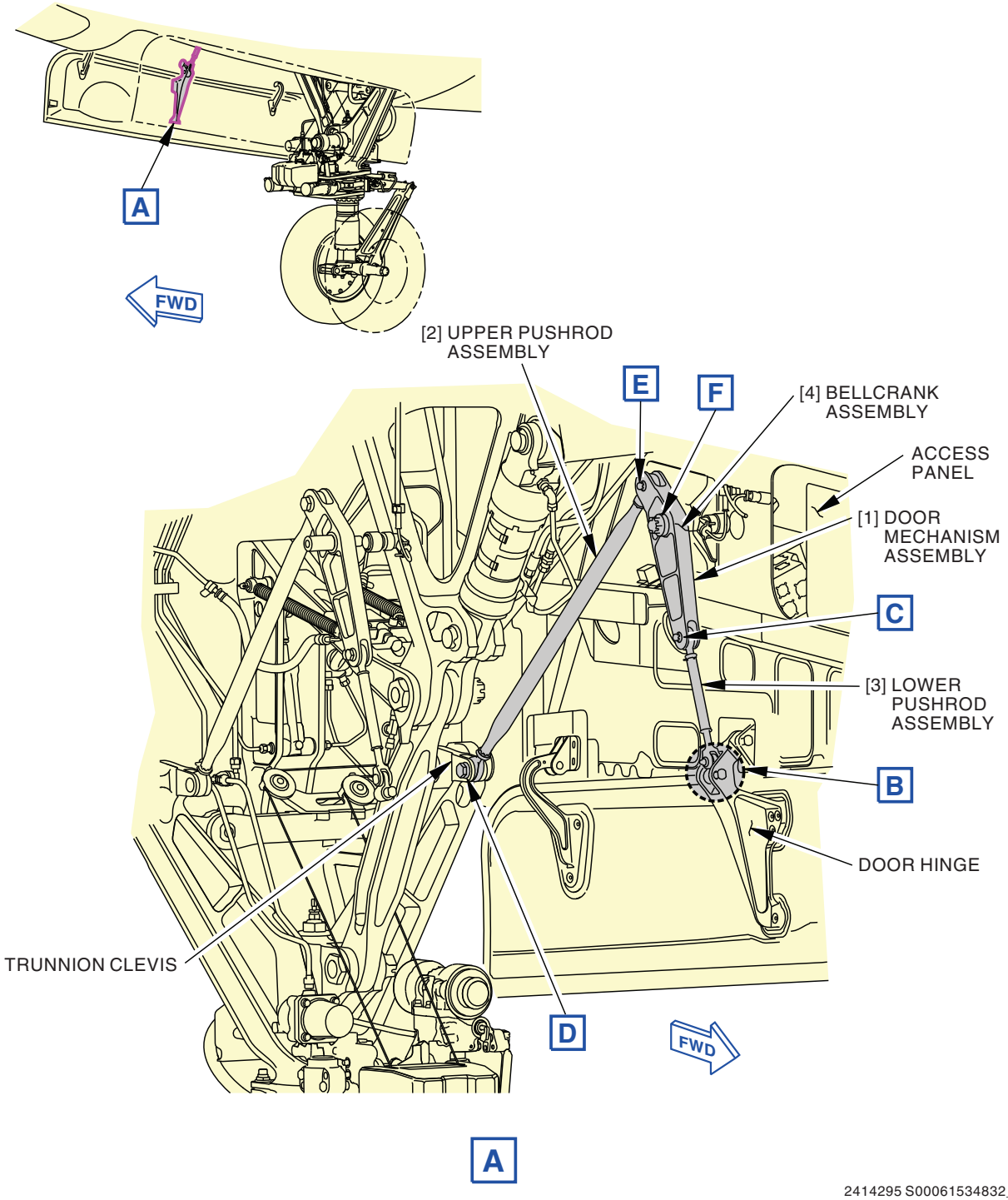
(a) Remove the aft access panel in the nose gear wheel well from the side on which you will remove the wheel well door linkage.

(b) Remove the nut [26], washer [27], and cross bolt [22] at the outboard end of the door operator bolt [23].

(c) Remove the door operator bolt [23] from the bulkhead fitting.

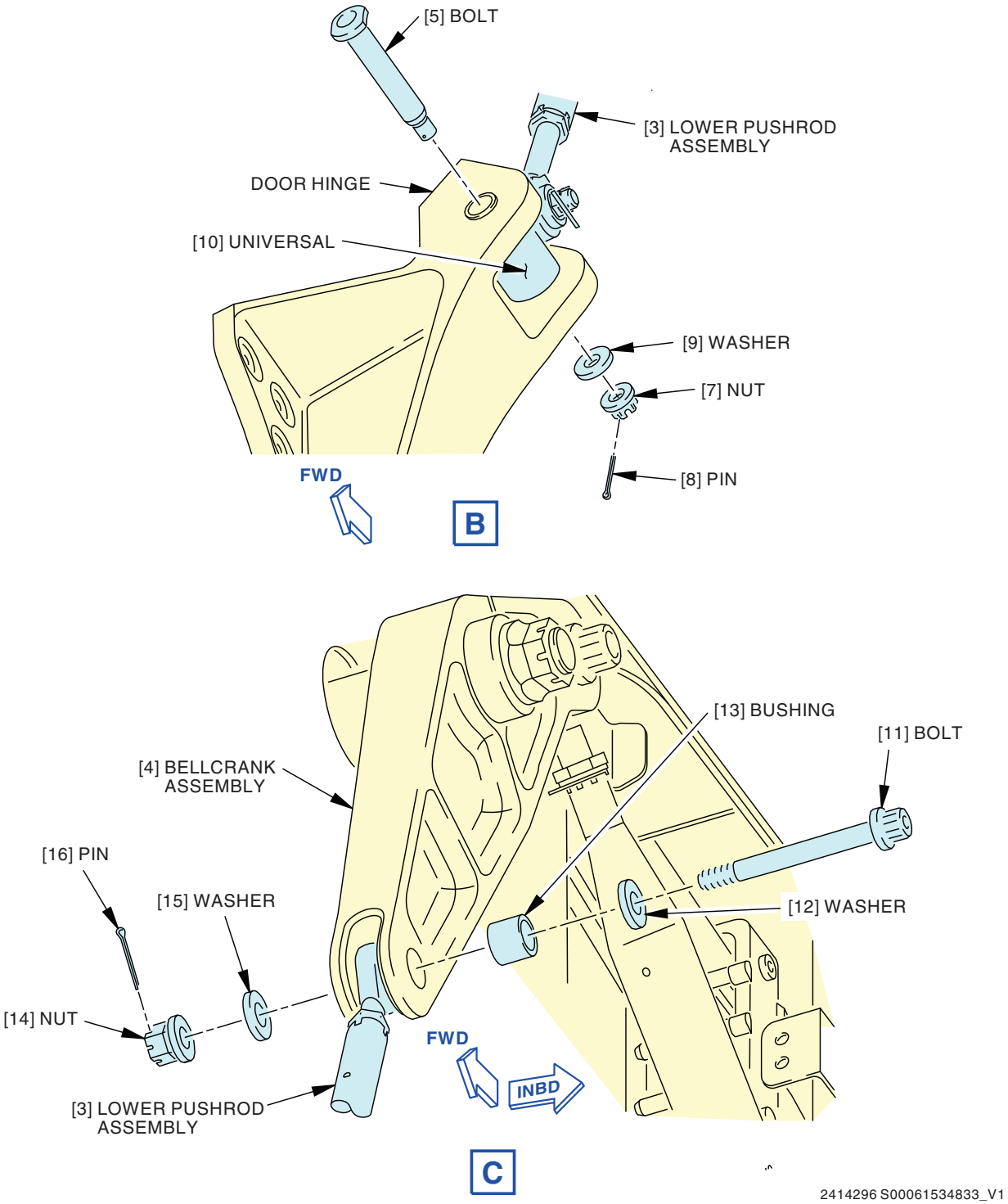
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



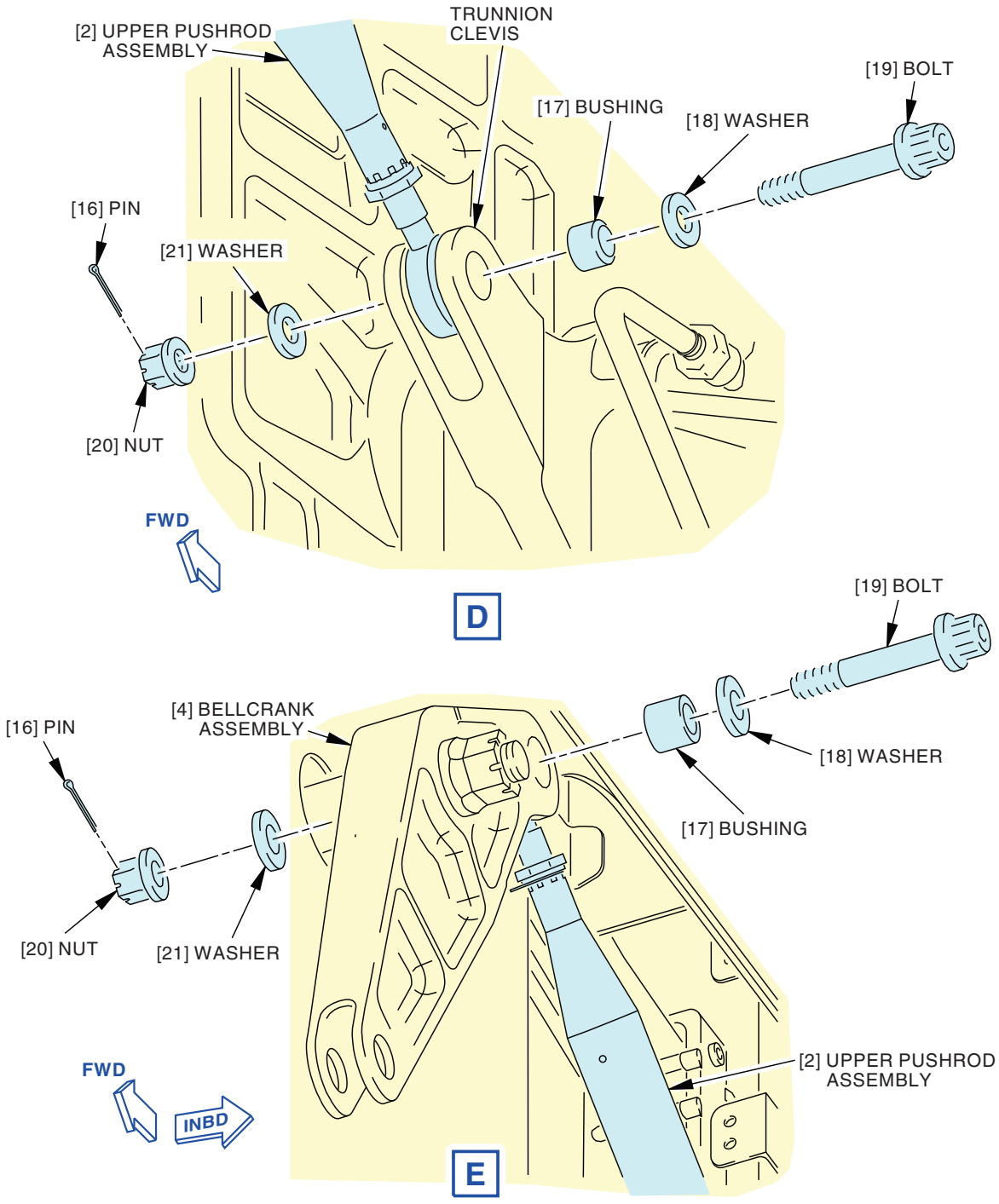
Nose Gear Wheel Well Door Linkage Installation
Figure 401/32-33-61-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



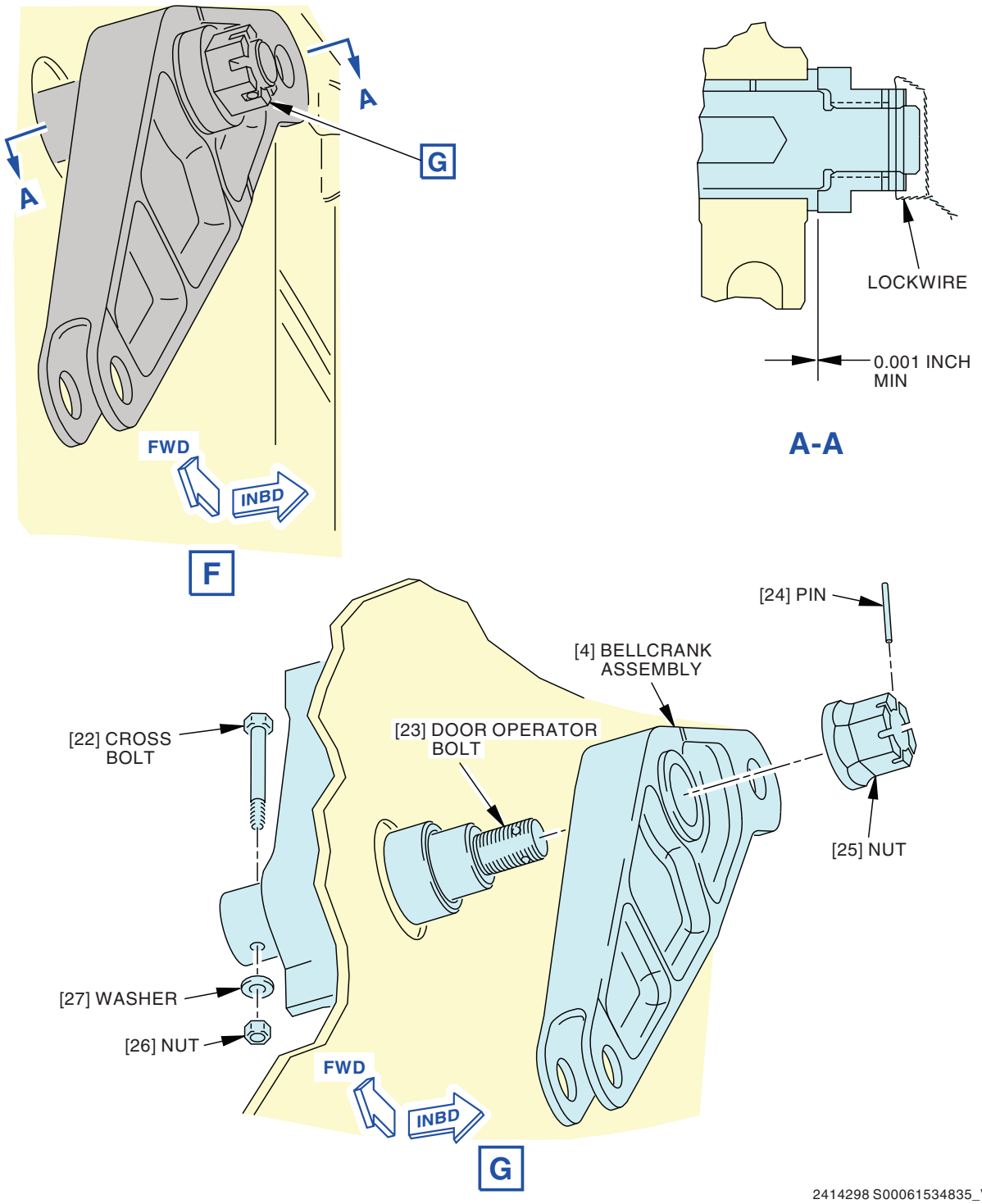
Nose Gear Wheel Well Door Linkage Installation
Figure 401/32-33-61-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Wheel Well Door Linkage Installation
Figure 401/32-33-61-990-801 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Wheel Well Door Linkage Installation
Figure 401/32-33-61-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-61-400-801

3. Nose Gear Wheel Well Door Linkage - Installation

(Figure 401)

A. References

Reference	Title
32-33-61-820-801	Nose Gear Wheel Well Door Linkage Adjustment and Check (P/B 501)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Upper pushrod assembly	32-33-61-01-040	PTF ALL
3	Lower pushrod assembly	32-33-61-01-180	PTF ALL
4	Bellcrank assembly	32-33-61-01-110	PTF ALL
8	Pin	32-33-61-01-160	PTF ALL
16	Pin	32-22-11-03-005	PTF ALL
		32-33-61-01-010	PTF ALL
		32-33-61-01-080	PTF ALL
24	Pin	32-22-11-03-050	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Nose Gear Wheel Well Door Linkage Installation

SUBTASK 32-33-61-620-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(1) Apply a thin layer of corrosion inhibiting material, G50136, to the thread relief and threads of the door operator bolt [23] and the nut [25] (View G, Figure 401).

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SUBTASK 32-33-61-640-001

(2) Apply a layer of grease, D00633, to the chrome plated surfaces of the door operator bolt [23].

SUBTASK 32-33-61-930-001

(3) Put a reference line on the unthreaded end of the door operator bolt [23] to align with the center of the hole for the cross bolt [22].

SUBTASK 32-33-61-420-001

(4) Install the door operator bolt [23] from the inboard side.

NOTE: Make sure that the reference line for the center of the door operator bolt [23] is aligned with the bolt hole in the bulkhead fitting.

SUBTASK 32-33-61-600-001

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Put a thick layer of corrosion inhibiting material, G50136, to the shank and threads of the cross bolt [22], and on the nut [26] and the face of the washer [27].

SUBTASK 32-33-61-420-002

(6) Align the bolt holes in the door operator bolt [23] and the bulkhead fitting and install the cross bolt [22].

(a) Remove the unwanted corrosion inhibiting material, G50136, from the cross bolt [22] after you install it.

SUBTASK 32-33-61-420-003

(7) Install the washer [27] and nut [26] on the cross bolt [22].

SUBTASK 32-33-61-420-004

(8) Install the bellcrank assembly [4] on the door operator bolt [23].

SUBTASK 32-33-61-420-005

(9) Install the nut [25] and tighten until a gap of 0.001 in. (0.025 mm) or more is measured between the nut [25] and the bushing in the bellcrank assembly [4].

SUBTASK 32-33-61-420-006

(10) Loosen the nut [25] only enough to align the hole in the door operator bolt [23] with the castellations in the nut [25].

SUBTASK 32-33-61-420-007

(11) Install the pin [24] to lock the nut [25].

SUBTASK 32-33-61-410-001

(12) Install the MS20995NC32 lockwire, G50347, on the pin [24].

SUBTASK 32-33-61-100-001

(13) Remove the excess corrosion inhibiting material, G50136, from the bellcrank assembly [4], door operator bolt [23], and nut [25].

SUBTASK 32-33-61-420-008

(14) Do these steps to connect the upper pushrod assembly [2] to the bellcrank assembly [4] (View E, Figure 401):

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(a) Apply grease, D00633, to the chrome plated surfaces of the bolt [19] and bushing [17].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the threads on the bolt [19] and nut [20], and the faces of the washer [18] and washer [21].

(c) Install the bolt [19], washer [18], bushing [17], washer [21], nut [20], and a new pin [16] to connect the upper pushrod assembly [2] to the bellcrank assembly [4].

SUBTASK 32-33-61-420-009

(15) Do these steps to connect the lower pushrod assembly [3] to the bellcrank assembly [4] (View C, Figure 401):

(a) Apply grease, D00633, to the chrome plated surfaces of the bolt [11] and the bushing [13].

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the threads on the bolt [11] and nut [14], and the faces of the washer [12] and washer [15].

(c) Install the bolt [11], washer [12], bushing [13], washer [15], nut [14], and a new pin [16] to connect the lower pushrod assembly [3] to the bellcrank assembly [4].

SUBTASK 32-33-61-620-002

(16) Do these steps to connect the upper pushrod assembly [2] to the trunnion clevis (View D, Figure 401):

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Put a thin layer of corrosion inhibiting material, G50136, to the threads and thread relief of the bolt [19] and the nut [20] and on the faces of the washer [18] and washer [21].

(b) Apply a layer of grease, D00633, to the wear surfaces of the bolt [19] and the bushing [17].

(c) Install the bolt [19], washer [18], bushing [17], washer [21], and nut [20] to connect the upper pushrod assembly [2] to the trunnion clevis.

(d) Torque the nut [20] to 325 in-lb (37 N·m) - 360 in-lb (41 N·m).

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(e) If necessary, loosen the nut [20] to the nearest castellation to align the holes for the pin [16].

(f) Install the new pin [16] in the nut [20].

SUBTASK 32-33-61-820-001

(17) Do the adjustment of the nose gear wheel well door linkage (TASK 32-33-61-820-801).

SUBTASK 32-33-61-420-010

(18) Do these steps to connect the lower pushrod assembly [3] to the door hinge (View B, Figure 401):

(a) Apply grease, D00633, to the chrome plated surfaces of the bolt [5].

WARNING

DO NOT GET THE CORROSION-INHIBITING COMPOUND ON YOUR SKIN, IN YOUR EYES, OR ON YOUR CLOTHES. THE COMPOUND IS POISONOUS AND FLAMMABLE. USE APPROVED GLOVES, AND EYE PROTECTION. MAKE SURE THAT THERE IS GOOD AIRFLOW IN THE WORK AREA. KEEP THE COMPOUND AWAY FROM SPARKS AND FLAMES. THE COMPOUND CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) Apply a thin layer of corrosion inhibiting material, G50136, to the threads and thread relief of the bolt [5] and the nut [7] and on the face of the washer [9].

(c) Install the bolt [5], washer [9], nut [7] , and a new pin [8] to connect the lower pushrod assembly [3] to the door hinge.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-61-480-005

(1) Remove the chocks from around the landing gear tires.

——— END OF TASK ———

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NOSE GEAR WHEEL WELL DOOR LINKAGE - ADJUSTMENT/TEST

1. General

A. This procedure includes the instructions to adjust the linkage for the nose gear wheel well doors.

TASK 32-33-61-820-801

2. Nose Gear Wheel Well Door Linkage Adjustment and Check

(Figure 501)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-755	Scale - Spring, Push/Pull Type, 0 - 150lbs (0 - 68kg) Capacity

C. Consumable Materials

Reference	Description	Specification
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Pin	32-33-61-01-160	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Prepare for the Adjustment

NOTE: A NLG door rigging check is not necessary if reinstalling the same NLG door that was removed and no changes to the door stops, door seals, door hinges, or door pushrods were made.

NOTE: The door seal installed incorrectly could prevent the wheel well retract to the normal close position.

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SUBTASK 32-33-61-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-61-580-002

(2) Lift the airplane on jacks (Lift the Airplane on Jacks, TASK 07-10-01-580-801).

SUBTASK 32-33-61-860-001

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

SUBTASK 32-33-61-080-003

(4) Remove the downlock pins from the main landing gear (Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

SUBTASK 32-33-61-860-008

(5) Retract the main landing gear.

SUBTASK 32-33-61-020-007

(6) Remove the lockwire from the jamnuts [5] on each end of the lower rod assembly.

SUBTASK 32-33-61-020-008

(7) Loosen the jamnut [5] on each end of the lower rod assemblies.

(a) Loosen the jamnut [5] only if you will turn the lower rod assembly, for adjustment.

NOTE: If you loosen the jamnut too much, it will disengage the rod end lock from the rod barrel.

(b) Do not adjust the length of upper rod assembly.

NOTE: Adjustment of the upper rod assembly can cause problems to the gear retraction or extension.

SUBTASK 32-33-61-020-009

(8) If the lower rod assemblies are not disconnected from the door hinges already, remove the pins [4], nuts [3], washers [2], and bolts [1] to disconnect the universal joints [6] on the lower rod assemblies from the hinges.

(a) Discard the pin [4].

SUBTASK 32-33-61-210-001

(9) Make sure that the tamper proof seals are on the upper rod assembly.

G. Nose Gear Wheel Well Door Linkage Adjustment

SUBTASK 32-33-61-080-001

(1) Remove the downlock pin from the nose landing gear (TASK 32-00-01-080-801).

SUBTASK 32-33-61-870-001

(2) Retract the nose landing gear.

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SUBTASK 32-33-61-040-001

WARNING BEFORE YOU DO AN ADJUSTMENT TASK, REMOVE THE HYDRAULIC PRESSURE FROM THE RETRACT ACTUATOR FOR THE NOSE LANDING GEAR. IF THERE IS HYDRAULIC PRESSURE, THE NOSE LANDING GEAR COULD MOVE AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Remove the power from hydraulic system A (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

SUBTASK 32-33-61-600-002

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(4) Apply a thin layer of corrosion inhibiting material, G50136, to these items:

(a) The threads and thread reliefs of the bolt [1], nut [3] and the faces of the washer [2].

SUBTASK 32-33-61-420-011

CAUTION MAKE SURE THAT THE HEAD OF THE BOLT THAT ATTACHES THE UNIVERSAL JOINT TO THE LOWER ROD ASSEMBLY IS OUTBOARD. IF YOU DO NOT OBEY, THERE WILL NOT BE SUFFICIENT CLEARANCE WITH THE DOOR HINGE AND DAMAGE CAN OCCUR.

(5) Adjust one door at a time, do these steps:

(a) Install the bolt [1], washer [2], and nut [3] to connect the lower rod assembly to the door hinge.

1) Adjust the length of the lower rod assembly to permit a free fit of the bolt [1] with no preload on the door.

NOTE: To get access, make sure that the lower rod assembly on the opposite door is disconnected.

NOTE: If it is necessary to adjust the opposite door, disconnect the first door after its adjustment.

SUBTASK 32-33-61-820-002

(6) Shorten the length of the lower rod assembly until the door touches the forward stop fitting.

SUBTASK 32-33-61-700-001

WARNING MAKE SURE THE SPRING SCALE IS SECURELY ATTACHED TO THE DOOR. THE SCALE COULD SLIP OFF THE DOOR AND CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(7) Use the spring scale, STD-755, to apply a 60 lb (27 kg) - 90 lb (41 kg) down load approximately 1.0 in. (25.4 mm) aft and 1.0 in. (25.4 mm) outboard of the forward inboard corner of the door or a 85 lb (39 kg) - 105 lb (48 kg) load at the forward stop fitting on the door with the nose landing gear retracted.

(a) Connect the scale to the inside of the landing gear door at the correct location.

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(b) Pull down on the scale to apply the correct load.

(c) Keep this load applied while you take the door measurements.

SUBTASK 32-33-61-820-003

(8) If the measured dimension of the door below the body contour at the forward inboard corner on the door is not 0.09 in. (2.29 mm) - 0.13 in. (3.30 mm), do these steps to adjust the lower rod assembly to get a 0.09 in. (2.29 mm) - 0.13 in. (3.30 mm) dimension:

NOTE: The lower rod assembly is adjustable as a turnbuckle.

(a) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

(b) Extend the nose landing gear slowly.

WARNING

BEFORE YOU DO AN ADJUSTMENT TASK, REMOVE THE HYDRAULIC PRESSURE FROM THE RETRACT ACTUATOR FOR THE NOSE LANDING GEAR. IF THERE IS HYDRAULIC PRESSURE, THE NOSE LANDING GEAR COULD MOVE AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(c) Remove the power from hydraulic system A (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

(d) Decrease the length of the lower rod assembly, one thread at a time, and by a maximum dimension of 0.40 in. (10.16 mm).

NOTE: This dimension will be from the initial length where the door touched the forward stop fitting.

NOTE: One half turn of the barrel or one turn of the clevis rod end will give one thread of adjustment.

(e) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

(f) Retract the nose landing gear.

(g) Remove the power from hydraulic system A (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

(h) Measure the dimension of the door below the body contour to get a dimension of 0.09 in. (2.29 mm) - 0.13 in. (3.30 mm).

(i) If it is necessary, do these steps again, until you get the correct dimension.

SUBTASK 32-33-61-080-002

(9) Remove the weight and spring scale, STD-755, from the door.

SUBTASK 32-33-61-860-009

(10) For hydraulic system A, do this task: Hydraulic System A or B Pressurization with a Portable Hydraulic Cart, TASK 29-11-00-860-802.

SUBTASK 32-33-61-860-002

(11) Extend the nose landing gear.

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SUBTASK 32-33-61-040-002

WARNING BEFORE YOU DO AN ADJUSTMENT TASK, REMOVE THE HYDRAULIC PRESSURE FROM THE RETRACT ACTUATOR FOR THE NOSE LANDING GEAR. IF THERE IS HYDRAULIC PRESSURE, THE NOSE LANDING GEAR COULD MOVE AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(12) Remove the power from hydraulic system A (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

SUBTASK 32-33-61-420-012

(13) Tighten the jamnut [5] on each end of the lower rod assembly.

(a) If the 10.47 in. (266 mm) dimension is met and threads are visible through the witness holes, then the adjustment values are reached.

NOTE: The threads of the rods ends do not need to cover the witness holes.

SUBTASK 32-33-61-420-013

(14) Install the lockwire on the jamnuts [5] on each end of the lower rod assembly.

SUBTASK 32-33-61-020-010

(15) If adjustment for the other door is necessary, do these steps:

(a) Disconnect the door that you adjusted to get access.

1) Remove the nut [3], washer [2], and bolt [1] to disconnect the lower rod assembly from the door hinge.

(b) Do the adjustment for the other door.

SUBTASK 32-33-61-420-014

CAUTION MAKE SURE THAT THE HEAD OF THE BOLT THAT ATTACHES THE UNIVERSAL JOINT TO THE LOWER ROD ASSEMBLY IS OUTBOARD. IF YOU DO NOT OBEY, THERE WILL NOT BE SUFFICIENT CLEARANCE WITH THE DOOR HINGE AND DAMAGE CAN OCCUR.

(16) Connect each door as necessary, do these steps:

(a) Install the bolt [1], washer [2], and nut [3] to connect the lower rod assembly to the door hinge, if not done.

(b) Tighten the nut [3] to 60 in-lb (6.8 N·m) - 90 in-lb (10.2 N·m).

(c) Align the nut [3] back to the nearest pin hole.

(d) Install the new pin [4] to the nut [3].

H. Check of the Door Operation

SUBTASK 32-33-61-860-003

(1) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-61-710-001

(2) Retract and extend the nose landing gear and look for correct operation of the nose gear doors.

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SUBTASK 32-33-61-860-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(3) Extend the nose and main landing gear.

SUBTASK 32-33-61-480-003

(4) Install the downlock pins to the nose and main landing gear (Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801).

SUBTASK 32-33-61-580-003

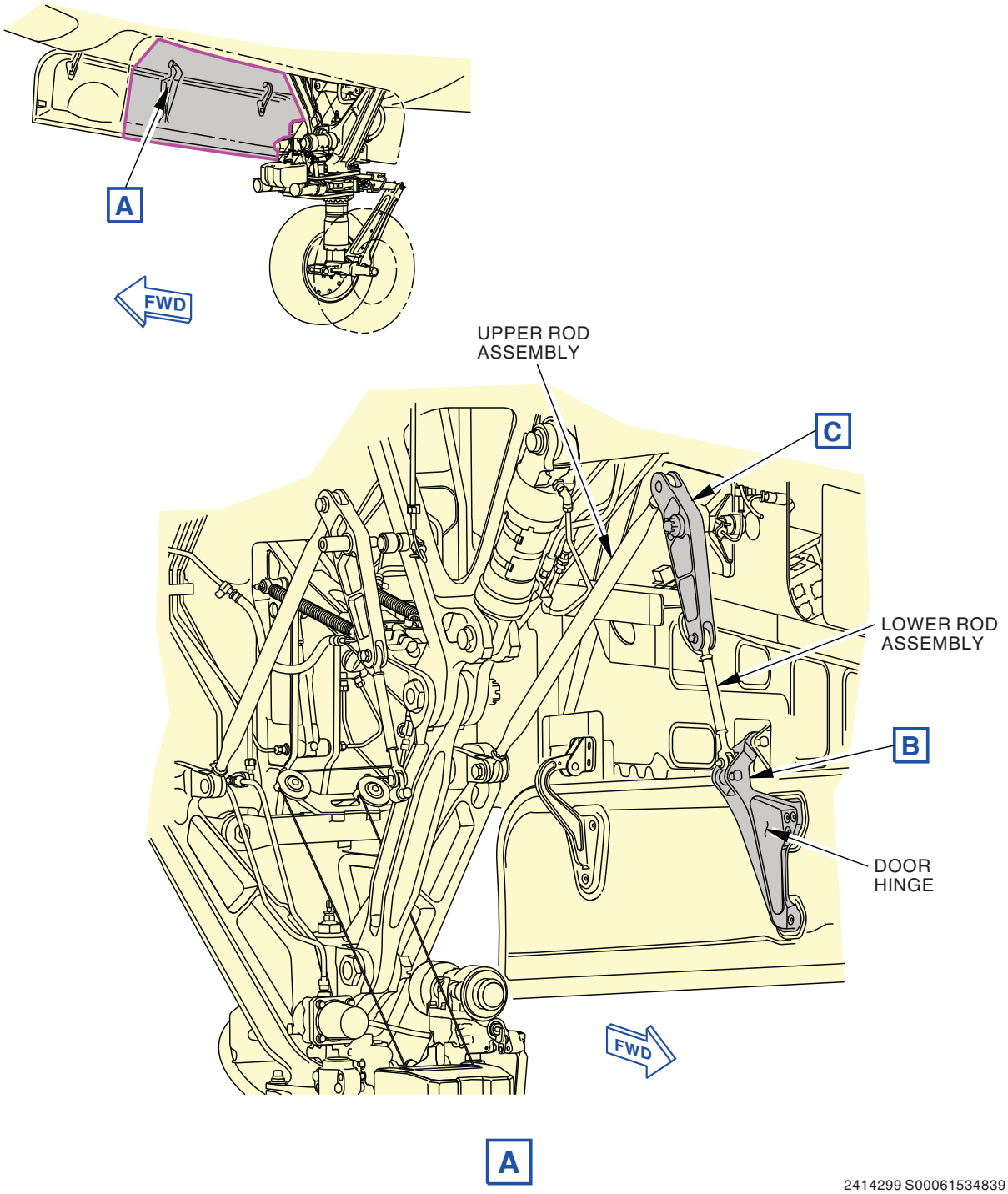
(5) Lower the airplane off the jacks (Lower the Airplane off Jacks, TASK 07-10-01-580-802).

SUBTASK 32-33-61-860-005

(6) If hydraulic power is not necessary at this time, remove the power from hydraulic system A (Hydraulic System A or B Power Removal, TASK 29-11-00-860-805).

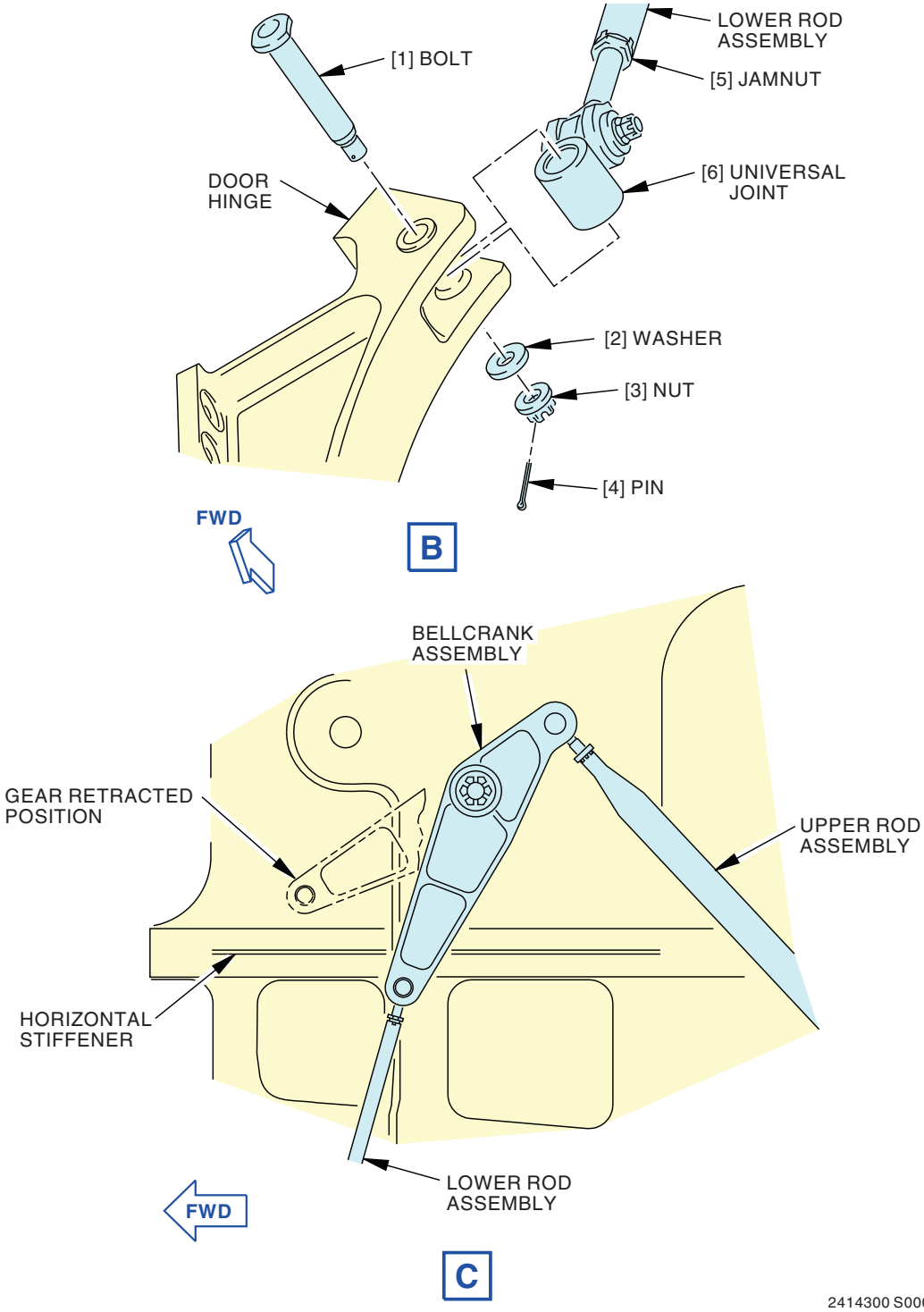
——— END OF TASK ———

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Nose Gear Wheel Well Door Linkage Adjustment
Figure 501/32-33-61-990-802 (Sheet 1 of 2)

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Nose Gear Wheel Well Door Linkage Adjustment
Figure 501/32-33-61-990-802 (Sheet 2 of 2)

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NOSE GEAR WHEEL WELL DOOR LINKAGE - INSPECTION/CHECK

1. General

A. This task has the data for a dimensional inspection of the linkage for the wheel well doors on the nose landing gear. No procedure is given in this section to get access to permit the inspection. The door linkage must be removed to do the inspection.

These are the tasks:

Nose Gear Wheel Well Door Linkage - Removal, TASK 32-33-61-000-801,

Nose Gear Wheel Well Door Linkage - Installation, TASK 32-33-61-400-801.

TASK 32-33-61-200-801

2. Nose Gear Wheel Well Door Linkage Inspection

(Figure 601)

A. General

(1) This task gives wear limits and repair data for the linkage for the wheel well doors on the nose landing gear.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-33-61-000-801	Nose Gear Wheel Well Door Linkage - Removal (P/B 401)
32-33-61-400-801	Nose Gear Wheel Well Door Linkage - Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-1096	Micrometer - Depth, 0-1 inch, Readable to 1/1000 inch
STD-1097	Caliper - Vernier, 0-6 inch, Readable to 1/1000 inch

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Nose Gear Wheel Well Door Linkage Inspection

SUBTASK 32-33-61-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-61-020-011

(2) Remove parts to get access to the linkage for the wheel well doors on the nose landing gear (Figure 601) (TASK 32-33-61-000-801).

SUBTASK 32-33-61-020-012

(3) Use these tools to do a check for wear on the parts of the linkage for the wheel well doors on the nose landing gear:

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(a) micrometer (0-1 inch, readable to 1/1000 inch), STD-1096

(b) vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

SUBTASK 32-33-61-020-013

(4) Examine the parts of the linkage for worn areas:

(a) Remove the parts to be checked.

(b) Measure the parts as necessary with a micrometer (0-1 inch, readable to 1/1000 inch), STD-1096, or a vernier caliper (0 - 6 inch, readable to 1/1000 inch), STD-1097.

(c) Compare the dimensions you measured with the permitted wear dimensions in Table 601, Table 602.

(d) If the parts are not in the tolerance, replace or repair the part as necessary.

Table 601/32-33-61-993-801

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
[1] BUSHING	I.D.	0.5000 in 12.7 mm	0.5007 in 112.717 mm	0.5031 in 12.778 mm	0.0041 in
[1] BOLT	O.D.	0.4980 in 12.649 mm	0.4990 in 12.674 mm	0.4966 in 12.613 mm	0.104 mm
[2] BUSHING	I.D.	0.5000 in 12.7 mm	0.5006 in 12.715 mm	0.5030 in 12.776 mm	0.004 in
[2] BOLT	O.D.	0.4980 in 12.649 mm	0.4990 in 12.674 mm	0.4966 in 12.613 mm	0.102 mm
[3] BEARING BALL	I.D.	in mm	in mm	in mm	in
[3] BOLT	O.D.	0.4935 in 12.534mm	0.4950 in 12.573mm	in mm	mm
[4] BUSHING	I.D.	0.5000 in 12.7mm	0.5005 in 12.712mm	0.5034 in 12.786mm	0.0084 in
[4] BOLT	O.D.	0.4935 in 12.534mm	0.4950 in 12.573mm	0.4921 in 12.499mm	00.213 mm
[5] BUSHING	I.D.	0.5000 in 12.7 mm	0.5005 in 12.712mm	0.5034 in 12.786mm	0.0084 in
[5] BOLT	O.D.	0.4935 in 12.534mm	0.4950 in 12.573mm	0.4921 in 12.499mm	0.213 mm

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Table 601/32-33-61-993-801 (Continued)

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
[6] BUSHING	I.D.	0.6870 in 17.499 mm	0.6875 in 17.462 mm	0.6897 in 17.518 mm	0.0647 in
[6] BUSHING	O.D.	0.5610 in 14.249 mm	0.5615 in 14.262 mm	0.5590 in 14.199 mm	1.643 mm
[7] BUSHING	I.D.	in mm	in mm	in mm	in
[7] BOLT	O.D.	1.122 in 28.50 mm	1.124 in 28.55 mm	in mm	mm
[8] BEARING BALL	I.D.	in mm	in mm	in mm	in
[8] BOLT	O.D.	0.6180 in 15.697 mm	0.6195 in 15.735 mm	in mm	mm

Table 602/32-33-61-993-802

NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
[9] BUSHING	I.D.	0.7502 in 19.055 mm	0.7507 in 19.067 mm	0.7529 in 19.123 mm	0.0029 in
[9] BUSHING	O.D.	0.7495 in 19.037 mm	0.7500 in 19.050 mm	0.7478 in 18.994 mm	0.073 mm
[10] BUSHING	I.D.	0.6250 in 15.875 mm	0.6255 in 15.887 mm	0.6286 in 15.966 mm	0.0091 in
[10] BOLT	O.D.	0.6180 in 15.697 mm	0.6195 in 15.735 mm	0.6164 in 15.656 mm	0.231 mm
[11] BUSHING	I.D.	0.6250 in 15.875 mm	0.6255 in 15.887 mm	0.6286 in 15.966 mm	0.0091 in
[11] BOLT	O.D.	0.6180 in 15.697 mm	0.6195 in 15.735 mm	0.6164 in 15.656 mm	0.231 mm

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Table 602/32-33-61-993-802 (Continued)

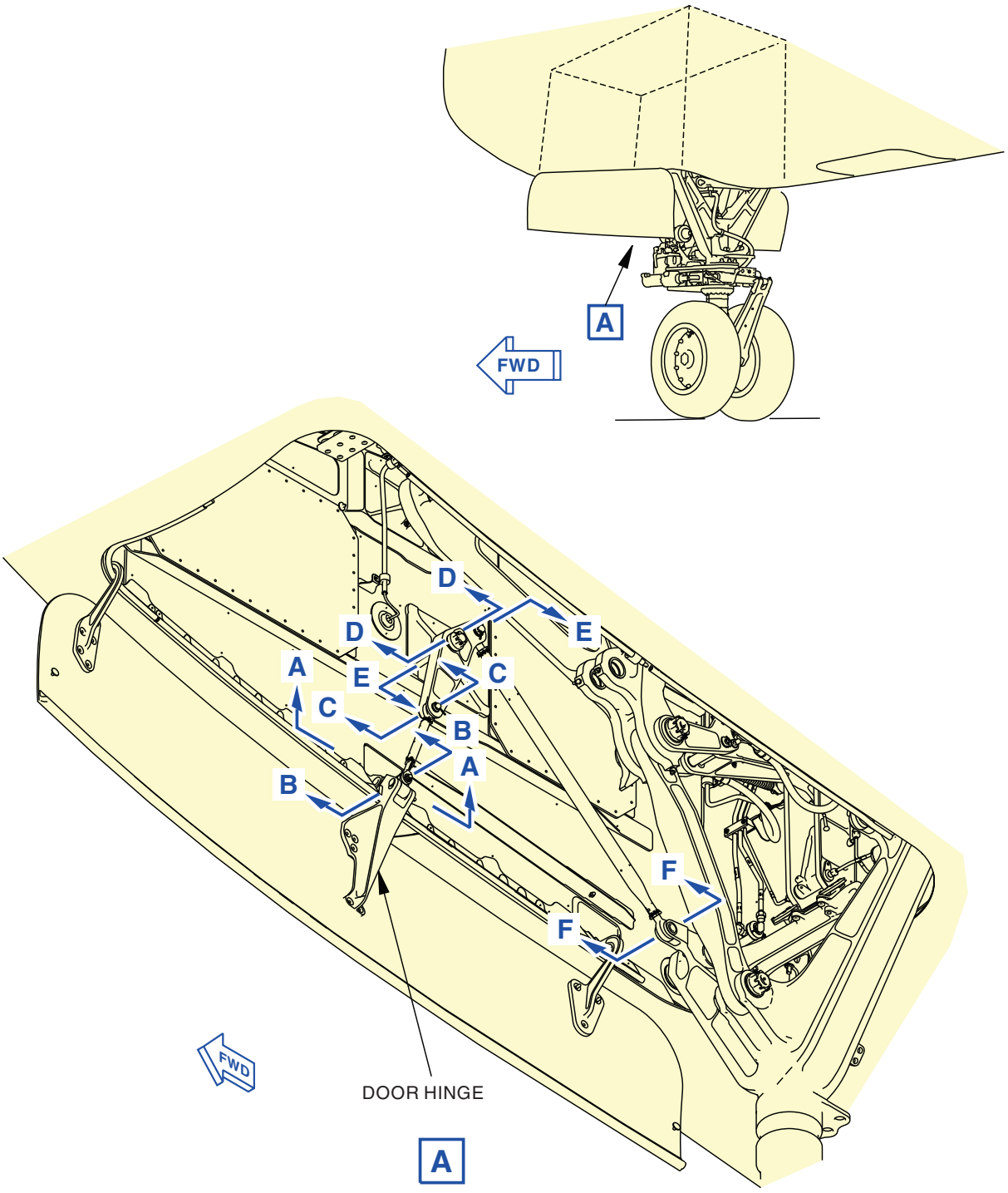
NAME OF THE PARTS THAT ARE IN CONTACT	DIM.	DIAMETER DESIGN LIMITS		WEAR LIMITS	
		MINIMUM inches/mm	MAXIMUM inches/mm	PERMITTED MAXIMUM I.D./ MINIMUM O.D. OF WORN PART	MAXIMUM SERVICE DIAMETRIC CLEARANCE
[12] BUSHING	I.D.	0.7514 in 19.085 mm	0.7519 in 19.098 mm	0.7533 in 19.133 mm	0.0032 in
[12] BUSHING	O.D.	0.7499 in 19.047 mm	0.7501 in 19.052 mm	0.7482 in 19.004 mm	0.081 mm
[13] BEARING BALL	I.D.	in mm	in mm	in mm	in
[13] BOLT	O.D.	0.6180 in 15.697 mm	0.6195 in 15.735 mm	in mm	mm

SUBTASK 32-33-61-020-014

(5) Install the parts on the linkage for the wheel well doors on the nose landing gear (TASK 32-33-61-400-801).

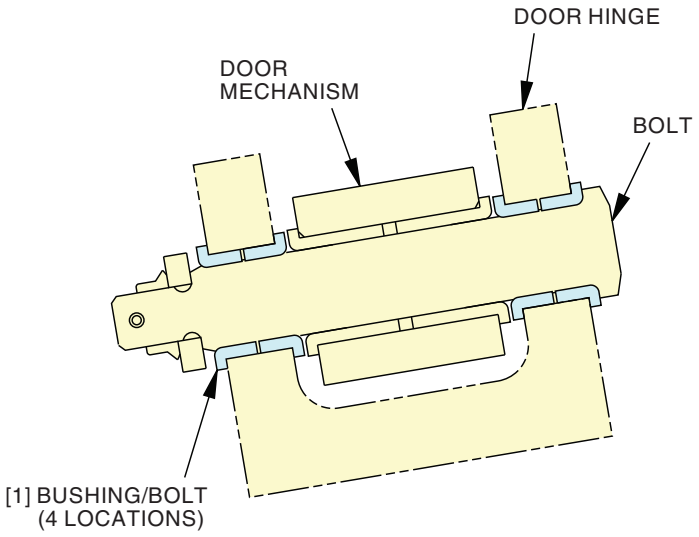
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Wheel Well Door Linkage Wear Limits
Figure 601/32-33-61-990-803 (Sheet 1 of 4)

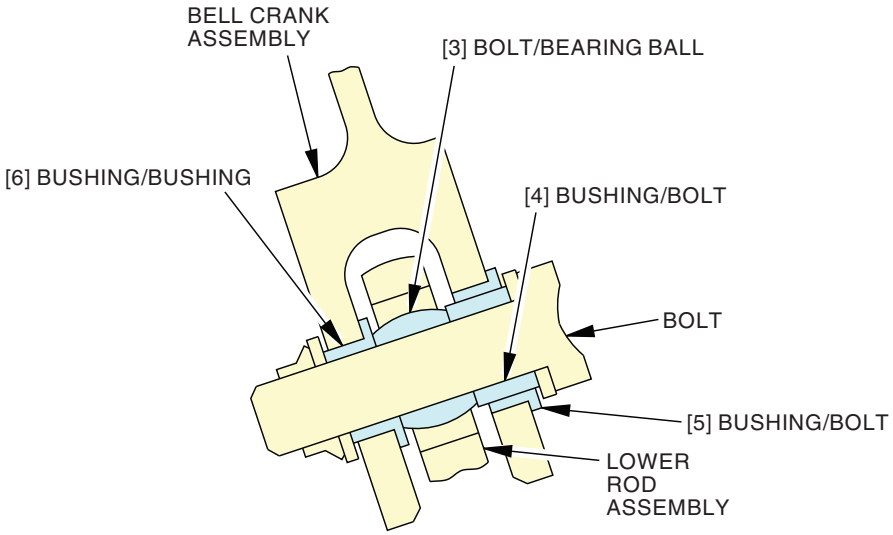
ECCN 9E991 BOEING PROPRIETARY - See title page for details



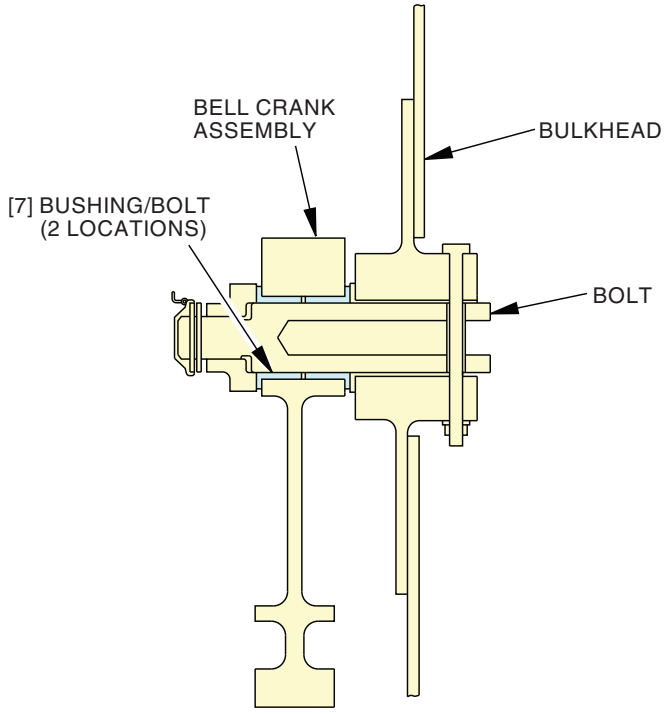
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Nose Gear Wheel Well Door Linkage Wear Limits
Figure 601/32-33-61-990-803 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



C-C

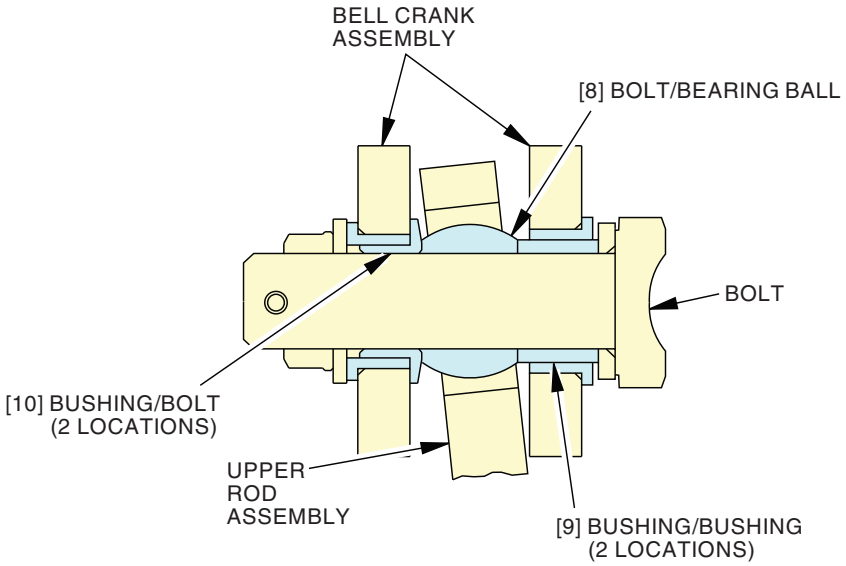


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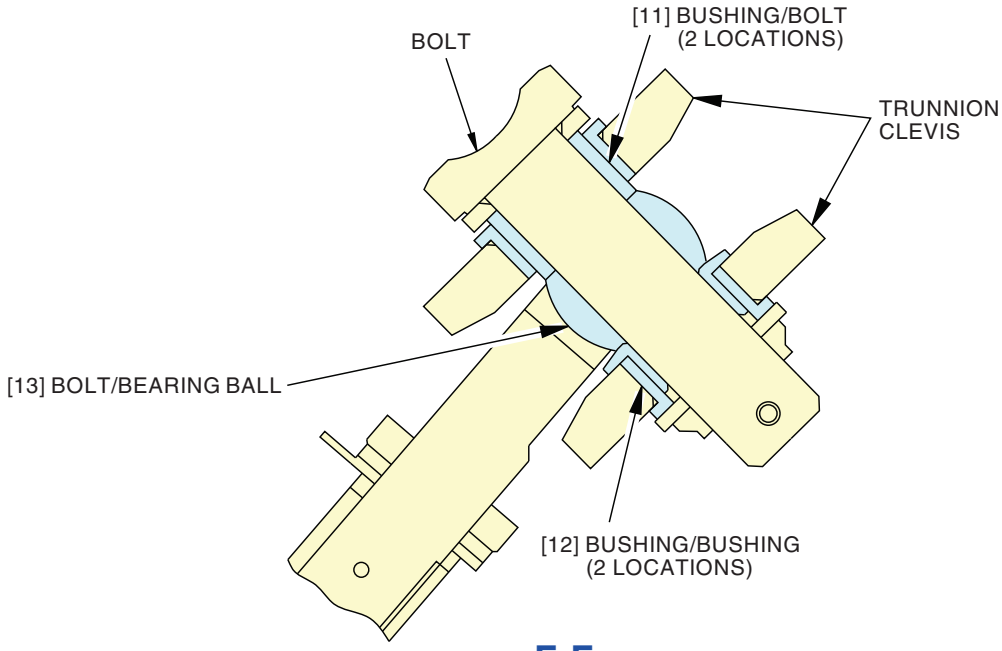
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Nose Gear Wheel Well Door Linkage Wear Limits
Figure 601/32-33-61-990-803 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



E-E



F-F

2414304 S00061534846_V1

Nose Gear Wheel Well Door Linkage Wear Limits
Figure 601/32-33-61-990-803 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR PRESSURE FUSE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the retract pressure fuse for the nose landing gear.

(2) Installation of the retract pressure fuse for the nose landing gear.

(a) The installation of the retract pressure fuse for the nose landing gear includes an installation test of the retract pressure fuse.

(3) Removal of the extend pressure fuse for the nose landing gear.

(4) Installation of the extend pressure fuse for the nose landing gear.

(a) The installation of the extend pressure fuse for the nose landing gear includes an installation test for the extend pressure fuse.

TASK 32-33-71-000-801

2. Nose Gear Retract Pressure Fuse Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-71-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-71-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-71-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Nose Gear Retract Pressure Fuse Removal

SUBTASK 32-33-71-020-001

(1) Loosen the B-nut [5] and B-nut [6] that connecting the hydraulic lines to the retract pressure fuse [4].

SUBTASK 32-33-71-020-002

(2) Remove the screws [1], washers [2], nuts [3], and clamps [7] that hold the hydraulic lines to the structure on the two ends of the retract pressure fuse [4].

SUBTASK 32-33-71-020-003

(3) Disconnect the hydraulic lines from the retract pressure fuse [4].

SUBTASK 32-33-71-480-002

(4) Install caps/plugs, STD-14686, on the hydraulic lines.

SUBTASK 32-33-71-020-004

(5) Remove the retract pressure fuse [4] from the airplane.

F. Nose Gear Retract Pressure Hydraulic Fuse Functional Test

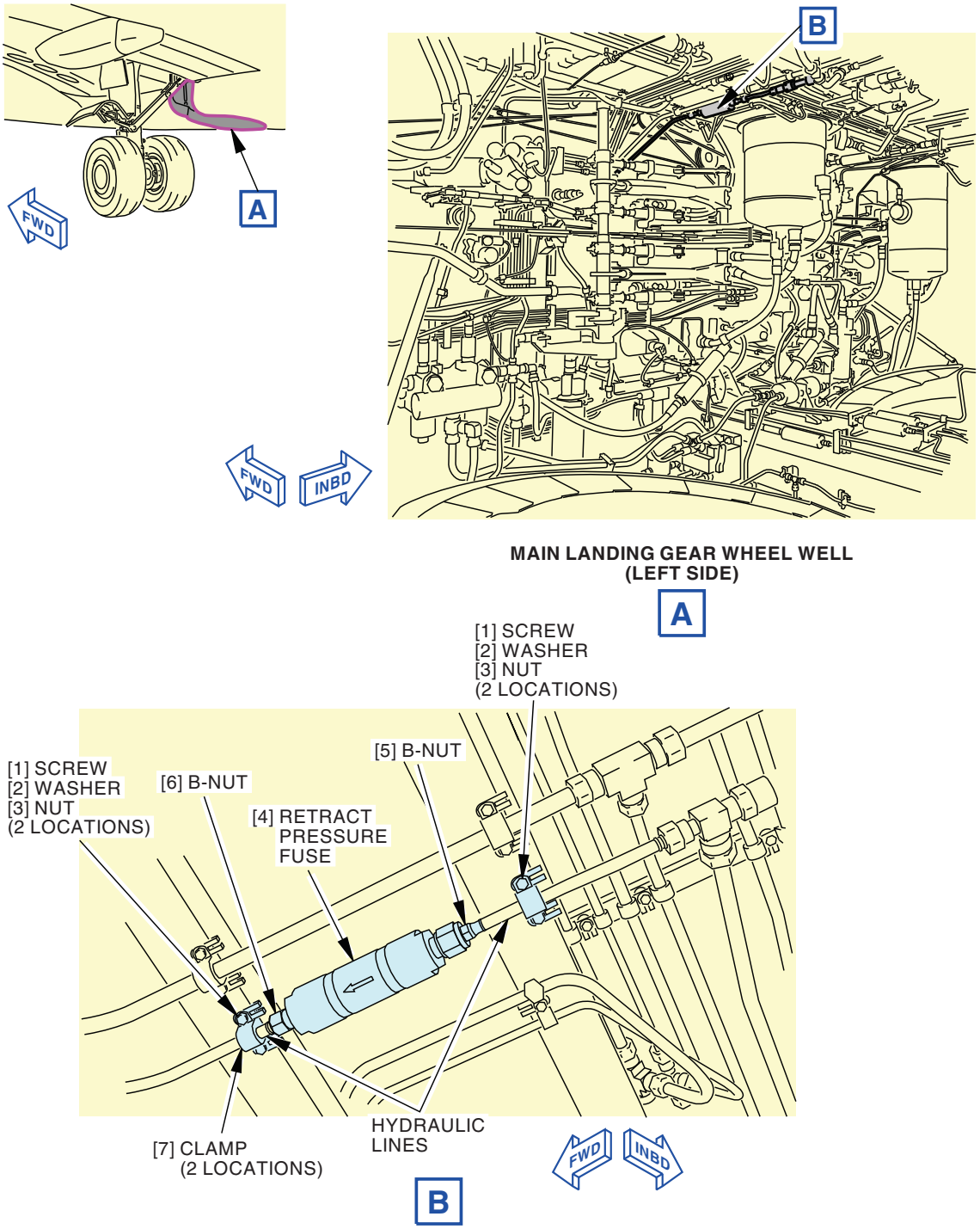
SUBTASK 32-33-71-720-001

(1) Do the functional test of the retract pressure fuse [4] with the supplier's recommended component maintenance test instructions and test equipment.

NOTE: The supplier's instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the retract pressure fuse [4]. This is an off-airplane bench test.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Retract Pressure Fuse Installation
Figure 401/32-33-71-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-71-400-801

3. Nose Gear Retract Pressure Fuse Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Retract pressure fuse	32-33-71-01-010	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Nose Gear Retract Pressure Fuse Installation

SUBTASK 32-33-71-420-001

(1) Put the retract pressure fuse [4] in position to align it with the hydraulic lines.

SUBTASK 32-33-71-080-001

(2) Remove the caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-33-71-420-002

(3) Connect the hydraulic lines to the retract pressure fuse [4]:

(a) For the B-nut [5] and B-nut [6], apply MCS 352B fluid, D00054, to the entire circumference of the external fitting threads, the shoulder of the sleeve in contact with the nut, and to the sealing surface of the sleeve.

(b) Tighten the B-nut [5] and B-nut [6] on the hydraulic lines to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-33-71-420-003

(4) Install the screws [1], washers [2], nuts [3], and clamps [7] to hold the hydraulic lines to the structure on the two ends of the retract pressure fuse [4].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Nose Gear Retract Pressure Fuse Installation Test

SUBTASK 32-33-71-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-71-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-71-700-001

(3) Use the LOCK OVRD button, on the P2-1 panel, to move the control lever for the landing gear from the DN position to the UP position.

SUBTASK 32-33-71-710-001

(4) Make sure that the UP hose of the retract actuator moves to show that hydraulic pressure is applied, when the control lever is moved to the UP position.

SUBTASK 32-33-71-700-002

(5) Move the control lever to the DN position.

SUBTASK 32-33-71-700-003

(6) Make sure that the DOWN hose on the retract actuator moves, when you move the control lever from the UP position to the DN position.

SUBTASK 32-33-71-210-001

(7) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-33-71-860-003

(8) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-71-210-002

(9) Make sure that there are no leaks in the hydraulic connections.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-71-840-001

(1) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Service them, if it is necessary.

END OF TASK

TASK 32-33-71-000-802

4. Nose Gear Extend Pressure Fuse Removal

(Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-71-480-004

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-71-860-004

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-33-71-580-002

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

E. Extend Pressure Fuse Removal

SUBTASK 32-33-71-020-008

(1) Remove the screw [9], washers [11], spacer [10], and nut [12] to remove the loop clamp [8] that holds the extend pressure fuse [25].

NOTE: The loop clamp [13] and loop clamp [8] are tied to each other.

SUBTASK 32-33-71-020-005

(2) Disconnect the extend pressure fuse [25] from the elbow.

SUBTASK 32-33-71-020-009

(3) Disconnect the hydraulic line from the fitting assembly [14] that attaches to the extend pressure fuse [25].

SUBTASK 32-33-71-020-006

(4) Disconnect the fitting assembly [14] from the extend pressure fuse [25].

SUBTASK 32-33-71-480-005

(5) Install cap/plug, STD-14686, on the hydraulic line.

SUBTASK 32-33-71-020-007

(6) Remove the extend pressure fuse [25] from the airplane.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Nose Gear Extend Pressure Fuse Functional Test

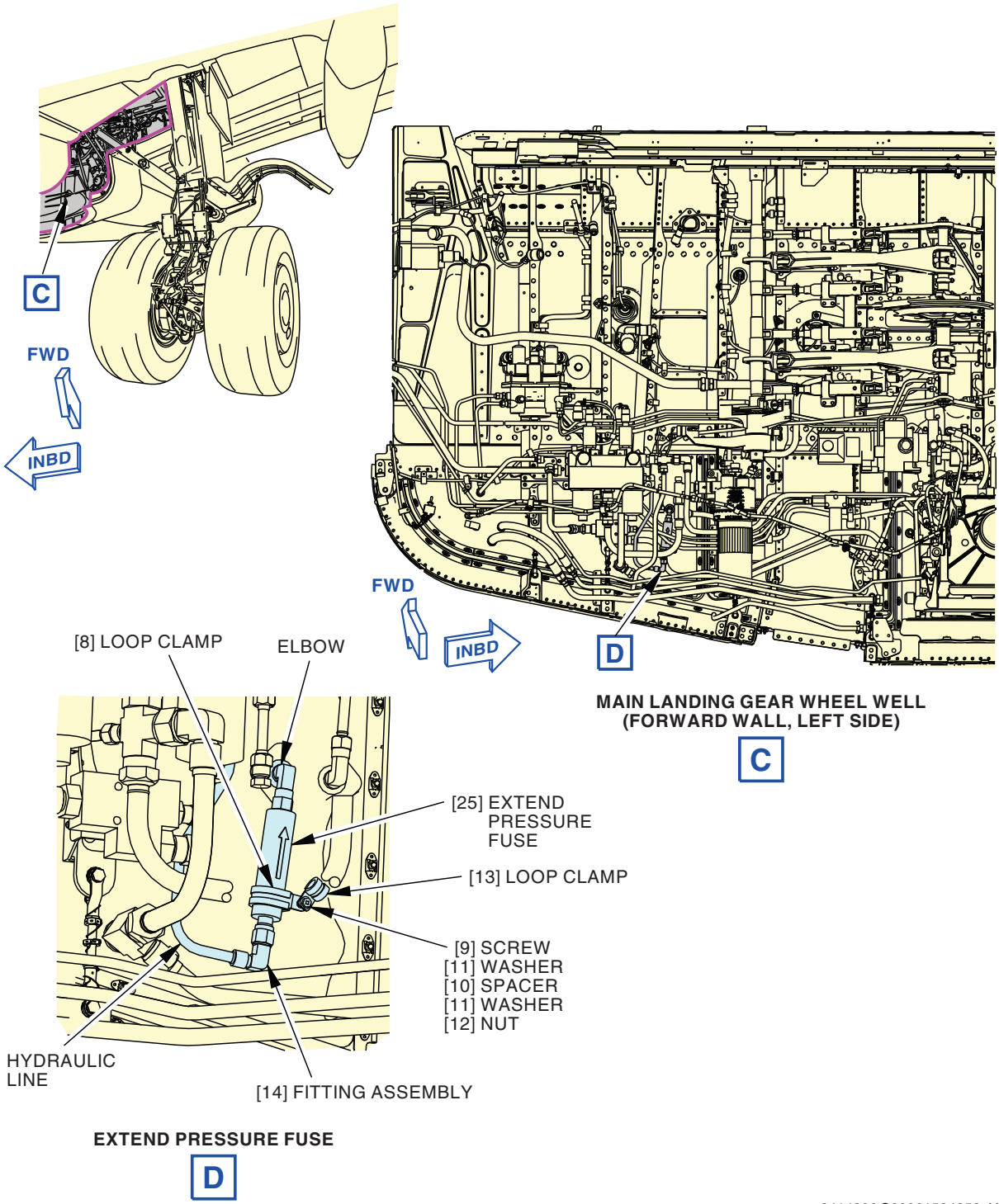
SUBTASK 32-33-71-720-002

(1) Do the functional test of the extend pressure fuse [25] with the supplier's recommended component maintenance test instructions and test equipment.

NOTE: The supplier's instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the extend pressure fuse [25]. This is an off-airplane bench test.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Extend Pressure Fuse Installation
Figure 402/32-33-71-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-71-400-802

5. Nose Gear Extend Pressure Fuse Installation

(Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
25	Extend pressure fuse	32-33-71-05-010	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Nose Gear Extend Pressure Fuse Installation

SUBTASK 32-33-71-420-004

(1) Put the extend pressure fuse [25] in position to align it with the hydraulic line and the elbow.

SUBTASK 32-33-71-420-006

(2) Connect the extend pressure fuse [25] to the elbow:

(a) Tighten the nut on the extend pressure fuse [25] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-33-71-420-007

(3) Connect the fitting assembly [14] to the extend pressure fuse [25]:

(a) Tighten the nut on the on the fitting assembly [14] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-33-71-080-002

(4) Remove the cap/plug, STD-14686, from the hydraulic line and connect the hydraulic line to the fitting assembly [14]:

(a) Tighten the B-nut on the hydraulic line that connects to the fitting assembly [14] to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-33-71-420-005

(5) Install the screw [9], washers [11], spacer [10], and nut [12] to attach the loop clamp [8] to hold the extend pressure fuse [25].

NOTE: The loop clamp [13] and loop clamp [8] are tied to each other.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Nose Gear Extend Pressure Fuse Installation Test

SUBTASK 32-33-71-480-006

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-71-860-005

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-71-700-004

(3) Use the LOCK OVRD button, on the P2-1 panel, to move the control lever for the landing gear from the DN position to the UP position.

SUBTASK 32-33-71-710-002

(4) Make sure that the UP hose of the retract actuator moves to show that hydraulic pressure is applied, when the control lever is moved to the UP position.

SUBTASK 32-33-71-700-005

(5) Move the control lever to the DN position.

SUBTASK 32-33-71-700-006

(6) Make sure that the DOWN hose on the retract actuator moves, when you move the control lever from the UP position to the DN position.

SUBTASK 32-33-71-210-003

(7) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-33-71-210-004

(8) Make sure that there are no leaks in the hydraulic connections.

SUBTASK 32-33-71-860-006

(9) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-33-71-840-002

(1) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Service them, if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR HYDRAULIC FLOW REGULATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the hydraulic flow regulator (referred to as the flow regulator) for the nose landing gear

(2) An installation of the hydraulic flow regulator for the nose landing gear.

(a) The installation of the hydraulic flow regulator for the nose landing gear includes an installation test of the hydraulic flow regulator.

TASK 32-33-72-000-801

2. Nose Gear Hydraulic Flow Regulator Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-7394	Cap - Dust, Electrical Connector

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-33-72-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-72-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Nose Gear Hydraulic Flow Regulator Removal

SUBTASK 32-33-72-020-001

(1) Loosen the hydraulic lines to the flow regulator [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-33-72-020-002

(2) Remove the screws [1], washers [2], and nuts [4] that hold the hydraulic lines to the structure on both ends of the flow regulator [5].

SUBTASK 32-33-72-020-005

(3) Remove the clamps [3].

SUBTASK 32-33-72-020-003

(4) Disconnect the hydraulic lines from the flow regulator [5].

SUBTASK 32-33-72-480-002

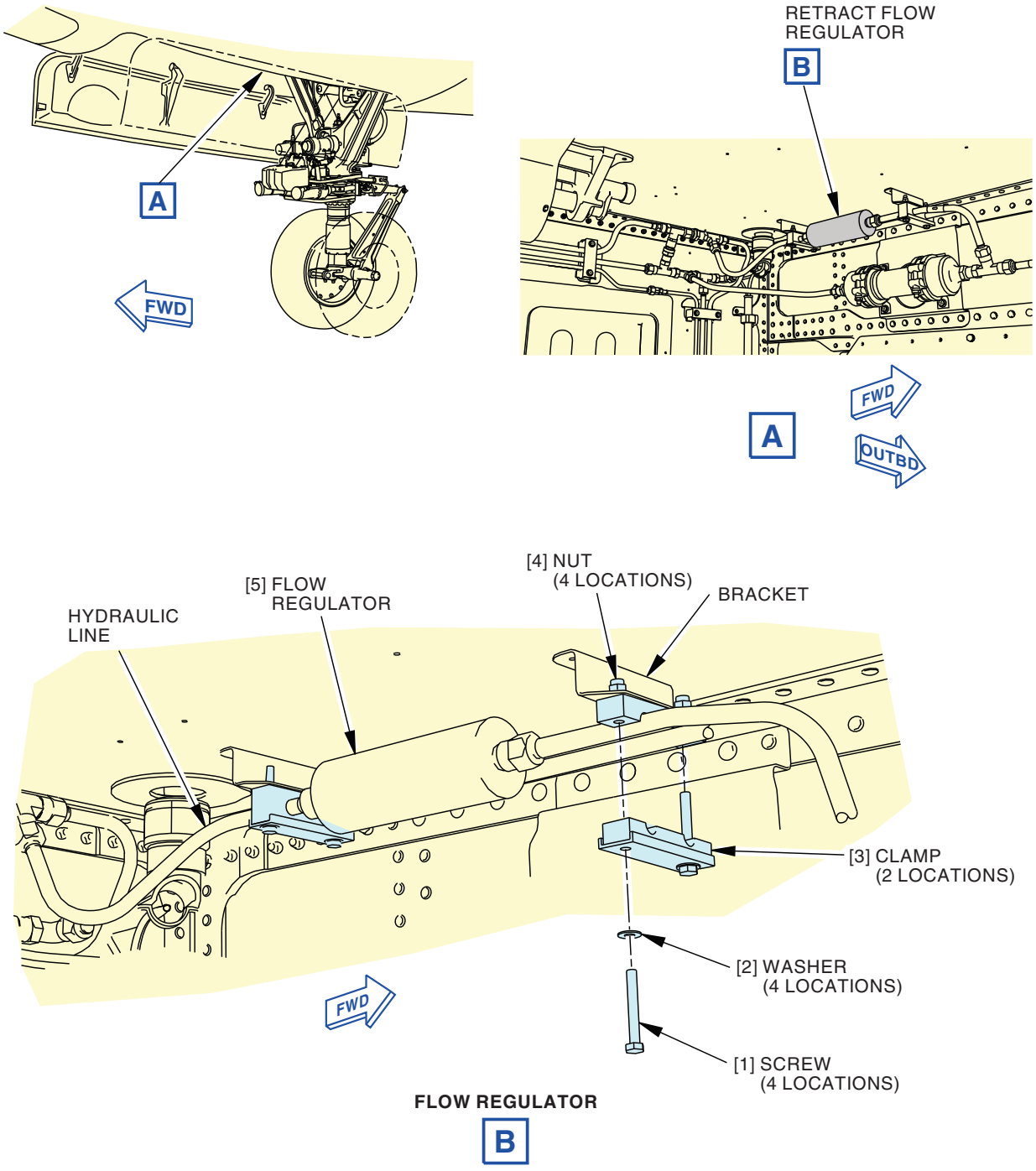
(5) Install dust caps, STD-7394, on the hydraulic lines.

SUBTASK 32-33-72-020-004

(6) Remove the flow regulator [5] from the airplane.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Nose Gear Flow Regulator Installation
Figure 401/32-33-72-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-33-72-400-801

3. Nose Gear Hydraulic Flow Regulator Installation

(Figure 401)

A. References

Reference	Title
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-7394	Cap - Dust, Electrical Connector

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Flow regulator	32-21-51-05-415	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

E. Nose Gear Hydraulic Flow Regulator Installation

SUBTASK 32-33-72-420-001

(1) Put the flow regulator [5] in position to align it with the hydraulic lines.

SUBTASK 32-33-72-080-001

(2) Remove the dust caps, STD-7394, from the hydraulic lines and connect the hydraulic lines to the flow regulator [5].

SUBTASK 32-33-72-420-002

(3) Install the screws [1], washers [2], and nuts [4] in the clamps [3] for the hydraulic lines.

SUBTASK 32-33-72-420-003

(4) Connect the hydraulic lines to the flow regulator [5].

(a) Tighten the B-nuts on the hydraulic lines to 270 ±14 in-lb (31 ±2 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Nose Gear Hydraulic Flow Regulator Installation Test

SUBTASK 32-33-72-480-003

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-33-72-860-002

(2) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-33-72-700-001

(3) Move the landing gear lever to the UP position, then to the DN position three times to cycle the gear.

NOTE: This will bleed the air from the regulator and the lines.

NOTE: Use the override for the landing gear lever lock to move the landing gear lever from the DN position to the UP position.

(a) Hold the landing gear lever in each position for three seconds before the landing gear lever is moved to the next position.

(b) Make sure that the UP hose of the retract actuator moves to show that hydraulic pressure is applied when the landing gear lever is moved to the UP position.

(c) Move the landing gear lever to the DN position.

(d) Make sure that the DOWN hose on the retract actuator moves when the landing gear lever is moved from the UP position to the DN position.

SUBTASK 32-33-72-210-001

(4) Do a check for leakage with the hydraulic pressure applied.

SUBTASK 32-33-72-860-003

(5) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-33-72-210-002

(6) Make sure that there are no leaks in the hydraulic connections.

SUBTASK 32-33-72-840-001

(7) Examine the hydraulic reservoirs for the correct servicing (TASK 12-12-00-610-801).

(a) Do the servicing if it is necessary.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR MANUAL EXTENSION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Main Landing Gear Manual Extension System Adjustment and Test

(2) Main Gear Manual Extension System Test (Airplane on Jacks)

(3) Main Gear Manual Extension System Test (Airplane not on Jacks)

(4) Manual Extension System Access Door Switch Adjustment.

TASK 32-34-00-820-801

2. Main Landing Gear Manual Extension System Adjustment and Test

(Figure 501, Figure 502)

A. General

(1) This task gives the instructions to adjust (rig) the left or right manual extension system for the main landing gears.

NOTE: Control cables may contact fairleads, grommets or rub strips after rigging at operating conditions. Refer to the task: Inspection of the Control Cable Fittings, TASK 20-20-31-200-802, for control cable damage limits.

(2) In this task the rig pin used for the left side main landing gear is LGEA2 and for the right side is LGEA3.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-20-31-200-802	Inspection of the Control Cable Fittings (P/B 601)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-4896	Adapter Assembly - Uplock, MLG Part #: C32048-1 Supplier: 81205

D. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Adjustment and Test

SUBTASK 32-34-00-860-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-34-00-860-002

(2) If necessary, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-34-00-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-34-00-480-001

(4) Install the rig pin LGEA2 (L/H) or LGEA3 (R/H) from rigging pin kit, SPL-1585, in the cable guard bracket and quadrant on the manual extension mechanism.

SUBTASK 32-34-00-010-001

(5) Remove the ceiling panels in the forward cargo compartment that are at approximately body station 420.

F. Main Landing Gear Manual Extension System Adjustment and Test

SUBTASK 32-34-00-020-001

(1) Remove the locking clips from the turnbuckle.

SUBTASK 32-34-00-020-002

(2) Loosen the MLGEA cable at the turnbuckle.

SUBTASK 32-34-00-820-003

(3) Make sure that the uplock hook is in the up and locked position with the MLG uplock adapter assembly, SPL-4896, or a suitable tool.

SUBTASK 32-34-00-020-003

(4) Remove the pin [1] and loosen the nut [2] and bolt [3].

SUBTASK 32-34-00-820-004

(5) Loosen the jamnut on the control rod on the manual extension mechanism.

SUBTASK 32-34-00-820-005

(6) Adjust the control rod to get the 0.0625 in. (1.59 mm) to 0.1500 in. (3.8100 mm) gap between the uplock kicker arm and the upper lock link (Figure 502).

SUBTASK 32-34-00-420-001

(7) Tighten the jamnut to 95 in-lb (10.73 N·m) - 160 in-lb (18.08 N·m).

SUBTASK 32-34-00-080-001

(8) Remove the rig pin LGEA2 or LGEA3 from the manual extension mechanism.

SUBTASK 32-34-00-860-034

(9) If necessary, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-34-00-980-001

(10) Put the uplock hook in the down and locked position with the MLG uplock adapter assembly, SPL-4896, or a suitable tool.

NOTE: You can remove the tool adapter after the Main Landing Gear (MLG) uplock hook is in its initial configuration.

(a) Make sure that the MLG is in its initial configuration before you do a gear retraction.

SUBTASK 32-34-00-700-001

(11) Turn the cable quadrant to make sure that the control rod does not rub against the uplock kicker arm.

(a) Install the rig pin LGEA2 or LGEA3 on the manual extension mechanism.

SUBTASK 32-34-00-420-002

(12) Do these steps to tighten the bolt [3] and nut [2] to 80 in-lb (9.04 N·m) plus run on torque to 240 in-lb (27.12 N·m) maximum:

(a) Begin at the lower torque value and increase as necessary to align the castellation in the nut [2] with the hole in the bolt [3].

(b) Do not use more than 240 in-lb (27.12 N·m) applied torque.

(c) Install the pin [1].

SUBTASK 32-34-00-700-023

(13) Make sure that the main gear manual extension handles in the flight compartment are securely fitted into their v-slots when the cables are adjusted.

SUBTASK 32-34-00-420-003



CAUTION

MAKE SURE THAT THE MAIN-GEAR MANUAL EXTENSION-HANDLES ARE IN THEIR V-SLOTS WHEN THE CABLES ARE ADJUSTED. WHEN YOU MOVE THE CONTROL HANDLE TO THE UP POSITION, DAMAGE CAN OCCUR WHEN THE HANDLES ARE NOT IN THEIR V-SLOTS.

(14) Adjust the turnbuckle on cable MLGEA until the rig pin LGEA 2 or LGEA 3 fits freely in the manual extension mechanism and the cable guard.

SUBTASK 32-34-00-080-002

(15) Remove the rig pin LGEA2 or LGEA3 from the manual extension mechanism and cable guard.

(a) Install the locking clips on the turnbuckle.

SUBTASK 32-34-00-710-001

(16) Do a test of the manual extension system for the main gear.

(a) If airplane is on jacks, do this task: Main Gear Manual Extension System Test (Airplane on Jacks), TASK 32-34-00-730-801.

(b) If airplane is not on jacks, do this task: Main Gear Manual Extension System Test (Airplane not on Jacks), TASK 32-34-00-730-802.

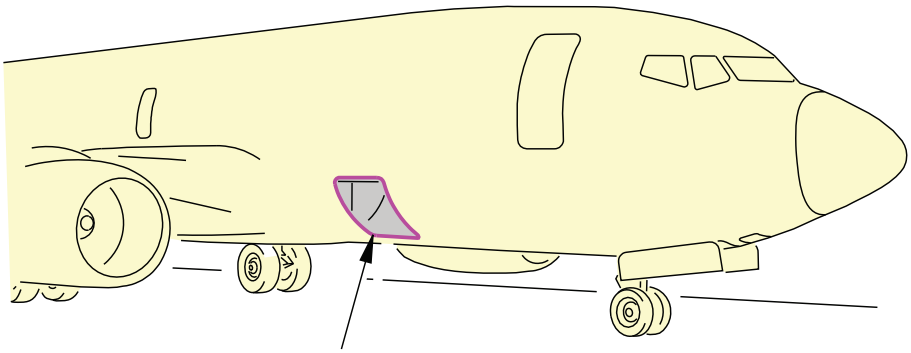
G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-00-410-001

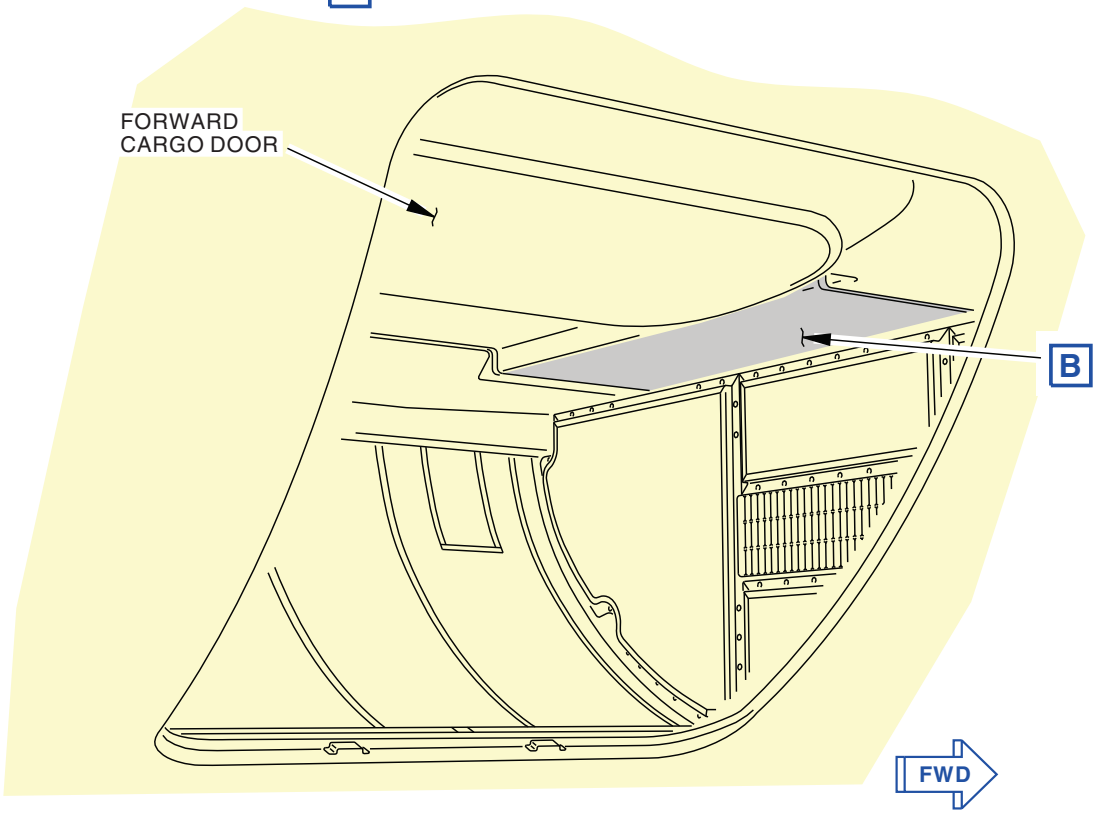
(1) Install the ceiling panels that you removed in the forward cargo compartment.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A



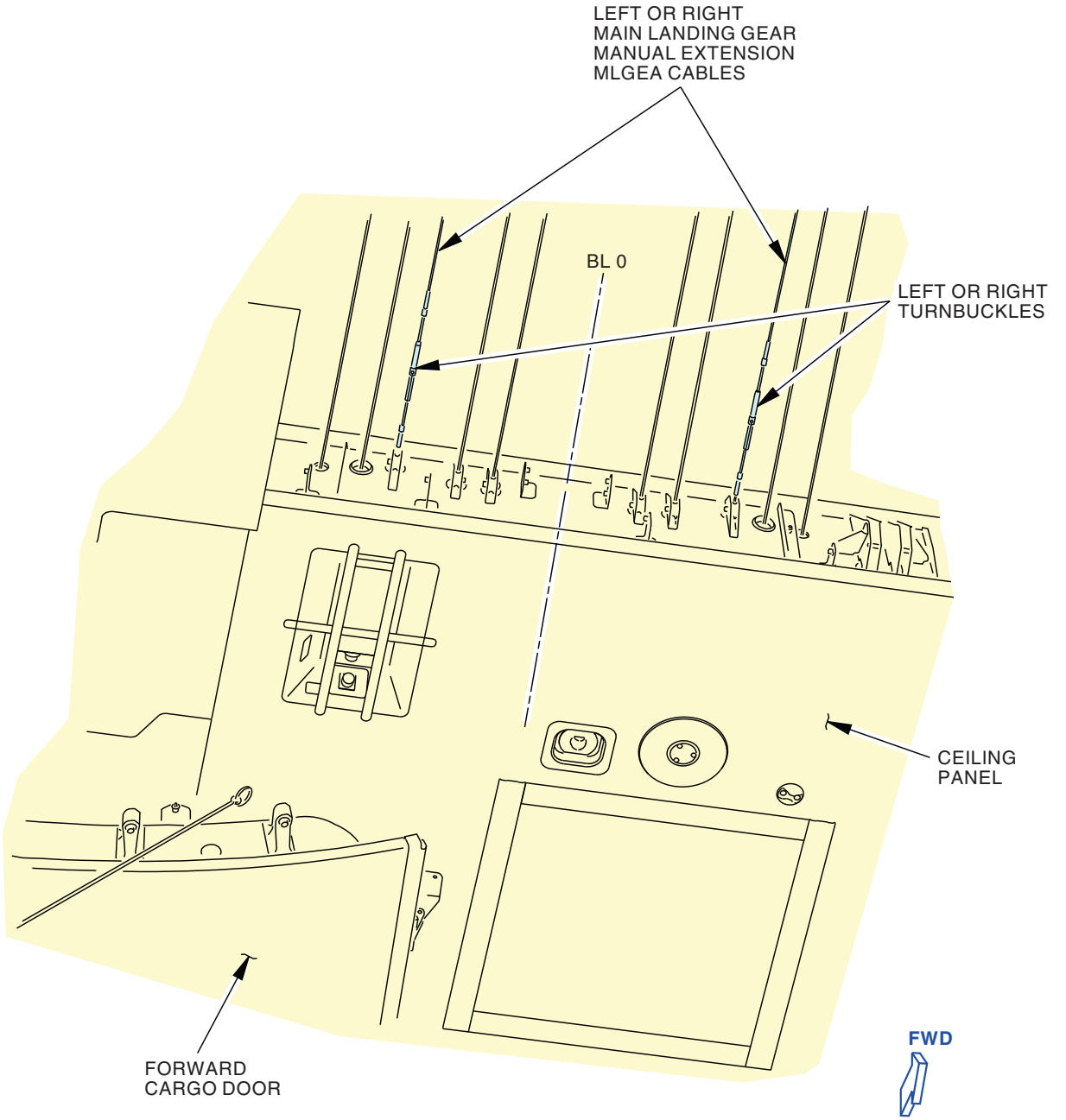
**FORWARD CARGO COMPARTMENT
(CEILING PANEL INSTALLED)**

A

2414308 S00061534864_V2

**Main Gear Manual Extension System - Turnbuckle Location
Figure 501/32-34-00-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



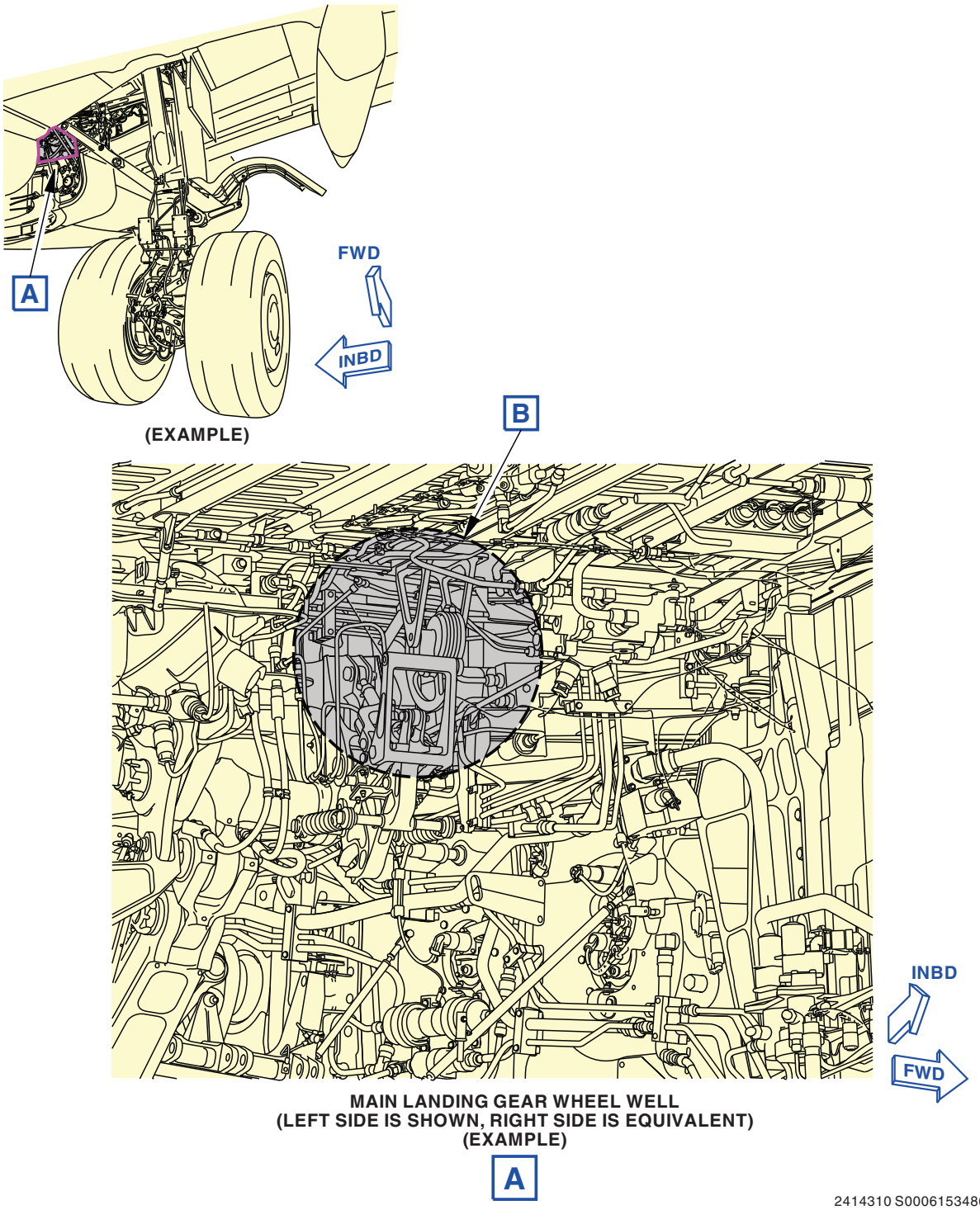
FORWARD CARGO COMPARTMENT
(CEILING PANEL REMOVED)

B

2414309 S00061534865_V2

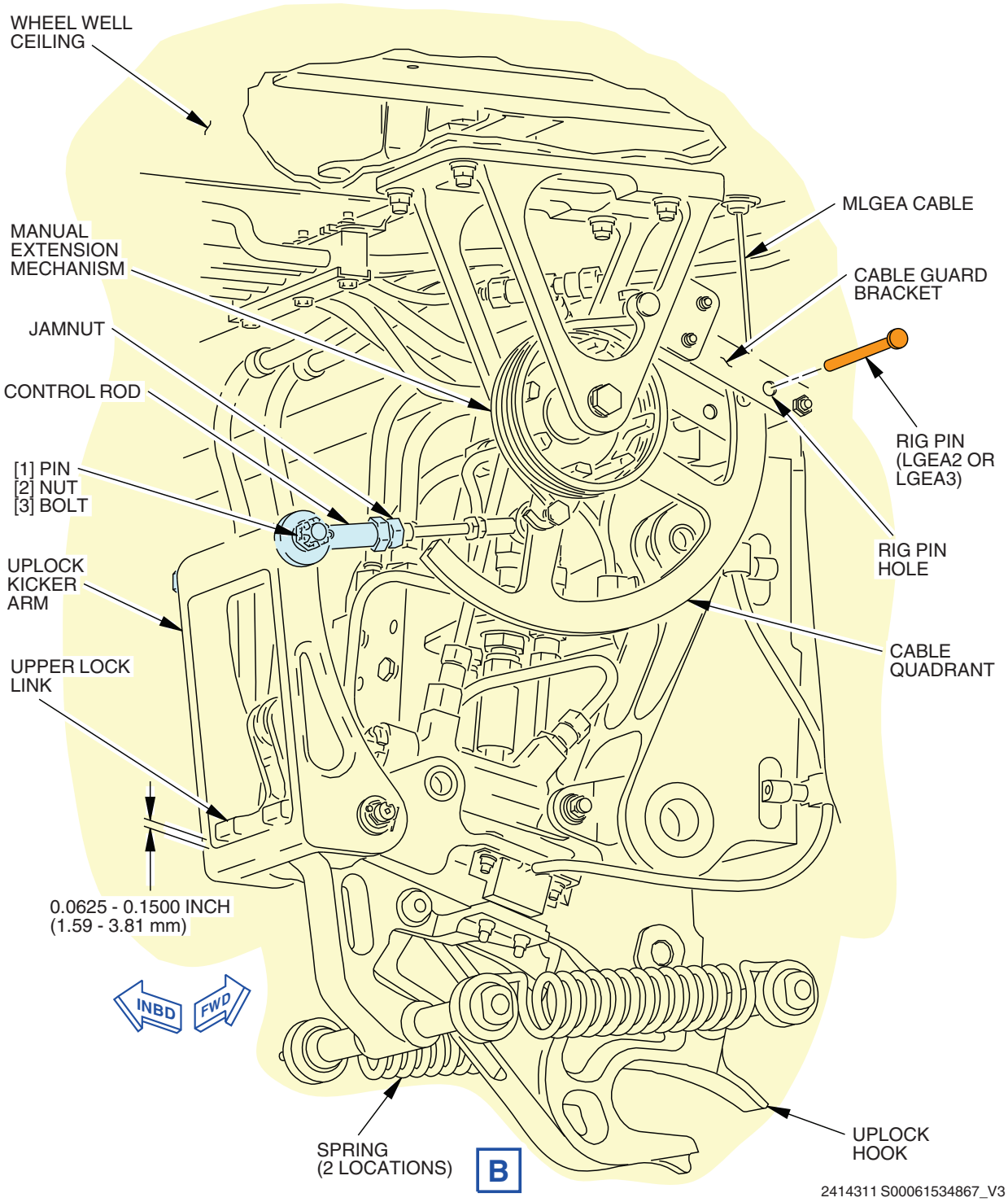
Main Gear Manual Extension System - Turnbuckle Location
Figure 501/32-34-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Manual Extension System - Adjustment
Figure 502/32-34-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Manual Extension System - Adjustment
Figure 502/32-34-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-00-730-801

3. Main Gear Manual Extension System Test (Airplane on Jacks)

(Figure 503)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The normal extension/retraction system will not operate with the access door for the alternate manual extension handle open; make sure that the manual extension access door is closed when you will extend or retract the landing gear with the landing gear control lever.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)

C. Tools/Equipment

Reference	Description
STD-1184	Scale - Spring, 0-100 Lbs, Tension

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 32-34-00-480-002

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-34-00-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(2) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

SUBTASK 32-34-00-860-005

(3) Make sure that these circuit breakers are closed:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	3	C01312	ENGINE 1 RUN/PWR
D	1	C04030	DSP CAPT EFIS CONT PNL
D	2	C04003	MULTI FUNCTION PANEL
D	3	C04026	DISPLAY LEFT INBOARD
D	4	C04027	DISPLAY LEFT OUTBOARD
D	5	C04023	DISPLAY DPC 1 PRI

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	9	C04024	DISPLAY DPC 2 HOLDUP
D	10	C04022	DISPLAY DPC 1 HOLDUP
E	10	C04029	DISPLAY RIGHT OUTBOARD
E	11	C04028	DISPLAY RIGHT INBOARD
E	12	C04002	MAINT/BITE PANEL
E	13	C04031	DSP F/O EFIS CONT PNL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	17	C00129	LANDING GEAR LATCH & PRESS WARN
C	15	C01355	LANDING GEAR AIR/GND SYS 2
C	16	C01356	LANDING GEAR AIR/GND SYS 1
D	1	C01399	PSEU PRI
D	2	C01400	PSEU ALTN
D	16	C01432	LANDING GEAR ALTN EXTEND SOL
E	12	C00314	INDICATOR MASTER DIM SECT 2
F	11	C00317	INDICATOR MASTER DIM SECT 5
F	13	C01179	INDICATOR MASTER DIM SECT 7

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
D	18	C04020	LANDING GEAR CONTROL 1
D	19	C04021	LANDING GEAR CONTROL 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-860-006

(4) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-34-00-860-007

(5) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-34-00-860-008

(6) Move the No. 1 and No. 2 throttle levers to the full forward position.

NOTE: Throttles in the advanced position prevent the landing gear warning (red) lights from remaining on after gear is up and locked.

SUBTASK 32-34-00-860-009

(7) Put the master dim and test switch in the bright position.

F. Main Gear Manual Extension System Test (Airplane on Jacks)

SUBTASK 32-34-00-700-002

(1) Open the manual extension access door for the main gear manual extension handles in the flight compartment floor.

(a) Make sure that the switch under the manual extension access door clicks audibly when the manual extension access door opens.

SUBTASK 32-34-00-700-003

(2) Pull the manual extension handle for the left main gear.

(a) Make sure that the full travel of the manual extension handle is 20.0 in. (508.0 mm) or more.

SUBTASK 32-34-00-700-004

(3) Release the manual extension handle for the left main gear.

(a) Make sure that the manual extension handle returns to the stowed position without hesitation or binding.

SUBTASK 32-34-00-480-003

(4) Attach a spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the left main gear.

SUBTASK 32-34-00-730-001

(5) Pull the manual extension handle for the left main gear.

(a) Make sure that the load to start the movement of the manual extension handle from the seat is 3.0 lbf (13.3 N) or more.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the load during the full travel of the manual extension handle is 18.0 lbf (80.1 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

(c) Make sure that the manual extension mechanism operates freely.

SUBTASK 32-34-00-700-005

(6) Pull the manual extension handle for the right main gear.

(a) Make sure that the full travel of the manual extension handle is 20.0 in. (508.0 mm) or more.

SUBTASK 32-34-00-700-006

(7) Release the manual extension handle for the right main gear.

(a) Make sure that the manual extension handle returns to the stowed position without hesitation or binding.

SUBTASK 32-34-00-860-010

(8) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the left main gear.

SUBTASK 32-34-00-480-004

(9) Attach the spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the right main gear.

SUBTASK 32-34-00-730-002

(10) Pull the manual extension handle for the right main gear.

(a) Make sure that the load to start the movement of the manual extension handle from the seat is 3.0 lbf (13.3 N) or more.

(b) Make sure that the load during the full travel of the manual extension handle is 18.0 lbf (80.1 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

(c) Make sure that the manual extension mechanism operates freely.

SUBTASK 32-34-00-860-011

(11) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the right main gear.

SUBTASK 32-34-00-860-012

(12) Make sure that the manual extension access door for the manual extension handle is closed.

NOTE: The extension/retraction system will not operate with the manual extension access door for the alternate manual extension handle open.

SUBTASK 32-34-00-080-003

(13) Remove the downlock pin from the left and right main gears (TASK 32-00-01-080-801).

NOTE: Do not remove the Nose Landing Gear (NLG) downlock pin.

SUBTASK 32-34-00-860-013

(14) Make sure that the control lever for the landing gear is still in the DN position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-860-004

(15) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-34-00-730-003

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(16) Move the control lever for the landing gear to UP and wait while the main gear retracts.

SUBTASK 32-34-00-210-001

(17) Make sure that both main gears green lights and red lights are off.

SUBTASK 32-34-00-860-014

(18) Open the manual extension access door to the manual extension handles.

SUBTASK 32-34-00-730-004

(19) Pull the manual extension handle of the left main gear.

NOTE: The manual extension handle must be pulled upward and forward. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

SUBTASK 32-34-00-700-007

(20) Make sure that the left main gear free falls to the down and locked position.

(a) Make sure that the red light for the left main gear is on.

NOTE: The red light for the main gear will stay on to show that the landing gear control lever and landing gear positions disagree.

(b) Make sure that the green light for the left main gear is on.

SUBTASK 32-34-00-700-008

(21) Release the manual extension handle for the left main gear.

SUBTASK 32-34-00-730-005

(22) Pull the manual extension handle of the right main gear.

NOTE: The manual extension handle must be pulled upward and forward. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

SUBTASK 32-34-00-700-009

(23) Make sure that the right main gear free falls to the down and locked position.

(a) Make sure that the red light for the right main gear is on.

NOTE: The red light for the main gear will stay on to show that the landing gear control lever and landing gear positions disagree.

(b) Make sure that the green light for the right main gear is on.

SUBTASK 32-34-00-700-010

(24) Release the manual extension handle for the right main gear.

SUBTASK 32-34-00-860-015

(25) Move the control lever for the landing gear to the DN position.

(a) Make sure that the red light for the main gear is off.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-860-016

(26) Close the manual extension access door for the manual extension handle.

SUBTASK 32-34-00-730-006

(27) Move the control lever for the landing gear to UP and wait while the main gear retracts.

SUBTASK 32-34-00-860-019

(28) Open the manual extension access door to the manual extension handles.

SUBTASK 32-34-00-700-011

(29) Attach a spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the left main gear.

SUBTASK 32-34-00-730-007

(30) Pull the manual extension handle for the left main gear.

(a) Make sure that the maximum load when you pull the manual extension handle is 50.0 lbf (222.4 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

SUBTASK 32-34-00-700-012

(31) Make sure that the left main gear free falls to the down and locked position.

(a) Make sure that the red light for the left main gear is on.

NOTE: The red light for the main gear will stay on to show that the landing gear control lever and landing gear positions disagree.

(b) Make sure that the green light for the left main gear is on.

SUBTASK 32-34-00-080-004

(32) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the left main gear.

SUBTASK 32-34-00-700-013

(33) Attach a spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the right main gear.

SUBTASK 32-34-00-700-014

(34) Pull the manual extension handle for the right main gear.

(a) Make sure that the maximum load when you pull the manual extension handle is 50.0 lbf (222.4 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

SUBTASK 32-34-00-700-015

(35) Make sure that the right main gear free falls to the down and locked position.

(a) Make sure that the red light for the right main gear is on.

NOTE: The red light for the main gear will stay on to show that the landing gear control lever and landing gear positions disagree.

(b) Make sure that the green light for the right main gear is on.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-860-024

WARNING MAKE SURE THAT ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE MAIN LANDING GEAR. FAST MOVEMENT OF THE MAIN LANDING GEAR CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(36) Move the control lever for the landing gear to the DN position.

NOTE: The left main gear will extend.

(a) Make sure that the red light for the right and left main gear are off.

SUBTASK 32-34-00-080-005

(37) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the right main gear.

SUBTASK 32-34-00-860-025

(38) Close the manual extension access door for the manual extension handles.

SUBTASK 32-34-00-710-002

(39) Perform a full main gear retract and extend cycle to verify normal hydraulic operation.

G. Put the Airplane Back to Its Usual Condition.

SUBTASK 32-34-00-840-001

(1) Close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-34-00-860-026

(2) For Hydraulic System A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-34-00-480-005

(3) Install the downlock pins on the main gear that you tested (TASK 32-00-01-480-801).

SUBTASK 32-34-00-860-027

(4) Move the No. 1 and No. 2 throttle levers back to the idle position (full aft).

SUBTASK 32-34-00-840-002

(5) Close this circuit breaker:

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-580-003

CAUTION

MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(6) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

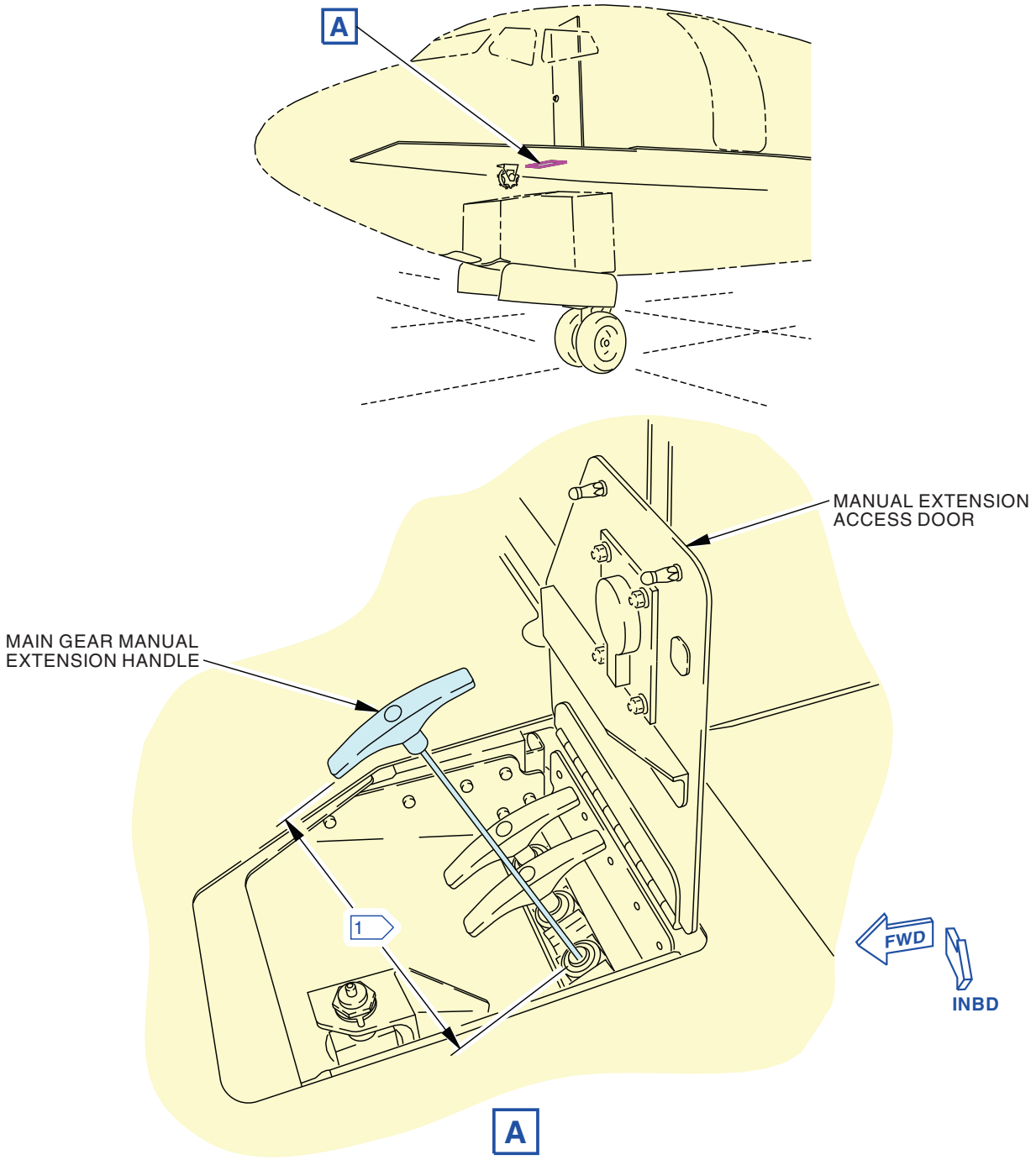
SUBTASK 32-34-00-200-001

(7) Do a check of the Proximity Switch Electronic Unit (PSEU) for stored faults using the SELF-TEST (TASK 32-09-10-740-801).

NOTE: When the SELF-TEST is completed, it is acceptable for FAILED to show. Refer to the applicable Maintenance Message reference in the FAULT GUIDANCE or select the maintenance message to see the recommended maintenance actions.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 FULL TRAVEL DISTANCE TO UNLOCK POSITION

2414312 S00061534869_V2

Main Gear Manual Extension System - System Test
Figure 503/32-34-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-00-730-802

4. Main Gear Manual Extension System Test (Airplane not on Jacks)

(Figure 503)

A. General

(1) The normal extension/retraction system will not operate with the access door for the alternate manual extension handle open. Make sure that the manual extension access door is closed when you will extend or retract the landing gear with the landing gear control lever.

B. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-1184	Scale - Spring, 0-100 Lbs, Tension

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Test

SUBTASK 32-34-00-480-006

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-34-00-860-029

(2) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-34-00-840-005

(3) For Hydraulic System A, make sure that the hydraulic power is off (TASK 29-11-00-860-805).

F. Main Gear Manual Extension System Test (Airplane not on Jacks)

SUBTASK 32-34-00-010-003

(1) Open the manual extension access door for the main gear manual extension handles in the flight compartment floor.

(a) Make sure that the switch under the manual extension access door clicks audibly when the manual extension access door opens.

SUBTASK 32-34-00-700-016

(2) Pull the manual extension handle for the left main gear.

(a) Make sure that the full travel of the manual extension handle is 20.0 in. (508.0 mm) or more.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-00-700-017

(3) Release the manual extension handle for the left main gear.

(a) Make sure that the manual extension handle returns to the stowed position without hesitation or binding.

SUBTASK 32-34-00-480-007

(4) Attach a spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the left main gear.

SUBTASK 32-34-00-730-009

(5) Pull the manual extension handle for the left main gear.

(a) Make sure that the load to start the movement of the manual extension handle from the seat is a minimum of 3.0 lbf (13.3 N).

(b) Make sure that the load during the full travel of the manual extension handle is 18.0 lbf (80.1 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

(c) Make sure that the manual extension mechanism operates freely.

SUBTASK 32-34-00-700-020

(6) Release the manual extension handle for the left main gear.

SUBTASK 32-34-00-080-007

(7) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the left main gear.

SUBTASK 32-34-00-700-018

(8) Pull the manual extension handle for the right main gear.

(a) Make sure that the full travel of the manual extension handle is 20.0 in. (508.0 mm) or more.

SUBTASK 32-34-00-700-019

(9) Release the manual extension handle for the right main gear.

(a) Make sure that the manual extension handle returns to the stowed position without hesitation or binding.

SUBTASK 32-34-00-480-008

(10) Attach a spring scale (0-100 Lbs), STD-1184, to the manual extension handle for the right main gear.

SUBTASK 32-34-00-730-010

(11) Pull the manual extension handle for the right main gear.

(a) Make sure that the load to start the movement of the manual extension handle from the seat is 3.0 lbf (13.3 N) or more.

(b) Make sure that the load during the full travel of the manual extension handle is 18.0 lbf (80.1 N) or less.

NOTE: The manual extension handle must be pulled upward and forward when you measure the load. This will simulate a person seated in the First Officer's seat. Do not pull the manual extension handle straight up.

(c) Make sure that the manual extension mechanism operates freely.

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SUBTASK 32-34-00-700-021

(12) Release the manual extension handle for the right main gear.

SUBTASK 32-34-00-080-008

(13) Remove the spring scale (0-100 Lbs), STD-1184, from the manual extension handle for the right main gear.

SUBTASK 32-34-00-700-022

(14) Make sure the left and right manual extension handles are return to the stowed position without hesitation or binding.

SUBTASK 32-34-00-410-003

(15) Close the manual extension access door for the manual extension handles.

——— END OF TASK ———

TASK 32-34-00-820-802

5. Manual Extension System Access Door Switch Adjustment

(Figure 504)

A. General

(1) This task gives the instructions to adjust (rig) the manual extension system access door switch (access door position switch) for the main landing gears.

(2) Use this task to get the access door position switch into the correct operating range for the access door position switch to operate correctly.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-122	Brush

D. Consumable Materials

Reference	Description	Specification
B00130	Alcohol - Isopropyl	TT-I-735

E. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Adjustment

SUBTASK 32-34-00-860-032

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-34-00-840-003

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

SUBTASK 32-34-00-010-002

(3) Open this access panel:

(TASK 53-14-01-020-801)

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

H. Manual Extension System Access Door Switch Adjustment

SUBTASK 32-34-00-480-009

(1) Open the access door [4] located in the flight compartment.

SUBTASK 32-34-00-160-001

(2) Clean the area around the access door position switch [6] with alcohol, B00130, and a dry brush, STD-122.

NOTE: The location of the access door makes it for dirt and residue found in the flight compartment to collect around the access door position switch. The location of the manual extension system access door will collect dirt and residue found in the cockpit.

SUBTASK 32-34-00-200-002

(3) Do a check of the access door position switch [6] adjustment:

(a) Make sure that the access door [4] stays fully closed with no external force.

(b) With the access door [4] closed, make sure that there is no continuity between Pins A and B of D15510 Connector.

(c) With the access door [4] open, make sure that there is continuity between Pins A and B of D15510 Connector.

(d) With the access door [4] fully closed, the access door position switch [6] should not be fully be bottomed out since this might hold the access door [4] slightly open.

NOTE: The access door position switch has 0.187 in. (4.750 mm) over-travel from the point of actuation (where there is an audible click) to fully bottomed out.

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SUBTASK 32-34-00-820-007

(4) If the access door position switch [6] does not have sufficient over-travel and the access door [4] does not fully close, or if too much force is necessary to close the access door [4], adjust the height of the access door position switch [6]:

NOTE: The access door position switch usually needs three washers below the bracket to correctly rig the access door position switch.

(a) If the access door position switch [6] is bottomed out and is holding the access door [4] open, add one washer [5] at a time below the bracket until the access door position switch [6] does not hold the access door [4] open.

1) Remove the two bolts [3] and bracket [7] to get access to the washers [5].

(b) If the access door position switch [6] does not actuate when the access door [4] is closed, remove one washer [5] at a time from below the bracket until the access door position switch [6] actuates.

1) Remove the two bolts [3] and bracket [7] to get access to the washers [5].

(c) If they were removed, install the two bolts [3] and bracket [7].

(d) Do a check of the access door position switch [6] adjustment again if the washers [5] are added or removed below the bracket.

SUBTASK 32-34-00-780-001

(5) Pressurize hydraulic system A (TASK 29-11-00-860-801).

SUBTASK 32-34-00-820-011

(6) Use the override trigger to move the landing gear lever from the DN position to the UP position three times.

(a) Hold the landing gear lever in each position for three seconds before you move the landing gear lever to the subsequent position.

SUBTASK 32-34-00-820-012

(7) Make sure that the rod end of the downlock actuator tries to extend when you move the landing gear lever from the DN position to the UP position.

SUBTASK 32-34-00-800-001

(8) Move the landing gear lever to the DN position.

SUBTASK 32-34-00-800-002

(9) Remove hydraulic power from system A (TASK 29-11-00-860-805).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-00-410-002

(1) Close this access panel:

(TASK 53-14-01-420-801)

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

SUBTASK 32-34-00-860-031

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

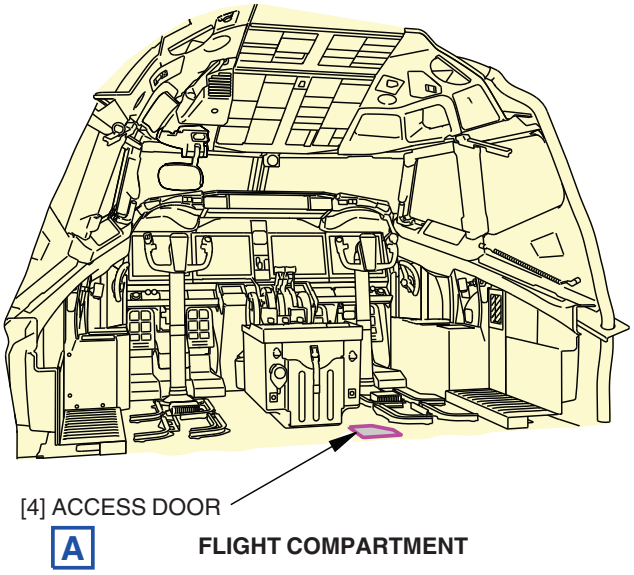
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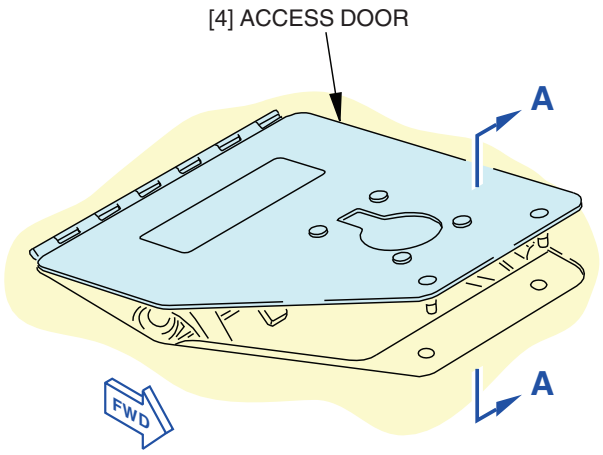
SUBTASK 32-34-00-800-003

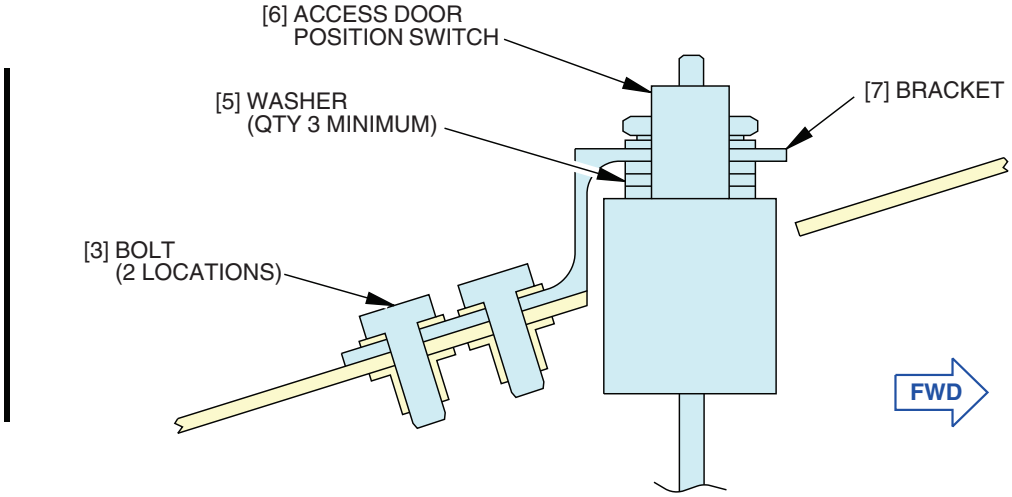
(3) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details







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Landing Gear Manual Extension Access Door Switch Adjustment
Figure 504/32-34-00-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR MANUAL EXTENSION CONTROL MECHANISM - REMOVAL/INSTALLATION

1. General

A. The manual extension control mechanism provides controls for the nose gear manual extension system and each main gear manual extension system. The complete mechanism must be removed and installed when you do maintenance on a part of the drum assembly for the nose or main gear manual extension systems.

B. This procedure has these tasks:

(1) A removal of the manual extension control mechanism (referred to as the manual extension control mechanism [38]) for the landing gear

(2) An installation of the manual extension control mechanism [38] for the landing gear.

(3) A removal of the manual extension system access door switch.

(4) An installation of the manual extension system access door switch.

TASK 32-34-11-000-801

2. Main Gear Manual Extension Control Mechanism Removal

(Figure 401, Figure 402)

A. General

(1) This task gives instructions to remove the Main Gear Manual Extension Control Mechanism.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
33-18-00-960-805	Annunciator And Dimming Module - Module Replacement (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
116	Nose Landing Gear Wheel Well - Right
122	Forward Cargo Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

D. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Removal

SUBTASK 32-34-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-34-11-860-001

(2) For hydraulic system A, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-34-11-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-34-11-010-001

(4) To get access to the turnbuckle locking clips, do this task: Passenger Cabin Floor Panel Removal, TASK 53-21-00-000-801.

SUBTASK 32-34-11-010-002

(5) Disconnect the MLGEA cable [3] at the turnbuckle (Figure 401).

SUBTASK 32-34-11-010-003

(6) Remove the cargo compartment ceiling panels at approximately body station 420 to remove the turnbuckle locking clips.

(a) Disconnect the MLGEA cable [1] and MLGEA cable [2] at the turnbuckles.

F. Main Gear Manual Extension Control Mechanism Removal

SUBTASK 32-34-11-010-004

(1) To get access to the manual extension control mechanism [38], do this step (Figure 402):

(a) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

1) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 32-34-11-020-001

(2) Remove the annunciator and dimming module (TASK 33-18-00-960-805).

NOTE: The annunciator and dimming module is removed to get access to the manual extension control mechanism [38].

SUBTASK 32-34-11-020-002

(3) Remove the cable retainers [34] and disconnect the manual extension handle cables [29] from the control drums [23].

SUBTASK 32-34-11-020-003

(4) Remove the cable retainer bolts [35] and disconnect the MLGEA cable [1], MLGEA cable [2], and MLGEA cable [3] from the control drums [23].

SUBTASK 32-34-11-020-004

(5) Remove the nuts [26], washers [25], bolts [22], and the cable guard standoffs [24] from the outboard cable guard [27] and inboard cable guard [21].

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SUBTASK 32-34-11-020-005

(6) Remove the pin [32], nut [33], and washer [28] from the shaft [30].

SUBTASK 32-34-11-020-006

(7) Remove the shaft [30] from the manual extension control mechanism [38].

(a) When you pull the shaft [30] from the manual extension control mechanism [38], make sure you catch the spacer [31], spacer [36], and spacer [37].

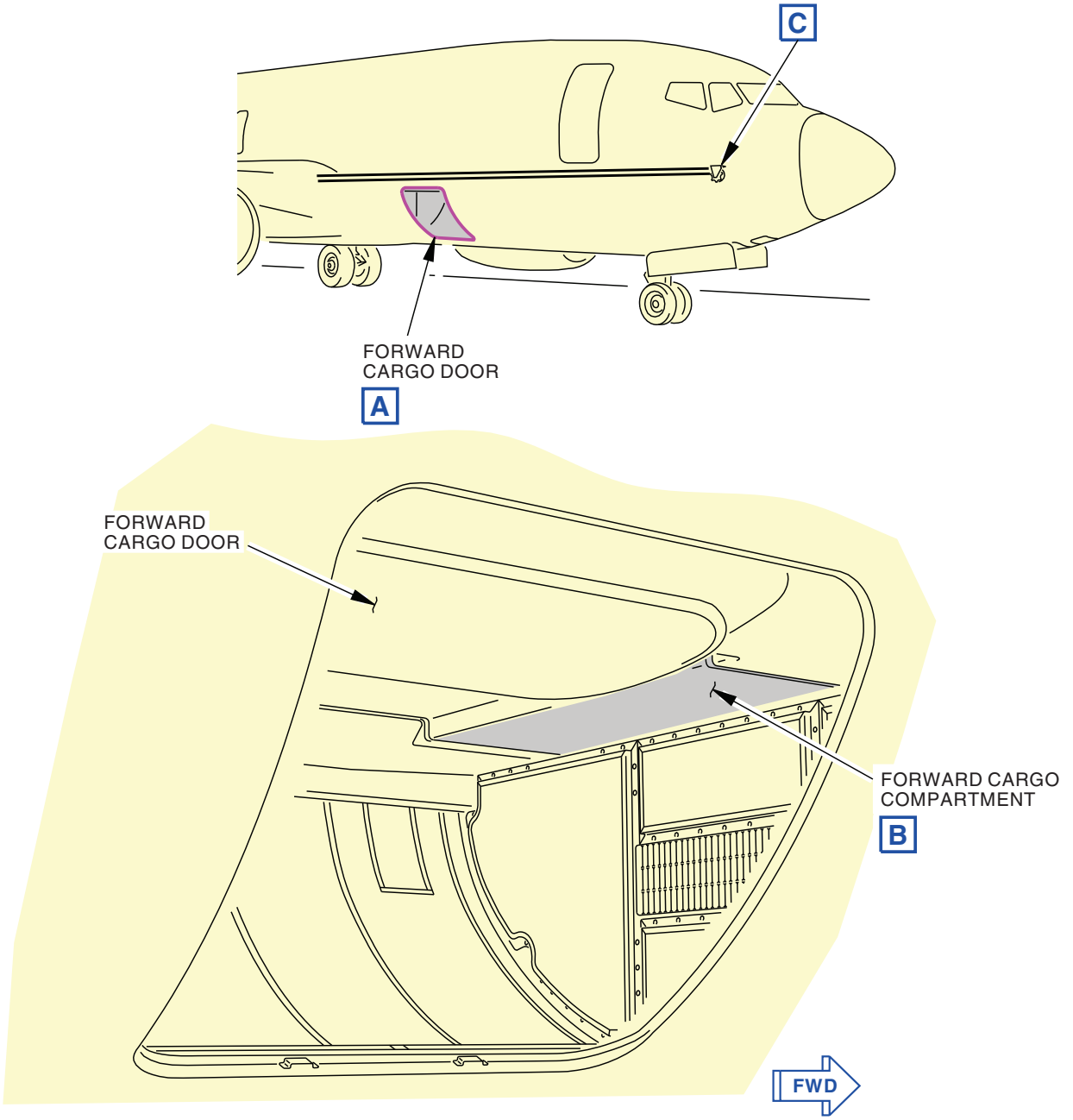
SUBTASK 32-34-11-000-001

(8) Remove the manual extension control mechanism [38], with the control drums [23], outboard cable guard [27] and inboard cable guard [21], from the airplane structure.

(a) Move the outboard end of the manual extension control mechanism [38] forward to remove it.

——— END OF TASK ———

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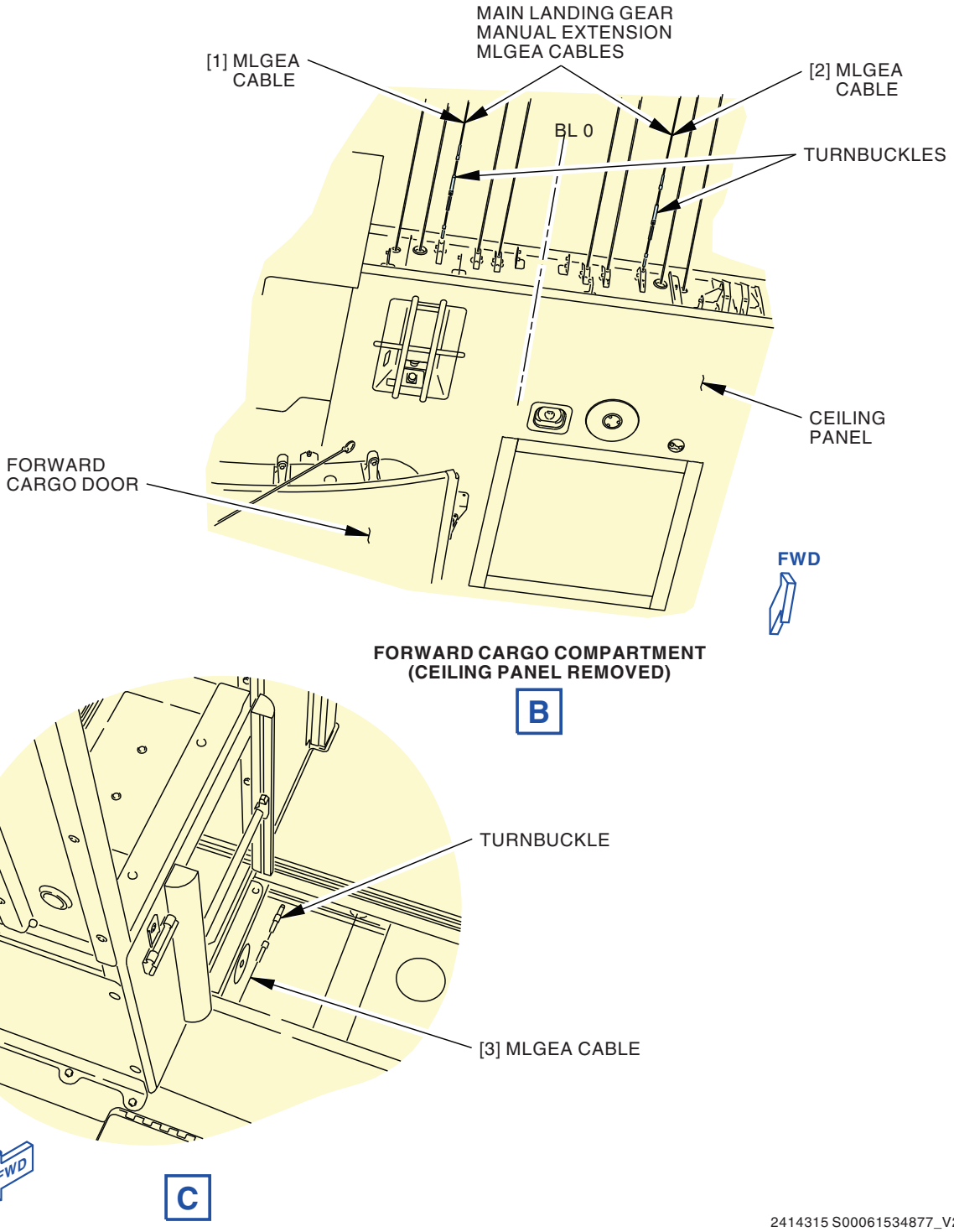
FORWARD CARGO COMPARTMENT
(CEILING PANEL INSTALLED)

A

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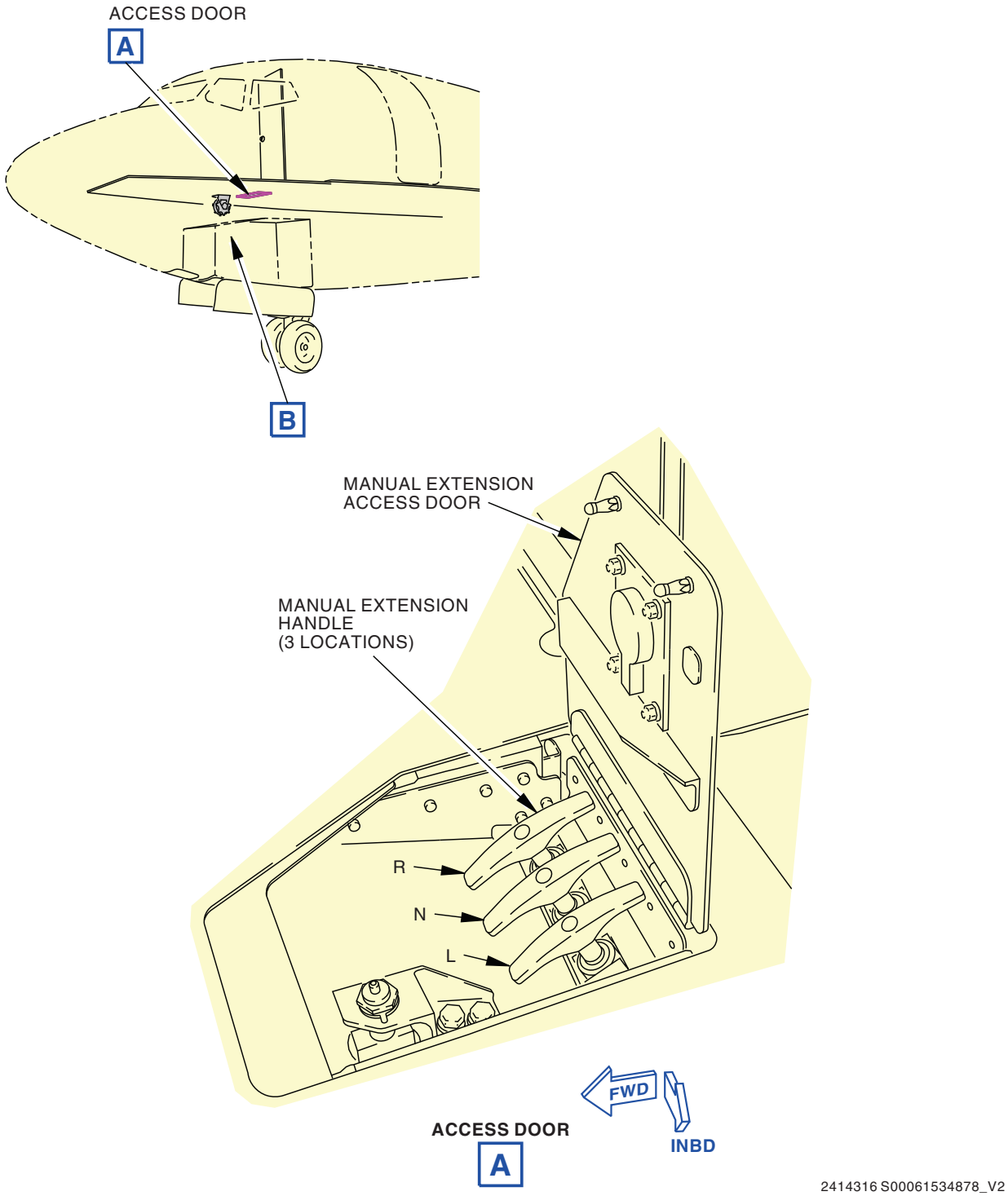
Manual Extension Cable Turnbuckle Locations
Figure 401/32-34-11-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



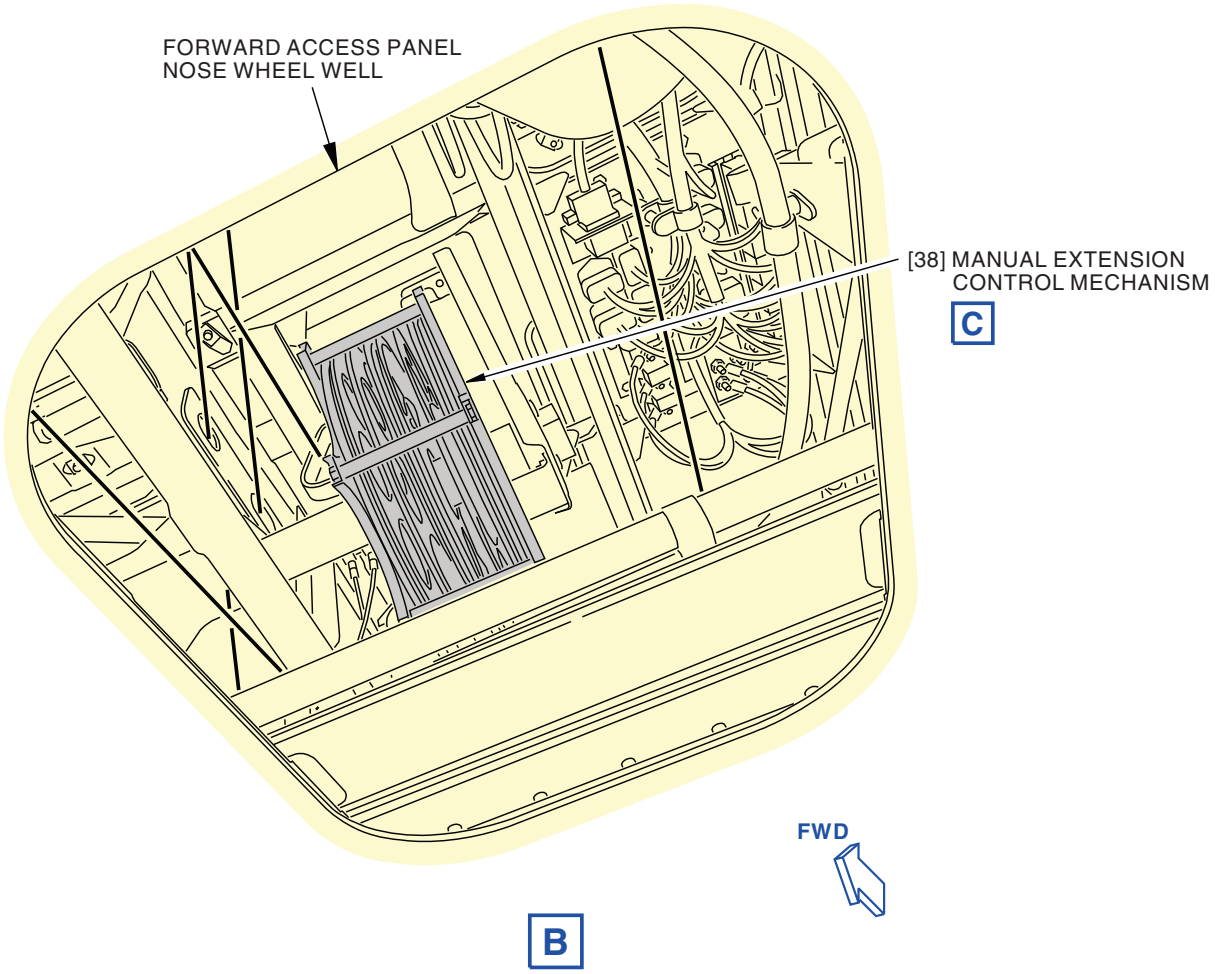
Manual Extension Cable Turnbuckle Locations
Figure 401/32-34-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Manual Extension Control Mechanism Installation
Figure 402/32-34-11-990-802 (Sheet 1 of 4)

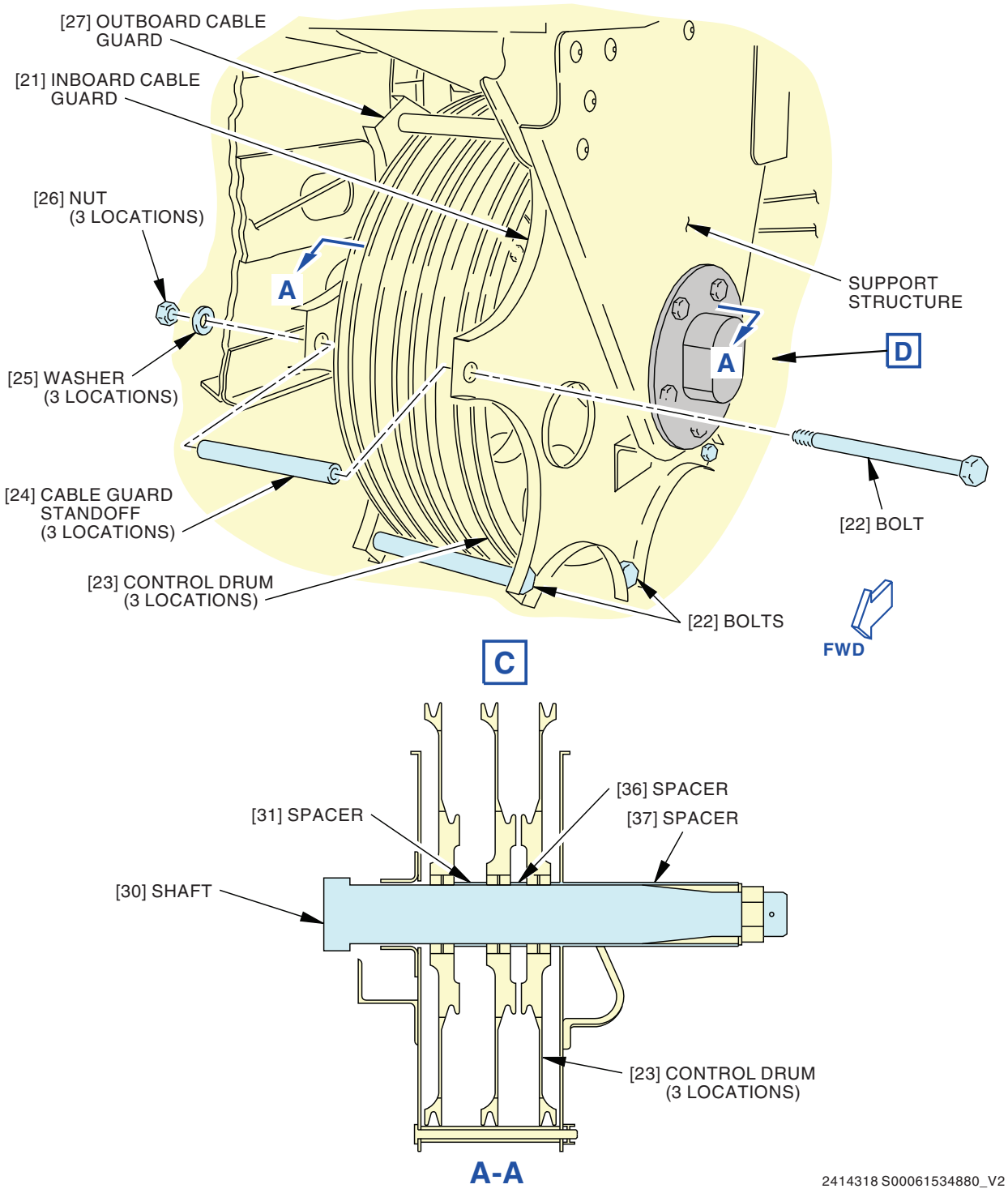
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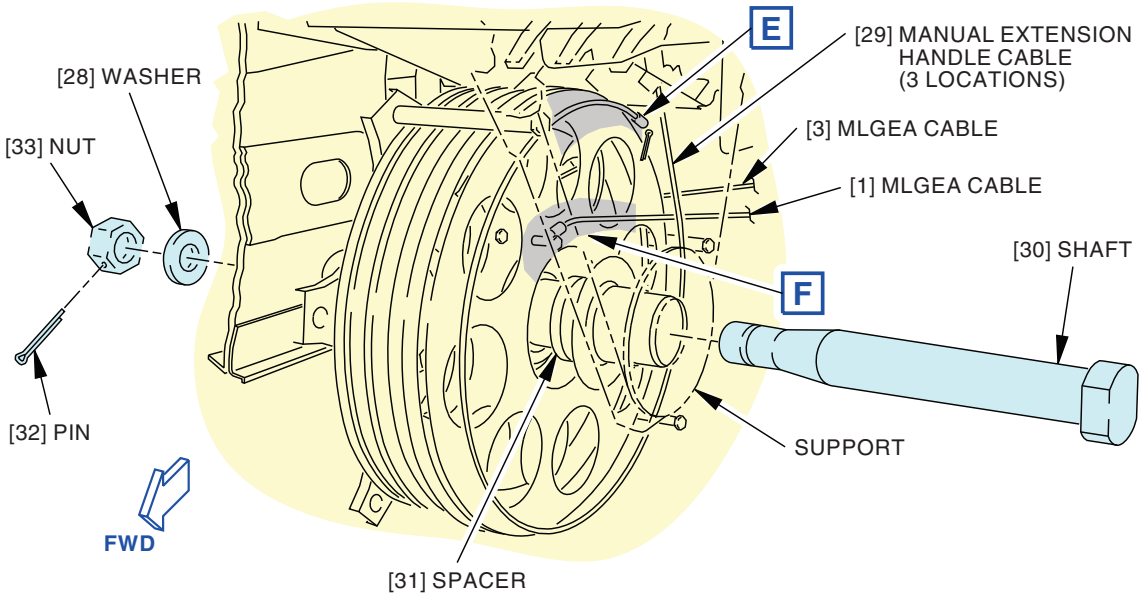
Manual Extension Control Mechanism Installation
Figure 402/32-34-11-990-802 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

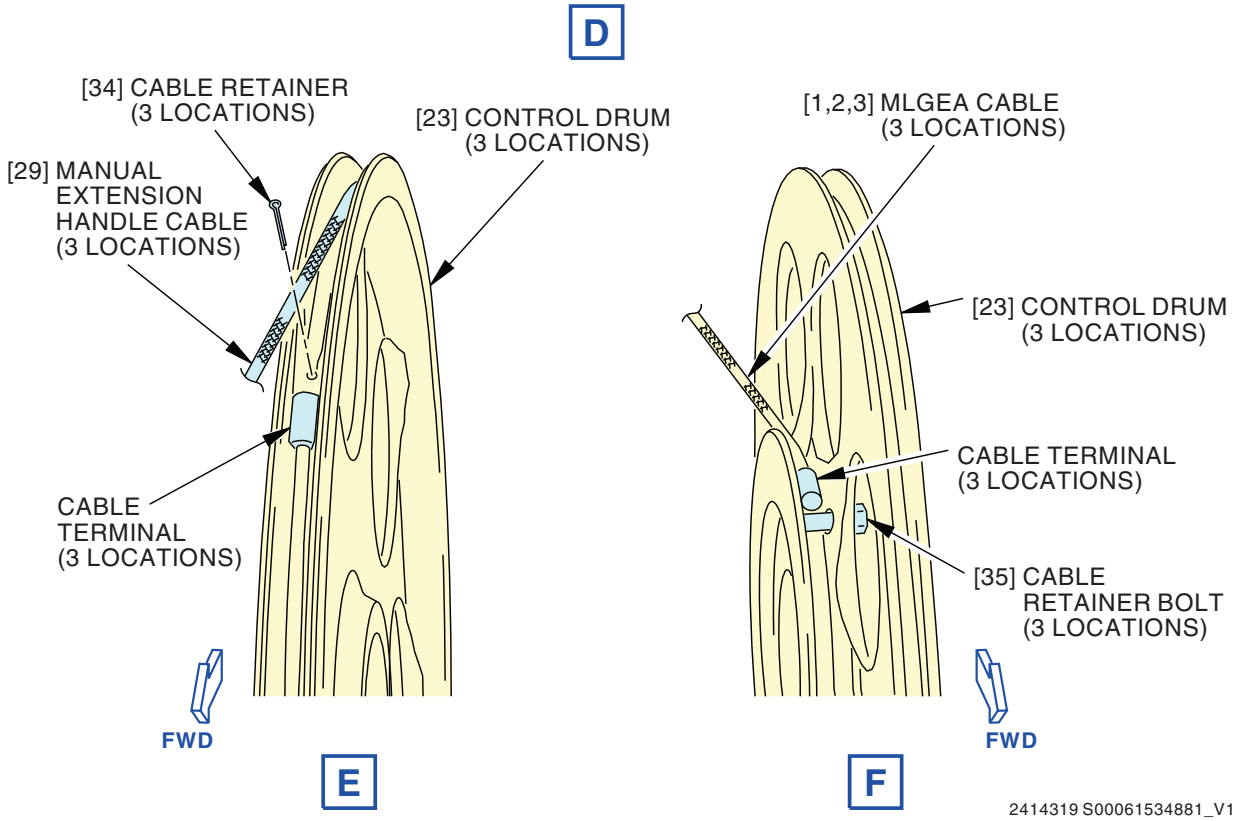


Manual Extension Control Mechanism Installation
Figure 402/32-34-11-990-802 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(INBOARD CABLE GUARD AND INBOARD CONTROL DRUM ARE NOT SHOWN)



Manual Extension Control Mechanism Installation
Figure 402/32-34-11-990-802 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-11-400-801

3. Main Gear Manual Extension Control Mechanism Installation

(Figure 401, Figure 402)

A. General

(1) This task gives instructions to install the Main Gear Manual Extension Control Mechanism.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
32-34-00-730-801	Main Gear Manual Extension System Test (Airplane on Jacks) (P/B 501)
32-34-00-820-801	Main Landing Gear Manual Extension System Adjustment and Test (P/B 501)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-35-00-820-801	Nose Landing Gear Manual Extension System Adjustment (P/B 501)
33-18-00-960-805	Annunciator And Dimming Module - Module Replacement (P/B 201)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
38	Manual extension control mechanism	32-34-11-01-095	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
116	Nose Landing Gear Wheel Well - Right
122	Forward Cargo Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

F. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Main Gear Manual Extension Control Mechanism Installation

SUBTASK 32-34-11-400-001

(1) Put the control drums [23], spacer [31], spacer [36] and spacer [37] in position between the inboard cable guard [21] and outboard cable guard [27] (Figure 402).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-34-11-420-001

(2) Put the shaft [30] through the manual extension control mechanism [38], control drum [23], spacer [31], control drum [23], spacer [36], control drum [23], outboard cable guard [27], spacer [37], and support structure.

SUBTASK 32-34-11-420-002

(3) Apply a thin coat of grease, D00633, to the threads of the shaft [30].

SUBTASK 32-34-11-420-010

(4) Install the washer [28] and nut [33] on the shaft [30].

SUBTASK 32-34-11-420-003

(5) Tighten the nut [33] to 50 in-lb (6 N·m) - 180 in-lb (20 N·m).

(a) Tighten to align with the pin [32] hole.

(b) Do not apply more than a maximum of 180 in-lb (20 N·m).

SUBTASK 32-34-11-420-004

(6) Install the pin [32].

SUBTASK 32-34-11-420-005

(7) Install the MLGEA cable [1], MLGEA cable [2], and MLGEA cable [3] on the control drums [23].

(a) Install the cable retainer bolt [35] to attach the MLGEA cable [1], MLGEA cable [2], and MLGEA cable [3] to the control drums [23].

SUBTASK 32-34-11-420-006

(8) Install the manual extension handle cable [29] on the control drums [23].

(a) Install the cable retainers [34] to attach the manual extension handle cables [29] to the control drums [23].

1) Bend the end of the cable retainers [34] into the control drums [23].

SUBTASK 32-34-11-420-007

(9) Connect the MLGEA cable [1] and MLGEA cable [2] at the turnbuckles, located at approximately body station 420.

SUBTASK 32-34-11-420-008

(10) Connect the MLGEA cable [3] at the turnbuckle (Figure 401).

SUBTASK 32-34-11-420-009

(11) Install the annunciator and dimming module (TASK 33-18-00-960-805).

H. Main Gear Manual Extension Control Mechanism Installation Test

SUBTASK 32-34-11-820-001

(1) Do the adjustment of the manual extension system for the main landing gear (TASK 32-34-00-820-801).

SUBTASK 32-34-11-710-001

(2) Do a test on the manual extension system for the main landing gear (TASK 32-34-00-730-801).

SUBTASK 32-34-11-820-002

(3) Do the adjustment of the manual extension system for the nose landing gear (TASK 32-35-00-820-801).

SUBTASK 32-34-11-710-002

(4) Do a test on the manual extension system for the nose landing gear (TASK 32-35-00-730-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-11-410-001

(1) Install this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 32-34-11-410-002

(2) Install the cargo compartment ceiling panels at approximately body station 420.

SUBTASK 32-34-11-410-003

(3) Install the cabin floor panel (TASK 53-21-00-400-801).

——— END OF TASK ———

TASK 32-34-11-820-801

4. Manual Extension System Access Door Switch Removal

(Figure 403)

A. General

(1) This procedure gives the instructions to remove the manual extension system access door switch (access door position switch [65]) for the main landing gears.

B. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
116	Nose Landing Gear Wheel Well - Right
122	Forward Cargo Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

D. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Procedure

SUBTASK 32-34-11-860-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-34-11-840-001

(2) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

SUBTASK 32-34-11-010-005

(3) Open this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

SUBTASK 32-34-11-010-006

(4) Open the access door [51].

SUBTASK 32-34-11-020-007

(5) Remove the access door position switch [65] from the airplane.

(a) Remove the screws [58] and washers [59] from the switch bracket [63].

(b) Remove the nut [60], lock washer [61], keyway washer [62], switch bracket [63] and the washers [64] from the access door position switch [65].

1) Make a record of the quantity of installed washers [64].

(c) Put the switch end of the access door position switch [65] through the hole in the floor.

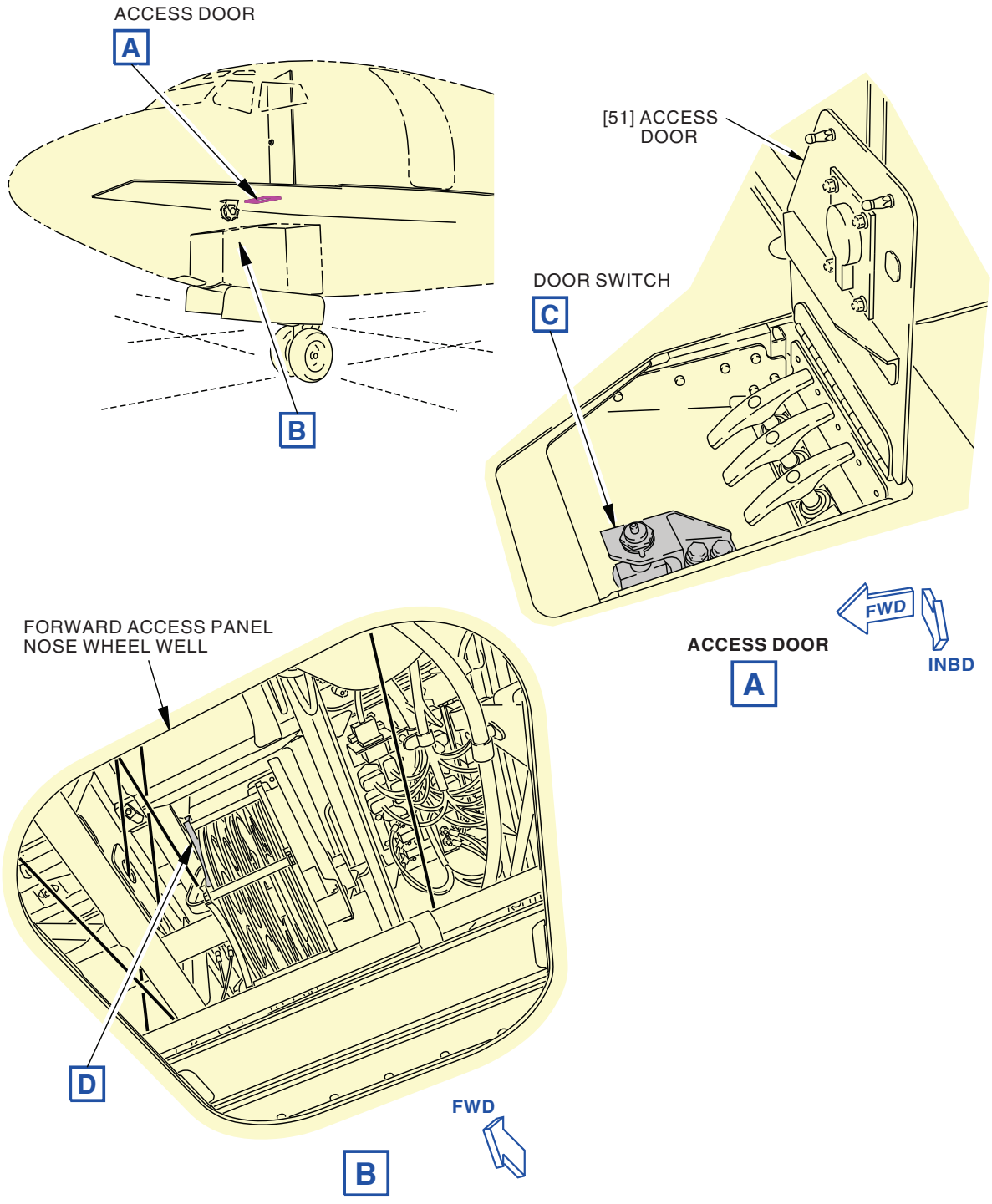
(d) Remove the screw [66], washer [67] and clamp [68] from the structure.

(e) Disconnect the access door position switch [65] from the wire [57] at the location of the clamp [69].

(f) Remove the access door position switch [65] from the airplane.

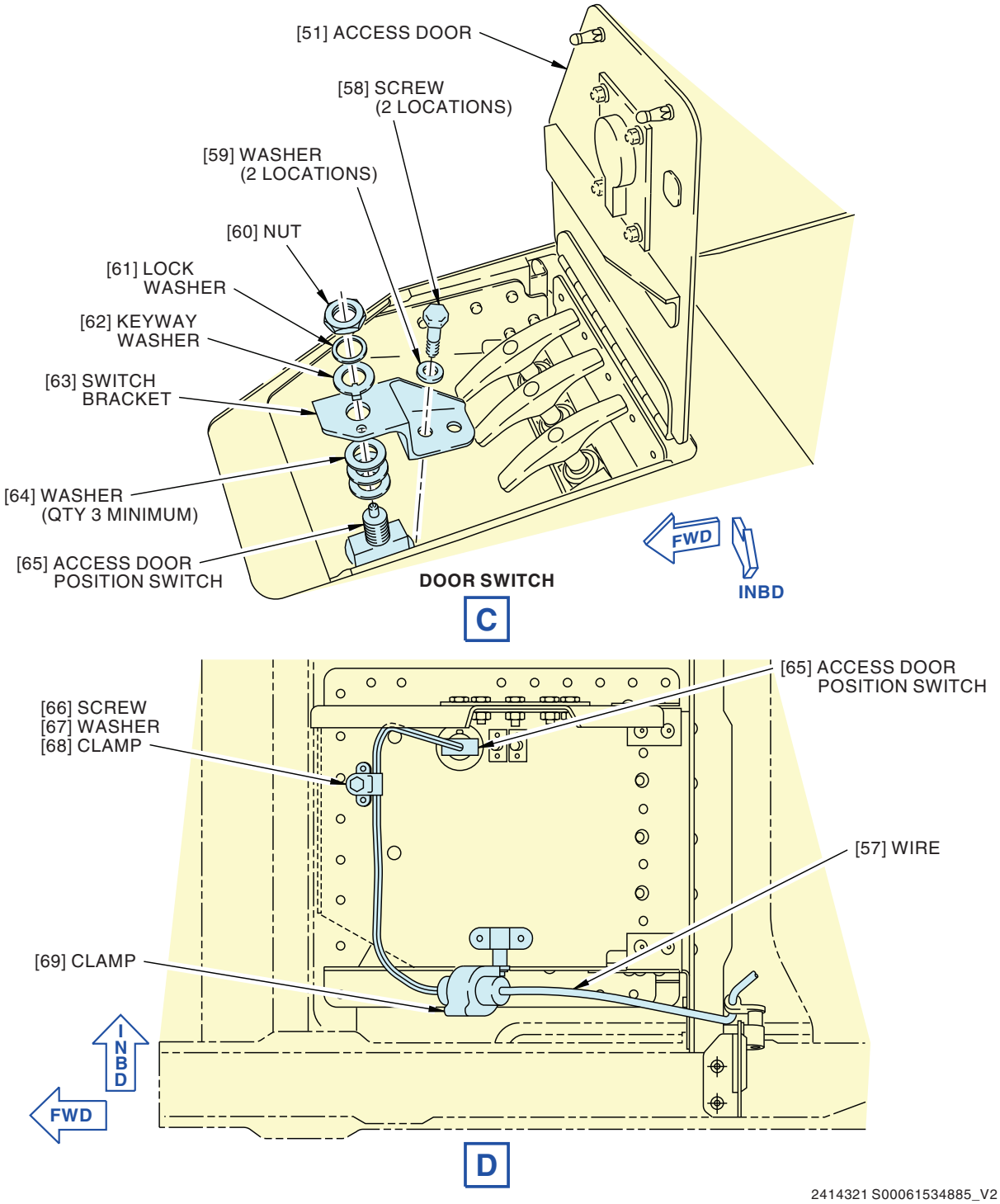
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Landing Gear Manual Extension Access Door Switch Installation
Figure 403/32-34-11-990-803 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Landing Gear Manual Extension Access Door Switch Installation
Figure 403/32-34-11-990-803 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-11-820-802

5. Manual Extension System Access Door Switch Installation

(Figure 403)

A. General

(1) This procedure gives the instructions to install the manual extension system access door switch (access door position switch [65]) for the main landing gears.

B. References

Reference	Title
05-51-91-790-801	Cabin Pressure Leak Test (P/B 201)
32-34-00-820-802	Manual Extension System Access Door Switch Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
116	Nose Landing Gear Wheel Well - Right
122	Forward Cargo Compartment - Right
210	Subzone - Control Compartment - Body Station 178.00 to Body Station 259.50
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

D. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

E. Manual Extension System Access Door Switch Installation Procedure

SUBTASK 32-34-11-400-002

(1) Connect the access door position switch [65] to the wire [57] at the clamp [69] location.

(a) Make sure the plug and socket are fully engaged, which is identified with two clicks.

(b) Put the clamp [68] on the wires of the access door position switch [65].

(c) Put the clamp [68] in position on the structure.

(d) Install the washer [67] and screw [66] to attach the clamp [68] to the structure.

(e) Put the switch end of the access door position switch [65] through the hole in the floor.

(f) Install the washers [64], switch bracket [63], keyway washer [62], lock washer [61] and nut [60] on the access door position switch [65].

NOTE: Use a minimum of three washers [64]. If necessary, use more washers [64] for adjustment.

1) Tighten the nut [60] to 13.0 ±2.0 in-lb (1.5 ±0.3 N·m).

(g) Put the switch bracket [63] in position on the structure.

(h) Install the washers [59] and screws [58] on the switch bracket [63].

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SUBTASK 32-34-11-410-004

(2) Close this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

(b) Do this task: Cabin Pressure Leak Test, TASK 05-51-91-790-801.

SUBTASK 32-34-11-840-002

(3) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	16	C01432	LANDING GEAR ALTN EXTEND SOL

SUBTASK 32-34-11-010-007

(4) Do the adjustment of the access door position switch [65] (TASK 32-34-00-820-802).

SUBTASK 32-34-11-410-006

(5) Close the manual extension access door [51].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-11-410-005

(1) Install the nose wheel well access panel (TASK 53-14-01-420-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR MANUAL EXTENSION MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks that apply to the left or right main landing gear:

(1) A removal of the manual extension mechanism (referred to as the manual extension mechanism [21]), for the main gear

(2) An installation of the manual extension mechanism [21] for the main gear.

TASK 32-34-21-000-801

2. Main Gear Manual Extension Mechanism Removal

(Figure 401, Figure 402)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
25-22-00-000-801	Passenger Seat Removal (P/B 401)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

D. Prepare for the Removal

SUBTASK 32-34-21-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-34-21-860-001

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-34-21-580-001

(3) Install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-34-21-480-002

WARNING MAKE SURE THAT YOU INSTALL THE RIG PIN IN THE MANUAL EXTENSION MECHANISM. IF YOU DO NOT OBEY, THE SPRING IN THE MECHANISM COULD UNWIND AND CAUSE INJURY TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Install the rig pin LGEA2 (L/H), or LGEA3 (R/H) from rigging pin kit, SPL-1585, through the rig pin holes in the manual extension mechanism [21].

SUBTASK 32-34-21-010-001

(5) Remove the ceiling panels in the forward cargo compartment that are at approximately body station 420.

SUBTASK 32-34-21-020-001

(6) Remove the locking clips from the turnbuckle [2].

SUBTASK 32-34-21-020-002

(7) Loosen the MLGEA cable [1] at the turnbuckle [2].

SUBTASK 32-34-21-010-002

(8) For the passenger seats that are at approximately body station 692 (Passenger Seat Removal, TASK 25-22-00-000-801).

SUBTASK 32-34-21-010-003

(9) Remove the floor panels at approximately body station 692.

E. Main Gear Manual Extension Mechanism Removal

SUBTASK 32-34-21-020-003

(1) Remove the pin [33], nut [32], washer [31], bolt [29], and washer [30] to disconnect the control rod from the uplock kicker arm.

SUBTASK 32-34-21-020-004

(2) Remove the nut [28], washers [26], bolt [25], and spacer [27] from the cable guard bracket on the manual extension mechanism [21].

SUBTASK 32-34-21-020-005

(3) Remove the washers [23], and bolt [22] to disconnect the manual extension mechanism [21] from the structure.

SUBTASK 32-34-21-020-006

(4) Remove the pin [34].

SUBTASK 32-34-21-020-007

(5) Disconnect the MLGEA cable [1] from the manual extension mechanism [21].

SUBTASK 32-34-21-020-008

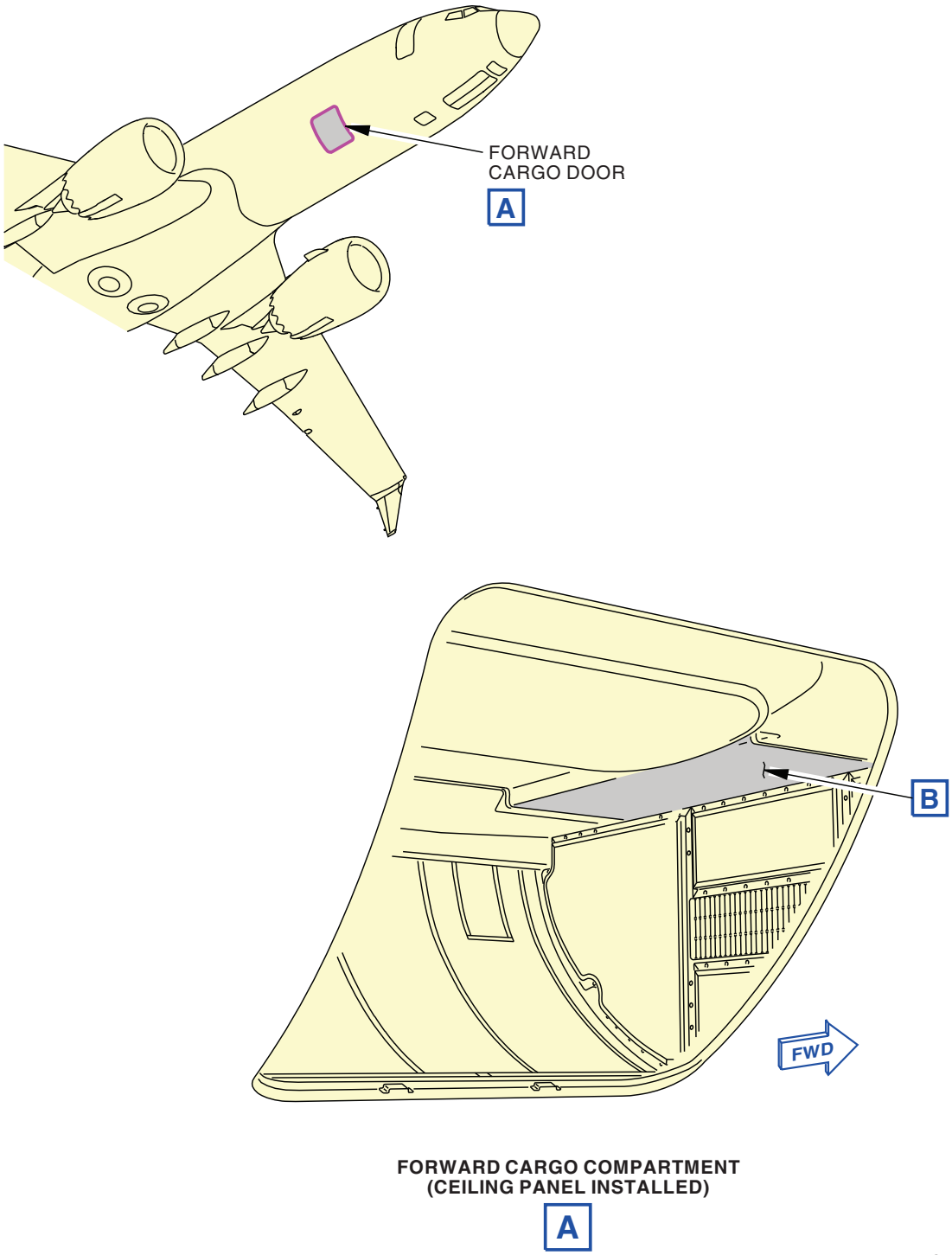
(6) Remove the manual extension mechanism [21] from the airplane.

SUBTASK 32-34-21-420-001

(7) Install the bolt [25], spacer [27], washers [26], and nut [28] on the cable guard bracket on the manual extension mechanism [21].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



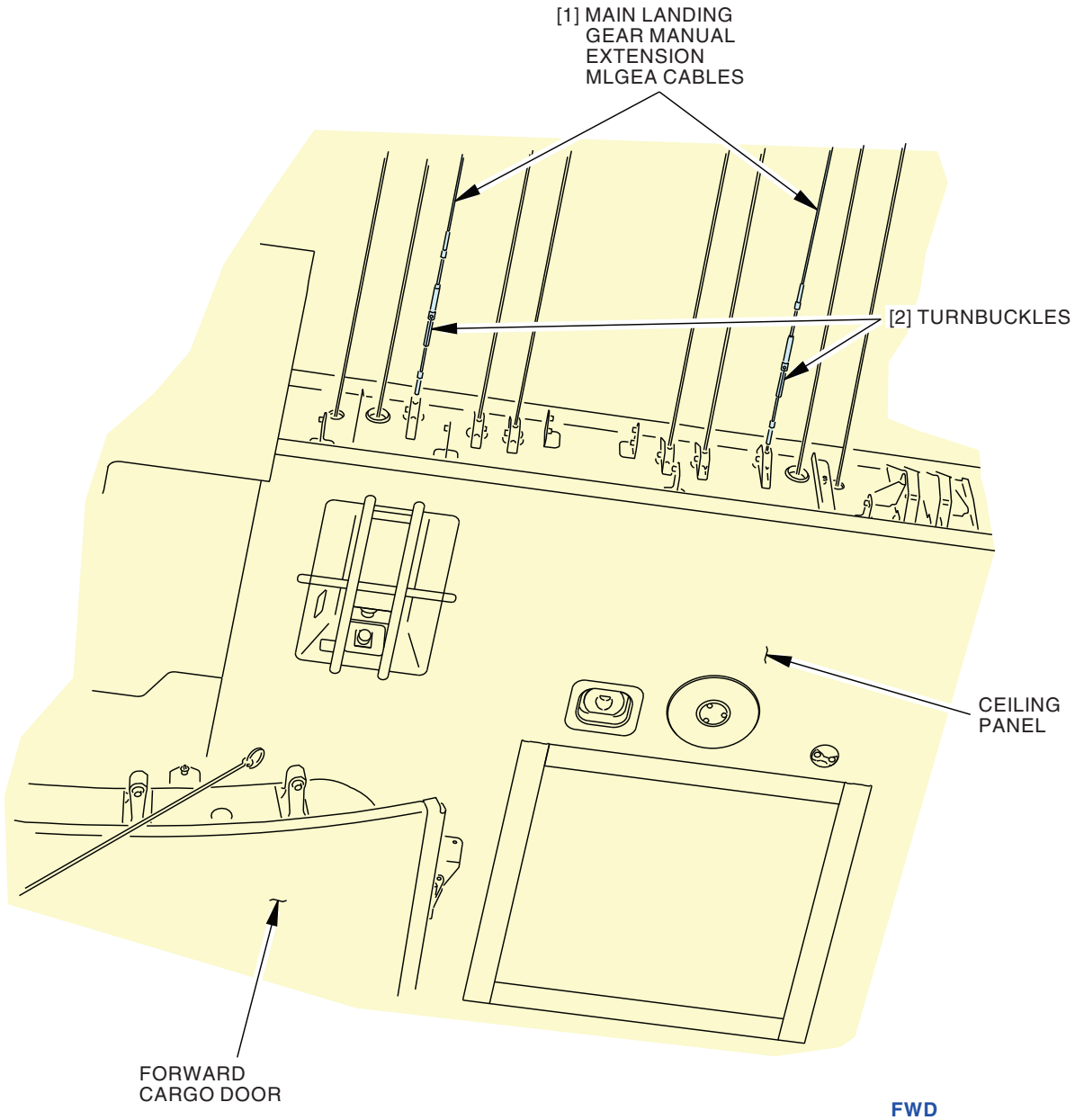
**FORWARD CARGO COMPARTMENT
(CEILING PANEL INSTALLED)**

A

2414322 S00061534890_V1

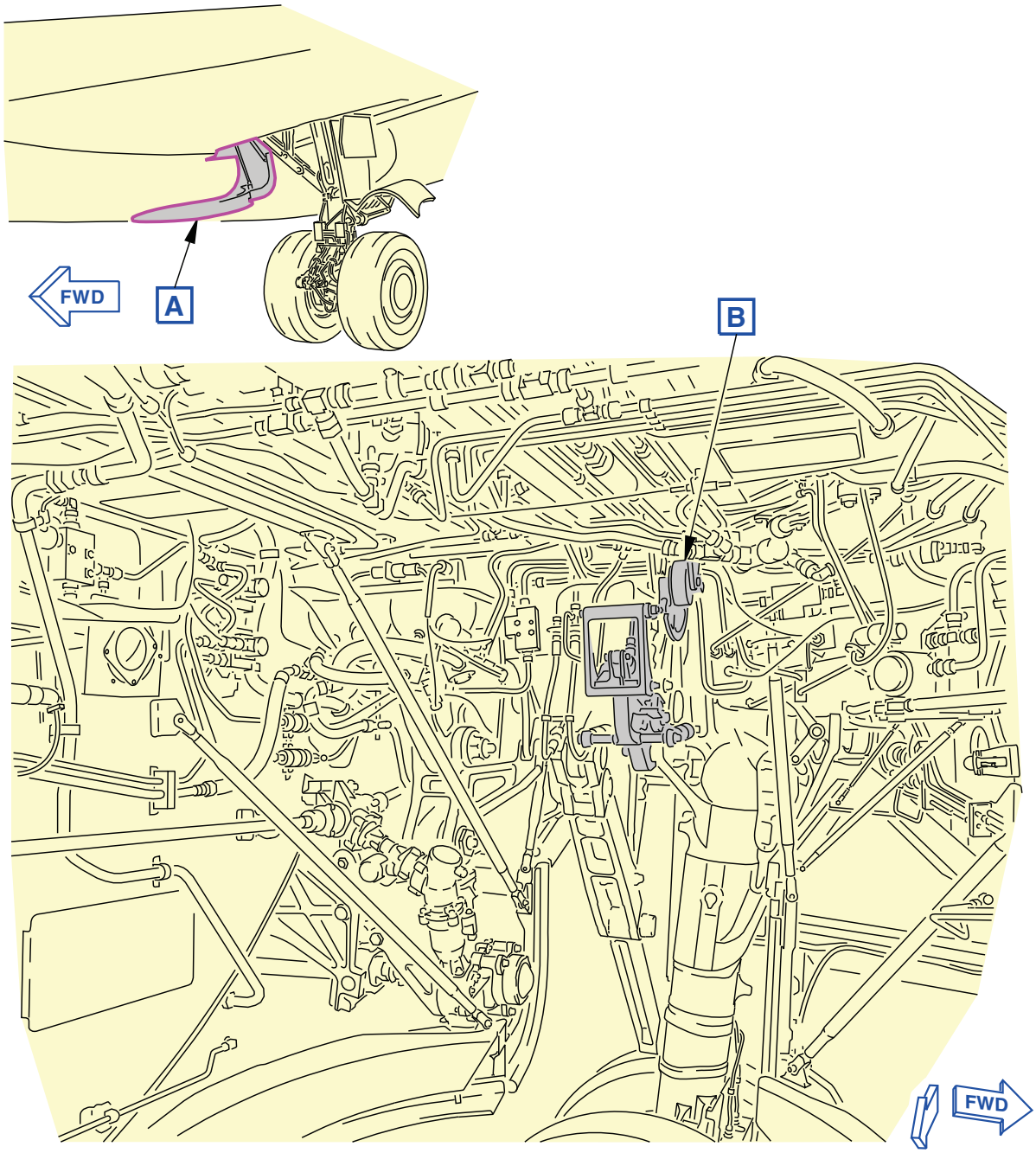
**Manual Extension Cable Turnbuckle Locations
Figure 401/32-34-21-990-801 (Sheet 1 of 2)**

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Manual Extension Cable Turnbuckle Locations
Figure 401/32-34-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



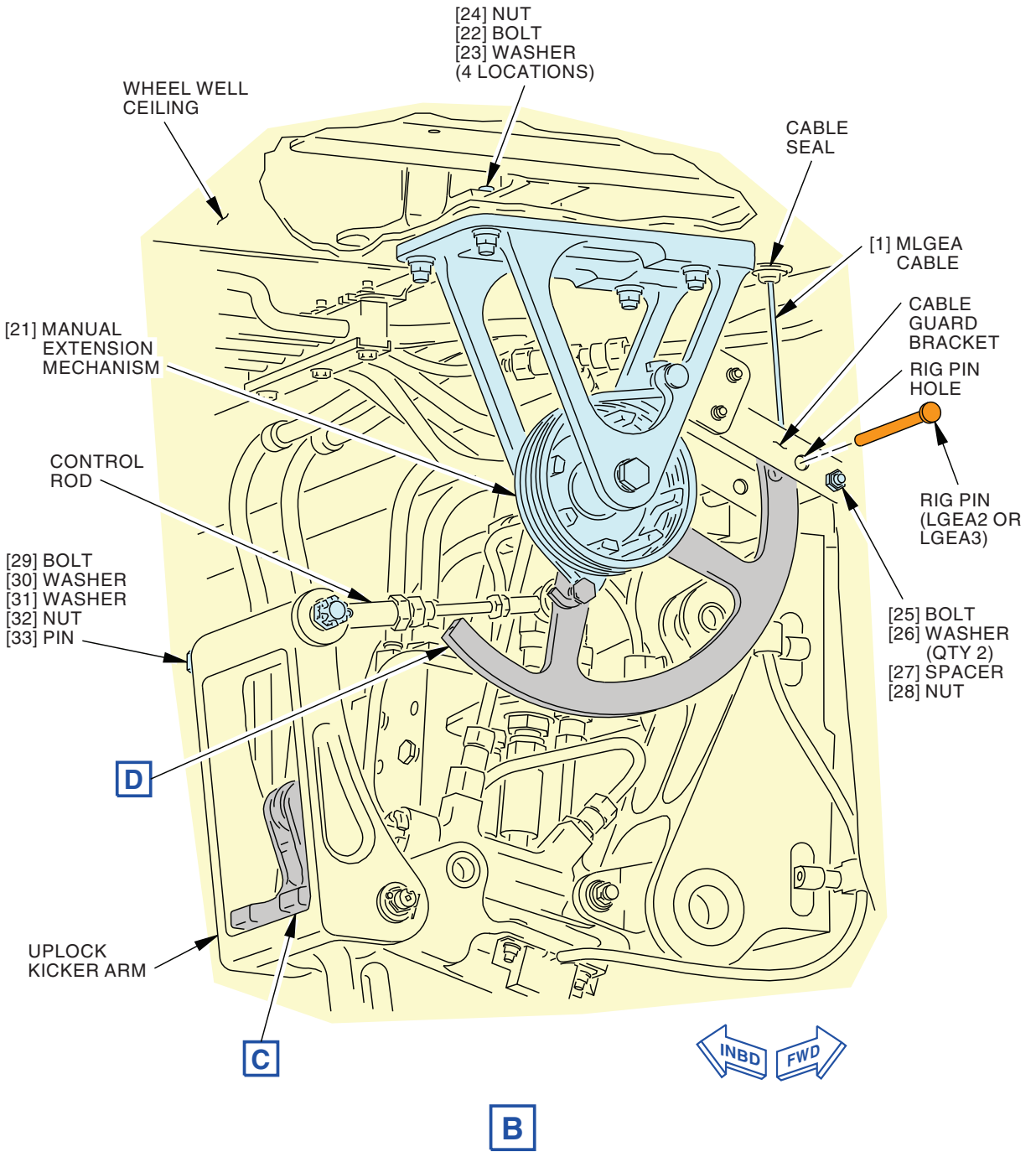
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2414324 S00061534892_V1

Main Landing Gear Manual Extension Mechanism Installation
Figure 402/32-34-21-990-802 (Sheet 1 of 3)

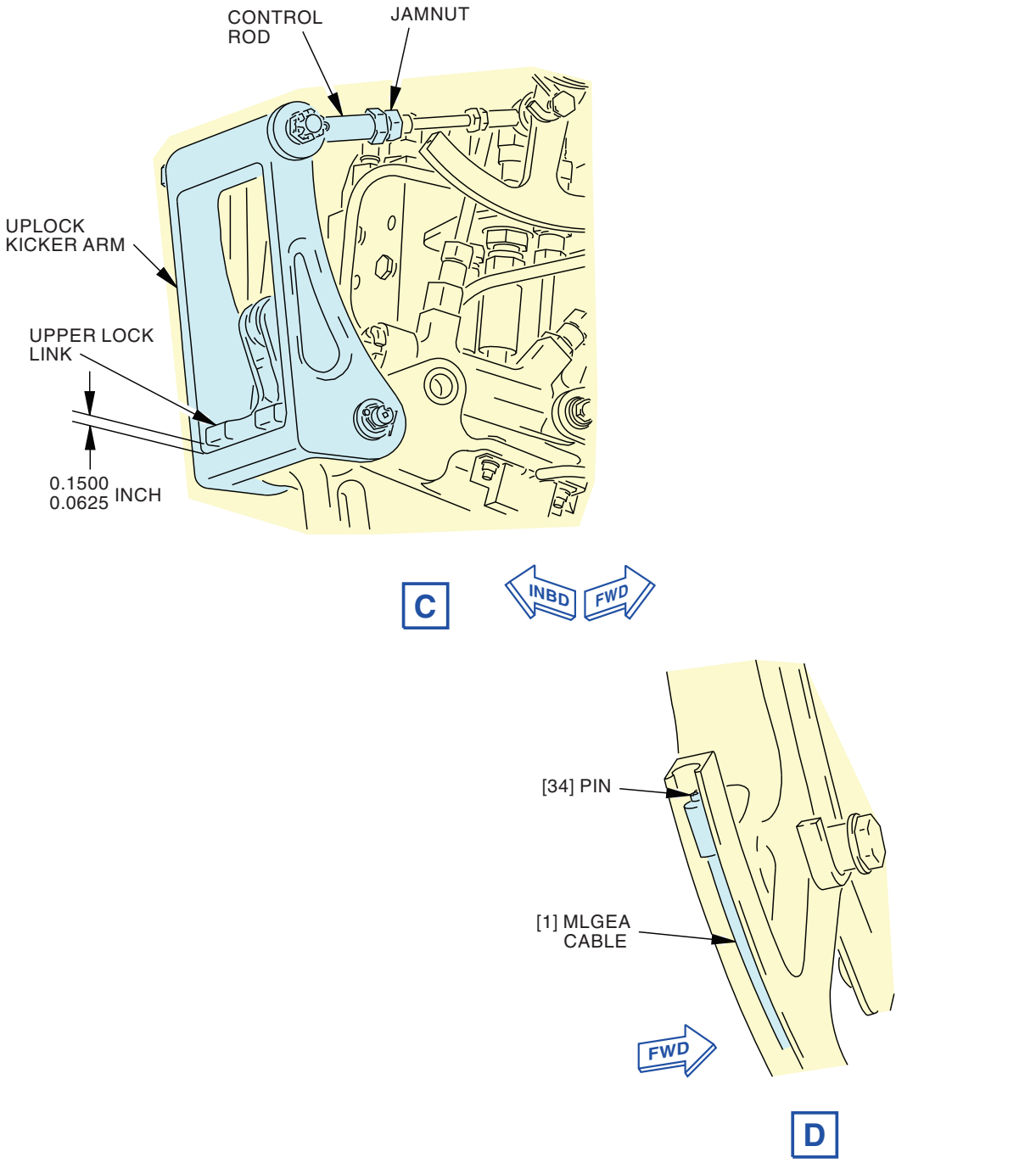
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414325 S00061534893_V3

Main Landing Gear Manual Extension Mechanism Installation
Figure 402/32-34-21-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414326 S00061534894_V2

Main Landing Gear Manual Extension Mechanism Installation
Figure 402/32-34-21-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-21-400-801

3. Main Gear Manual Extension Mechanism Installation

(Figure 401, Figure 402)

A. References

Reference	Title
25-22-00-400-801	Passenger Seat Installation (P/B 401)
32-32-31-400-801	Main Gear Uplock Assembly Installation (P/B 401)
32-34-00-730-801	Main Gear Manual Extension System Test (Airplane on Jacks) (P/B 501)
32-34-00-730-802	Main Gear Manual Extension System Test (Airplane not on Jacks) (P/B 501)
32-34-00-820-801	Main Landing Gear Manual Extension System Adjustment and Test (P/B 501)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
21	Manual extension mechanism	32-34-21-01-020	PTF ALL

E. Location Zones

Zone	Area
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
241	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Left
242	Aft Passenger Compartment - Station 663.75 to Aft Pressure Bulkhead - Right

F. Main Gear Manual Extension Mechanism Installation

SUBTASK 32-34-21-600-001

(1) Apply sealant, A00247, to the entire top surface of the bracket on the manual extension mechanism [21].

SUBTASK 32-34-21-420-002

(2) Put the manual extension mechanism [21] in its position on the main wheel well ceiling.

SUBTASK 32-34-21-600-002

(3) Apply sealant, A00247, to the threads and below the head of the bolts [22], and washers [23].

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SUBTASK 32-34-21-420-003

(4) Install the bolts [22], and washers [23].

SUBTASK 32-34-21-020-009

(5) Connect the MLGEA cable [1] to the manual extension mechanism [21].

SUBTASK 32-34-21-020-010

(6) Install the pin [34].

SUBTASK 32-34-21-420-004

(7) Install the bolt [29], and washer [30] to connect the control rod of the manual extension mechanism [21] to the uplock kicker arm.

(a) Put the countersink on the washer [30] against the head of the bolt [29].

SUBTASK 32-34-21-420-005

(8) Install the washer [31], and nut [32].

(a) Do not tighten the nut [32] fully.

SUBTASK 32-34-21-820-001

(9) Make sure the uplock assembly is correctly adjusted to engage the uplock roller on the main gear when the gear is retracted (Main Gear Uplock Assembly Installation, TASK 32-32-31-400-801).

SUBTASK 32-34-21-980-001

(10) Put the uplock hook in the up and locked position.

SUBTASK 32-34-21-820-002

(11) Install the rig pin LGEA2 (L/H), or LGEA3 (R/H) from rigging pin kit, SPL-1585, through the rig pin holes in the manual extension mechanism [21].

SUBTASK 32-34-21-820-003

(12) Loosen the jamnut on the control rod on the manual extension mechanism [21].

SUBTASK 32-34-21-820-004

(13) Adjust the control rod to get the 0.0625 in. (1.59 mm) to 0.1500 in. (3.8100 mm) gap between the uplock kicker arm and the upper lock link (Figure 402).

SUBTASK 32-34-21-420-006

(14) Tighten the jamnut on the control rod to 127.5 +32.5 / -32.6 in-lb (14 +4 / -4 N·m).

SUBTASK 32-34-21-420-007

(15) Install a lockwire on the jamnut on the control rod.

SUBTASK 32-34-21-080-001

(16) Remove the rig pin LGEA2, or LGEA3 from the manual extension mechanism [21].

SUBTASK 32-34-21-980-002

(17) Put the uplock hook in the down and locked position before performing a gear retraction.

SUBTASK 32-34-21-700-001

(18) Turn the cable quadrant to make sure the control rod does not rub against the uplock kicker arm.

(a) Install the rig pin LGEA2, or LGEA3 from rigging pin kit, SPL-1585, through the rig pin holes in the manual extension mechanism [21].

SUBTASK 32-34-21-420-008

(19) Do these steps to tighten the bolt [29] and nut [32] to a torque of 80 in-lb (9 N·m) plus run-on torque (minimum) to 240 in-lb (27 N·m) (maximum):

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(a) Begin at the minimum torque value and increase the torque as necessary to align the castellation in the nut [32] with the hole in the bolt [29].

(b) Make sure not to exceed more than 240 in-lb (27 N·m) applied torque.

(c) Install the pin [33].

SUBTASK 32-34-21-420-009

(20) Tighten the MLGEA cable [1] at the turnbuckle [2].

SUBTASK 32-34-21-820-005

(21) Do the adjustment of the manual extension system for the main gear (Main Landing Gear Manual Extension System Adjustment and Test, TASK 32-34-00-820-801).

G. Main Gear Manual Extension Mechanism Installation Test

SUBTASK 32-34-21-710-001

(1) Do a test on the manual extension system for the main gear.

(a) If the airplane is on jacks, do this task: Main Gear Manual Extension System Test (Airplane on Jacks), TASK 32-34-00-730-801.

(b) If the airplane is not on jacks, do this task: Main Gear Manual Extension System Test (Airplane not on Jacks), TASK 32-34-00-730-802.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-21-410-001

(1) Install the ceiling panels that you removed in the forward cargo compartment.

SUBTASK 32-34-21-410-002

(2) Install the floor panels that you removed at approximately body station 692.

SUBTASK 32-34-21-410-003

(3) Install the passenger seats that you removed at approximately body station 692 (TASK 25-22-00-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN GEAR MANUAL EXTENSION SYSTEM CABLES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the manual extension system cables (referred to as cable D (MLGEA-L) [1], cable C (MLGEA-L) [3], cable A (MLGEA-R) [4], or cable B (MLGEA-R) [5]) for the main landing gear

(2) Installation of the manual extension system cables for the main landing gear.

TASK 32-34-31-000-801

2. Main Gear Manual Extension System Cables Removal

(Figure 401)

A. References

Reference	Title
20-10-04-000-802	Control Cable Half Ball Pressure Seal Removal (P/B 401)
20-10-91-000-801	Control Cables Removal (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

C. Main Gear Manual Extension System Cables Removal

SUBTASK 32-34-31-010-001

(1) Remove the cargo compartment ceiling panels [6] at approximately body station 420 to remove the turnbuckle locking clips [7].

(a) Disconnect the cable D (MLGEA-L) [1] from the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] from the cable B (MLGEA-R) [5] at the turnbuckle [2].

SUBTASK 32-34-31-010-002

(2) Remove the forward access panel from the ceiling of the nose wheel well.

SUBTASK 32-34-31-010-003

(3) Remove the nuts [11], washers [10], cable guard standoffs [9], and bolts [8] from the manual extension control mechanism.

SUBTASK 32-34-31-020-001

(4) Remove the cotter pins [12] to disconnect the manual extension handle cable from the MLGEA control drum on the manual extension control mechanism (Detail E).

SUBTASK 32-34-31-020-002

(5) Remove the cable retainer bolts [13] (Detail F).

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SUBTASK 32-34-31-020-003

(6) Disconnect the cable C (MLGEA-L) [3] or cable A (MLGEA-R) [4] from the MLGEA control drum on the manual extension control mechanism.

SUBTASK 32-34-31-020-004

(7) Remove the cable seal [14] on the wheel well of the main landing gear (TASK 20-10-04-000-802).

SUBTASK 32-34-31-020-005

(8) Remove the pin [15].

SUBTASK 32-34-31-020-006

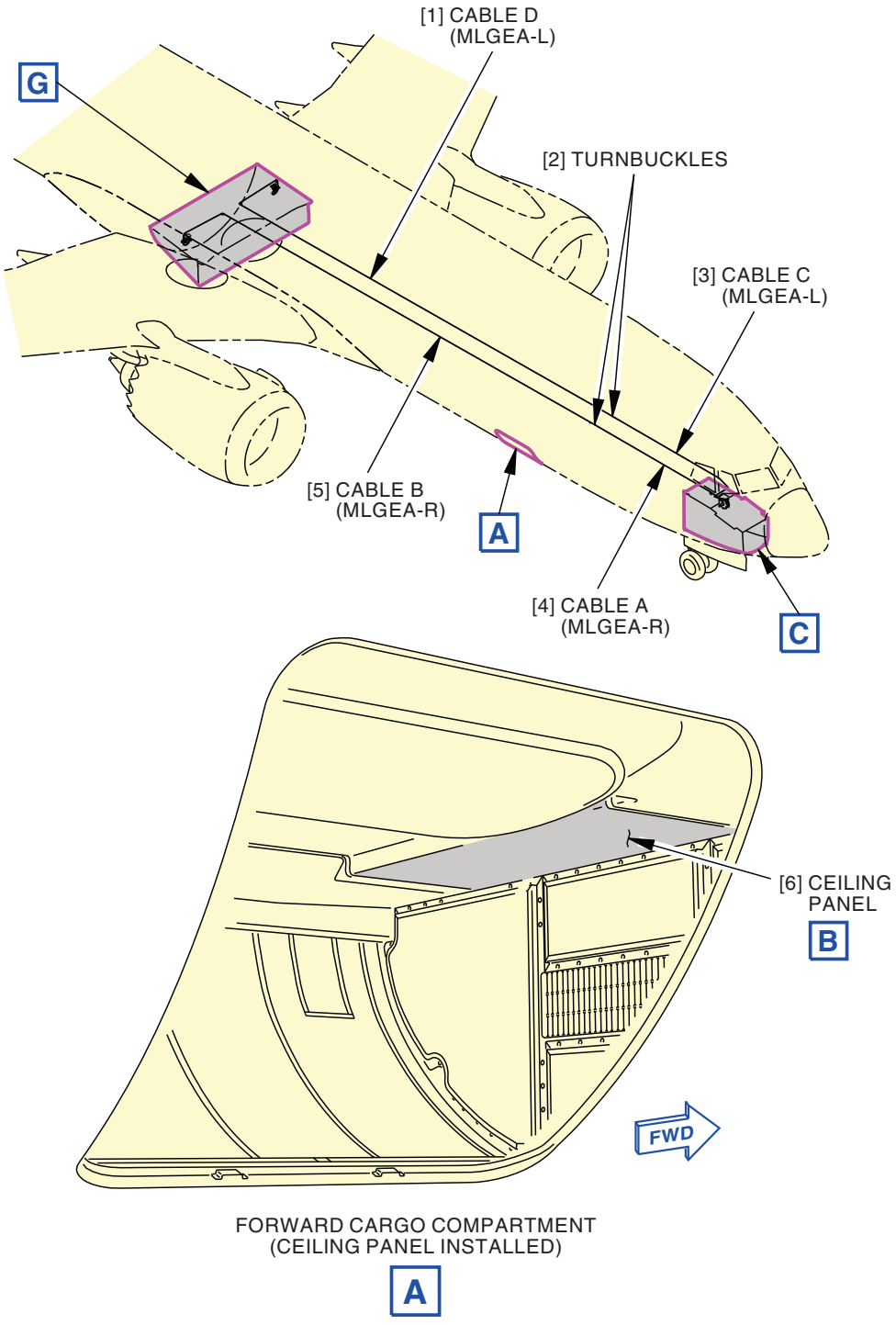
(9) Disconnect the cable D (MLGEA-L) [1] or the cable B (MLGEA-R) [5] from the quadrant assembly on the main landing gear manual extension mechanism.

SUBTASK 32-34-31-000-001

(10) Remove the cable D (MLGEA-L) [1] from the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] from the cable B (MLGEA-R) [5] (TASK 20-10-91-000-801).

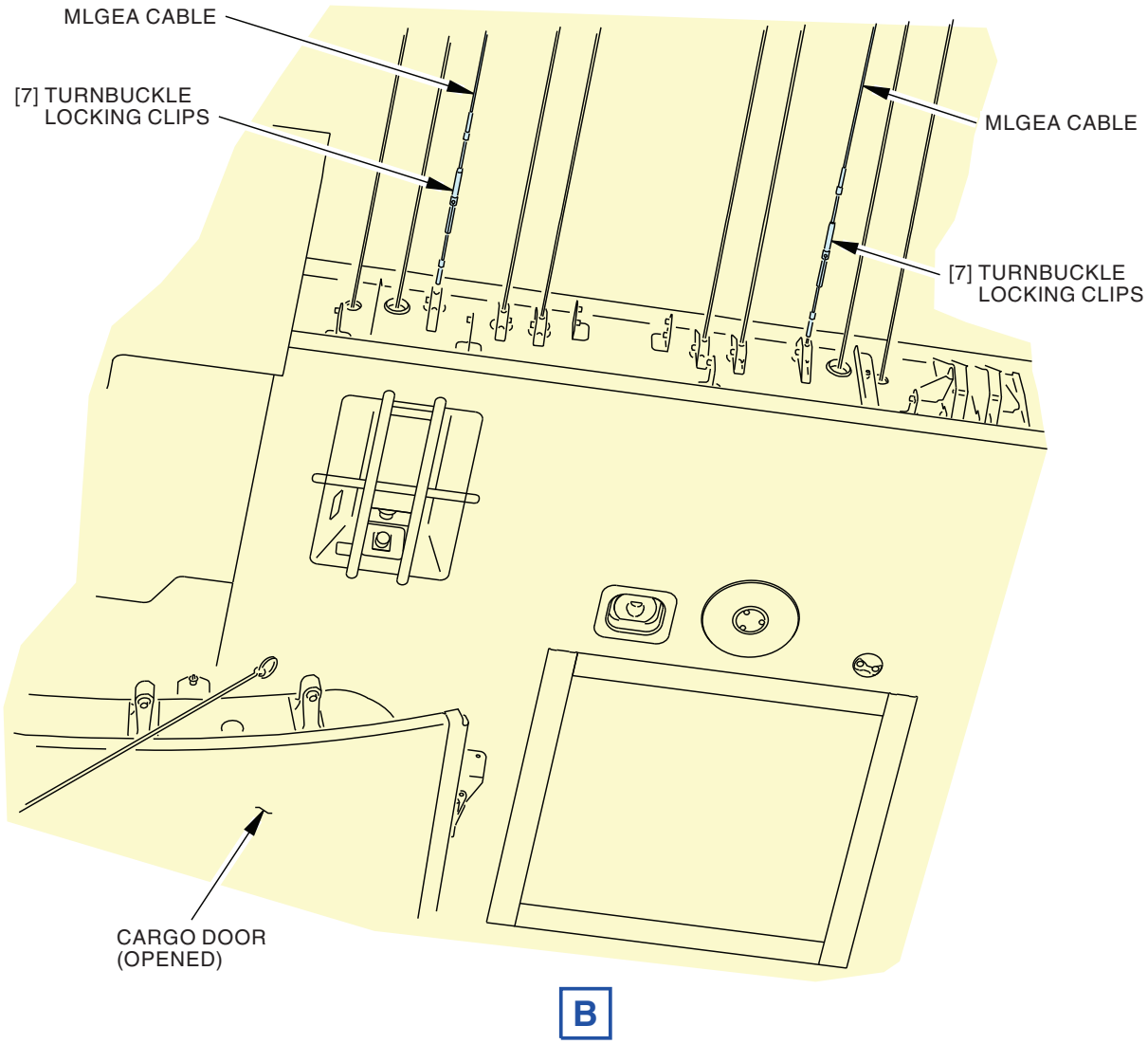
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 1 of 6)

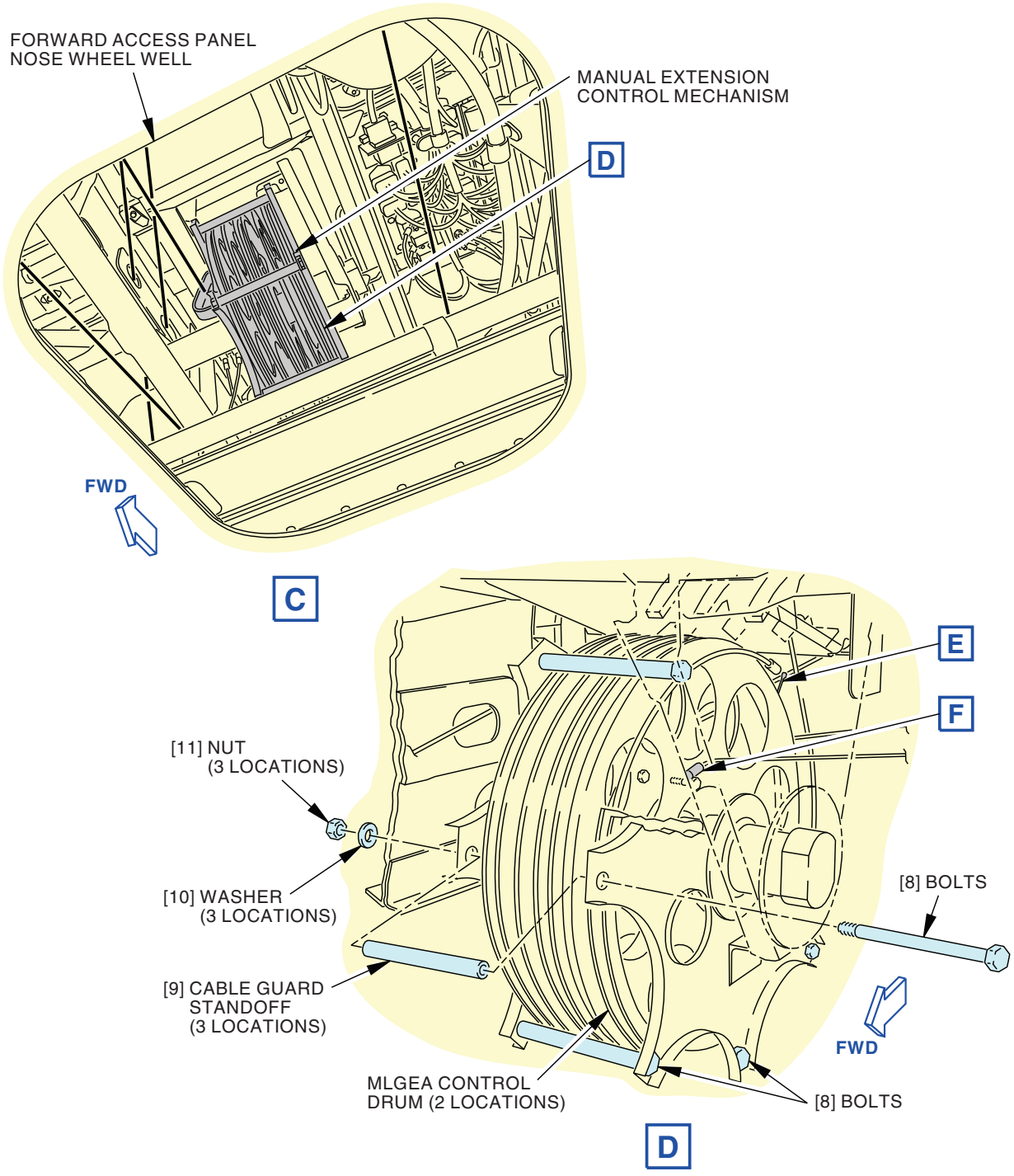
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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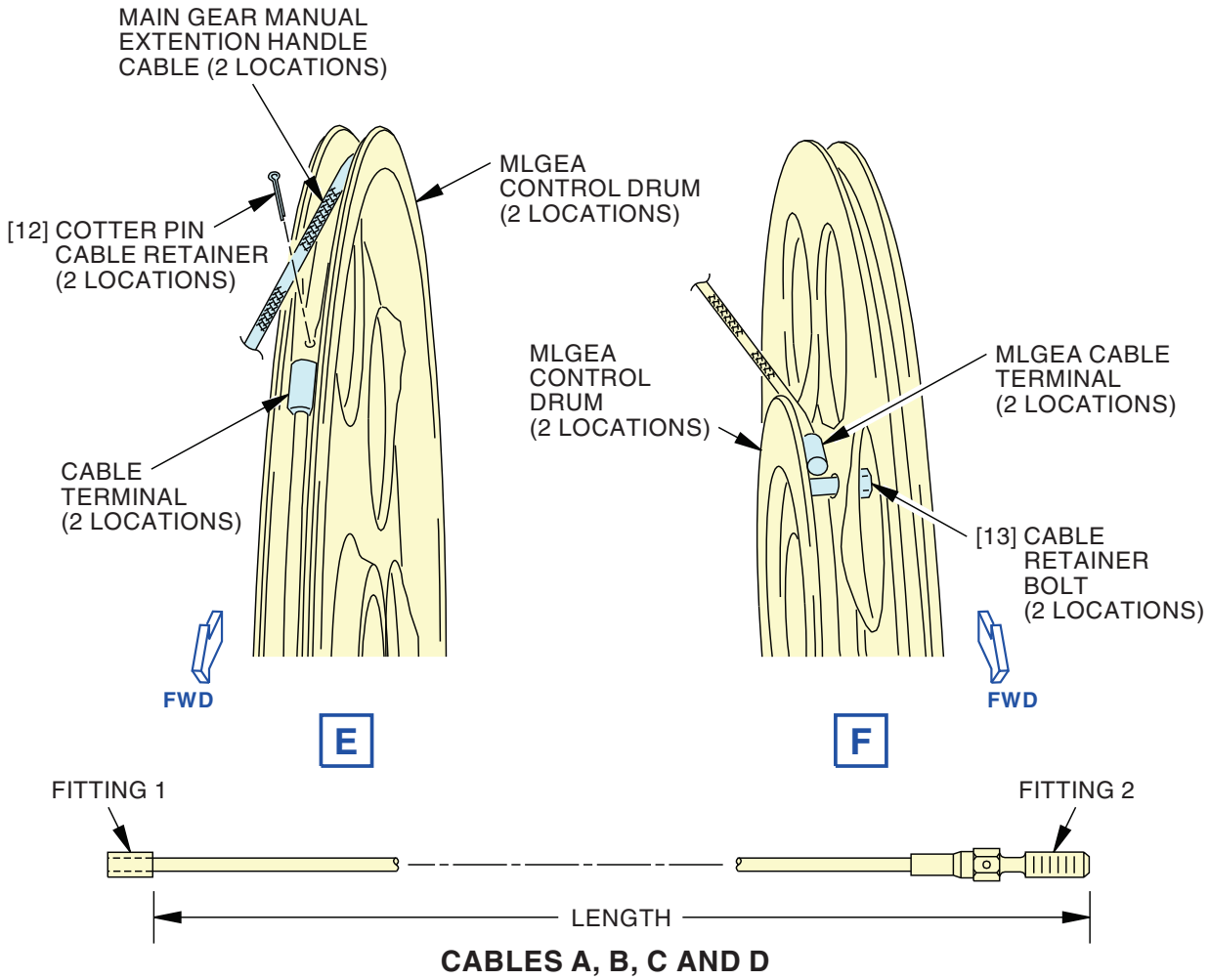
Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



CABLES A, B, C AND D

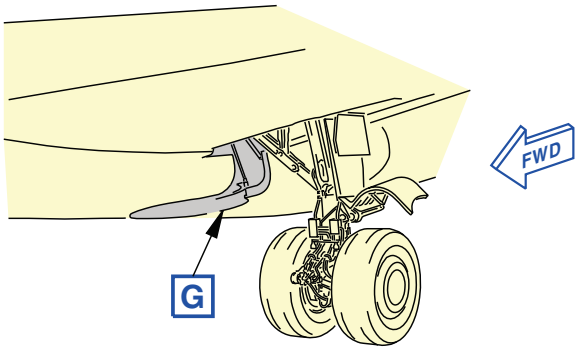
CABLE REF	FUNCTION	NO. REQ	LENGTH INCHES (METERS)	CABLE SIZE	FITTINGS	
					1	2
A	MLGEA-R	1	163.7 (4.158)	3/32 7x7	BACT14A3	MS21260L3 LH
B	MLGEA-R	1	543.9 (13.815)	3/32 7x7	BACT14A3	MS21260L3 RH
C	MLGEA-L	1	182.4 (4.633)	3/32 7x7	BACT14A3	MS21260L3 LH
D	MLGEA-L	1	543.9 (13.815)	3/32 7x7	BACT14A3	MS21260L3 RH

MATERIAL: CABLE - CARBON STEEL PER BMS 7-265, TYPE I, COMPOSITION A, TIN-ZINC (TZ).

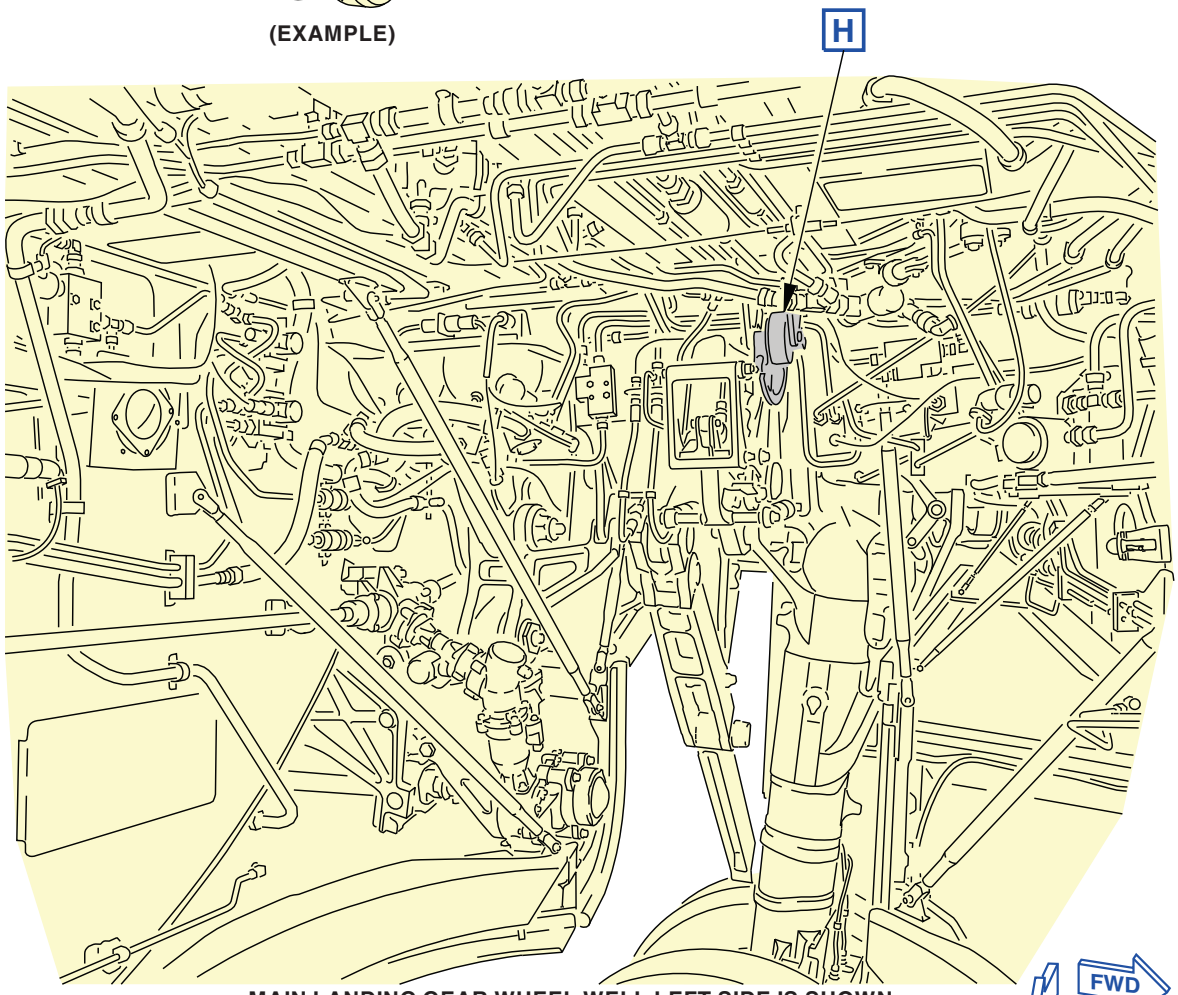
2411025 S00061526764_V1

Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 4 of 6)

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(EXAMPLE)



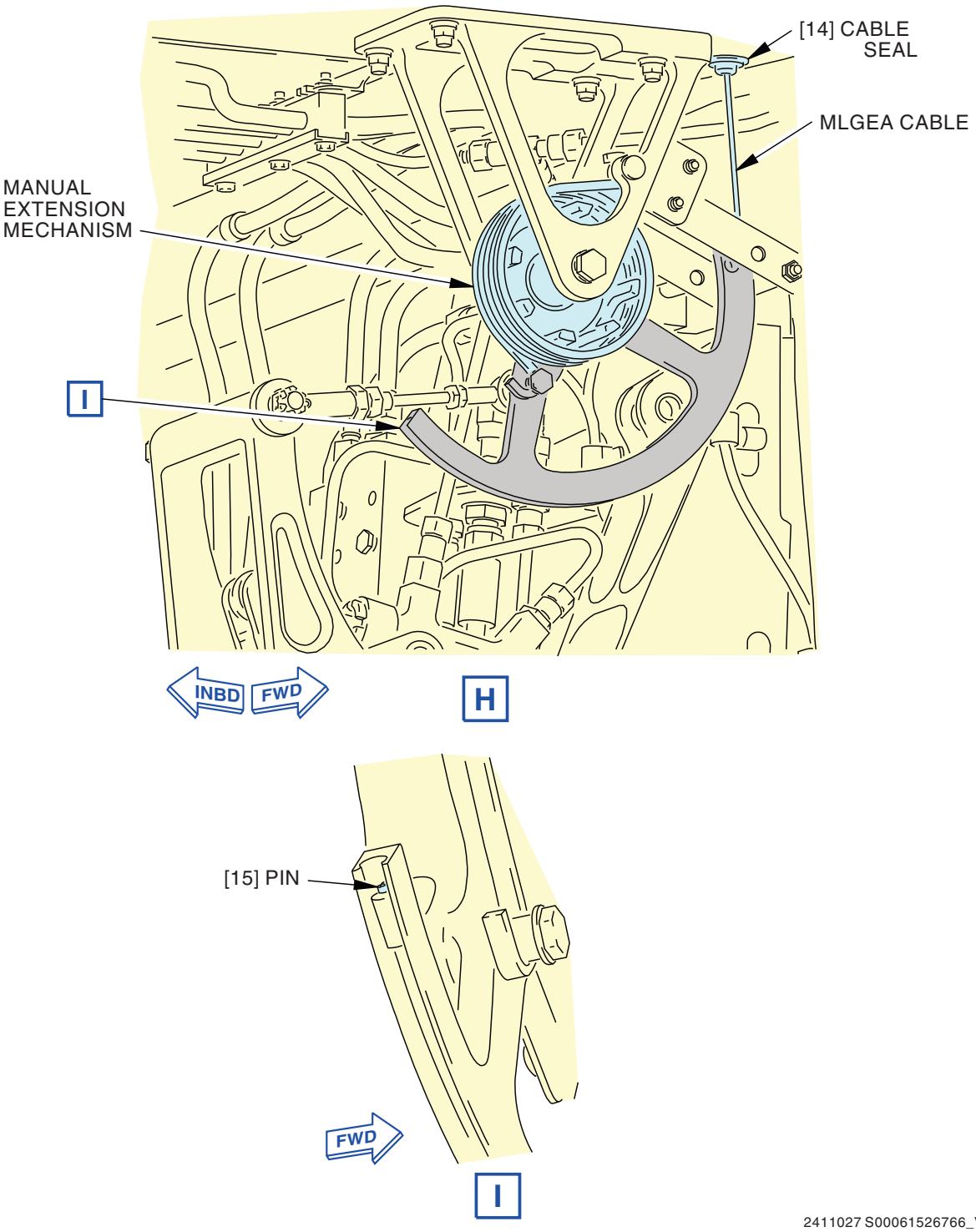
MAIN LANDING GEAR WHEEL WELL LEFT SIDE IS SHOWN,
(RIGHT SIDE IS EQUIVALENT)
(EXAMPLE)

G

2411026 S00061526765_V2

Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Gear Manual Extension System Cable Installation
Figure 401/32-34-31-990-801 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-34-31-400-801

3. Main Gear Manual Extension System Cables Installation

(Figure 401)

A. References

Reference	Title
20-10-04-400-802	Control Cable Half Ball Pressure Seal Installation (P/B 401)
20-10-91-400-801	Control Cables Installation (P/B 401)
32-34-00-730-802	Main Gear Manual Extension System Test (Airplane not on Jacks) (P/B 501)
32-34-00-820-801	Main Landing Gear Manual Extension System Adjustment and Test (P/B 501)

B. Consumable Materials

Reference	Description	Specification
C00174	Compound - Corrosion Preventive, Solvent Cutback, Cold Application	MIL-PRF-16173 (Supersedes MIL-C-16173)
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Cable D (MLGEA-L)	32-34-31-01-045	PTF ALL
3	Cable C (MLGEA-L)	32-34-31-01-040	PTF ALL
4	Cable A (MLGEA-R)	32-34-31-01-030	PTF ALL
5	Cable B (MLGEA-R)	32-34-31-01-045	PTF ALL
12	Cotter pin	32-34-11-01-040	PTF ALL
15	Pin	32-34-31-01-005	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
121	Forward Cargo Compartment - Left
122	Forward Cargo Compartment - Right
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

E. Main Gear Manual Extension System Cables Installation

SUBTASK 32-34-31-420-001

(1) Install the cable D (MLGEA-L) [1] and the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] and the cable B (MLGEA-R) [5] (TASK 20-10-91-400-801).

NOTE: The turnbuckles [2] are located at approximately body station 420.

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(a) Apply a thin layer of compound, C00174, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckles [2].

SUBTASK 32-34-31-420-002

(2) Install the cable D (MLGEA-L) [1] or the cable B (MLGEA-R) [5] on the quadrant assembly of the main landing gear extension mechanism.

(a) Install the pin [15].

SUBTASK 32-34-31-420-003

(3) Connect the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] on the control drum on the control mechanism.

(a) Install the bolts [8] that connect the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] to the control drum on the manual extension control mechanism.

SUBTASK 32-34-31-410-001

(4) Install the cable guard standoffs [9], washers [10], and nuts [11] on the control drum on the manual extension control mechanism.

SUBTASK 32-34-31-420-004

(5) Put the terminal end of the manual extension handle cable in its position on the MLGEA control drum.

(a) Attach the manual extension handle cable on the control drum of the control mechanism and install the new cotter pins [12].

(b) Bend one end of the new cotter pins [12] into the drum.

(c) Connect the cable D (MLGEA-L) [1] to the cable C (MLGEA-L) [3] or the cable A (MLGEA-R) [4] to the cable B (MLGEA-R) [5] at the turnbuckle [2].

F. Main Gear Manual Extension System Cables Installation Test

SUBTASK 32-34-31-820-001

(1) Do the adjustment of the manual extension system for the main landing gear (TASK 32-34-00-820-801).

SUBTASK 32-34-31-710-001

(2) Do a test on the manual extension system for the main landing gear (TASK 32-34-00-730-802).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-34-31-420-005

(1) Install the cable seal [14] on the wheel well ceiling of the main landing gear (TASK 20-10-04-400-802).

SUBTASK 32-34-31-410-002

(2) Install the cargo compartment ceiling panels [6].

NOTE: The cargo compartment ceiling panels [6] are located at approximately body station 420.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

NOSE GEAR MANUAL EXTENSION SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure contains these three tasks:

(1) Nose landing gear manual extension system adjustment.

(2) System test of the manual extension system for the nose gear, with the airplane on jacks.

(3) System test of the manual extension system for the nose gear, with the airplane not on jacks.

TASK 32-35-00-820-801

2. Nose Landing Gear Manual Extension System Adjustment

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-20-31-200-801	Inspection of the Control Cable Wire Rope (P/B 601)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Procedure

SUBTASK 32-35-00-480-008

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-35-00-860-033

(2) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-35-00-580-001

(3) Make sure that you install chocks around the tires of the main landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-35-00-010-001

(4) Remove the screws [2] and cover [1] from the nose gear manual release mechanism (Figure 501, View A).

SUBTASK 32-35-00-980-001

(5) Align the rig pin holes on the nose gear manual release mechanism.

SUBTASK 32-35-00-480-001

(6) Install the rig pin LGEA1 from rigging pin kit, SPL-1585, through the quadrant and the side cover on the nose gear manual release mechanism.

(a) Move the spring if necessary.

SUBTASK 32-35-00-010-002

(7) Remove the cabin floor panel at approximately body station 270 (TASK 53-21-00-000-801).

SUBTASK 32-35-00-700-008

(8) Do this task: Inspection of the Control Cable Wire Rope, TASK 20-20-31-200-801.

NOTE: Control cables can touch fairleads, grommets or rub strips after rigging while in operation.

SUBTASK 32-35-00-020-001

(9) Remove the turnbuckle locking clips from the cable MNGEA (Figure 501, View B).

SUBTASK 32-35-00-820-004

(10) Adjust the turnbuckle on cable MNGEA until the rig pin LGEA 1 fits freely in the manual extension mechanism.

NOTE: When you do this, make sure you correctly set the release handle in the V-slot of the manual extension control. When the cable tension is equal, rig pin LGEA1 will fit freely in the manual extension mechanism.

SUBTASK 32-35-00-080-001

(11) Remove the rig pin LGEA1 from the nose gear manual release mechanism.

NOTE: You can let the rig pin touch the spring.

SUBTASK 32-35-00-410-001

(12) Install the screws [2] and the cover [1] on the nose gear manual release mechanism.

SUBTASK 32-35-00-420-001

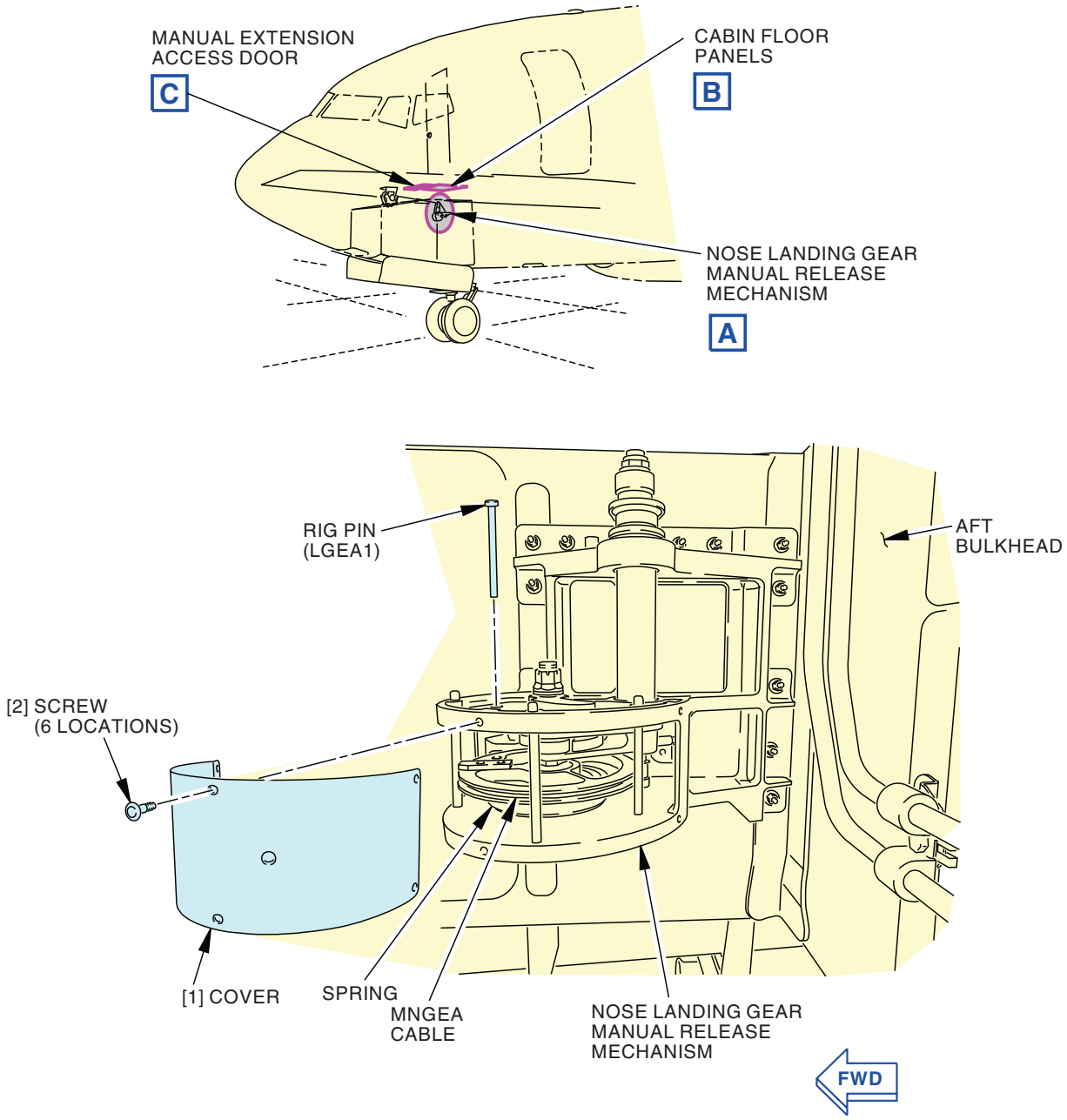
(13) Install the turnbuckle locking clips on the cable MNGEA.

SUBTASK 32-35-00-410-002

(14) Install the cabin floor panel at approximately body station 270 (TASK 53-21-00-400-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



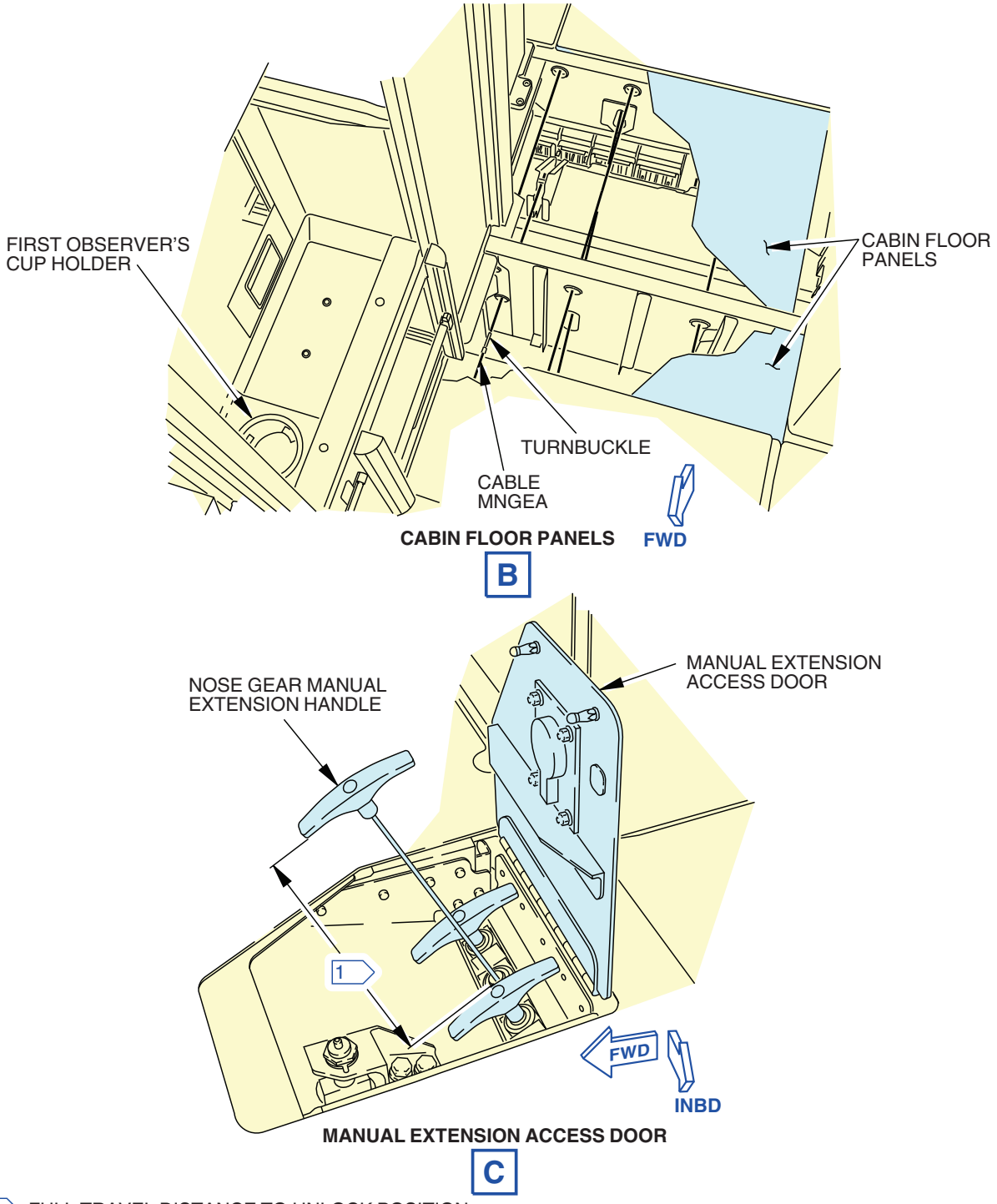
NOSE LANDING GEAR MANUAL RELEASE MECHANISM (VIEW IN THE UP DIRECTION)

A

2414327 S00061534904_V1

Nose Gear Manual Extension System
Figure 501/32-35-00-990-801 (Sheet 1 of 2)

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Nose Gear Manual Extension System
Figure 501/32-35-00-990-801 (Sheet 2 of 2)

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TASK 32-35-00-730-801

3. Nose Gear Manual Extension System Test - Airplane on Jacks

(Figure 501)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) The usual extension/retraction system will not operate with the manual extension access door open. Make sure that you close the access door when you extend or retract the landing gear with the landing gear control handle.

B. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
07-10-21-580-801	Lift the Airplane Nose with the Nose Jack at Jack Point D (P/B 201)
07-10-21-580-802	Lower the Airplane Nose, Jack Point D (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-801	Put the Airplane in the Air Mode (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
32-09-10-740-801	Proximity Switch Electronics Unit - Operational Test (P/B 501)
32-33-00-710-802	Nose Landing Gear Operational Test (P/B 501)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1690	Actuators/Deactuators Set - Proximity Sensor Part #: 8-758-01 Supplier: 08748 Part #: A27092-139 Supplier: 81205 Part #: A27092-146 Supplier: 81205 Opt Part #: A27092-106 Supplier: 81205
STD-5505	Spring Scale

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
118	Electrical and Electronics Compartment - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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E. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

F. Prepare for the Test

SUBTASK 32-35-00-480-009

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-35-00-580-002

(2) Do one of these tasks to lift the airplane nose:

(a) Lift the Airplane Nose with the Nose Jack at Jack Point D, TASK 07-10-21-580-801.

(b) Lift the Airplane on Jacks, TASK 07-10-01-580-801.

NOTE: If you lift the main gear with the jacks, it is not necessary to put the airplane in the air mode. The airplane will be in the air mode if you lift the main landing gear off the ground.

SUBTASK 32-35-00-860-034

(3) For hydraulic system A, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-35-00-860-006

(4) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
C	4	C00236	HEATERS ELEV PITOT LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT
D	4	C00237	HEATERS ELEV PITOT RIGHT
D	5	C00525	HEATERS F/O PITOT
D	6	C00524	HEATERS AUX PITOT
E	4	C00700	HEATERS DRAIN MAST AIR

F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

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F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-35-00-010-004

(5) To get access to the P92 panel, open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-35-00-860-039

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(6) Open this circuit breaker and install safety tag:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-35-00-860-007

(7) Make sure that the control lever for the landing gear is in the DN position.

SUBTASK 32-35-00-860-008

(8) If you did not lift the main landing gear with the jacks, make sure that the airplane is in the air mode (TASK 32-09-00-860-801).

(a) Install deactuators on a minimum of 5 of the 6 Air/Ground proximity sensors on the landing gear.

NOTE: Use the deactuator plate that is provided with the proximity sensor test set, SPL-1690, to put the airplane in the air mode.

NOTE: As an option, deactuator plate for rectangular sensor can be made from 5052, 6061, 7075 or 2024 aluminum with a dimension of 1.5 in. (38 mm) x 0.75 in. (19 mm) x 0.05 in. (1.27 mm). Deactuator plate for round sensor can be made from copper sheet with a dimension of 1.0 in. (25 mm) x 1.0 in. (25 mm) x 0.025 in. (0.64 mm).

SUBTASK 32-35-00-080-002

(9) For the nose gear, do this task: Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801.

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SUBTASK 32-35-00-860-009

(10) Move the No. 1 and No. 2 throttle levers to the full forward position.

NOTE: Throttles in the forward position do not let the landing gear warning (red) lights stay on after gear is up-and-locked.

SUBTASK 32-35-00-860-010

(11) Put the master dim and test switch in the bright position.

G. Nose Gear Manual Extension System Test

SUBTASK 32-35-00-860-038

(1) Open the manual extension access door for the nose gear manual extension handle in the flight compartment floor.

(a) Make sure that the switch under the manual extension access door clicks audibly when the manual extension access door opens.

SUBTASK 32-35-00-700-009

(2) Pull the nose gear manual extension handle.

(a) Make sure that the full travel of the nose gear manual extension handle is a minimum of 23.0 in. (584.2 mm) (View C, Figure 501).

SUBTASK 32-35-00-700-002

(3) Release the nose gear manual extension handle.

(a) Make sure that the nose gear manual extension handle goes back to the stowed position freely and smoothly.

SUBTASK 32-35-00-480-003

(4) Attach a spring scale, STD-5505, to the nose gear manual extension handle.

SUBTASK 32-35-00-730-001

(5) Pull the nose gear manual extension handle.

(a) Make sure that the load to start the nose gear manual extension handle from the seat is a minimum of 3.0 lbf (13.3 N) and a maximum of 7.0 lbf (31.1 N).

(b) Make sure that the load during the full travel of the nose gear manual extension handle is 15.0 lbf (66.7 N) or less.

1) Pull the nose gear manual extension handle up and forward when you measure the load.

NOTE: This will simulate a person in the First Officer's seat.

2) Do not pull the nose gear manual extension handle directly up.

(c) Make sure that the manual extension mechanism operates freely.

SUBTASK 32-35-00-860-012

(6) Remove the spring scale, STD-5505, from the nose gear manual extension handle.

SUBTASK 32-35-00-860-013

(7) Make sure that you close the manual extension access door.

NOTE: The extension retraction system will not operate with the manual extension access door open.

SUBTASK 32-35-00-860-014

(8) Make sure that the control lever for the landing gear is in the DN position.

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SUBTASK 32-35-00-730-002

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE AREA AROUND THE NOSE GEAR. THE MOVEMENT OF THE NOSE GEAR CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) Move the control lever for the landing gear to UP.

(a) Wait while the nose gear retracts.

SUBTASK 32-35-00-860-015

(10) Open the access door to the nose gear manual extension handle.

SUBTASK 32-35-00-730-003

(11) Make sure that you pull the nose gear manual extension handle up and forward.

NOTE: This will simulate a person in the First Officer's seat.

(a) Do not pull the nose gear manual extension handle directly up.

SUBTASK 32-35-00-700-003

(12) Make sure that the nose gear free falls to the down-and-locked position.

(a) Make sure that the red light for the nose gear is on.

NOTE: The red light for the nose gear will stay on to show that the landing gear control lever and landing gear positions disagree.

(b) Make sure that the green light for the nose gear is on.

NOTE: The green light for the nose gear will stay on to show that the nose landing gear is in the down-and-locked position.

SUBTASK 32-35-00-700-004

(13) Release the nose gear manual extension handle.

SUBTASK 32-35-00-860-016

(14) Move the control lever for the landing gear to the DN position.

(a) Make sure that the red light for the nose gear is off.

SUBTASK 32-35-00-860-017

(15) Make sure that the manual extension access door is closed.

SUBTASK 32-35-00-730-005

WARNING KEEP ALL PERSONNEL AND EQUIPMENT CLEAR OF THE AREA AROUND THE NOSE GEAR. THE MOVEMENT OF THE NOSE GEAR CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(16) Move the control lever for the landing gear to the UP position.

(a) Wait while the nose gear retracts.

SUBTASK 32-35-00-860-018

(17) Open the access door to the nose gear manual extension handle.

SUBTASK 32-35-00-820-003

(18) Attach a spring scale, STD-5505, to the nose gear manual extension handle.

SUBTASK 32-35-00-730-006

(19) Pull the nose gear manual extension handle.

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(a) Make sure that the maximum load when you pull the nose gear manual extension handle is 40.0 lbf (177.9 N) or less.

1) Pull the nose gear manual extension handle up and forward when you measure the load.

NOTE: This will simulate a person in the First Officer's seat.

2) Do not pull the nose gear manual extension handle directly up.

SUBTASK 32-35-00-700-005

(20) Make sure that the nose gear free falls to the down-and-locked position.

(a) While the nose landing gear is in free fall, the red light must be on, and the green light must be off.

NOTE: The red light for the nose gear will stay on to show that the landing gear control lever and the landing gear positions disagree.

(b) When the nose landing gear locks in the down position, the green light must be on, and the red light must stay on.

(c) Move the landing gear control lever to the DN position.

1) Make sure that the red light for the nose gear is off.

SUBTASK 32-35-00-080-003

(21) Remove the spring scale, STD-5505, from the nose gear manual extension handle.

SUBTASK 32-35-00-860-019

(22) Close the access door to nose gear manual extension handles.

SUBTASK 32-35-00-860-041

(23) Do an operational test of the nose landing gear release mechanism (TASK 32-33-00-710-802).

NOTE: The usual operation of the nose landing gear mechanism makes sure that the release roller of the nose gear manual extension system correctly resets to its retracted position.

SUBTASK 32-35-00-860-020

(24) As an alternative, you can do the steps below to make sure that the release roller of nose landing gear manual extension system has reset to its retracted position:

(a) For the nose gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

(b) Open the access door to the nose gear manual extension handle.

(c) Pull the nose gear manual extension handle.

(d) It is necessary for one person to be in the nose landing gear wheel well.

(e) Release the nose gear manual extension handle.

1) Monitor the manual extension mechanism to make sure that the release roller correctly resets to its retracted position.

(f) Make sure that the nose gear manual extension handle goes back to the stowed position freely and smoothly.

SUBTASK 32-35-00-860-021

(25) Close the access door to the nose gear manual extension handle.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-35-00-480-004

(1) For the nose gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-35-00-860-022

(2) Do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

(a) Remove the deactuators from the Air/Ground proximity sensors.

SUBTASK 32-35-00-860-023

(3) Move the No. 1 and the No. 2 throttle levers to the idle (full aft) position.

SUBTASK 32-35-00-860-040

WARNING

WHEN YOU OPEN THE P91 AND P92 PANELS, MAKE SURE THAT THE OUTER DOOR STAYS AS OPEN AS POSSIBLE. IF THE OUTER DOOR TURNS IN, THE ATTACHED DOOR COMPONENTS COULD TOUCH THE INNER DOOR COMPONENTS. THIS CAN CAUSE AN ARC CONDITION WHEN YOU SUPPLY POWER. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

WARNING

DO NOT TOUCH THE CONDUCTORS IN THE P91 AND P92 PANELS. BE CAREFUL WHEN YOU GET ACCESS TO THE CIRCUIT BREAKERS ON THE INNER SIDE OF THE P91 AND P92 PANELS (ROW F). IF IT IS POSSIBLE, REMOVE AIRPLANE ELECTRICAL POWER FIRST. THE P91 AND P92 PANELS HAVE HIGH VOLTAGES AND CURRENTS. ELECTRICAL VOLTAGE AND CURRENT CAN KILL YOU OR CAUSE INJURIES.

(4) Remove the safety tag and close this circuit breaker:

Power Distribution Panel Number 2, P92

Row	Col	Number	Name
F	2	C01449	STANDBY HYDRAULIC PUMP

SUBTASK 32-35-00-410-003

(5) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-35-00-860-024

(6) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-3

Row	Col	Number	Name
C	1	C00523	HEATERS CAPT PITOT
C	2	C00238	HEATERS TEMP PROBE
C	3	C01072	HEATERS ALPHA VANE LEFT
C	4	C00236	HEATERS ELEV PITOT LEFT
D	3	C01071	HEATERS ALPHA VANE RIGHT
D	4	C00237	HEATERS ELEV PITOT RIGHT
D	5	C00525	HEATERS F/O PITOT
D	6	C00524	HEATERS AUX PITOT
E	4	C00700	HEATERS DRAIN MAST AIR

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F/O Electrical System Panel, P6-1

Row	Col	Number	Name
D	13	C00120	WEATHER RADAR RT

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	15	C00799	HYD SYS LDG GR SYS XFR VALVE SEC
C	16	C00781	HYD SYS LDG GR SYS XFR VALVE PRI

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
D	18	C00451	LANDING GEAR AURAL WARN

SUBTASK 32-35-00-200-002

(7) Do this task: Proximity Switch Electronics Unit - Operational Test, TASK 32-09-10-740-801.

SUBTASK 32-35-00-860-035

(8) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-35-00-580-003

(9) Do the applicable task to lower the airplane:

(a) Lower the Airplane Nose, Jack Point D, TASK 07-10-21-580-802

(b) Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

TASK 32-35-00-730-802

4. Nose Gear Manual Extension System Test - Airplane Not on Jacks

(Figure 501)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-5505	Spring Scale

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

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D. Prepare for the Test

SUBTASK 32-35-00-480-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-35-00-860-027

(2) Make sure that the hydraulic power is off (TASK 29-11-00-860-805).

SUBTASK 32-35-00-860-028

(3) Make sure that the control lever for the landing gear is in the DN position.

E. Procedure

SUBTASK 32-35-00-700-010

(1) Pull the nose gear manual extension handle.

(a) Make sure that the full travel of the nose gear manual extension handle is a minimum of 23.0 in. (584.2 mm) (Figure 501, View C).

SUBTASK 32-35-00-700-007

(2) Release the nose gear manual extension handle.

(a) Make sure that the nose gear manual extension handle goes back to the stowed position freely and smoothly.

SUBTASK 32-35-00-480-006

(3) Attach a spring scale, STD-5505, to the nose gear manual extension handle.

SUBTASK 32-35-00-730-007

(4) Pull the nose gear manual extension handle.

(a) Make sure that the load to start the nose gear manual extension handle from the seat is a minimum of 3.0 lbf (13.3 N) and a maximum of 7.0 lbf (31.1 N).

(b) Make sure that the load during the full travel of the nose gear manual extension handle is 15.0 lbf (66.7 N) or less.

1) Pull the nose gear manual extension handle up and forward when you measure the load.

NOTE: This will simulate a person in the First Officer's seat.

2) Do not pull the nose gear manual extension handle directly up.

(c) Make sure that the manual extension mechanism operates freely.

SUBTASK 32-35-00-080-004

(5) Remove the spring scale, STD-5505, from the nose gear manual extension handle.

SUBTASK 32-35-00-860-030

(6) Pull the nose gear manual extension handle.

(a) It is necessary for one person to be in the nose landing gear wheel well.

(7) Release the nose gear manual extension handle.

(a) Monitor the manual extension mechanism to make sure that the release roller correctly resets to its retracted position.

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(b) Make sure that the nose gear manual extension handle goes back to the stowed position freely and smoothly.

SUBTASK 32-35-00-860-031

(8) Close the access door to the nose gear manual extension handle.

——— END OF TASK ———

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NOSE GEAR MANUAL EXTENSION RELEASE MECHANISM - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the manual extension release mechanism (referred to as the cable assembly [2]) for the nose landing gear.

(2) An installation of the manual extension release mechanism (referred to as the cable assembly [2]) for the nose landing gear.

TASK 32-35-11-000-801

2. Nose Gear Manual Extension Release Mechanism Removal

(Figure 401)

A. References

Reference	Title
20-10-04-000-801	Control Cable Air Seal Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

C. Manual Extension Release Mechanism Removal

SUBTASK 32-35-11-010-001

(1) Remove the cabin floor panel, near body station 270, to get access to the turnbuckle (TASK 53-21-00-000-801).

SUBTASK 32-35-11-010-002

(2) Disconnect the MNGEA cable assembly [2] at the turnbuckle [1].

SUBTASK 32-35-11-020-001

(3) Remove the nut [5], washer [4], and bolt [3].

NOTE: This will let you move the pulley when you remove the MNGEA cable assembly [2] that is attached to the release mechanism [7].

SUBTASK 32-35-11-020-002

(4) Remove the washers [9] and bolts [8], which attach the release mechanism [7] to the wheel well ceiling.

SUBTASK 32-35-11-020-003

(5) Remove the release mechanism [7] from the wheel well ceiling.

SUBTASK 32-35-11-020-004

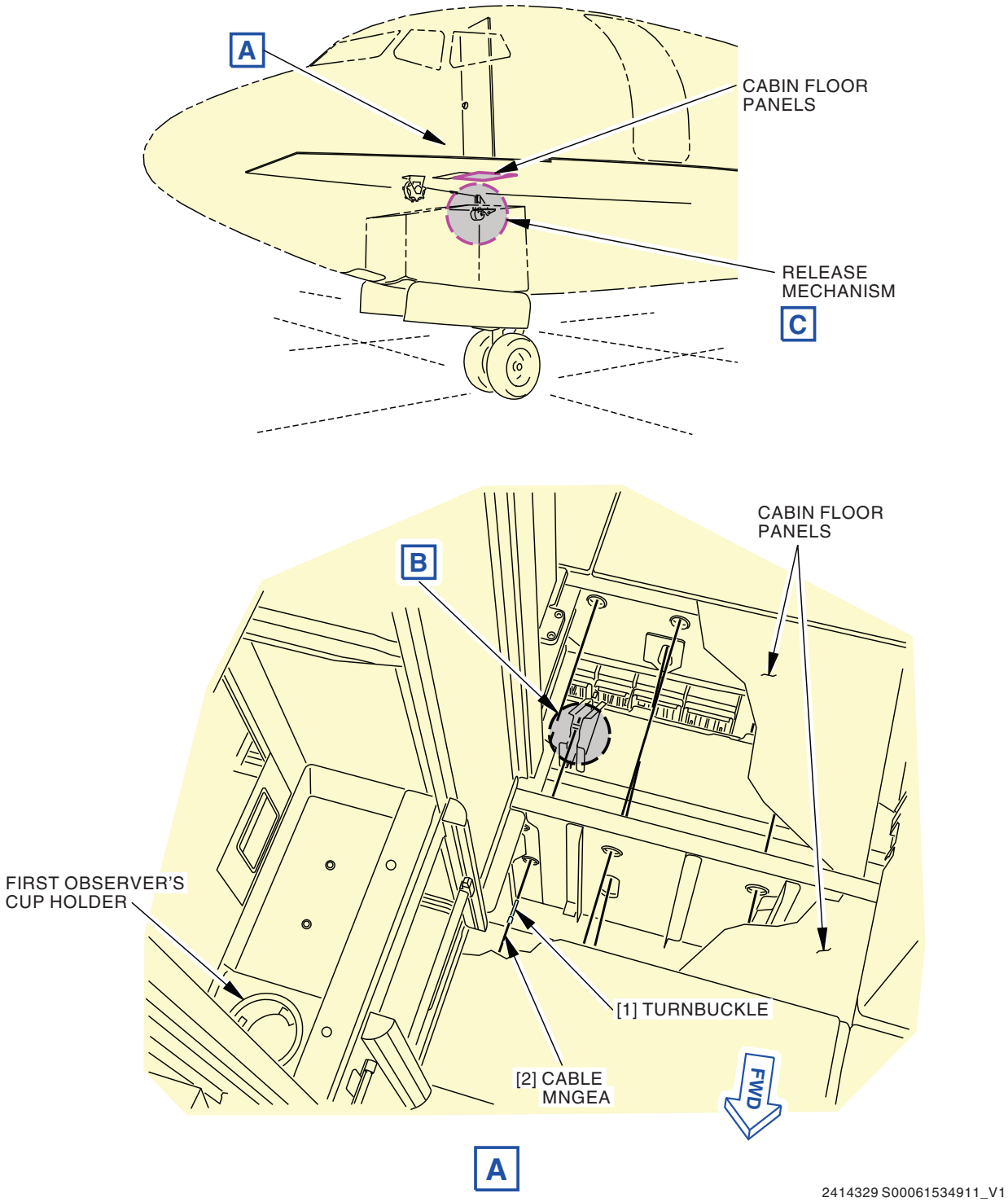
(6) Remove the seal [6] from the ceiling of the nose wheel well (TASK 20-10-04-000-801).

SUBTASK 32-35-11-020-005

(7) Remove the release mechanism [7], with the MNGEA cable assembly [2], from the airplane.

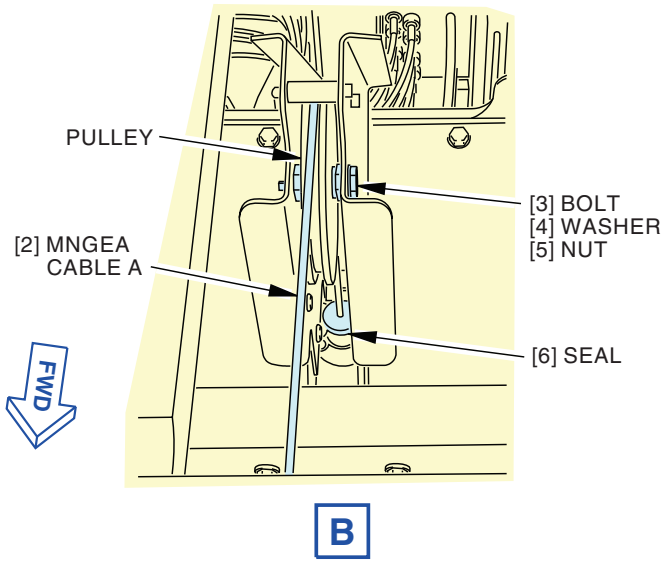
END OF TASK

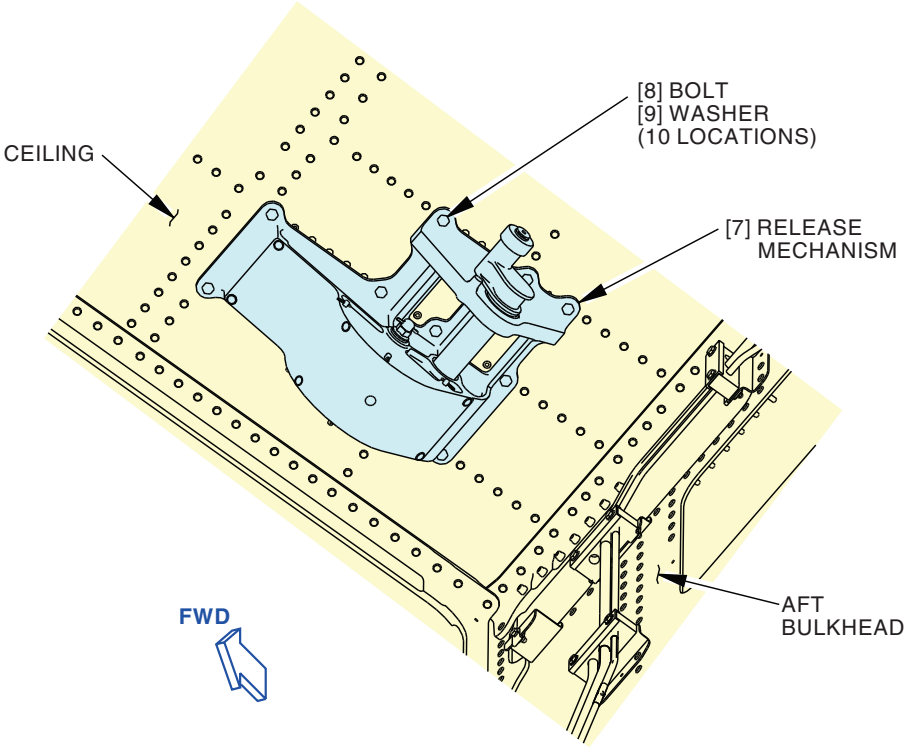
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Nose Landing Gear Manual Extension Release Mechanism Installation
Figure 401/32-35-11-990-801 (Sheet 1 of 2)

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Nose Landing Gear Manual Extension Release Mechanism Installation
Figure 401/32-35-11-990-801 (Sheet 2 of 2)

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TASK 32-35-11-400-801

3. Nose Gear Manual Extension Release Mechanism Installation

(Figure 401)

A. References

Reference	Title
20-10-04-400-801	Control Cable Air Seal Installation (P/B 401)
20-10-91-400-801	Control Cables Installation (P/B 401)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-35-00-820-801	Nose Landing Gear Manual Extension System Adjustment (P/B 501)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00174	Compound - Corrosion Preventive, Solvent Cutback, Cold Application	MIL-PRF-16173 (Supersedes MIL-C-16173)
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Release mechanism	32-35-11-01-290	PTF ALL

D. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

E. Manual Extension Release Mechanism Installation

SUBTASK 32-35-11-100-001

(1) Clean the surface where the release mechanism [7] will contact the ceiling of the nose wheel well.

SUBTASK 32-35-11-600-001

(2) Apply sealant, A00247, to the entire top surface of the release mechanism [7] to fay seal where it will contact the ceiling of the nose wheel well.

SUBTASK 32-35-11-420-001

(3) Install the MNGEA cable assembly [2] (TASK 20-10-91-400-801).

NOTE: The turnbuckle [1] is located at approximately body station 270.

(a) Apply a thin layer of compound, C00174, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckle [1].

SUBTASK 32-35-11-420-002

(4) Install the bolt [3], washer [4], and nut [5] to attach the pulley, with the MNGEA cable assembly [2] installed, to the bracket.

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SUBTASK 32-35-11-420-003

(5) Install the seal [6] on the ceiling of the nose wheel well (TASK 20-10-04-400-801).

SUBTASK 32-35-11-420-004

(6) Put the release mechanism [7] in its position on the wheel well ceiling.

SUBTASK 32-35-11-600-002

(7) Apply sealant, A00247, to the bolts [8] and washers [9].

SUBTASK 32-35-11-420-005

(8) Install the bolts [8] and washers [9] to attach the release mechanism [7] to the wheel well ceiling.

SUBTASK 32-35-11-420-006

(9) Connect the MNGEA cable assembly [2] at the turnbuckle [1].

SUBTASK 32-35-11-410-001

(10) Install the passenger cabin floor panel near body station 270 (TASK 53-21-00-400-801).

SUBTASK 32-35-11-820-001

(11) Do this task: Nose Landing Gear Manual Extension System Adjustment, TASK 32-35-00-820-801.

F. Manual Extension Release Mechanism Installation Test

SUBTASK 32-35-11-710-001

(1) Do a test on the manual extension system (TASK 32-35-00-730-801).

——— END OF TASK ———

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NOSE GEAR MANUAL EXTENSION SPRING - REMOVAL/INSTALLATION

1. General

A. This procedure these tasks:

(1) Removal of the nose landing gear manual extension spring.

(2) Installation of the nose landing gear manual extension spring.

TASK 32-35-20-000-801

2. Nose Gear Manual Extension Spring Removal

(Figure 401)

A. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

B. Manual Extension Spring Removal

SUBTASK 32-35-20-010-001

(1) Remove screws [3] and cover [2] from the housing assembly [1].

SUBTASK 32-35-20-000-001

(2) Remove cotter pins [4], washers [5] and guard pins [6] from the housing assembly [1].

SUBTASK 32-35-20-000-002

(3) Pull the arm [9] away from the cam assembly [8] to remove the spring [7] preload.

SUBTASK 32-35-20-000-003

(4) Remove the cotter pin [10], pin [11], washer [12], and spacer [13].

SUBTASK 32-35-20-000-004

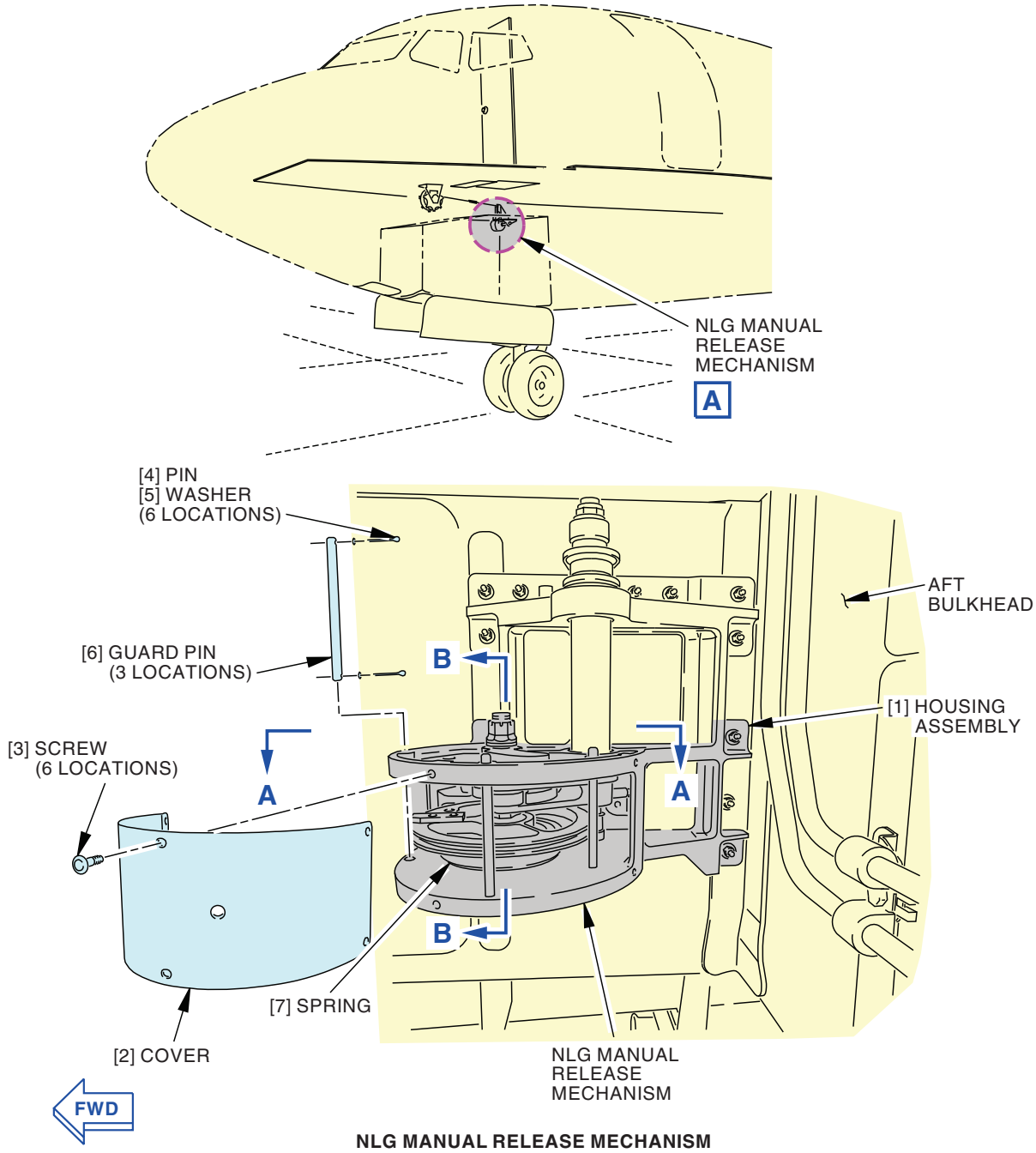
(5) Remove the cotter pin [14], nut [15], washer [16], and bolt [17].

SUBTASK 32-35-20-020-001

(6) Remove the cam assembly [8] with the spring [7].

——— END OF TASK ———

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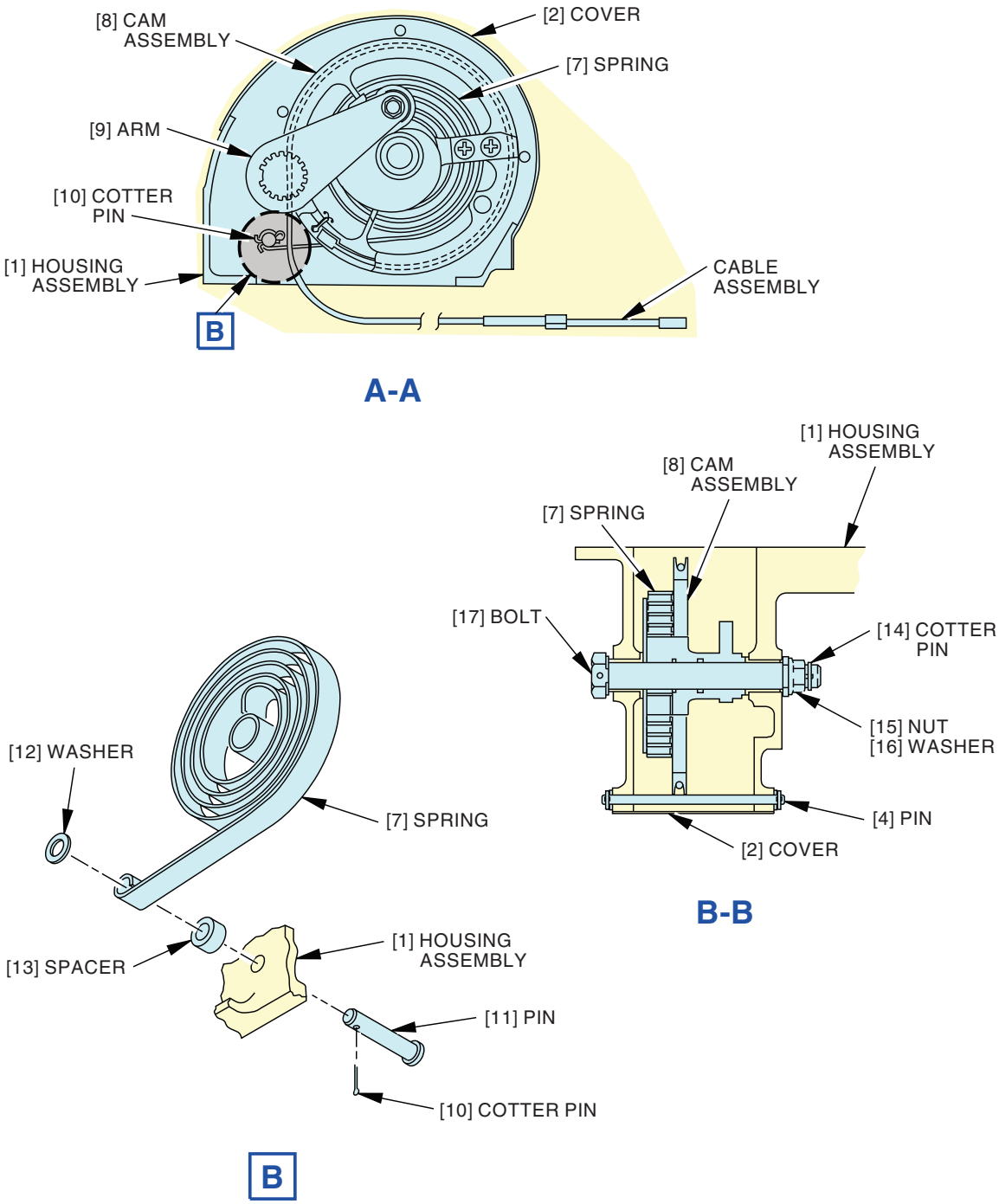
NLG MANUAL RELEASE MECHANISM
(VIEW IN THE UP DIRECTION)

A

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Nose Gear Manual Extension Spring - Removal/Installation
Figure 401/32-35-20-990-801 (Sheet 1 of 2)

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Nose Gear Manual Extension Spring - Removal/Installation
Figure 401/32-35-20-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-35-20-400-801

3. Nose Gear Manual Extension Spring Installation

(Figure 401)

A. References

Reference	Title
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-35-00-730-802	Nose Gear Manual Extension System Test - Airplane Not on Jacks (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Pin	32-35-11-01-045	PTF ALL
7	Spring	32-35-11-01-105	PTF ALL
10	Cotter pin	32-35-11-01-080	PTF ALL

C. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors

D. Nose Gear Manual Extension Spring Installation

SUBTASK 32-35-20-420-001

(1) Install the cam assembly [8] with the spring [7] into the housing assembly [1].

SUBTASK 32-35-20-400-001

(2) Install the bolt [17], washer [16], nut [15], and the cotter pin [14].

SUBTASK 32-35-20-400-002

(3) Install one end of the spring [7] onto the housing assembly [1] with the pin [11], spacer [13], washer [12], and the new cotter pin [10].

SUBTASK 32-35-20-400-003

(4) Install the guard pins [6], washers [5], and the new cotter pins [4] onto the housing assembly [1].

SUBTASK 32-35-20-700-001

(5) Do a check of the spring [7]:

(a) Make sure that the full range of motion is possible when you pull the cable.

(b) Make sure that the spring [7] is not solid when the cable is fully pulled out.

(c) Make sure that the spring [7] fully returns the cam to the initial position when the cable is released.

SUBTASK 32-35-20-410-001

(6) Install the cover [2] with the screws [3] onto the housing assembly [1].

E. Nose Gear Manual Extension Spring Installation Test

SUBTASK 32-35-20-700-002

(1) Do one of these tests of the nose landing gear manual release mechanism:

(a) Nose Gear Manual Extension System Test - Airplane on Jacks, TASK 32-35-00-730-801,

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(b) Nose Gear Manual Extension System Test - Airplane Not on Jacks, TASK 32-35-00-730-802.

——— END OF TASK ———

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NOSE GEAR MANUAL EXTENSION SYSTEM CABLES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the manual extension system cables for the nose landing gear

(2) Installation of the manual extension system cables for the nose landing gear.

TASK 32-35-21-000-801

2. Nose Gear Manual Extension System Cables Removal

(Figure 401)

A. References

Reference	Title
20-10-04-000-801	Control Cable Air Seal Removal (P/B 401)
20-10-91-000-801	Control Cables Removal (P/B 401)
53-14-01-020-801	Nose Wheel Well Access Panels Removal (P/B 401)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

C. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

D. Manual Extension Cable Removal

SUBTASK 32-35-21-010-001

(1) Remove the cabin floor panel [1] to get access to the pulley on the Nose Landing Gear (NLG) manual release mechanism (TASK 53-21-00-000-801).

SUBTASK 32-35-21-010-002

(2) Remove the flight deck floor panel [18] to get access to the turnbuckle [2].

SUBTASK 32-35-21-020-001

(3) Disconnect the MNGEA cable B [16] and the MNGEA cable A [3] at the turnbuckle [2].

SUBTASK 32-35-21-020-002

(4) Remove the nut [6], washers [5], and bolt [4] that attach the pulley for the MNGEA cable A [3] to the bracket.

SUBTASK 32-35-21-010-003

(5) Remove this access panel from the ceiling of the nose wheel well:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Removal, TASK 53-14-01-020-801.

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SUBTASK 32-35-21-020-003

(6) Remove the pin [15] to disconnect the nose gear manual extension handle cable from the MNGEA control drum on the manual extension control mechanism.

SUBTASK 32-35-21-020-004

(7) Open the manual extension access door and remove the nose gear manual extension handle [14].

NOTE: The nose gear manual extension handle [14] includes the nose gear manual extension handle and cable.

SUBTASK 32-35-21-020-008

(8) Remove six bolts [20], washers [21], spacers [22], washers [23] and nuts [24] to get access to the MNGEA control drum.

SUBTASK 32-35-21-020-005

(9) Remove the cable retainer bolt [17] and disconnect the MNGEA cable B [16] from the MNGEA control drum on the manual extension control mechanism.

SUBTASK 32-35-21-010-004

(10) Remove the screws [10] and the cover [9] from the NLG manual release mechanism.

SUBTASK 32-35-21-010-005

(11) Remove the pins [12], washers [13], and guide pins [11] from the NLG manual release mechanism to give clearance for cable removal.

SUBTASK 32-35-21-020-006

(12) Remove the pressure seal [7] on the ceiling of the nose wheel well (TASK 20-10-04-000-801).

SUBTASK 32-35-21-020-007

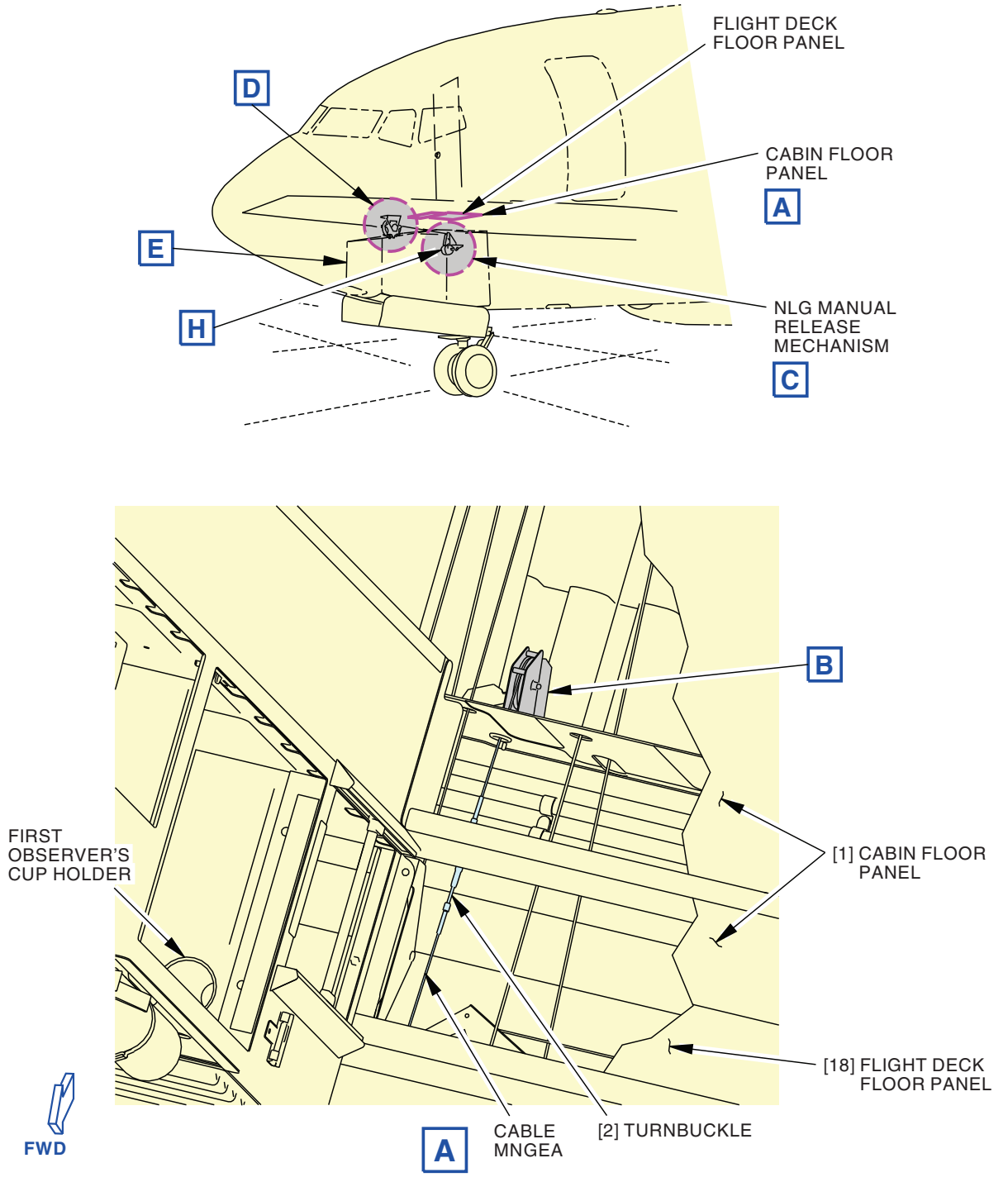
(13) Remove the pin [8] and disconnect the MNGEA cable A [3] from the quadrant on the NLG manual release mechanism.

SUBTASK 32-35-21-000-001

(14) Remove the MNGEA cable B [16] (TASK 20-10-91-000-801).

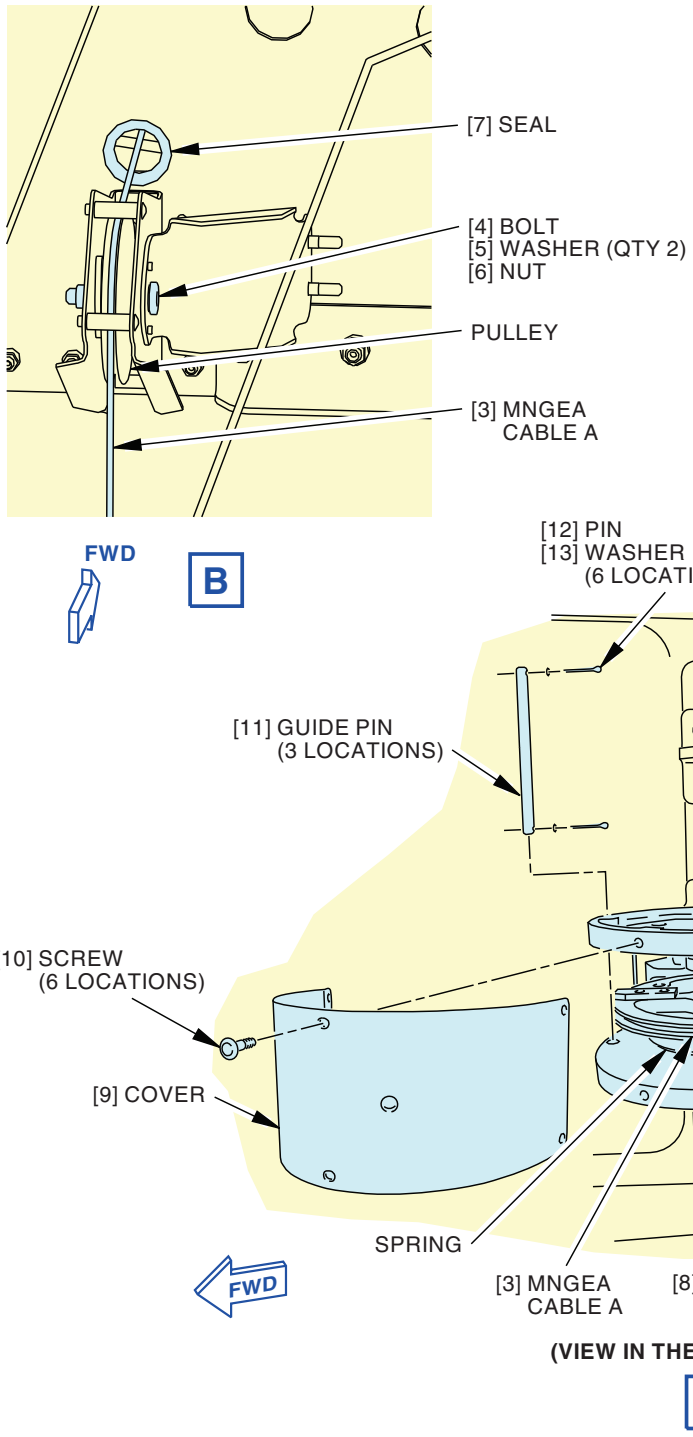
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 1 of 6)

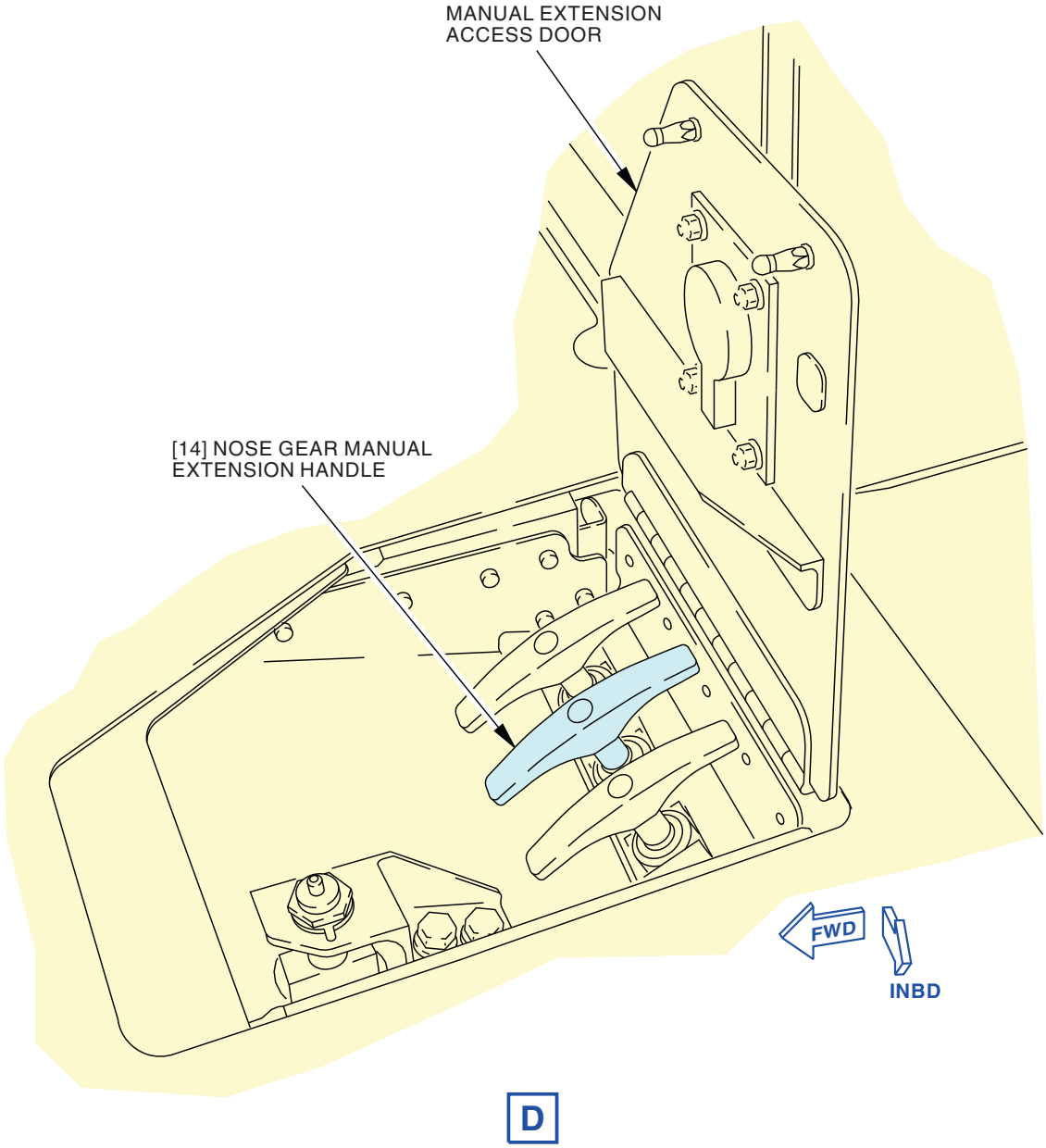
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Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 2 of 6)

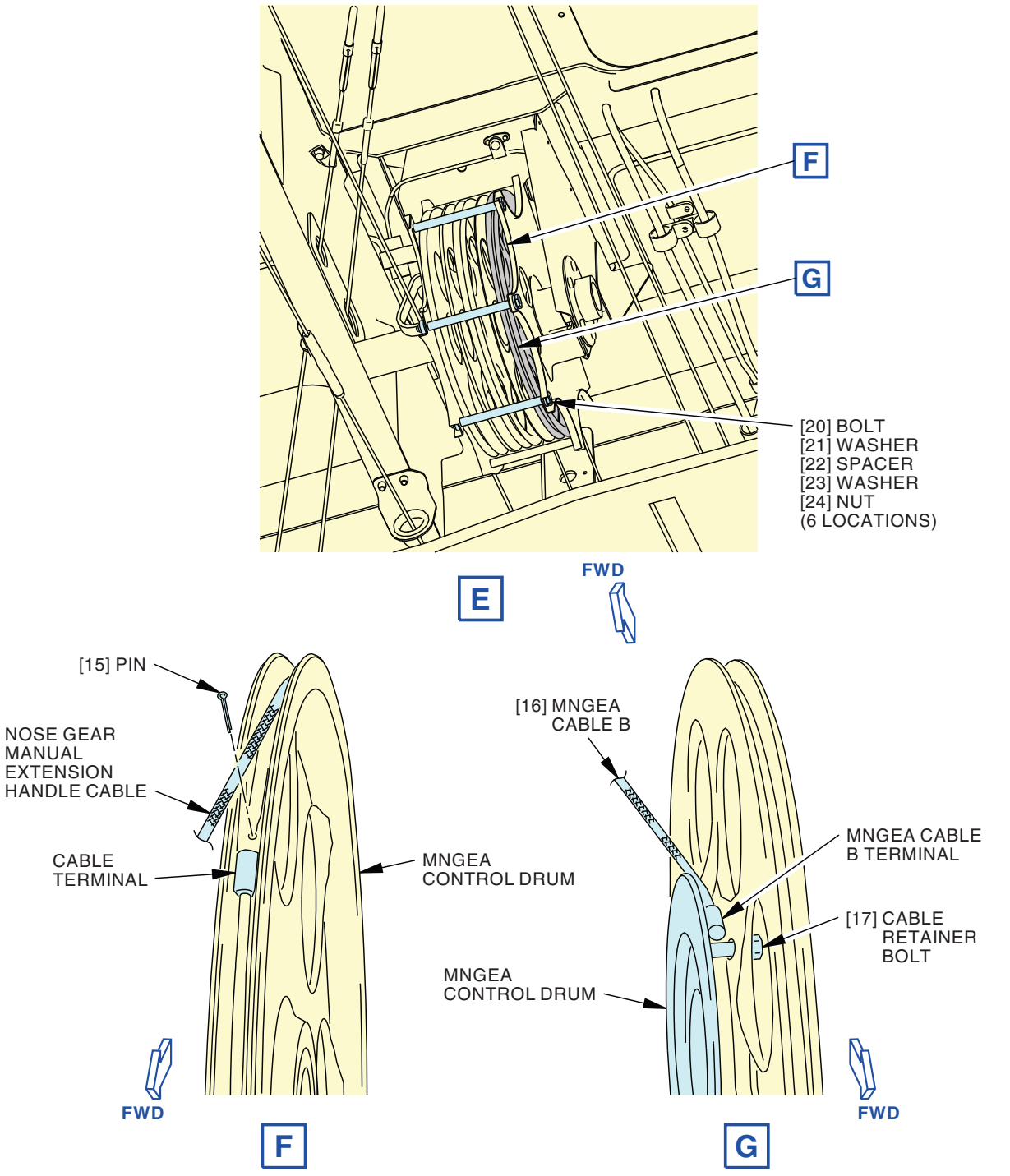
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Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 3 of 6)

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Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 4 of 6)

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CABLES A AND B

CABLE NAME	FUNCTION	NO. REQ	LENGTH (INCHES)	CABLE SIZE	FITTINGS	
					1	2
A	MNGEA	1	30.2	3/32 7x7	BACT14A3	MS21260S3(RH)
B	MNGEA	1	21.7	3/32 7x7	BACT14A3	MS21260S3(LH)

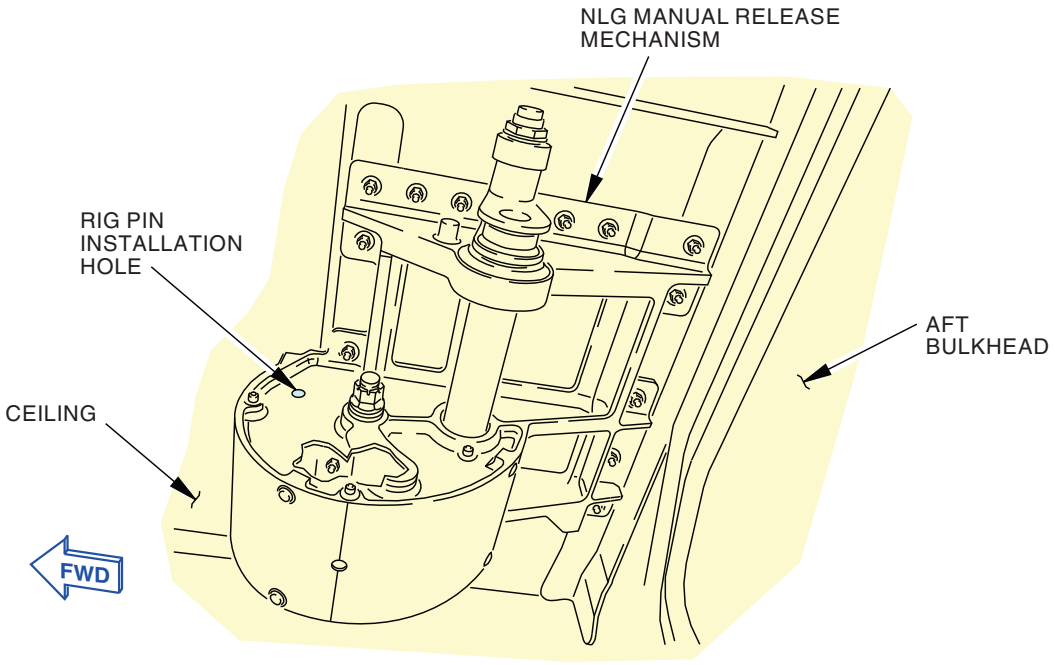
MATERIAL: CABLE A - C0RROSION RESISTANT STEEL PER BMS 7-265, TYPE I, COMPOSITION B

CABLE B - CARBON STEEL PER BMS 7-265, TYPE I, COMPOSITION A (TIN OVER ZINC)

3020229 S0000792649_V1

Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(VIEW FROM NOSE LANDING GEAR WHEEL WHEEL)

H

2411033 S00061526772_V1

Nose Gear Manual Extension System Cable Installation
Figure 401/32-35-21-990-801 (Sheet 6 of 6)

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TASK 32-35-21-400-801

3. Nose Gear Manual Extension System Cables Installation

(Figure 401)

A. References

Reference	Title
20-10-04-400-801	Control Cable Air Seal Installation (P/B 401)
20-10-91-400-801	Control Cables Installation (P/B 401)
32-35-00-730-801	Nose Gear Manual Extension System Test - Airplane on Jacks (P/B 501)
32-35-00-820-801	Nose Landing Gear Manual Extension System Adjustment (P/B 501)
53-14-01-420-801	Nose Wheel Well Access Panels Installation (P/B 401)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
C00174	Compound - Corrosion Preventive, Solvent Cutback, Cold Application	MIL-PRF-16173 (Supersedes MIL-C-16173)
C00528	Compound - Corrosion Preventive, Petroleum Hot Application (Soft Film)	MIL-C-11796 Class III

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	MNGEA cable A	32-35-11-01-040	PTF ALL
8	Pin	32-35-11-01-035	PTF ALL
12	Pin	32-35-11-01-045	PTF ALL
16	MNGEA cable B	32-34-31-01-035	PTF ALL

E. Location Zones

Zone	Area
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right
115	Nose Landing Gear Wheel Well - Left
116	Nose Landing Gear Wheel Well - Right
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right
220	Subzone - Passenger Compartment - Body Station 259.50 to 360.00

F. Access Panels

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

G. Nose Gear Manual Extension System Cables Installation

SUBTASK 32-35-21-420-001

(1) Install the MNGEA cable B [16] (TASK 20-10-91-400-801).

NOTE: The turnbuckle [2] is located at approximately body station 270.

SUBTASK 32-35-21-420-002

(2) Apply a thin layer of compound, C00174, or compound, C00528, to the threads on the internal and external surfaces of the mating adjustable parts on the turnbuckle [2].

SUBTASK 32-35-21-420-003

(3) Install the bolt [4], washers [5] and nut [6] to attach the pulley, with the MNGEA cable A [3] installed, to the bracket.

SUBTASK 32-35-21-480-001

(4) Install the rig pin LGEA1 from rigging pin kit, SPL-1585, in the NLG manual release mechanism.

SUBTASK 32-35-21-420-004

(5) Install the MNGEA cable A [3] on the quadrant on the NLG manual release mechanism.

(a) Install the new cotter pin [8] to attach the MNGEA cable A [3] to the quadrant.

SUBTASK 32-35-21-420-005

(6) Install the pressure seal [7] on the ceiling of the nose wheel well (TASK 20-10-04-400-801).

SUBTASK 32-35-21-410-001

(7) Install the guide pins [11], washers [13], and new cotter pins [12] on the NLG manual release mechanism.

SUBTASK 32-35-21-410-002

(8) Install the screws [10] and the cover [9] on the NLG manual release mechanism.

SUBTASK 32-35-21-420-006

(9) Attach the MNGEA cable B [16] on the MNGEA control drum on the manual extension control mechanism.

(a) Install the cable retainer bolt [17] that connects the MNGEA cable B [16] to the MNGEA control drum on the manual extension control mechanism.

SUBTASK 32-35-21-420-009

(10) Install the six bolts [20], washers [21], spacers [22], washers [23], and nuts [24].

SUBTASK 32-35-21-420-007

(11) Put the terminal end of the nose gear manual extension handle [14] through the hole in the V-slot on the MNGEA control drum.

(a) Attach the nose gear manual extension cable on the MNGEA control drum of the manual extension control mechanism and install the pin [15].

(b) Bend one end of the pin [15] into the MNGEA control drum.

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SUBTASK 32-35-21-420-008

(12) Connect the MNGEA cable A [3] and the MNGEA cable B [16] at the turnbuckle [2]t.

SUBTASK 32-35-21-820-001

(13) Use the low tension cable tensiometer, COM-1553, to adjust the tension in the MNGEA cable to 100 lbf (444.8 N).

(a) Make sure there are an equal number of each MNGEA cable threaded ends engaged in the turnbuckle [2].

(b) Hold this tension for a minimum of 3 minutes.

SUBTASK 32-35-21-820-002

(14) Reduce the tension in the MNGEA cable until you can easily remove the rig pin LGEA1 from rigging pin kit, SPL-1585, from the NLG manual release mechanism.

SUBTASK 32-35-21-080-001

(15) Remove the rig pin LGEA1 from rigging pin kit, SPL-1585, from the NLG manual release mechanism.

SUBTASK 32-35-21-820-003

(16) Do this task: Nose Landing Gear Manual Extension System Adjustment, TASK 32-35-00-820-801.

H. Nose Gear Manual Extension System Cables Installation Test

SUBTASK 32-35-21-710-001

(1) Do a test on the manual extension system for the nose landing gear (TASK 32-35-00-730-801).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-35-21-410-003

(1) Close this access panel:

Number	Name/Location
114AC	Fwd Nose Wheel Well Upper Access Panel

(a) Do this task: Nose Wheel Well Access Panels Installation, TASK 53-14-01-420-801.

SUBTASK 32-35-21-410-004

(2) Install the flight deck floor panel [18].

SUBTASK 32-35-21-410-005

(3) Install the cabin floor panel [1] at approximately body station 270 (TASK 53-21-00-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HYDRAULIC BRAKE SYSTEM - MAINTENANCE PRACTICES

1. General

A. This procedure has these tasks:

(1) Bleed the brake system upstream of the antiskid valves.

NOTE: Use this procedure when a brake system component upstream of the antiskid valves is removed or replaced. This procedure will not remove the air that may be in the system downstream of the antiskid valves.

(2) Bleed the normal (System B) hydraulic brake system.

NOTE: Use this procedure when an antiskid valve, or a component downstream of an antiskid valve in the normal brake system is removed or replaced. This procedure can also be used if there are signs of air in the brake system. This will be shown by slow brake response or chattering.

(3) Bleed the alternate (System A) hydraulic brake system.

NOTE: Use this procedure when an antiskid valve, or a component downstream of an antiskid valve in the alternate brake system is removed or replaced. This procedure can also be used in there are signs of air in the brake system. This will be shown by slow brake response or chattering.

(4) Test of the normal hydraulic system brake fuses (System B).

(5) Test of the alternate hydraulic system brake fuses (System A).

B. It is not usually necessary to bleed the brakes during a routine brake replacement since the hydraulic disconnect fitting at the brake prevents air from getting into the brakes.

C. When you bleed the hydraulic systems, the hydraulic fuses can set. You can reset the fuses one of these two ways:

(1) You can use the fuse bypass lever on each fuse to reset the fuse.

(2) Or, you can release and reset the parking brake to reset the fuse.

TASK 32-41-00-870-801

2. Hydraulic Brake System Upstream of the Antiskid Valves Bleeding

A. General

(1) This task has instructions to bleed the brake system upstream of the antiskid valves. When the system is bled, the air in the brake lines or a component of the brake system is sent through the brake metering valves to the hydraulic reservoir by the return lines.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Bleeding

SUBTASK 32-41-00-480-028

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-480-003

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-480-002

(3) Make sure that the electrical power is on (TASK 24-22-00-860-801).

E. Normal (System B) Hydraulic Brake System Upstream of the Antiskid Valves Bleeding

SUBTASK 32-41-00-860-001

(1) Release the parking brake.

SUBTASK 32-41-00-790-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-00-840-001

(3) Do the steps that follow to bleed the normal brake system through the normal brake metering valves to return:

(a) Push the brake pedals slowly through approximately one-half of their total travel and slowly release them.

1) Repeat this step several times.

(b) Push both of the brake pedals fully to the stops and hold them there for 2 to 4 seconds and then release them.

1) Repeat this step five more times.

(c) Make sure that the brakes apply when you push on the brakes.

(d) Push the brake pedals fully to the stops and set the parking brake.

SUBTASK 32-41-00-860-002

(4) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Alternate (System A) Hydraulic Brake System Upstream of the Antiskid Valves Bleeding

SUBTASK 32-41-00-860-028

(1) Release the parking brake.

SUBTASK 32-41-00-790-002

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the alternate (System A) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-00-840-002

(3) Do the steps that follow to bleed the alternate brake system through the alternate brake metering valves to return:

(a) Push the brake pedals slowly through approximately one-half of their total travel and slowly release them.

1) Repeat this step several times.

(b) Push both of the brake pedals fully to the stops and hold them there for 2 to 4 seconds and then release them.

1) Repeat this step five more times.

(c) Make sure that the brakes apply when you push on the brakes.

(d) Push the brake pedals fully to the stops and set the parking brake.

SUBTASK 32-41-00-860-003

(4) For the alternate (System A) hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

TASK 32-41-00-870-802

3. Normal (System B) Hydraulic Brake System Bleeding

(Figure 201, Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-15035	Bleeder Hose, Brake Part #: J29006-1 Supplier: 81205 Opt Part #: J32096-1 Supplier: 81205
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Normal (System B) Hydraulic Brake System Bleeding

SUBTASK 32-41-00-480-029

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-480-005

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-860-004

(3) Release the parking brake.

F. Normal (System B) Hydraulic Brake System Bleeding

SUBTASK 32-41-00-710-001

(1) To pressurize the normal (System B) hydraulic brake system, do these steps:

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WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Use the hydraulic service cart to pressurize the normal (System B) hydraulic system to 800 psig (5516 kPa) (TASK 29-11-00-860-801).

NOTE: It is recommended that you use the hydraulic service cart because of the large volume of hydraulic fluid this procedure removes from the normal hydraulic system.

NOTE: If you use the airplane's hydraulic supply, monitor the quantity of hydraulic fluid in the reservoir when you bleed the system.

(b) Push the brake pedals slowly through approximately one-half of their total travel and slowly release them.

1) Repeat this step several times to remove the air from the brake lines upstream of the antiskid valves.

(c) Push the two brake pedals fully to the stops and hold them there for 2 to 4 seconds and then release them.

1) Repeat this step five more times.

(d) Push the brake pedals fully to the stops and set the parking brake.

SUBTASK 32-41-00-710-002

(2) To bleed the air from each brake, do these steps for all four brakes (Figure 201):

NOTE: It is recommended that you do all of the steps below for one brake before you start the steps again for the next brake.

NOTE: There are two bleeder valves on each brake, use the bleeder valve to which you can get easy access.

(a) Remove the screw [2] and washer [3] from the center of the bleeder valve [1].

(b) Install the bleeder hose, SPL-15035, on the end of the bleeder screw [4].

(c) Put the free end of the bleeder hose, SPL-15035, into an oil resistant container (5 gal)(19 Liter), STD-1055.

(d) Slowly loosen the bleeder screw [4] approximately one half turn.

NOTE: This will let hydraulic fluid flow out of the bleeder hose, SPL-15035.

(e) Let hydraulic fluid flow out of the bleeder hose, SPL-15035, until there are no air bubbles in the flow of hydraulic fluid.

NOTE: The fluid can flow out of the hose very quickly and intermittently.

(f) If the fluid flow stops (the brake fuse has set), do the steps that follow to reset the brake fuse:

1) Close the bleeder valve [1].

2) On the normal system brake fuse for the brake being bled, pull the fuse bypass lever and hold it in this bypass position for 5 seconds minimum (Figure 202).

3) Open the bleeder valve [1] and continue to bleed the brake until there are no air bubbles in the flow of hydraulic fluid.

(g) If you will not bleed the alternate (System A) hydraulic system, do the steps that follow:

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1) Remove the bleeder hose, SPL-15035, from the bleeder screw [4].
2) Tighten the bleeder screw [4] to 60 in-lb (6.8 N·m).
3) Apply a thin layer of hydraulic fluid, D00153, to the threads of the screw [2].
4) Install the washer [3] and screw [2] in the center of the bleeder screw [4].

SUBTASK 32-41-00-870-001

(3) Release the brake pedals or the parking brake if it is set.

SUBTASK 32-41-00-870-002

(4) Make sure that all of the normal (System B) hydraulic fuses are not set:
   (a) Operate the brake pedals several times.
   (b) Look for movement of the brake wear indicator pins to make sure that all of the brakes operate.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-00-860-005

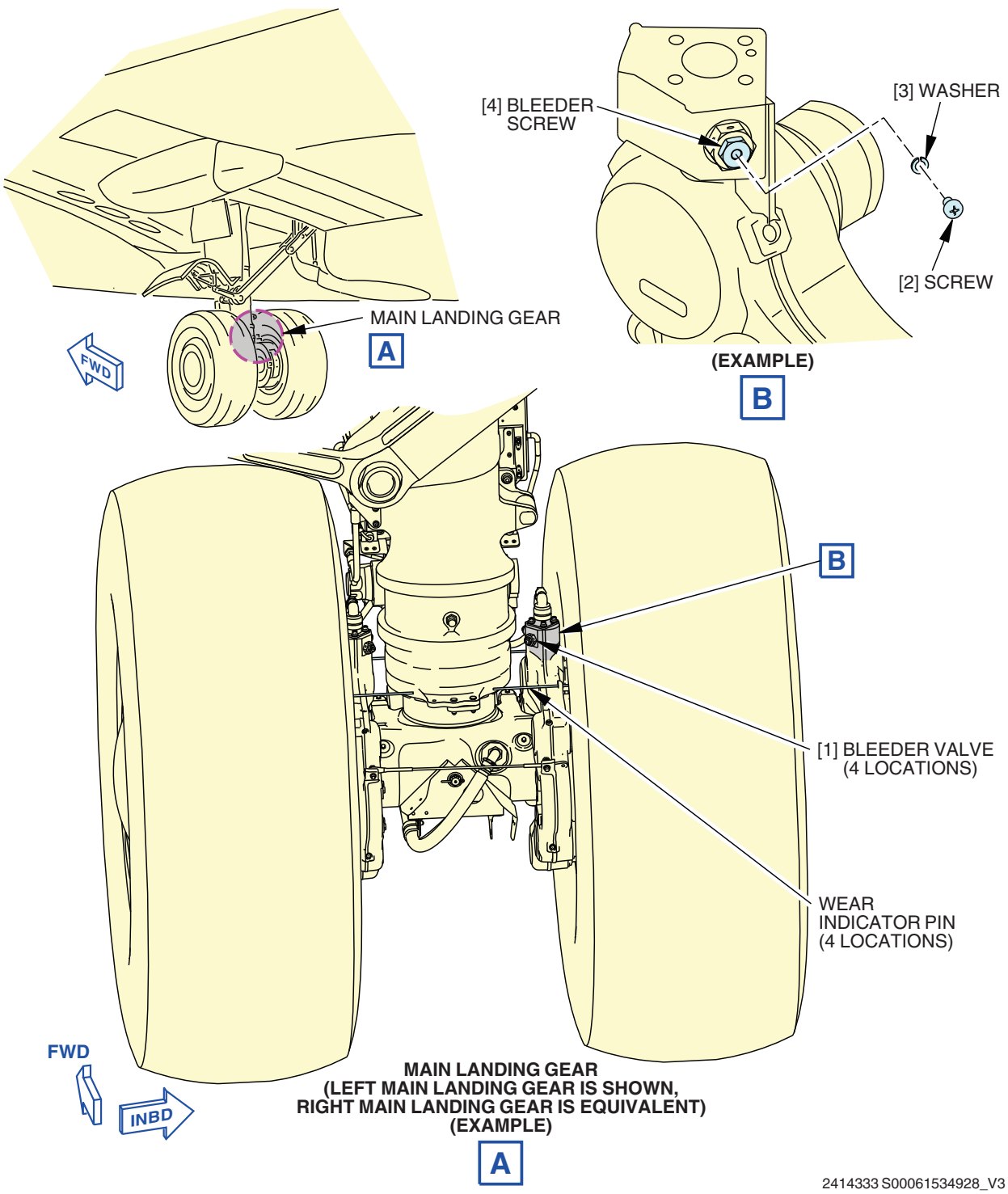
(1) Remove the external hydraulic pressure for the normal (System B) hydraulic system (TASK 29-11-00-860-805).

SUBTASK 32-41-00-870-004

(2) Make sure that the fluid in the normal (System B) hydraulic reservoir is at the correct level (TASK 12-12-00-610-801).
   (a) If the normal (System B) hydraulic reservoir needs to be filled, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

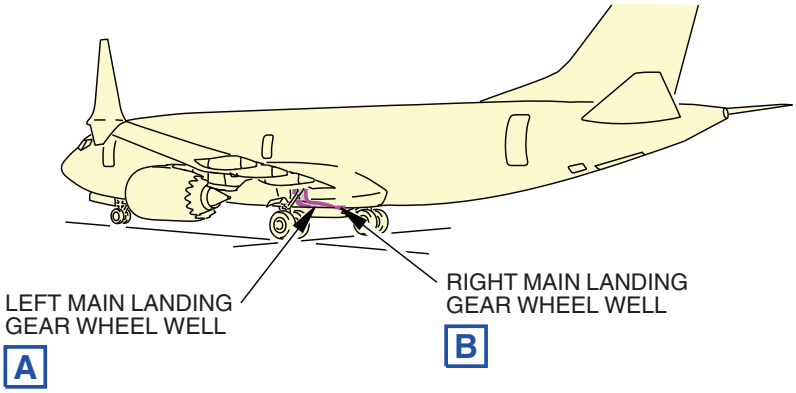
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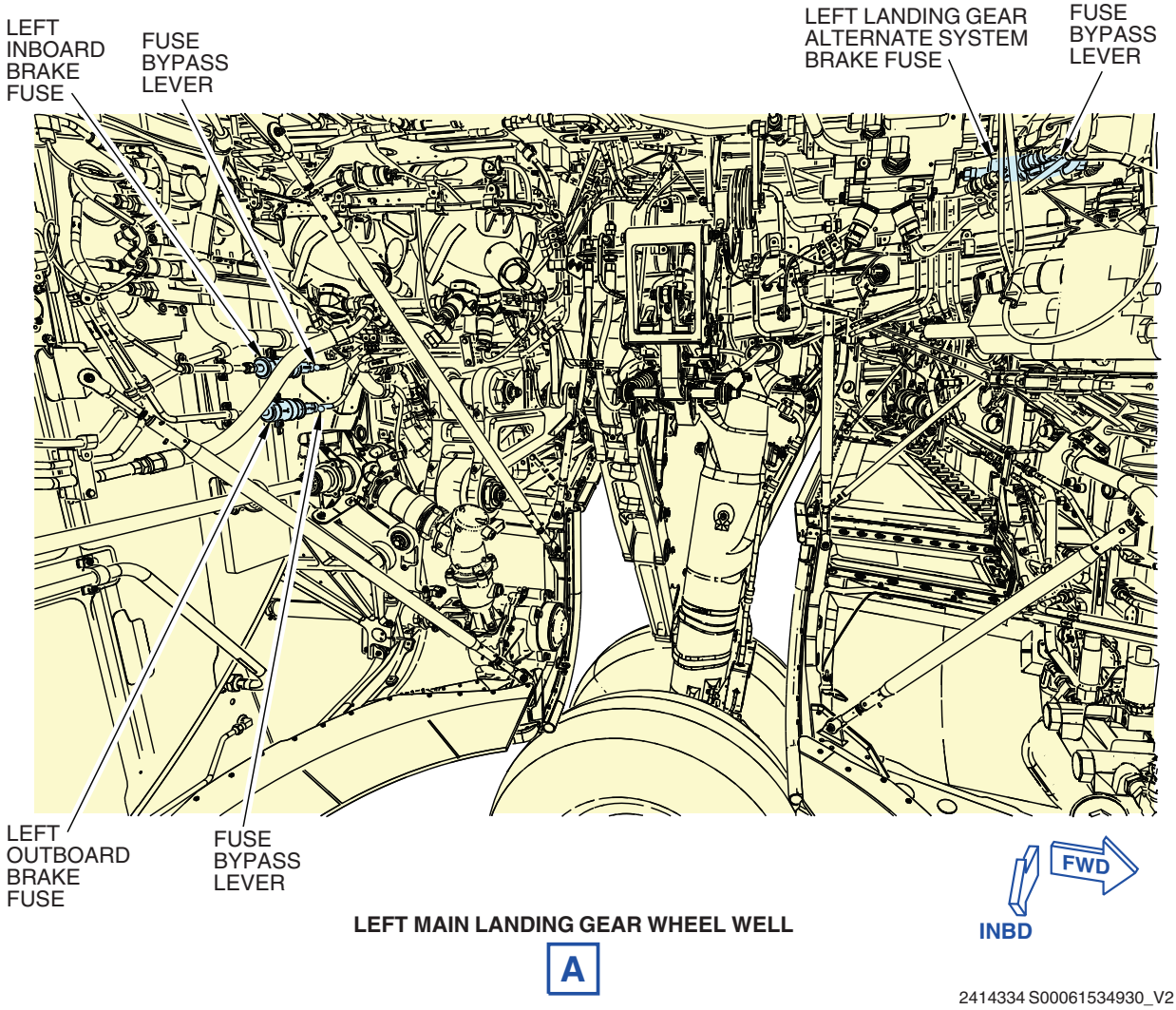
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Hydro-Mechanical Control System - Bleeding
Figure 201/32-41-00-990-806

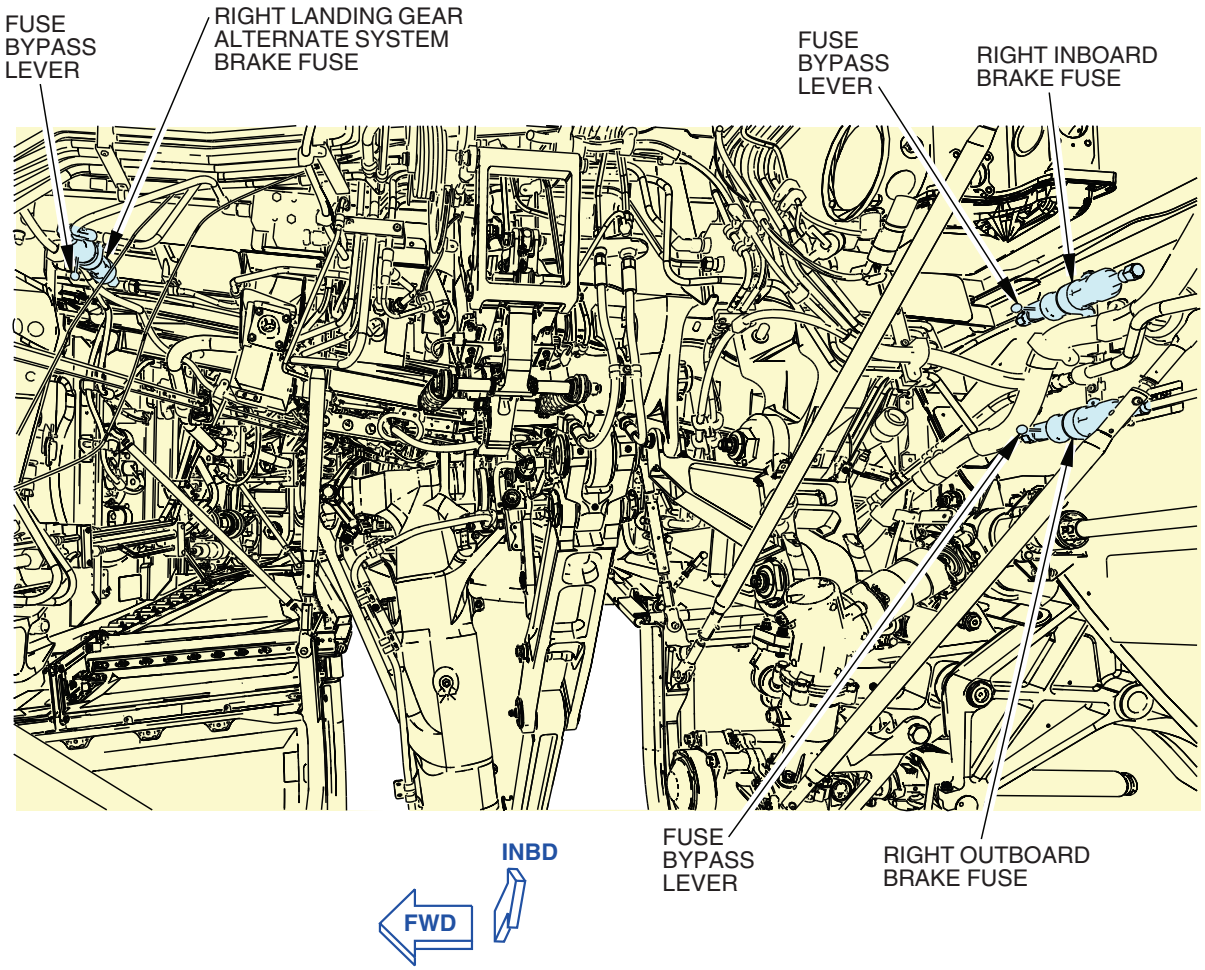
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Brake Hydro-Mechanical Control System - Fuse Locations
Figure 202/32-41-00-990-802 (Sheet 1 of 2)

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Brake Hydro-Mechanical Control System - Fuse Locations
Figure 202/32-41-00-990-802 (Sheet 2 of 2)

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TASK 32-41-00-870-803

4. Alternate (System A) Hydraulic Brake System Bleeding

(Figure 201, Figure 202)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
29-11-00-860-802	Hydraulic System A or B Pressurization with a Portable Hydraulic Cart (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15035	Bleeder Hose, Brake Part #: J29006-1 Supplier: 81205 Opt Part #: J32096-1 Supplier: 81205
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Bleeding

SUBTASK 32-41-00-480-030

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-480-007

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

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SUBTASK 32-41-00-860-006

(3) Release the parking brake.

F. Alternate (System A) Hydraulic Brake System Bleeding

SUBTASK 32-41-00-710-003

(1) To prepare to bleed the alternate (System A) hydraulic brake system, do these steps:

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Use the hydraulic service cart to pressurize the alternate (System A) hydraulic system to 3000 psig (20,684 kPa) (TASK 29-11-00-860-802).

NOTE: It is recommended that you use the hydraulic service cart because of the large volume of hydraulic fluid this procedure removes from the alternate hydraulic system.

NOTE: If you use the airplane's hydraulic supply, monitor the quantity of hydraulic fluid in the reservoir when you bleed the system.

(b) Push the brake pedals slowly through approximately one-half of their total travel and slowly release them.

1) Repeat this step several times to remove the air from the brake lines upstream of the antiskid valve.

(c) Push the two brake pedals fully to the stops and hold them there for 2 to 4 seconds and then release them.

1) Repeat this step five more times.

(d) Push the brake pedals fully to the stops and set the parking brake.

SUBTASK 32-41-00-710-004

(2) To bleed the air from each brake, do these steps for all four brakes (Figure 201):

NOTE: It is recommended you do all of the steps below for one brake before you start the steps again for the next brake.

NOTE: There are bleeder valves on each brake, use the bleeder valve to which you can get easy access.

(a) Remove the screw [2] and washer [3] from the center of the bleeder valve [1].

(b) Install the bleeder hose, SPL-15035, on the end of the bleeder screw [4].

(c) Put the free end of the bleeder hose, SPL-15035, into an oil resistant container (5 gal)(19 Liter), STD-1055.

(d) Slowly loosen the bleeder screw [4] about one half turn.

NOTE: This will let hydraulic fluid flow out of the bleeder hose, SPL-15035.

(e) Let hydraulic fluid flow out of the bleeder hose, SPL-15035, until there are no air bubbles in the flow of hydraulic fluid.

NOTE: The fluid can flow out of the hose very quickly and intermittently.

(f) If the fluid flow stops (the brake fuse has set), do the steps that follow to reset the brake fuse:

1) Close the bleeder valve [1].

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2) On the alternate system brake fuse for the landing gear brake pair being bled, pull the fuse bypass lever and hold it in this bypass position for 5 seconds minimum (Figure 202).

3) Open the bleeder valve [1] and continue to bleed the brake until there are no air bubbles in the flow of hydraulic fluid.

(g) If you will not bleed the normal (System B) hydraulic system, do the steps that follow:

1) Remove the bleeder hose, SPL-15035, from the bleeder screw [4].

2) Tighten the bleeder screw [4] to 60 in-lb (6.8 N·m).

3) Apply a thin layer of hydraulic fluid, D00153, to the threads of the screw [2].

4) Install the washer [3] and screw [2] in the center of the bleeder screw [4].

SUBTASK 32-41-00-870-005

(3) Release the brake pedals or the parking brake if it is set.

SUBTASK 32-41-00-870-006

(4) Make sure that all of the alternate (System A) hydraulic fuses are not set:

(a) Operate the brake pedals several times.

(b) Look for movement of the brake wear indicator pins to make sure that all of the brakes operate.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-00-860-007

(1) Remove the external hydraulic pressure for the alternate (System A) hydraulic system (TASK 29-11-00-860-805).

SUBTASK 32-41-00-870-007

(2) Make sure that the fluid in the alternate (System A) hydraulic reservoir is at the correct level (TASK 12-12-00-610-801).

——— END OF TASK ———

TASK 32-41-00-870-804

5. Normal (System B) Hydraulic System Brake Fuse Test

(Reference Not Currently Available, Figure 202)

A. General

(1) This task gives instructions to do a test of each hydraulic brake fuse for the normal hydraulic brake system. There is one hydraulic brake fuse for each brake for the normal hydraulic system. Each hydraulic brake fuse must pass a specific quantity of hydraulic fluid before it closes. For this task, each brake for the main landing gear must be bled to do a check the hydraulic brake fuse for that brake.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

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C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15035	Bleeder Hose, Brake Part #: J29006-1 Supplier: 81205 Opt Part #: J32096-1 Supplier: 81205
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Test

SUBTASK 32-41-00-480-008

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-860-008

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 32-41-00-860-009

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Use the hydraulic service cart to pressurize the normal (System B) hydraulic system to 800 psig (5516 kPa) (TASK 29-11-00-860-801).

NOTE: It is recommended that you use the hydraulic service cart because of the large volume of hydraulic fluid this procedure removes from the normal hydraulic system.

NOTE: If you use the airplane's hydraulic supply, monitor the quantity of hydraulic fluid in the reservoir when you bleed the system.

SUBTASK 32-41-00-480-009

(4) Set the parking brake.

SUBTASK 32-41-00-480-010

(5) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

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F. Normal (System B) Hydraulic Brake System Fuse Test

SUBTASK 32-41-00-710-005

(1) To do a check of the fuse for each brake, do these steps for all four brakes:

(a) Remove the screw [2] from the center of the bleeder valve [1].

(b) Remove the lockwire, if it is installed, from the bleeder valve [1].

(c) Install the bleeder hose, SPL-15035, on the end of the bleeder valve [1].

(d) Put the free end of the bleeder hose, SPL-15035, into an oil resistant container (5 gal)(19 Liter), STD-1055.

(e) Slowly loosen the bleeder valve [1] about one half turn.

NOTE: This will let hydraulic fluid flow out of the bleeder hose, SPL-15035.

(f) Let the hydraulic fluid flow until the fuse closes and the flow stops.

(g) Make sure that the quantity collected in the oil resistant container (5 gal)(19 Liter), STD-1055, is 60 - 95 cubic inches or approximately 0.26 gal (0.98 l) - 0.41 gal (1.55 l).

(h) Close the bleeder valve [1].

(i) Remove the bleeder hose, SPL-15035, from the bleeder valve [1].

(j) Tighten the bleeder valve [1] to 45 ±10 in-lb (5 ±1 N·m).

(k) Install the screw [2] in the center of the bleeder valve [1].

(l) Install a lockwire, if previously installed, on the bleeder valve [1].

(m) On the normal system fuse for the brake being bled, pull the fuse bypass lever and hold it in this bypass position for 5 seconds minimum.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-00-870-008

(1) Do the steps that follow when you have checked all of the fuses for the normal (System B) hydraulic system:

(a) Make sure all of the normal (System B) hydraulic fuses are not set:

1) Operate the brake pedals several times.

2) Look for movement of the brake wear indicator pins to make sure all of the brakes operate.

(b) Remove the external hydraulic pressure for the normal (System B) hydraulic system. To do this, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(c) Make sure the fluid in the normal (System B) hydraulic reservoir is at the correct level (TASK 12-12-00-610-801).

(d) If the normal (System B) hydraulic reservoir needs to be filled, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

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TASK 32-41-00-870-805

6. Alternate (System A) Hydraulic System Brake Fuse Test

(Reference Not Currently Available, Figure 202)

A. General

(1) This task provides instructions to do a test of each hydraulic brake fuse for the alternate hydraulic brake system. There is one hydraulic brake fuse for the two brakes on each main landing gear for the alternate hydraulic system. Each hydraulic brake fuse must pass a specific quantity of hydraulic fluid before it closes. For this task, only one brake for the each main landing gear must be bled to do a check of the hydraulic brake fuse for that brake.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-12-00-610-801	Hydraulic Reservoir Servicing (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-15035	Bleeder Hose, Brake Part #: J29006-1 Supplier: 81205 Opt Part #: J32096-1 Supplier: 81205
STD-1055	Container - Oil Resistant, 5 Gallon (19 Liter)

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Test

SUBTASK 32-41-00-480-011

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-860-010

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

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SUBTASK 32-41-00-860-011

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Use the hydraulic service cart to pressurize the alternate (System A) hydraulic system to 800 psig (5516 kPa) (TASK 29-11-00-860-801).

NOTE: It is recommended that you use the hydraulic service cart because of the large volume of hydraulic fluid this procedure removes from the alternate hydraulic system.

NOTE: If you use the airplane's hydraulic supply, monitor the quantity of hydraulic fluid in the reservoir when you bleed the system.

SUBTASK 32-41-00-480-012

(4) Set the parking brake.

SUBTASK 32-41-00-480-013

(5) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

F. Alternate (System A) Hydraulic System Brake Fuse Test

SUBTASK 32-41-00-710-006

(1) To do a check of the fuse for both brakes on each main landing gear, do these steps for one brake on each main landing gear:

(a) Remove the screw [2] from the center of the bleeder valve [1].

(b) Remove the lockwire, if it is installed, from the bleeder valve [1].

(c) Install the bleeder hose, SPL-15035, on the end of the bleeder valve [1].

(d) Put the free end of the bleeder hose, SPL-15035, into an oil resistant container (5 gal)(19 Liter), STD-1055.

(e) Slowly loosen the bleeder valve [1] approximately one half turn.

NOTE: This will let hydraulic fluid flow out of the bleeder hose, SPL-15035.

(f) Let the hydraulic fluid flow until the fuse closes and the flow stops.

(g) Make sure that the quantity collected in the oil resistant container (5 gal)(19 Liter), STD-1055, is 60 - 95 cubic inches or approximately 0.26 gal (0.98 l) - 0.41 gal (1.55 l).

(h) Close the bleeder valve [1].

(i) Remove the bleeder hose, SPL-15035, from the bleeder valve [1].

(j) Tighten the bleeder valve [1] to 45 ±10 in-lb (5 ±1 N·m).

(k) Install the screw [2] in the center of the bleeder valve [1].

(l) Install a lockwire, if previously installed, on the bleeder valve [1].

(m) On the alternate system fuse for the main landing gear being bled, pull the fuse bypass lever and hold it in this bypass position for 5 seconds minimum.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-00-870-009

(1) Do the steps that follow after you have done a check of the fuses for the alternate (System A) hydraulic system:

(a) Make sure the two alternate (System A) hydraulic fuses are not set:

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1) Operate the brake pedals several times.
2) Look for movement of the brake wear indicator pins to make sure all of the brakes operate.

(b) Remove the external hydraulic pressure for the alternate (System A) hydraulic system (TASK 29-11-00-860-805).

(c) Make sure the fluid in the alternate (System A) hydraulic reservoir is at the correct level (TASK 12-12-00-610-801).

(d) If the alternate (System A) hydraulic reservoir needs to be filled, do this task: Hydraulic Reservoir Servicing, TASK 12-12-00-610-801.

——— END OF TASK ———

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HYDRAULIC BRAKE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An adjustment of the hydraulic brake system

NOTE: After repair or replacement of the linkage or cables, the mechanical portion of the brake control system must be adjusted (rigged).

(2) An functional test of the normal and alternate brake application.

(3) An operational test of the hydraulic brake accumulator.

(4) An operational test of the alternate brake system.

B. To do a functional test of the hydraulic brake system, do all of the tasks in this procedure.

TASK 32-41-00-820-801

2. Hydraulic Brake System Adjustment

(Figure 501)

A. General

(1) The landing gear brake system can be adjusted with the airplane on or off jacks.

(2) The hydraulic pressure can be 'On' or 'Off' during adjustment.

(3) The landing gear brake system must be adjusted with the brakes off.

(4) If rig pin tools are not available, then use pins that are 0.309 in. (7.849 mm) - 0.311 in. (7.899 mm) in diameter.

(5) One system cycle is defined as alternately depressing the toe pad portion of the pedals to their stop (full on) and fully releasing (full off).

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-10-10-000-801	Turnbuckle Lock Removal (P/B 401)
20-10-10-400-801	Turnbuckle Lock Installation (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4

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(Continued)

Reference	Description
COM-1554	Tensiometer - Cable, High Tension (200 lbs and above) Part #: 102-03108 Supplier: 21844 Part #: 102-03120 Supplier: 21844 Part #: ACM-300 Supplier: 13331 Part #: ACM-600 Supplier: 13331 Part #: ACX-500-1 Supplier: 13331 Part #: T5-8008-106-00 Supplier: 0N8U4 Part #: T60-1001-C9-1A Supplier: 0N8U4
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
SPL-17778	Rig Pin, F70207-129 (included in F70207 Kit) Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Adjustment

SUBTASK 32-41-00-480-014

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-480-015

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-860-012

(3) Release the parking brake.

SUBTASK 32-41-00-860-014

(4) Make sure that the cable system is rigged so that the rig pin LGB 1 and rig pin LGB 2 can be inserted and removed without binding in the rig pin holes.

(a) If the rig pin LGB 1 and rig pin LGB 2 move freely through the rig pin holes, no further adjustment of the forward pushrod [2] and aft pushrod [2] is necessary.

(b) If the rig pin LGB 1 and rig pin LGB 2 do not move freely through the rig pin holes, do the adjustment of the forward pushrod [2] and aft pushrod [2].

SUBTASK 32-41-00-860-029

(5) Make sure that the rig pin LGB 1 and rig pin LGB 2 are removed.

F. Brake Control System Adjustment

SUBTASK 32-41-00-820-001

(1) Do the steps that follow to adjust the brake pedal bus [1]:

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(a) Reduce the cable tension in all four cables to zero:

1) Alternate the adjustment between the cables to make sure that the tension in each cable does not differ by more than 50 lbf (222 N) at any time.

2) Remove the turnbuckle locking clips [5] from the turnbuckles [6].

3) Turn the turnbuckles [6] to reduce the tension.

(b) Adjust the aft pushrod [2] on the brake pedal bus [1]:

1) Loosen the jamnut [7] on the right end.

2) Remove the nut [13], washer [12], bolt [8], and bushing [9] to disconnect the aft pushrod [2] from the crank arm clevis [11].

(c) Adjust the forward pushrod [2] on the brake pedal bus [1]

1) Create a cross hair with a fine tip felt pen at the junction of the two rods:

NOTE: This will be a reference point for future adjustments.

a) Do not remove the screws [28] before the cross hair is created.

b) Put one line horizontally across both tubes, halfway between the screw [28] head and nut [30].

c) Put the other line around the forward pushrod [2] where the smaller tube meets the larger tube to complete the cross hair.

2) Loosen the jamnut [7] on the right end of the forward pushrod [2].

3) Remove the screws [28], washers [29], and nuts [30].

(d) Install rig pin LGB1, F70207-14, from the rigging pin kit, SPL-1585, through the left beam structure and brake control bellcranks [4].

NOTE: It is necessary to remain installed the rig pin for the cable adjustment.

(e) Install rig pin LGB2, F70207-14, from the rigging pin kit, SPL-1585, through the right beam structure and brake control bellcranks [4].

NOTE: It is necessary to remain installed the rig pin for the cable adjustment.

(f) Twist the right half of the forward pushrod [2] to get the original cross hair position.

(g) Adjust the forward pushrod [2] to make sure the rig pins are evenly weighted:

1) Let the pedals rest on the rig pins.

2) Twist the right half of the forward pushrod [2] until there is a small gap under each rig pin in each right brake crank arm rig pin hole.

NOTE: This ensures that the brake pedals are on their correct rig pins, and that the weight does not move across the forward rod to the other rig pin.

3) Install only one screw [28], washer [29], and nut [30].

(h) Do a check for clearance between the first officer crank arm and the first officer brake pedal stop:

1) Remove the two rig pins to let the pedals rest against the brake pedal stops.

2) Make sure that there is a 0.02 in. (0.51 mm) - 0.04 in. (1.02 mm) gap between the first officer right brake crank arm and the first officer brake pedal stop.

3) Make sure that there is no gap between the captain right brake crank arm and the captain brake pedal stop.

4) Reinstall the two rig pins.

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(i) If the clearances are not correct, adjust the forward pushrod [2] again until the clearances are correct.

(j) Adjust the aft pushrod [2] to make sure the rig pins are evenly weighted.

1) Twist the right rod end [10] to align it with the right side crank arm clevis [11]:

2) Temporarily install the bolt [8], and bushing [9] to hold the adjusted rod end [10].

3) Let the pedals rest on the rig pins.

4) Adjust the length of the aft pushrod [2] until there is a small gap under each rig pin in each right brake crank arm rig pin hole.

NOTE: This ensures that the brake pedals are on their correct rig pins, and that the weight does not move across the forward rod to the other rig pin.

(k) Do a check for clearance between the first officer crank arm and the first officer brake pedal stop:

1) Remove the two rig pins to let the pedals rest against the brake pedal stops.

2) Make sure that there is a 0.02 in. (0.51 mm) - 0.04 in. (1.02 mm) gap between the first officer left brake crank arm and the first officer brake pedal stop.

3) Make sure that there is no gap between the captain left brake crank arm and the captain brake pedal stop.

4) Reinstall the two rig pins.

(l) If the clearances are not correct, adjust the aft pushrod [2] again until the clearances are correct.

(m) Do a check of the rig pin fit.

NOTE: The rig pins will be biased in the hole. This means the rig pins will support the weight of the pedals and bus structure and not be not be entirely free.

1) Make sure that the rig pin does not bind or need excessive force to install or remove it.

(n) Install the remaining fasteners to complete the installation of the forward pushrod [2] and aft pushrod [2]:

1) For the forward pushrod [2], install the remaining screw [28], washer [29], and nut [30].

a) Tighten the nuts [30] to snug plus 1/4 turn.

2) For the forward pushrod [2], tighten jamnuts [7] to snug plus 1/4 turn.

3) For the aft pushrod [2], install the bolt [8], bushing [9], washer [12], and nut [13] to the crank arm clevis [11].

4) For the aft pushrod [2], tighten the jamnuts [7].

5) Make sure the threads of the rod ends [10] can be seen for at least half of the inspection holes, or there are no more than 15 exposed threads between the jamnut [7] and bearing.

NOTE: This will make sure that the threads of the rod ends correctly engage the pushrods [2].

6) If no further adjustments are necessary, increase the tension in the cables per the tension value in Table 502.

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a) When the tension is increased with the rig pins in position, alternate the adjustment between the cables to make sure that the tension in each cable does not differ by more than 50 lbf (222 N) at any time.

NOTE: This will keep the rig pins from bending.

(o) If the brake cables will not be adjusted, remove the rig pins LGB1 and LGB2 from the brake control bellcrank [4].

SUBTASK 32-41-00-820-002

(2) Do the steps that follow to adjust the left brake metering valve [21] and right brake metering valve [27]:

(a) Disconnect the control rod [20] from the crank [25] for the left brake metering valve [21] and right brake metering valve [27].

NOTE: The left brake metering valve and right brake metering valve are spring loaded to the brakes off position. This will position the left brake metering valve and right brake metering valve.

1) Remove the nut [18], washer [26], and bolt [22] to disconnect the control rod [20] from the crank [25] for the left brake metering valve [21] and right brake metering valve [27].

2) Loosen the jamnuts [19] on the two ends of the control rod [20].

(b) With the brake pedals in the off position, install the rigging pin kit, SPL-1585, between the crank [25] and rig pin structure hole.

1) Install the rig pin LGB3 with rig pin, F70207-129, SPL-17778, from the rigging pin kit, SPL-1585, through the rig pin hole for the left brake metering valve [21].

2) Install the rig pin LGB4 with rig pin, F70207-129, SPL-17778, from the rigging pin kit, SPL-1585, through the rig pin hole for the right brake metering valve [27].

3) Adjust the control rod [20] to get a rig pin slip fit.

(c) Install the bushing [24] in the crank [25].

(d) Install the retainer [23] to hold the bushing [24].

(e) Install the bolt [22], washer [26], and nut [18] to connect the control rod [20] to the crank [25] for the left brake metering valve [21] and right brake metering valve [27].

(f) Make sure the rig pins LGB3 or LGB4 moves freely through the rig pin holes.

(g) Do these steps to position the crank [25]:

NOTE: This will position the rig pin in the neutral position in the crank [25].

1) Install the bushing [24] in the crank [25].

2) Install the retainer [23] to hold the bushing [24].

3) Install the bolt [22], washer [26], and nut [18] to connect the control rod [20] to the crank [25] for the left brake metering valve [21] and right brake metering valve [27].

4) Insert the rig pin.

5) Adjust the control rod [20] by turning the threaded control rod [20] only, all the way in one direction until the rig pin contacts the crank [25].

6) Reverse the adjustment direction and count the turns until the rig pin contacts the other side of the crank [25].

7) Divide the number of turns by 2 and reverse the control rod [20] adjustment direction again.

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(h) Tighten the jamnuts [19] on the two ends of the control rod [20].

(i) If the brake cables will not be adjusted, remove rig pin LGB3 or LGB4 from the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

SUBTASK 32-41-00-820-003

(3) Do these steps to adjust the brake cables for the left side and the right side:

NOTE: It is recommended that airplanes supplied with the cables that contain higher zinc coating (Grade D or Grade K) be re-tensioned 3 to 4 months into service. This will bring the tension of the control cables within the desired range. After the cables are re-tensioned, the high zinc cables (Grade D or Grade K) will continue to relax at a rate that is equivalent to the Grade B cables. To identify the grade of the cable, refer to the table below:

Table 501/32-41-00-993-803 Control Cable Grades and Diameters

CONTROL CABLE DIAMETER (TERMINAL SIZE REF)	CONSTRUCTION FIRST DASH NUMBER *[1]									
	GRADE A					GRADE B		GRADE D		GRADE K
	ZINC			TIN OVER ZINC		ZINC	TIN OVER ZINC	ZINC	TIN OVER ZINC	ZINC
	7 X 7	7 X 7 (RT)	7 X 19	7 X 7	7 X 7 (RT)	7 X 19	7 X 19	7 X 19	7 X 19	7 X 19
.094 (3/32)	3B	3G*[2]	3A	3D	3E *[2]	- — —	- — —	- — —	- — —	- — —
.125 (1/8)	- — —	- — —	4A	- — —	- — —	4F	4C	4J*[3]	4H*[3]	4K*[3]
.188 (3/16)	- — —	- — —	- — —	- — —	- — —	- — —	- — —	6J*[4]	- — —	- — —

*[1] First dash number in the part number of the cable. Example: BACC2C-3D-01085-CE is Grade A, Tin Over Zinc, 7 x 7.

*[2] Code 3E and 3G are specified only where there is a specific requirement for REDUCED TOLERANCE (RT) percent elongation wire rope, 7 X 7 RT, per BMS7-265.

*[3] Only Grade D and Grade K .125 (1/8) cables (4J, 4H and 4K) are high zinc cables.

*[4] It is not necessary to adjust the Grade D 0.188 (3/16) cables (6J) 3 to 4 months after installation.

(a) Make sure that the rig pins LGB1 and LGB2, are installed in the bellcrank rig pin holes [3], and the rig pins LGB3 and LGB4 with rig pin, F70207-129, SPL-17778, are installed in the brake metering valve rig pin holes.

(b) For the locking clips [5] on the brake cables, do this task: Turnbuckle Lock Removal, TASK 20-10-10-000-801.

(c) Turn the turnbuckle [6] with the low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to adjust the brake cable tension to the value shown in Table 502.

1) Alternate the adjustment between the LGB1A cable and LGB1B cable for the left side, and the LGB2A cable and the LGB2B cable for the right side.

Table 502/32-41-00-993-801 Brake Cable Rigging Loads

Ambient Temperature	Rigging Load (±10 lbf (44 N))
-20°F (-29°C)	150 lbf (667 N)
-10°F (-23°C)	155 lbf (689 N)
0°F (-18°C)	160 lbf (712 N)

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Table 502/32-41-00-993-801 Brake Cable Rigging Loads (Continued)

Ambient Temperature	Rigging Load (±10 lbf (44 N))
10°F (-12°C)	165 lbf (734 N)
20°F (-7°C)	170 lbf (756 N)
30°F (-1°C)	175 lbf (778 N)
40°F (4°C)	180 lbf (801 N)
50°F (10°C)	185 lbf (823 N)
60°F (16°C)	190 lbf (845 N)
70°F (21°C)	195 lbf (867 N)
80°F (27°C)	200 lbf (890 N)
90°F (32°C)	205 lbf (912 N)
100°F (38°C)	210 lbf (934 N)
110°F (43°C)	215 lbf (956 N)
120°F (49°C)	220 lbf (979 N)

a) Make sure that the tension in each cable does not differ by more than 50 lbf (222 N) at any time.

NOTE: This will keep the rig pins from bending.

(d) Make sure that the rig pins LGB1 and LGB2 move freely in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4 move freely in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

(e) Do these steps new brake cables were installed:

1) Increase the tension of the brake cables to 350 ±10 lbf (1557 ±45 N) with the high tension cable tensiometer, COM-1554.

a) Alternate the adjustment between the LGB1A cable and LGB1B cable for the left side, and the LGB2A cable and LGB2B cable for the right side.

<1> Make sure that the tension in each cable does not differ by more than 50 lbf (222 N) at any time.

NOTE: This will keep the rig pins from bending.

2) Make sure that the rig pins LGB1 and LGB2 move freely in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4 move freely in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

3) Remove the rig pins LGB1 and LGB2 from the brake pedal bus [1] and the rig pins LGB3 and LGB4 from the left brake metering valve [21] and right brake metering valve [27].

4) Push the left brake pedals to their stops and then fully release them 25 times.

5) Push the right brake pedals to their stops and then fully release them 25 times.

6) Install the rig pins LGB1 and LGB2, in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4, in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

NOTE: LGB1 is for the captain' brake pedal bus and LGB2 is for the first officer's brake pedal bus.

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7) Adjust the brake cable tension with the low tension cable tensiometer, COM-1553, or high tension cable tensiometer, COM-1554, to the value shown in the brake cable tension vs temperature chart (Table 502).

8) Make sure that the rig pins LGB1 and LGB2 move freely in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4 move freely in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

(f) Remove the rig pins LGB1 and LGB2 from the brake pedal bus [1] and the rig pins LGB3 and LGB4 from the left brake metering valve [21] and right brake metering valve [27].

(g) Push the left brake pedals to their stops and then fully release them 3 times.

(h) Push the right brake pedals to their stops and then fully release them 3 times.

(i) Install the rig pins LGB1 and LGB2, in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4, in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

NOTE: LGB1 is for the captain' brake pedal bus and LGB2 is for the first officer's brake pedal bus.

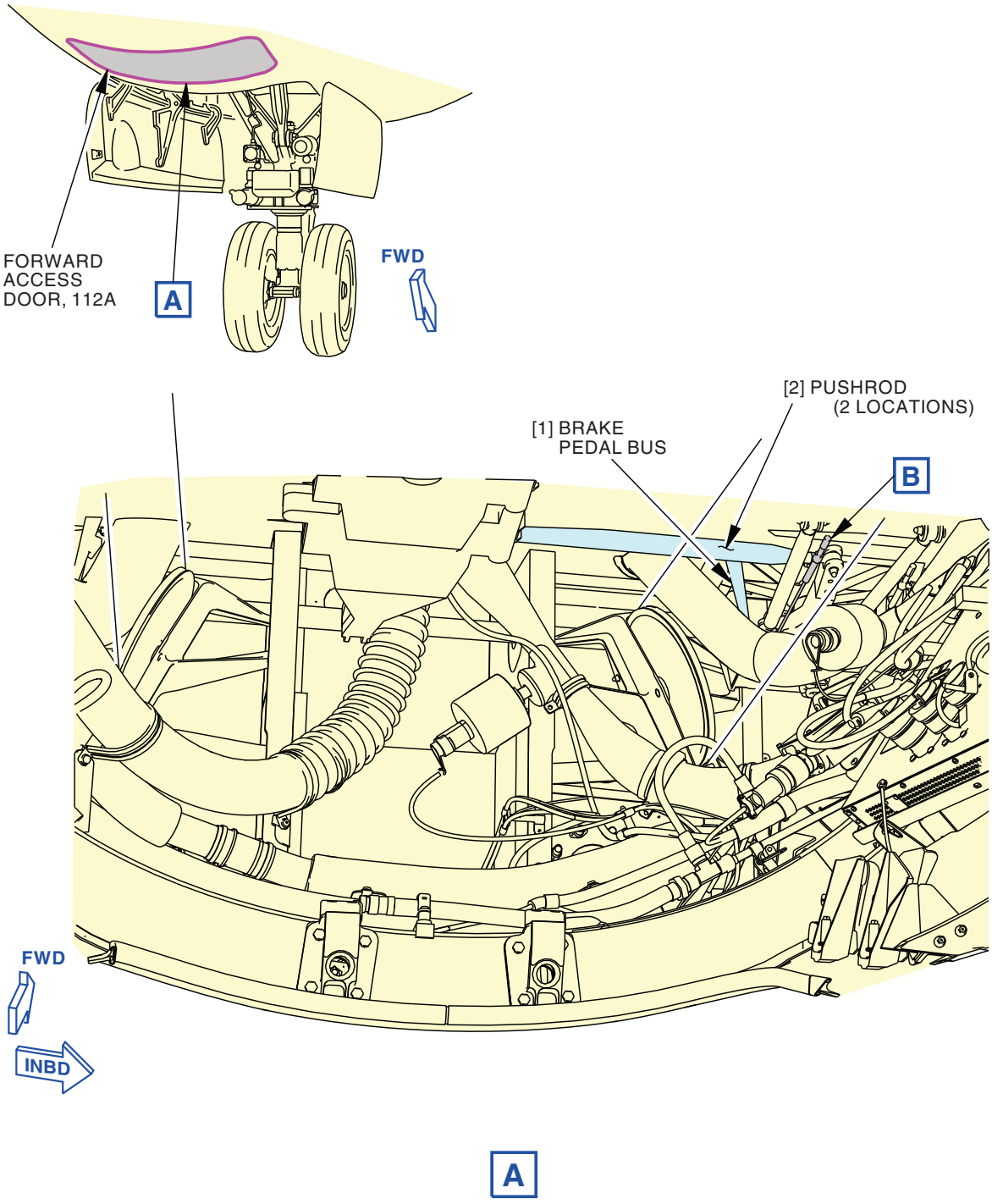
(j) Make sure the rig pins LGB1 and LGB2 move freely in the rig pin holes for the brake pedal bus [1] and the rig pins LGB3 and LGB4 move freely in the rig pin holes for the left brake metering valve [21] and right brake metering valve [27].

(k) Remove the rig pins LGB1 and LGB2 from the brake pedal bus [1] and the rig pins LGB3 and LGB4 rig pin, F70207-129, SPL-17778 from the left brake metering valve [21] and right brake metering valve [27].

(l) For the lock clips on the brake cables, do this task: Turnbuckle Lock Installation, TASK 20-10-10-400-801.

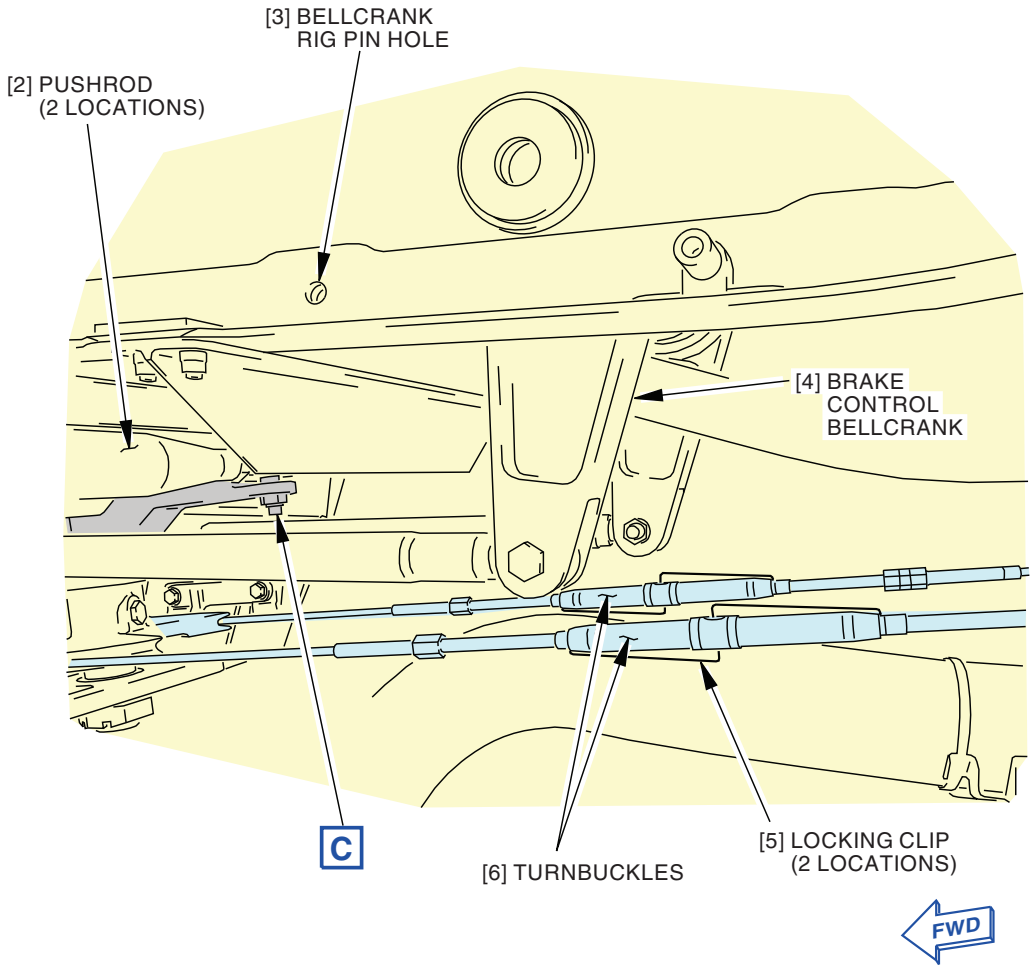
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



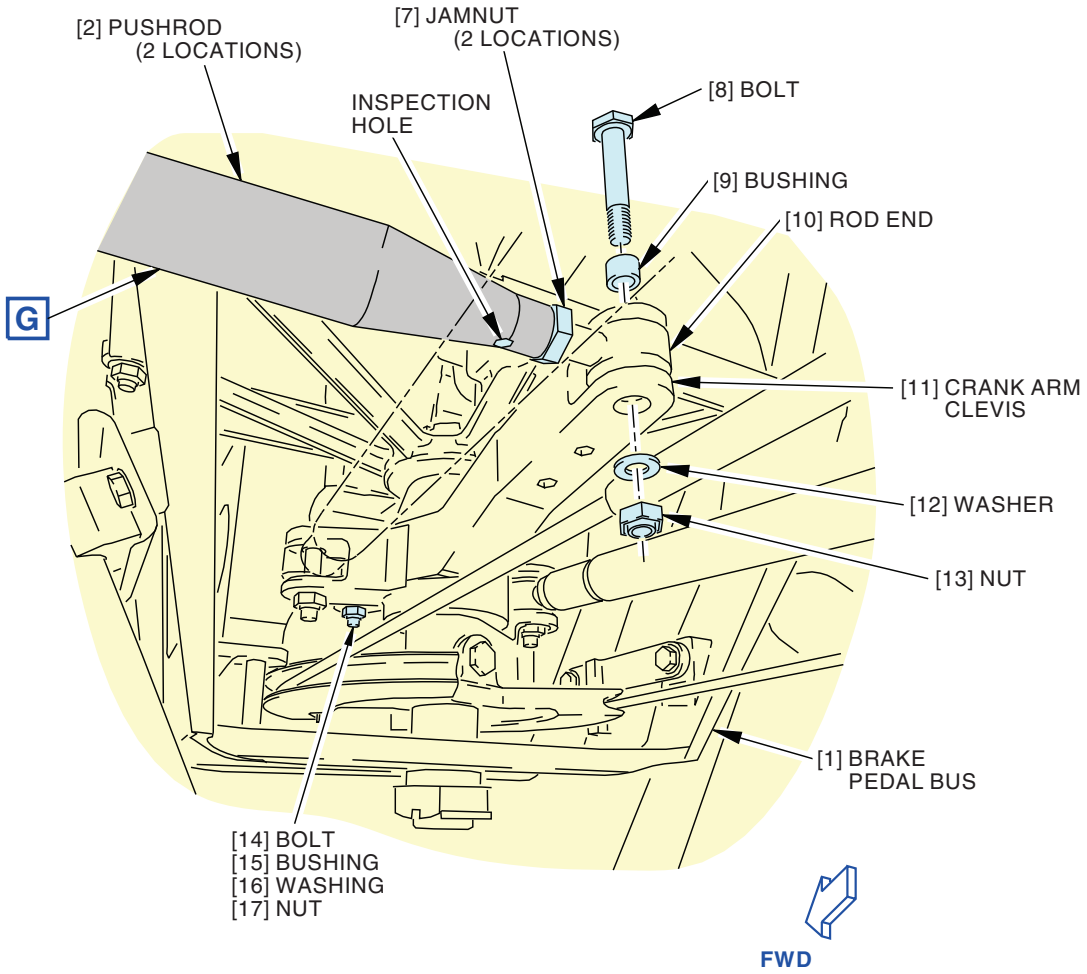
RIGHT CABLE INSTALLATION
(LEFT CABLE INSTALLATION IS EQUIVALENT)

B

2414337 S00061534938_V1

Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



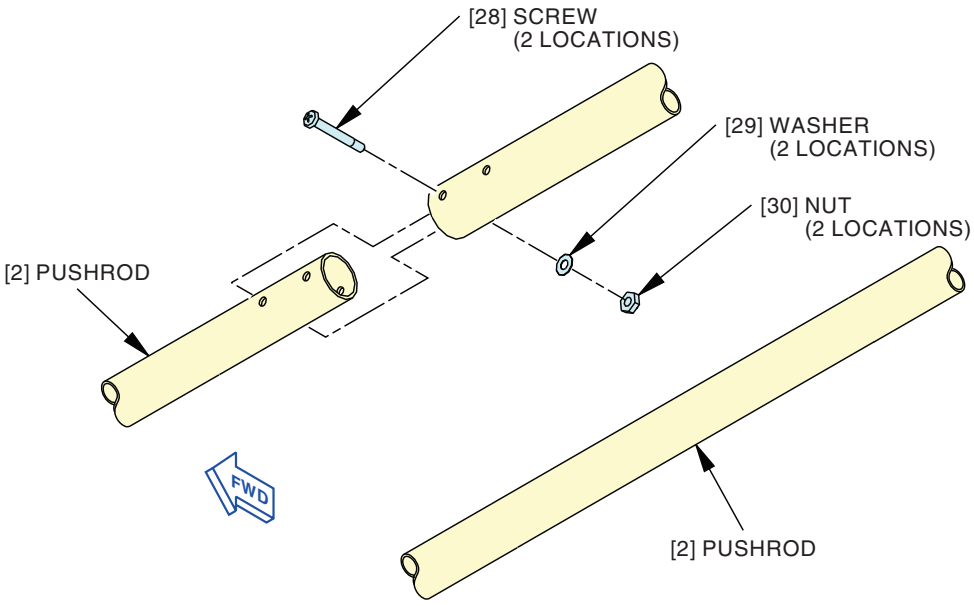
RIGHT BRAKE PEDAL BUS
(LEFT BRAKE PEDAL BUS IS EQUIVALENT)

C

2414338 S00061534939_V2

Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



**BRAKE PEDAL BUS PUSHRODS
(BOTTOM VIEW)**

G

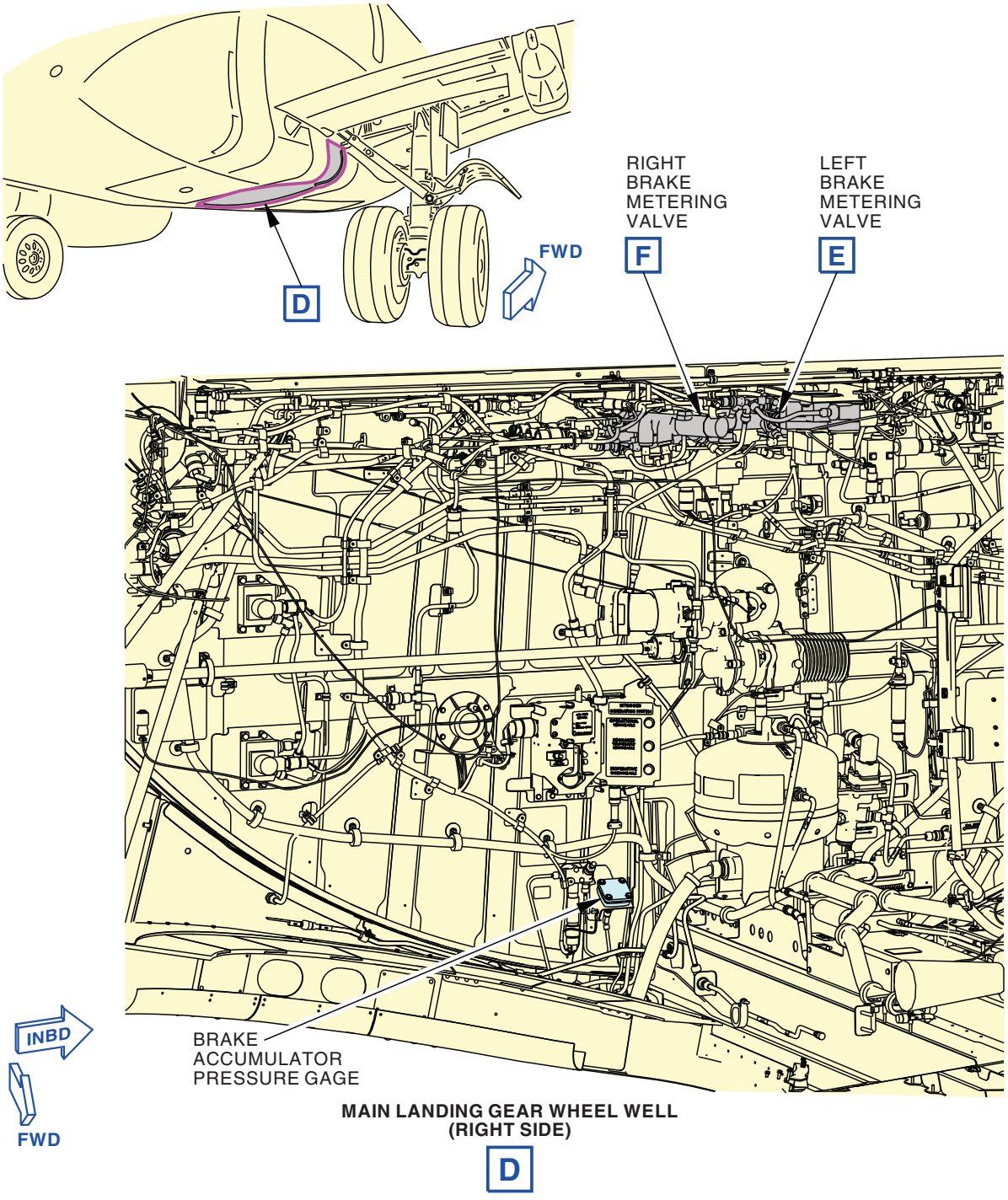
NOTE:

SURROUNDING STRUCTURE AND COMPONENTS ARE REMOVED FOR CLARITY.

2995979 S0000772600_V1

**Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 4 of 6)**

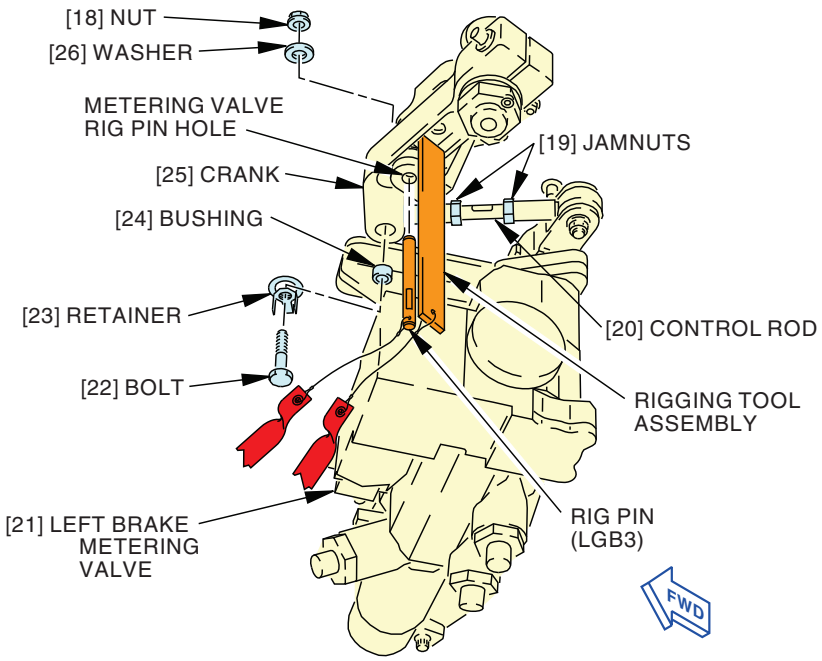
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414339 S00061534940_V1

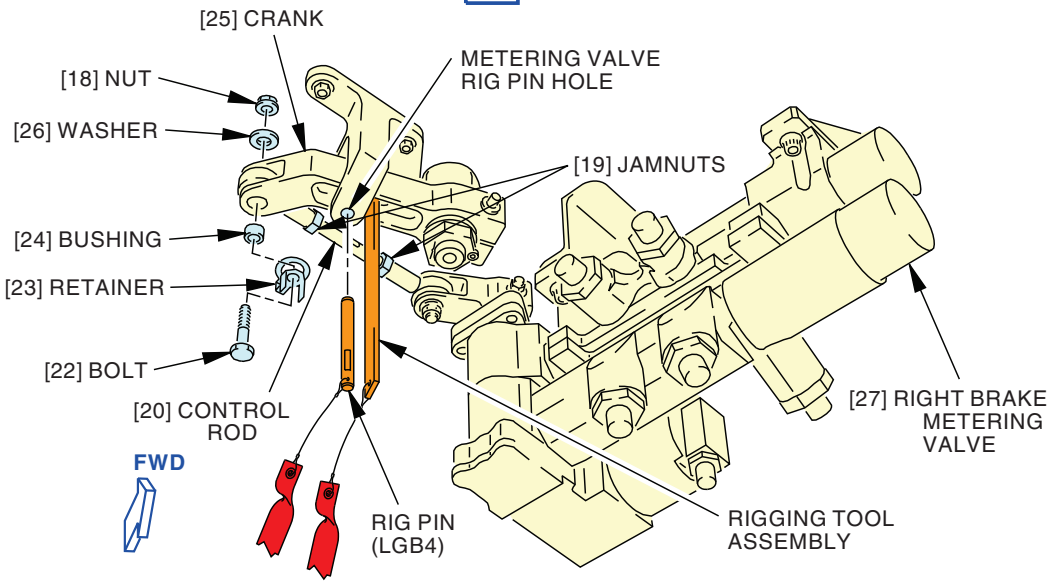
Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEFT BRAKE METERING VALVE

E



RIGHT BRAKE METERING VALVE

F

2414340 S00061534941_V2

Hydraulic Brake System Adjustment
Figure 501/32-41-00-990-803 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-00-710-801

3. Normal and Alternate Brake Application Functional Test

A. General

(1) This procedure does a functional test of these systems:

(a) The operation of the brakes with hydraulic power supplied by the NORMAL (System B) brake hydraulic system.

(b) The operation of the brakes with hydraulic power supplied by the ALTERNATE (System A) brake hydraulic system.

(2) The brake pedal force and angle will also be checked in this task.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1822	Gauge - Digital Force, Push-Pull (100 Lb Capacity, 0.125 Lb Resolution or better) Part #: DFS3-100 Supplier: 11710 Part #: FDIX 100 Supplier: 0BFD9 Opt Part #: DFGS100 Supplier: 11710 Opt Part #: FDI 100 Supplier: 0BFD9 Opt Part #: FDV 100 Supplier: 0BFD9
SPL-2555	Measuring Equipment - Brake Pedal (Single Pedal) Part #: J32001-67 Supplier: 81205 Opt Part #: J32001-63 Supplier: 81205 Opt Part #: J32001-64 Supplier: 81205
SPL-10813	Gauge Set, Pressure Check - Hydraulic Brake System Part #: C32051-1 Supplier: 81205 Opt Part #: F72977-45 Supplier: 81205 Opt Part #: F72977-46 Supplier: 81205 Opt Part #: F72977-47 Supplier: 81205 Opt Part #: F72977-62 Supplier: 81205
SPL-11464	Measuring Equipment - Brake Pedal (Dual Pedal) Part #: J32001-68 Supplier: 81205
SPL-15031	Test Equipment - Brake Bleed Part #: G32075-16 Supplier: 81205
STD-1232	Protractor - Digital

D. Location Zones

Zone	Area
211	Flight Compartment - Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Procedure

SUBTASK 32-41-00-480-016

(1) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-480-017

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation,(TASK 32-00-01-480-801).

SUBTASK 32-41-00-860-015

(3) Release the parking brake.

SUBTASK 32-41-00-480-019

(4) Install one of these tools, or equivalent, on the captain's and the first officer's brake pedals.

(a) push-pull digital force gauge, COM-1822, or

(b) brake pedal measuring equipment, SPL-2555, or

(c) brake pedal measuring equipment, SPL-11464.

SUBTASK 32-41-00-480-020

(5) Install the pressure gauge, SPL-15031, or the gauge set, SPL-10813, on each brake.

SUBTASK 32-41-00-710-007

(6) Use a digital protractor, STD-1232, to measure the angle of the brake pedals.

SUBTASK 32-41-00-790-003

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) For the hydraulic systems A or B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(a) To do a test of the normal (System B) brake system, only pressurize the system B hydraulic system.

(b) To do a test of the alternate (System A) brake system, only pressurize the system A hydraulic system.

NOTE: The normal hydraulic system (System B) must not be pressurized for this test.

F. Normal and Alternate Brake Application Functional Test

NOTE: The steps that follow are applicable for the normal and alternate brake systems. The system that is tested depends on which hydraulic system is pressurized.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-00-710-008

(1) Measure the brake pedal force and travel for each brake pedal.

(a) Make sure that each pedal meets the requirements in Table 503.

(b) Attach a digital protractor, STD-1232, to the brake pedal to measure the travel (position).

1) Make sure the digital protractor, STD-1232, is set to zero.

(c) Attach a spring scale or brake pedal measuring equipment, SPL-2555, to measure pedal forces.

Table 503/32-41-00-993-802 Brake Pedal

	*[1]	*[2]	
Brake Pedal Travel (Deg)	Brake Pedal Force	Brake Pedal Force	Brake Pressure
4.5 +/-1.5	28 ±3 lbf (125 ±14 N)	42 ±3 lbf (187 ±14 N)	400 ±50 psig (2758 ±345 kPa)

*[1] The pedal force must be measured by the load that you apply perpendicular to the tip of the pedal, 8.25 in. (209.55 mm) from the brake pedal pivot. The brake pressure is the value shown at each brake gage.

*[2] The pedal force must be measured with the brake pedal measuring equipment, SPL-2555. The load on the brake pedal measuring equipment, SPL-2555, is located 5.48 in. (139.19 mm) from the brake pedal pivot. The brake pressure is the value shown at each brake gage.

1) Make sure the force gage is set to zero.

(d) Push the captain’s left and right brake pedals until the brake force gage at each pedal shows the value in Table 503.

(e) Hold the brake pedals in this position.

(f) Make sure the hydraulic pressure at each brake is within the specified pressure.

(g) Make sure the angle of the captain’s left and right brake pedals are 4.5 ± 1.5 degrees.

(h) Release the captain’s brake pedals.

(i) Make sure the brakes fully release by observing that the brake pistons retract away from the heat sink.

(j) Make sure the angle of the captain's left and right brake pedals are 0.0 ± 0.50 degrees.

(k) Push the captain’s brake pedals fully down.

1) Make sure that the hydraulic pressure at each brake is 3000 ±150 psi (20,684 ±1034 kPa).

(l) Release the captain’s brake pedals.

(m) Make sure the brakes fully release by observing that the brake pistons retract away from the brake heat sink.

(n) Repeat the above steps with the first officer’s brake pedals.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-00-840-004

(1) If it is necessary, set the parking brakes.

SUBTASK 32-41-00-040-001

(2) For the B hydraulic systems A or B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-00-080-001

(3) Remove the pressure gauge, SPL-15031, or the gauge set, SPL-10813, from each brake.

END OF TASK

TASK 32-41-00-720-801

4. Accumulator Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task has instructions to do an operational test of the brake accumulator system. This will do a test of the operation of the accumulator charging valve, the pressure transmitter, the accumulator isolation valve, and the accumulator pressure gauge.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
12-15-11-610-801	Check of the Brake Accumulator Pre-charge Pressure (P/B 301)
24-22-00-860-801	Supply Electrical Power (P/B 201)
24-22-00-860-802	Remove Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Accumulator Operational Test

SUBTASK 32-41-00-480-025

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-00-480-026

(2) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 32-41-00-480-027

(3) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-860-019

(4) Release the parking brake.

SUBTASK 32-41-00-860-020

(5) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-00-840-003

(6) Fully push the brake pedals 12 times to remove hydraulic pressure from the brake accumulator.

NOTE: You can look at the wear pin indicators on each brake to see that the brakes no longer actuate when the brake pedals are pushed.

SUBTASK 32-41-00-710-012

(7) Wait for approximately 5 minutes to let the accumulator hydraulic pressure stabilize.

SUBTASK 32-41-00-700-001

(8) To make sure the brake accumulator has the correct pressure precharge, do this task: Check of the Brake Accumulator Pre-charge Pressure, TASK 12-15-11-610-801.

SUBTASK 32-41-00-790-005

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) For the A and the B hydraulic systems, for at least 2 minutes, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-00-710-013

(10) Make sure the pressure shown by the HYD BRAKE PRESS gauge on the First Officer's panel reads within 100 psig (689 kPa) of the hydraulic system 'B' pressure.

(a) Write down the hydraulic system 'B' steady state pressure value.

SUBTASK 32-41-00-710-014

(11) Make sure the HYD BRAKE PRESS gauge on the First Officer's panel reads within 100 psig (689 kPa) of the brake accumulator pressure gauge on the rear bulkhead in the right main gear wheel well.

SUBTASK 32-41-00-860-021

(12) For the B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-00-710-015

(13) Wait approximately one minute until the B hydraulic system pressure drops to less than 1000 psi (6895 kPa)i.

SUBTASK 32-41-00-710-016

(14) Push the brake pedals to the stops 7 times while you observe the brake accumulator pressure.

(a) Wait approximately 3 seconds after each time you apply the brakes.

(b) Make sure the brake accumulator pressure does not decrease with each brake application.

SUBTASK 32-41-00-860-022

(15) For the A hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-00-710-017

(16) Make sure the HYD BRAKE PRESS gauge on the First Officer's panel reads within 150 psig (1034 kPa) of hydraulic system 'B' steady state pressure previously written down.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-00-710-018

(17) Push the brake pedals to the stops 7 times while you observe the brake accumulator pressure.

(a) Wait approximately 3 seconds after each time you apply the brakes.

(b) Make sure the brake accumulator pressure decreases with each brake application.

SUBTASK 32-41-00-700-002

(18) To make sure the brake accumulator has the correct pressure precharge, do this task: Check of the Brake Accumulator Pre-charge Pressure, TASK 12-15-11-610-801.

SUBTASK 32-41-00-860-023

(19) Set the parking brake.

SUBTASK 32-41-00-860-024

(20) If it is not necessary, remove electrical power (TASK 24-22-00-860-802).

——— END OF TASK ———

TASK 32-41-00-710-803

5. Alternate Brake System Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Alternate Brake System Operational Test

SUBTASK 32-41-00-480-034

(1) Make sure that the chocks are installed at the wheels (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-00-860-027

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-00-840-006

(3) Fully push the brake pedals 12 times to remove hydraulic pressure from the brake accumulator.

NOTE: You can look at the wear pin indicators on each brake to see that the brakes no longer actuate when the brake pedals are pushed.

SUBTASK 32-41-00-790-006

(4) Do these steps that follow to do an operational test of the alternate (System A) brake system:

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WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) For the system A (alternate), do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

NOTE: At least one person needs to be stationed near the main landing gear brakes.

1) Repeat brake pedal application and release as needed to verify operation of each brake.

(b) Fully push the captain's brake pedals.

(c) Observe the movement of each brake’s pistons or wear pins to make sure that the brakes engaged.

(d) Release the brake pedals.

(e) Observe the movement of each brake’s pistons or wear pins to make sure that the brakes released.

(f) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE DISCONNECT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake disconnect

(2) An installation of the brake disconnect.

B. The brake disconnects are installed on each main landing gear brake.

TASK 32-41-11-000-801

2. Brake Disconnect Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-41-11-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-11-860-001

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-11-480-002

(3) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-11-860-002

(4) Release the parking brake.

SUBTASK 32-41-11-860-003

(5) Fully push the captain's brake pedals twelve times to remove the pressure from the brake accumulator.

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SUBTASK 32-41-11-020-001

(6) Disconnect the hose half [1] of the quick disconnect from the brake half [7] of the quick disconnect:

(a) Push the barrel [2] toward the brake and turn the barrel [2] clockwise to disengage the brake half [7] from the hose half [1].

(b) Pull up on the barrel [2] of the hose half [1] of the quick disconnect until the threads of the hose half [1] and brake half [7] are engaged.

(c) Turn the barrel [2] counterclockwise to disconnect the hose half [1] from the brake half [7] of the quick disconnect.

E. Brake Disconnect Removal (Hose Half)

SUBTASK 32-41-11-020-002

(1) Disconnect the hydraulic hose [8] from the hose half [1] of the quick disconnect.

SUBTASK 32-41-11-480-003

(2) Install a cap/plug, STD-14686, on the hydraulic hose [8].

SUBTASK 32-41-11-020-006

(3) If necessary remove the elbow fitting [9] from the hose half [1] of the quick disconnect.

F. Brake Disconnect Removal (Brake Half)

SUBTASK 32-41-11-020-007

(1) Cut and remove the lockwire or safety cable.

SUBTASK 32-41-11-020-003

(2) Remove the bolts [3] and washers [4] that hold the brake half [7] of the quick disconnect to the brake.

SUBTASK 32-41-11-020-004

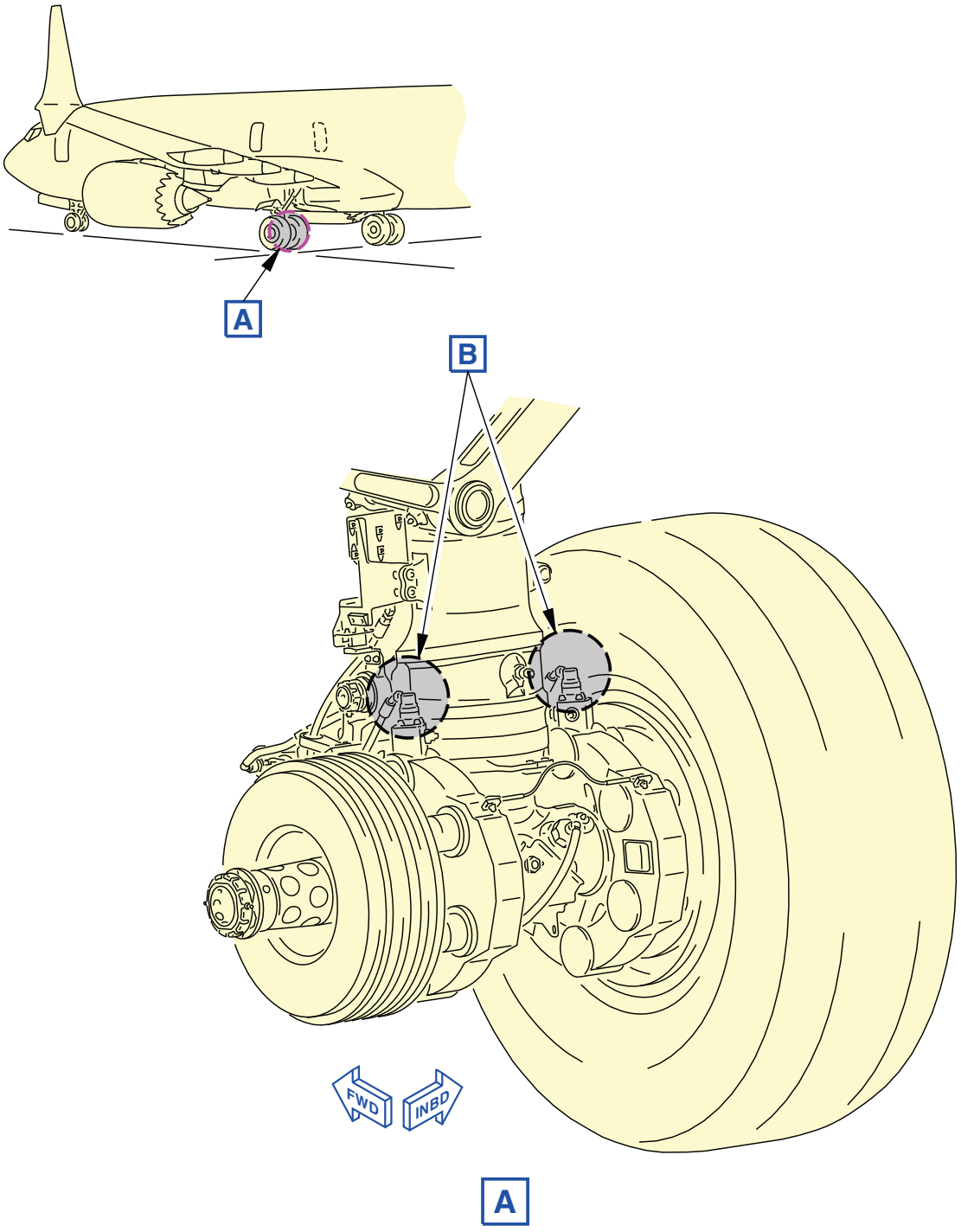
(3) Remove the brake half [7] of the quick disconnect from the brake.

SUBTASK 32-41-11-020-005

(4) Remove and discard the packing [6] and the rings [5] from the brake half [7] of the quick disconnect.

——— END OF TASK ———

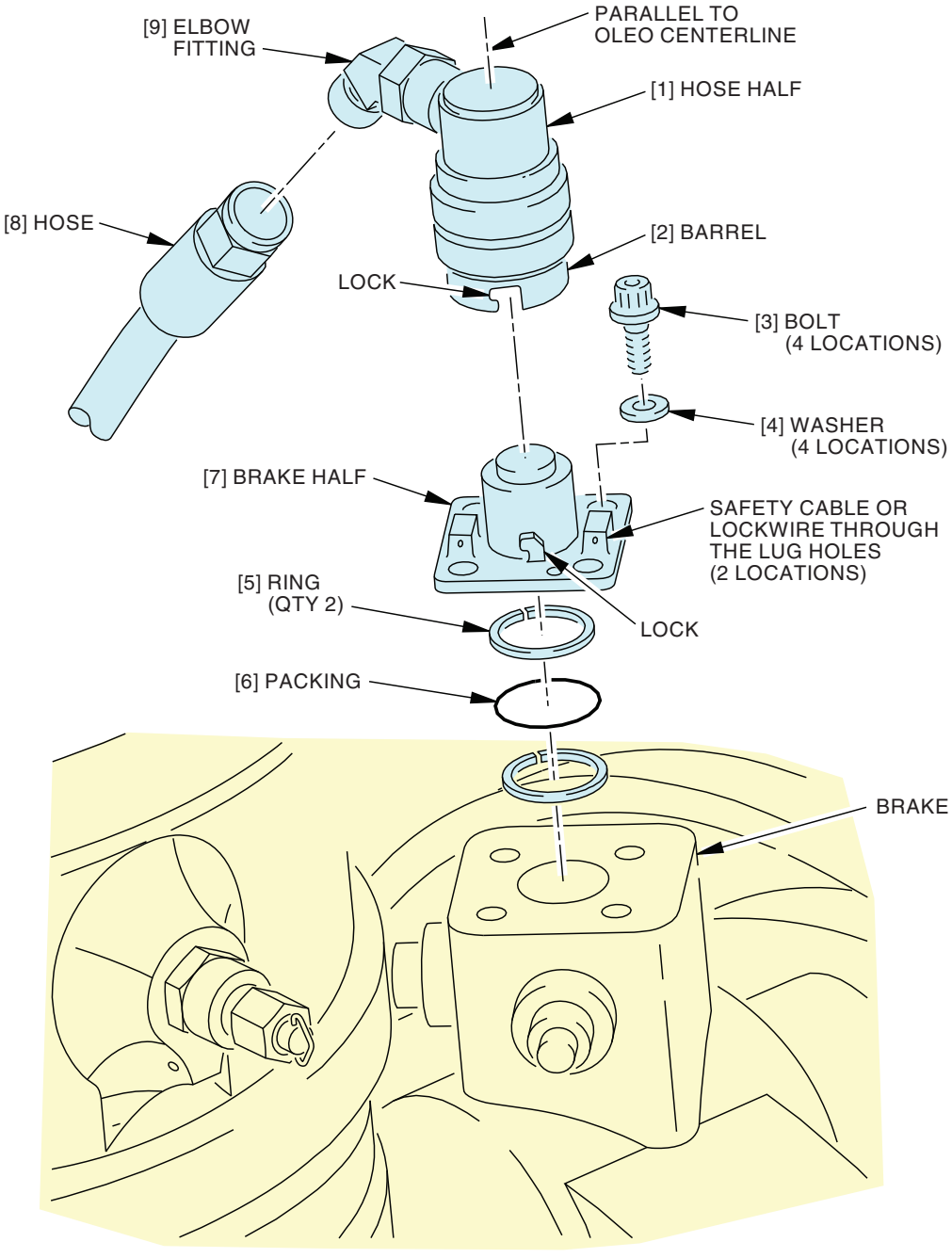
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Brake Disconnect Installation
Figure 401/32-41-11-990-801 (Sheet 1 of 2)

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(EXAMPLE)

B

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Brake Disconnect Installation
Figure 401/32-41-11-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-11-400-801

3. Brake Disconnect Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00571	Grease - BATCO X8401-2	
G01048	Lockwire - MS20995C32, Corrosion Resistant Steel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50411	Torque Stripe - EC-1252 [pink or yellow (as required per BAC5001-10)]	BMS8-45 Type I
G50466	Safety Cable (Bergen Safety Cable - 0.032 Diameter - P/N C30B18)	AMS 5689 (321 CRES)

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Ring	32-41-41-01-040	PTF ALL
6	Packing	32-41-41-01-045	PTF ALL
7	Brake half	32-41-11-01-015	PTF ALL

E. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

F. Brake Disconnect Installation

SUBTASK 32-41-11-420-004

(1) If it is necessary install the elbow fitting [9] to the hose half [1] of the quick disconnect.

(a) Apply a thin layer of hydraulic fluid, D00153, to the threads of the hose half [1] of the quick disconnect.

(b) Install the elbow fitting [9] to the hose half [1] of the quick disconnect.

(c) Do not tighten the B-nut on the hose half [1] of the quick disconnect at this time.

SUBTASK 32-41-11-420-001

(2) Brake Disconnect Installation (Hose Half):

(a) Remove the cap/plug, STD-14686, from the hydraulic hose [8].

(b) Apply a thin layer of hydraulic fluid, D00153, to the hydraulic hose [8] threads and elbow fitting [9] threads (hose half).

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(c) Connect the hydraulic hose [8] to the elbow fitting [9], do not tighten at this time.

(d) If it is necessary, loosen the B-nut of the hose half [1] of the quick disconnect.

(e) Point the elbow fitting [9] on the hose half [1] of the quick disconnect so that it is parallel to the oleo centerline, or a maximum of ten degrees in the direction of the oleo centerline (away from tire).

CAUTION

USE TWO WRENCHES TO TORQUE THE HOSE B-NUT. USE ONE TO HOLD THE ELBOW FITTING, AND THE OTHER TO TORQUE THE HOSE B-NUT. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU DO NOT TWIST OR MAKE KINKS IN THE HOSE WHEN YOU TIGHTEN THE NUT. IF YOU DO NOT OBEY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(f) Tighten the B-nut on the hydraulic hose [8]:

1) Apply initial torque to 270 ±14 in-lb (30.5 ±1.6 N·m).

2) Loosen the B-nut on the hydraulic hose [8] to decrease the torque to zero.

3) Tighten the B-nut on the hydraulic hose [8] to 400 ±20 in-lb (45.2 ±2.3 N·m).

(g) Tighten the B-nut on the elbow fitting [9] connection to hose half [1] quick disconnect.

1) Apply initial torque to 270 ±14 in-lb (30.5 ±1.6 N·m).

2) Loosen the B-nut on the hose half [1] of the quick disconnect to decrease the torque to zero.

3) Tighten the B-nut on the hose half [1] of the quick disconnect to 400 ±20 in-lb (45.2 ±2.3 N·m).

(h) Apply a torque stripe, G50411, or other special identification mark to the B-nuts.

SUBTASK 32-41-11-420-002

(3) Brake Disconnect Installation (Brake Half):

(a) Lubricate the new packing [6] and two new rings [5] with hydraulic fluid, D00153.

(b) Install the new packing [6] and two new rings [5] on the brake half [7] of the quick disconnect.

(c) Apply a thin layer of BATCO X8401-2 grease, D00571, to the brake half [7] of the disconnect.

(d) Make sure that the BATCO X8401-2 grease, D00571, is on the diametral surface above the upper retaining ring and lower surface of the mounting flange.

(e) Install the brake half [7] of the quick disconnect on the brake.

1) Make sure that the alignment pin on the brake goes into the alignment hole in the brake half [7] of the quick disconnect.

(f) Apply BATCO X8401-2 grease, D00571, to the bolts [3].

(g) Install the bolts [3] and washers [4] that hold the brake half [7] of the quick disconnect to the brake.

(h) Tighten the bolts [3] to 77 ±5 in-lb (8.7 ±0.6 N·m).

(i) Install MS20995C32 lockwire, G01048, or Safety Cable, G50466, on the bolts [3] in pairs.

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1) Make sure that MS20995C32 lockwire, G01048, or Safety Cable, G50466, passes through the lug hole to the other bolt head on each side of the quick disconnect (Figure 401).

SUBTASK 32-41-11-420-003

(4) Install the hose half [1] of the quick disconnect to the brake half [7] of the quick disconnect on the brake:

(a) Install the hose half [1] on the brake half [7]:

1) Turn the hose half [1] until the internal tangs align with the slots inside the brake half [7] and you can push the hose half [1] down.

2) Push down on the barrel [2] and turn the barrel [2] clockwise to engage the threads.

3) Turn the barrel [2] clockwise until the lock in the bottom of the barrel [2] snaps onto the external lock on the brake half [7].

4) Make sure that the elbow fitting [9] is approximately ten degrees from parallel to the oleo centerline towards the shock strut.

NOTE: This is to help prevent worn areas on the hoses.

(b) Do a visual check to make sure that the lock on hose half [1] is fully and correctly engaged with the lock on the brake half [7].

G. Brake Disconnect Installation Test

SUBTASK 32-41-11-790-001

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) For the A and the B hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-11-860-004

(2) Fully push and hold the captain's brake pedals.

NOTE: You can set the parking brake to hold the pedals.

SUBTASK 32-41-11-790-002

(3) Do a check of the hose half [1] and brake half [7] for leaks.

SUBTASK 32-41-11-790-003

(4) Release the captain's brake pedals (or parking brake).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-11-860-005

(1) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE METERING VALVE CONTROL QUADRANT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake metering valve control quadrant

(2) An installation of the brake metering valve control quadrant.

B. There are two brake metering valve control quadrants on the airplane. One brake metering control quadrant is for the captain's brake system. The other brake metering control quadrant is for the first officer's brake system. This procedure is applicable to the two brake metering control quadrants.

C. The brake metering valve control quadrant is referred to as quadrant in this procedure.

TASK 32-41-21-000-801

2. Brake Metering Valve Control Quadrant Removal

(Figure 401)

A. References

Reference	Title
25-22-00-000-801	Passenger Seat Removal (P/B 401)
25-27-15-000-801	Carpet Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205

C. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Access Panels

Number	Name/Location
112A	Forward Access Door

E. Prepare for the Removal

SUBTASK 32-41-21-480-002

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-41-21-840-001

(2) Open the access panel to get access to the forward cables [3] and turnbuckles [42].

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-21-840-002

(3) Remove the applicable passenger seats (TASK 25-22-00-000-801).

SUBTASK 32-41-21-840-003

(4) Remove the applicable carpet (TASK 25-27-15-000-801).

SUBTASK 32-41-21-840-004

(5) Remove the applicable floor panels (TASK 53-21-00-000-801).

F. Brake Metering Valve Control Quadrant Removal

SUBTASK 32-41-21-020-001

(1) Install a control cable clamp, SPL-1569, on the cables [39] that are connected to the quadrant [1] (View G, Figure 401).

NOTE: This will keep light tension on the cables. Light tension on the cables will prevent wind off on the cable drums and also make sure they do not move out from the pulley guides.

SUBTASK 32-41-21-020-002

(2) Do these steps to disconnect the cables [39] from the quadrant [1] (View B and G, Figure 401):

(a) Remove the locking clips [41] from the turnbuckles [42] on the brake forward cables [3].

(b) Loosen the turnbuckles [42] for the forward cables [3] to release the tension in the forward cables [3].

(c) Remove the pins [38], that hold the ends of the cables [39], from the quadrant [1].

(d) Pull the fittings on the end of the cables [39] from the quadrant [1].

SUBTASK 32-41-21-020-004

(3) Do these steps to disconnect the control rod [6] from the crank [5] (View D, Figure 401):

(a) Remove the nut [4], bolt [9], washer [12], retainer [10], and bushing [11].

(b) Discard the retainer [10].

SUBTASK 32-41-21-020-010

(4) Do these steps to remove the crank [5] from the shaft [45] (View E and G, Figure 401):

(a) Remove the pin [30], nut [29], washer [28], washer [27], and spacer [26] from the shaft [45].

(b) Remove the bolt [20], washer [21], and nut [23] from the crank [5].

(c) Remove the crank [5] and spacer [25] from the shaft [45].

(d) Make sure that the two bearings, long spacer, and seal stay in the support [46].

(e) Install the spacer [25], spacer [26], washer [27], washer [28], and nut [29] to the shaft [45].

NOTE: These parts are reinstalled to help keep the quadrant assembly in the support when you remove it.

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SUBTASK 32-41-21-020-005

(5) For the right quadrant [1], do these steps to remove the support [46] with quadrant [1] from the pressure deck (View E and G, Figure 401).

(a) Remove the bolts [14], washers [15], bolts [17], bolt [43], and washer [22].

(b) Remove the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], washers [34], nuts [31], nuts [33], and rig pin bracket [13].

(c) Remove the bolt [16] and washer [19].

(d) Remove the support [46] with quadrant [1] from the pressure deck.

SUBTASK 32-41-21-020-011

(6) For the left quadrant [1], do these steps to remove the support [46] with quadrant [1] from the pressure deck (View E and G, Figure 401).

(a) Remove the bolts [14], washers [15], bolts [17], bolt [50], and rig pin bracket [49].

(b) Remove the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], washers [34], nuts [31], and nuts [33].

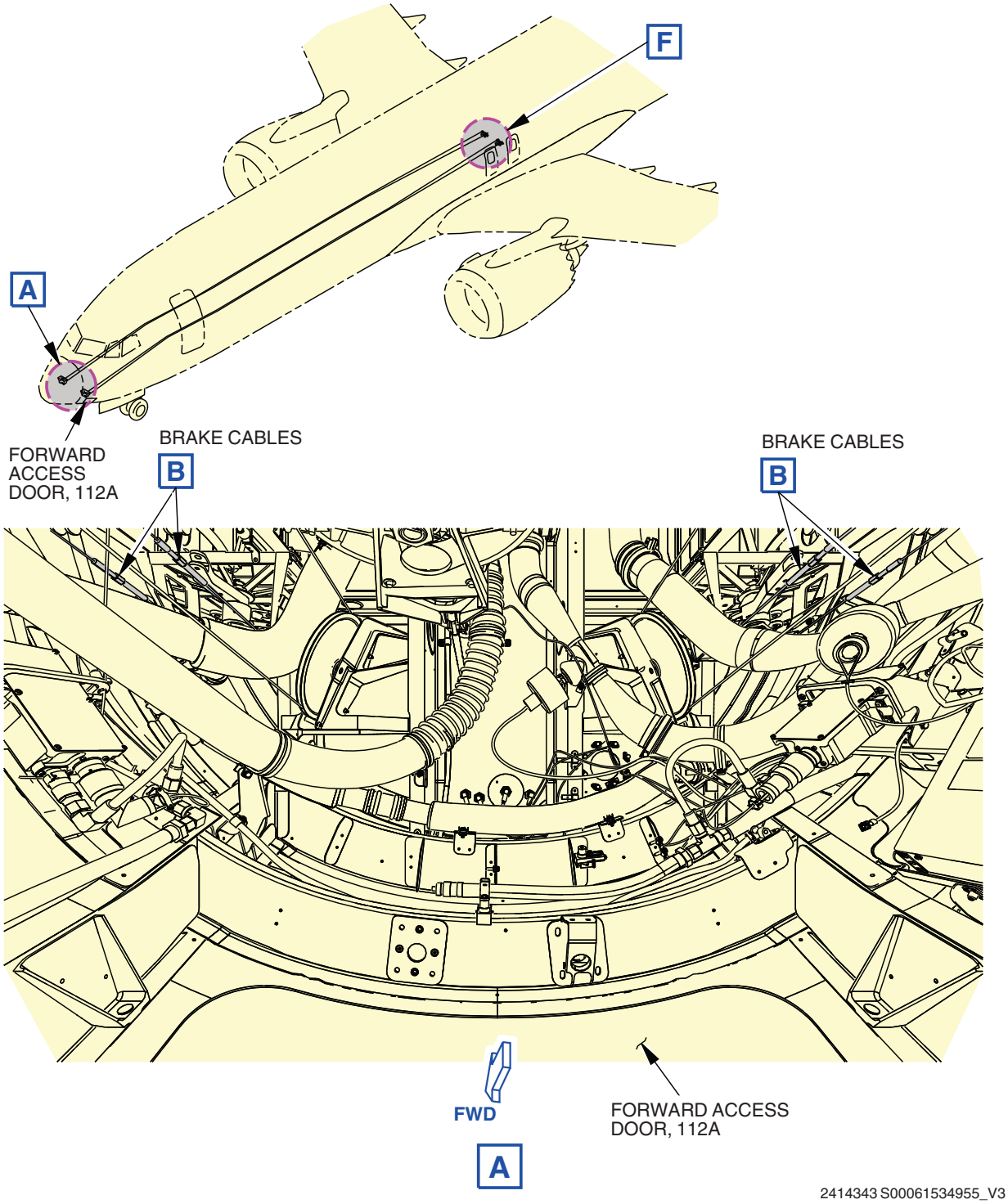
(c) Remove the support [46] with quadrant [1] from the pressure deck.

SUBTASK 32-41-21-020-012

(7) Remove the shims [35].

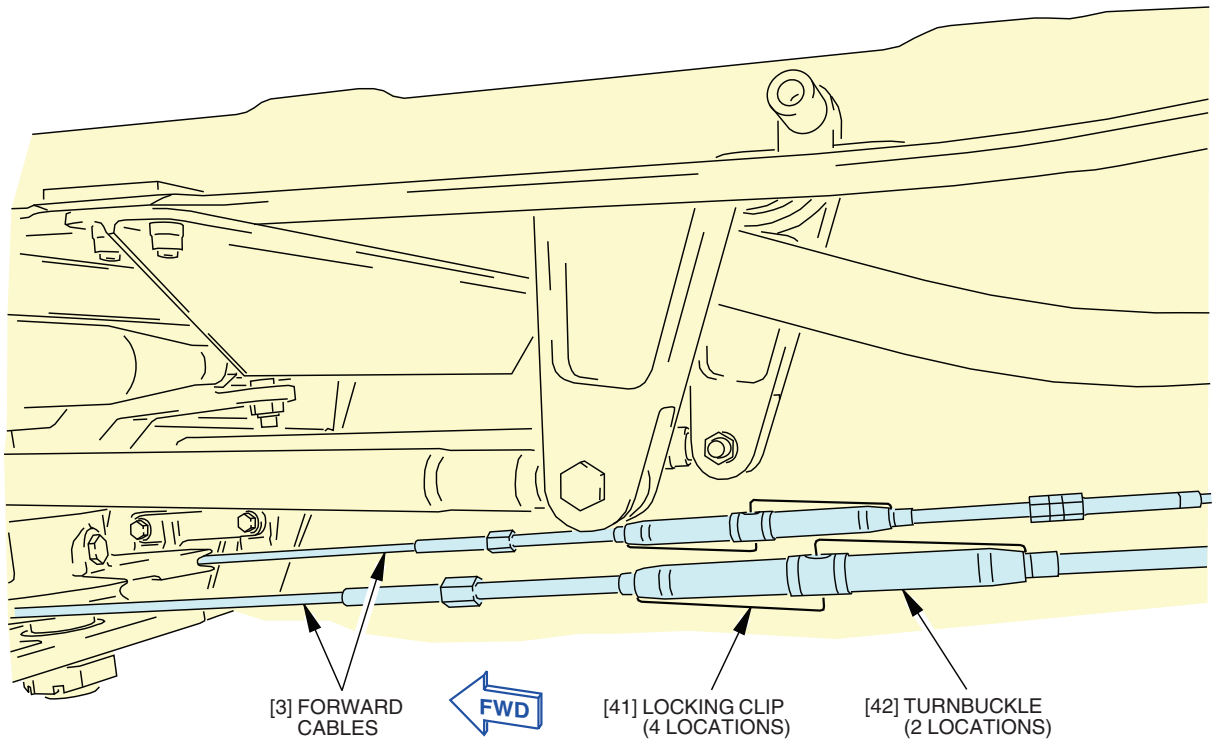
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 1 of 7)

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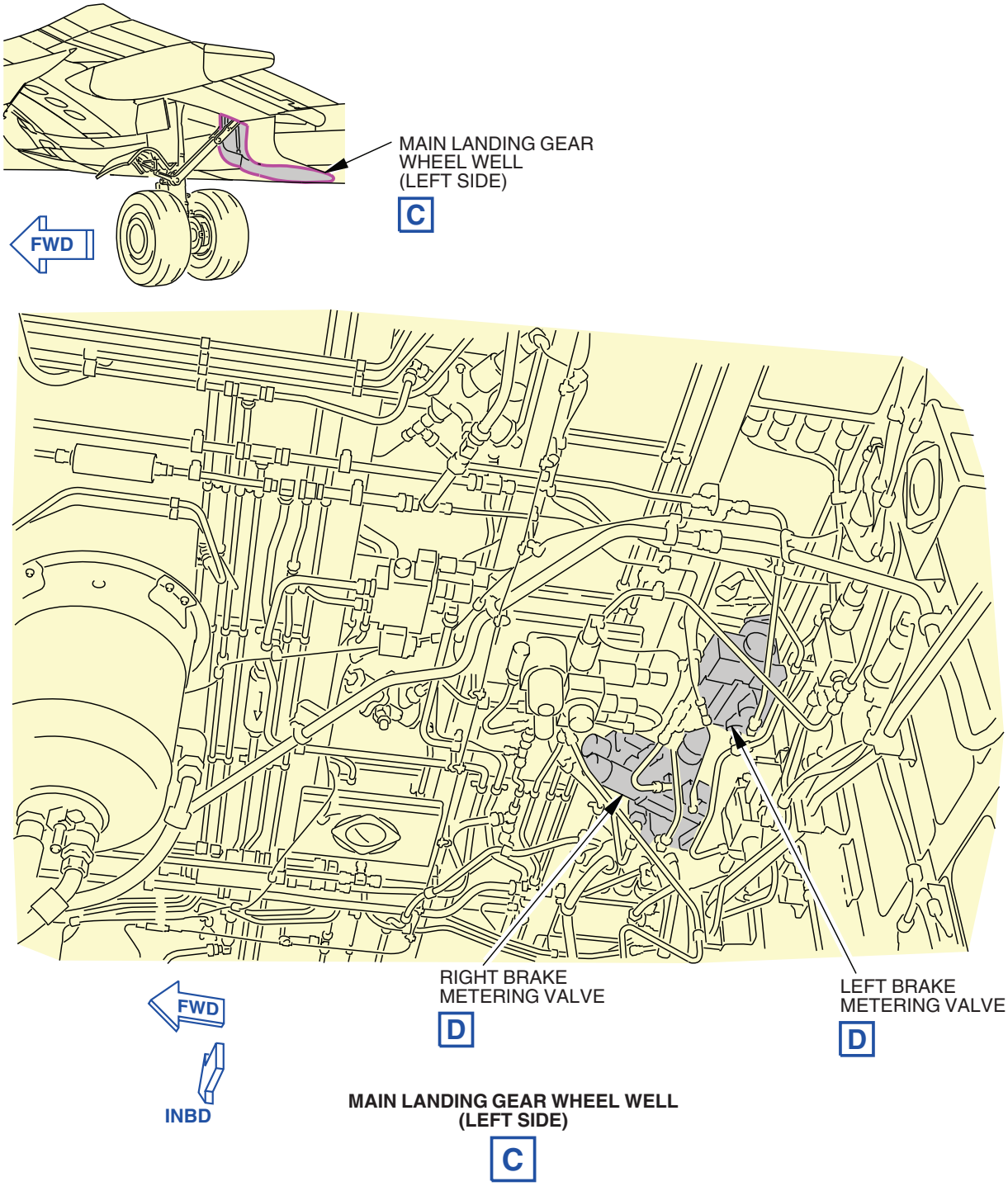
RIGHT BRAKE CABLES
(LEFT BRAKE CABLES ARE EQUIVALENT)

B

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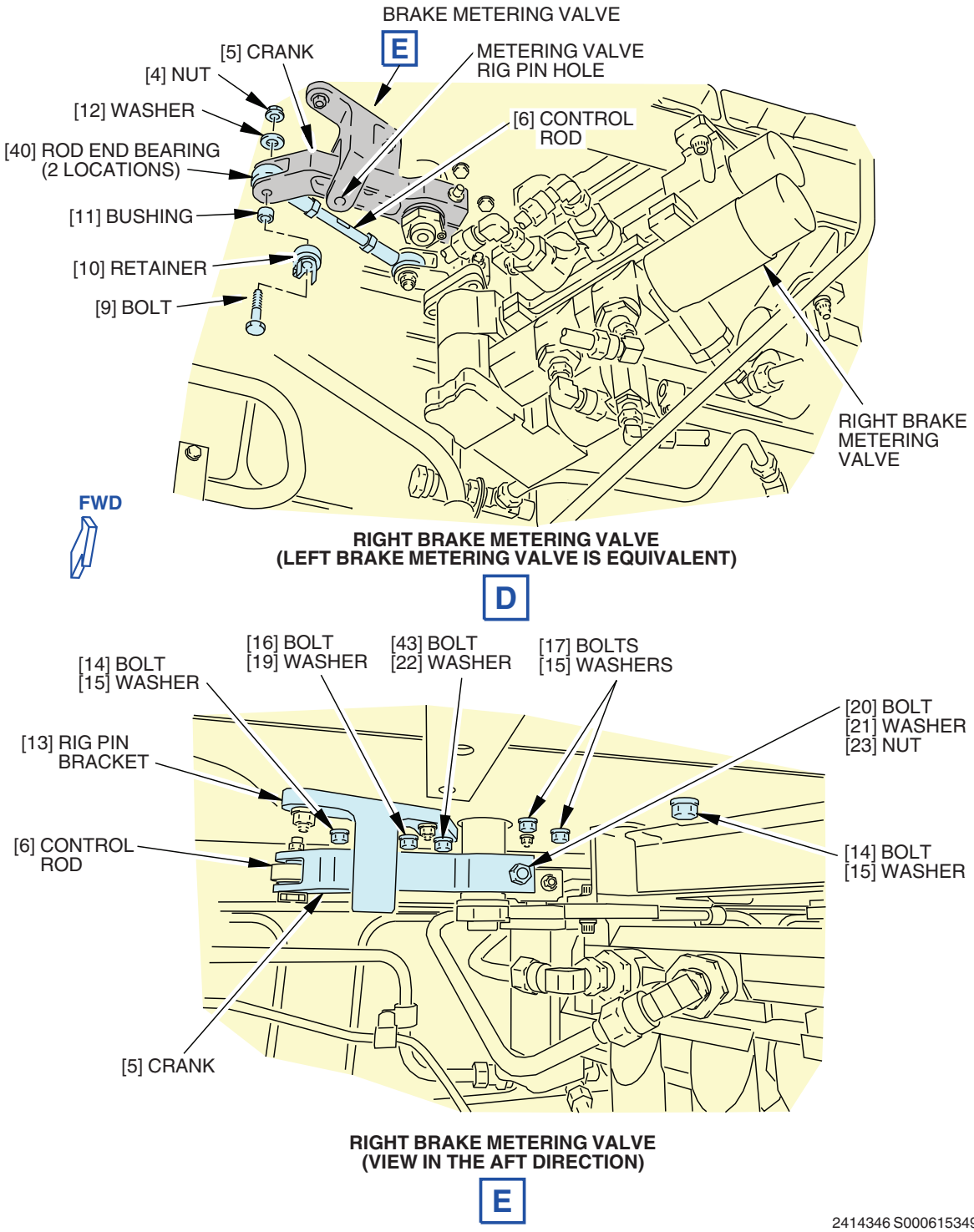
Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 2 of 7)

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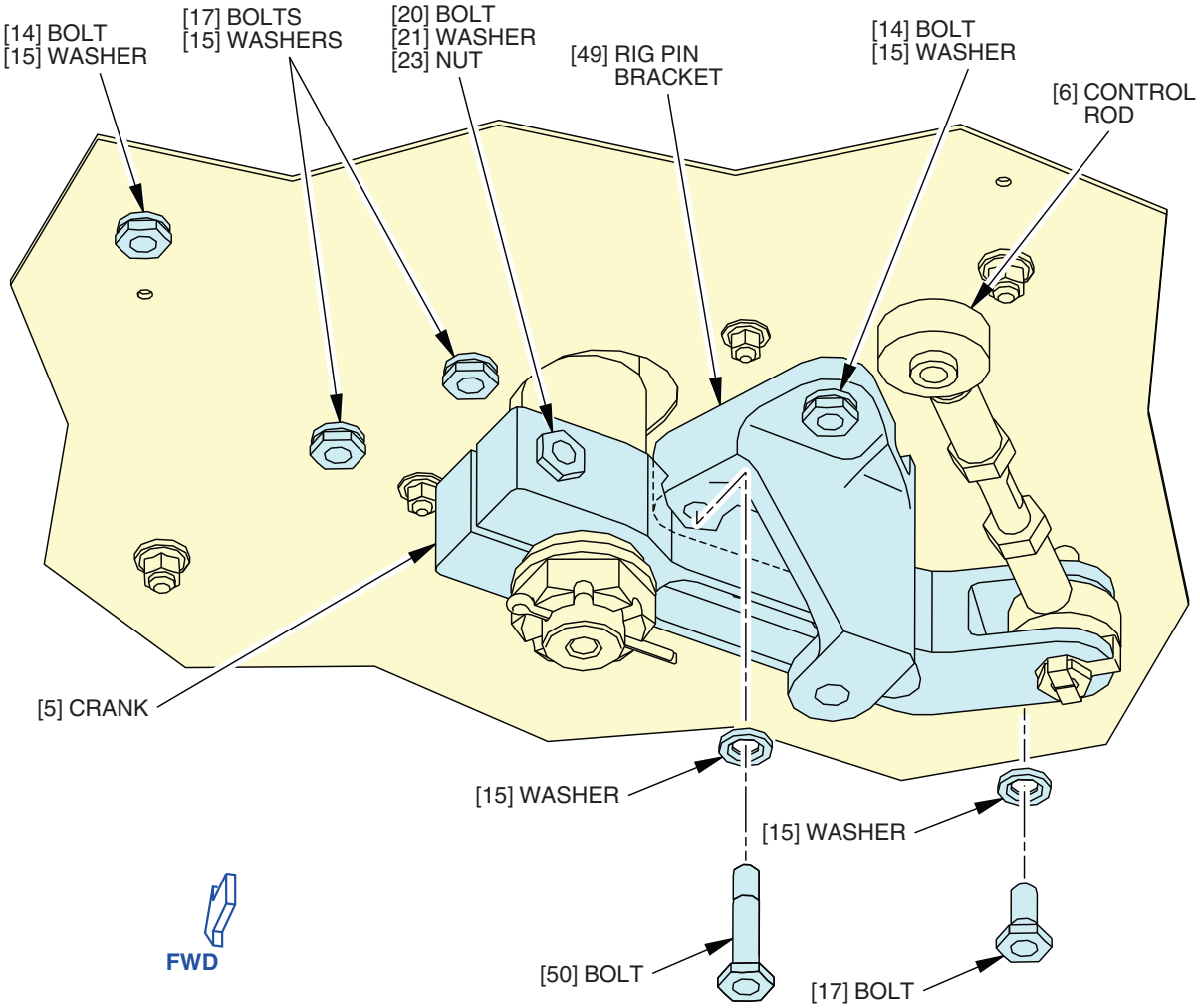
Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 3 of 7)

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Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



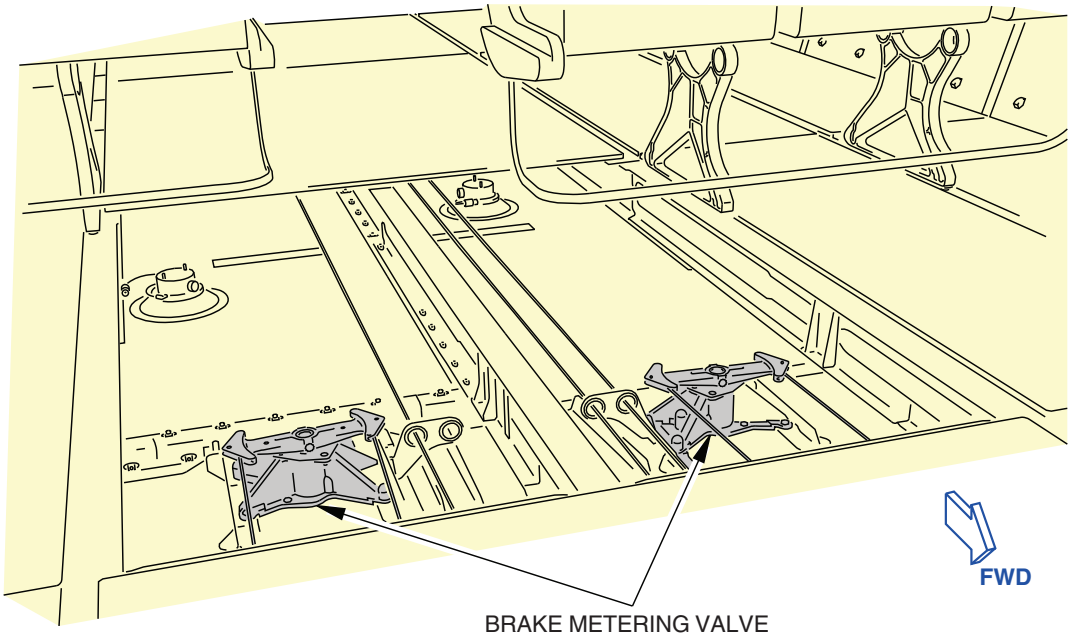
LEFT BRAKE METERING VALVE

E

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Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 5 of 7)

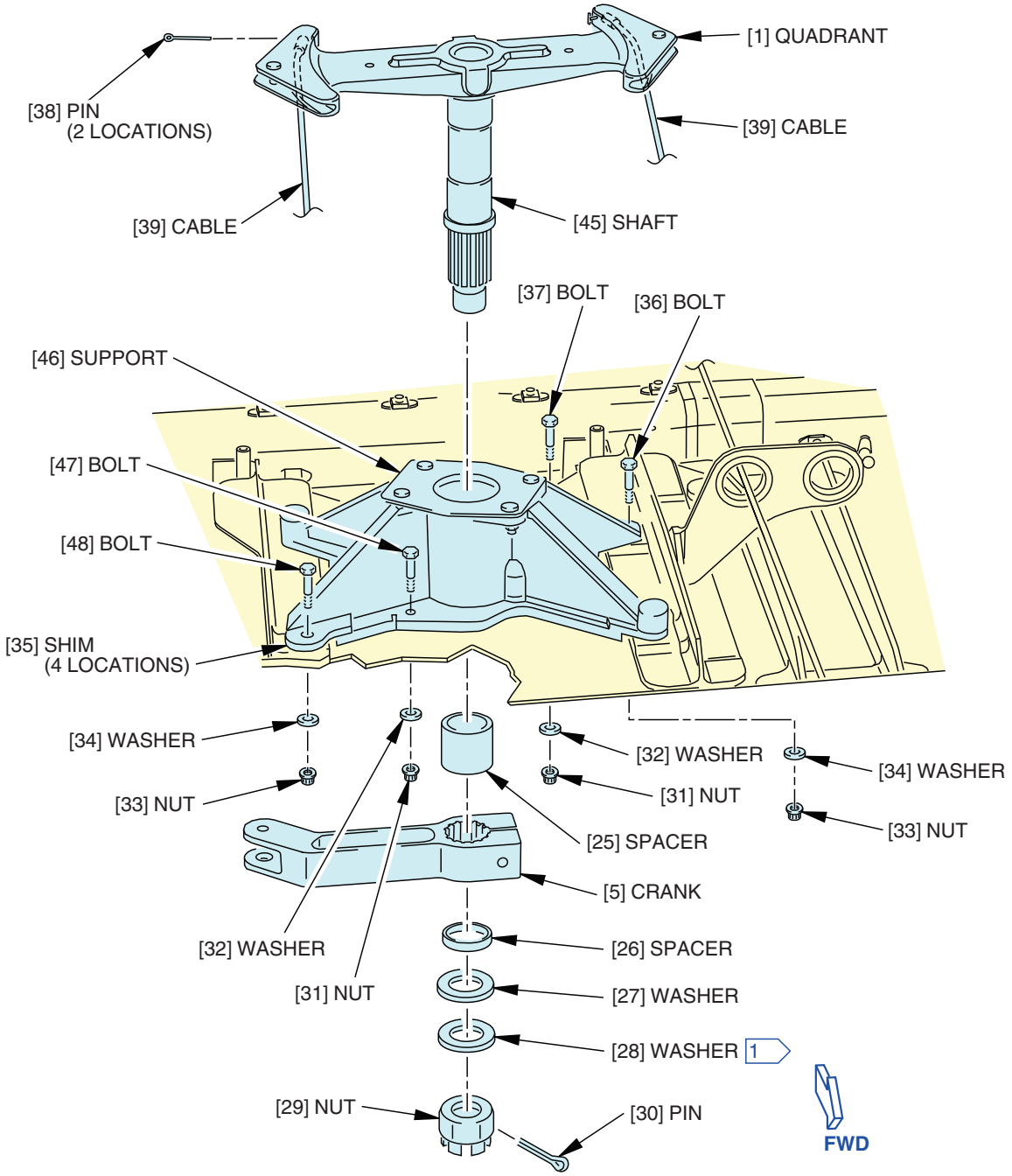
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Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 6 of 7)

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RIGHT BRAKE METERING VALVE CONTROL QUADRANT
(LEFT SIDE IS OPPOSITE)

1 AS REQUIRED

G

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Brake Metering Valve Control Quadrant Installation
Figure 401/32-41-21-990-801 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-21-400-801

3. Brake Metering Valve Control Quadrant Installation

(Figure 401)

A. References

Reference	Title
25-22-00-400-801	Passenger Seat Installation (P/B 401)
25-27-15-400-801	Carpet - Installation (P/B 401)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Quadrant	32-41-21-03-090	PTF ALL
		32-41-21-03-095	PTF ALL
10	Retainer	32-41-21-05-005	PTF ALL
30	Pin	32-41-21-03-070	PTF ALL
38	Pin	32-41-81-01-005	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

F. Access Panels

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Brake Metering Valve Control Quadrant Installation

SUBTASK 32-41-21-160-001

(1) Make sure that the pressure deck is clean in the area where the quadrant [1] is installed.

SUBTASK 32-41-21-020-006

(2) Do these steps to install the shims [35] at the attach points on the structure stiffeners (View G, Figure 401):

(a) Put the shims [35] in their positions on the pressure deck.

(b) Put the quadrant [1] in its position on the shims [35] and pressure deck.

(c) Make sure that the clearance between each shim [35] and the quadrant [1] is 0.005 in. (0.127 mm) or less.

1) If the clearance at a shim [35] is too large, remove laminations from the other shims [35] to get the tolerance at that shim [35].

(d) Remove the quadrant [1] from its position.

(e) Install the shims [35] with wet primer, C00259.

SUBTASK 32-41-21-390-001

(3) Use sealant, A00247, to apply a fay surface seal between the support [46], pressure deck, and shims [35].

(a) Make sure that is no blockage of sealant in the drain hole for the control quadrant assembly.

SUBTASK 32-41-21-420-001

(4) Put the quadrant [1] in its position on the shims [35] and pressure deck.

SUBTASK 32-41-21-390-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) For the right quadrant [1], do these steps to install the support [46] with quadrant [1] to the pressure deck (View E and G, Figure 401):

(a) Apply corrosion inhibiting material, G50136, to the bolt [16] and washer [19].

(b) Install the bolt [16] and washer [19].

(c) Use sealant, A00247, to apply a fay surface seal between the rig pin bracket [13] and pressure deck.

(d) Apply corrosion inhibiting material, G50136, to the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], and washers [34].

(e) Install the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], washers [34], nuts [31], nuts [33], and rig pin bracket [13].

(f) Apply corrosion inhibiting material, G50136, to the bolts [14], washers [15], bolts [17], bolt [43], and washer [22].

(g) Install the bolts [14], washers [15], bolts [17], bolt [50], and washer [22].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-21-420-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(6) For the left quadrant [1], do these steps to install the support [46] with quadrant [1] to the pressure deck (View E and G, Figure 401):

(a) Apply corrosion inhibiting material, G50136, to the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], and washers [34].

(b) Install the bolt [36], bolt [37], bolt [47], bolt [48], washers [32], washers [34], nuts [31], and nuts [33].

(c) Use sealant, A00247, to apply a fay surface seal between the rig pin bracket [49] and pressure deck.

(d) Apply corrosion inhibiting material, G50136, to the bolts [14], washers [15], bolts [17], and bolt [50].

(e) Install the bolts [14], washers [15], bolts [17], bolt [50], and rig pin bracket [49].

SUBTASK 32-41-21-420-003

(7) Do these steps to install the crank [5] (View E and G, Figure 401):

(a) Remove the nut [29], washer [28], washer [27], spacer [25], and spacer [26] from the shaft [45].

1) Make sure that the two bearings, long spacer, and seal stay in the support [46].

(b) Apply Cor-Ban 27L Compound, G50237, to the mating surfaces of the crank [5] and shaft [45].

(c) Install the crank [5], spacer [26], washer [27], washer [28], and nut [29] on the shaft [45].

1) Make sure that the chamfer side of the crank [5] is against the spacer [25].

(d) Tighten the nut [29] to 22.5 ±12.5 in-lb (2.5 ±1.5 N·m).

(e) Install the new pin [30] in the nut [29] and shaft [45].

(f) Apply corrosion inhibiting material, G50136, to the bolt [20] and washer [21].

(g) Install the bolt [20], washer [21], and nut [23].

SUBTASK 32-41-21-420-004

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(8) Do these steps to connect the control rod [6] to the crank [5] (View D, Figure 401).

(a) Put the control rod [6] in its position in the crank [5].

(b) Install the bushing [11] and new retainer [10] on the crank [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Apply corrosion inhibiting material, G50136, to the bolt [9] and washer [12].

(d) Install the bolt [9], washer [12], and nut [4] to connect the control rod [6] to the crank [5].

1) Bend the tabs of the retainer [10].

(e) Lubricate the rod end bearings [40] with grease, D00633.

SUBTASK 32-41-21-420-005

(9) Do these steps to connect the cables [39] to the quadrant [1] (View B and G, Figure 401):

(a) Put the fittings on the end of the cables [39] in their position in the quadrant [1].

(b) Install the new pins [38] that hold the ends of the cables [39] in the quadrant [1].

(c) Tighten the turnbuckles [42] for the forward cables [3] to put the cables in light tension.

(d) Remove the control cable clamp, SPL-1569, from the cables [39].

SUBTASK 32-41-21-020-009

(10) Do this task: Hydraulic Brake System Adjustment, TASK 32-41-00-820-801.

SUBTASK 32-41-21-400-001

(11) Make sure that the locking clips [41] are installed on the turnbuckles [42] (View B, Figure 401).

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-21-840-005

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-21-840-006

(2) Install the floor panels (TASK 53-21-00-400-801).

SUBTASK 32-41-21-840-007

(3) Install the carpet (TASK 25-27-15-400-801).

SUBTASK 32-41-21-840-008

(4) Install the passenger seats (TASK 25-22-00-400-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE PEDAL BUS CRANK ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the brake pedal bus crank assembly

(2) Installation of the brake pedal bus crank assembly

B. There are two brake pedal bus crank assemblies on the airplane. One brake pedal bus crank assembly is for the captain's brake system. The other brake pedal bus crank assembly is for the first officer's brake system. This procedure is applicable to the two brake pedal bus crank assemblies.

TASK 32-41-22-000-801

2. Brake Pedal Bus Crank Assembly Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
27-21-31-000-801	Rudder Pedals Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-81-000-801	Hydraulic Brake Control Cable Removal (P/B 401)

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Removal

SUBTASK 32-41-22-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed in the nose and main landing gear (Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801).

SUBTASK 32-41-22-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-22-860-001

(3) Release the parking brake.

SUBTASK 32-41-22-010-001

(4) Open this access panel:

Number	Name/Location
112A	Forward Access Door

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E. Brake Pedal Bus Crank Assembly Removal

SUBTASK 32-41-22-020-001

(1) Remove the brake control cables [31] from the crank [24] (TASK 32-41-81-000-801).

SUBTASK 32-41-22-020-002

(2) Disconnect the aft ends of the rod assemblies [5] from the bellcrank assemblies [6] (TASK 27-21-31-000-801).

SUBTASK 32-41-22-020-003

(3) Remove the bolts [1], bushings [2], washers [3], and nuts [4] to disconnect the rod assemblies [7] from the crank assembly [21] and crank assembly [23].

SUBTASK 32-41-22-020-004

(4) Remove the bolts [8], washers [9], washers [10], and nuts [11] to remove the brake pedal bus crank assembly [30] from the structure.

SUBTASK 32-41-22-020-006

(5) Remove the bolts [1], bushings [2], washers [3], and nuts [4] to disconnect the forward ends of the rod assemblies [5] from the crank assembly [23] and crank [24].

SUBTASK 32-41-22-020-005

(6) Disassemble the brake pedal bus crank assembly [30].:

(a) Remove the bolt [12], washer [13], and nut [14] from the crank [24].

(b) Remove the pin [15], nut [16], if installed washer [17], and spacer [18] from the shaft [26].

1) Discard the pin [15].

(c) Remove the pin [15], nut [29], if installed washer [28], and washer [27] from the shaft [26].

1) Discard the pin [15].

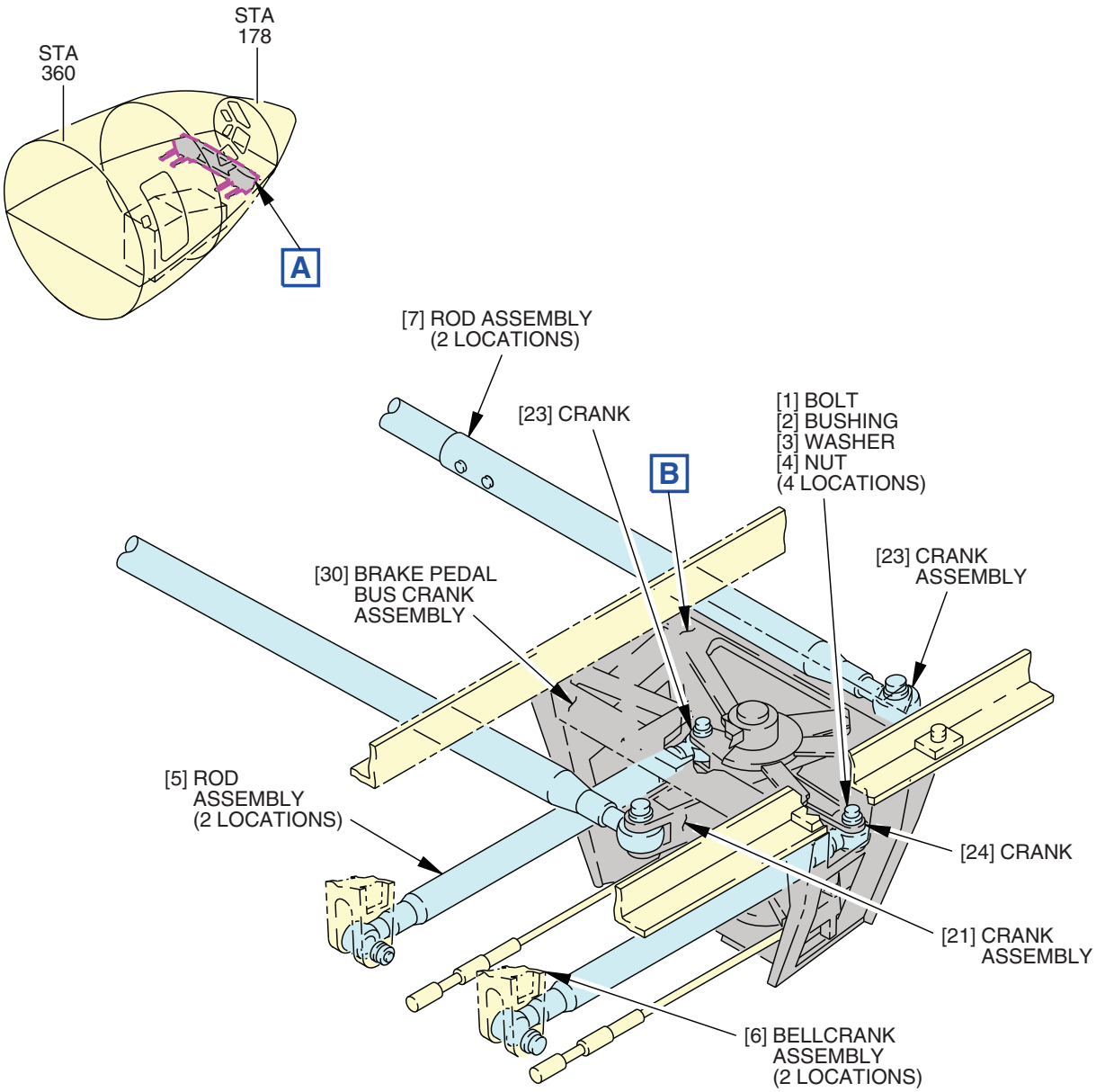
(d) Remove the shaft [26], bearing [19], crank assembly [21], bearing [20], spacer [22], crank assembly [23], crank [24], and spacer [25] from the bus support assembly [33].

1) If necessary for the replacement, remove the bearing [32] from the bus support assembly [33].

2) If necessary for the replacement, remove the bearing [20] from the crank assembly [23].

——— END OF TASK ———

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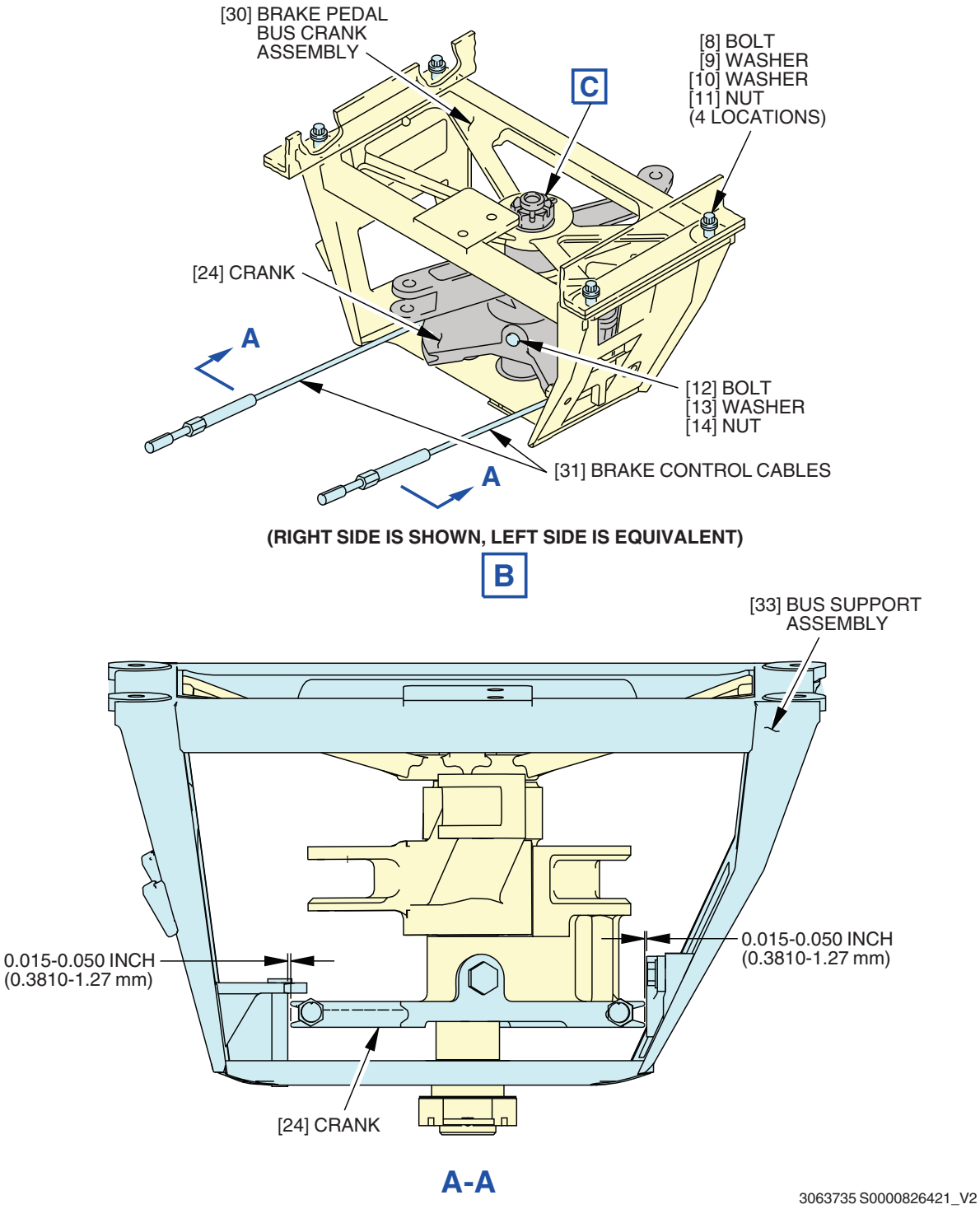
(RIGHT SIDE IS SHOWN, LEFT SIDE IS EQUIVALENT)

A

3063728 S0000826419_V2

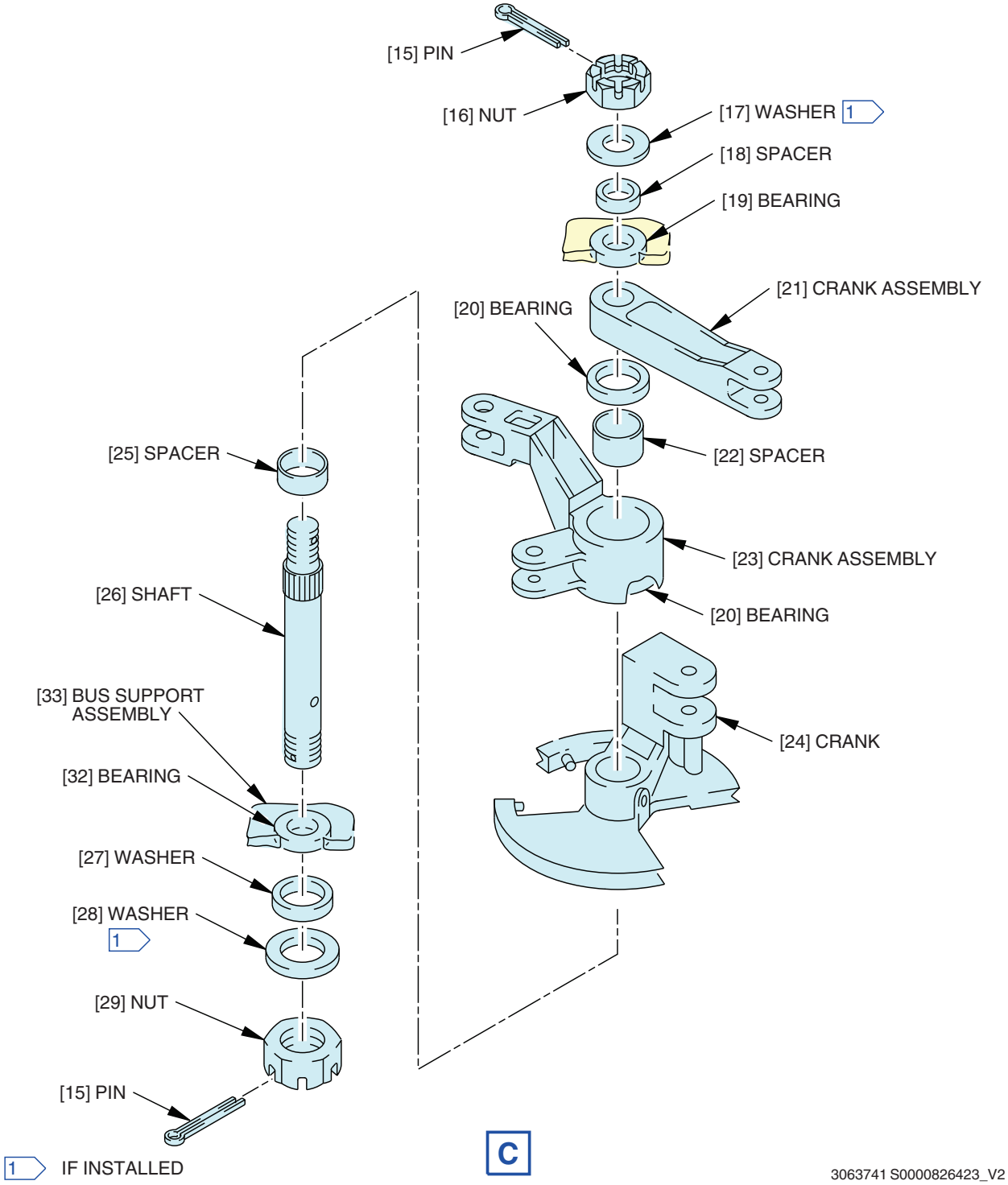
Brake Pedal Bus Crank Assembly Installation
Figure 401/32-41-22-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Pedal Bus Crank Assembly Installation
Figure 401/32-41-22-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Pedal Bus Crank Assembly Installation
Figure 401/32-41-22-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-22-400-801

3. Brake Pedal Bus Crank Assembly Installation

(Figure 401)

A. References

Reference	Title
06-41-00-800-801	Finding an Access Door or Panel on the Lower Half of the Fuselage (P/B 201)
20-10-02-900-801	Bearings and Bushings - Removal/Installation (P/B 401)
27-21-31-400-801	Rudder Pedals Installation (P/B 401)
32-00-01-080-801	Landing Gear Downlock Pins Removal (P/B 201)
32-41-00-710-801	Normal and Alternate Brake Application Functional Test (P/B 501)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)
32-41-81-400-801	Hydraulic Brake Control Cable Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D50180	Grease - Aircraft General Purpose (AeroShell Grease 33)	BMS3-33, MIL-PRF-23827
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G50961	Coating - Corrosion Inhibiting Epoxy Primer - (BAC 377 Yellow Color)	BMS10-11 Type I

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
15	Pin	32-41-21-01-055	PTF ALL
		32-41-21-02-055	PTF ALL
20	Bearing	32-41-21-01-122	PTF ALL
32	Bearing	32-41-21-01-135	PTF ALL

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
113	Area Above and Outboard of Nose Landing Gear Wheel Well - Left
114	Area Above and Outboard of Nose Landing Gear Wheel Well - Right

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Brake Pedal Bus Crank Assembly Installation

SUBTASK 32-41-22-420-001

(1) If removed for replacement, install new bearing [20] and new bearing [32] as follows:

(a) Apply coating, G50961 to the bearing [20]. Roller swage to the crank assembly [21] (TASK 20-10-02-900-801).

(b) Apply coating, G50961 to the bearing [32]. Roller swage to the bus support assembly [33] (TASK 20-10-02-900-801).

(2) Install the shaft [26] to the bus support assembly [33] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Install the spacer [25] onto the shaft [26]. Install shaft [26] into the crank [24] and the bus support assembly [33] with the washer [27] and, if removed, washer [28].

(b) Apply Cor-Ban 27L Compound, G50237 to the nut [29]. Install the nut [29] to the shaft [26].

1) Tighten the nut [29] 50 in-lb (5.6 N·m) - 100 in-lb (11.3 N·m).

(c) Install new pin [15] to nut [29].

(d) Install the crank assembly [23] and spacer [22] to the shaft [26].

(e) Apply AeroShell Grease 33, D50180 to bearing [19] and bearing [20].

(f) Install bearing [20], crank assembly [21], bearing [19], spacer [18], and, if removed, washer [17] to shaft [26].

(g) Apply Cor-Ban 27L Compound, G50237 to nut [16] and install to shaft [26].

1) Tighten the nut [16] 50 in-lb (5.6 N·m) - 100 in-lb (11.3 N·m).

(h) Install new pin [15] to the nut [16].

(i) Apply Cor-Ban 27L Compound, G50237 to the bolt [12]. Install the bolt [12], washer [13], and nut [14] to the crank [24].

1) Tighten the nut [14] 20 in-lb (2.3 N·m) above running torque.

SUBTASK 32-41-22-820-001

(3) Make sure that the clearance between the crank [24] and bus support assembly [33] is 0.015 in. (0.3810 mm) - 0.050 in. (1.27 mm) (View A-A, Figure 401).

SUBTASK 32-41-22-420-003

(4) Attach the forward end of one rod assembly [5] to the short arm of the crank assembly [23]. Attach the other rod assembly [5] to the crank [24]. Install with the bolts [1], bushings [2], washers [3], and nuts [4] to attach the rod assemblies [5].

SUBTASK 32-41-22-420-004

(5) Put the brake pedal bus crank assembly [30] into the structure. Install with the bolts [8], washers [9], washers [10], and nuts [11].

NOTE: Install the washers [9] with the countersunk adjacent to the heads of the bolts [8].

SUBTASK 32-41-22-420-006

(6) Install the rod assemblies [7] to the crank assembly [23] and crank assembly [21] using the bolts [1], bushings [2], washers [3], and nuts [4].

SUBTASK 32-41-22-420-007

(7) Connect the aft ends of the rod assemblies [5] to the bellcrank assemblies [6] (Rudder Pedals Installation, TASK 27-21-31-400-801).

SUBTASK 32-41-22-420-008

(8) Install the brake control cables [31] to the brake pedal bus crank assembly [30] (Hydraulic Brake Control Cable Installation, TASK 32-41-81-400-801).

SUBTASK 32-41-22-820-002

(9) Do this task: Hydraulic Brake System Adjustment, TASK 32-41-00-820-801.

SUBTASK 32-41-22-710-001

(10) Do this task: Normal and Alternate Brake Application Functional Test, TASK 32-41-00-710-801.

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G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-22-410-001

(1) Close this access panel:

(TASK 06-41-00-800-801)

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-22-860-002

(2) If it is necessary, set the parking brake.

SUBTASK 32-41-22-860-003

(3) If necessary, remove the chocks installed around the tires.

SUBTASK 32-41-22-860-004

(4) If necessary, remove the downlock pins installed in the nose and main landing gear (ref Landing Gear Downlock Pins Removal, TASK 32-00-01-080-801).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE METERING VALVE ASSEMBLY - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake metering valve

(2) An installation of the brake metering valve.

B. There are two brake metering valves. There is one for each main landing gear. Each brake metering valve supplies normal system braking and alternate system braking.

C. This procedure gives instructions to remove one brake metering valve, but it is applicable for the two brake metering valves.

D. The brake metering valves are in the aft section of the main landing gear wheel well on the ceiling.

TASK 32-41-31-000-801

2. Brake Metering Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 32-41-31-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-31-480-002

(2) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-31-860-001

(3) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-31-860-002

(4) To release the pressure from the normal brake (System B) and the alternate brake (System A) hydraulic system, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-31-840-001

(5) Fully push the brake pedals 12 times to remove hydraulic pressure from the brake accumulator.

E. Brake Metering Valve Removal

SUBTASK 32-41-31-020-001

(1) Install labels on the hydraulic lines to make the installation easier.

SUBTASK 32-41-31-020-002

(2) Disconnect the hydraulic lines from the brake metering valve [1]:

SUBTASK 32-41-31-480-003

(3) Install caps/plugs, STD-14686, in the hydraulic lines and hydraulic fittings on the brake metering valve [1].

SUBTASK 32-41-31-020-003

(4) Remove the nut [10], washer [9] and bolt [8] that hold the crank to the shaft [11] on the brake metering valve [1].

SUBTASK 32-41-31-020-004

(5) Remove the nuts [12], washers [13], washers [5], and bolts [3] that hold the brake metering valve [1] to the wheel well ceiling.

SUBTASK 32-41-31-020-005

(6) If the replacement brake metering valve [1] does not have hydraulic fittings installed, do these steps:

(a) Remove the adapters [20], adapter [24], adapters [29], packings [14], packings [16], packings [17], packings [19], packing [21], packing [23], and the rings [15], rings [18], ring [22] from the removed brake metering valve [1].

(b) Remove the filters [25], packings [26], packings [28], and rings [27] from the removed brake metering valve [1].

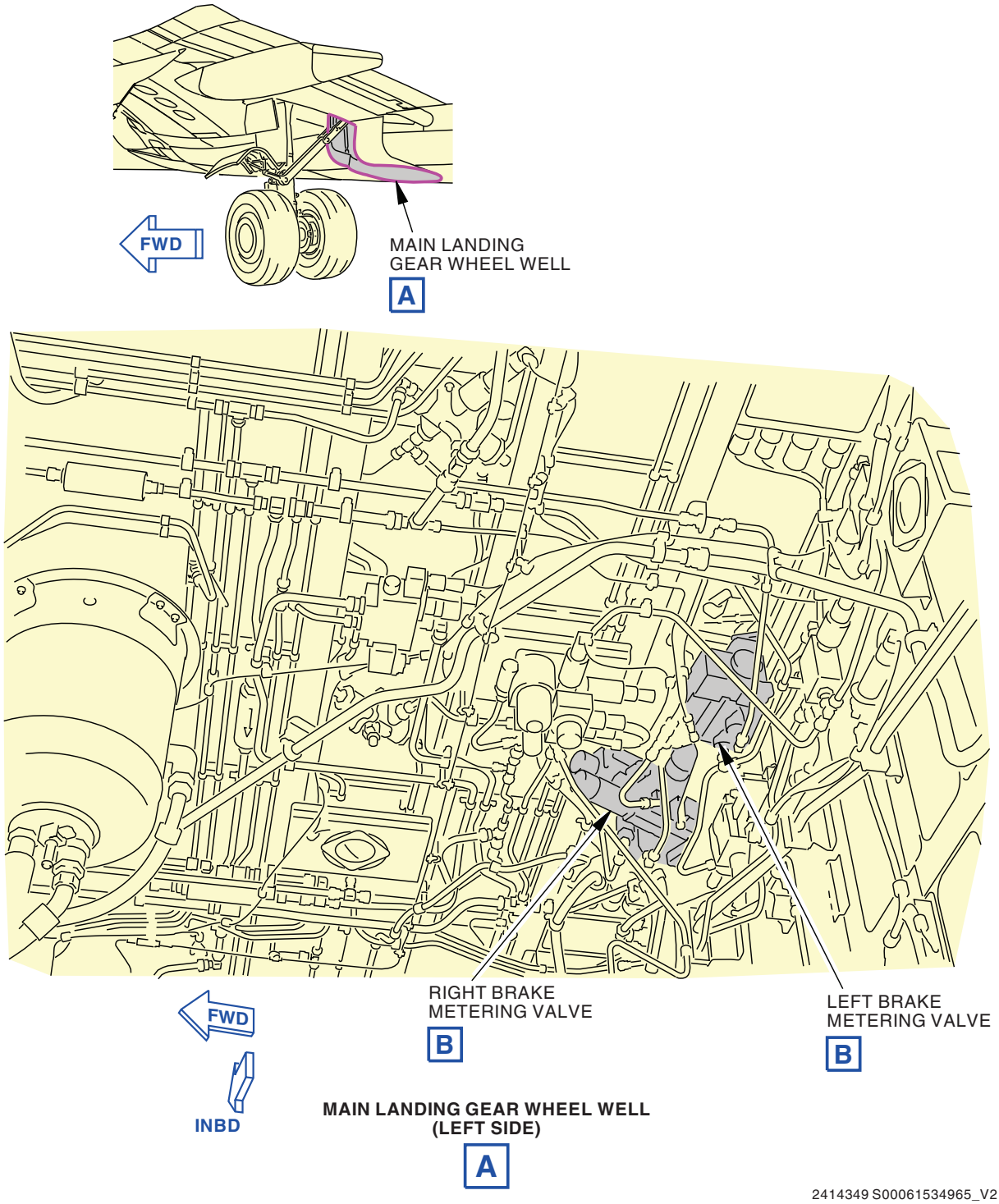
(c) Discard the packings [14], packings [16], packings [17], packings [19], packing [21], packing [23], packings [26], packings [28] and the rings [15], rings [18], ring [22], rings [27].

SUBTASK 32-41-31-480-004

(7) Install caps/plugs, STD-14686, in the ports of the brake metering valve [1].

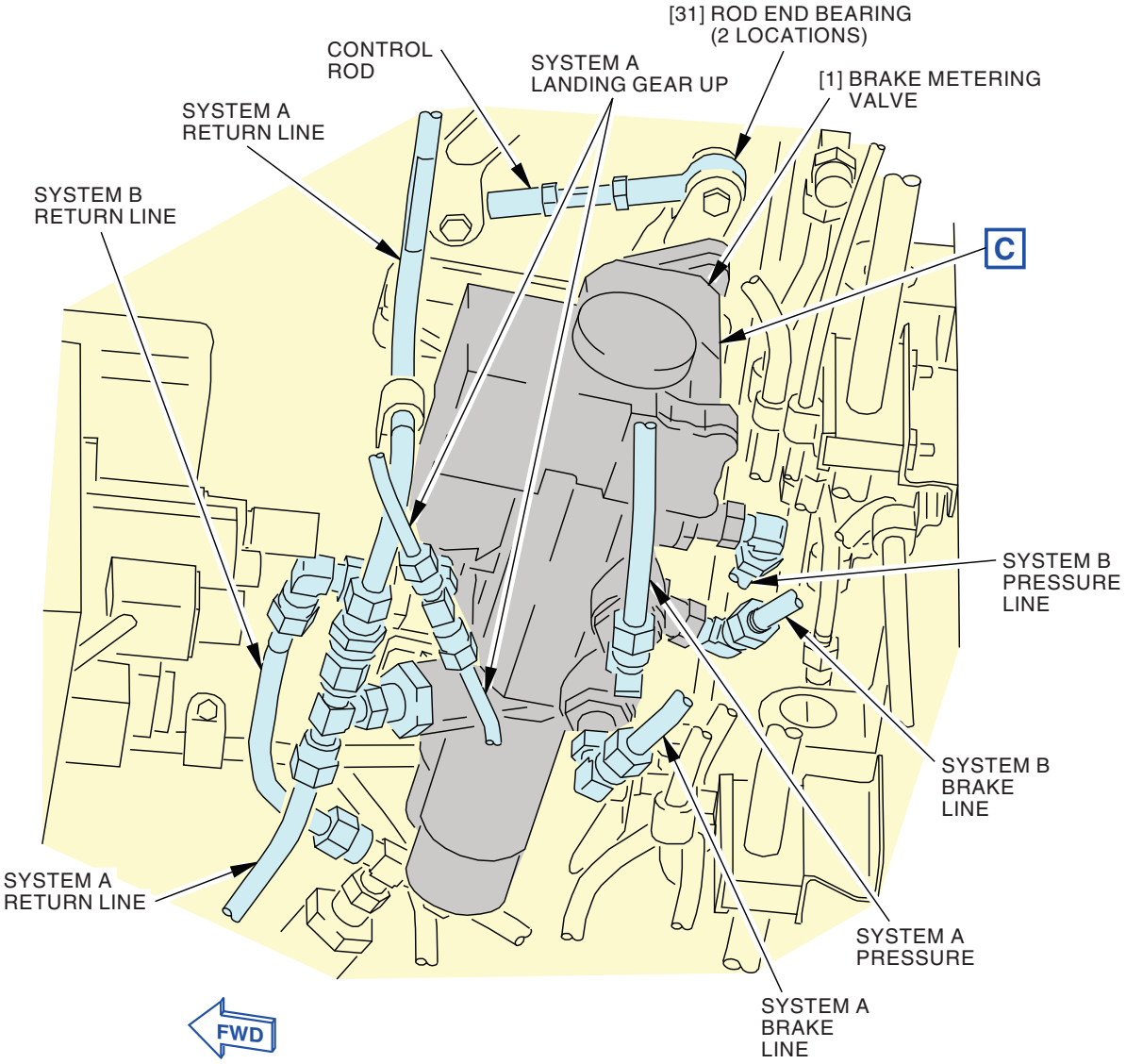
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Metering Valve Installation
Figure 401/32-41-31-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



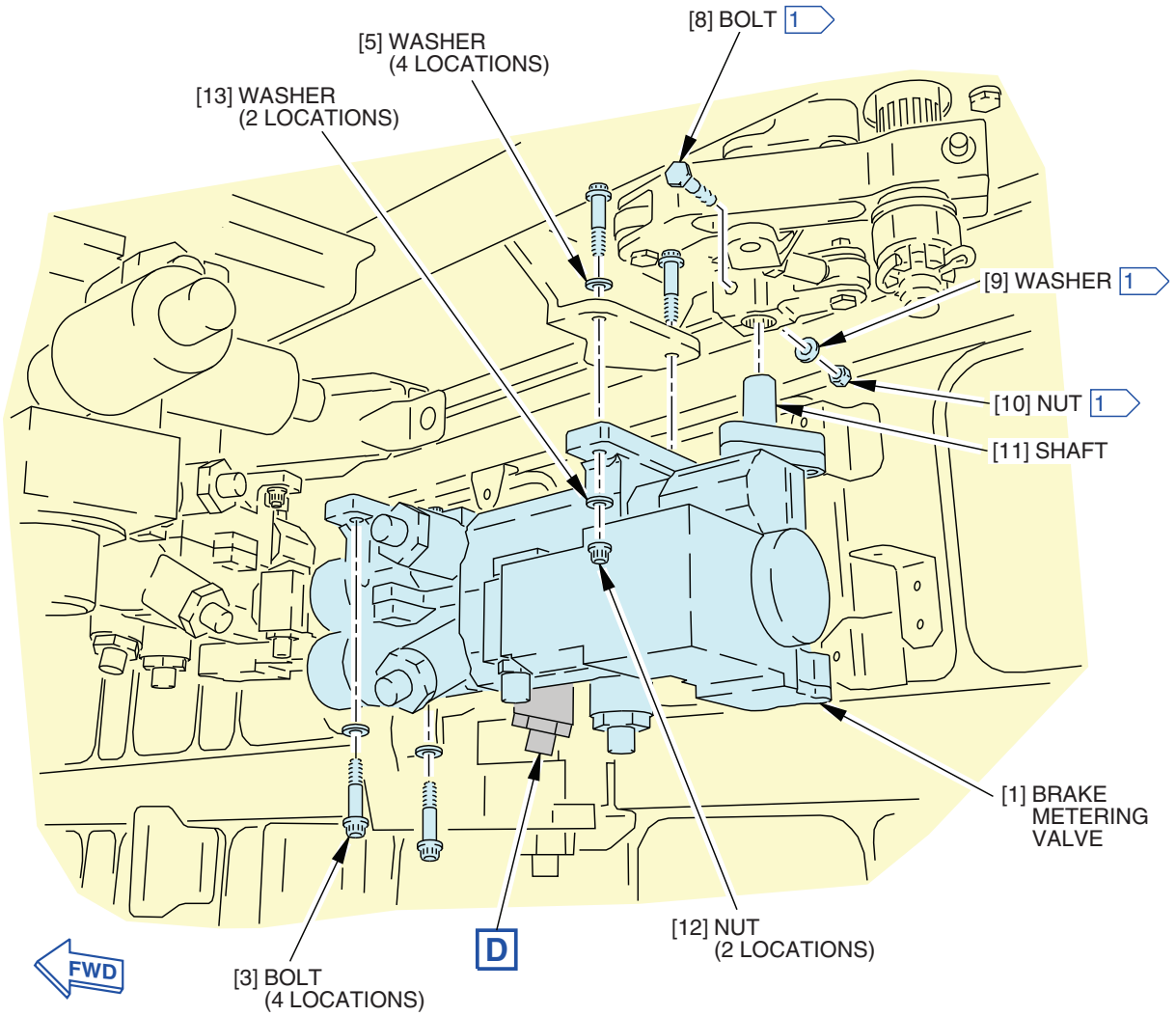
LEFT BRAKE METERING VALVE
(RIGHT BRAKE METERING VALVE IS EQUIVALENT)

B

2414350 S00061534966_V3

Brake Metering Valve Installation
Figure 401/32-41-31-990-801 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

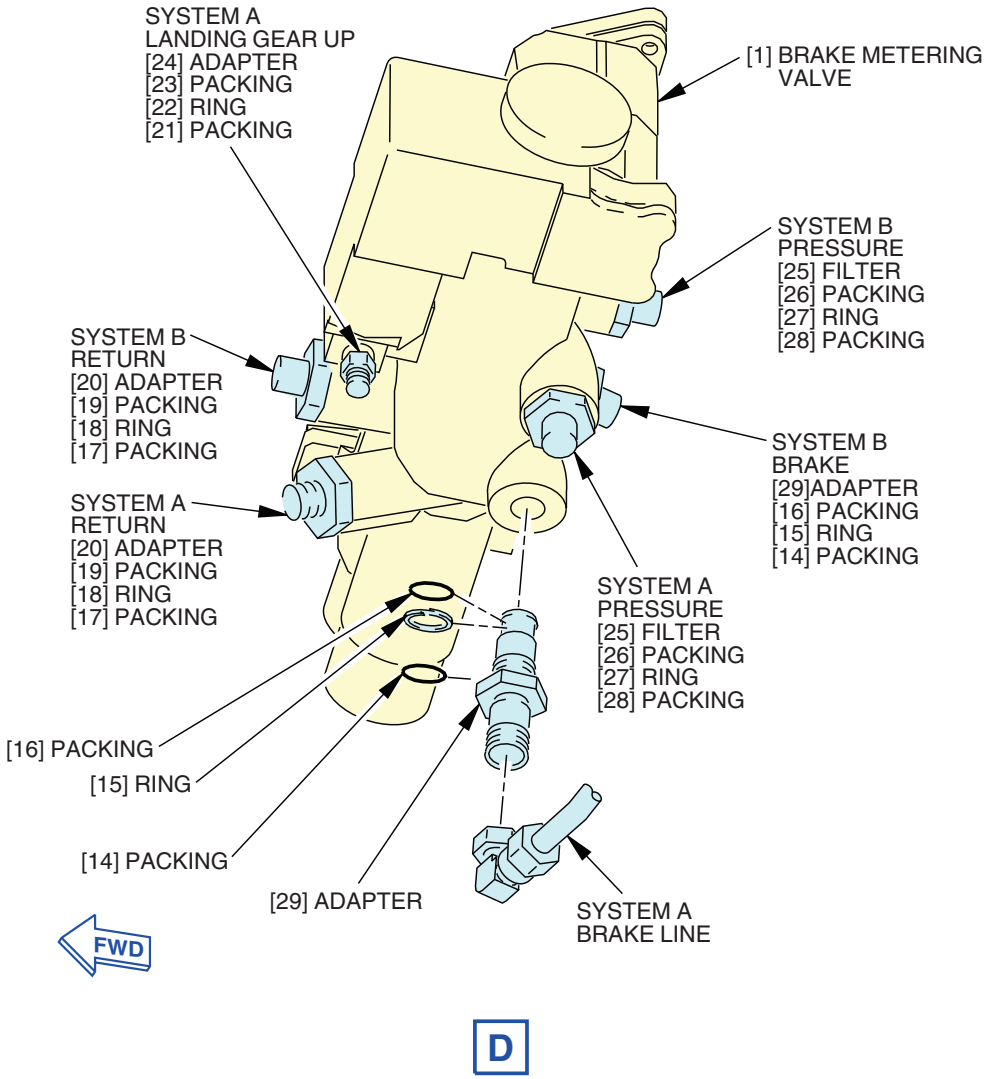


1 LEFT SIDE BOLT HEAD FACES FORWARD AS SHOWN, RIGHT SIDE BOLT HEAD FACES AFT

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Brake Metering Valve Installation
Figure 401/32-41-31-990-801 (Sheet 3 of 4)

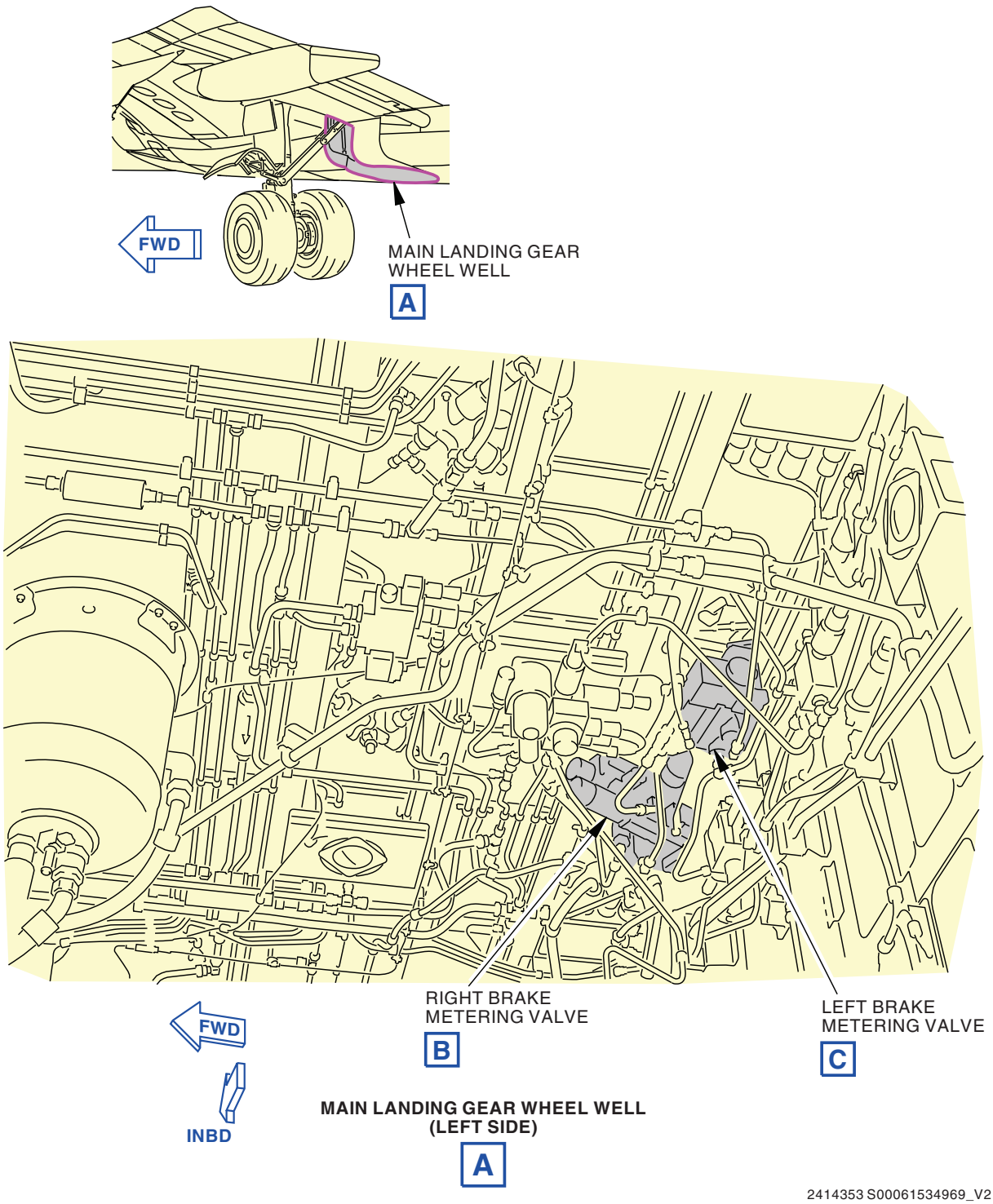
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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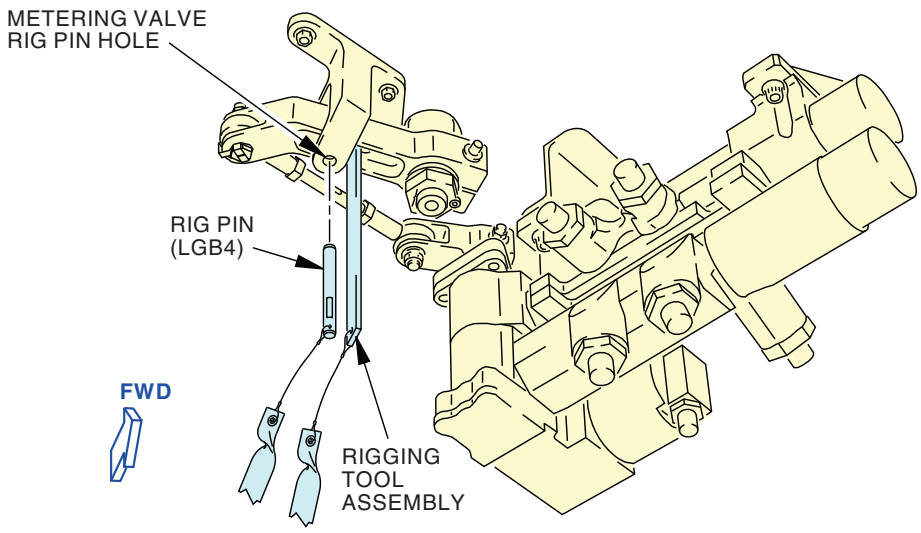
Brake Metering Valve Installation
Figure 401/32-41-31-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



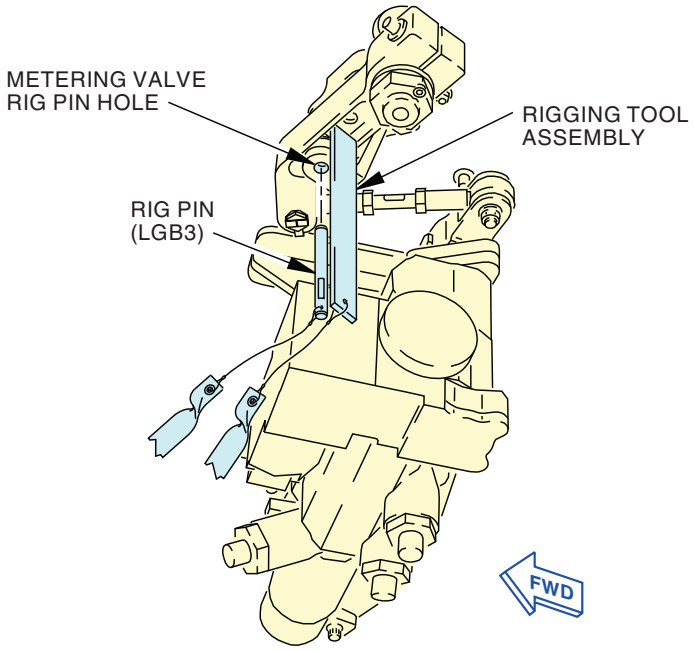
Brake Metering Valve Rig Pin Installation
Figure 402/32-41-31-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RIGHT BRAKE METERING VALVE

B



LEFT BRAKE METERING VALVE

C

2414354 S00061534970_V1

Brake Metering Valve Rig Pin Installation
Figure 402/32-41-31-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-31-400-801

3. Brake Metering Valve Installation

(Figure 401, Figure 402)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
C50056	Compound - Corrosion Inhibiting Material, Nondrying Resin Mix (Use until stock depleted)	BMS3-27
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D00633	Grease - Aircraft General Purpose	BMS3-33
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Brake metering valve	32-41-31-03-090	PTF ALL
14	Packing	32-41-31-03-060	PTF ALL
15	Ring	32-41-31-03-055	PTF ALL
16	Packing	32-41-31-03-060	PTF ALL
17	Packing	32-41-31-03-045	PTF ALL
18	Ring	32-41-31-03-035	PTF ALL
19	Packing	32-41-31-03-040	PTF ALL
21	Packing	32-41-31-03-025	PTF ALL
22	Ring	32-41-31-03-015	PTF ALL
23	Packing	32-41-31-03-020	PTF ALL
26	Packing	32-41-31-03-085	PTF ALL
27	Ring	32-41-31-03-075	PTF ALL

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(Continued)

AMM Item	Description	AIPC Reference	AIPC Effectivity
28	Packing	32-41-31-03-080	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Brake Metering Valve Installation

SUBTASK 32-41-31-080-001

(1) Remove all caps/plugs, STD-14686, that are in the ports of the brake metering valve [1].

SUBTASK 32-41-31-420-001

(2) If the replacement brake metering valve [1] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the adapters [20], adapter [24], adapters [29], filters [25], new packings [14], new packings [16], new packings [17], new packings [19], new packing [21], new packing [23], new packings [26], new packings [28], new rings [15], new rings [18], new ring [22], and new rings [27] with hydraulic fluid, D00153.

(b) Install the ring [18], packing [17], and packing [19] on each of the adapters [20].

(c) Install the ring [22], packing [21], and packing [23] on the adapter [24].

(d) Install the ring [15], packing [14], and packing [16] on each of the adapters [29].

(e) Install the ring [27], packing [26], and packing [28] on each of the filters [25].

(f) Install the adapters [20], adapter [24], adapters [29], and filters [25] in the brake metering valve [1].

1) Tighten the adapters [20] to 392.5 ±12.5 in-lb (44.3 ±1.4 N·m).

2) Tighten the adapter [24] to 100.0 ±5.0 in-lb (11.3 ±0.6 N·m).

3) Tighten the adapters [29] to 292.5 ±12.5 in-lb (33.0 ±1.4 N·m).

4) Tighten the filters [25] to 392.5 ±12.5 in-lb (44.3 ±1.4 N·m).

SUBTASK 32-41-31-420-002

(3) Install rig pin LGB3 (left side) or rig pin LGB4 (right side), F70207-88, from rigging pin kit, SPL-1585, through the rig pin hole for the brake metering valve [1].

SUBTASK 32-41-31-420-003

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR COMPOUND. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF IT GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONS.

(4) Apply corrosion inhibiting material, G50136 (preferred), or Mastinox 6856 K, C50056 (optional), to the bolts [3], washers [13], washers [5], and nuts [12] for the brake metering valve [1].

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SUBTASK 32-41-31-420-004

(5) Hold the brake metering valve [1] in its position on the airplane such that the shaft [11] goes into the crank.

SUBTASK 32-41-31-020-006

(6) Install the bolts [3], washers [13], washers [5], and nuts [12], to hold the brake metering valve [1] to the wheel well ceiling.

SUBTASK 32-41-31-420-005

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR COMPOUND. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF IT GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONS.

(7) Apply corrosion inhibiting material, G50136 (preferred), or Mastinox 6856 K, C50056 (optional), to the bolt [8], washer [9], and nut [10] that hold the crank to the shaft [11].

SUBTASK 32-41-31-020-007

(8) Install the nut [10], washer [9], and bolt [8] to hold the crank to the shaft [11] on the brake metering valve [1].

(a) For the right brake metering valve installation, make sure that the bolt [8] head points in the aft direction.

SUBTASK 32-41-31-820-001

(9) Make sure that the rig pin LGB3 or LGB4 moves freely through the rig pin hole.

(a) If you can not move the rig pin freely, do the adjustment of the brake metering valve.

1) Do this task: Hydraulic Brake System Adjustment, TASK 32-41-00-820-801.

SUBTASK 32-41-31-080-002

(10) Remove the rig pin LGB3 or LGB4.

SUBTASK 32-41-31-080-003

(11) Remove the caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-41-31-420-006

(12) Connect the hydraulic lines to the brake metering valve [1].

NOTE: Each port on the metering valve may have marking of designated hydraulic system, and it may not correspond to the hydraulic system used on aircraft. Refer to Figure 401 for the correct connection.

(a) Tighten the B-nuts on the System A Pressure and Brake hydraulic lines to 270.0 ±14.0 in-lb (30.5 ±1.6 N·m).

(b) Tighten the B-nuts on the System B Pressure and Brake hydraulic lines to 270.0 ±14.0 in-lb (30.5 ±1.6 N·m).

(c) Tighten the B-nut on the System A Return hydraulic line to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(d) Tighten the B-nut on the System B Return hydraulic line to 170.0 ±8.5 in-lb (19.2 ±1.0 N·m).

(e) Tighten the B-nut on the System A Landing Gear Up hydraulic line to 140.0 ±7.0 in-lb (15.8 ±0.8 N·m).

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SUBTASK 32-41-31-640-001

(13) If it is necessary, lubricate the rod end bearings [31] with grease, D00633.

G. Brake Metering Valve Installation Test

SUBTASK 32-41-31-790-001

(1) Do these steps to do a leak check and an operational check of the brake metering valve for each hydraulic system:

(a) For the normal (System B) hydraulic system:

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.
2) Fully push the captain's brake pedals twelve times.
3) Make sure that there are no leaks around the brake metering valve.
4) Examine the brake wear indicators on the brakes to make sure that the brakes engaged.

(b) For the alternate (System A) hydraulic system:

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

1) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.
2) Fully push the captain's brake pedals twelve times.
3) Make sure that there are no leaks around the brake metering valve.
4) Examine the brake wear indicators on the brakes to make sure that the brakes engaged.

SUBTASK 32-41-31-790-002

(2) Do a gear retract braking actuator check:

WARNING MAKE SURE ALL PERSONS AND EQUIPMENT ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) For the normal (System B) and the alternate (System A) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

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WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(b) Move the control lever for the landing gear to the UP position.

NOTE: Each gear door connects to each shock strut and only moves when that landing gear extends or retracts.

(c) Examine the brake wear indicators on the brakes to make sure that the brakes engaged.

(d) Make sure there that the valve assembly does not produce noise or unusual sound.

(e) Make sure that there are no leaks around the brake metering valve.

(f) Move the control lever for the landing gear to the DN position.

(g) Examine each brake's pistons to make sure that all brakes are released.

SUBTASK 32-41-31-710-001

(3) For each hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-31-710-002

(4) Bleed the hydraulic systems:

(a) For the normal brake metering valves, do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

(b) For the alternate brake metering valves, do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE METERING VALVE ASSEMBLY - INSPECTION/CHECK

1. General

A. This procedure has this task:

(1) A detailed inspection of the brake metering valve.

TASK 32-41-31-200-801

2. Brake Metering Valve Detailed Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

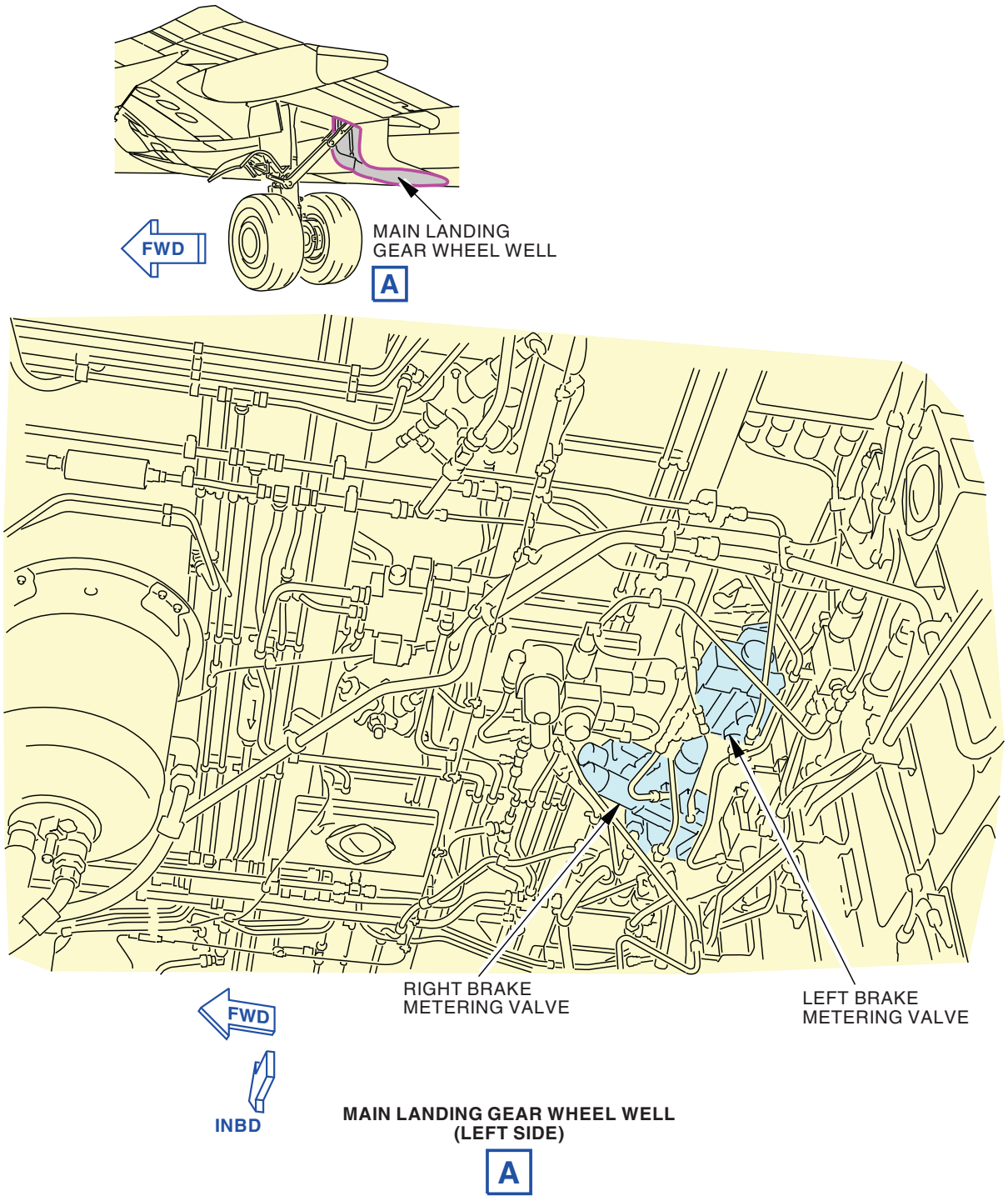
B. Brake Metering Valve Detailed Inspection

SUBTASK 32-41-31-210-002

(1) Do a detailed visual inspection of the brake metering valve.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Metering Valve - Detailed Inspection
Figure 601/32-41-31-990-804

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR BRAKE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the main landing gear brake

(2) An installation of the main landing gear brake.

B. For wheel, brake and tire interchangeability, refer to 277A6001, 297W5005 and 737-SL-32-101.

C. There are four main landing gear brakes (two on the left main landing gear and two on the right main landing gear). This procedure contains the removal and installation tasks for one of the main landing gear brakes. The removal and installation tasks for the all main landing gear brakes are the same.

D. In this procedure the main landing gear brake will be referred to as the brake assembly.

TASK 32-41-41-000-801

2. Main Landing Gear Brake Removal

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to remove the Main Landing Gear (MLG) brake.

B. References

Reference	Title
05-51-07-210-801	High Energy Stop (P/B 201)
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-11-000-801	Brake Disconnect Removal (P/B 401)
32-45-11-000-801	Main Landing Gear Wheel and Tire Assembly Removal (P/B 401)
32-46-21-000-801	Brake Temperature Sensor Removal (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1816	Sling - Hot Brake, 6 Inch Mesh Width Part #: 6T112X6 (6 ft. length) Supplier: 23755 Opt Part #: SLG35-6X72-002-01 Supplier: 10380

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(Continued)

Reference	Description
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535
COM-1823	Brake Cradle (Malabar Wheel Dolly only) Part #: BC400P Supplier: 94861 Opt Part #: BC400 Supplier: 94861
COM-1874	Bar - Tow, Dolly, Wheel and Brake Part #: 9092F Supplier: 00994
COM-13594	Dolly - Brake Change Part #: 15F3702 Supplier: 56535 Part #: STC90474-101 Supplier: 07132
SPL-1876	Thread Protector Equipment - Axle, MLG Part #: C32022-38 Supplier: 81205 Opt Part #: C32022-15 Supplier: 81205 Opt Part #: C32022-17 Supplier: 81205 Opt Part #: C32022-23 Supplier: 81205
SPL-14320	Thread Protector Eqpt - MLG Axle (1/8 inch undersized threads) Part #: C32022-39 Supplier: 81205 Opt Part #: C32022-32 Supplier: 81205

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

E. Prepare for the Removal

SUBTASK 32-41-41-840-001

(1) For ground safety precautions necessary in the handling of hot brakes, refer to this task: High Energy Stop, TASK 05-51-07-210-801.

SUBTASK 32-41-41-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(2) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

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SUBTASK 32-41-41-020-001

(3) Do this task: Main Landing Gear Wheel and Tire Assembly Removal, TASK 32-45-11-000-801.

NOTE: This task will install the equipment to protect the axle. Leave the equipment installed for the brake removal in subsequent steps.

SUBTASK 32-41-41-860-001

(4) Make sure that the hydraulic system A and B power is off (TASK 29-11-00-860-805).

SUBTASK 32-41-41-480-002

(5) Make sure that the tires for the nose landing gear and for the other main landing gear have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-41-860-002

(6) Make sure that the parking brake is released.

SUBTASK 32-41-41-860-003

(7) Fully push the captain's or first officer's brake pedals twelve times to remove the pressure from the brake accumulator.

SUBTASK 32-41-41-580-001

(8) Make sure that the wheel and tire assemblies are lifted with jacks until they are clear of the ground (TASK 07-10-03-580-801).

F. Main Landing Gear Brake Removal

SUBTASK 32-41-41-020-002

(1) Disconnect the brake hose [12] from the brake assembly [1] at the brake disconnect [3], do these steps (View E, Figure 401):

(a) Push the sleeve [11] toward the brake and turn the sleeve [11] in the clockwise direction to disengage the brake half [10] from the hose half [9].

(b) Pull up on the sleeve [11] of the brake disconnect [3], away from the brake assembly [1], until the threads of the brake half [10] and hose half [9] are engaged.

(c) Turn the sleeve [11] in the counterclockwise direction to disconnect the brake hose [12] from the brake assembly [1].

SUBTASK 32-41-41-020-003

(2) If you will use the brake half [10] of the brake disconnect [3] from the old brake assembly [1] on a new brake assembly [1], remove the brake half [10] of the brake disconnect [3] (TASK 32-41-11-000-801).

SUBTASK 32-41-41-020-004

(3) Remove the bolt [4], washers [5], and nut [6] to disconnect the retention cable [2] from the brake assembly [1] (View C, Figure 401).

SUBTASK 32-41-41-020-008

(4) Disconnect the electrical connector [7] from the brake temperature sensor [8].

(a) If the brake temperature sensor [8] from the old brake will be used, remove the brake temperature sensor [8] (TASK 32-46-21-000-801).

SUBTASK 32-41-41-480-003

(5) Make sure that the axle thread protector, SPL-1876, and thread protector, SPL-14320 (if the axle has undersized threads), are installed.

NOTE: This equipment was installed during the wheel and tire removal steps.

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SUBTASK 32-41-41-020-005

CAUTION BE VERY CAREFUL WHEN YOU DO MAINTENANCE ON CARBON BRAKES. CARBON BRAKES ARE MORE EASILY DAMAGED THAN STEEL BRAKES. YOU CAN NOT REPAIR THEM. IF YOU LET THEM FALL OR HIT THEM, DAMAGE CAN OCCUR.

CAUTION DO NOT APPLY FORCE TO THE CARBON BRAKE HEAT SHIELDS WHEN YOU REMOVE OR INSTALL THE CARBON BRAKES. BENT HEAT SHIELDS CANNOT BE REPAIRED. A CARBON BRAKE WITH A BENT HEAT SHIELD WILL CAUSE SCORING ON THE WHEEL INNER SURFACE WHEN THE WHEEL TURNS.

CAUTION DO NOT PUT THE BRAKE DOWN ON THE OUTER DIAMETER OF THE ROTORS AFTER YOU REMOVE IT. CARBON BRAKES ARE NOT SERVICEABLE IF THEY HAVE NICKS, OR CRACKS. YOU CANNOT REPAIR THEM. THE WEIGHT OF THE BRAKE CAN CAUSE DAMAGE TO CARBON BRAKE ROTORS.

(6) Do these steps to remove the brake assembly [1] from the axle:

WARNING DO NOT PUT FINGERS IN THE LIGHTENING HOLES OF THE WHEEL RETAINING NUT SOCKET TOOL WHEN YOU REMOVE OR INSTALL THE WHEEL RETAINING NUT. THIS CAN CAUSE INJURIES TO PERSONS.

(a) Use one of the subsequent sets of equipment and remove the brake assembly [1] from the axle.

1) The wheel/brake changer dolly, COM-1818, with the optional tow bar, COM-1874, or
2) The cradle, COM-1823, and dolly, 175S-2, or
3) The dolly, PF32-002-1, and hot brake sling, COM-1816, or
4) The dolly, TB900, with the cradle, COM-1823, or
5) The brake dolly, COM-13594.

SUBTASK 32-41-41-160-001

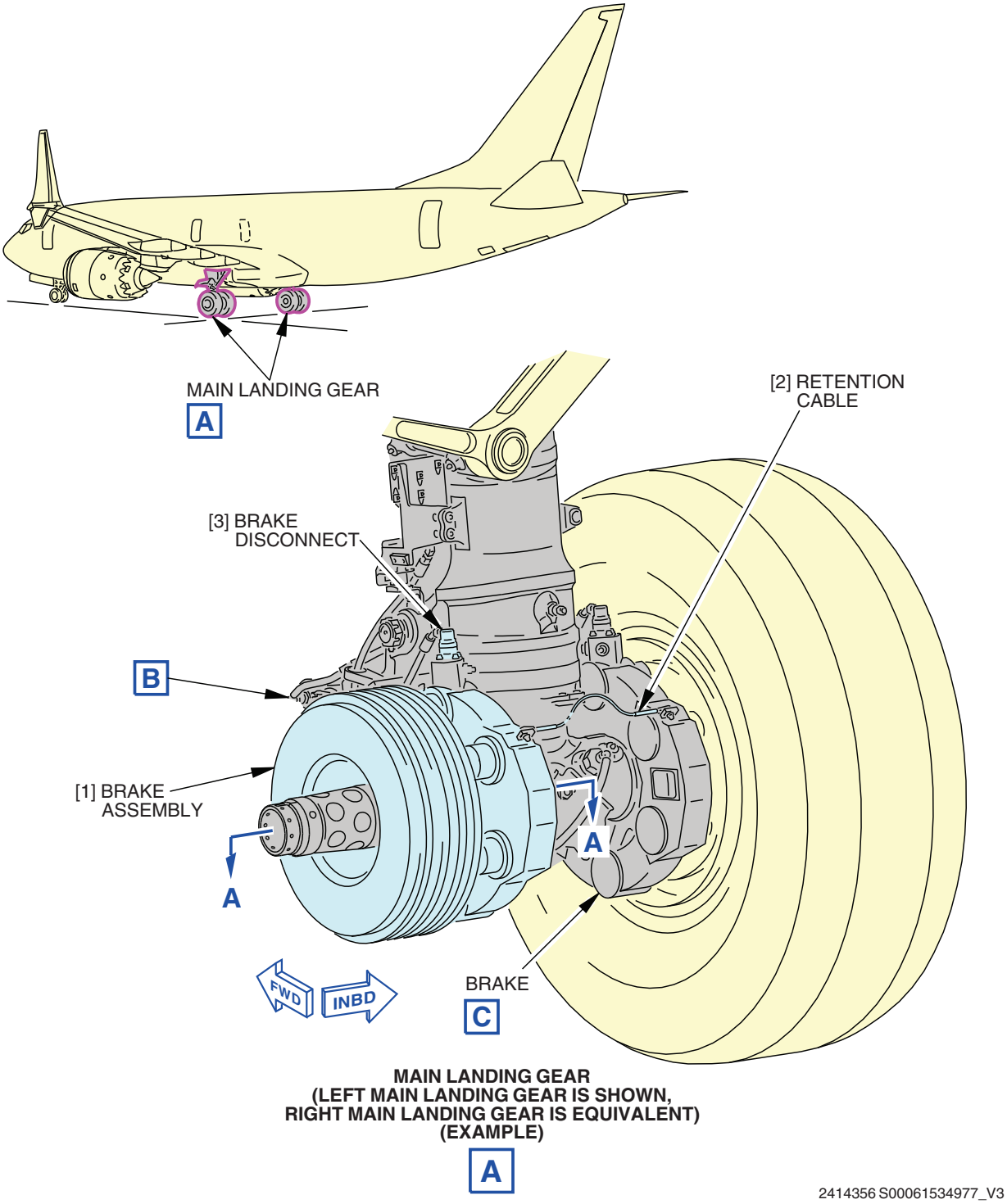
(7) Remove the old grease from the axle and brake sleeve [14].

SUBTASK 32-41-41-020-006

(8) If the brake sleeve [14] remained in the brake assembly [1], remove the brake sleeve [14] from the brake assembly [1] and install it back on the axle.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



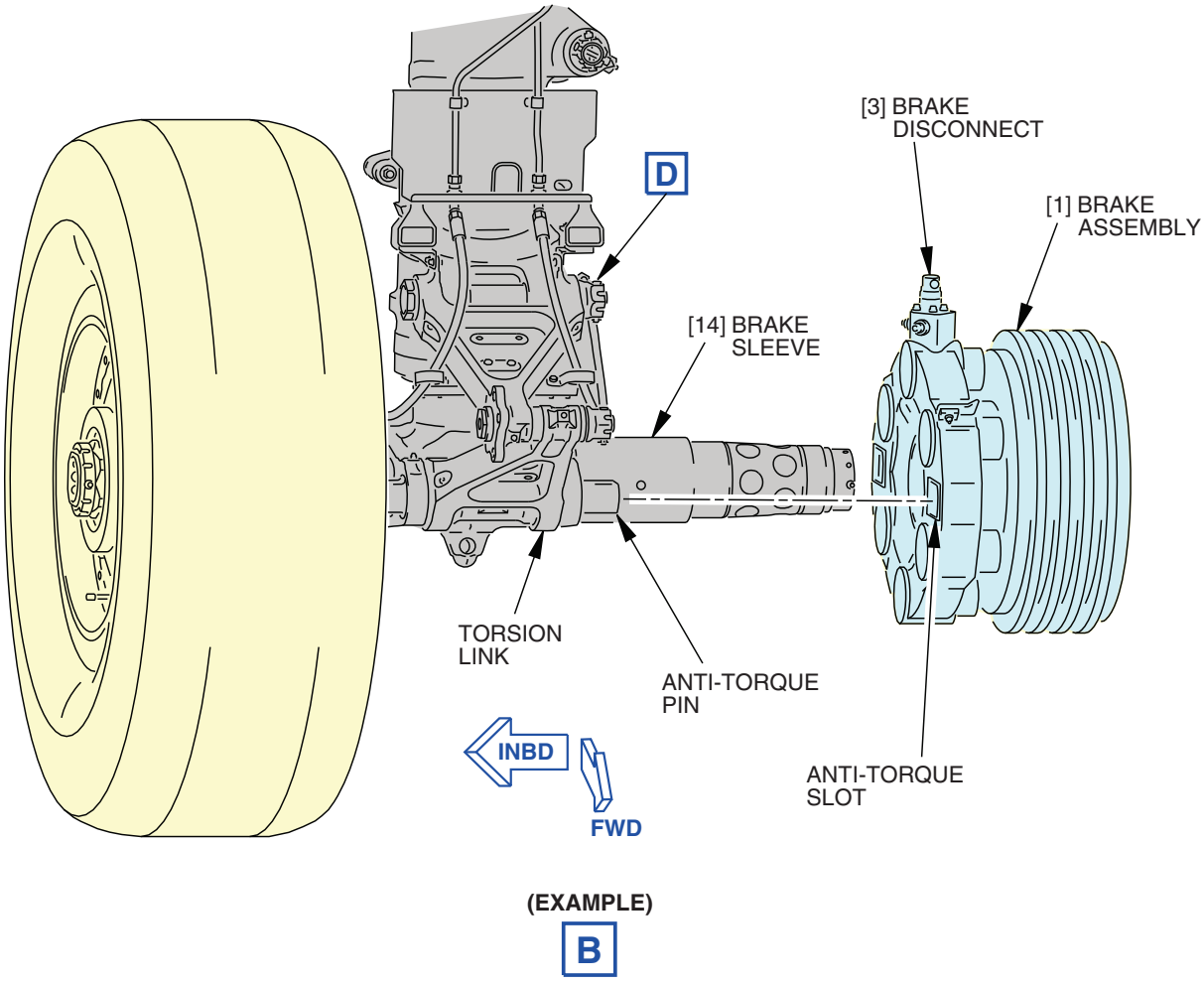
MAIN LANDING GEAR
(LEFT MAIN LANDING GEAR IS SHOWN,
RIGHT MAIN LANDING GEAR IS EQUIVALENT)
(EXAMPLE)

A

2414356 S00061534977_V3

Main Landing Gear Brake Installation
Figure 401/32-41-41-990-801 (Sheet 1 of 5)

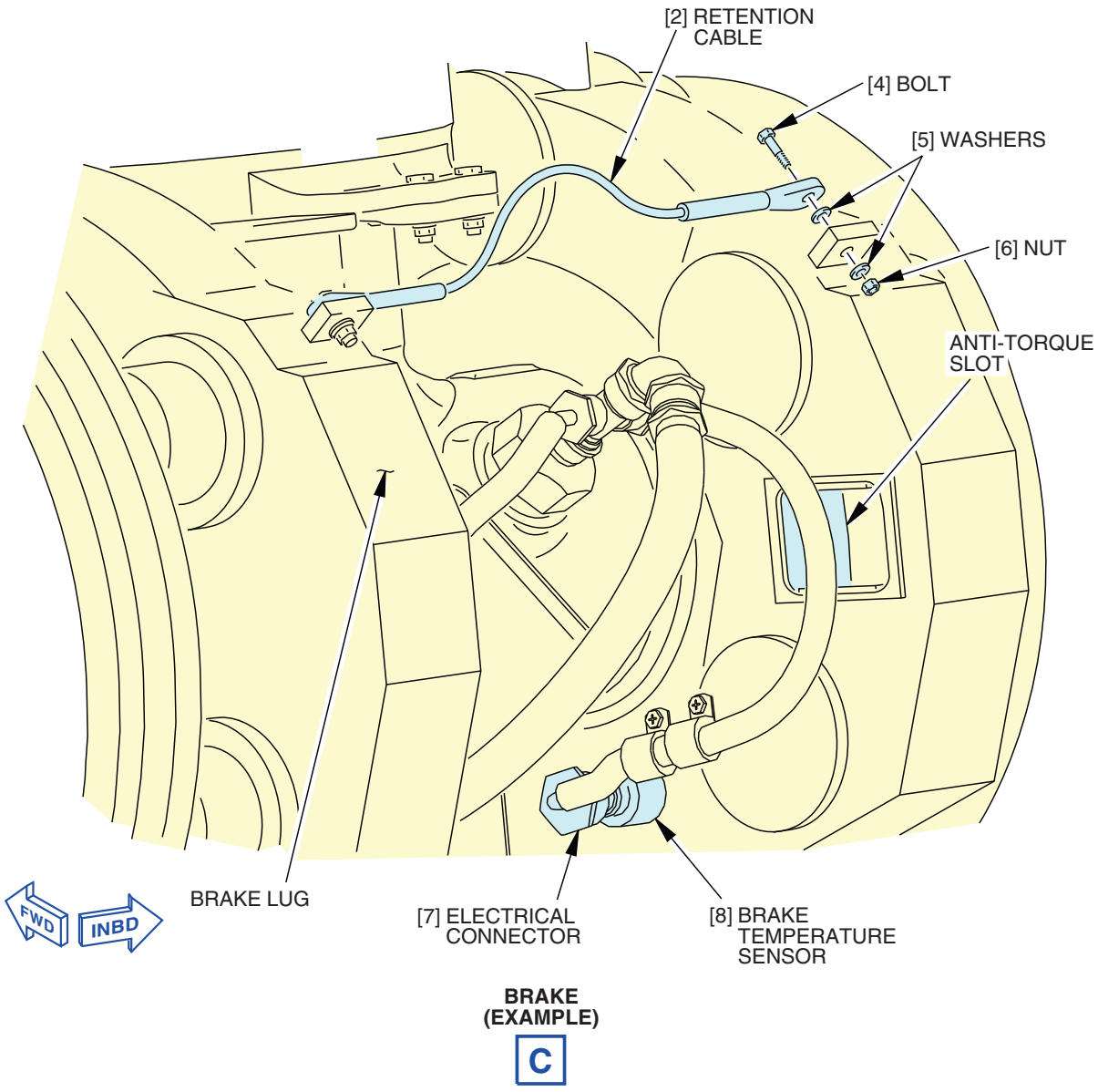
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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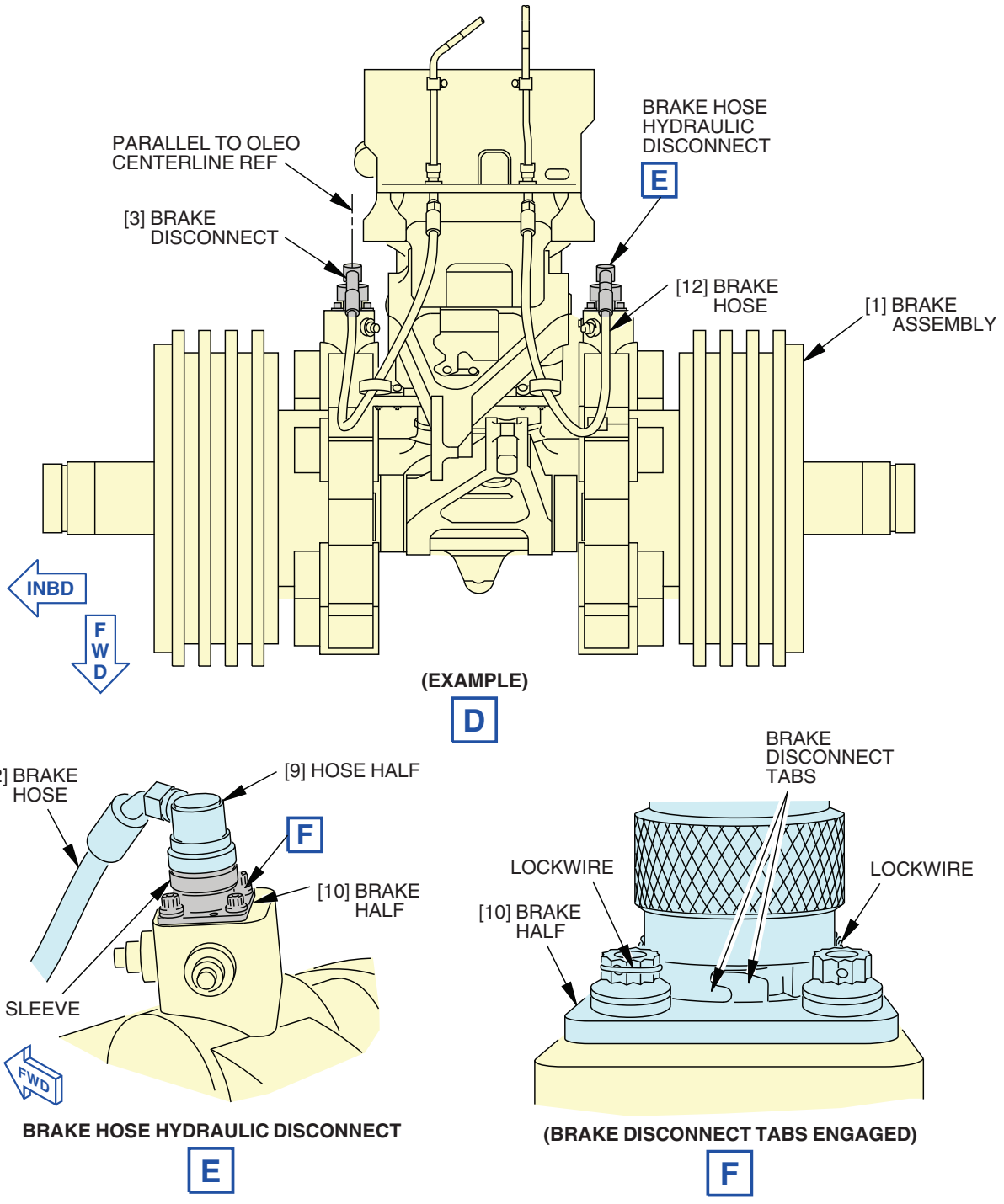
Main Landing Gear Brake Installation
Figure 401/32-41-41-990-801 (Sheet 2 of 5)

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Main Landing Gear Brake Installation
Figure 401/32-41-41-990-801 (Sheet 3 of 5)

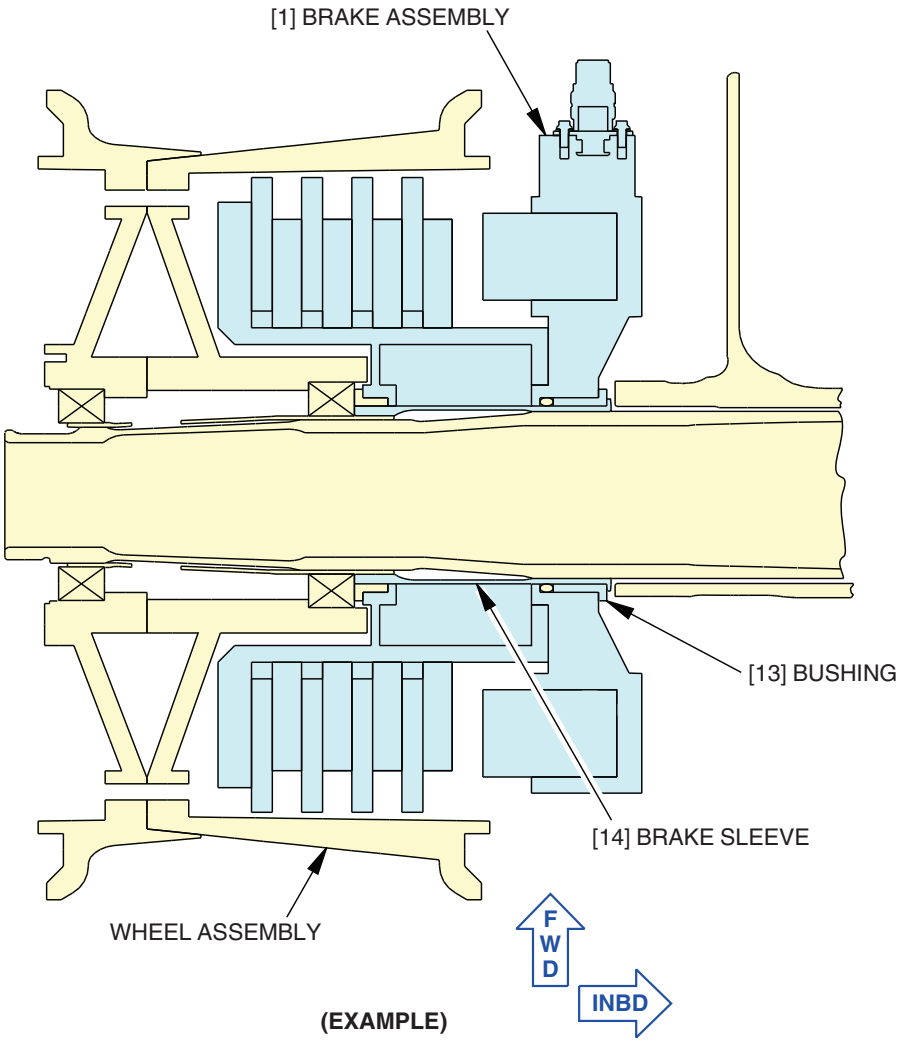
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414359 S00061534980_V2

Main Landing Gear Brake Installation
Figure 401/32-41-41-990-801 (Sheet 4 of 5)

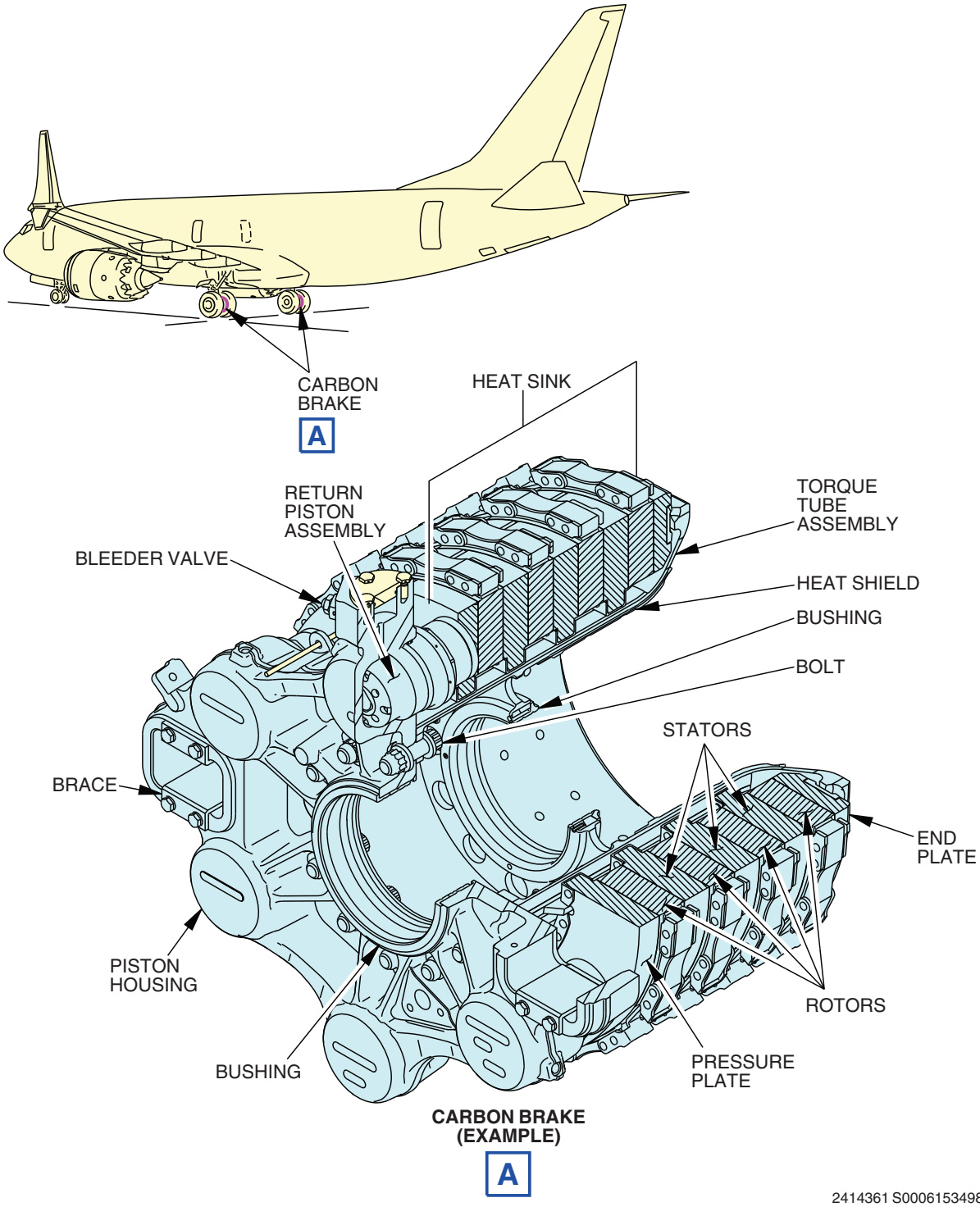
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414360 S00061534981_V2

Main Landing Gear Brake Installation
Figure 401/32-41-41-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Carbon Brake
Figure 402/32-41-41-990-802

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-41-400-801

3. Main Landing Gear Brake Installation

(Figure 401 and Figure 402)

A. General

(1) This task gives the instructions to install the Main Landing Gear (MLG) brake.

(2) It is only necessary to lubricate the bushing for the MLG brake housing with grease at the brake installation. There is no scheduled requirement to lubricate the MLG brake housing and there is no grease fitting on the brake.

B. References

Reference	Title
07-10-03-580-801	Lift the Airplane on Axle Jacks-Handling (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)
32-41-11-400-801	Brake Disconnect Installation (P/B 401)
32-45-11-400-801	Main Landing Gear Wheel and Tire Assembly - Installation (P/B 401)
32-46-00-730-801	Tire and Brake Monitoring System - System Test (P/B 501)
32-46-21-400-801	Brake Temperature Sensor Installation (P/B 401)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1816	Sling - Hot Brake, 6 Inch Mesh Width Part #: 6T112X6 (6 ft. length) Supplier: 23755 Opt Part #: SLG35-6X72-002-01 Supplier: 10380
COM-1818	Dolly - Wheel/Brake Change Part #: 15F590 Supplier: 56535 Part #: 17500M Supplier: 94861 Part #: 175S-2 Supplier: 94861 Part #: 227A-JMS Supplier: CD856 Part #: 9092-016 Supplier: 00994 Part #: IA3201A-500 Supplier: 053H3 Part #: MH13-003 Supplier: D2029 Part #: STC90475-101 Supplier: 07132 Opt Part #: 175M Supplier: 94861 Opt Part #: 8436-012 Supplier: 00994 Opt Part #: 9092-010 Supplier: 00994 Opt Part #: 9092-015 Supplier: 00994 Opt Part #: PF32-002-1 Supplier: 3D5B2 Opt Part #: TB900 Supplier: 56535
COM-1823	Brake Cradle (Malabar Wheel Dolly only) Part #: BC400P Supplier: 94861 Opt Part #: BC400 Supplier: 94861

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(Continued)

Reference	Description
COM-1874	Bar - Tow, Dolly, Wheel and Brake Part #: 9092F Supplier: 00994
COM-13594	Dolly - Brake Change Part #: 15F3702 Supplier: 56535 Part #: STC90474-101 Supplier: 07132
SPL-1876	Thread Protector Equipment - Axle, MLG Part #: C32022-38 Supplier: 81205 Opt Part #: C32022-15 Supplier: 81205 Opt Part #: C32022-17 Supplier: 81205 Opt Part #: C32022-23 Supplier: 81205
SPL-14320	Thread Protector Eqpt - MLG Axle (1/8 inch undersized threads) Part #: C32022-39 Supplier: 81205 Opt Part #: C32022-32 Supplier: 81205

D. Consumable Materials

Reference	Description	Specification
D00233	Grease - Aircraft, General Purpose, Wide Temperature - Mobil 28	MIL-PRF-81322
D00378	Grease - Aircraft, General Purpose, Wide Temperature - Aeroshell 22	MIL-PRF-81322
D00633	Grease - Aircraft General Purpose	BMS3-33
D50005	Grease - Wheel Bearing - Mobil Aviation Grease SHC 100	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Brake assembly	32-41-41-01-060	PTF ALL

F. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

G. Prepare for the Installation

SUBTASK 32-41-41-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-41-020-007

CAUTION

MAKE SURE THAT THE BRAKE SLEEVE IS INSTALLED BEFORE YOU INSTALL THE BRAKE. THE BRAKE SLEEVE CAN COME OFF WHEN YOU REMOVE THE BRAKE. IF YOU INSTALL THE BRAKE AND THE BRAKE SLEEVE IS NOT ON THE AXLE, DAMAGE CAN OCCUR TO THE AIRCRAFT.

(2) Make sure that the brake sleeve [14] is installed on the axle.

SUBTASK 32-41-41-480-005

(3) Make sure that the axle thread protector, SPL-1876, and thread protector, SPL-14320 (if the axle has undersized threads), are installed on the axle.

SUBTASK 32-41-41-640-001

WARNING

APPLY GREASE TO THE BRAKE AXLE BUSHINGS. APPLY A THIN LAYER OF GREASE TO THE MATING LANDS OF THE BRAKE SLEEVE. DO NOT APPLY GREASE IN THE SPACE BETWEEN THE AXLE BUSHINGS ON THE BRAKE ASSEMBLY. REMOVE ALL UNWANTED GREASE. IF YOU APPLY TOO MUCH GREASE, A FIRE CAN OCCUR WHEN THE BRAKE BECOMES HOT. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL CAN OCCUR.

(4) Apply a thin layer of Mobil 28 grease, D00233, or Aeroshell 22 grease, D00378, or Mobil Aviation Grease SHC 100 grease, D50005, to the bushing [13] and mating lands of the brake sleeve [14] on the brake assembly [1].

NOTE: This will let the brake assembly slide easily onto the axle.

(a) Make sure that the grooves in the bushing [13] are completely filled with the grease.

(b) Remove all unwanted grease.

H. Main Landing Gear Brake Installation

SUBTASK 32-41-41-420-001

(1) If it is necessary, install the brake half [10] of the brake disconnect [3] (TASK 32-41-11-400-801).

SUBTASK 32-41-41-420-006

(2) If it is not installed, install the brake temperature sensor [8] (TASK 32-46-21-400-801).

SUBTASK 32-41-41-410-001

WARNING

DO NOT APPLY TOO MUCH GREASE TO THE LANDING GEAR BRAKES. IF YOU APPLY TOO MUCH GREASE, A FIRE CAN OCCUR WHEN THE BRAKES BECOME HOT.

(3) Apply a thin layer of grease, D00633, to the anti-torque pin of the lower torsion link and anti-torque slot on the brake assembly [1].

SUBTASK 32-41-41-420-002

CAUTION

BE VERY CAREFUL WHEN YOU DO MAINTENANCE ON CARBON BRAKES. CARBON BRAKES ARE MORE EASILY DAMAGED THAN STEEL BRAKES. YOU CAN NOT REPAIR THEM. IF YOU LET THEM FALL OR HIT THEM, DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT APPLY FORCE TO THE CARBON-BRAKE HEAT-SHIELDS WHEN YOU REMOVE OR INSTALL THE CARBON-BRAKES. YOU CAN NOT REPAIR BENT HEAT SHIELDS. A CARBON BRAKE WITH A BENT HEAT SHIELD WILL CAUSE SCORING ON THE WHEEL INNER SURFACE WHEN THE WHEEL TURNS. THIS CAN CAUSE DAMAGE.

CAUTION

DO NOT PUT THE BRAKE DOWN ON THE OUTER DIAMETER OF THE ROTORS AFTER YOU REMOVE IT. CARBON BRAKES ARE NOT SERVICEABLE IF THEY HAVE NICKS, OR CRACKS. YOU CANNOT REPAIR THEM. THE WEIGHT OF THE BRAKE CAN CAUSE DAMAGE TO CARBON BRAKE ROTORS.

(4) Do this step to install the brake assembly [1] on the axle:

(a) Make sure that the anti-torque pin on the lower torsion link goes into the anti-torque slot in the brake assembly [1].

NOTE: There is no requirement to install a grease seal.

WARNING

DO NOT PUT FINGERS IN THE HOLES YOU USE TO LIGHTEN THE WHEEL-RETAINING NUT SOCKET-TOOL. DO NOT PUT THEM IN WHEN YOU REMOVE/INSTALL THE WHEEL RETAINING NUT. THIS CAN CAUSE INJURIES TO PERSONS.

(b) Use one of the subsequent sets of equipment to install the brake assembly [1] in its position on the axle:

1) The wheel/brake changer dolly, COM-1818, with the optional tow bar, COM-1874, or
2) The cradle, COM-1823, and dolly, 175S-2, or
3) The dolly, PF32-002-1, and hot brake sling, COM-1816, or
4) The dolly, TB900, with the cradle, COM-1823, or
5) The brake dolly, COM-13594.

SUBTASK 32-41-41-410-002

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(5) Apply a layer of Cor-Ban 27L Compound, G50237, to the bolt [4], washers [5], and nut [6] for the retention cable [2] (View C, Figure 401).

SUBTASK 32-41-41-410-003

(6) Install the bolt [4], washers [5], and nut [6] to connect the retention cable [2] to the brake assembly [1].

SUBTASK 32-41-41-820-001

(7) Make sure that the axial free play is not more 0.03 in. (0.76 mm) from the bottom of the nut [6] to the brake lug on the brake assembly [1] that are not reinstalled.

(a) If the axial free play is more 0.03 in. (0.76 mm), tighten the nut [6].

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SUBTASK 32-41-41-420-003

(8) Connect the brake hose [12] to the brake assembly [1] at the brake disconnect [3] (View E, Figure 401).

(a) Install the hose half [9] of the brake disconnect [3] on the brake half [10] of the brake disconnect [3].

1) Turn the hose half [9] of the brake disconnect [3] until the internal tangs align with the slots inside the brake half [10] of the brake disconnect [3] and hose half [9] down can be pushed.

2) Push down on the sleeve [11] and turn it in the clockwise direction to engage the threads.

3) Turn the sleeve [11] in the clockwise direction until the slots in the bottom of the sleeve [11] snap onto the external tangs on the brake half [10] of the brake disconnect [3].

4) Make sure that the elbow fitting on the brake disconnect [3] is parallel to the oleo centerline or up to ten degrees toward the shock strut.

NOTE: This will make sure that the hoses do not rub on the brake housing.

5) Make sure that the lockwires, installed on the screws, do not contact the outer diameter of the hose half [9] (View F, Figure 401).

SUBTASK 32-41-41-210-001

(9) Do a visual check to make sure that the slots on the sleeve [11] are fully and correctly engaged with the tangs on the brake half [10] of the brake disconnect [3] (View F, Figure 401).

SUBTASK 32-41-41-420-008

(10) Make sure that the brake hose B-nut is tightened.

(a) If the brake hose B-nut is not tightened, tighten the brake hose B-nut (TASK 32-41-11-400-801).

SUBTASK 32-41-41-420-004

(11) Visually check the hose half [9] and brake half [10] of the brake disconnect [3] to make sure that they are correctly engaged.

SUBTASK 32-41-41-420-007

(12) If not already done, connect the electrical connector [7] to the brake temperature sensor [8] (TASK 32-46-21-400-801).

SUBTASK 32-41-41-420-005

(13) Do this task: Main Landing Gear Wheel and Tire Assembly - Installation, TASK 32-45-11-400-801.

I. Main Gear Wheel Brakes Bleeding and Installation Test

SUBTASK 32-41-41-860-016

(1) To do a test of the brake temperature sensors, do this task: Tire and Brake Monitoring System - System Test, TASK 32-46-00-730-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-41-860-005

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Activate the normal (System B) hydraulic system (TASK 29-11-00-860-801).

SUBTASK 32-41-41-790-001

(3) Do these steps to do an operational check of all four brakes:

NOTE: At least one person needs to be stationed near the main landing gear brakes. Repeat brake pedal application and release as needed to verify operation of each brake.

(a) Push one set of brake pedals to the limit of travel to fully operate each brake assembly [1].

(b) Set the parking brake.

(c) Monitor the movement of each brake's pistons or wear pins to make sure that each brake assembly [1] engaged.

(d) Wait two minutes.

(e) Look for leaks around each brake assembly [1] and at each brake disconnect [3].

(f) Release the parking brake.

(g) Monitor the movement of each brake's pistons or wear pins to make sure that each brake assembly [1] released.

SUBTASK 32-41-41-870-001

(4) If the brakes do not feel normal, do this task: Normal (System B) Hydraulic Brake System Bleeding, TASK 32-41-00-870-802.

J. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-41-860-006

(1) Deactivate the normal (System B) hydraulic system (TASK 29-11-00-860-805).

SUBTASK 32-41-41-580-002

WARNING MAKE SURE THAT THE AREA BELOW THE TIRES IS CLEAR OF PERSONS AND EQUIPMENT. IF THE AREA IS NOT CLEAR, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) If it is necessary, lower the wheel and tire assemblies and remove the jacks (TASK 07-10-03-580-801).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR BRAKE - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Examine the Main Landing Gear Brakes for Wear

(2) Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane)

(3) Main Landing Gear Brake Check (Wheel Removed from the Airplane).

TASK 32-41-41-700-801

2. Examine the Main Landing Gear Brakes for Wear

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This procedure has steps to examine the main landing gear brakes for wear.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Procedure

SUBTASK 32-41-41-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-41-41-480-008

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-41-860-007

(3) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-41-790-002

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Make sure to activate the normal (System B) hydraulic system (TASK 29-11-00-860-801).

E. Main Landing Gear Brakes Inspection

SUBTASK 32-41-41-210-002

(1) Do these steps to examine the brakes for wear:

(a) Fully push the captain's or first officer's brake pedals and hold.

NOTE: You can set the parking brake to hold the brake pedals.

(b) For each of the brakes [3], make sure the wear indicator pins [1] extend out of the guides [2].

NOTE: For accuracy, do a check of the wear indicators pins after the brakes have cooled.

(c) Replace each of the brakes [3], if the end of the wear indicator pins [1] is level with, or below the face of a guide [2].

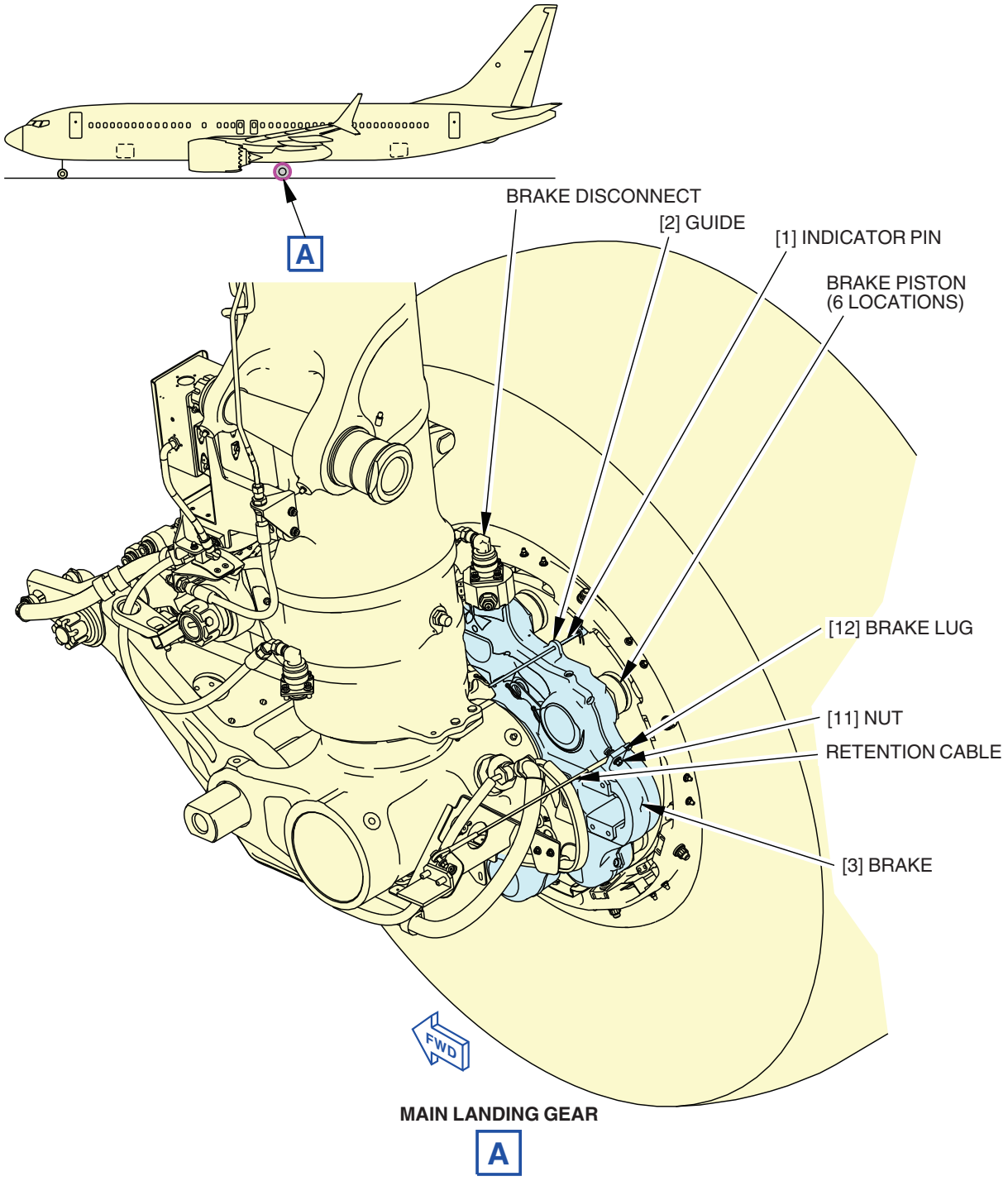
F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-41-860-008

(1) Make sure to deactivate the normal (System B) hydraulic system (TASK 29-11-00-860-805).

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Carbon Brake Inspection (Wheel Installed)
Figure 601/32-41-41-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-41-700-802

3. Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane)

(Figure 601)

A. General

(1) This procedure has steps to do a fast check of the main landing gear brakes with the wheels installed.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Check

SUBTASK 32-41-41-480-010

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-41-41-480-011

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-41-860-009

(3) Release the parking brake.

SUBTASK 32-41-41-790-003

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Make sure to activate the normal (System B) hydraulic system (TASK 29-11-00-860-801).

E. Main Landing Gear Brake Fast Check (Wheel Installed on the Airplane)

SUBTASK 32-41-41-860-010

(1) Fully apply and release the captain's or first officers brake pedals five times.

SUBTASK 32-41-41-860-011

(2) With the brake pedals not applied, do a check of the brakes for fluid leaks at these locations:

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(a) brake pistons

(b) brake disconnect

(c) brake housing ports

(d) inlet and drain ports

(e) bleed ports.

SUBTASK 32-41-41-790-004

(3) Do a check of the brake.

(a) If the total leakage per brake at the above locations is greater than 1 drop/minute with the brake pedals not applied, repair the leak(s) or replace the brake prior to dispatch.

SUBTASK 32-41-41-720-001

(4) Slowly apply the brake pedals to the stops.

NOTE: You can set the parking brake to hold the brake pedals.

SUBTASK 32-41-41-790-005

(5) While you apply the brake pedals, do a check of the brakes for fluid leaks at these locations.

(a) brake pistons

(b) brake disconnect

(c) brake housing ports

(d) inlet and drain ports

(e) bleed ports.

SUBTASK 32-41-41-790-006

(6) Do a check of the brake.

(a) If the total leakage per brake at the above locations is greater than 1 drops per 5 minutes while the brakes are being applied, repair the leak(s) or replace the brake prior to dispatch.

(b) With leaks below these limits, the brake must be rechecked prior to each flight, and should be repaired or replaced at the next opportunity that manpower and material will permit.

SUBTASK 32-41-41-790-008

(7) Do a check of the brake for contamination.

NOTE: Signs of contamination include a wet or oily appearance, buildup of charred residue, or heavy smoke after landing.

(a) Remove, inspect and clean the brake in accordance with the brake supplier component maintenance manual if it is suspected that a brake has been exposed to significant levels of contamination.

SUBTASK 32-41-41-790-009

(8) Release the captain's or first officers brake pedals (or parking brake).

SUBTASK 32-41-41-911-001

(9) Do a check for the wear indicator pin length:

NOTE: Each brake has two brake wear indicator pins.

(a) If either wear indicator pin is no longer extended through the piston housing assembly, you must replace the brakes prior to the next flight.

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SUBTASK 32-41-41-210-003

(10) Examine all of the brakes for these conditions:

(a) chips

(b) cracks or broken parts

(c) severe oxidation exhibited by porosity

(d) inconsistent appearance/form, distortion

(e) softness and/or exposed fibers

(f) parts that show a large amount of wear

(g) other damage that can cause a malfunction or a leak.

SUBTASK 32-41-41-860-014

(11) Make sure that the axial free play is not more 0.03 in. (0.76 mm) from the bottom of the nut [11] to the brake lug [12].

(a) If the axial free play is more 0.03 in. (0.76 mm), tighten the nut [11].

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-41-860-012

(1) Make sure to deactivate the normal (System B) hydraulic system (TASK 29-11-00-860-805).

——— END OF TASK ———

TASK 32-41-41-700-803

4. Main Landing Gear Brake Check (Wheel Removed from the Airplane)

(Figure 602, Figure 603, and Figure 604)

A. General

(1) This task has steps to check the brake on the main landing gear with the wheels removed.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41 P/B 401	MAIN LANDING GEAR BRAKE - REMOVAL/INSTALLATION

C. Tools/Equipment

Reference	Description
STD-765	Scraper - Plastic

D. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Prepare for the Check

SUBTASK 32-41-41-480-012

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-41-41-480-013

(2) Make sure that the tires have chocks installed around them (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-41-860-013

(3) Release the parking brake.

F. Main Landing Gear Brake Check

SUBTASK 32-41-41-210-004

WARNING MAKE SURE THAT THE BRAKE TEMPERATURES DECREASE TO WHERE IT DOES NOT CAUSE INJURY. YOU CAN PUT FANS ADJACENT TO THE BRAKE ASSEMBLY TO DECREASE TEMPERATURES QUICKLY. HOT BRAKES CAN CAUSE DAMAGE TO EQUIPMENT AND INJURY TO PERSONNEL.

(1) Visually examine the brake for these types of damage once the temperature of the brake has cool down sufficiently (Figure 603).

(a) Rotate each rotor by hand to visually inspect each rotor clip for cracks and missing rivets.

1) Make sure that there is no binding.

(b) Broken and damaged parts.

(c) Parts that show a large quantity of worn areas.

(d) Other damage that can cause worn areas, or leaks of hydraulic fluid.

(e) Remove the brake if these conditions exist.

SUBTASK 32-41-41-210-005

(2) Do a check for missing brake wear indicator pins (Figure 602).

NOTE: Each brake has two brake wear indicator pins.

(a) If the two wear indicator pins are missing, replace the brake before the next flight.

(b) If one wear indicator pin is missing, the brake can stay in service, if the remaining wear indicator pin operation is satisfactory.

SUBTASK 32-41-41-210-006

(3) Do a check for corrosion on or around the brake.

(a) Make sure that the surface finishes are in good condition.

NOTE: Corrosion can occur quickly in some climates.

SUBTASK 32-41-41-210-007

(4) Do a check for approximately the same brake running clearance between each brake piston and pressure plate (View A-A, Figure 603).

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(a) With the brake released, visually check to make sure that there is clearance between each brake piston and pressure plate.

(b) Make sure that the running clearance between each brake piston and pressure plate is not less than 0.050 in. (1.270 mm).

(c) If the running clearance is less than 0.050 in. (1.270 mm), replace the main landing gear brake to check the condition of the brake assembly springs (PAGEBLOCK 32-41-41/401).

(d) Make sure that the running clearance is from 0.071 in. (1.8 mm) to 0.213 in. (5.4 mm).

SUBTASK 32-41-41-210-008

(5) Check the visible areas of the brake heat sink (rotors, stators, pressure plate, and end plate) for oxidation damage (Figure 602).

(a) Replace the carbon brakes if these conditions exist (Figure 602 and Figure 603).

1) Examine the brakes for indentations or flaking carbon where the hydraulic brake pistons touch the carbon pressure plate.

2) Examine the brake and surrounding areas for unusual carbon debris such as chips or small pieces of carbon broken off of the rotors and stators.

3) Make sure that the metal rotor drive clips on the outside diameter of each carbon rotor do not show excessive movement and there is no loose flaking carbon near the clips.

(b) Do a check for loose rotor drive clips and rivets.

(c) Do a check for pieces of carbon that have broken off the carbon disks.

(d) Do a check of the outside diameter of the stator disks for the correct 12 o'clock positions of their alignment grooves (Figure 604).

NOTE: Most carbon brakes are configured with worn stator indicators. A worn stator indicator is a line machined in the stator at the 12 o'clock position of the heatsink. If this indicator is available on the heatsink, the indicator can be used as a visual reference to make sure that the stator does not rotate more than 0.25 in. (0.64 cm).

NOTE: A stator groove that is not aligned is an indication of possible stator disk drive failure.

NOTE: Manufacturing of alignment markings vary from brake supplier to brake supplier, not all brake assemblies have this indicator.

1) With one hand at the 3 o'clock position and the other hand at the 9 o'clock position, rotate each stator individually and see that the stator does not move more than 0.25 in. (0.64 cm) as measured on the outside diameter of the adjacent stators.

2) Replace the brake if any of the stators can be moved by hand beyond 0.25 in. (0.64 cm).

(e) If the carbon material in a damaged area is soft, flaking, or can be easily removed with a fingernail or plastic scraper, STD-765, the brake should be replaced.

NOTE: Carbon brake oxidation causes softness, flaking, and pieces to break off of the rotors and stators. Stains that are brown or orange are initial indications of possible oxidation. Oxidation damage normally occurs on the outside diameters of rotors and stators and non-wear surfaces of pressure plate and end plate and is easy to see. Oxidation damage can also occur on the inner diameter of the rotors and stators, end plate and pressure plate.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-41-210-009

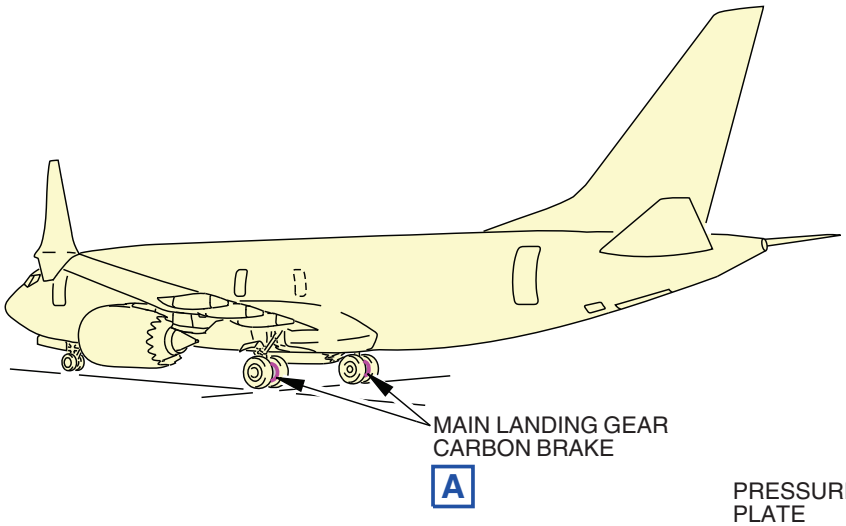
(6) Damage such as nicks, scratches, and gouges in the carbon that do not have cracks are acceptable provided that the carbon is not soft or flaking and cannot be scraped off with a fingernail or plastic scraper, STD-765.

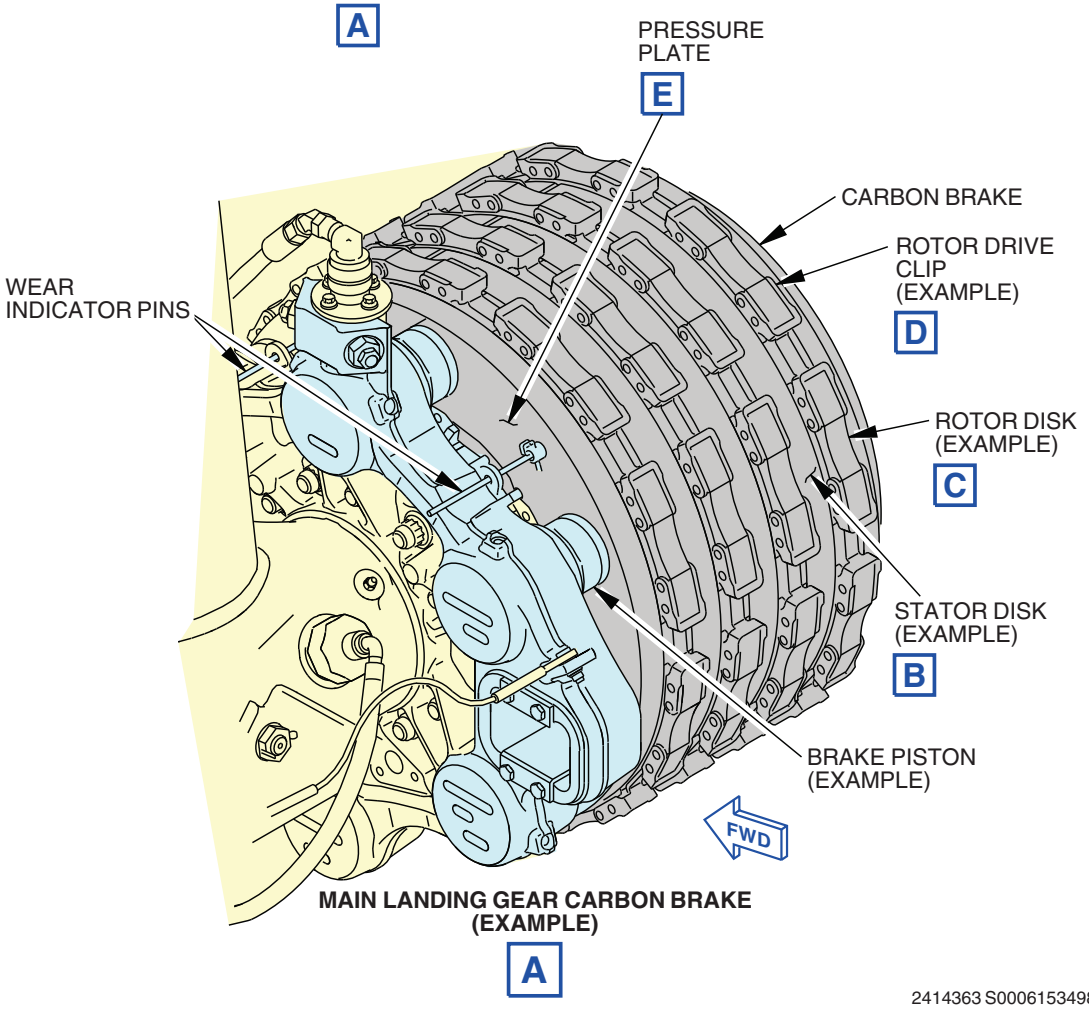
SUBTASK 32-41-41-960-001

(7) If the degree of the conditions that are seen on the brake during the above inspections are significant enough then replacement of the brake is necessary.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

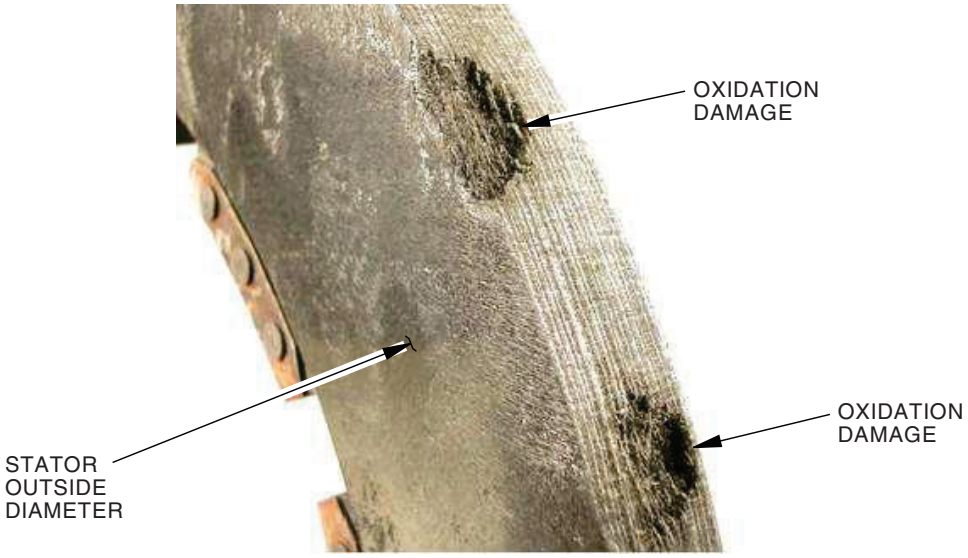




2414363 S00061534989_V6

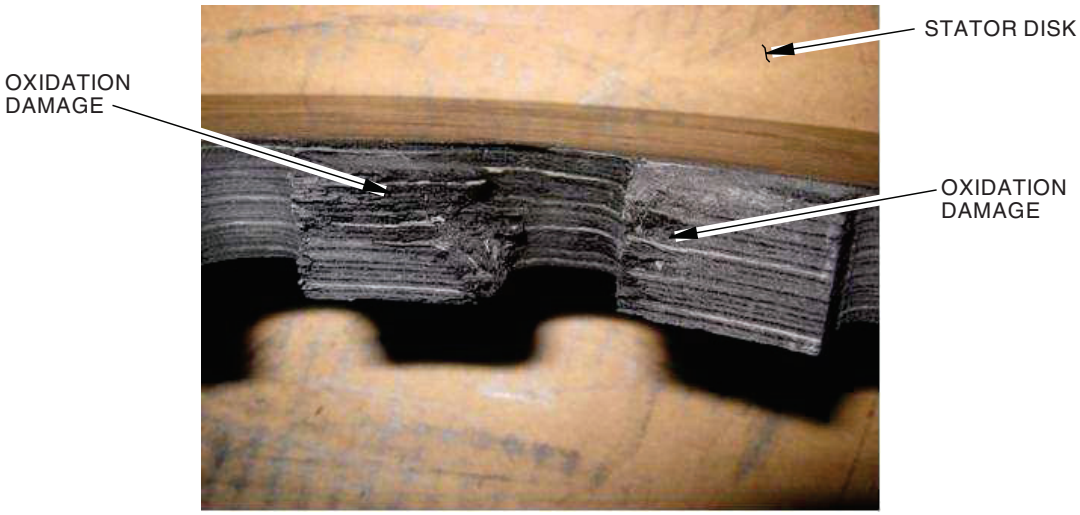
Main Landing Gear Carbon Brakes Inspection (Wheel Removed)
Figure 602/32-41-41-990-804 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



STATOR DISK
(OXIDATION DAMAGE ON OUTSIDE DIAMETER)
(EXAMPLE)

B



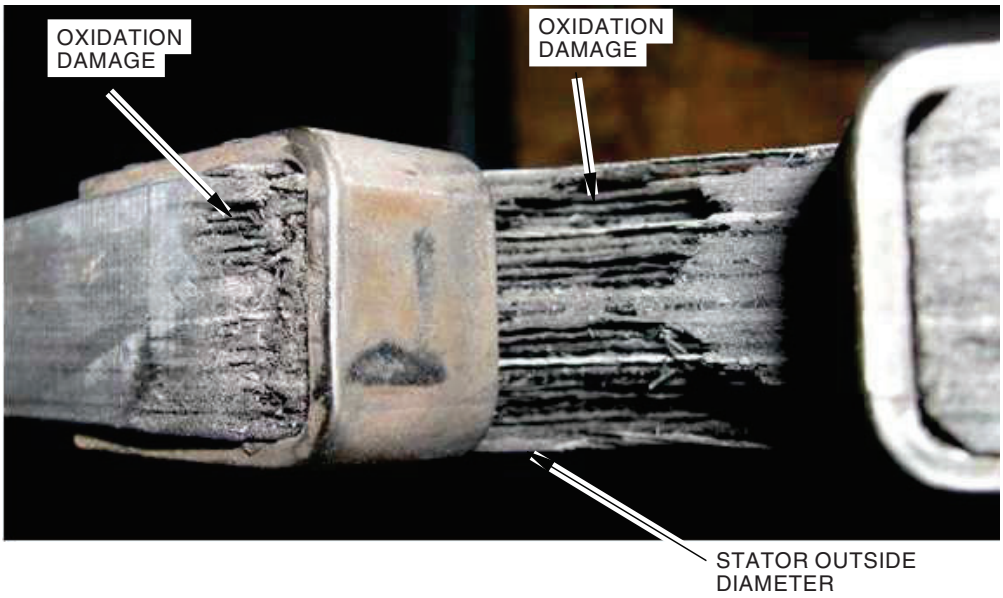
STATOR DRIVE LUGS
(OXIDATION DAMAGE ON INSIDE DIAMETER)
(EXAMPLE)

B

2414364 S00061534990_V1

Main Landing Gear Carbon Brakes Inspection (Wheel Removed)
Figure 602/32-41-41-990-804 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ROTOR DISK
(OXIDATION DAMAGE ON OUTSIDE DIAMETER)
(EXAMPLE)

C



ROTOR DRIVE CLIP
(OXIDATION DAMAGE ON OUTSIDE DIAMETER)
(EXAMPLE)

D

2414365 S00061534991_V2

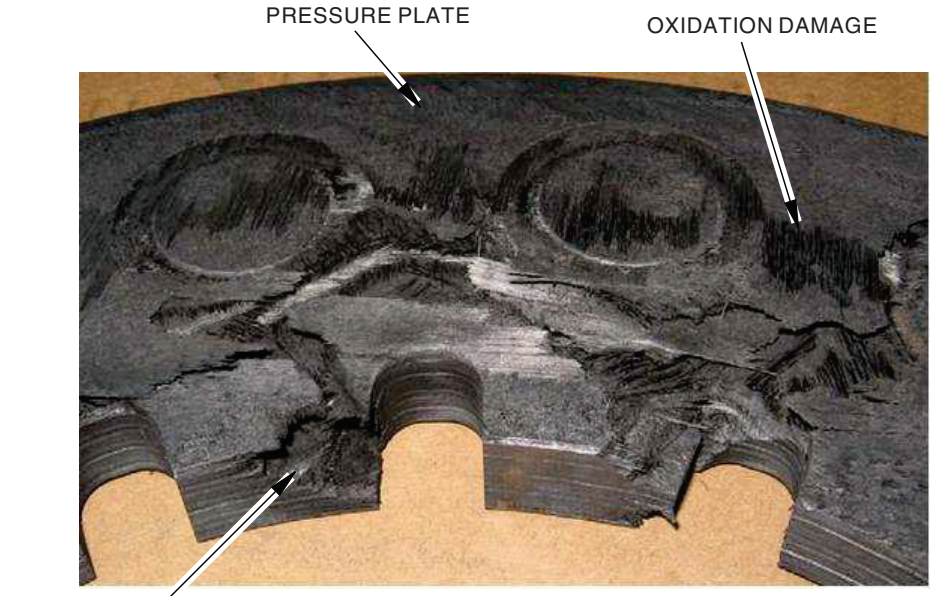
Main Landing Gear Carbon Brakes Inspection (Wheel Removed)
Figure 602/32-41-41-990-804 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ROTOR DRIVE CLIP
(ROTOR CLIP BROKEN, MISSING, OR LOOSE)
(EXAMPLE)

D



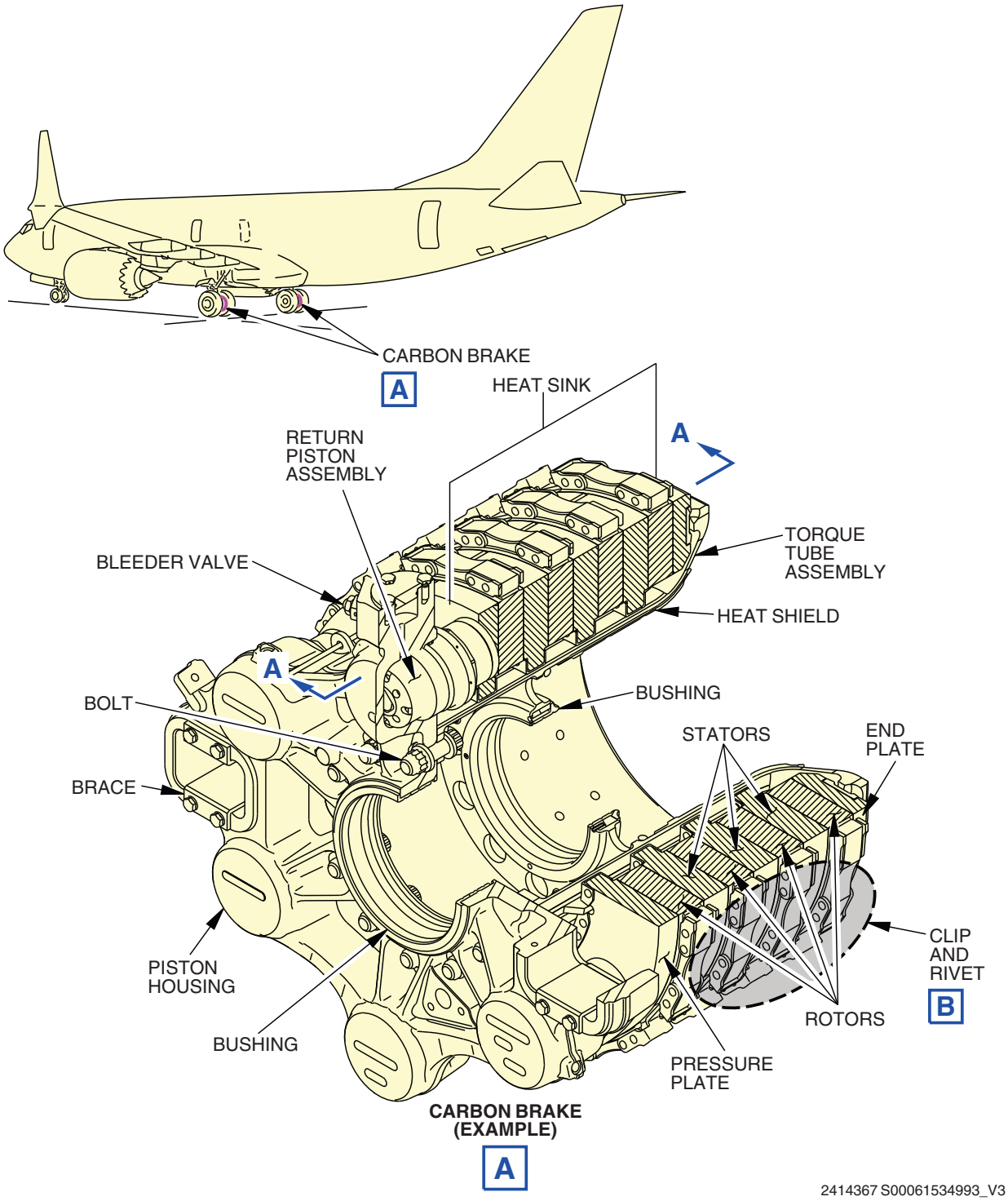
PRESSURE PLATE
(OXIDATION DAMAGE ON THRUST FACE)
(EXAMPLE)

E

2414366 S00061534992_V2

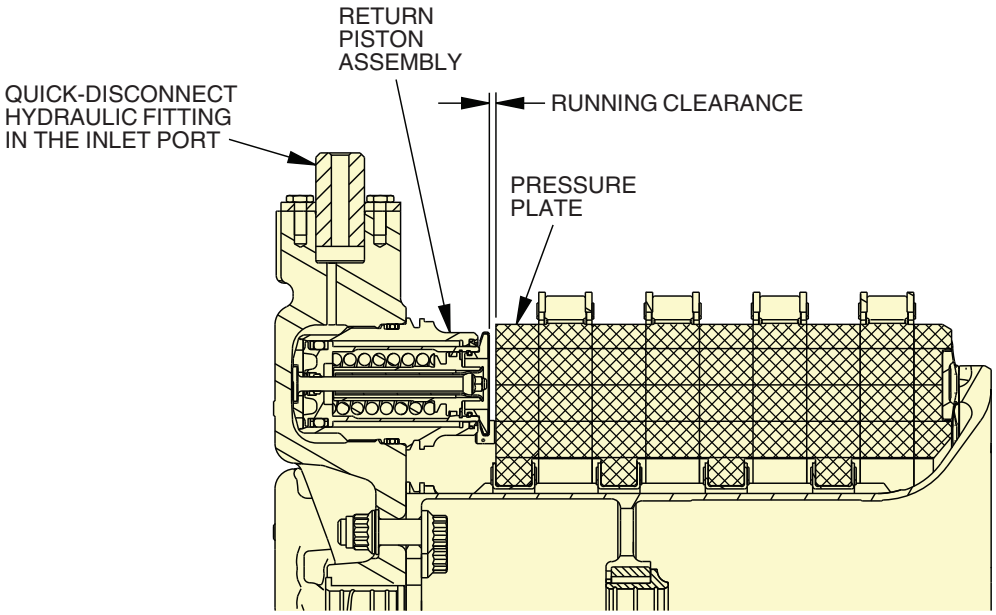
Main Landing Gear Carbon Brakes Inspection (Wheel Removed)
Figure 602/32-41-41-990-804 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



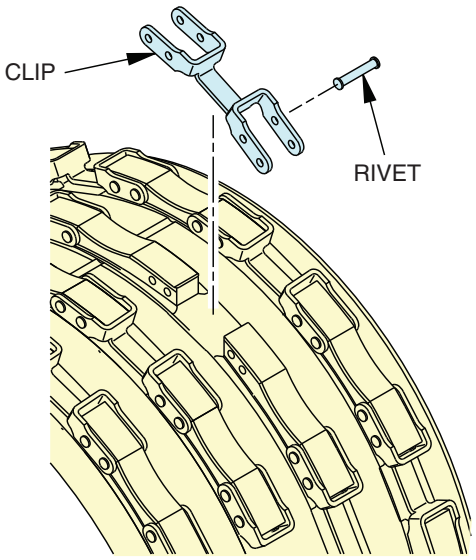
Carbon Brake
Figure 603/32-41-41-990-809 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



RUNNING CLEARANCE (EXAMPLE)

A-A



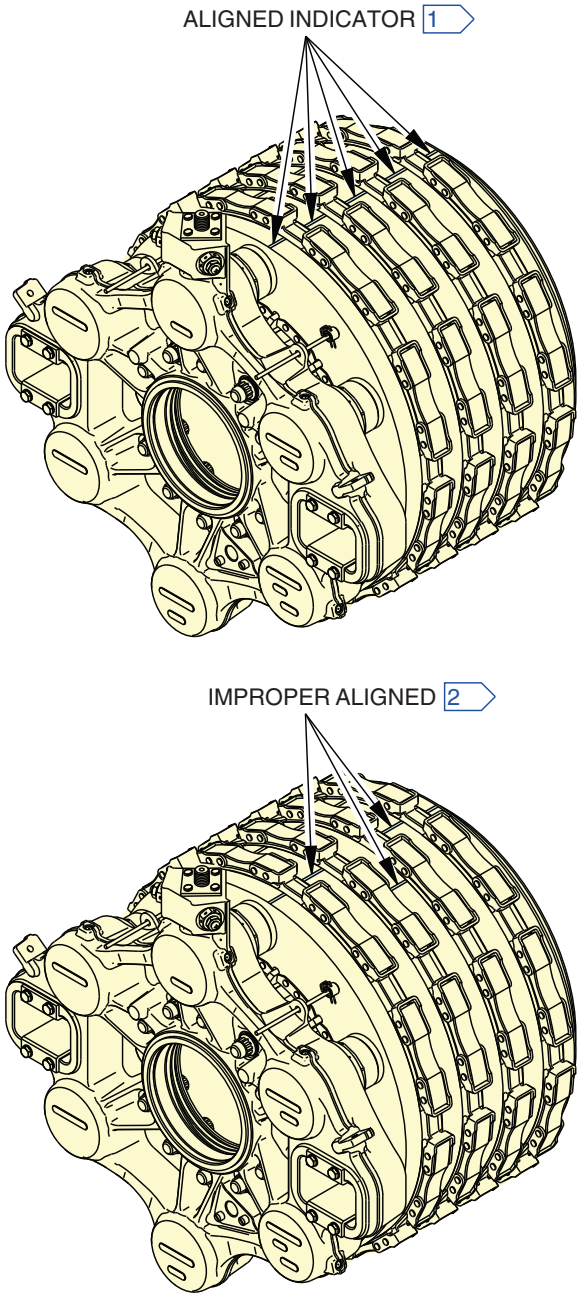
CLIP AND RIVET (EXAMPLE)

B

3069312 S0000827977_V2

Carbon Brake
Figure 603/32-41-41-990-809 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 STATOR GROOVES THAT ARE PROPERLY ALIGNED INDICATE A GOOD STATOR LUG.
2 STATOR GROOVES THAT DO NOT ALIGN INDICATE A DAMAGED STATOR LUG.

3069348 S0000827978_V1

Stator Indicator
Figure 604/32-41-41-990-810

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN LANDING GEAR BRAKE SLEEVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake sleeve for the main landing gear

(2) An installation of the brake sleeve for the main landing gear.

TASK 32-41-42-000-801

2. Main Landing Gear Brake Sleeve Removal

(Figure 401)

A. References

Reference	Title
07-10-01-580-801	Lift the Airplane on Jacks (P/B 201)
32-41-41-000-801	Main Landing Gear Brake Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1864	Puller Equipment - Axle Sleeve Assembly, MLG Part #: C32032-125 Supplier: 81205 Opt Part #: C32032-98 Supplier: 81205

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Removal

SUBTASK 32-41-42-580-001

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Do this task: Lift the Airplane on Jacks, TASK 07-10-01-580-801.

E. Main Landing Gear Brake Sleeve Removal

SUBTASK 32-41-42-020-001

(1) Do this task: Main Landing Gear Brake Removal, TASK 32-41-41-000-801.

NOTE: This procedure will also remove the wheel and tire assemblies.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-42-020-002

CAUTION

DO NOT APPLY TOO MUCH FORCE WHEN YOU REMOVE THE BRAKE SLEEVE. IF YOU USE TOO MUCH FORCE, DAMAGE TO THE AIRPLANE AND THE SUPPORT EQUIPMENT CAN OCCUR.

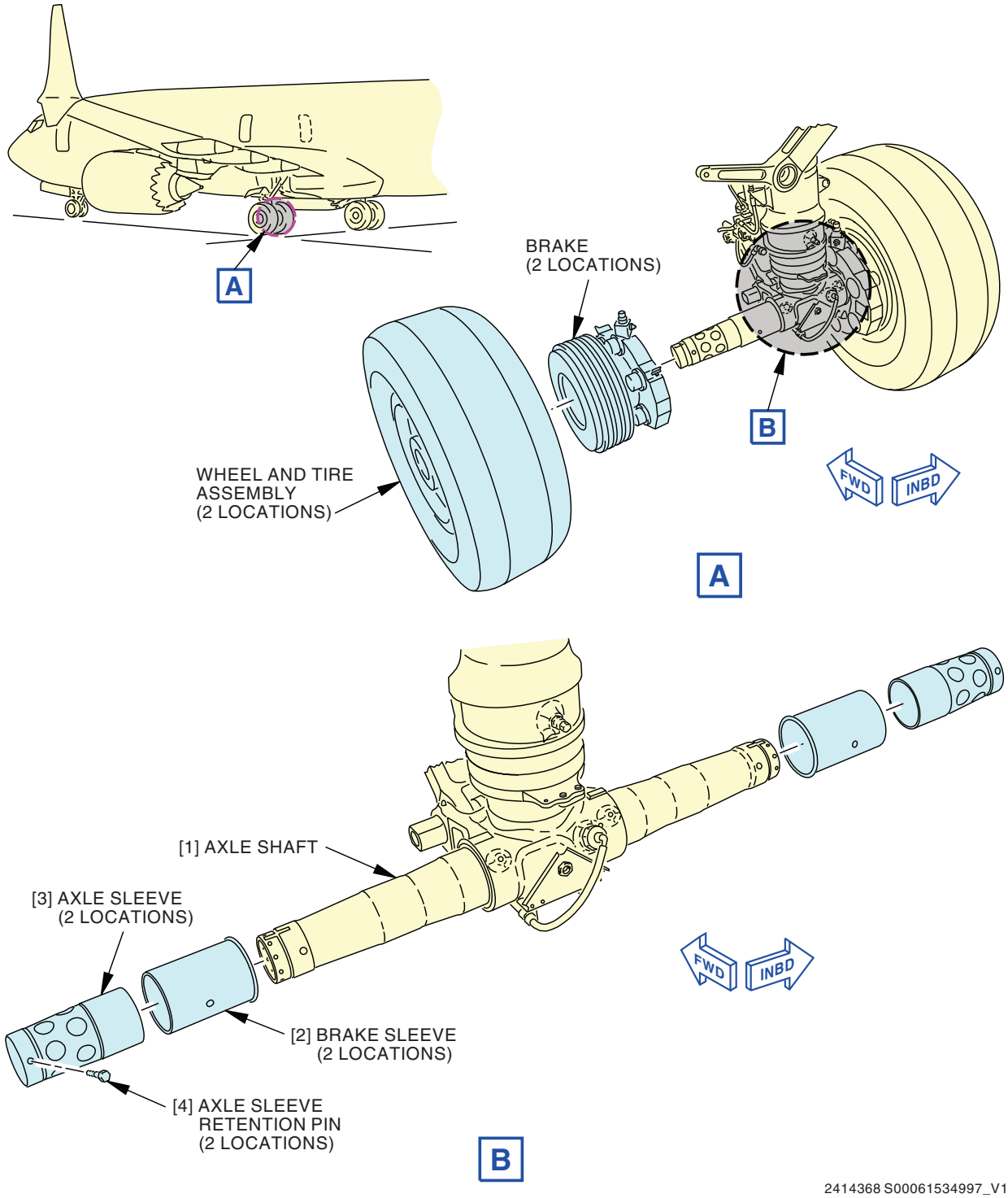
(2) Do these steps to remove the brake sleeve [2] from the axle shaft [1]:

(a) Use puller equipment, SPL-1864, to remove the brake sleeve [2].

NOTE: Applied heat can decrease the force necessary for the removal.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Brake Sleeve Installation
Figure 401/32-41-42-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-42-420-801

3. Main Landing Gear Brake Sleeve Installation

(Figure 401)

A. References

Reference	Title
07-10-01-580-802	Lower the Airplane off Jacks (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-41-41-400-801	Main Landing Gear Brake Installation (P/B 401)

B. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

C. Location Zones

Zone	Area
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Installation

SUBTASK 32-41-42-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on all the landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

E. Main Landing Gear Brake Sleeve Installation

SUBTASK 32-41-42-420-001

(1) Do the steps that follow to install the brake sleeve [2]:

(a) Apply a light coating of grease, D00633, to the lands of the axle shaft [1] and axle sleeve [3], where the brake sleeve [2] inside diameters set.

(b) Heat the brake sleeve [2] to 375 ±25°F (191 ±14°C).

(c) Install the brake sleeve [2] on the axle shaft [1] and the axle sleeve [3].

NOTE: The brake sleeve [2] will rapidly shrink to normal size and seize the axle, which prevents additional movement.

SUBTASK 32-41-42-420-002

(2) Do this task: Main Landing Gear Brake Installation, TASK 32-41-41-400-801.

NOTE: This procedure will also install the wheel and tire assemblies.

NOTE: Make sure you connect the correct wires to the correct in-axle assembly.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-42-580-002

CAUTION MAKE SURE THAT YOU RELEASE THE PARKING BRAKE BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT OBEY, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION	MAKE SURE THAT YOU REMOVE THE WHEEL CHOCKS BEFORE YOU LIFT OR LOWER THE AIRPLANE WITH JACKS. IF YOU DO NOT REMOVE BLOCKAGES AROUND THE MAIN LANDING GEAR, DAMAGE TO THE AIRPLANE AND EQUIPMENT CAN OCCUR.

(1) Do this task: Lower the Airplane off Jacks, TASK 07-10-01-580-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HYDRAULIC BRAKE ACCUMULATOR - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the hydraulic brake accumulator

(2) An installation of the hydraulic brake accumulator

(3) A removal of the brake accumulator pressure gage

(4) An installation of the brake accumulator pressure gage.

B. The brake accumulator is above the 194DR aft wing to body fairing panel, that is aft of the right wheel well.

C. The brake accumulator pressure gage is installed aft of the right wheel well.

TASK 32-41-51-000-801

2. Hydraulic Brake Accumulator Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

D. Prepare for the Removal

SUBTASK 32-41-51-480-006

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-51-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-51-860-001

(3) Release the parking brake.

SUBTASK 32-41-51-010-001

(4) To get access to the brake accumulator [1], do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

SUBTASK 32-41-51-860-002

(5) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-51-870-001

(6) Fully push the brake pedals 12 times to remove the hydraulic pressure from the brake accumulator [1].

SUBTASK 32-41-51-870-002

(7) Do these steps to remove the gas pressure from the brake accumulator [1]:

(a) Remove the cap [6] from the charging valve [8].

WARNING DO NOT LET THE VALVE TURN WITH THE SWIVEL NUT. THE PRESSURE IN THE ACCUMULATOR WILL DISENGAGE THE VALVE WITH EXPLOSIVE FORCE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Loosen the swivel nut [7] on the charging valve [8] a maximum of one complete turn in the counterclockwise direction to release the gas pressure from the brake accumulator [1].

E. Brake Accumulator Removal

SUBTASK 32-41-51-020-001

(1) Do these steps to remove the brake accumulator [1]:

(a) Disconnect the hydraulic line [23] from the forward end of the brake accumulator [1].

(b) Install a plug on the hydraulic line [23] and a cap on the hydraulic port of the brake accumulator [1].

(c) Disconnect the gas line [21] from the aft end of the brake accumulator [1].

(d) Install a plug on the gas line [21] and a cap in the gas port of the brake accumulator [1].

(e) Loosen the clamp nuts [22] for the clamps on the brake accumulator [1].

(f) Open the clamps.

(g) Remove the brake accumulator [1] from the airplane.

SUBTASK 32-41-51-020-002

(2) If the brake accumulator [1] that will be installed, does not have the reducer [3] and reducer [5] installed, do these steps:

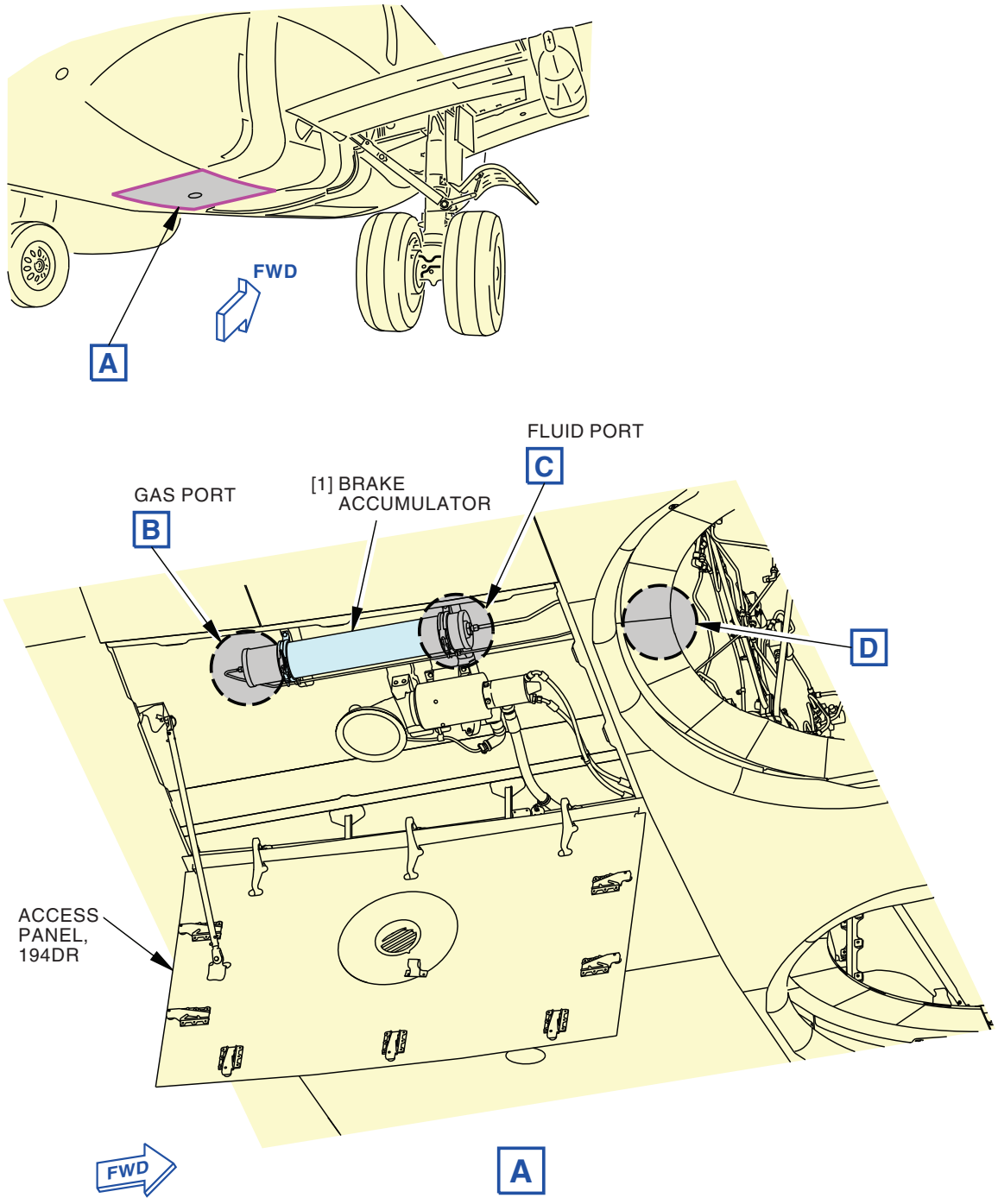
(a) Remove the reducer [3] and reducer [5] from the brake accumulator [1].

(b) Remove and discard the packing [2] and packing [4] from the applicable fitting.

(c) Install plugs in the ports of the brake accumulator [1].

END OF TASK

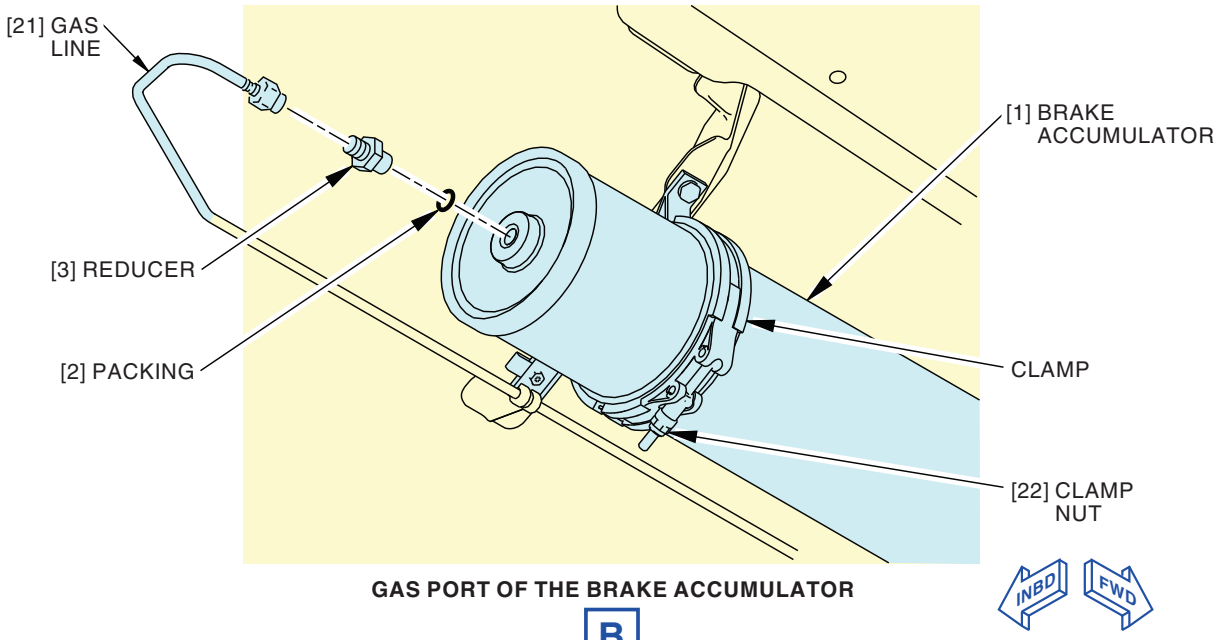
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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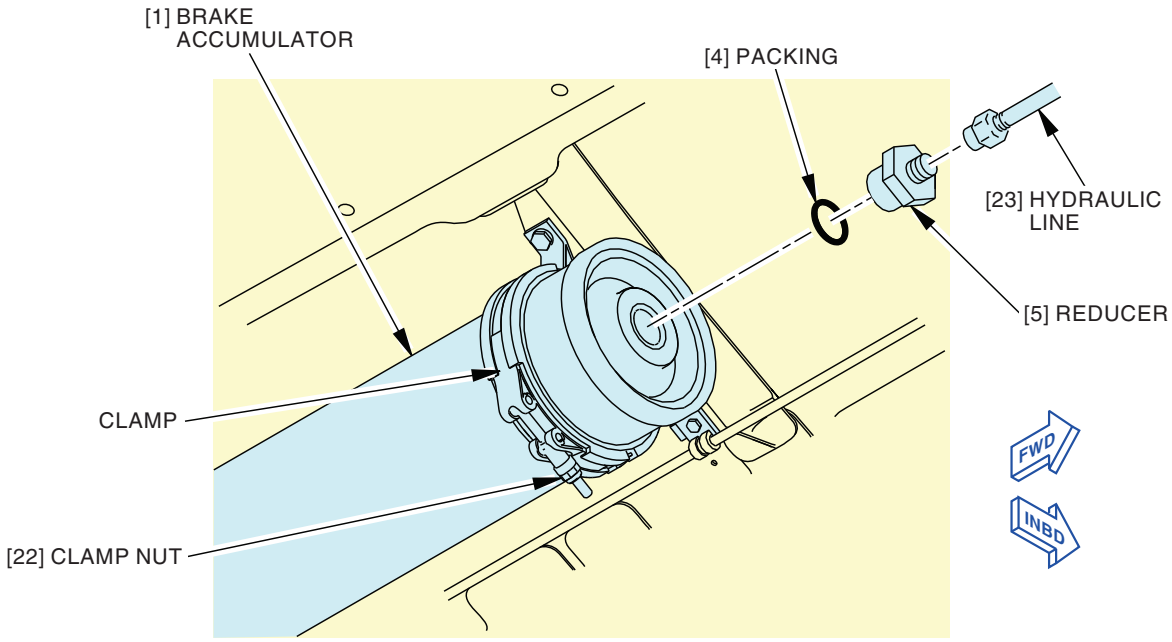
Hydraulic Brake Accumulator Installation
Figure 401/32-41-51-990-801 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



GAS PORT OF THE BRAKE ACCUMULATOR

B



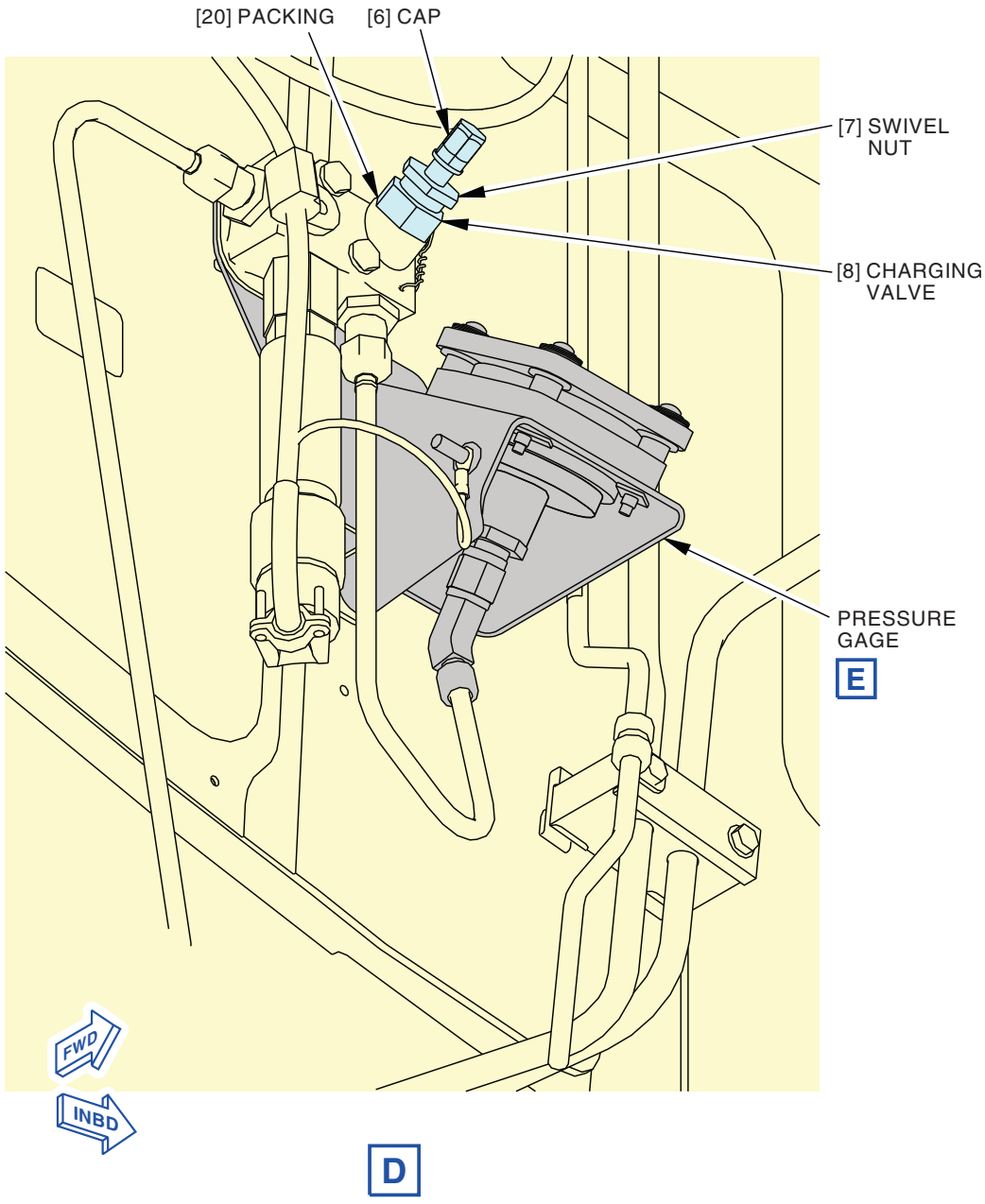
FLUID PORT OF THE BRAKE ACCUMULATOR

C

2414370 S00061535004_V3

Hydraulic Brake Accumulator Installation
Figure 401/32-41-51-990-801 (Sheet 2 of 4)

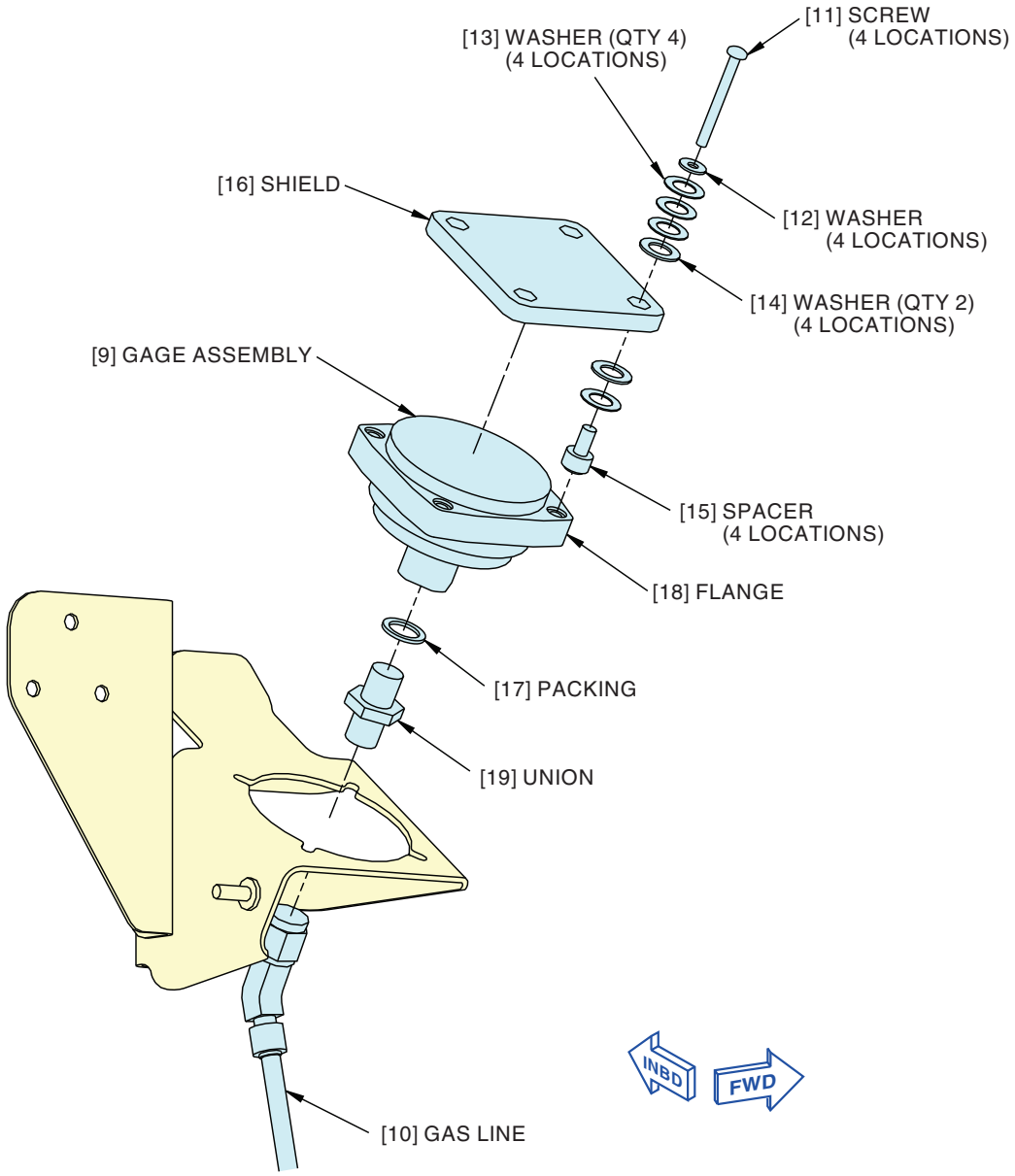
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414371 S00061535005_V2

Hydraulic Brake Accumulator Installation
Figure 401/32-41-51-990-801 (Sheet 3 of 4)

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BRAKE ACCUMULATOR PRESSURE GAGE

E

2414372 S00061535006_V1

Hydraulic Brake Accumulator Installation
Figure 401/32-41-51-990-801 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-51-420-801

3. Hydraulic Brake Accumulator Installation

(Figure 401)

A. References

Reference	Title
12-15-11-420-801	Hydraulic Brake Accumulator Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50356	Lockwire - MS20995C47, Corrosion Resistant Steel - 0.047 Inch (1.1938 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Brake accumulator	32-41-51-01-045	PTF ALL
2	Packing	32-41-51-01-035	PTF ALL
4	Packing	32-41-51-01-040	PTF ALL
8	Charging valve	32-41-53-01-050	PTF ALL
20	Packing	32-41-53-01-052	PTF ALL

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Access Panels

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

F. Brake Accumulator Installation

SUBTASK 32-41-51-020-003

(1) Remove the caps or plugs from the ports of the brake accumulator [1].

SUBTASK 32-41-51-410-001

(2) If the reducer [3] and reducer [5] are not installed in the brake accumulator [1], do these steps:

(a) Lubricate the new packing [2], new packing [4], reducer [3], and reducer [5] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) For the GAS end of the brake accumulator [1], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Install the packing [2] on the reducer [3].
2) Install the reducer [3] in the GAS PORT of the brake accumulator [1].
3) Tighten the reducer [3] to 190 ±10 in-lb (21 ±1 N·m).

(c) For the HYDRAULIC end of the brake accumulator [1], do these steps:
1) Install the packing [4] on the reducer [5].
2) Install the reducer [5] in the FLUID PORT of the brake accumulator [1].
3) Tighten the reducer [5] to 900 ±45 in-lb (102 ±5 N·m).

SUBTASK 32-41-51-420-001

(3) To install the brake accumulator [1], do these steps:
(a) Align the gas line with the reducer [3] on the brake accumulator [1] when you put the brake accumulator [1] in its position in the clamps.
(b) Tighten the clamp nuts.
(c) Connect the gas line from the brake accumulator charging valve [8] to the GAS PORT of the brake accumulator [1].
(d) Connect the hydraulic line to the FLUID PORT of the brake accumulator [1].
(e) Tighten the B-nut on the hydraulic line to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-41-51-420-002

(4) To put the brake accumulator [1] back to its usual condition, do these steps:
(a) Do this task: Hydraulic Brake Accumulator Servicing, TASK 12-15-11-420-801.
1) Make sure that the cap [6] is installed on the charging valve [8].
2) Make sure that MS20995C47 lockwire, G50356, is installed on the charging valve [8] (TASK 20-10-44-400-801).

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(b) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.
(c) Make sure that the gages for normal (System B) hydraulic system and brake accumulator [1] show approximately 3000 psig.
(d) Make sure that there is no leak at the connection for the hydraulic line on the brake accumulator [1].
(e) Apply a soap solution to the connection at the charging line.
1) Examine the connection for air bubbles.
(f) If leakage is detected at the charging valve [8], replace the charging valve [8], do these steps:
1) Make sure that the hydraulic reservoirs have been depressurized before removing the charging valve [8] (TASK 29-09-00-860-802).
2) Remove the cap [6] on the charging valve [8].
3) Slowly unscrew the swivel nut [7] to release any residual air pressure.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

4) Remove MS20995C47 lockwire, G50356, from the charging valve [8] and manifold assembly.

5) Remove the charging valve [8] from the manifold assembly.

6) Remove and discard the packing [20] from the charging valve [8].

7) Lubricate the new packing [20] with MCS 352B fluid, D00054, or hydraulic fluid, D00153.

8) Install the packing [20] onto the charging valve [8].

9) Install the charging valve [8] in the manifold assembly port.

a) Tighten the charging valve [8] to 100 in-lb (11.3 N·m) - 110 in-lb (12.4 N·m).

10) Install a new MS20995C47 lockwire, G50356, to the charging valve [8] and manifold assembly.

11) Install the cap [6] on the charging valve [8].

12) Tighten the swivel nut [7] on the charging valve [8].

13) Pressurize the pneumatic crossover manifold duct.

14) Use a solution of soap and water to make sure that no air leaks are at the charging valve [8].

15) Release pressure from the pneumatic cross manifold duct.

(g) Bleed the normal (System B) hydraulic system, do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

G. Brake Accumulator Installation Test

SUBTASK 32-41-51-420-003

(1) Do these steps to do an operational check of the brake accumulator [1]:

(a) Depressurize the normal (System B) hydraulic system (TASK 29-09-00-860-802).

(b) Push and release the brake pedals fully a minimum of seven times.

(c) Monitor the movement of the brake-wear indicators on each brake to make sure that the brakes work correctly.

SUBTASK 32-41-51-010-002

(2) Close this access panel:

Number	Name/Location
194DR	Aft Wing To Body Fairing Panel

SUBTASK 32-41-51-860-003

(3) Set the parking brake.

END OF TASK

TASK 32-41-51-000-802

4. Brake Accumulator Pressure Gage Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

B. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Prepare for the Removal

SUBTASK 32-41-51-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-51-480-005

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-51-860-004

(3) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-51-860-005

(4) Release the parking brake.

SUBTASK 32-41-51-860-006

(5) Fully push the captain's brake pedals 12 times to remove the hydraulic pressure from the brake accumulator [1].

SUBTASK 32-41-51-840-001

(6) Do these steps to remove the gas pressure from the brake accumulator [1]:

(a) Remove the cap [6] from the charging valve [8] for the brake accumulator [1].

WARNING DO NOT LET THE VALVE TURN WITH THE SWIVEL NUT. THE PRESSURE IN THE ACCUMULATOR WILL DISENGAGE THE VALVE WITH EXPLOSIVE FORCE. INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

(b) Loosen the swivel nut [7] on the charging valve [8] a maximum of one complete turn in the counterclockwise direction to release the gas pressure from the brake accumulator [1].

(c) Make sure that the gage assembly [9] for the brake accumulator [1] goes to zero.

SUBTASK 32-41-51-020-004

(7) Do these steps to remove the gage assembly [9]:

(a) Disconnect the gas line [10] from the gage assembly [9].

(b) Install a plug on the gas line [10].

(c) Remove the screws [11], washers [12], washers [13], and washers [14] from the airplane.

(d) Remove the shield [16] from the airplane.

(e) Remove the washers [13], washers [14], and spacers [15].

(f) Remove the gage assembly [9] from the airplane.

1) Remove the flange [18] from the gage assembly [9].

(g) Put a cap on the union [19] on the gage.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-51-020-005

(8) If the replacement gage assembly [9] does not have a union [19] installed, do these steps:

(a) Remove the union [19] and packing [17] from the port of the gage.

(b) Discard the packing [17].

——— END OF TASK ———

TASK 32-41-51-420-802

5. Brake Accumulator Pressure Gage Installation

(Figure 401)

A. References

Reference	Title
12-15-11-420-801	Hydraulic Brake Accumulator Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50356	Lockwire - MS20995C47, Corrosion Resistant Steel - 0.047 Inch (1.1938 mm) Diameter	NASM20995

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
9	Gage assembly	32-41-51-07-035	PTF ALL
17	Packing	32-41-51-07-038	PTF ALL

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Brake Accumulator Pressure Gage Installation

SUBTASK 32-41-51-080-001

(1) Remove the cap or plug from the port of the gage assembly [9] and gas line [10].

SUBTASK 32-41-51-420-004

(2) If the replacement gage assembly [9] does not have a union [19] installed in the port, do these steps:

(a) Lightly lubricate the new packing [17], threads of gage assembly [9], and threads of union [19] with hydraulic fluid, D00153, or MCS 352B fluid, D00054 (alternative).

(b) Install the packing [17] on the union [19].

(c) Install the union [19] on the port of the gage assembly [9].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-51-420-005

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Do these step to install the gage assembly [9]:

(a) Install the flange [18] on the gage assembly [9].

(b) Install the gage assembly [9] on the bracket in the right wheel well.

(c) Apply a thin layer of corrosion inhibiting material, G50136, to the faces of the washers [13], washers [14], and spacers [15].

1) Remove the excess corrosion inhibiting material, G50136.

(d) Install the spacers [15], washers [14], washers [13] on the flange [18].

NOTE: The rubber washers [14] should always be against the shield [16].

(e) Install the shield [16] on the spacers [15].

(f) Apply a thin layer of corrosion inhibiting material, G50136, to the thread reliefs and threads of the screws [11], and faces of the washers [12], washers [13], and washers [14].

1) Remove the excess corrosion inhibiting material, G50136.

(g) Install the washers [14], washers [13], washers [12], and screws [11].

1) Install washers [13] until a maximum of one washer [13] is above the spacer [15] to prevent movement of the shield [16].

2) Tighten, but do not damage, the washers [14].

(h) Connect the gas line [10] to the union [19] on the gage assembly [9].

F. Brake Accumulator Pressure Gage Installation Test

SUBTASK 32-41-51-710-001

(1) For each pressure reading, compare the pressure measured by the brake accumulator pressure gage in the wheel well with the pressure measured by the brake accumulator pressure gage in the flight compartment.

(a) Make sure that the compared pressure values are approximately the same.

SUBTASK 32-41-51-790-001

(2) Do these steps to do an operational test of the gage assembly [9]:

(a) Do this task: Hydraulic Brake Accumulator Servicing, TASK 12-15-11-420-801.

1) Make sure that the cap [6] is installed on the charging valve [8].

2) Make sure that MS20995C47 lockwire, G50356, is installed on the charging valve [8] (TASK 20-10-44-400-801).

(b) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(c) Make sure that the gages for the normal (System B) hydraulic system and brake accumulator show approximately 3000 psig (20,684 kPa).

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(d) Make sure that there is no leak at the connection for the gas line [10] on the gage assembly [9].

(e) Apply a soap solution to the connection at the union [19].

1) Examine the connection for air bubbles.

(f) Bleed the normal (System B) hydraulic system, do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE ACCUMULATOR PRESSURE TRANSMITTER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake accumulator pressure transmitter

(2) An installation of the brake accumulator pressure transmitter.

B. In this procedure the brake accumulator pressure transmitter will be referred to as the pressure transmitter.

TASK 32-41-53-000-801

2. Brake Accumulator Pressure Transmitter Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the pressure transmitter.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 32-41-53-480-003

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-53-860-001

(2) Make sure that the hydraulic power is off (TASK 29-11-00-860-805).

SUBTASK 32-41-53-480-002

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-53-860-002

(4) Release the parking brake.

SUBTASK 32-41-53-860-003

(5) Fully push the brake pedals twelve times to remove the hydraulic pressure from the brake accumulator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-53-860-006

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	13	C00306	HYDRAULIC SYSTEM BRAKE PRESS IND

E. Brake Accumulator Pressure Transmitter Removal

SUBTASK 32-41-53-840-001

(1) Do these steps to remove the gas pressure from the brake accumulator:

(a) Remove the cap [4] from the brake accumulator charging valve [6].

WARNING

DO NOT LOOSEN THE BODY OF THE BRAKE ACCUMULATOR CHARGING VALVE. THE PRESSURE IN THE BRAKE ACCUMULATOR CAN QUICKLY PUSH THE GAS CHARGING VALVE OFF THE MANIFOLD. THIS CAN CAUSE INJURY TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Turn the swivel nut [5] of the brake accumulator charging valve [6] one turn counterclockwise.

NOTE: This will release the gas pressure from the brake accumulator.

(c) Make sure that the brake accumulator pressure gage [7] goes to zero.

SUBTASK 32-41-53-020-001

(2) Do these steps to remove the pressure transmitter [1]:

(a) Disconnect the electrical connector [8] from the pressure transmitter [1].

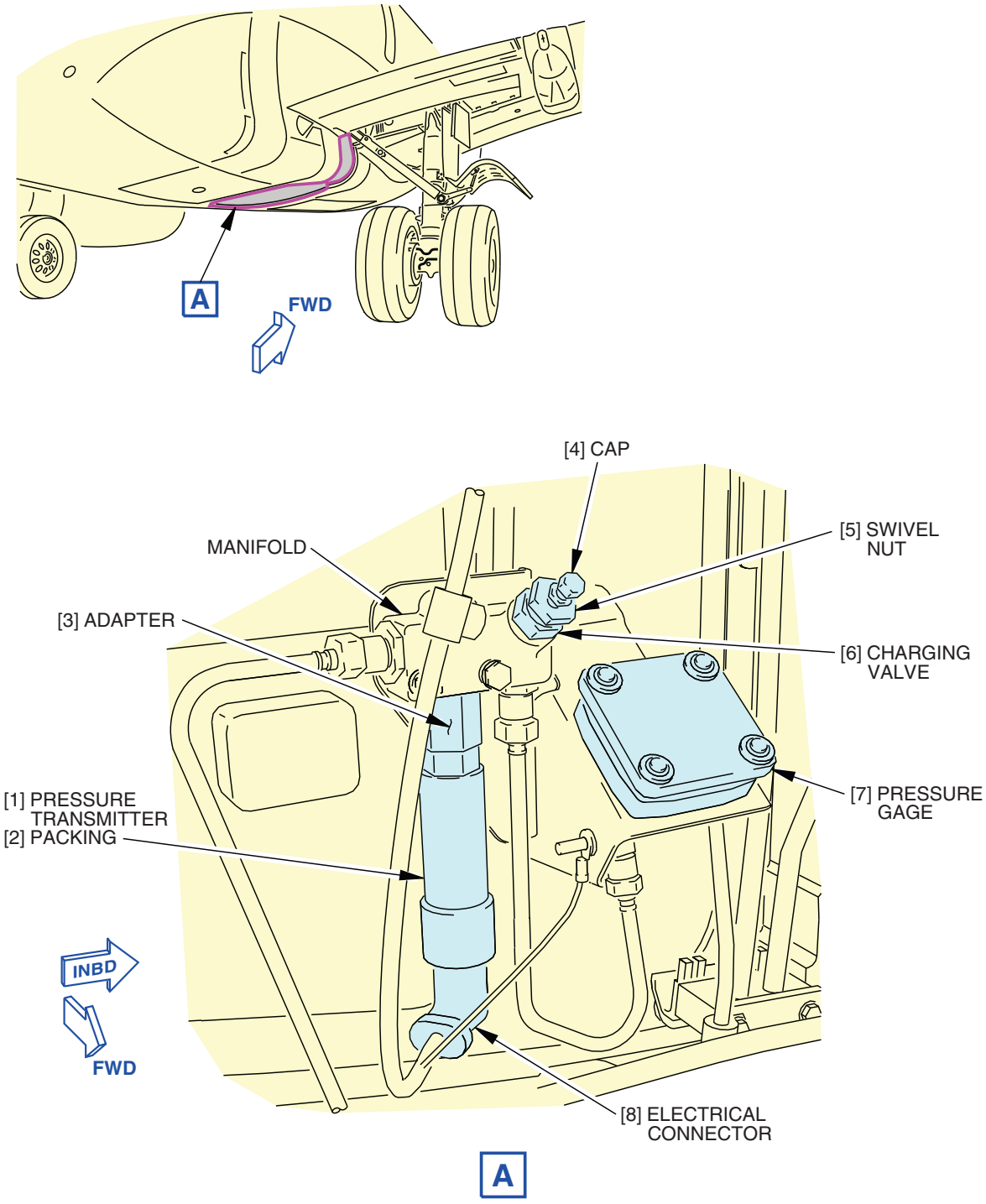
(b) Remove the pressure transmitter [1] from the adapter [3].

(c) Remove the packing [2] from the pressure transmitter [1].

1) Discard the packing [2].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hydraulic Brake Accumulator Pressure Transmitter Installation
Figure 401/32-41-53-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-53-400-801

3. Brake Accumulator Pressure Transmitter Installation

(Figure 401)

A. References

Reference	Title
12-15-11-420-801	Hydraulic Brake Accumulator Servicing (P/B 301)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Pressure transmitter	32-41-53-01-075	PTF ALL
2	Packing	32-41-53-01-070	PTF ALL

D. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Brake Accumulator Pressure Transmitter Installation

SUBTASK 32-41-53-420-001

(1) Do these steps to install the pressure transmitter [1]:

(a) Lubricate a new packing [2] with hydraulic fluid, D00153.

(b) Install the packing [2] on the pressure transmitter [1].

(c) Install the pressure transmitter [1] in its location on the accumulator manifold [10].

(d) Before you connect the electrical connector [8] to the pressure transmitter [1], examine the electrical connector [8] for corrosion.

WARNING

DO THE STEPS BELOW WHEN YOU OPERATE THE AIRPLANE WHERE YOU USE DEICING FLUID (THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE). ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT. IF YOU DO NOT DO THESE STEPS, YOU CAN CAUSE DAMAGE TO EQUIPMENT OR INJURY TO PERSONNEL.

(e) If there was corrosion, correct the problem (SWPM 20-60-03).

(f) Apply D5026NS or ZC-026 compound, G50171, to the connector (SWPM 20-60-03).

(g) Connect the electrical connector [8] to the pressure transmitter [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-53-610-001

(2) Do this task: Hydraulic Brake Accumulator Servicing, TASK 12-15-11-420-801.

SUBTASK 32-41-53-410-001

(3) Make sure that the cap [4] is installed after you service the brake accumulator.

F. Brake Accumulator Pressure Transmitter Installation Test

SUBTASK 32-41-53-860-007

(1) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	13	C00306	HYDRAULIC SYSTEM BRAKE PRESS IND

SUBTASK 32-41-53-860-004

(2) For the B hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-53-720-001

(3) Measure the brake accumulator pressure with the direct reading gage (pressure gage [7]).

NOTE: The direct reading gage (pressure gage) is in the wheel well for the right main landing gear, next to the pressure transmitter.

SUBTASK 32-41-53-720-002

(4) Measure the brake accumulator pressure with the BRAKE ACCUM PRESS gage.

NOTE: The BRAKE ACCUM PRESS gage is in the flight compartment on the P3 first officer's instrument panel.

SUBTASK 32-41-53-710-001

(5) Make sure that the pressure shown by the BRAKE ACCUM PRESS gage is approximately the same as the pressure shown by the direct reading pressure gage (pressure gage [7]).

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-53-860-005

(1) For the B hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE ACCUMULATOR PRESSURE GAGE (FLIGHT COMPARTMENT) - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake accumulator pressure gage (flight compartment).

(2) An installation of the brake accumulator pressure gage (flight compartment).

TASK 32-41-54-000-801

2. Brake Accumulator Pressure Gage (Flight Compartment) Removal

(Figure 401)

A. Location Zones

Zone	Area
212	Flight Compartment - Right

B. Prepare for the Removal

SUBTASK 32-41-54-860-001

(1) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	13	C00306	HYDRAULIC SYSTEM BRAKE PRESS IND

C. Brake Accumulator Pressure Gage Removal

SUBTASK 32-41-54-020-001

(1) Loosen the quick release fasteners [5] that hold the P9-82 panel [4] to the airplane.

SUBTASK 32-41-54-020-002

(2) Pull the P9-82 panel [4] away from the right forward panel.

SUBTASK 32-41-54-020-003

(3) Disconnect the electrical connector [7] from the brake accumulator pressure gage [3].

SUBTASK 32-41-54-020-004

(4) Loosen the clamp [1] that holds the brake accumulator pressure gage [3]:

NOTE: The clamp has two fasteners: the mounting screw that holds the clamp to the P9-82 panel, and the adjusting screw that loosens the clamp.

(a) Loosen the adjusting screw [2] that holds the clamp [1] to the brake accumulator pressure gage [3] until the brake accumulator pressure gage [3] can be removed from the P9-82 panel [4].

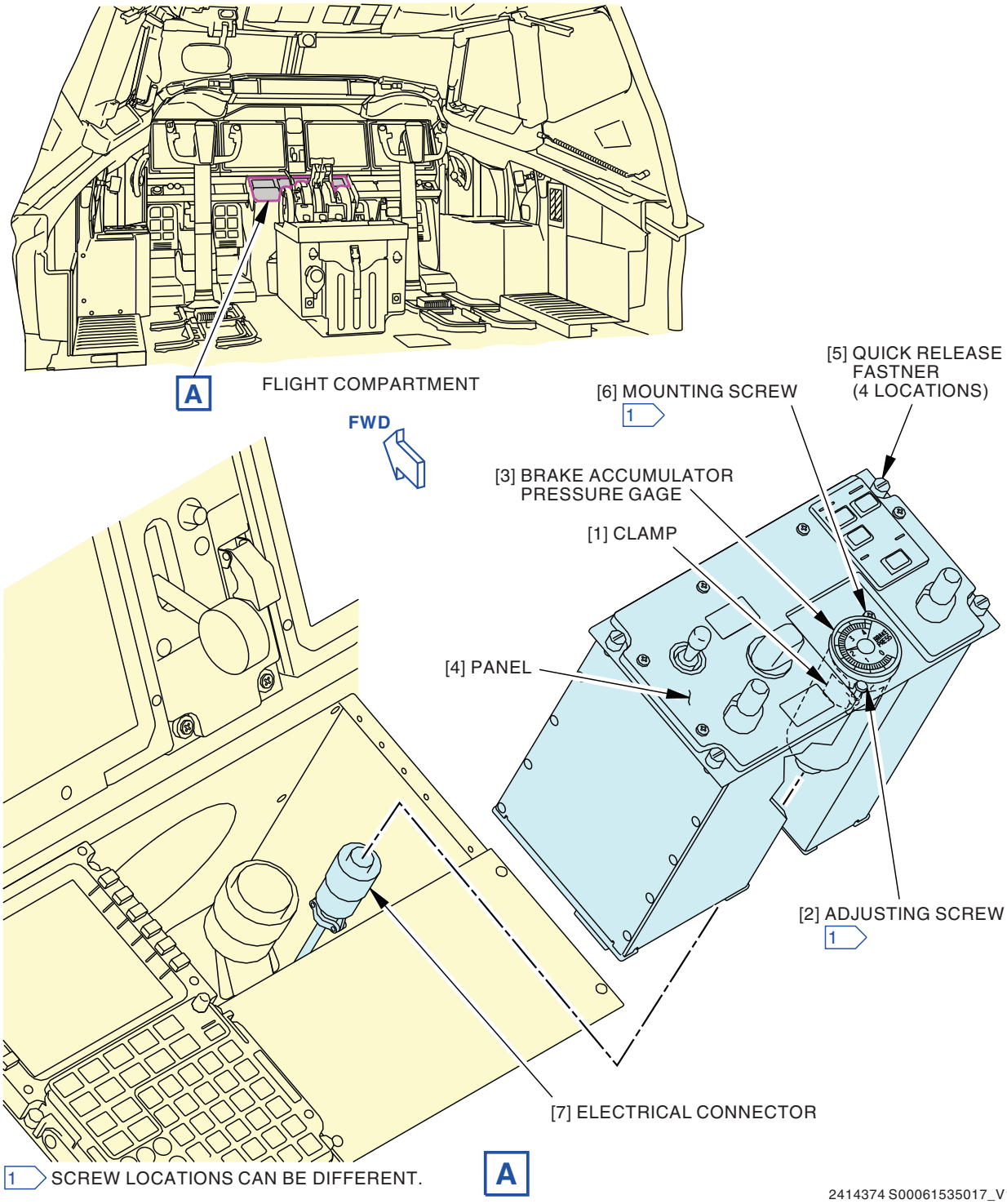
1) Do not loosen the mounting screw [6] that holds the clamp [1] to the P9-82 panel [4].

SUBTASK 32-41-54-020-005

(5) Remove the brake accumulator pressure gage [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Accumulator Pressure Gage Installation
Figure 401/32-41-54-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-54-400-801

3. Brake Accumulator Pressure Gage (Flight Compartment) Installation

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
24-22-00-860-801	Supply Electrical Power (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Consumable Materials

Reference	Description	Specification
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Brake accumulator pressure gage	32-41-54-01-020	PTF ALL

D. Location Zones

Zone	Area
212	Flight Compartment - Right

E. Brake Accumulator Pressure Gage Installation

SUBTASK 32-41-54-420-001

(1) Put the brake accumulator pressure gage [3] in its location in the P9-82 panel [4].

SUBTASK 32-41-54-020-006

(2) Tighten the adjusting screw [2] on the clamp [1] that holds the brake accumulator pressure gage [3]:

(a) Tighten the mounting screw [6] on the clamp [1] until the brake accumulator pressure gage [3] is firmly held to the P9-82 panel [4].

SUBTASK 32-41-54-420-002

(3) Before you connect the electrical connector [7] to the brake accumulator pressure gage [3], examine the electrical connector [7] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there is corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [7] (SWPM 20-60-03).

(4) Connect the electrical connector [7] to the brake accumulator pressure gage [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-54-420-003

(5) Hold the P9-82 panel [4] on the forward panel.

SUBTASK 32-41-54-420-004

(6) Tighten the quick release fasteners [5] that hold the P9-82 panel [4] to the airplane.

SUBTASK 32-41-54-860-002

(7) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
A	13	C00306	HYDRAULIC SYSTEM BRAKE PRESS IND

F. Brake Accumulator Pressure Gage Installation Test

SUBTASK 32-41-54-480-001

(1) Do this task: Supply Electrical Power, TASK 24-22-00-860-801.

SUBTASK 32-41-54-790-001

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the B hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-54-710-001

(3) Make sure that the brake accumulator pressure gage [3] shows approximately 3000 psig (20,684 kPa).

SUBTASK 32-41-54-860-003

(4) For the B hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-54-480-002

(5) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-54-860-004

(6) Release the parking brake.

SUBTASK 32-41-54-860-005

(7) Fully push the captain's brake pedals twelve times to remove the pressure from the brake accumulator.

SUBTASK 32-41-54-710-002

(8) Make sure that the brake accumulator pressure gage [3] (flight compartment) shows approximately 1000 psig (6895 kPa).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE ACCUMULATOR PRESSURE RELIEF VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake accumulator pressure relief valve

(2) An installation of the brake accumulator pressure relief valve

B. The brake system relief valve [1] is on the aft wall of the right wheel well.

TASK 32-41-71-000-801

2. Brake Accumulator Pressure Relief Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 32-41-71-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-71-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-71-860-001

(3) Release the parking brake.

SUBTASK 32-41-71-860-002

(4) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-71-870-001

(5) Fully push the brake pedals twelve times to remove the hydraulic pressure from the brake accumulator (Figure 401).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Brake Accumulator Pressure Relief Valve Removal

SUBTASK 32-41-71-020-001

(1) Do these steps to remove the brake accumulator pressure relief valve (brake system relief valve [1]):

(a) Remove the bolts [4] and the washers [3] and the clamp [2] from the airplane.

(b) Disconnect the outboard hydraulic line from the brake system relief valve [1].

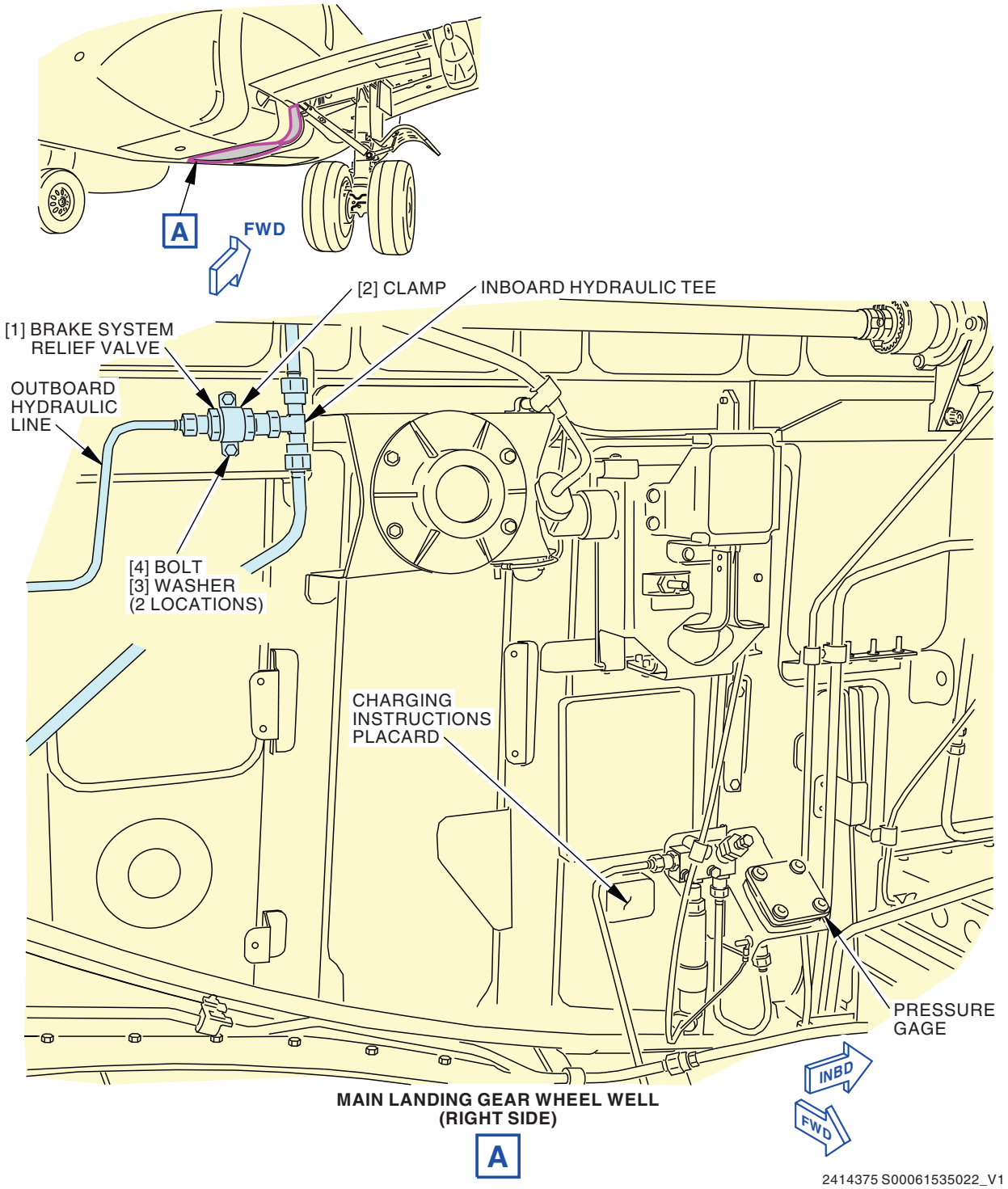
(c) Disconnect the inboard hydraulic tee from the brake system relief valve [1].

(d) Install caps/plugs, STD-14686, on the hydraulic lines.

(e) Remove the brake system relief valve [1] from the airplane.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake System Relief Valve Installation
Figure 401/32-41-71-990-801

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-71-420-801

3. Brake Accumulator Pressure Relief Valve Installation

(Figure 401)

A. References

Reference	Title
12-15-11-420-801	Hydraulic Brake Accumulator Servicing (P/B 301)
20-10-44-400-801	Lockwire, Cotter Pins, and Lockrings - Installation (P/B 401)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
G01505	Lockwire - Safety And Lock	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Brake system relief valve	32-41-71-01-040	PTF ALL

E. Location Zones

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

F. Brake Accumulator Pressure Relief Valve Installation

SUBTASK 32-41-71-420-001

(1) Do these steps to install the brake accumulator pressure relief valve (brake system relief valve [1]):

(a) Remove the caps/plugs, STD-14686, from the hydraulic lines.

(b) Put the brake system relief valve [1] in its position in the hydraulic line.

(c) Connect the inboard hydraulic tee to the brake system relief valve [1].

(d) Connect the outboard hydraulic line to the brake system relief valve [1].

(e) Tighten the B-nut on the hydraulic line to 140 ±7 in-lb (16 ±1 N·m).

(f) Put the clamp [2] in its position on the brake system relief valve [1].

(g) Install the bolts [4] and the washers [3] to connect the clamp [2] to the airplane.

SUBTASK 32-41-71-420-002

(2) Do the steps that follow to make sure the brake accumulator is in its usual condition:

(a) Make sure that the brake accumulator charge pressure is correct.

NOTE: The brake accumulator charge pressure can be found on the placard located next to the brake accumulator pressure gage.

(b) If the brake accumulator needs to be serviced, do this task: Hydraulic Brake Accumulator Servicing, TASK 12-15-11-420-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the cap is installed on the charging valve and it has lockwire, G01505, on it (TASK 20-10-44-400-801).

G. Brake Accumulator Pressure Relief Valve Leakage Test

SUBTASK 32-41-71-870-002

(1) Do the steps that follow to do a leakage check of the brake accumulator pressure relief valve:

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Fully push the brake pedals twelve times to bleed the air out through the brake metering valves.

NOTE: Wait approximately 5 seconds between each time that you push the brake pedals.

(c) Make sure that there are no leaks from the brake system relief valve [1] when you bleed the brake system.

(d) For the normal (System B) hydraulic system, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE HYDRAULIC FUSES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the hydraulic fuses for the alternate brake system for the main landing gear

(2) An installation of the hydraulic fuses for the alternate brake system for the main landing gear

(3) A removal of the hydraulic fuses for the normal brake system for the main landing gear

(4) An installation of the hydraulic fuses for the normal brake system for the main landing gear.

B. There is one fuse for the left alternate brake hydraulic system and one fuse for the right alternate hydraulic system.

(1) The two fuses are in the wheel well of the main landing gear.

(2) This procedure shows one hydraulic fuse for the alternate brake system.

(3) The example fuse is for the left main landing gear.

(4) This procedure also applies to the other hydraulic fuse for the alternate brake system.

C. There are two fuses for the left normal brake hydraulic system and two fuses for the right normal hydraulic system.

(1) The four fuses are in the wheel well of the main landing gear.

(2) This procedure shows one hydraulic fuse for the normal brake system.

(3) The example fuse is for the left main landing gear.

(4) This procedure also applies to the other three hydraulic fuses for the normal brake system.

TASK 32-41-72-020-801

2. Brake Hydraulic Fuse (Alternate Brake System) Removal

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare for the Removal

SUBTASK 32-41-72-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-72-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-72-860-001

(3) Release the parking brake.

SUBTASK 32-41-72-860-002

(4) For the A hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-72-870-001

(5) Fully push the brake pedals 12 times to remove the hydraulic pressure from the brake accumulator.

E. Hydraulic Fuse Removal

SUBTASK 32-41-72-020-001

(1) To remove each fuse (alternate brake system) [1] do these steps:

(a) Disconnect each hydraulic line [5] from the fuse (alternate brake system) [1].

(b) Install caps/plugs, STD-14686, in each hydraulic line [5] and the fittings of the fuse (alternate brake system) [1].

(c) Remove the bolts [2], washers [3] and clamp [4] that hold the fuse (alternate brake system) [1] to the airplane.

(d) Remove the fuse (alternate brake system) [1] from the airplane.

F. Brake Hydraulic Fuses Functional Test

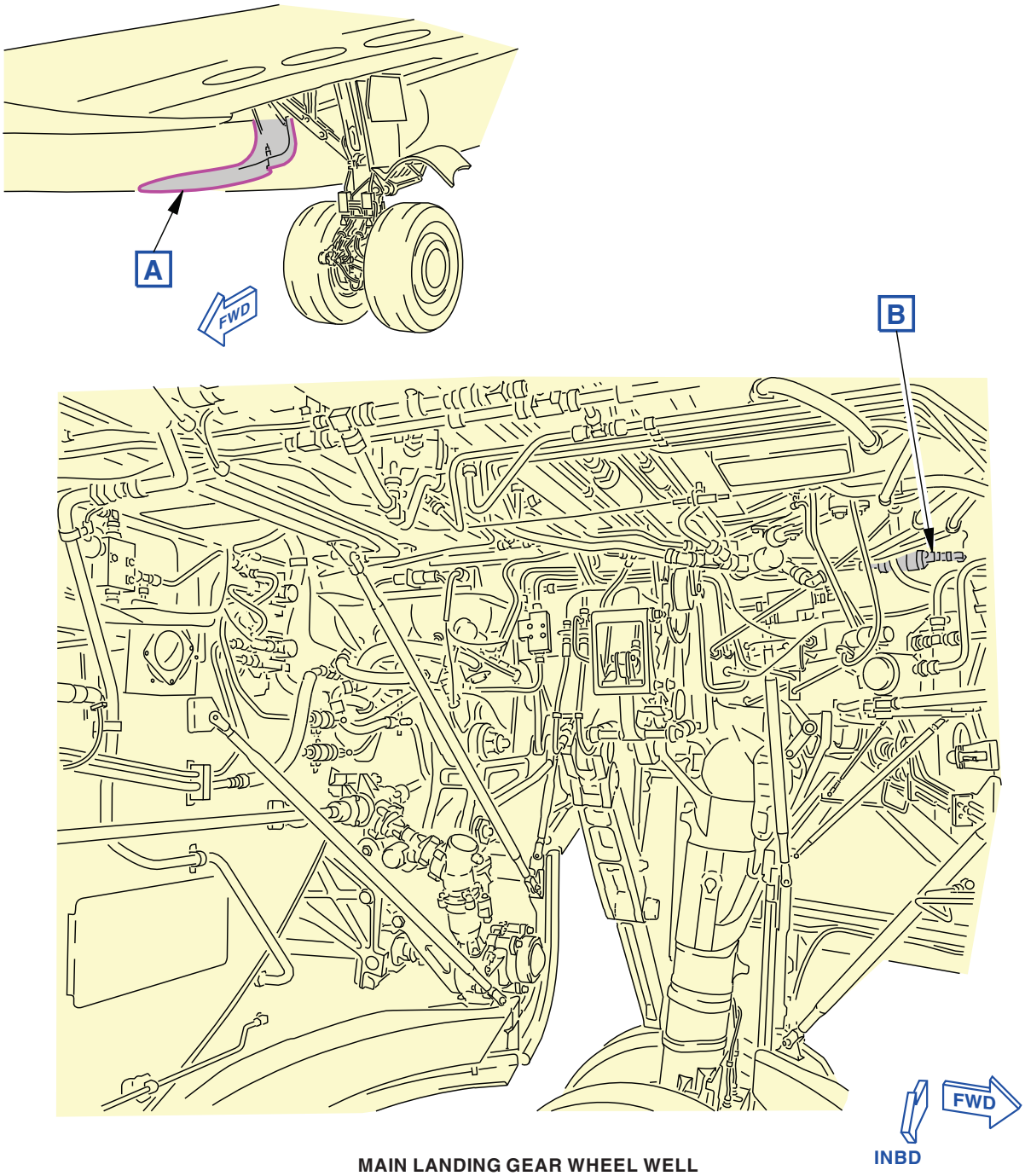
SUBTASK 32-41-72-720-001

(1) Do the functional test of the brake hydraulic fuses with the suppliers recommended component maintenance test instructions and test equipment.

NOTE: The suppliers instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the fuses. This is an off-airplane bench test.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



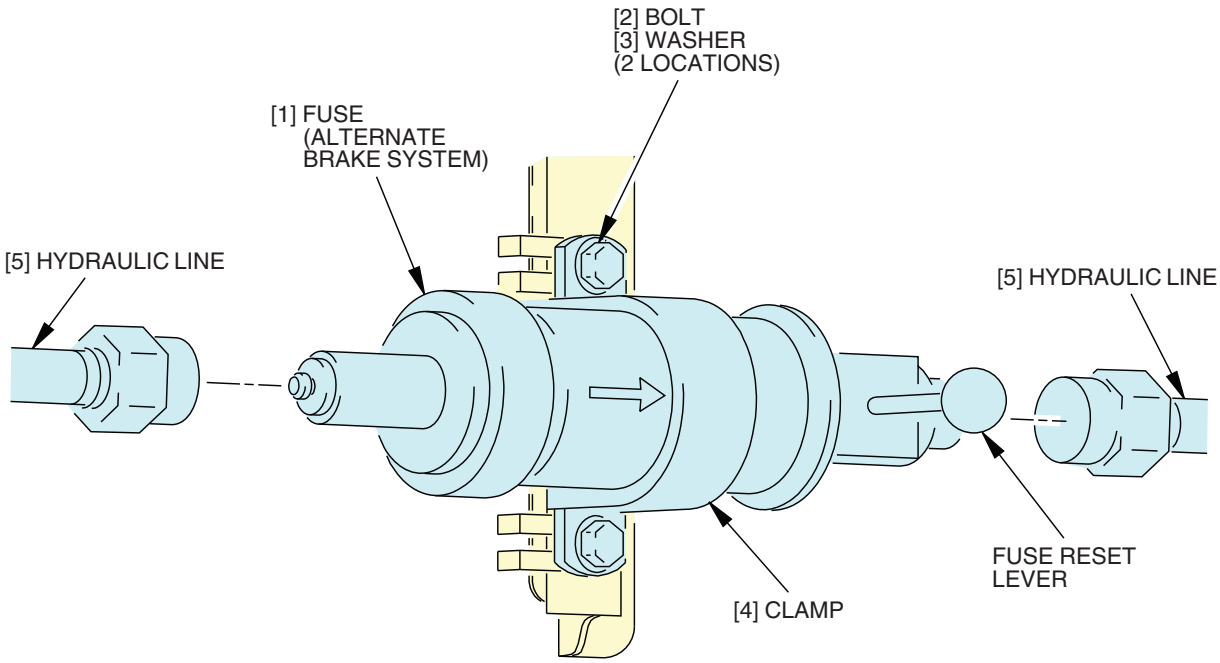
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

A

2414376 S00061535027_V1

Brake Hydraulic Fuse (Alternate Brake System) Installation
Figure 401/32-41-72-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



A SYSTEM - ALTERNATE BRAKE HYDRAULIC FUSE (EXAMPLE)

B

2414377 S00061535028_V2

Brake Hydraulic Fuse (Alternate Brake System) Installation
Figure 401/32-41-72-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-72-420-801

3. Brake Hydraulic Fuse (Alternate Brake System) Installation

(Figure 401)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-803	Alternate (System A) Hydraulic Brake System Bleeding (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuse (alternate brake system)	32-41-72-05-030	PTF ALL
		32-41-72-15-030	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Hydraulic Fuse Installation

SUBTASK 32-41-72-420-001

(1) To install each fuse (alternate brake system) [1], do these steps:

(a) Hold the fuse (alternate brake system) [1] in its position.

NOTE: Boeing recommends that you put the hydraulic fuse into a position that lets you see the reset lever from the ground.

(b) Install the bolts [2], washers [3], and clamp [4] to attach the fuse (alternate brake system) [1] to the airplane.

(c) Remove the caps/plugs, STD-14686, from each hydraulic line [5] and fittings of the fuse (alternate brake system) [1].

(d) Connect the hydraulic lines [5] to the fuse (alternate brake system) [1].

(e) Tighten the B-nuts on the hydraulic lines [5] to 270 ±14 in-lb (30.5 ±1.6 N·m).

F. Hydraulic Fuse Installation Test

SUBTASK 32-41-72-790-001

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) For the alternate brake system, pressurize the hydraulic system A (TASK 29-11-00-860-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-72-790-002

(2) Do a visual check of each fuse (alternate brake system) [1] for hydraulic leaks.

SUBTASK 32-41-72-870-002

(3) To bleed the alternate (System A) hydraulic system, do this task: Alternate (System A) Hydraulic Brake System Bleeding, TASK 32-41-00-870-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-72-860-003

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-72-860-004

(2) Set the parking brakes.

——— END OF TASK ———

TASK 32-41-72-000-801

4. Brake Hydraulic Fuse (Normal Brake System) Removal

(Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 32-41-72-860-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-72-860-006

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-72-860-007

(3) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-72-860-008

(4) For the B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-72-870-003

(5) Fully push the brake pedals 12 times to remove the hydraulic pressure from the brake accumulator.

E. Brake Hydraulic Fuse Removal

SUBTASK 32-41-72-020-002

(1) To remove each fuse (normal brake system) [6], do these steps:

(a) Disconnect each hydraulic line [5] from the fuse (normal brake system) [6].

(b) Install caps/plugs, STD-14686, in each hydraulic line [5] and the fittings of the fuse (normal brake system) [6].

(c) Remove the bolts [2], washers [3] and clamp [4] that attaches the fuse (normal brake system) [6] to the airplane.

(d) Remove the fuse (normal brake system) [6] from the airplane.

F. Brake Hydraulic Fuses Functional Test

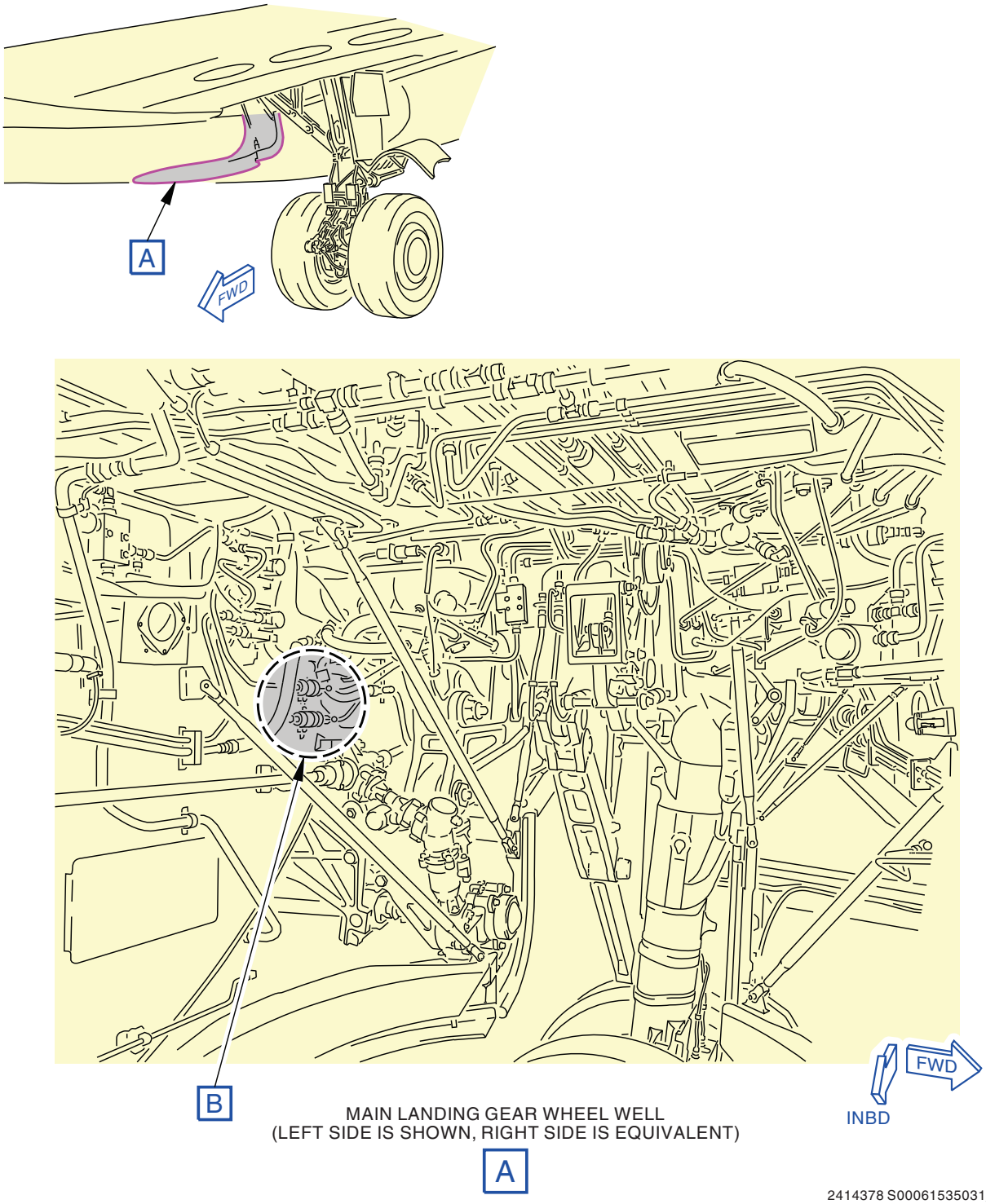
SUBTASK 32-41-72-720-002

(1) Do the functional test of the brake hydraulic fuses with the suppliers recommended component maintenance test instructions and test equipment.

NOTE: The suppliers instructions give a reverse flow test, internal leakage test, pressure drop test, volumetric capacity test and reset test for the fuses. This is an off-airplane bench test.

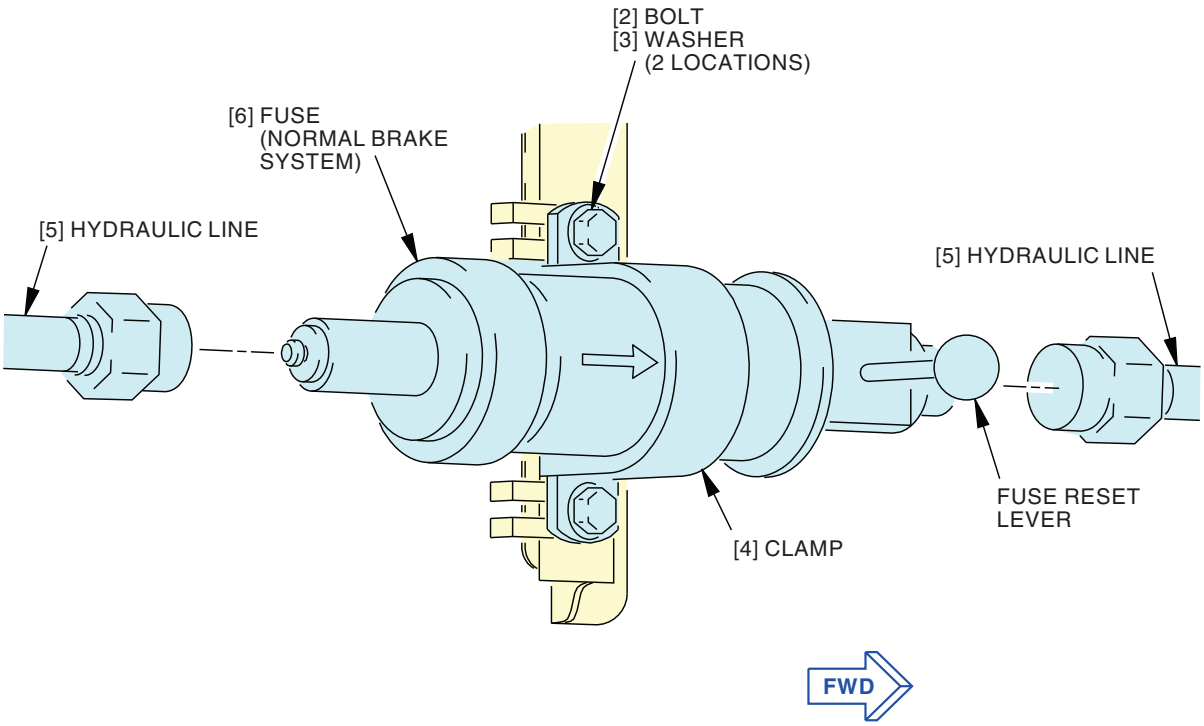
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Hydraulic Fuse (Normal Brake System) Installation
Figure 402/32-41-72-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



B SYSTEM - NORMAL BRAKE HYDRAULIC FUSE
(EXAMPLE)

B

2414379 S00061535032_V1

Brake Hydraulic Fuse (Normal Brake System) Installation
Figure 402/32-41-72-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-72-400-801

5. Brake Hydraulic Fuse (Normal Brake System) Installation

(Figure 402)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
6	Fuse (normal brake system)	32-41-72-01-030	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Brake Hydraulic Fuse Installation

SUBTASK 32-41-72-420-002

(1) To install each fuse (normal brake system) [6], do these steps:

(a) Hold the fuse (normal brake system) [6] in its position.

NOTE: Boeing recommends that you put the hydraulic fuse into a position that lets you see the reset lever from the ground.

(b) Install the bolts [2], washers [3], and clamp [4] to attach the fuse (normal brake system) [6] to the airplane.

(c) Remove the caps/plugs, STD-14686, from each hydraulic line [5] and fittings of the fuse (normal brake system) [6].

(d) Connect the hydraulic lines [5] to the fuse (normal brake system) [6].

(e) Tighten the B-nuts on the hydraulic lines [5] to 270 ±14 in-lb (30.5 ±1.6 N·m).

F. Brake Hydraulic Fuse Installation Test

SUBTASK 32-41-72-790-003

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) For the normal brake system, pressurize the hydraulic system B (TASK 29-11-00-860-801).

SUBTASK 32-41-72-790-004

(2) Do a visual check of each fuse (normal brake system) [6] for hydraulic leaks.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-72-870-004

(3) To bleed the normal (System B) hydraulic system, do this task: Normal (System B) Hydraulic Brake System Bleeding, TASK 32-41-00-870-802.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-72-860-009

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-72-860-010

(2) Set the parking brakes.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HYDRAULIC BRAKE CONTROL CABLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these three tasks:

(1) Removal of the hydraulic brake control cables.

(2) Installation of the hydraulic brake control cables.

TASK 32-41-81-000-801

2. Hydraulic Brake Control Cable Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
20-10-10-000-801	Turnbuckle Lock Removal (P/B 401)
25-22-00-000-801	Passenger Seat Removal (P/B 401)
25-27-15-000-801	Carpet Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
53-21-00-000-801	Passenger Cabin Floor Panel Removal (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1569	Clamp - Control Cable Part #: A20005-9 Supplier: 81205
STD-837	String

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Removal

SUBTASK 32-41-81-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-81-480-002

(2) Make sure the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-81-860-001

(3) Release the parking brake.

SUBTASK 32-41-81-840-001

(4) To get access to the aft cables [2], forward cables [4], turnbuckles [3], and quadrant [5] (forward of nose wheel well), do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-81-840-002

(5) Remove the passenger seats from station 706 through station 727 (TASK 25-22-00-000-801).

NOTE: Do this step to get access to the control quadrant assemblies of the brake metering valves.

SUBTASK 32-41-81-840-003

(6) Do this task: Carpet Removal, TASK 25-27-15-000-801.

SUBTASK 32-41-81-840-004

(7) Remove the center floor panels to get access to the control quadrant assemblies of the brake metering valve (TASK 53-21-00-000-801).

G. Hydraulic Brake Control Cable Removal

NOTE: The steps that follow are applicable for all of the brake control cables (aft cables [2], forward cable [4]).

SUBTASK 32-41-81-020-001

(1) In the forward compartment, forward of the nose wheel well, do these steps:

(a) Install a control cable clamp, SPL-1569, on the cable (aft cable [2] or forward cable [4]) opposite the one you will remove to keep light tension on the cable (aft cable [2] or forward cable [4]) .

NOTE: Light tension on the cable (aft cable [2] or forward cable [4]) will prevent wind off on the cable drums. Light tension will also make sure the cables (aft cable [2] or forward cable [4]) do not move out of the quadrant [5] or quadrant [8] guides.

(b) Remove the locking clip [1] from the turnbuckle [3] on the aft cables [2], forward cables [4] (TASK 20-10-10-000-801).

(c) Loosen the turnbuckle [3] for the aft cable [2], forward cable [4] to release the tension in the cable.

NOTE: Make sure you keep light tension on the cable (aft cable [2] or forward cable [4]) to keep the cable in the quadrant [5] or quadrant [8] guides.

(d) For the forward cable [4], disconnect the forward cable [4] from the quadrant [5]:

1) Remove the pins [7], that hold the ends of the forward cable [4], from the quadrants [5].

2) Pull the fittings on the ends of the forward cable [4] from the quadrant [5].

3) Remove the forward cable [4] from the quadrant [5].

SUBTASK 32-41-81-020-002

(2) In the passenger compartment, do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the aft cable [2] from the quadrant [8]:

1) Remove the pins [9] that holds the ends of the aft cable [2], from the quadrant [8].

2) Pull the fittings on the ends of the aft cable [2] from the quadrant [8].

SUBTASK 32-41-81-020-003

(3) Attach the new aft cable [2] or forward cable [4], or a piece of string, STD-837, to the old aft cables [2].

NOTE: The new aft cable [2] or forward cable [4] must be pulled through the system to make the new aft cable [2] or forward cable [4] installation easy. If you remove the old aft cable [2] or forward cable [4] without attaching anything to it, installation of the new aft cable [2] or forward cable [4] will be very difficult.

SUBTASK 32-41-81-020-004

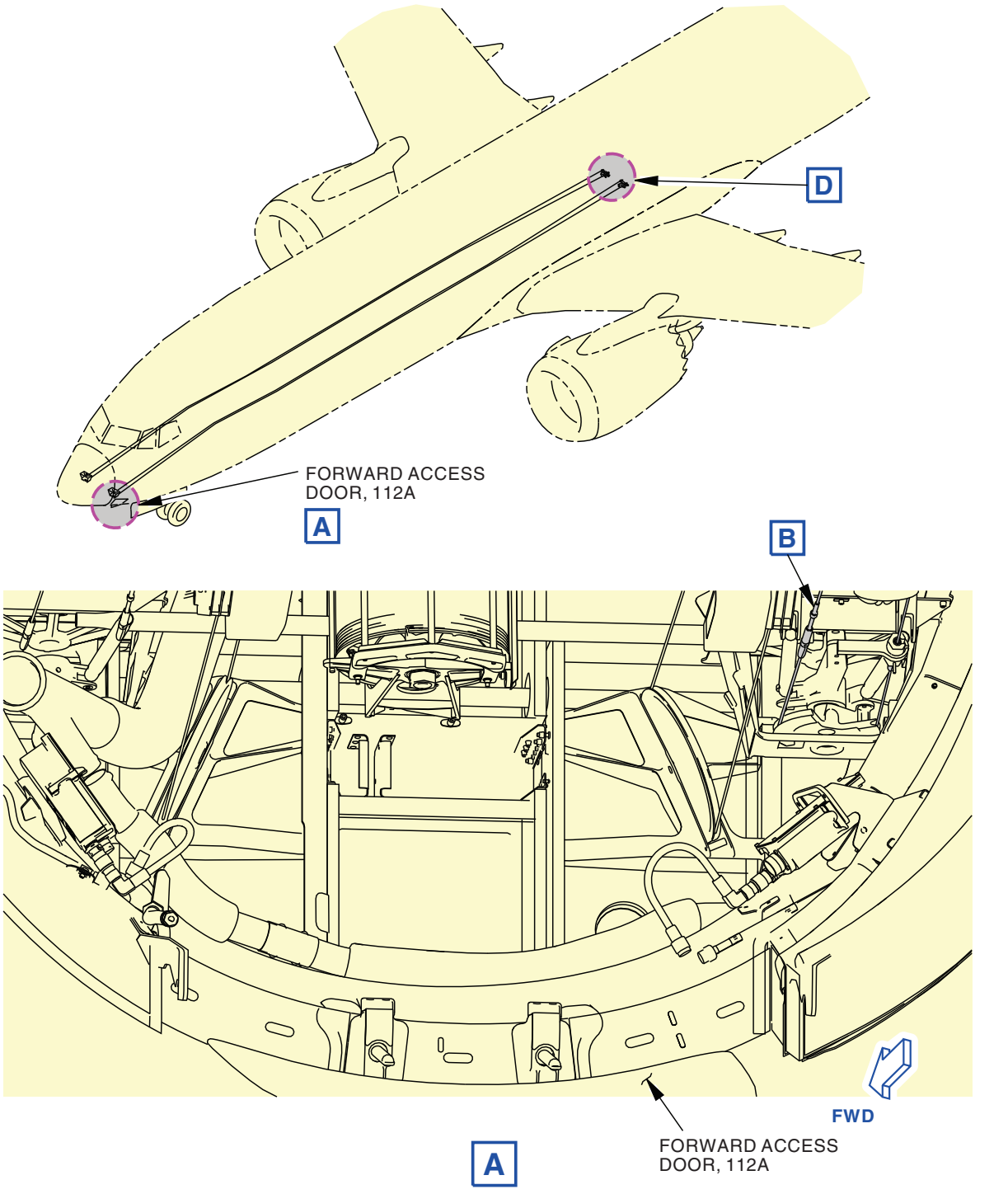
(4) Remove the aft cable [2] or forward cable [4]:

(a) Pull the old aft cable [2] or forward cable [4] out, which will pull the new aft cable [2] or forward cable [4], or string, STD-837, through the cable system at the same time.

NOTE: Make sure the new aft cable [2] or forward cable [4] has grease, D00633, applied to it before it is installed.

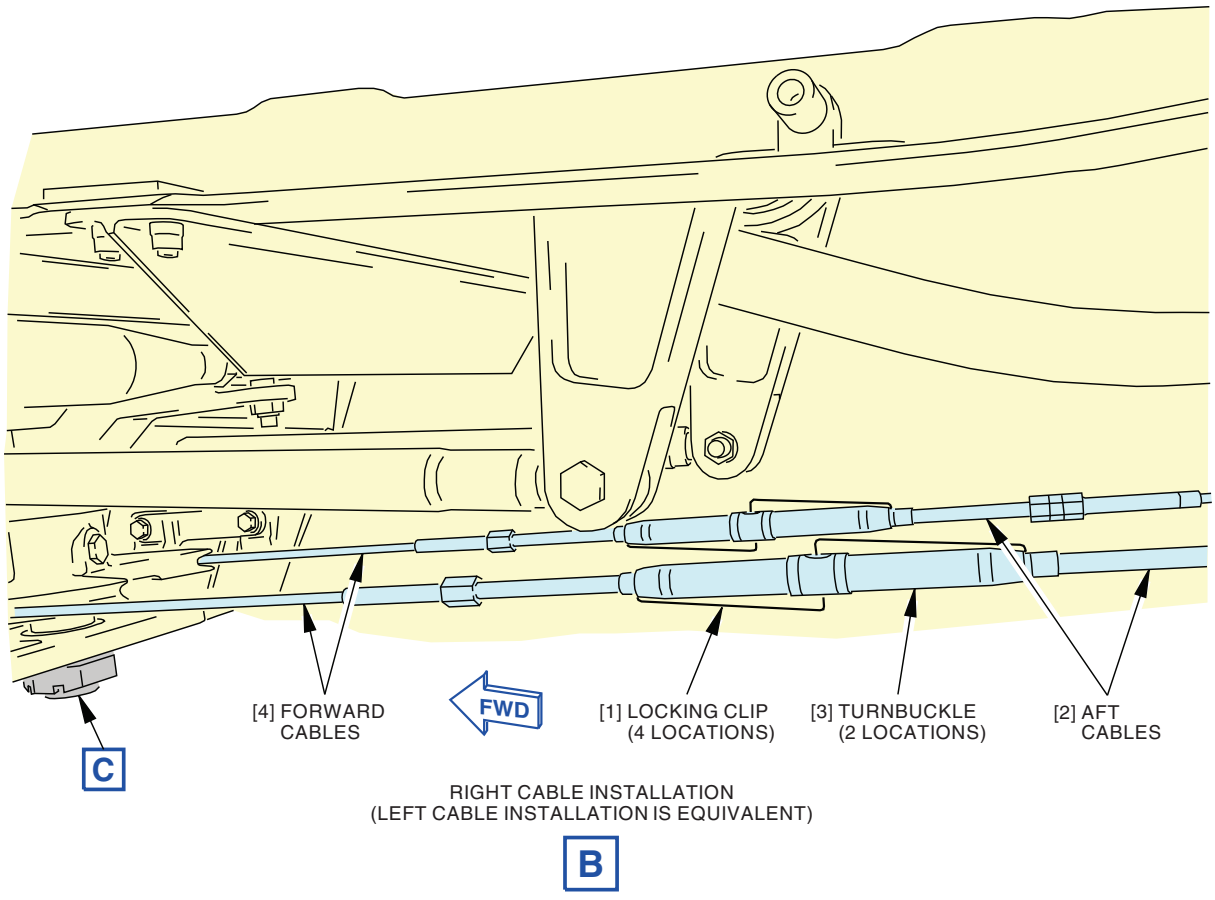
——— END OF TASK ———

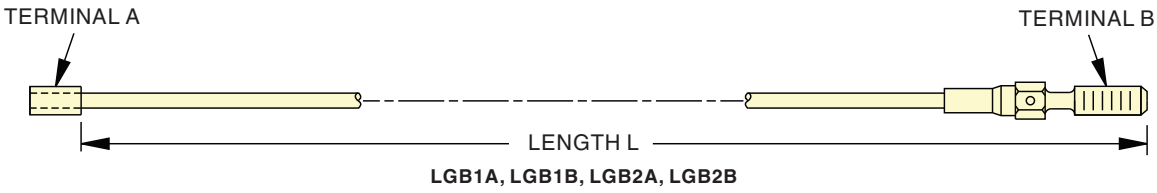
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hydraulic Brake Control Cable Installation
Figure 401/32-41-81-990-801 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

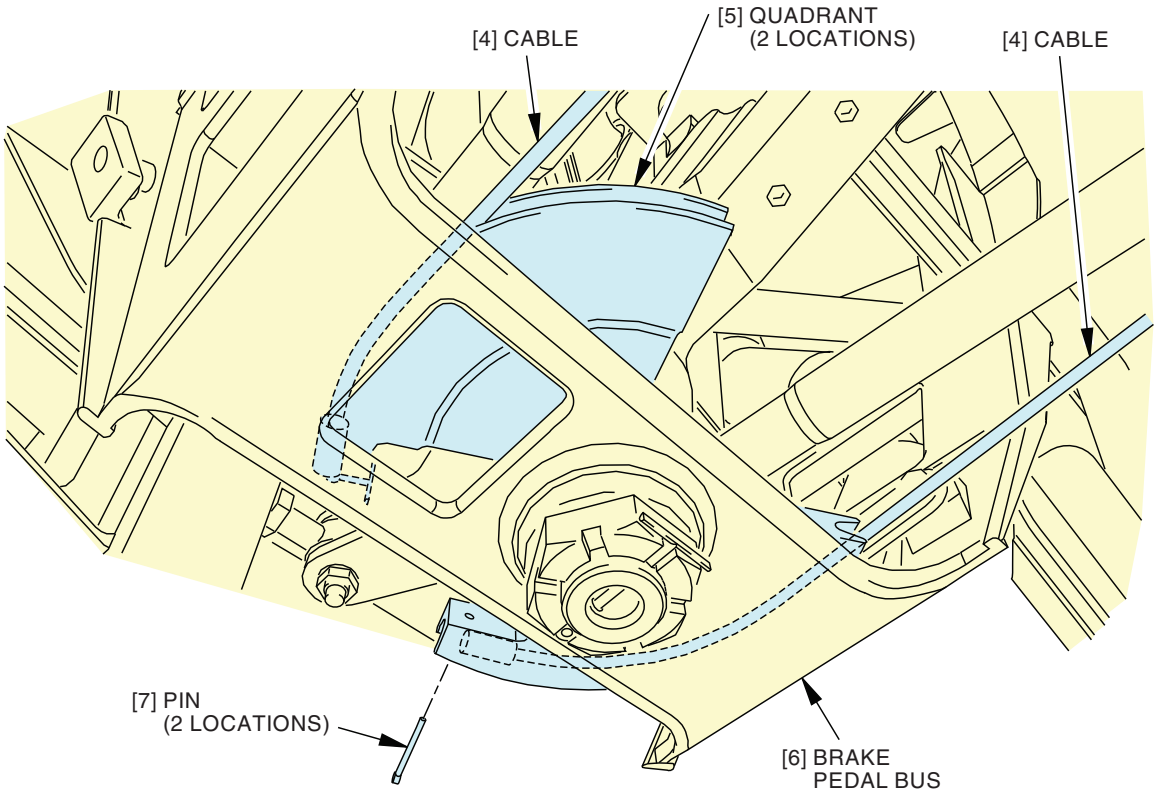




2411035 S00061526774_V2

Hydraulic Brake Control Cable Installation
Figure 401/32-41-81-990-801 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



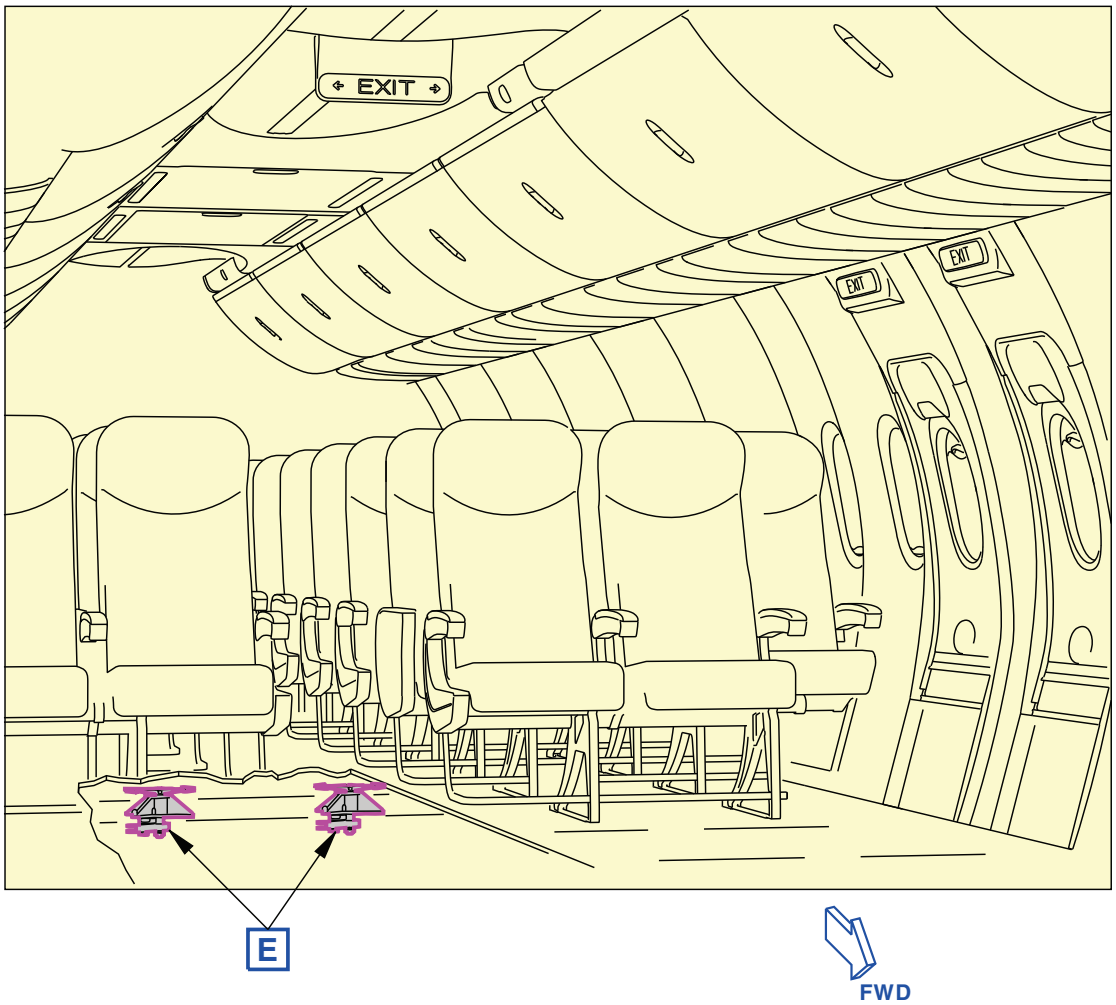
RIGHT BUS INSTALLATION
(LEFT BUS INSTALLATION IS EQUIVALENT)

C

2411036 S00061526775_V1

Hydraulic Brake Control Cable Installation
Figure 401/32-41-81-990-801 (Sheet 3 of 5)

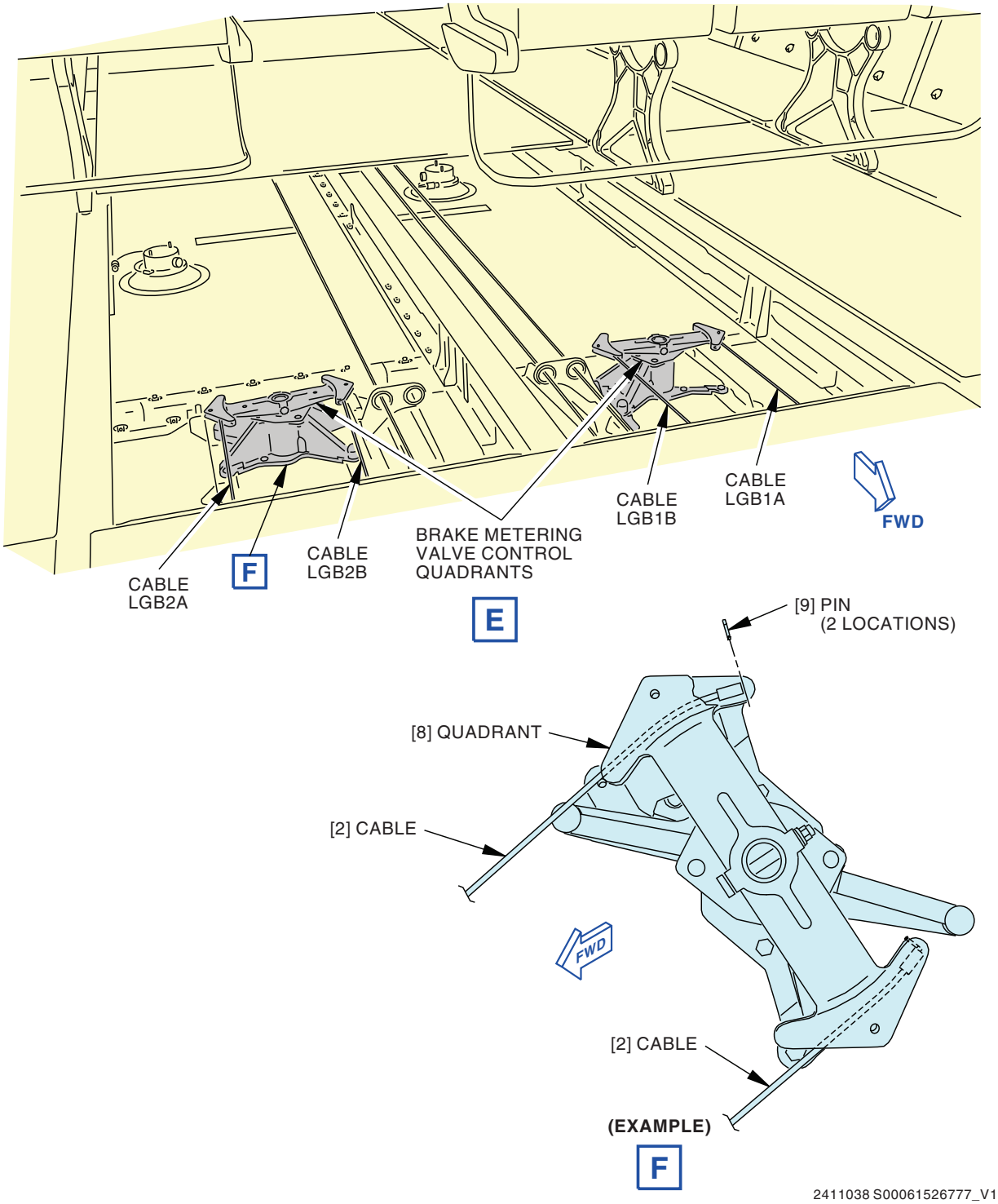
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2411037 S00061526776_V1

Hydraulic Brake Control Cable Installation
Figure 401/32-41-81-990-801 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Hydraulic Brake Control Cable Installation
Figure 401/32-41-81-990-801 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-81-400-801

3. Hydraulic Brake Control Cable Installation

(Figure 401)

A. References

Reference	Title
20-10-10-400-801	Turnbuckle Lock Installation (P/B 401)
20-10-91-400-801	Control Cables Installation (P/B 401)
20-20-31-200-801	Inspection of the Control Cable Wire Rope (P/B 601)
25-22-00-400-801	Passenger Seat Installation (P/B 401)
25-27-15-400-801	Carpet - Installation (P/B 401)
32-41-00-710-801	Normal and Alternate Brake Application Functional Test (P/B 501)
32-41-00-820-801	Hydraulic Brake System Adjustment (P/B 501)
53-21-00-400-801	Passenger Cabin Floor Panel Installation (P/B 401)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1553	Tensiometer - Cable, Low Tension (200 lbs and below) Part #: 102-03120 Supplier: 21844 Part #: ACM-200 Supplier: 13331 Part #: ACX-250-1 Supplier: 13331 Part #: T60-1001-C8-1A Supplier: 0N8U4
STD-837	String

C. Consumable Materials

Reference	Description	Specification
D00633	Grease - Aircraft General Purpose	BMS3-33

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Aft cable	32-41-81-01-035	PTF ALL
		32-41-81-01-040	PTF ALL
4	Forward cable	32-41-21-01-035	PTF ALL
		32-41-21-02-035	PTF ALL
7	Pin	32-41-21-01-030	PTF ALL
		32-41-21-02-030	PTF ALL
9	Pin	32-41-81-01-005	PTF ALL

E. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well

F. Access Panels

Number	Name/Location
112A	Forward Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Hydraulic Brake Control Cable Installation

SUBTASK 32-41-81-020-005

(1) Apply grease, D00633, to the new aft cable [2] or forward cable [4] before it is installed.

SUBTASK 32-41-81-420-001

(2) If the new aft cable [2] or new forward cable [4] was not pulled through the cable system with the old aft cable [2] or new forward cable [4], do these steps:

(a) Attach the new aft cable [2] or new forward cable [4] to the string, STD-837, that was pulled through the cable system.

(b) Pull the string, STD-837, out, which will pull the new aft cable [2] or forward cable [4] through the cable system.

(c) Make sure that you keep light tension on the aft cable [2] or forward cable [4] to keep the aft cable [2] or forward cable [4] in the quadrant [8] or quadrant [5] guides.

SUBTASK 32-41-81-020-006

(3) In the passenger compartment, do these steps:

(a) Connect the aft cable [2] to the quadrant [8] (TASK 20-10-91-400-801).

1) Put the aft cable [2] in its position in the quadrant [8].

2) Install the new pins [9] to hold the end of the aft cable [2] in the quadrant [8].

SUBTASK 32-41-81-020-007

(4) In the forward compartment, forward of the nose wheel well, do these steps:

(a) Connect the forward cable [4] to the quadrant [5] (TASK 20-10-91-400-801).

1) Put the forward cable [4] in its position in the quadrant [5].

2) Install the new pins [7] to hold the end of the forward cable [4] in the quadrant [5].

(b) Install the turnbuckles [3].

1) Connect the forward cable [4] to the aft cable [2].

SUBTASK 32-41-81-420-002

(5) Adjust the brake cable tension with the low tension cable tensiometer, COM-1553, to 200 ±10 lbf (890 ±45 N) tension.

SUBTASK 32-41-81-200-001

(6) Do this task: Inspection of the Control Cable Wire Rope, TASK 20-20-31-200-801.

SUBTASK 32-41-81-420-003

(7) Operate the aft cable [2] and forward cable [4] through full travel 25 times.

SUBTASK 32-41-81-420-004

(8) Do this task: Hydraulic Brake System Adjustment, TASK 32-41-00-820-801.

SUBTASK 32-41-81-420-005

(9) Make sure that a locking clip [1] is installed on the turnbuckle [3] when you have adjusted the aft cables [2] and forward cables [4] (TASK 20-10-10-400-801).

SUBTASK 32-41-81-420-006

(10) Do this task: Normal and Alternate Brake Application Functional Test, TASK 32-41-00-710-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-81-840-005

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-81-840-006

(2) Install the floor panels (TASK 53-21-00-400-801).

SUBTASK 32-41-81-840-007

(3) Do this task: Carpet - Installation, TASK 25-27-15-400-801.

SUBTASK 32-41-81-840-008

(4) Do this task: Passenger Seat Installation, TASK 25-22-00-400-801.

SUBTASK 32-41-81-840-009

(5) Set the parking brakes.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

HYDRAULIC BRAKE CONTROL CABLE - INSPECTION/CHECK

1. General

A. This procedure contains scheduled maintenance task data.

B. This procedure has this task:

(1) Brake Control System Detailed Inspection

TASK 32-41-81-211-801

2. Brake Control System Detailed Inspection

(Figure 601)

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION

B. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Access Panels

Number	Name/Location
112A	Forward Access Door

D. Prepare for the Inspection

SUBTASK 32-41-81-480-003

(1) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-81-860-002

(2) If necessary, release the parking brake.

E. Forward Brake Control Mechanism Detailed Inspection

SUBTASK 32-41-81-840-014

(1) To get access to the pilots and copilots brake/rudder pedals and visible brake control components (forward of nose wheel well), do this step:

(a) Open this access panel:

Number	Name/Location
112A	Forward Access Door

SUBTASK 32-41-81-210-003

(2) Do a detailed inspection of the brake cable quadrants, bell cranks, control rods.

F. Aft Brake Control Mechanism Detailed Inspection

SUBTASK 32-41-81-840-015

(1) Go to the main landing gear wheel well.

SUBTASK 32-41-81-210-004

(2) Do a detailed inspection of the visible part of the brake control mechanism to the brake metering valve.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Put the Airplane Back to Its Usual Condition

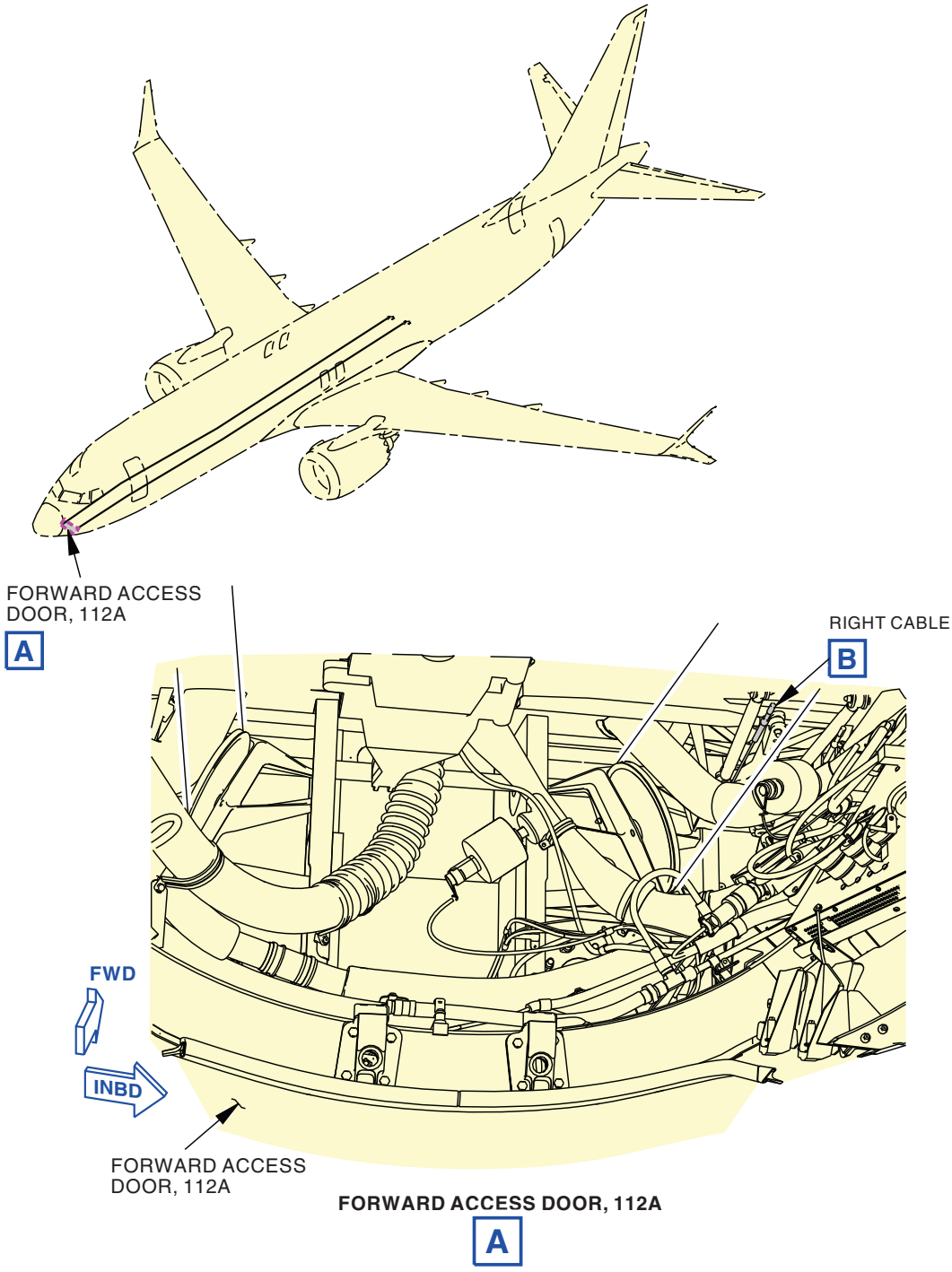
SUBTASK 32-41-81-840-016

(1) Close this access panel:

Number	Name/Location
112A	Forward Access Door

——— END OF TASK ———

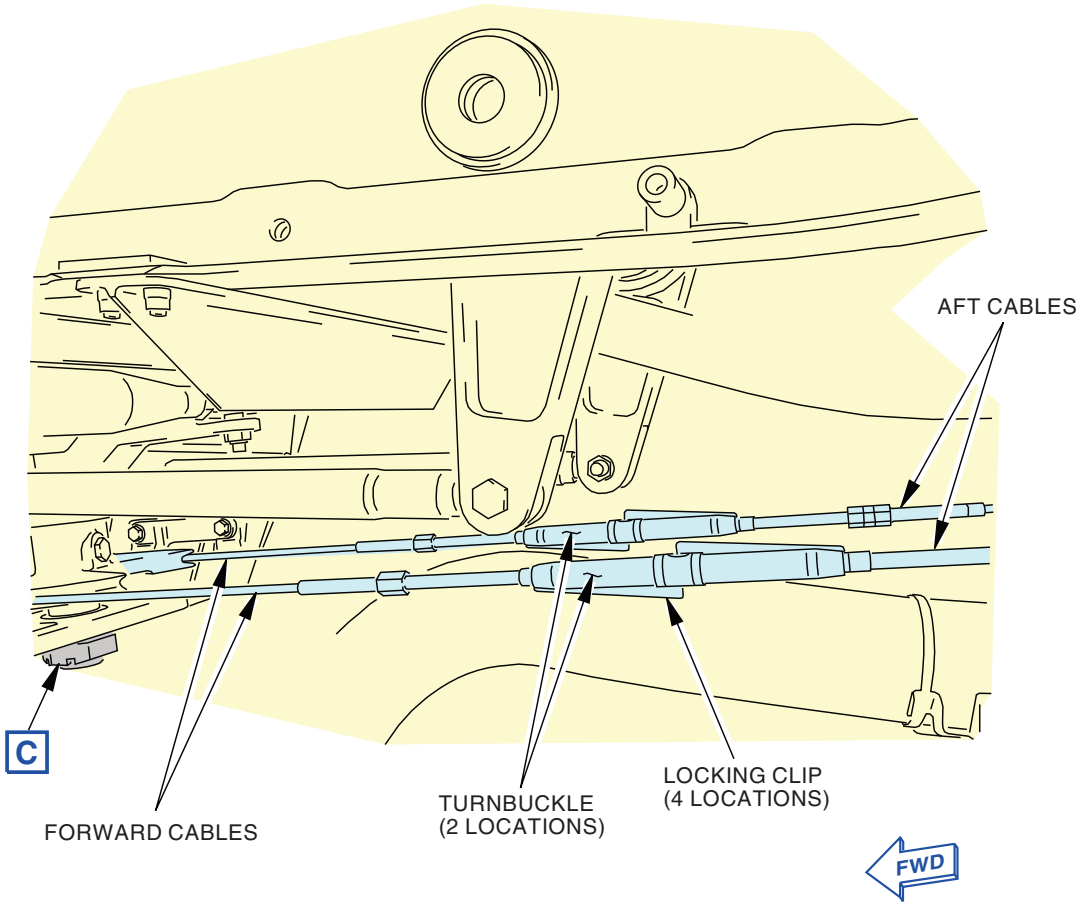
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414380 S00061535039_V1

Brake Control System Detailed Inspection
Figure 601/32-41-81-990-803 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



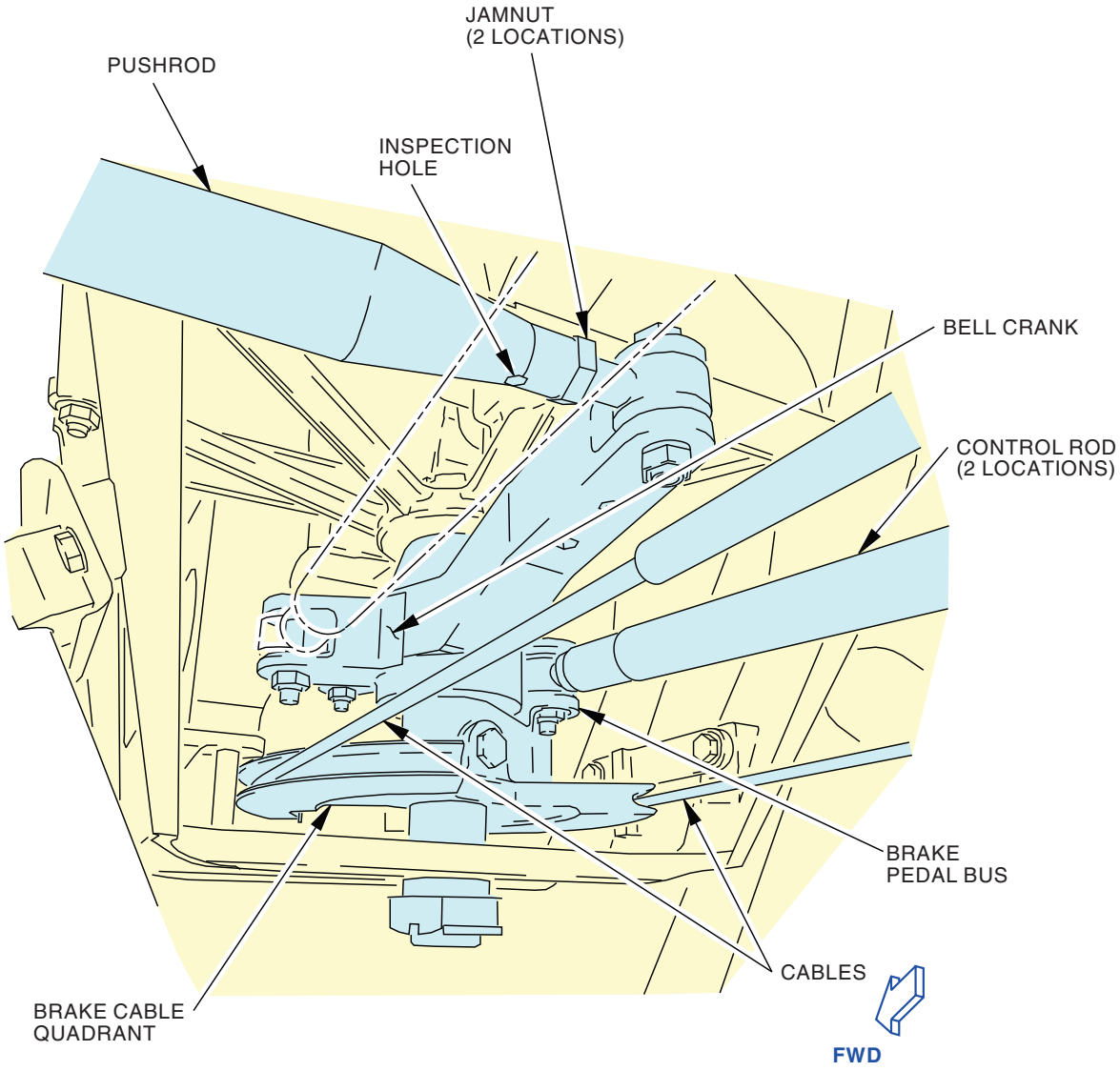
RIGHT CABLE
(LEFT CABLE IS EQUIVALENT)

B

2414381 S00061535040_V1

Brake Control System Detailed Inspection
Figure 601/32-41-81-990-803 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



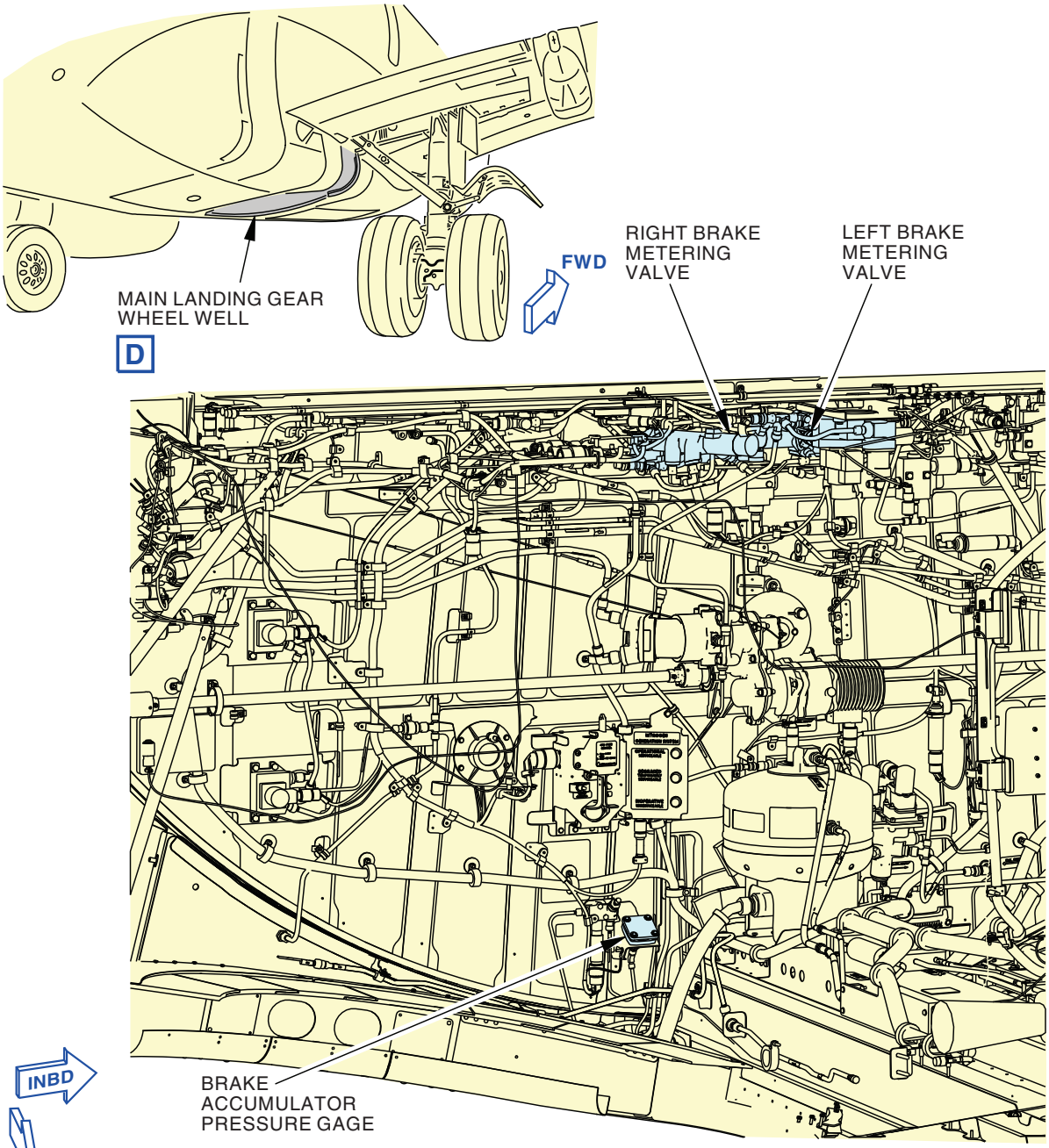
RIGHT BRAKE PEDAL BUS
(LEFT BRAKE PEDAL BUS IS EQUIVALENT)

C

2414382 S00061535041_V2

Brake Control System Detailed Inspection
Figure 601/32-41-81-990-803 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414383 S00061535042_V1

Brake Control System Detailed Inspection
Figure 601/32-41-81-990-803 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE ACCUMULATOR ISOLATION VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the brake accumulator isolation valve (referred to as the accumulator isolation valve [13]).

(2) Installation of the brake accumulator isolation valve.

TASK 32-41-91-000-801

2. Accumulator Isolation Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-41-91-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-91-480-002

(2) Put chocks around the tires of all the landing gear (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-91-860-001

(3) Release the parking brake.

SUBTASK 32-41-91-860-002

(4) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-91-870-001

(5) Bleed the pressure from the brake accumulator:

(a) Operate the brake pedals for approximately 12 times or until the gage on the brake accumulator shows no change in the pressure.

NOTE: This step will make sure that the pressure is fully released from the brake accumulator.

E. Accumulator Isolation Valve Removal

SUBTASK 32-41-91-020-001

(1) Do these steps to remove the accumulator isolation valve [13]:

(a) Disconnect the hydraulic line [1], hydraulic line [8], and elbow [11] on the hydraulic line [12] from the accumulator isolation valve [13].

(b) Install caps/plugs, STD-14686, in the hydraulic line [1], hydraulic line [8], and elbow [11].

(c) Install caps/plugs, STD-14686, on the reducer [2], reducer [7], and reducer [10] in the ports of the accumulator isolation valve [13].

(d) Remove the bolts [4], and washers [5] that attach the accumulator isolation valve [13] to the mating surface.

(e) Remove the accumulator isolation valve [13] from the airplane.

SUBTASK 32-41-91-020-002

(2) If the replacement accumulator isolation valve [13] does not have hydraulic fittings installed, do these steps:

(a) Remove the reducer [2], reducer [7], reducer [10], and plugs [14] from the ports of the accumulator isolation valve [13].

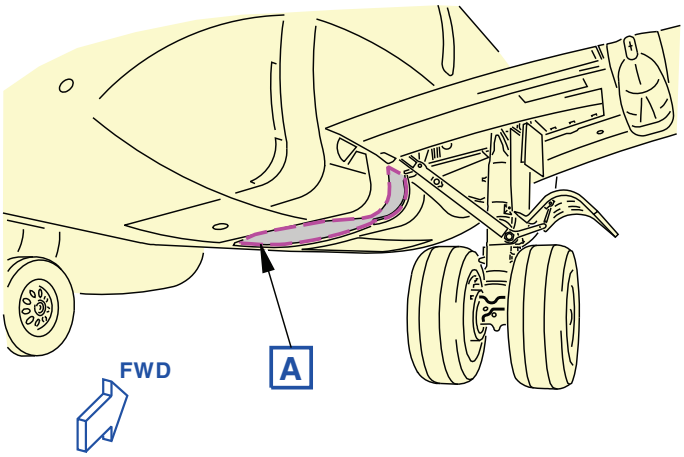
1) Remove the lockwire from the plugs [14].

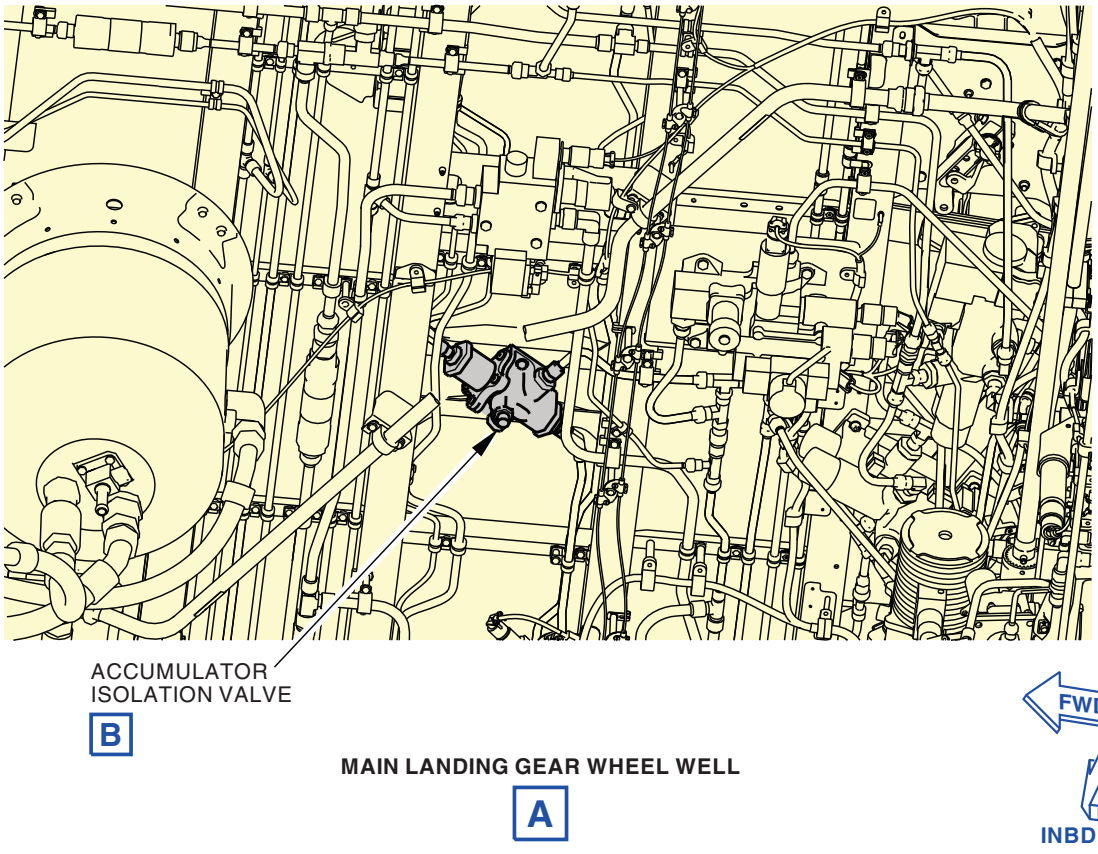
(b) Remove and discard the packings [3], packing [6], and packing [9].

(c) Install caps/plugs, STD-14686, in the hydraulic ports of the accumulator isolation valve [13].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

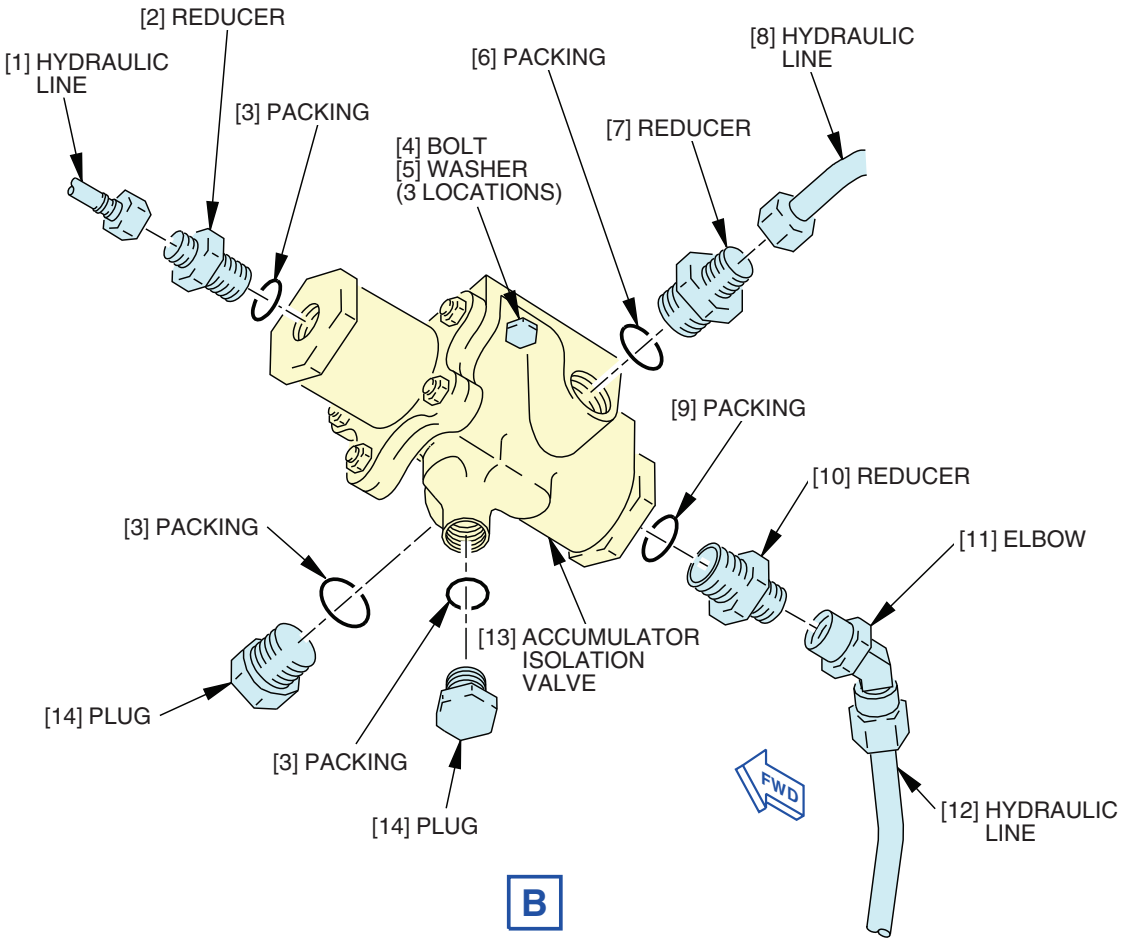




2414384 S00061535046_V1

Accumulator Isolation Valve Installation
Figure 401/32-41-91-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414385 S00061535047_V2

Accumulator Isolation Valve Installation
Figure 401/32-41-91-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-91-420-801

3. Accumulator Isolation Valve Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Packing	32-41-91-01-035	PTF ALL
6	Packing	32-41-91-01-030	PTF ALL
9	Packing	32-41-91-01-030	PTF ALL
13	Accumulator isolation valve	32-41-91-01-045	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Prepare for the Installation

SUBTASK 32-41-91-110-001

(1) Prepare the mating surfaces for installation of the accumulator isolation valve [13].

(a) Clean the mating surface of the accumulator isolation valve [13] with the solvent, B00083.

(b) Clean the mating surface that the accumulator isolation valve [13] attaches to with the solvent, B00083.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Accumulator Isolation Valve Installation

SUBTASK 32-41-91-640-001

(1) If the reducer [2], reducer [7], reducer [10], plugs [14] , and the packings [3], packing [6], and packing [9] are not installed, do these steps:

(a) Lubricate the reducer [2], reducer [7], reducer [10], plugs [14], and the new packings [3], new packing [6], and new packing [9] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the reducer [2], reducer [7], and reducer [10], plugs [14], and the new packing [3], new packing [6], and new packing [9] in their locations on the accumulator isolation valve [13].

PTF ALL; BRAKE ACCUMULATOR ISOLATION VALVE WITH PLUG P/N 274T4629-1

1) Tighten the reducer [2] to 270 ±14 in-lb (30.5 ±1.6 N·m).

NOTE: The reducer is made of steel.

PTF ALL; BRAKE ACCUMULATOR ISOLATION VALVE WITHOUT PLUG P/N 274T4629-1

2) Tighten the reducer [2] to 170 ±9 in-lb (19.2 ±1.0 N·m).

NOTE: The reducer is made of aluminum alloy.

PTF ALL

3) Tighten the reducer [7] to 280 ±14 in-lb (31.6 ±1.6 N·m).

4) Tighten the reducer [10] to 280 ±14 in-lb (31.6 ±1.6 N·m).

(c) Install MS20995NC32 lockwire, G01912, on the plugs [14] with the double-twist method.

SUBTASK 32-41-91-420-001

(2) Do these steps to install the accumulator isolation valve [13]:

(a) Put the accumulator isolation valve [13] on the mating surface.

(b) Install the bolts [4] and washers [5] to attach the accumulator isolation valve [13] on the mating surface.

(c) Remove the caps/plugs, STD-14686, from the hydraulic ports of the accumulator isolation valve [13].

(d) Remove the caps/plugs, STD-14686, from the hydraulic line [1], hydraulic line [8], and elbow [11] on the hydraulic line [12].

(e) Connect hydraulic line [1] to the accumulator isolation valve [13].

1) Tighten the B-nut on hydraulic line [1] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(f) Connect hydraulic line [8] to the accumulator isolation valve [13].

1) Tighten the B-nut on hydraulic line [8] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(g) Connect elbow [11] to the accumulator isolation valve [13].

1) Tighten the B-nut on elbow [11] to 270 ±14 in-lb (30.5 ±1.6 N·m).

H. Accumulator Isolation Valve Installation Test

SUBTASK 32-41-91-870-002

(1) Do the steps that follow to make sure that the accumulator isolation valve assembly does not leak:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) For the normal and alternate brake hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Make sure that the PRESS A, PRESS B, and BRAKE hydraulic connections on the accumulator isolation valve [13] do not leak.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-91-870-003

(1) Bleed the brake hydraulic system:

(a) Operate the brake pedals 6 times.

1) Operate the pedals slowly and wait a minimum of 5 seconds between each time you push the pedals.

SUBTASK 32-41-91-860-003

(2) For the normal and alternate hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-91-840-001

(3) Set the parking brakes.

SUBTASK 32-41-91-840-002

(4) Remove the chocks from around the tires of all the landing gear.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ALTERNATE BRAKE SELECTOR VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the alternate brake selector valve

(2) An installation of the alternate brake selector valve.

TASK 32-41-93-000-801

2. Alternate Brake Selector Valve Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

D. Prepare for the Removal

SUBTASK 32-41-93-860-018

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-93-860-002

(2) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-93-870-001

(3) Bleed the pressure from the brake accumulator:

(a) Operate the brake pedals for approximately 12 times or until the gage on the brake accumulator shows no change in the pressure.

NOTE: This step will make sure that the pressure is fully released from the brake accumulator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Alternate Brake Selector Valve Removal

SUBTASK 32-41-93-020-001

(1) Do these steps to remove the alternate brake selector valve [1]:

(a) Disconnect the electrical connector [3] from the pressure switch [2] on the alternate brake selector valve [1].

(b) Disconnect the hydraulic line [6] from the normal port on the alternate brake selector valve [1].

(c) Disconnect the elbow [7] from the return port on the alternate brake selector valve [1].

(d) Disconnect the T-fitting [10] from the brake port on the alternate brake selector valve [1].

(e) Disconnect the elbow [13] from the alt pressure port on the alternate brake selector valve [1].

(f) Install the caps/plugs, STD-14686, on the hydraulic line [6], elbow [7], T-fitting [10], and elbow [13].

(g) Install the caps/plugs, STD-14686, in the hydraulic ports of the alternate brake selector valve [1].

(h) Remove the nuts [16] and washers [17] that attach the alternate brake selector valve [1] to the mating surface.

(i) Remove the alternate brake selector valve [1] from the airplane.

SUBTASK 32-41-93-020-002

(2) If you will install a new alternate brake selector valve [1] without the unions or reducers, do these steps:

(a) Remove the union [8], reducer [5], reducer [11], and reducer [14] from the alternate brake selector valve [1].

(b) Remove and discard the packing [4], packing [9], packing [12], and packing [15].

SUBTASK 32-41-93-020-003

(3) Do the steps that follow to remove the pressure switch [2] from the alternate brake selector valve [1]:

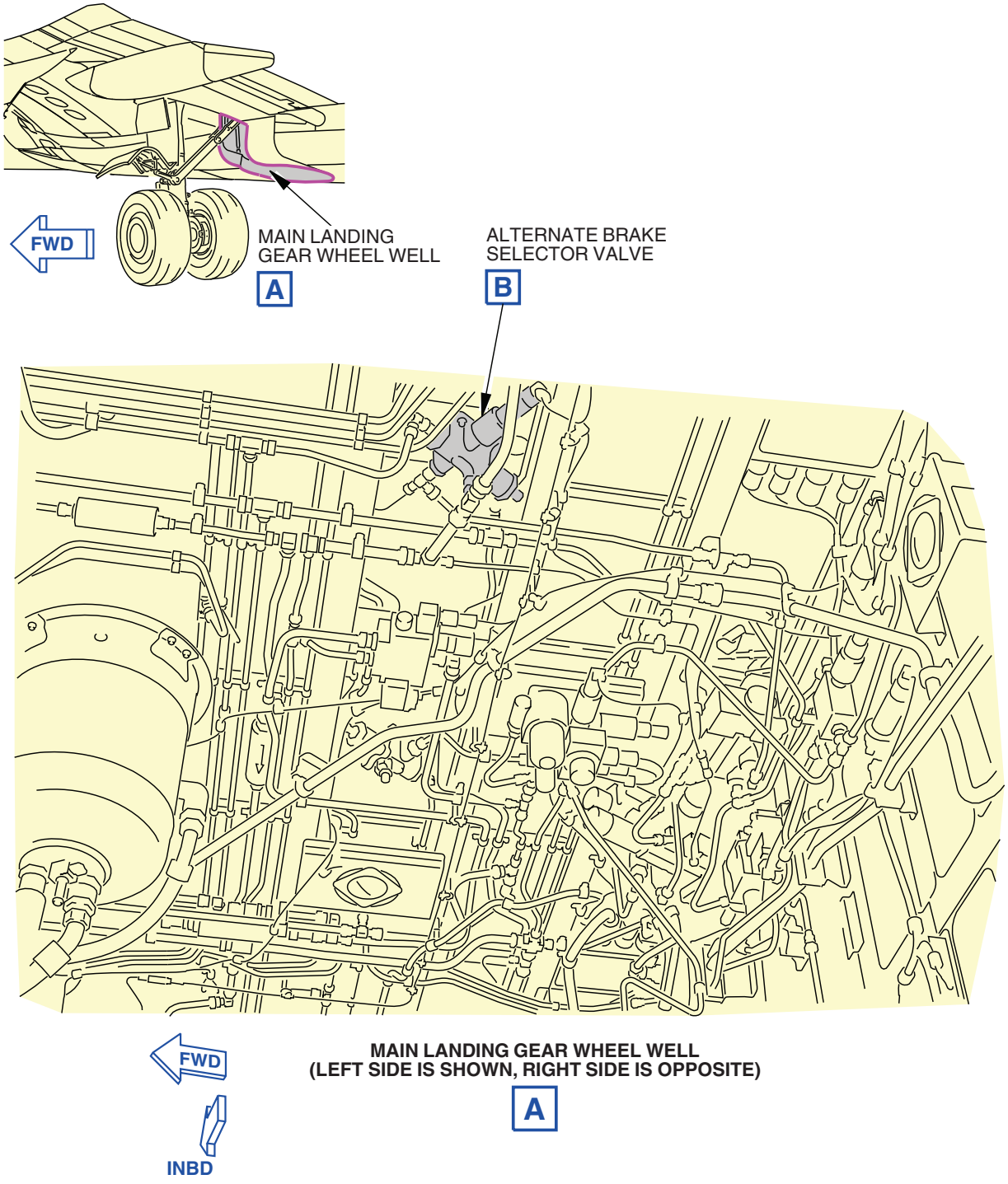
(a) Remove the lockwire.

(b) Remove the pressure switch [2] from the alternate brake selector valve [1].

(c) Remove and discard the packing [21] from the pressure switch [2].

——— END OF TASK ———

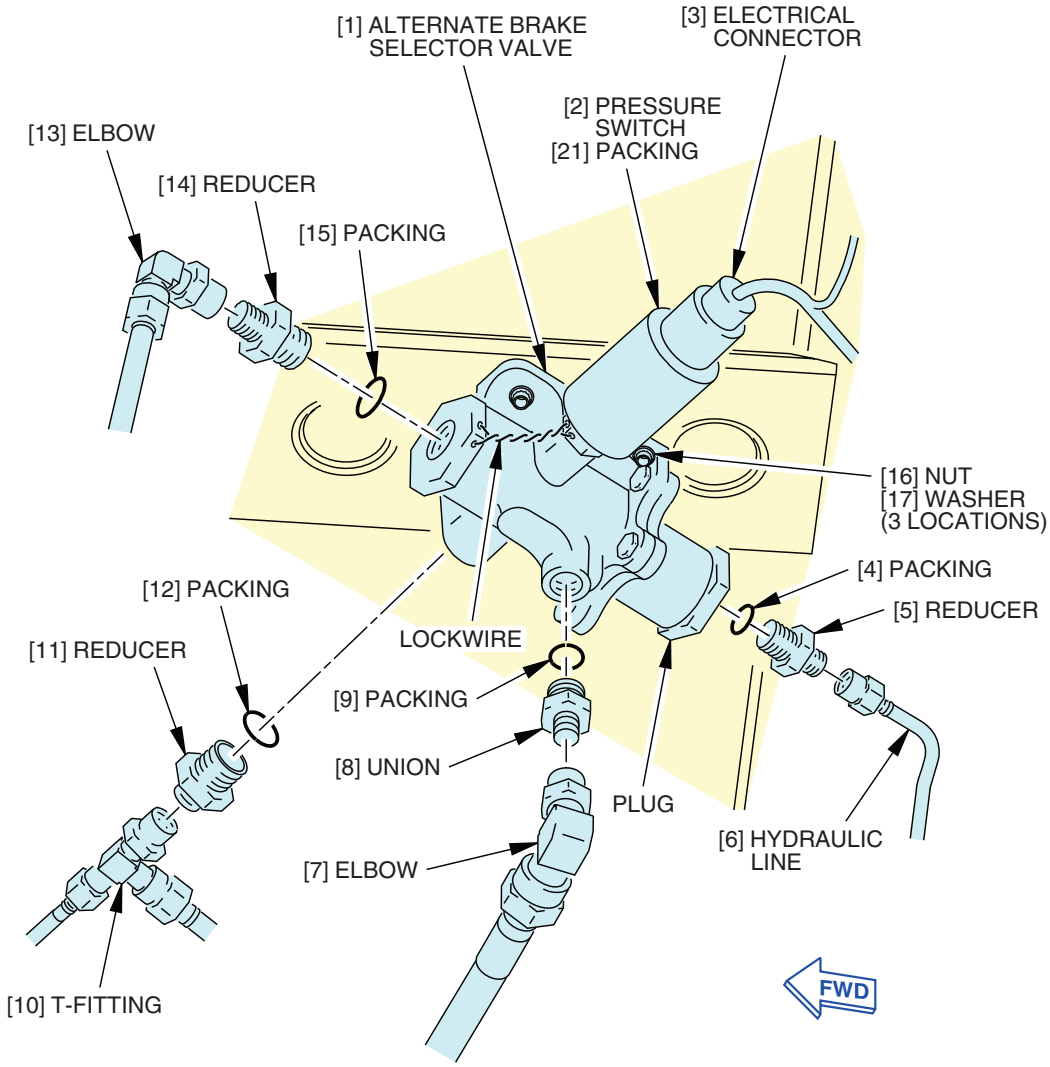
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414386 S00061535052_V2

Alternate Brake Selector Valve Installation
Figure 401/32-41-93-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ALTERNATE BRAKE SELECTOR VALVE

B

3083138 S0000838292_V3

Alternate Brake Selector Valve Installation
Figure 401/32-41-93-990-801 (Sheet 2 of 2)

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TASK 32-41-93-420-801

3. Alternate Brake Selector Valve Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-93-710-801	Alternate Brake Selector Valve Operational Test (P/B 501)
32-41-93-710-802	Gear Retract Braking Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00 Paragraph 14.D	Category 2 Fay Sealed Fay Surface Bonds
SWPM 20-20-00, Paragraph 2	CLEANING OF FAY SURFACES
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-614	Bonding Meters - Non-Intrinsically Safe (For use in outside Class I, Divisions I & II non-hazardous locations. For hazardous locations, use COM-1550). Part #: 247000 Supplier: 00426 Part #: 620LK Supplier: 1CRL2 Part #: BLR-0003-XX Supplier: KC432 Part #: BT51 Supplier: 00426 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659 Opt Part #: 247001 Supplier: 00426
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A50110	Sealant - Fuel Tank	BMS5-45 Class B-2
A50155	Sealant - Fuel Tank	BMS5-45 Class C
A50231	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class B
A50296	Sealant - Pressure And Environmental - Chromate Type	BMS5-95 Class C
B00083	Solvent - VM&P Naphthas	TT-N-95 Type II, ASTM D-3735 Type III
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
C00259	Coating - Chemical And Solvent Resistant Finish, Corrosion Inhibiting Primer	BMS10-11 Type I

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(Continued)

Reference	Description	Specification
C50178	Coating - Chemical And Solvent Resistant Finish - (BAC 707 Gray Color)	BMS10-11 Type II
C50475	Coating - Chemical Conversion - Bonderite M-CR 600 Aero	BAC5719 Type II Class D
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
D50305	Compound - Organic Corrosion Inhibiting	BMS3-23 Type II
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50347	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Alternate brake selector valve	32-41-93-01-050	PTF ALL
4	Packing	32-41-93-01-035	PTF ALL
9	Packing	32-41-93-01-035	PTF ALL
12	Packing	32-41-93-01-030	PTF ALL
15	Packing	32-41-93-01-030	PTF ALL
21	Packing	32-41-93-01-035	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

F. Prepare for the Installation

SUBTASK 32-41-93-110-001

(1) Prepare the mating surfaces for installation of the alternate brake selector valve [1].

(a) Clean the mating surface of the alternate brake selector valve [1] with solvent, B00083.

G. Alternate Brake Selector Valve Installation

SUBTASK 32-41-93-640-001

(1) If the union [8], reducer [5], reducer [11], reducer [14], packing [4], packing [9], packing [12], and packing [15] are not installed, do these steps:

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(a) Lubricate the union [8], reducer [5], reducer [11], reducer [14], new packing [4], packing [9], packing [12], and packing [15] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the union [8], reducer [5], reducer [11], reducer [14], new packing [4], packing [9], packing [12], and packing [15], in their locations on the alternate brake selector valve [1].

PTF ALL; AIRPLANES WITH ALTERNATE BRAKE SELECTOR VALVE WITH PLUG P/N 274T4629-1

1) Tighten the reducer [5] to 270 ±14 in-lb (30.5 ±1.6 N·m).

PTF ALL; AIRPLANES WITH ALTERNATE BRAKE SELECTOR VALVE WITH PLUG P/N 274T4629-4

2) Tighten the reducer [5] to 170 ±9 in-lb (19.2 ±1.0 N·m).

PTF ALL

3) Tighten the reducer [11] to 280 ±14 in-lb (31.6 ±1.6 N·m).

4) Tighten the reducer [14] to 280 ±14 in-lb (31.6 ±1.6 N·m).

5) Tighten the union [8] to 170 ±9 in-lb (19.2 ±1.0 N·m).

SUBTASK 32-41-93-420-001

(2) If the pressure switch [2] is not installed in the alternate brake selector valve [1], do these steps:

(a) Lubricate a new packing [21] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the packing [21] on the pressure switch [2].

(c) Install the pressure switch [2] on the alternate brake selector valve [1].

1) Tighten the pressure switch [2] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(d) Install MS20995NC32 lockwire, G50347, on the pressure switch [2] and alternate brake selector valve [1] with the double twist method.

SUBTASK 32-41-93-080-001

(3) Do these steps to install the alternate brake selector valve [1]:

(a) Clean the alternate brake selector valve [1] and bracket per abrasive cleaning method (SWPM 20-20-00, Paragraph 2).

(b) Apply a layer of sealant, A50110, or sealant, A50155, or sealant, A50231, or sealant, A50296, to the bracket or alternate brake selector valve [1] (SWPM 20-20-00 Paragraph 14.D).

(c) Apply corrosion inhibiting material, G50136, to all areas of the hole in the alternate brake selector valve [1].

(d) Immediately install the washers [17] and nuts [16] to attach the alternate brake selector valve [1] to the mating surface.

(e) Make sure that the squeeze-out is continuous and visible along edges of bonding interface.

(f) Measure the electrical resistance between the alternate brake selector valve [1] and bracket (SWPM 20-20-00).

1) Use a non-intrinsically safe bonding meter, COM-614.

2) Make sure that the electrical bonding resistance is 0.0025 ohm (2.5 milliohm) or less.

(g) Manually apply coating, C00064, or Bonderite M-CR 600 Aero coating, C50475.

1) Apply one coat of primer, C00259, and coating, C50178.

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2) Apply corrosion inhibiting compound, D50305.

(h) Remove the caps/plugs, STD-14686, from the hydraulic ports of the alternate brake selector valve [1].

(i) Remove the caps/plugs, STD-14686, from the hydraulic line [6], elbow [7], T-fitting [10], and elbow [13].

(j) Connect the hydraulic line [6] to the normal port on the alternate brake selector valve [1].

1) Tighten the B-nut on hydraulic line [6] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(k) Connect the elbow [7] to the return port on the alternate brake selector valve [1].

1) Tighten the B-nut on the elbow [7] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(l) Connect the T-fitting [10] to the brake port on the alternate brake selector valve [1].

1) Tighten the B-nut on the T-fitting [10] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(m) Connect the elbow [13] to the alt pressure port on the alternate brake selector valve [1].

1) Tighten the B-nut on the elbow [13] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(n) Before you connect the electrical connector [3] to the pressure switch [2] on the alternate brake selector valve [1], examine the connector for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(o) If there was corrosion, correct the problem (SWPM 20-60-03).

(p) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [3] (SWPM 20-60-03).

(q) Connect the electrical connector [3] to the pressure switch [2] on the alternate brake selector valve [1].

SUBTASK 32-41-93-790-001

WARNING

KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) For the A and the B hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-93-870-002

(5) Bleed the brake hydraulic system:

(a) Operate the brake pedals 6 times.

1) Operate the pedals slowly and wait a minimum of 5 seconds between each operation.

SUBTASK 32-41-93-710-001

(6) Do this task: Alternate Brake Selector Valve Operational Test, TASK 32-41-93-710-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-93-710-006

(7) Do this task: Gear Retract Braking Operational Test, TASK 32-41-93-710-802.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-93-860-011

(1) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-93-860-014

(2) Set the FLT CONTROL A and B switches, on the P5 forward overhead panel, to the ON position.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ALTERNATE BRAKE SELECTOR VALVE - ADJUSTMENT/TEST

1. General

A. This procedure contains these tasks:

(1) Alternate Brake Selector Valve Test

(2) Gear Retract Braking Operational Test.

TASK 32-41-93-710-801

2. Alternate Brake Selector Valve Operational Test

NOTE: This procedure is a scheduled maintenance task.

A. References

Reference	Title
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right
710	Subzone - Landing Gear: Nose Landing Gear and Landing Gear Doors
730	Subzone - Left Main Landing Gear and Landing Gear Doors
740	Subzone - Right Main Landing Gear and Landing Gear Doors

C. Prepare for the Test

SUBTASK 32-41-93-480-004



WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-93-790-002



WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Make sure that the A and the B hydraulic systems are pressurized (TASK 29-11-00-860-801).

D. Alternate Brake Selector Valve Operational Test (Hydraulic System B)

SUBTASK 32-41-93-720-001

(1) Push the brake pedals 5 times to apply and release the brakes.

(a) Wait approximately 3 seconds each time the brakes are applied.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-93-720-002

(2) Make sure that the brake pistons move to show that the brakes operate correctly.

E. Alternate Brake Selector Valve Operational Test (Hydraulic System A)

SUBTASK 32-41-93-860-003

(1) For hydraulic system B, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-93-210-001

(2) Make sure that the BRAKE PRESS indicator gage on the First Officer's panel shows 3000 ±100 psi (20,684 ±689 kPa) and stays constant during the brake operation when you do the steps that follow.

SUBTASK 32-41-93-720-003

(3) Push the brake pedals 5 times to apply and release the brakes.

SUBTASK 32-41-93-720-004

(4) Make sure that the brake pistons move to show that the brakes operate correctly.

SUBTASK 32-41-93-710-002

(5) Do this test of the ANTISKID INOP light:

(a) Remove the connector from one of the alternate antiskid valves.

(b) Make sure that the ANTISKID INOP light, on the P2 panel, comes on.

(c) Attach the connector to the alternate antiskid valve.

SUBTASK 32-41-93-860-004

(6) After five to ten minutes, do a check for leaks around the alternate brake selector valve.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-93-860-013

(1) For hydraulic system A, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-41-93-710-802

3. Gear Retract Braking Operational Test

(Figure 501)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

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Reference	Description
SPL-10813	Gauge Set, Pressure Check - Hydraulic Brake System Part #: C32051-1 Supplier: 81205 Opt Part #: F72977-45 Supplier: 81205 Opt Part #: F72977-46 Supplier: 81205 Opt Part #: F72977-47 Supplier: 81205 Opt Part #: F72977-62 Supplier: 81205
SPL-15031	Test Equipment - Brake Bleed Part #: G32075-16 Supplier: 81205

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Prepare for the Operational Test

SUBTASK 32-41-93-480-009

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that the nose and main landing gear downlock pins are installed (TASK 32-00-01-480-801).

SUBTASK 32-41-93-860-015

(2) For the normal (system B) and alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

E. Gear Retract Braking Operational Test

SUBTASK 32-41-93-480-010

(1) Do these steps to install a pressure gage at each brake:

(a) Disconnect the brake at the brake disconnect:

1) Push the barrel toward the brake and turn the barrel in the clockwise direction to disengage the brake half from the hose half.

2) Pull up on the barrel of the brake disconnect, away from the brake, until the threads of the two halves are engaged.

3) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the brake bleeder assembly from the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(c) Install the pressure gauge, SPL-15031, or gauge set, SPL-10813, on the brake bleed port.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

SUBTASK 32-41-93-790-003

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the A and the B hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-93-860-006

WARNING

MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR BEFORE YOU MOVE THE CONTROL LEVER. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Put the landing gear control lever in the UP position.

SUBTASK 32-41-93-870-003

(4) Wait 2 seconds.

SUBTASK 32-41-93-710-003

(5) Make sure that the hydraulic pressure at all brakes is 537.5 ±37.5 psig (3705.9 ±258.6 kPa).

SUBTASK 32-41-93-710-004

(6) Put the landing gear control lever in the DN position.

SUBTASK 32-41-93-710-005

(7) Make sure that the hydraulic pressure at all of the brakes is less than 60 psig (414 kPa).

SUBTASK 32-41-93-860-016

(8) For the normal (system B) and alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-93-080-002

(9) Do these steps to remove the pressure gages at each brake:

(a) Disconnect the brake at the brake disconnect:

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the pressure gauge, SPL-15031, or gauge set, SPL-10813, from the brake bleed port.

(c) Install the brake bleeder assembly on the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(d) Connect the brake at the brake disconnect:

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1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.
2) Push down on the barrel and turn it in the clockwise direction to engage the threads.
3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-93-860-017

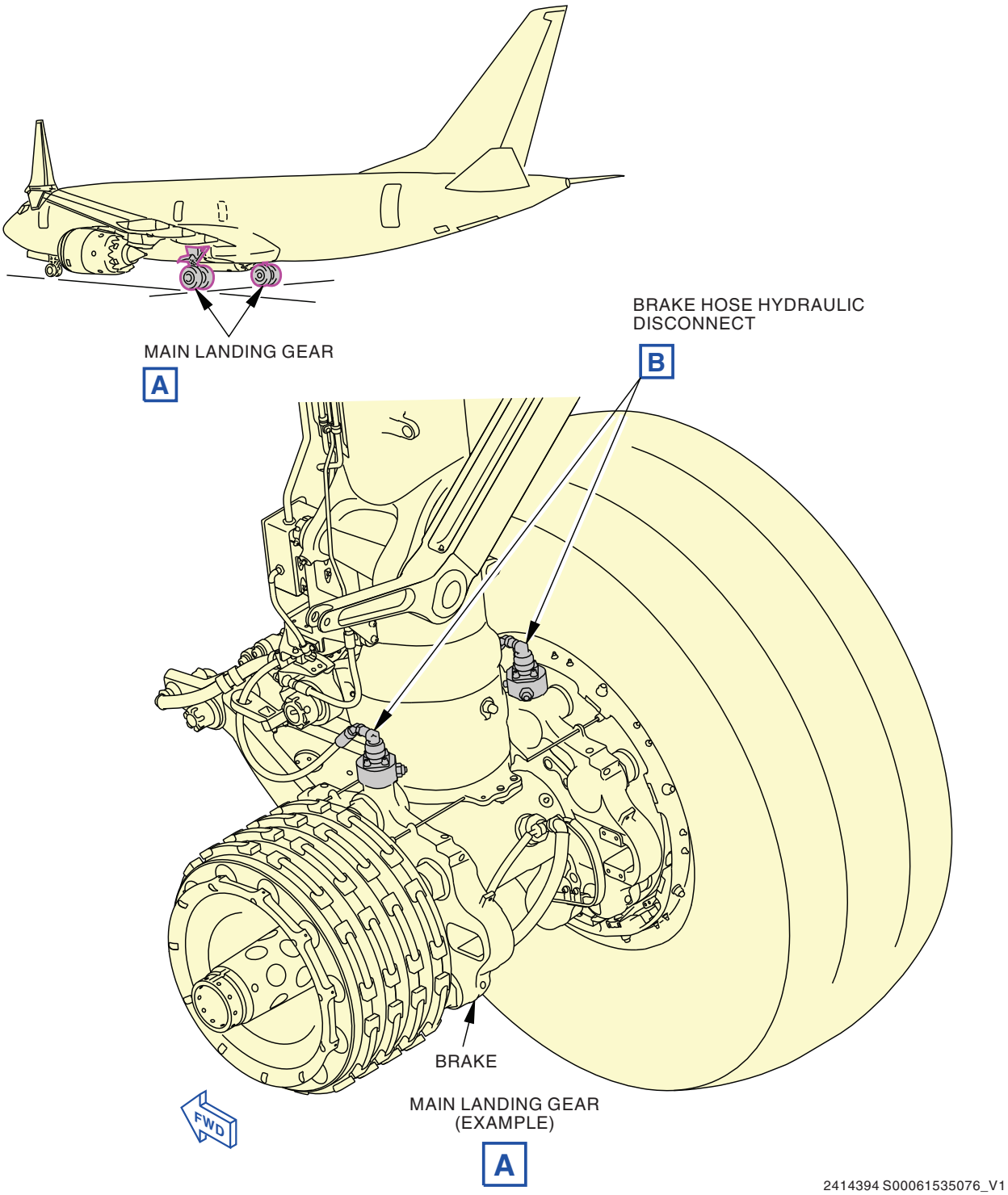
(1) Pressurize hydraulic systems A and B, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-41-93-860-008

(2) For the A and the B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

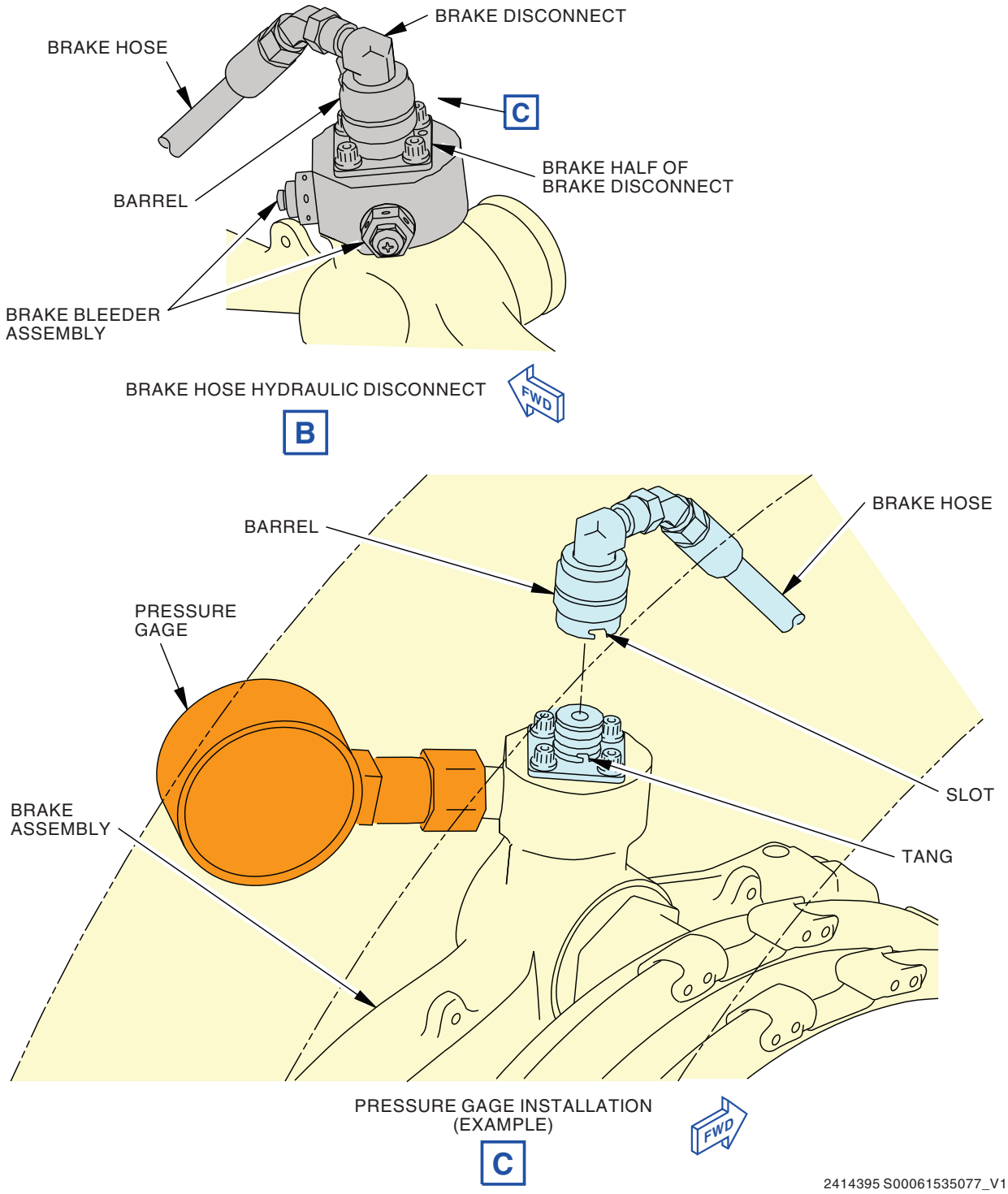
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Line Disconnection and Pressure Gage Installation
Figure 501/32-41-93-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Line Disconnection and Pressure Gage Installation
Figure 501/32-41-93-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE SHUTTLE VALVES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the brake shuttle valves

(2) An installation of the brake shuttle valves.

B. There are four brake shuttle valves (two on the right and the two on the left) on the sides of the main gear wheel well. This procedure contains the removal and installation tasks for each of the brake shuttle valves.

C. The brake shuttle valve is referred to as the shuttle valve in this procedure.

TASK 32-41-95-000-801

2. Brake Shuttle Valve - Removal

(Figure 401)

A. References

Reference	Title
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 32-41-95-480-004

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-95-860-001

(2) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-41-95-840-001

(3) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Brake Shuttle Valve Removal

SUBTASK 32-41-95-020-001

(1) Do these steps to remove the shuttle valve [1]:

(a) Disconnect the hydraulic lines from the shuttle valve [1].

(b) Install caps/plugs, STD-14686, on the hydraulic lines.

(c) Remove the bolts [3] or bolts [7] and washers [4].

(d) Remove the shuttle valve [1] from the airplane.

SUBTASK 32-41-95-020-002

(2) If the shuttle valve [1] you will install does not have the unions [5] installed, do these steps:

(a) Remove the unions [5] from the old shuttle valve [1].

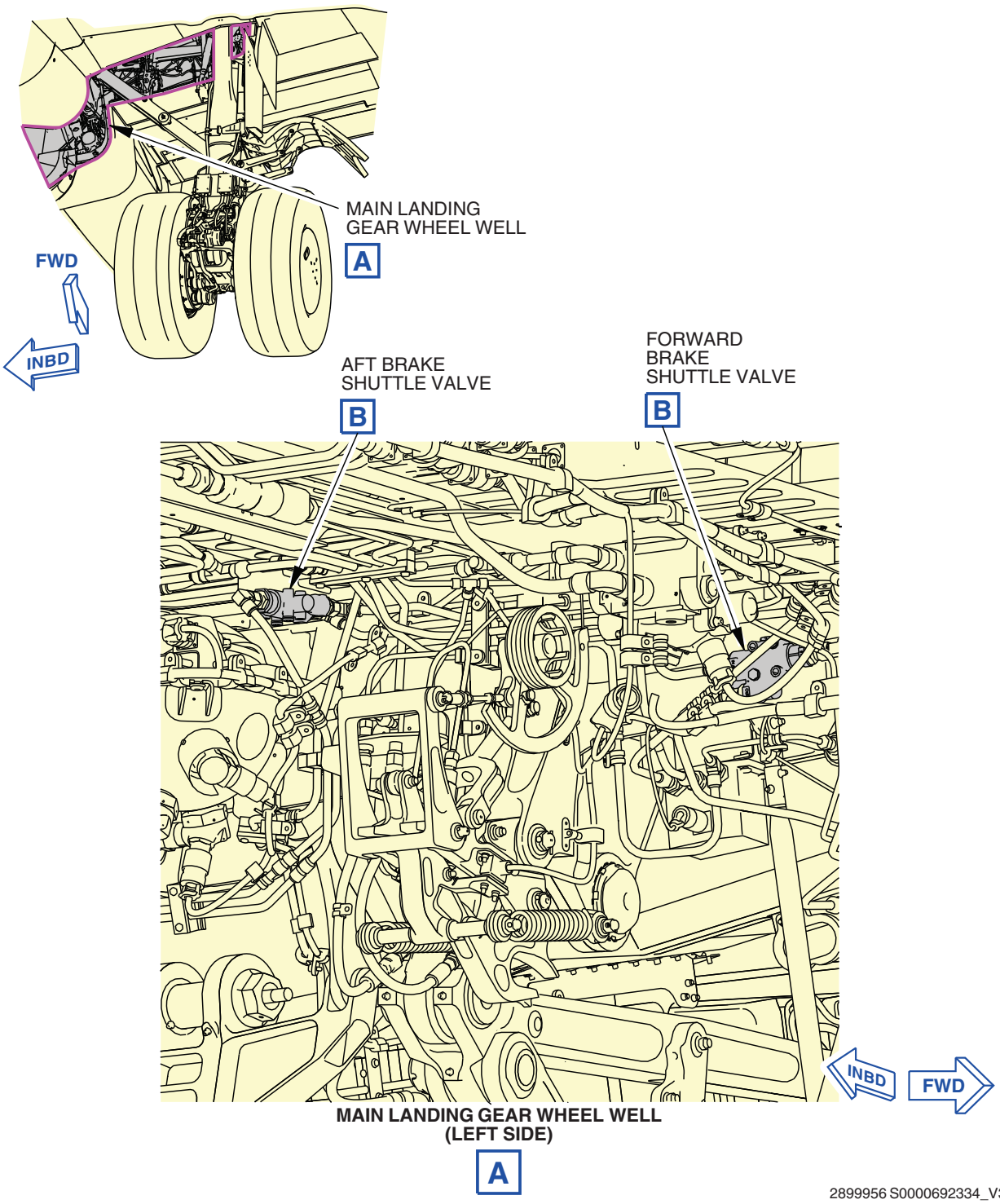
(b) Remove and discard the packings [2] from the applicable fitting.

SUBTASK 32-41-95-480-003

(3) Install caps/plugs, STD-14686, on the ports of the shuttle valve [1].

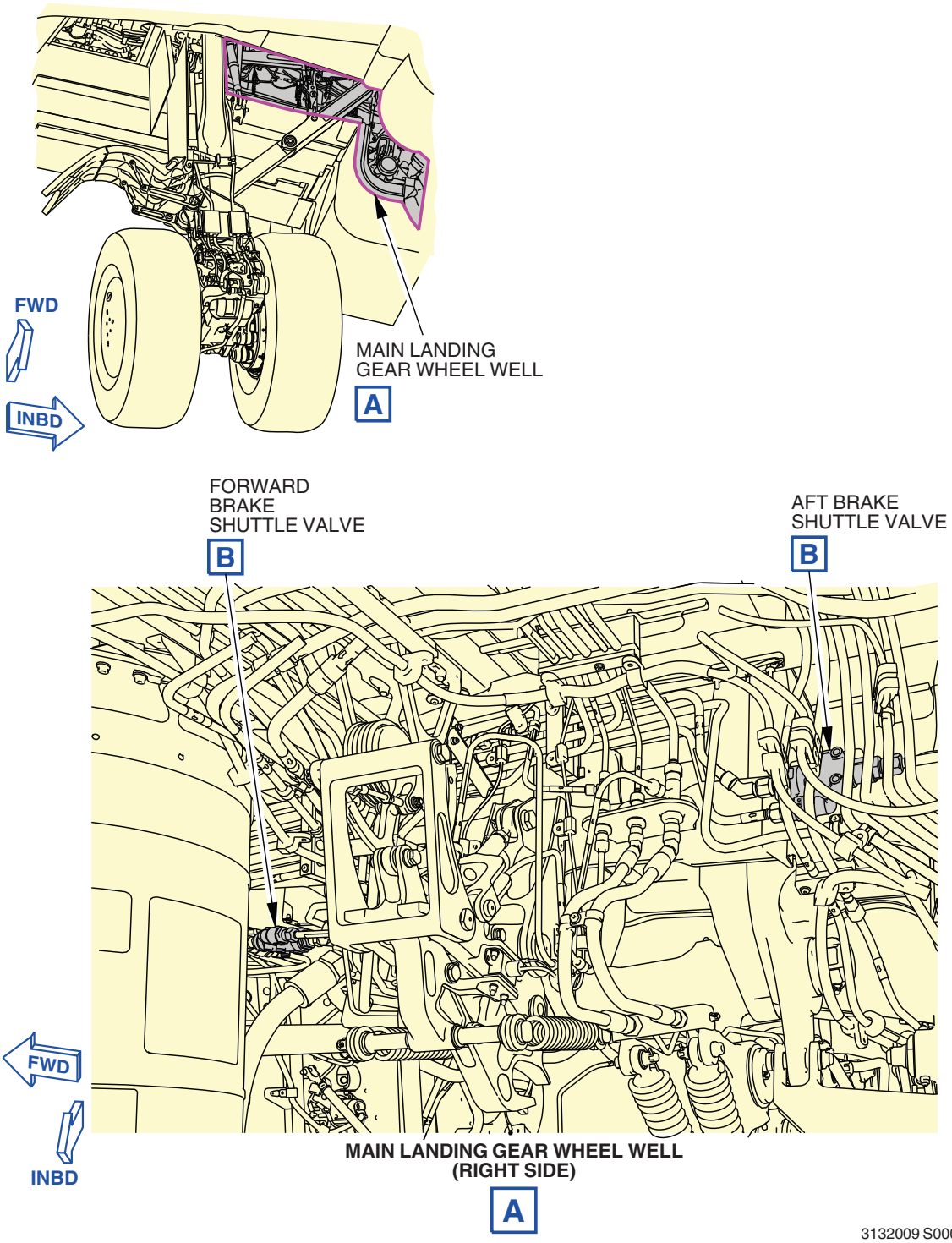
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Shuttle Valves Installation
Figure 401/32-41-95-990-801 (Sheet 1 of 3)

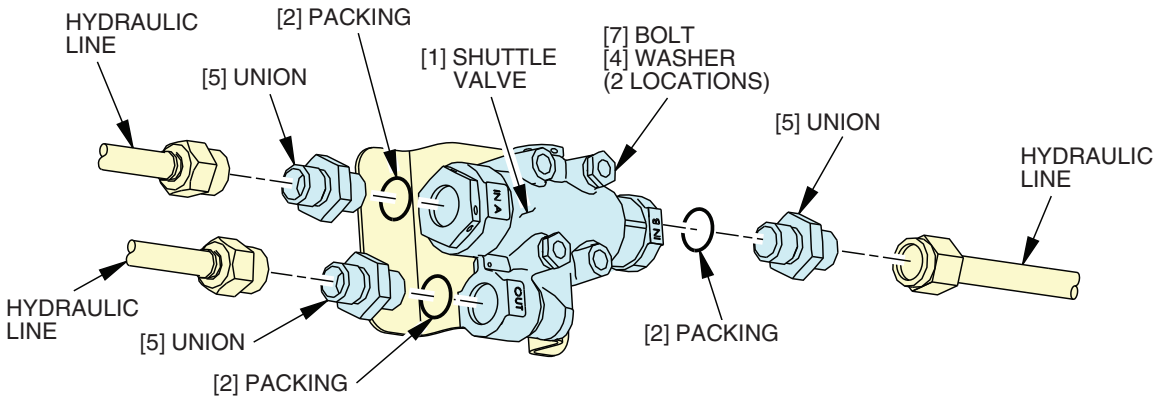
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Brake Shuttle Valves Installation
Figure 401/32-41-95-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



BRAKE SHUTTLE VALVE (EXAMPLE)

B

2414389 S00061535061_V6

Brake Shuttle Valves Installation
Figure 401/32-41-95-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-95-400-801

3. Brake Shuttle Valve Installation

(Figure 401)

A. References

Reference	Title
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)
32-41-00-870-803	Alternate (System A) Hydraulic Brake System Bleeding (P/B 201)

B. Consumable Materials

Reference	Description	Specification
D00054	Fluid - Hydraulic, Assembly Lubricant, MCS 352B	
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Shuttle valve	32-41-95-02-010	PTF ALL
		32-41-95-03-010	PTF ALL

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Brake Shuttle Valve Installation

SUBTASK 32-41-95-020-003

(1) Remove the caps/plugs from the ports of the shuttle valve [1].

SUBTASK 32-41-95-410-001

(2) If the unions [5] are not installed in the shuttle valve [1], do these steps:

(a) Lubricate the new packings [2] and the unions [5] with hydraulic fluid, D00153, or MCS 352B fluid, D00054.

(b) Install the packings [2] on the unions [5].

(c) Install the unions [5] in the ports of the shuttle valve [1].

1) Tighten the unions [5] to 162 in-lb (18.3 N·m) - 178 in-lb (20.1 N·m).

SUBTASK 32-41-95-420-001

CAUTION MAKE SURE THAT YOU POINT AND CONNECT THE BRAKE SHUTTLE VALVE CORRECTLY. THE INCORRECT CONNECTION OF THE VALVE CAN CAUSE DAMAGE TO THE EQUIPMENT AND BRAKE SYSTEM MALFUNCTION.

(3) Do the steps that follow to install the shuttle valve [1]:

(a) Make sure that the shuttle valve [1] and the mounting bracket are clean.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Hold the shuttle valve [1] on the mounting bracket.

(c) Install the bolts [3] or bolts [7] and washers [4].

(d) Remove the caps/plugs from the applicable hydraulic lines.

(e) Connect the hydraulic lines to the shuttle valve [1].

1) Tighten the B-nuts on the hydraulic lines to 270 ±14 in-lb (31 ±2 N·m).

SUBTASK 32-41-95-790-001

(4) To bleed the brake hydraulic systems, do these tasks: Normal (System B) Hydraulic Brake System Bleeding, TASK 32-41-00-870-802 and Alternate (System A) Hydraulic Brake System Bleeding, TASK 32-41-00-870-803.

(a) Make sure that there is no leakage from the shuttle valve [1] as you bleed the brake hydraulic system.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-95-860-003

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

BRAKE FLOW LIMITERS - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the brake flow limiters.

(2) Installation of the brake flow limiters.

B. There are four brake flow limiters, one for each main landing gear brake. The procedure that follows gives the removal and installation for each of the brake flow limiters.

TASK 32-41-96-000-801

2. Brake Flow Limiter Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

D. Prepare for Removal

SUBTASK 32-41-96-480-001

WARNING MAKE SURE THAT YOU INSTALL THE DOWNLOCKS ON THE NOSE AND MAIN LANDING GEAR. WITHOUT THE DOWNLOCKS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE THE EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-41-96-860-001

(2) For the A and B hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802 .

SUBTASK 32-41-96-480-002

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-41-96-860-002

(4) Release the parking brake.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-41-96-840-001

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the brake accumulator.

E. Brake Flow Limiter Removal

SUBTASK 32-41-96-020-001

(1) Do the steps that follow to remove the flow limiter [1].

(a) Loosen the coupling nut [4] and disconnect the hydraulic line [3] from the flow limiter [1].

(b) Loosen the coupling nut [5] and disconnect the high pressure flexible hose [2] from the outboard end of the flow limiter [1].

(c) Install caps/plugs, STD-14686, on the hydraulic lines [3].

(d) Loosen and remove the nut [6] and washers [7] from the flow limiter [1].

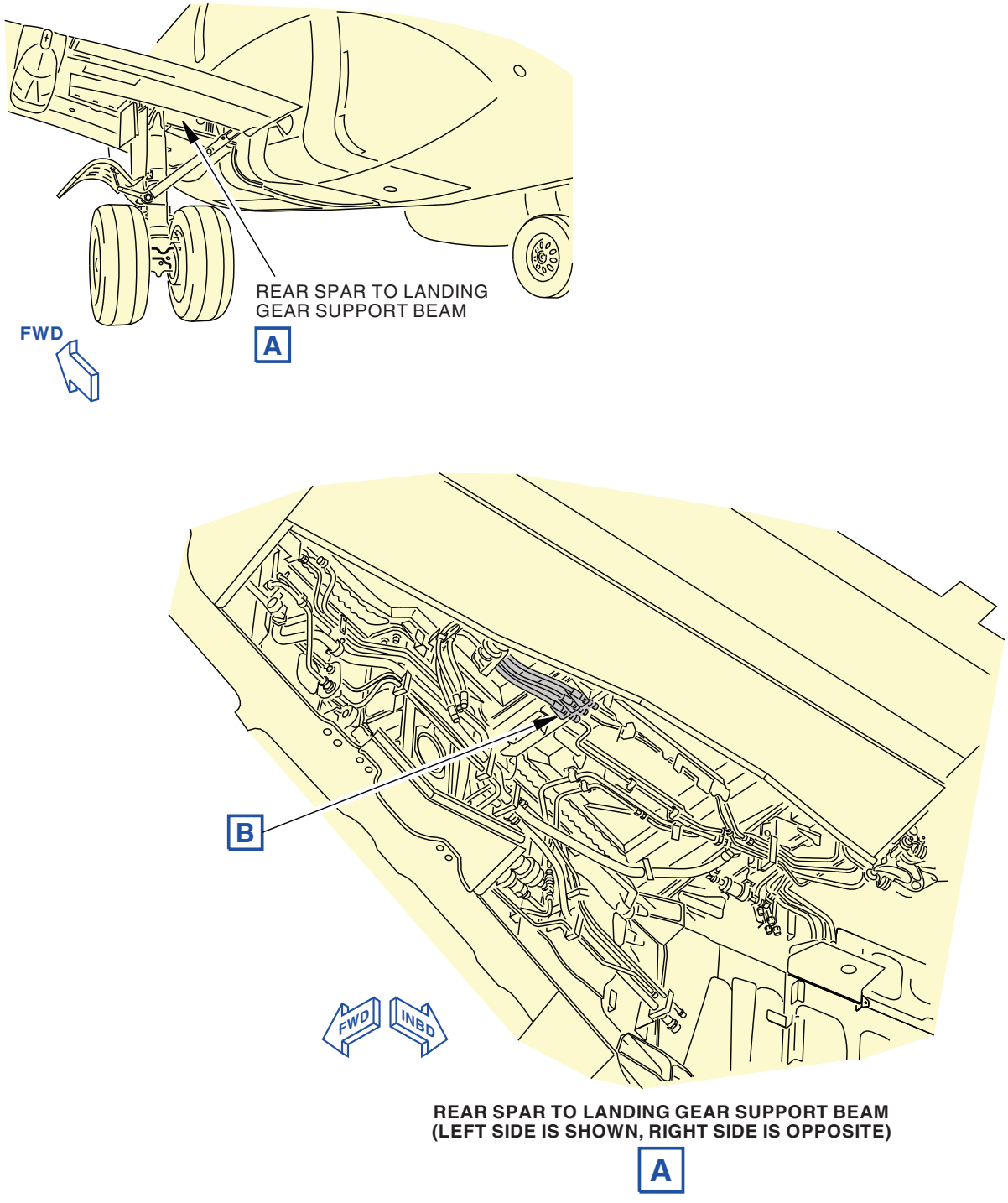
(e) Remove the flow limiter [1] from the bulkhead.

SUBTASK 32-41-96-480-003

(2) Install caps/plugs, STD-14686, on the ports of the flow limiter [1].

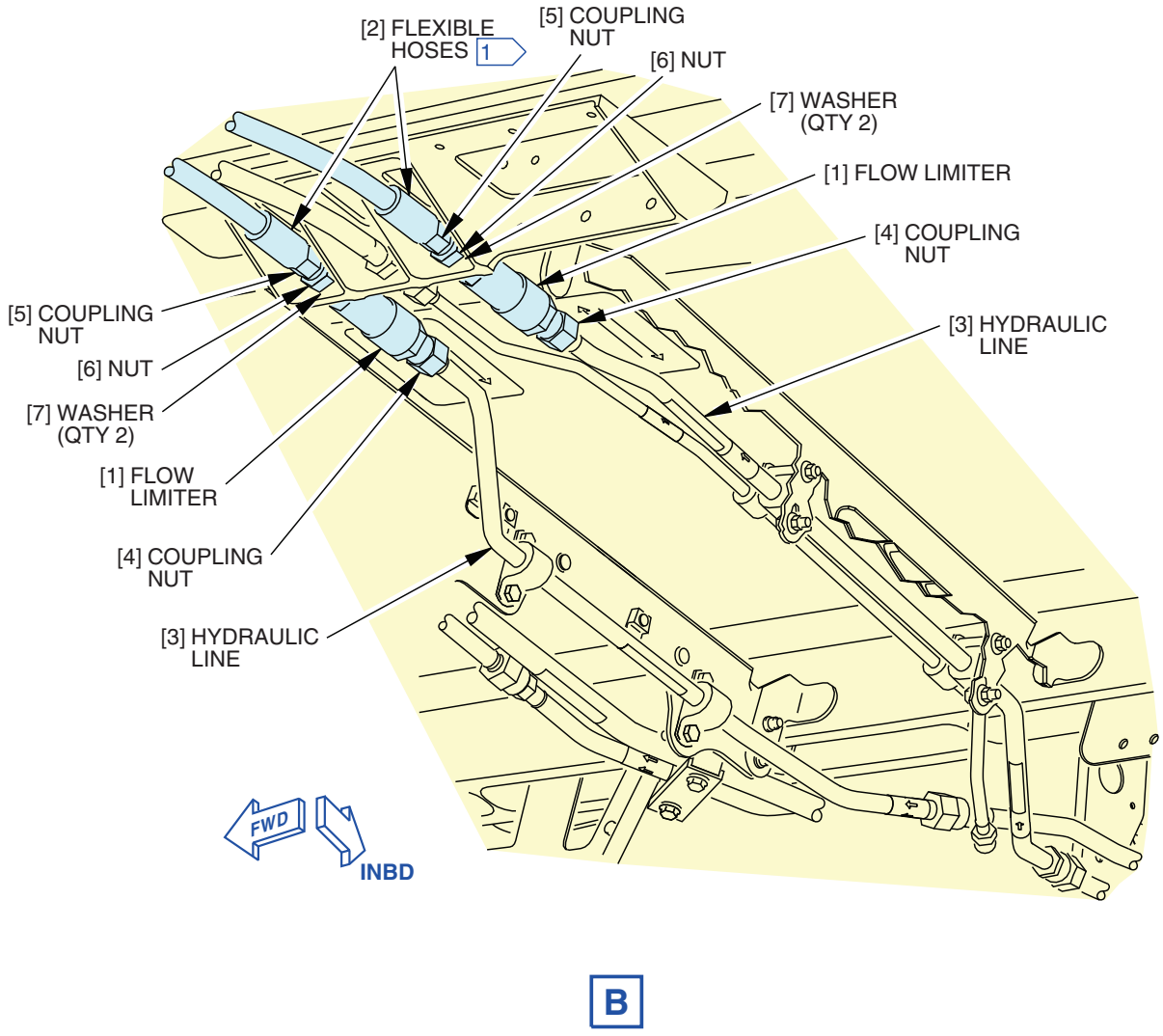
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Flow Limiter Installation
Figure 401/32-41-96-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 MAKE SURE THAT THE HOSES ARE CONNECTED TO THE CORRECT BRAKE POSITIONS (THEY ARE NOT CROSSED).

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Brake Flow Limiter Installation
Figure 401/32-41-96-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-41-96-400-801

3. Brake Flow Limiter Installation

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)
32-41-00-870-803	Alternate (System A) Hydraulic Brake System Bleeding (P/B 201)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Flow limiter	32-41-96-01-050	PTF ALL
		32-41-96-02-050	PTF ALL

D. Location Zones

Zone	Area
551	Left Wing - Rear Spar To Landing Gear Support Beam
651	Right Wing - Rear Spar to Landing Gear Support Beam

E. Brake Flow Limiter Installation

SUBTASK 32-41-96-020-002

(1) Remove the caps/plugs, STD-14686, from the ports of the flow limiter [1].

SUBTASK 32-41-96-030-001

(2) Remove the caps/plugs, STD-14686, from the ends of the hydraulic lines [3].

SUBTASK 32-41-96-420-001

(3) Do these steps to install the flow limiter [1]:

(a) Make sure that the flow limiter [1] and bulkhead are clean.

(b) Install the washer [7] on the end of the flow limiter [1] that will fit into the opening in the bulkhead.

(c) Install the flow limiter [1] in the bulkhead.

(d) Install the washer [7] and nut [6] on the flow limiter [1] on the opposite side of the bulkhead and tighten the nut [6].

(e) Install and connect the hydraulic line [3] to the inboard end of the flow limiter [1].

1) Tighten the coupling nut [4] on the hydraulic line [3] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(f) Install and connect the high pressure flexible hose [2] to the outboard end of the flow limiter [1].

1) Make sure that the hose is connected to the correct brake position and not cross-connected.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Tighten the coupling nut [5] on the high pressure flexible hose [2] to 140 ±7 in-lb (15.8 ±0.8 N·m).

F. Brake Flow Limiter Installation Test

SUBTASK 32-41-96-790-001

(1) Do the leak check of the flow limiter [1].

(a) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Push the brake pedals to the limit 10 or more times.

(c) After five minutes, make sure that there is no leakage from the ends of the flow limiter [1].

(d) Do these tasks: Normal (System B) Hydraulic Brake System Bleeding, TASK 32-41-00-870-802 and Alternate (System A) Hydraulic Brake System Bleeding, TASK 32-41-00-870-803.

(e) Do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-41-96-860-003

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-41-96-860-004

(2) Make sure that the chocks installed around tires (PAGEBLOCK 10-11-05/201).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTISKID/AUTOBRAKE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) An Antiskid/Autobrake Control Unit Operational Test

(2) A Normal and Alternate Antiskid Valve Functional Test

(3) An Antiskid Transducer Functional Test

(4) A Normal and Alternate Antiskid Valve Operational Test

(5) An Autobrake Pressure Control Module Functional Test

(6) An Autobrake Shuttle Valve Operational Test

(7) An Antiskid Transducer Operational Test

(8) An Antiskid/Autobrake Control Unit BITE Procedure

TASK 32-42-00-720-801

2. Antiskid/Autobrake Control Unit Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Test

SUBTASK 32-42-00-480-001

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-002

(3) Release the parking brake.

SUBTASK 32-42-00-790-001

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) For the A and B hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-003

(5) Make sure that the two thrust levers are in the IDLE position.

SUBTASK 32-42-00-860-004

(6) Make sure that the landing gear control lever is in the DN position.

SUBTASK 32-42-00-860-005

(7) Make sure that the spoilers are stowed.

SUBTASK 32-42-00-860-006

(8) Make sure that the right and left Air Data Inertial Reference System (ADIRS)s are in the NAV mode.

(a) Align the ADIRS (TASK 34-21-00-820-802 or TASK 34-21-00-820-801 or TASK 34-21-00-820-803).

SUBTASK 32-42-00-010-003

(9) To get access to the Antiskid/Autobrake Control Unit (AACU) on the E1-3 shelf, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-860-007

(10) Make sure that the BRAKE TEST selector switch on the AACU is in the NORM position.

E. Antiskid/Autobrake Control Unit Operational Test

SUBTASK 32-42-00-860-008

(1) Do the following steps for AACU:

(a) Push and then release the RESET switch on the AACU.

(b) Make sure that the display on the front panel of the AACU shows MEM CLR and then clears.

(c) Set the AUTOBRAKE selector switch on the P2-2 panel to "OFF" then to position "1".

NOTE: If the AUTOBRAKE DISARM light is on when you do the three subsequent steps, the test will not successfully complete and one or more faults will show on the front panel of the AACU.

(d) Push and hold the ENABLE/VERIFY switch on the AACU.

(e) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will flash WAIT during the test.

(g) Make sure that you see the following indications:

1) Make sure that the AACU display shows TEST END after 10 - 20 seconds.

PTF ALL; AIRPLANES WITH AACU P/N 142-147

2) Make sure that the ANTISKID INOP light on the P2-2 panel flashes two times.

PTF ALL; AIRPLANES WITH AACU P/N 142-219-X

3) Make sure that the ANTISKID INOP light on the P2-2 panel flashes four times.

PTF ALL

4) Make sure that the AUTOBRAKE DISARM light on the P2-2 panel is ON, then OFF, and then remains ON at the end of the test.

(h) Set the autobrake selector switch to the OFF position.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-410-002

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-860-009

(2) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-42-00-860-119

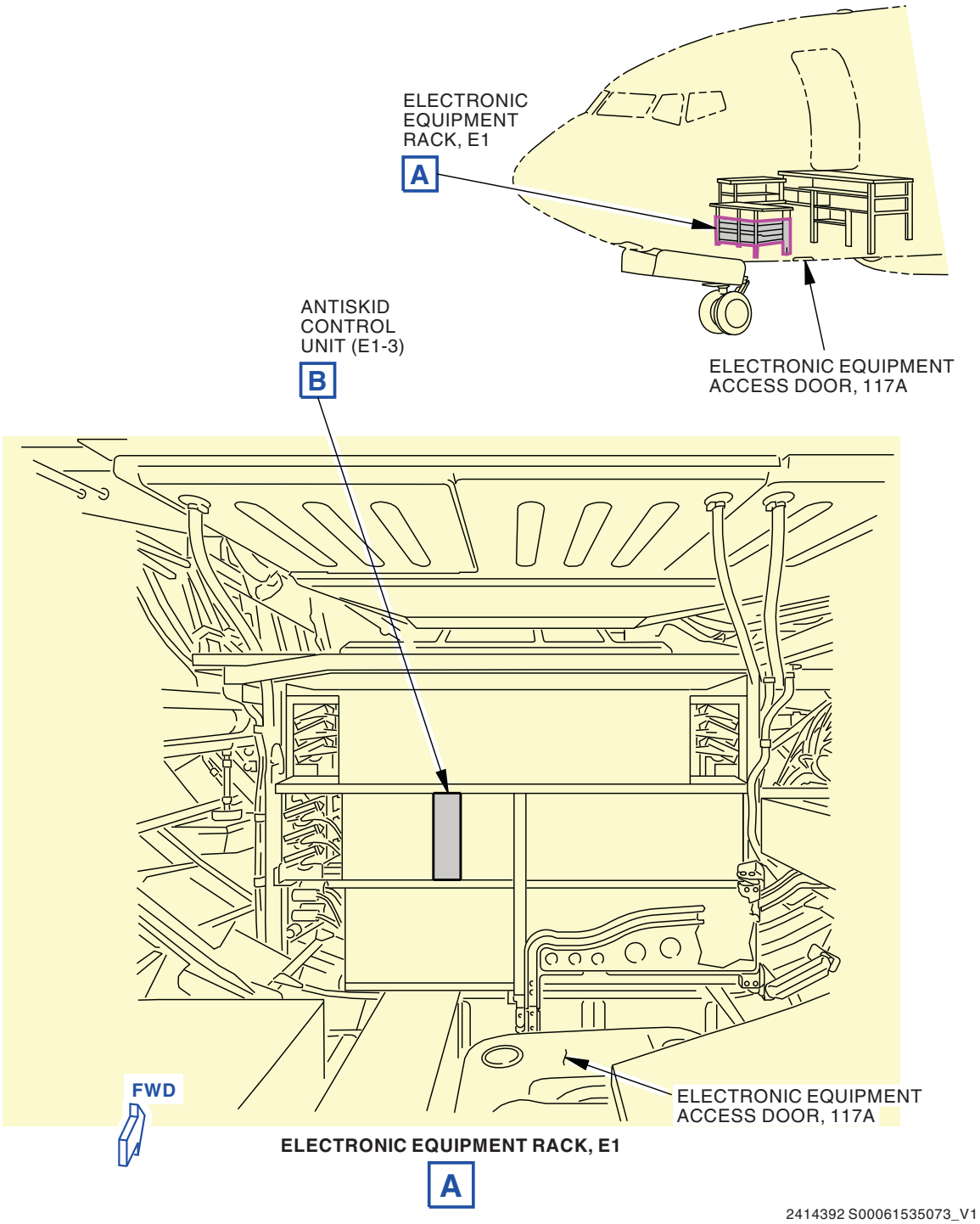
(3) Set the L and R Inertial Reference System (IRS) mode select switches on the Mode Select Unit on the P5 aft overhead panel to the OFF position.

SUBTASK 32-42-00-860-010

(4) Set the parking brake.

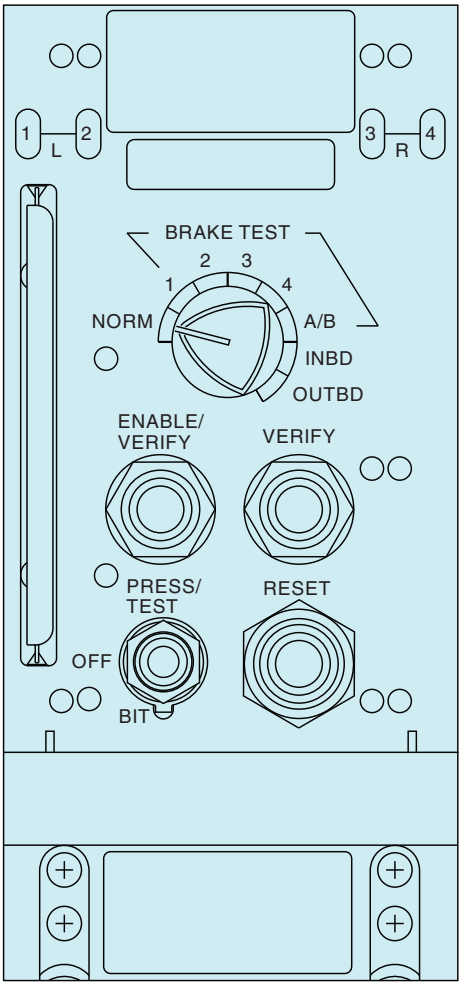
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Antiskid/Autobrake Control Unit Test
Figure 501/32-42-00-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ANTISKID CONTROL UNIT

B

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Antiskid/Autobrake Control Unit Test
Figure 501/32-42-00-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-00-720-802

3. Antiskid Valve Functional Test

(Figure 501, Figure 502)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-801	Hydraulic Reservoirs Pressurization (P/B 201)
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-10813	Gauge Set, Pressure Check - Hydraulic Brake System Part #: C32051-1 Supplier: 81205 Opt Part #: F72977-45 Supplier: 81205 Opt Part #: F72977-46 Supplier: 81205 Opt Part #: F72977-47 Supplier: 81205 Opt Part #: F72977-62 Supplier: 81205
SPL-15031	Test Equipment - Brake Bleed Part #: G32075-16 Supplier: 81205

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Functional Test

SUBTASK 32-42-00-860-115

(1) For the normal (system B) and alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-480-021

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(2) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-00-480-003

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-011

(4) If necessary, release the parking brake.

SUBTASK 32-42-00-480-004

(5) Do these steps to install a pressure gage at each brake:

(a) Disconnect the brake at the brake disconnect:

1) Push the barrel toward the brake and turn the barrel in the clockwise direction to disengage the brake half from the hose half.

2) Pull up on the barrel of the brake disconnect, away from the brake, until the threads of the two halves are engaged.

3) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the brake bleeder assembly from the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(c) Install the pressure gauge, SPL-15031, or the gauge set, SPL-10813, on the brake bleed port.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

SUBTASK 32-42-00-860-013

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) For the normal (system B) hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-014

(7) Disable the parking brake indication:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-015

(8) Set the parking brake.

SUBTASK 32-42-00-010-004

(9) To get access to the antiskid/autobrake control unit (AACU) on the E1-3 shelf, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Normal Antiskid Valve Functional Test

SUBTASK 32-42-00-860-016

(1) Make sure that the hydraulic pressure at the controlled brake is within 200 psi (1379 kPa) of B hydraulic supply pressure.

SUBTASK 32-42-00-860-017

(2) Do these steps for each antiskid valve that you removed:

(a) Move the BRAKE TEST selector switch on the Antiskid/Autobrake Control Unit to the wheel position for the brake valve that you will test.

NOTE: See the top of the Antiskid/Autoskid Control Unit for the brake number.

(b) Push and hold the ENABLE/VERIFY switch on the Antiskid/Autobrake Control Unit.

(c) Push and hold the VERIFY switch on the Antiskid/Autobrake Control Unit.

(d) Release the ENABLE/VERIFY and the VERIFY switches at the same time.

NOTE: The Antiskid/Autobrake Control Unit display will show BRK X during the test, where X = the wheel number selected by the BRAKE TEST selector switch.

(e) Make sure that the hydraulic pressure at the controlled brake is less than or equal to 150 psi (1034 kPa) for 5 to 10 seconds and then applies to within 200 psi (1379 kPa) of B hydraulic supply pressure.

(f) Move the BRAKE TEST selector switch to the NORM position.

G. Alternate Antiskid Valve Functional Test

SUBTASK 32-42-00-860-018

(1) For the normal (system B) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-42-00-860-019

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the alternate (system A) hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-860-020

(3) Do these steps for each alternate antiskid valve that you removed:

(a) Move the BRAKE TEST selector switch on the Antiskid/Autobrake Control Unit to the position for the brake valve that you will test.

NOTE: Either 1 or 2 for the left alternate antiskid control valve, or 3 or 4 for the right alternate antiskid control valve.

(b) Push and hold the ENABLE/VERIFY switch on the Antiskid/Autobrake Control Unit.

(c) Push and hold the VERIFY switch on the Antiskid/Autobrake Control Unit.

(d) Release the ENABLE/VERIFY and the VERIFY switches at the same time.

NOTE: The Antiskid/Autobrake Control Unit display will show BRK X during the test, where X = the wheel number selected by the BRAKE TEST selector switch.

(e) Make sure that the hydraulic pressures at the controlled brakes are less than or equal to 150 psi (1034 kPa) for 5 to 10 seconds and then applies to within 200 psi (1379 kPa) of A hydraulic supply pressure.

(f) Move the BRAKE TEST selector switch to the NORM position.

SUBTASK 32-42-00-860-116

(4) For the alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-42-00-080-001

(5) Do these steps to remove the pressure gages at each brake:

(a) Disconnect the brake at the brake disconnect:

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the pressure gauge, SPL-15031, or the gauge set, SPL-10813, from the brake bleed port.

(c) Install the brake bleeder assembly on the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-410-003

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-860-117

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Pressurize hydraulic systems A and B, do this task: Hydraulic Reservoirs Pressurization, TASK 29-09-00-860-801.

SUBTASK 32-42-00-710-001

(3) Enable the parking brake indication:

(a) Release the parking brake.

(b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

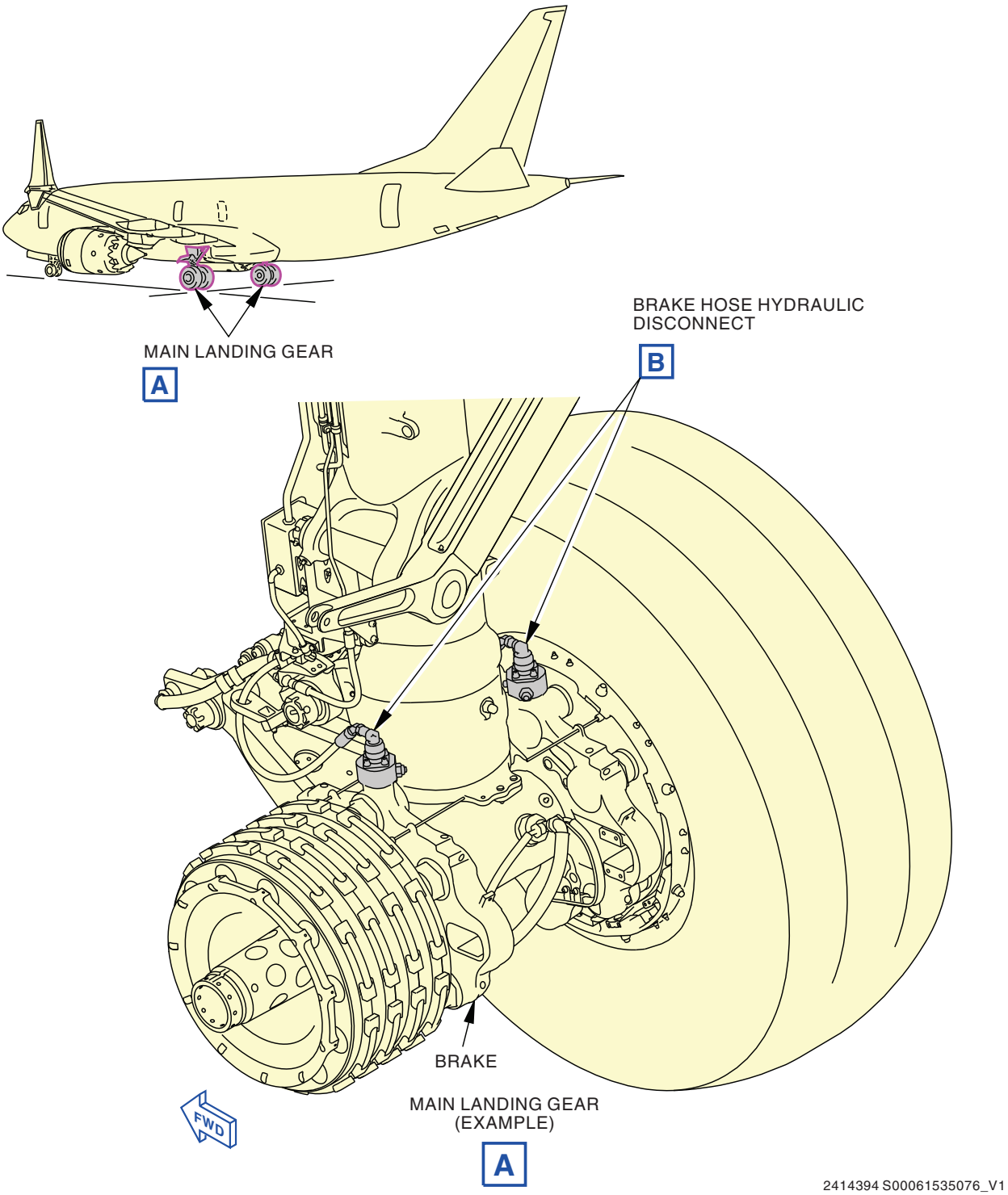
(c) Set the parking brake.

SUBTASK 32-42-00-860-021

(4) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

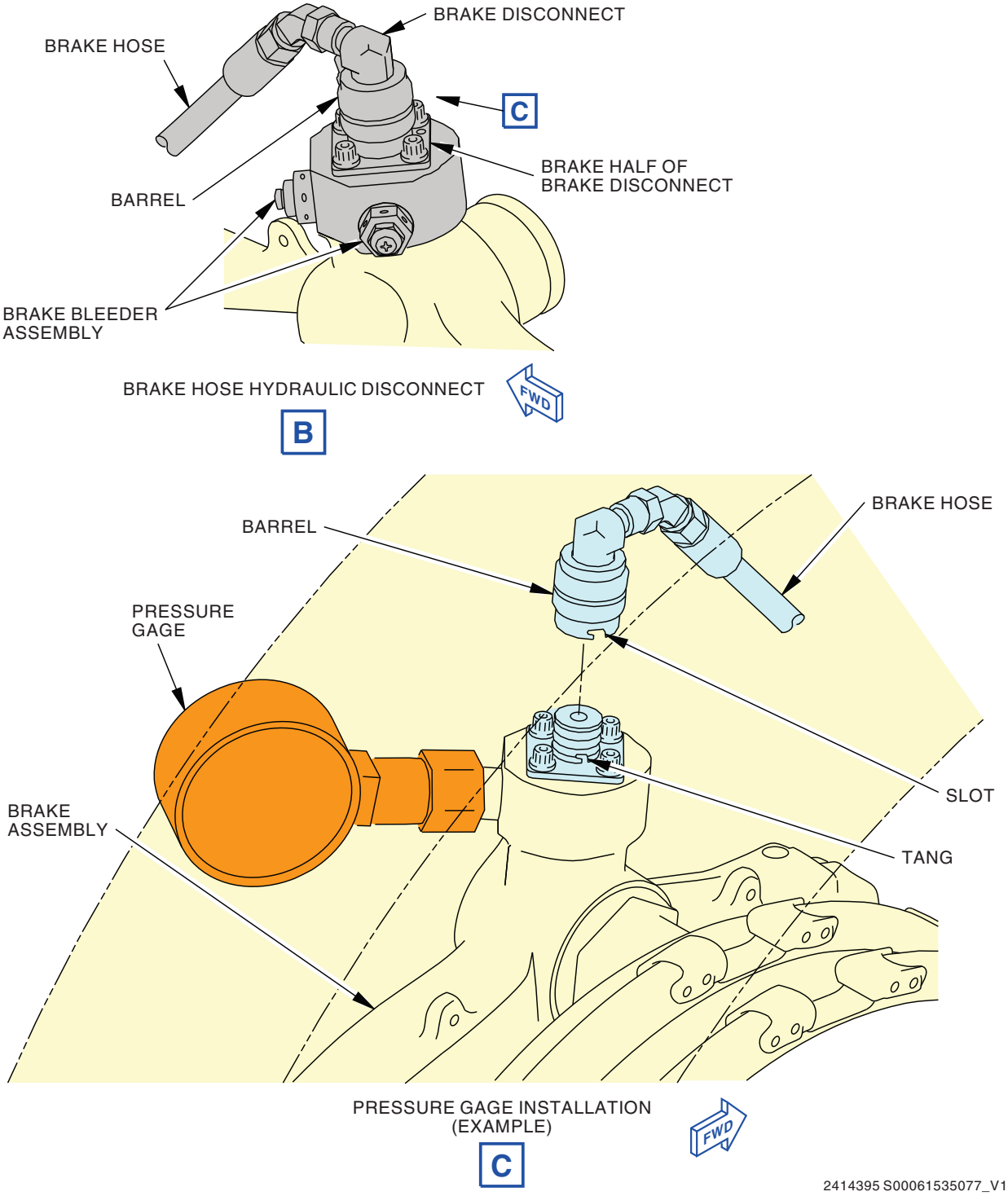
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Line Disconnection and Pressure Gage Installation
Figure 502/32-42-00-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Brake Line Disconnection and Pressure Gage Installation
Figure 502/32-42-00-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-00-710-801

4. Transducer Functional Test

(Figure 502, Figure 503)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-49-03-000-802	Main Landing Gear Hubcap Assembly Removal (P/B 401)
32-49-03-400-802	Main Landing Gear Hubcap Assembly Installation (P/B 401)
46-13-02-700-805	Existing Faults Check (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1831	Spin Up Adapter - Wheel Speed Transducer, Anti-Skid System Part #: A32075-10 Supplier: 81205 Opt Part #: A32075-1 Supplier: 81205 Opt Part #: A32075-9 Supplier: 81205
SPL-10813	Gauge Set, Pressure Check - Hydraulic Brake System Part #: C32051-1 Supplier: 81205 Opt Part #: F72977-45 Supplier: 81205 Opt Part #: F72977-46 Supplier: 81205 Opt Part #: F72977-47 Supplier: 81205 Opt Part #: F72977-62 Supplier: 81205
SPL-15031	Test Equipment - Brake Bleed Part #: G32075-16 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Functional Test

SUBTASK 32-42-00-860-118

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) Make sure that the downlock pins are installed on the nose and main landing gear (TASK 32-00-01-480-801).

SUBTASK 32-42-00-480-006

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-860-023

(3) Release the parking brake.

SUBTASK 32-42-00-480-007

(4) Do these steps to install a pressure gage at each brake (Figure 502):

(a) Disconnect the brake at the brake disconnect:

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the brake bleeder assembly from the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(c) Install the pressure gauge, SPL-15031, or the gauge set, SPL-10813, on the brake bleed port.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

SUBTASK 32-42-00-790-002

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-024

(6) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-025

(7) Set the parking brake.

E. Transducer Functional Test

SUBTASK 32-42-00-020-004

(1) Remove the hubcap [1] or hubcap [10], do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

SUBTASK 32-42-00-860-026

(2) Install the spin up adapter, SPL-1831, in a drill.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-860-027

(3) Make sure that the hydraulic pressure at all brakes is equal to the system B pressure ± 200 psi (1379 kPa).

NOTE: This is to make sure that the antiskid system does not inadvertently release the brakes.

SUBTASK 32-42-00-710-002

(4) Do these steps for each wheel speed transducer [5] that you will test:

NOTE: A minimum of one person must be near the main landing gear brakes.

WARNING

KEEP PERSONS AND EQUIPMENT AWAY FROM THRUST REVERSERS. THE OPERATION OF THE WHEEL SPEED TRANSDUCER CAN CAUSE AUTOMATIC OPERATION OF THE STANDBY HYDRAULIC PUMP AND SUDDEN THRUST REVERSER MOVEMENTS. IF YOU DO NOT DO THIS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Use the spin up adapter, SPL-1831, and a variable speed drill to turn the wheel speed transducer [5] between 250 and 2200 Revolutions Per Minute (RPM).

NOTE: It is recommend to keep the wheel speed below 60 knots. If the aircraft is not in this condition, the standby hydraulic pump can operate automatically.

1) If the wheel speed transducer [5] is spun for more than 30 seconds, then the SPEEDBRAKE DO NOT ARM Light can come on with a correlated Maintenance Message.

a) Do this task, if a correlated Maintenance Message occurs: Existing Faults Check, TASK 46-13-02-700-805.

(b) Visually confirm that the brake at the paired wheel is less than or equal to 150 psi (1034 kPa) (Table 501).

NOTE: The wheels are numbered from the left to the right.

Table 501/32-42-00-993-801 ANTISKID TRANSDUCER PAIRED WHEELS

TRANSDUCER UNDER TEST	PAIRED WHEEL TO CHECK
1	4
2	3
3	2
4	1

(c) Stop the movement of the wheel speed transducer [5] as quickly as possible.

(d) Visually confirm that the brake at the controlled wheel is less than or equal to 150 psi (1034 kPa) (Table 502).

Table 502/32-42-00-993-803 ANTISKID TRANSDUCER CONTROLLED WHEELS

TRANSDUCER UNDER TEST	CONTROLLED WHEEL TO CHECK
1	1
2	2
3	3
4	4

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Wait 5 to 10 seconds.

(f) Make sure that the hydraulic pressure at all brakes is equal to the system B pressure ± 200 psi (1379 kPa).

SUBTASK 32-42-00-420-003

(5) Install the hubcap [1] or hubcap [10], do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

SUBTASK 32-42-00-080-004

(6) Do these steps to remove the pressure gauge at each brake:

(a) Disconnect the brake at the brake disconnect.

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel counterclockwise, to disconnect the brake hose from the brake assembly.

(b) Remove the pressure gauge, SPL-15031, or gauge set, SPL-10813.

(c) Install the brake bleeder assembly on the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(d) Connect the brake at the brake disconnect.

1) Turn the hose half of the brake disconnect, until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it clockwise to engage the threads.

3) Turn the barrel clockwise, until the slots in the bottom of the barrel snap into place on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure the slots on the barrel are fully and correctly engaged with the tangs, on the brake half of the brake disconnect.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-860-028

(1) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-42-00-860-029

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

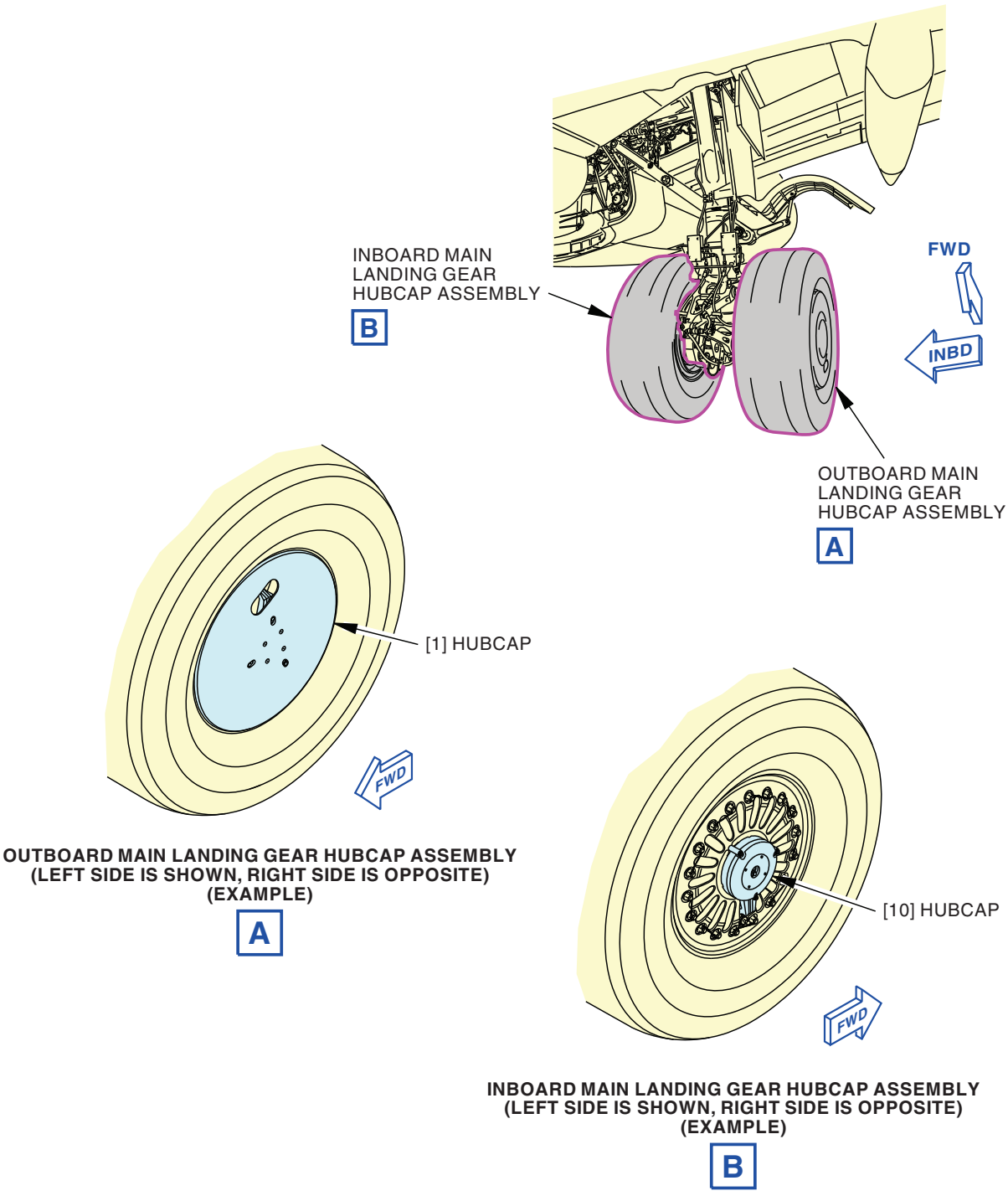
Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-030

(3) Release the parking brake.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Landing Gear Hubcap Installation
Figure 503/32-42-00-990-803

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-00-720-803

5. Antiskid Valve Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

D. Prepare for the Operational Test

SUBTASK 32-42-00-480-008

(1) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-031

(2) Release the parking brake.

SUBTASK 32-42-00-860-033

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) For the normal (system B) hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-034

(4) Disable the parking brake indication:

(a) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-035

(5) Set the parking brake.

SUBTASK 32-42-00-010-005

(6) To get access to the Antiskid/Autobrake Control Unit (AACU) on the E1-3 shelf, do this step:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Normal Antiskid Valve Operational Test

SUBTASK 32-42-00-860-036

(1) Do these steps for each antiskid valve that you removed or will test:

NOTE: One additional person is necessary at the Main Landing Gear (MLG) brakes area.

(a) Move the BRAKE TEST selector switch on the AACU to the wheel position for the brake valve that you test.

NOTE: See the top of the AACU for the brake number.

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) Push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY and the VERIFY switches at the same time.

NOTE: The AACU display will show BRK X during the test, where X = the wheel number selected by the BRAKE TEST selector switch.

(e) Check the movement of the brake pistons or wear pins to make sure that the selected brake releases and then applies again.

(f) Make sure that there are no fault codes on the AACU display.

(g) Move the BRAKE TEST selector switch to the NORM position.

F. Alternate Antiskid Valve Operational Test

SUBTASK 32-42-00-860-037

(1) For the normal (system B) hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-42-00-860-038

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) For the alternate (system A) hydraulic systems, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-039

(3) Do these steps for each alternate antiskid valve that you removed or will test:

NOTE: One additional person is necessary at the MLG brakes area.

(a) Move the BRAKE TEST selector switch on the AACU to the position for the brake valve that you test.

NOTE: Either 1 or 2 for the left alternate antiskid control valve, or 3 or 4 for the right alternate antiskid control valve.

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) Push and hold the VERIFY switch on the AACU.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(d) Release the ENABLE/VERIFY and the VERIFY switches at the same time.

NOTE: The AACU display will show BRK X during the test, where X = the wheel number selected by the BRAKE TEST selector switch.

(e) Check the movement of the brake pistons or wear pins to make sure that the selected brake releases and then applies again.

(f) Make sure that there are no fault codes on the AACU display.

(g) Move the BRAKE TEST selector switch to the NORM position.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-410-004

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-710-003

(2) Enable the parking brake indication:

(a) Release the parking brake.

(b) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

(c) Set the parking brake.

SUBTASK 32-42-00-860-040

(3) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-42-00-400-801

6. Autobrake Pressure Control Module Functional Test

(Figure 501, Figure 502)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

Reference	Description
SPL-10813	Gauge Set, Pressure Check - Hydraulic Brake System Part #: C32051-1 Supplier: 81205 Opt Part #: F72977-45 Supplier: 81205 Opt Part #: F72977-46 Supplier: 81205 Opt Part #: F72977-47 Supplier: 81205 Opt Part #: F72977-62 Supplier: 81205
SPL-15031	Test Equipment - Brake Bleed Part #: G32075-16 Supplier: 81205

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Autobrake Pressure Control Module Functional Test

SUBTASK 32-42-00-860-041

(1) Make sure that the two thrust levers are in the IDLE position.

SUBTASK 32-42-00-860-042

(2) Make sure that the LANDING GEAR control lever is in the DN position.

SUBTASK 32-42-00-860-043

(3) Make sure that the spoilers are stowed (SPD BRK lever in DN position).

SUBTASK 32-42-00-860-044

(4) Make sure that the right and left Air Data Inertial Reference System (ADIRS) are in the NAV mode.

(a) Align the ADIRS (TASK 34-21-00-820-802 or TASK 34-21-00-820-801 or TASK 34-21-00-820-803).

SUBTASK 32-42-00-480-023

WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(5) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-00-480-009

(6) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-045

(7) Release the PARKING BRAKE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-00-860-046

(8) Install the pressure gauge, SPL-15031, or the gauge set, SPL-10813, on each brake for the two Main Landing Gear (MLG).

SUBTASK 32-42-00-480-010

(9) Do these steps to install a pressure gage at each brake:

(a) Disconnect the brake at the brake disconnect:

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the brake bleeder assembly from the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(c) Install the pressure gauge on the brake bleed port.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

SUBTASK 32-42-00-860-047

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(10) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-010-007

(11) To get access to the Antiskid/Autobrake Control Unit (AACU), on the E1-3 shelf, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-860-049

(12) Set the BRAKE TEST selector switch, on the AACU, to A/B.

NOTE: In this step and the next four steps, if the AUTO BRAKE DISARM light is ON when you push the ENABLE/VERIFY switches, the test will not successfully complete and the display on the front panel of the AACU will show "BRK ####."

SUBTASK 32-42-00-860-050

(13) Do the steps that follow for AUTO BRAKE selector position "1".

(a) Set the AUTO BRAKE selector switch, on the P9-82 pane, l to "OFF" then to "1" position.

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(b) Push and then release the RESET switch on the AACU.

NOTE: The display on the front panel of the AACU will show "MEM CLR".

(c) Push and hold the ENABLE/VERIFY switch on the AACU.

(d) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(e) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B1 during the test.

(f) Make sure that the hydraulic pressure at a brake is 1250 ±250 psi (8618 ±1724 kPa) for approximately 10 to 30 seconds, then to 300 ±250 psi (2068 ±1724 kPa) for approximately 5 to 20 seconds, and then to less than 150 psi (1034 kPa).

(g) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(h) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-051

(14) Do the steps that follow for AUTO BRAKE selector position "2".

(a) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "2".

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B2 during the test.

(e) Make sure that the hydraulic pressure at a brake is 1500 ±250 psi (10,342 ±1724 kPa) for approximately 10 to 30 seconds, then to 300 ±250 psi (2068 ±1724 kPa) for approximately 5 to 20 seconds, and then to less than 150 psi (1034 kPa).

(f) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(g) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-052

(15) Do the steps that follow for AUTO BRAKE selector position "3".

(a) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "3".

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B3 during the test.

(e) Make sure that the hydraulic pressure at a brake is 2000 ±250 psi (13,790 ±1724 kPa) for approximately 10 to 30 seconds, then to 300 ±250 psi (2068 ±1724 kPa) for approximately 5 to 20 seconds, and then to less than 150 psi (1034 kPa).

(f) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(g) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-053

(16) Do the steps that follow for AUTO BRAKE selector position "MAX".

(a) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "MAX".

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

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(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B4 during the test.

(e) Make sure that the hydraulic pressure at a brake is 3000 ±300 psi (20,684 ±2068 kPa) for approximately 10 to 30 seconds, then to 300 ±250 psi (2068 ±1724 kPa) for approximately 5 to 20 seconds, and then to less than 150 psi (1034 kPa).

(f) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(g) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-054

(17) Do the steps that follow for AUTO BRAKE selector position "RTO".

(a) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "RTO".

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK RTO during the test.

(e) Make sure that the hydraulic pressure at any brake is 3000 ±300 psi (20,684 ±2068 kPa) for approximately 15 to 45 seconds, then decays to less than 150 psi (1034 kPa).

(f) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(g) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-055

(18) Do the steps that follow to test the brake metered pressure switches.

(a) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "1".

(b) Push and hold the left brake pedal to the stop.

(c) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

(d) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF" and then set it back to "1".

(e) Push and hold the right brake pedal to the stop.

(f) Make sure that the AUTO BRAKE DISARM light, on the P9-82 panel, comes on.

SUBTASK 32-42-00-860-056

(19) Set the AUTO BRAKE selector switch, on the P9-82 panel, to "OFF".

SUBTASK 32-42-00-860-057

(20) Set the BRAKE TEST selector switch, on the AACU, to NORM.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-410-007

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-080-003

(2) Do these steps to remove the pressure gages at each brake:

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(a) Disconnect the brake at the brake disconnect:

1) Pull up on the barrel of the brake disconnect.

2) Turn the barrel in the counterclockwise direction to disconnect the brake hose from the brake assembly.

(b) Remove the pressure gauge, SPL-15031, or the gauge set, SPL-10813, from the brake bleed port.

(c) Install the brake bleeder assembly on the brake.

NOTE: Each brake bleeder assembly has a bleeder valve installed in an adapter. The adapter is installed in the brake.

(d) Connect the brake at the brake disconnect:

1) Turn the hose half of the brake disconnect until the internal tangs align with the slots inside the brake half of the brake disconnect and you can push the hose half down.

2) Push down on the barrel and turn it in the clockwise direction to engage the threads.

3) Turn the barrel in the clockwise direction until the slots in the bottom of the barrel snap into position on the external tangs on the brake half of the brake disconnect.

(e) Do a visual check to make sure that the slots on the barrel are fully and correctly engaged with the tangs on the brake half of the brake disconnect.

SUBTASK 32-42-00-860-114

(3) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-42-00-710-802

7. Autobrake Shuttle Valve Operational Test

(Figure 501)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

C. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Prepare For the Operational Test

SUBTASK 32-42-00-860-060

(1) Make sure that these circuit breakers are closed:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-00-480-011

(2) Make sure that the tires have chocks installed around them CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-061

(3) Release the parking brake.

SUBTASK 32-42-00-860-062

WARNING

KEEP PERSONS AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES. THE AILERONS, ELEVATORS, RUDDER, FLAPS, SLATS, SPOILERS, STABILIZER, AND NOSE GEAR CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(4) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-010-006

(5) To get access to the antiskid/autobrake control unit (AACU) on the E1-3 shelf, do this step:

(a) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

E. Operational Test of the Autobrake Shuttle Valve

SUBTASK 32-42-00-860-064

(1) Set the BRAKE TEST selector switch on the Antiskid/Autobrake Control Unit to A/B.

SUBTASK 32-42-00-860-065

(2) Set the autobrake selector switch on the P2-2 panel to RTO.

SUBTASK 32-42-00-860-066

(3) Push and then release the RESET switch on the Antiskid/Autobrake Control Unit.

NOTE: The display on the front panel of the Antiskid/Autobrake Control Unit will show "MEM CLR".

SUBTASK 32-42-00-860-067

(4) Push and hold the ENABLE/VERIFY switch on the Antiskid/Autobrake Control Unit.

SUBTASK 32-42-00-860-068

(5) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the Antiskid/Autobrake Control Unit.

SUBTASK 32-42-00-860-069

(6) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

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SUBTASK 32-42-00-480-020

(7) Make sure that all of the brakes apply and then release after approximately 10 to 45 seconds.

SUBTASK 32-42-00-860-070

(8) Set the autobrake selector switch on the P2-2 panel to OFF.

SUBTASK 32-42-00-860-071

(9) Set the BRAKE TEST switch on the Antiskid/Autobrake Control Unit to NORM.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-410-005

(1) Close this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-860-072

(2) Push the brake pedals fully to the stops and then release them after approximately 10 seconds.

(a) Make sure that the brakes apply and release with brake pedal movement.

SUBTASK 32-42-00-860-073

(3) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

TASK 32-42-00-710-803

8. Transducer Operational Test

(Figure 503)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-49-03-000-802	Main Landing Gear Hubcap Assembly Removal (P/B 401)
32-49-03-400-802	Main Landing Gear Hubcap Assembly Installation (P/B 401)
46-13-02-700-805	Existing Faults Check (P/B 201)

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1831	Spin Up Adapter - Wheel Speed Transducer, Anti-Skid System Part #: A32075-10 Supplier: 81205 Opt Part #: A32075-1 Supplier: 81205 Opt Part #: A32075-9 Supplier: 81205

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left

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(Continued)

Zone	Area
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Operational Test

SUBTASK 32-42-00-480-022

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed on the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-00-480-014

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-074

(3) Release the parking brake.

SUBTASK 32-42-00-790-003

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) For the normal (system B) hydraulic system, do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

SUBTASK 32-42-00-860-075

(5) Open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-076

(6) Set the parking brake.

E. Transducer Operational Test

SUBTASK 32-42-00-020-005

(1) Remove the hubcap [1] or hubcap [10], do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

SUBTASK 32-42-00-020-003

(2) Do the steps that follow to do a check of the mechanical condition of the wheel speed transducer [5] and drive coupling:

(a) Turn the transducer drive coupling while you do a check to make sure that you do not have these conditions:

1) Free play in the bearing for the transducer drive coupling.

2) A drive shaft for the transducer drive coupling that is bent.

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SUBTASK 32-42-00-860-077

(3) Install the spin up adapter, SPL-1831, in a drill.

SUBTASK 32-42-00-710-004

(4) Do these steps for each wheel speed transducer [5] that you will test:

NOTE: At least one person needs to be stationed near the main landing gear brakes.

WARNING

KEEP PERSONS AND EQUIPMENT AWAY FROM THRUST REVERSERS. THE OPERATION OF THE WHEEL SPEED TRANSDUCER CAN CAUSE AUTOMATIC OPERATION OF THE STANDBY HYDRAULIC PUMP AND SUDDEN THRUST REVERSER MOVEMENTS. IF YOU DO NOT DO THIS, INJURY TO PERSONS OR DAMAGE TO EQUIPMENT CAN OCCUR.

(a) Use the spin up adapter, SPL-1831, and a variable speed drill to turn the wheel speed transducer [5] between 250 and 2200 Revolutions Per Minute (RPM).

NOTE: We recommend that you keep the wheel speed below 60 knots. If the aircraft is not in this condition, the standby hydraulic pump can operate automatically.

NOTE: If the wheel speed transducer is spun for more than 30 seconds, then the SPEEDBRAKE DO NOT ARM Light can come on with a correlated Maintenance Message.

1) If a correlated Maintenance Message occurs, do this task: Existing Faults Check, TASK 46-13-02-700-805

(b) Check the movement of the brake pistons or wear pins to make sure that the brake at the paired wheel releases (Table 503).

NOTE: The wheels are numbered from the left to the right.

Table 503/32-42-00-993-802 ANTISKID TRANSDUCER PAIR WHEELS

TRANSDUCER UNDER TEST	PAIRED WHEEL TO CHECK
1	4
2	3
3	2
4	1

(c) Stop the movement of the wheel speed transducer [5] as quickly as possible.

(d) Check the movement of the brake pistons or wear pins to make sure that the brake at the controlled wheel releases momentarily and then reapplies again in 5 to 10 seconds (Table 504).

Table 504/32-42-00-993-804 ANTISKID TRANSDUCER CONTROLLED WHEELS

TRANSDUCER UNDER TEST	CONTROLLED WHEEL TO CHECK
1	1
2	2
3	3
4	4

SUBTASK 32-42-00-400-001

(5) Install the hubcap [1] or hubcap [10], do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

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F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-00-860-078

(1) For the A and B hydraulic systems, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

SUBTASK 32-42-00-860-079

(2) Remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	16	C01346	LANDING GEAR PARKING BRAKE

SUBTASK 32-42-00-860-080

(3) Release the parking brake.

——— END OF TASK ———

TASK 32-42-00-740-801

9. Antiskid/Autobrake Control Unit BITE Procedure

(Figure 504)

A. General

(1) The antiskid/autobrake control unit (AACU) supplies skid control, automatic braking signals and monitors operation of the brake system.

(2) The AACU is on the E1-3 shelf in the electronic equipment bay.

(3) This Built-In-Test Equipment (BITE) procedure contains four tasks:

(a) A memory recall test (fault history)

(b) A display test

(c) An Antiskid/Autobrake control test

(d) A reset test

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-09-00-860-802	Return the Airplane to the Ground Mode (P/B 201)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

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D. Access Panels

Number	Name/Location
117A	Electronic Equipment Access Door

E. Prepare for the Procedure(s)

SUBTASK 32-42-00-010-001

(1) Open this access panel:

Number	Name/Location
117A	Electronic Equipment Access Door

F. Memory Recall Test (Fault History)

SUBTASK 32-42-00-740-001

(1) Do these BIT Test steps on the AACU front panel for the memory recall:

(a) Momentarily push the PRESS/TEST switch to the BIT position.

(b) If there are no faults, then the display will show TEST END.

(c) If there are faults in the memory, then the display will show the first fault message.

1) Make a record of the faults.

2) Push the PRESS/TEST-BIT switch to the BIT position again to look for more faults until the display show TEST END.

NOTE: If you do not set the PRESS/TEST-BIT switch to the BIT position in 45 seconds, the test will end and the fault will be removed from the display. The AACU will go back to its monitor mode.

G. Display Test

SUBTASK 32-42-00-740-005

(1) Do the steps that follow for the display test:

(a) Set the PRESS/TEST switch on the AACU front panel to the PRESS/TEST position.

(b) Make sure that all display segments on the AACU display come on in a checkboard pattern.

(c) Make sure that the ANTI SKID INOP light in the flight compartment also comes on with the display test.

NOTE: The display test function is automatically stopped after 5 seconds.

H. Antiskid/Autobrake Control Test

SUBTASK 32-42-00-480-017



WARNING OBEY THE PROCEDURE FOR THE INSTALLATION OF THE DOWNLOCK PINS. IF YOU MOVE THE CONTROL LEVER FOR THE LANDING GEAR TO THE UP POSITION, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONNEL, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-00-480-018

(2) Make sure that the tires have chocks installed around them (CHOCK INSTALLATION, PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-00-860-091

(3) Release the parking brake.

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SUBTASK 32-42-00-790-005

WARNING KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Prepare hydraulic systems A and B. Do this task: (Hydraulic System A or B Pressurization, TASK 29-11-00-860-801).

SUBTASK 32-42-00-860-092

(5) Make sure that the two thrust levers are in the IDLE position.

SUBTASK 32-42-00-860-093

(6) Make sure that the landing gear control lever is in the DN position.

SUBTASK 32-42-00-860-094

(7) Make sure that the spoilers are stowed.

SUBTASK 32-42-00-860-095

(8) Make sure that the left and right Air Data Inertial Reference System (ADIRS)s are in the NAV mode.

(a) Align the ADIRS. To do this, do this task: (Air Data Inertial Reference System - Alignment from the ISDU, TASK 34-21-00-820-802 or Air Data Inertial Reference System - Alignment from the FMC CDU, TASK 34-21-00-820-801 or Air Data Inertial Reference System - Automatic Alignment from the IRS MSU, TASK 34-21-00-820-803).

SUBTASK 32-42-00-860-097

(9) Make sure that the speedbrake lever is in the DOWN position.

SUBTASK 32-42-00-860-098

(10) Make sure that the airplane is in the ground mode. To put the airplane in the ground mode, do this task: Return the Airplane to the Ground Mode, TASK 32-09-00-860-802.

SUBTASK 32-42-00-740-006

(11) Do these steps for the antiskid/autobrake control test:

(a) Make sure that BRAKE TEST selector switch on the AACU is in the NORM position.

(b) Push and release the RESET switch on the AACU.

(c) Set the autobrake selector switch as follows:

1) Set the autobrake switch to the OFF position.

2) Set the autobrake switch to position 1.

NOTE: The AUTO BRAKE DISARM amber lights will stay on if the autobrake is disabled or the above arming conditions are not met.

(d) Push and hold the ENABLE/VERIFY switch on the AACU.

(e) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

NOTE: The ANTISKID INOP and AUTO BRAKE DISARM amber lights in the flight compartment must be on briefly during this test.

(f) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

(g) Make sure that the display shows TEST END. Do this step to complete the task:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Remove hydraulic power from hydraulic systems A and B. Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

(h) If the display shows BRK ####, then the test was started but not correctly executed. You must do the test again and make sure that all the initial conditions are completed.

(i) If there are existing faults, then the display on the AACU will show the first fault message. Do these steps:

1) Make a record of the faults.

2) Push the VERIFY switch to look for more faults until the display shows TEST END.

NOTE: The display shows TEST END when the test is completed. If you do not push the VERIFY switch in 45 seconds, the test will end and the fault will be removed from the display.

3) Remove hydraulic power from hydraulic systems A and B. To remove it, do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

4) If necessary, go to the Fault Isolation Manual and do the Fault Isolation Procedure for each Maintenance Message.

I. Reset Test

SUBTASK 32-42-00-740-007

(1) Do this step to erase all faults in the fault storage memory:

(a) Press push-button RESET on the AACU front panel.

(b) Make sure that a message MEM CLR shows for 2 seconds.

NOTE: The AACU will go through its Memory Clear procedure to make sure that there is no active fault. If the Line Replaceable Unit (LRU) is serviceable correctly, the AACU will then erase the fault from its storage memory.

J. Put the Airplane Back to its Usual Condition

SUBTASK 32-42-00-410-001

(1) Close this access panel:

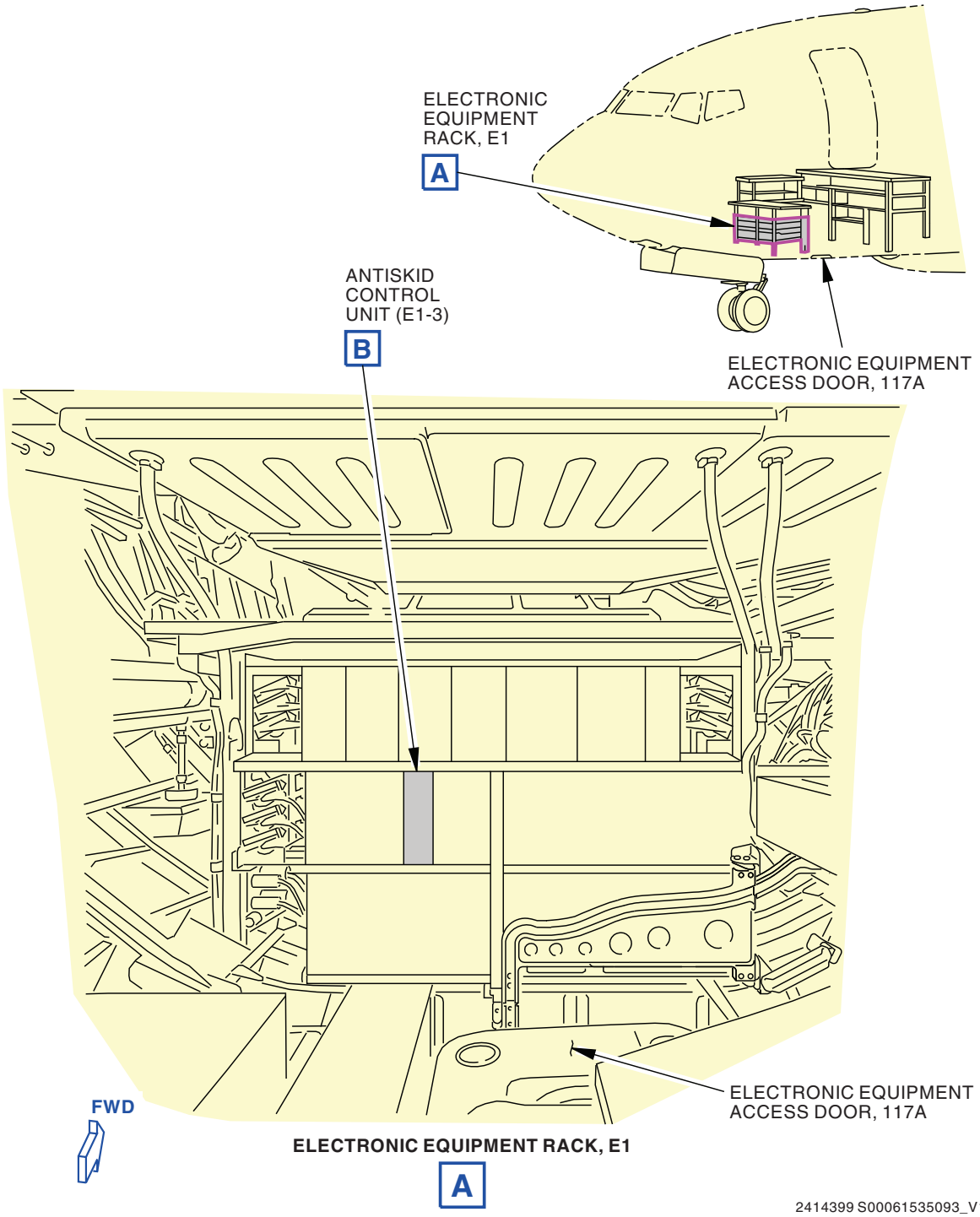
Number	Name/Location
117A	Electronic Equipment Access Door

SUBTASK 32-42-00-860-081

(2) Set the parking brake.

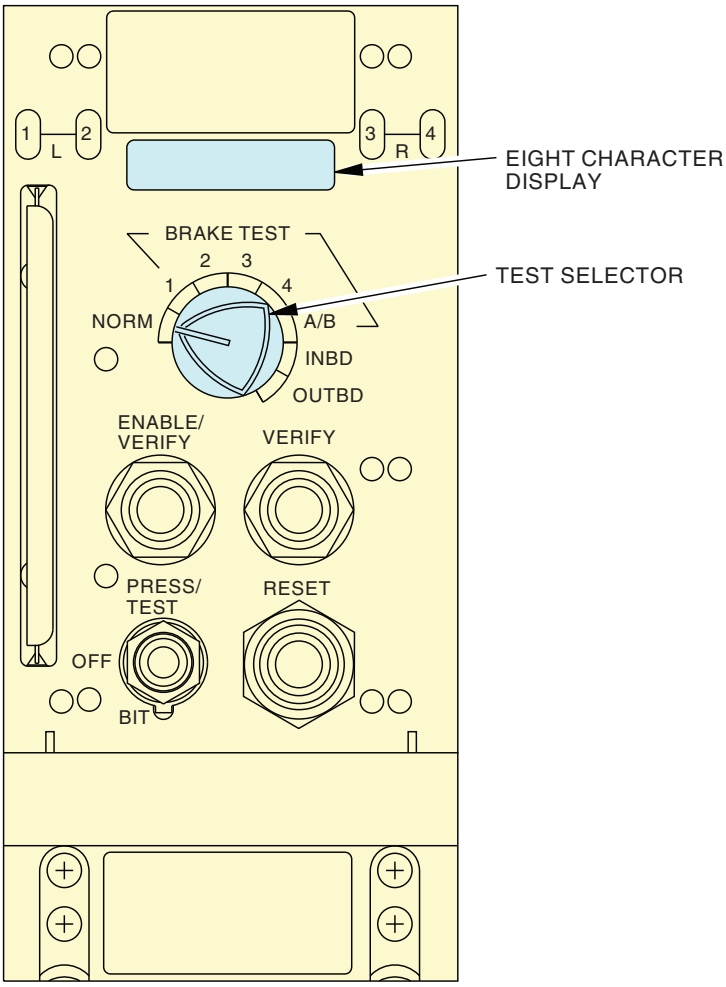
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Antiskid/Autobrake Control Unit
Figure 504/32-42-00-990-804 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



[1] ANTISKID CONTROL UNIT

B

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Antiskid/Autobrake Control Unit
Figure 504/32-42-00-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTISKID WHEEL SPEED TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the antiskid wheel speed transducer

(2) An installation of the antiskid wheel speed transducer.

TASK 32-42-11-000-801

2. Antiskid Wheel Speed Transducer Removal

(Figure 401)

A. References

Reference	Title
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)
32-49-03-000-802	Main Landing Gear Hubcap Assembly Removal (P/B 401)

B. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

C. Prepare for the Removal

SUBTASK 32-42-11-480-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-11-860-001

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

D. Antiskid Wheel Speed Transducer Removal

SUBTASK 32-42-11-020-004

(1) Do this task: Main Landing Gear Hubcap Assembly Removal, TASK 32-49-03-000-802.

SUBTASK 32-42-11-020-005

(2) Remove the in-axle adapter [17] as follows:

(a) Remove the nuts [8], washers [9], and bolts [10] that hold the in-axle adapter [17] to the axle.

(b) Carefully pull the in-axle adapter [17] out from the wheel assembly [2].

(c) Disconnect the electrical connectors [12] from the in-axle adapter [17] and transducer [1].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Identify each electrical connector [12] and wire with a label to record its initial position from the inboard or the outboard transducer [1].

SUBTASK 32-42-11-020-006

(3) Do these steps to remove the transducer [1] from the in-axle adapter [17]:

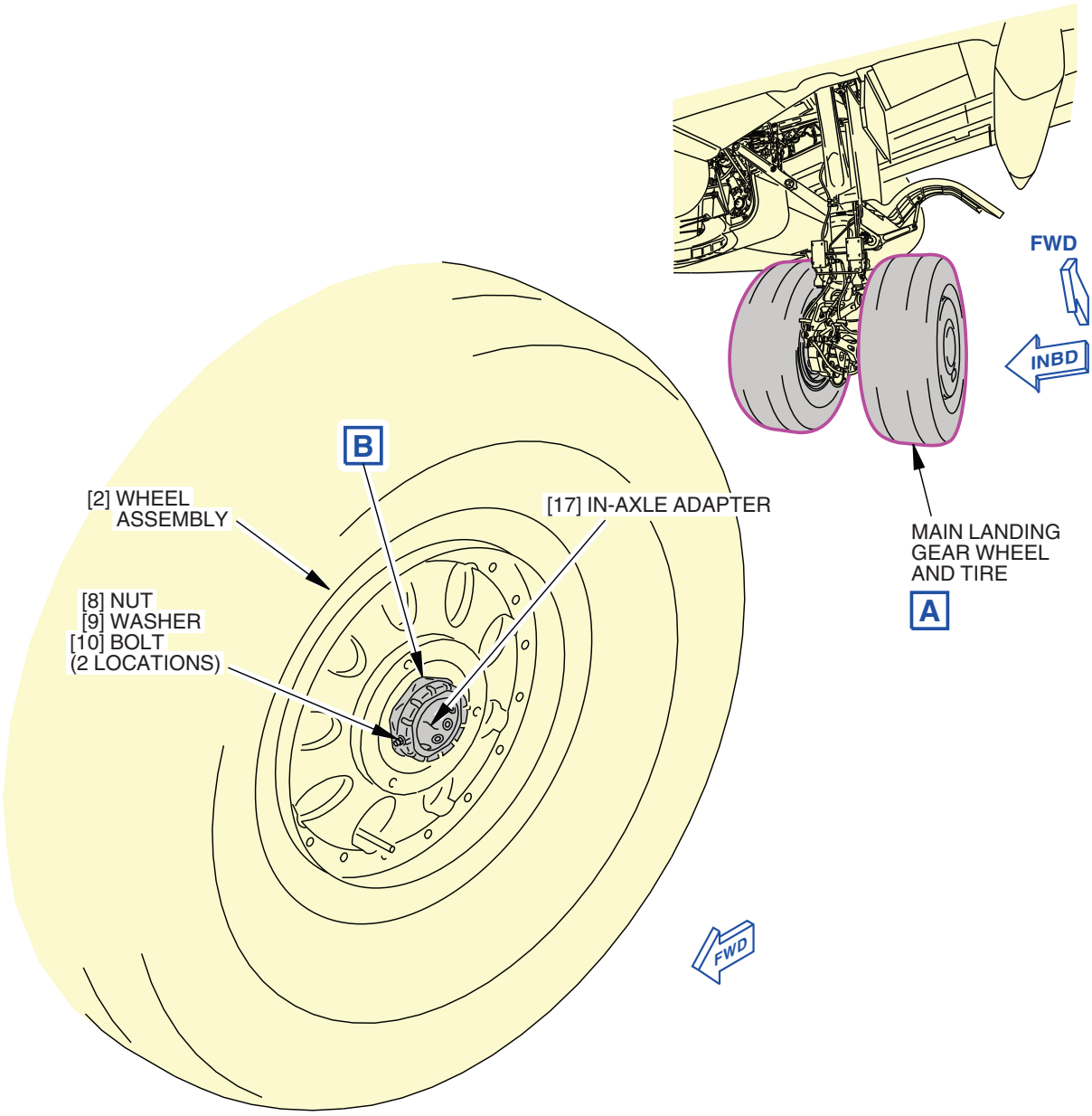
NOTE: The transducer must be removed from the back side of the in-axle adapter.

(a) Remove the screws [15], washers [14], and locking [16] that hold the transducer [1] to the in-axle adapter [17].

(b) Remove the transducer [1] from the back side of the in-axle adapter [17].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MAIN LANDING GEAR WHEEL AND TIRE (EXAMPLE, 4 LOCATIONS)

A

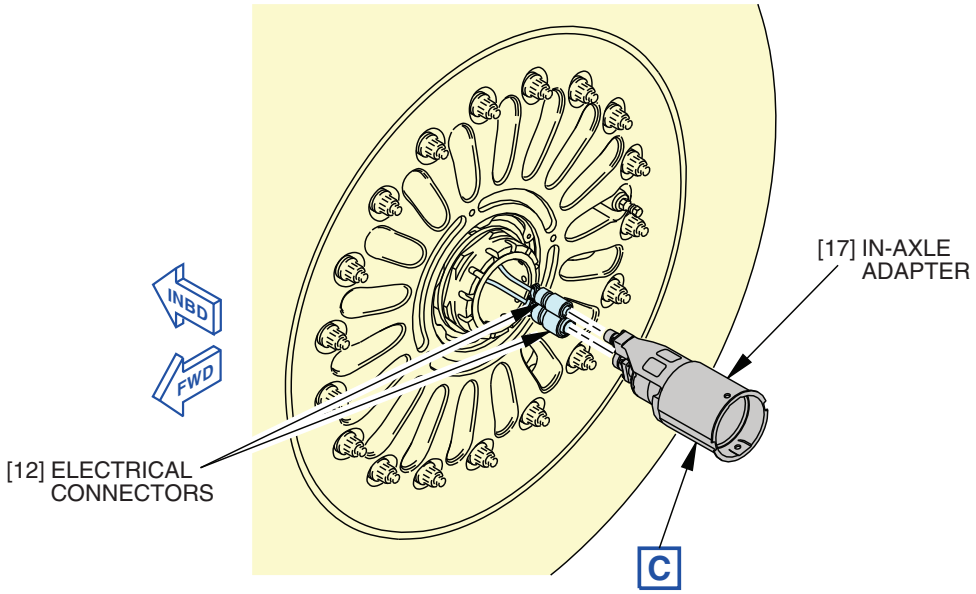
NOTE:

HUBCAP ASSEMBLY IS REMOVED

2717675 S0000627999_V3

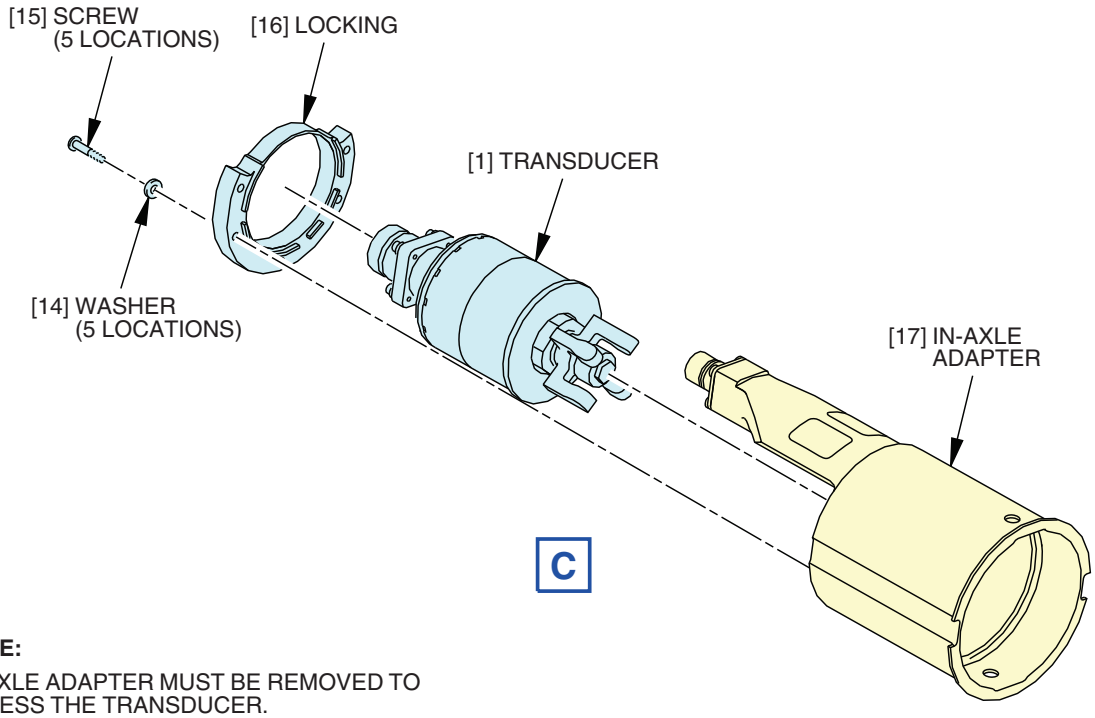
Antiskid Wheel Speed Transducer Installation
Figure 401/32-42-11-990-802 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

B



NOTE:

IN-AXLE ADAPTER MUST BE REMOVED TO ACCESS THE TRANSDUCER.

2717676 S0000628000_V3

Antiskid Wheel Speed Transducer Installation
Figure 401/32-42-11-990-802 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-11-400-801

3. Antiskid Wheel Speed Transducer Installation

(Figure 401)

A. References

Reference	Title
20-50-11-910-801	Standard Torque Values (P/B 201)
20-60-07-910-802	Inspection, Cleaning and Sealing of Electrical Connectors (P/B 201)
32-42-00-710-803	Transducer Operational Test (P/B 501)
32-49-03-400-802	Main Landing Gear Hubcap Assembly Installation (P/B 401)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Transducer	32-42-11-01-060	PTF ALL

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right
734	Left Main Landing Gear
744	Right Main Landing Gear

D. Antiskid Wheel Speed Transducer Installation

SUBTASK 32-42-11-860-006

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-11-420-007

(2) Install the transducer [1] in the in-axle adapter [17].

(a) Put the transducer [1] in its position in the in-axle adapter [17].

NOTE: The transducer is installed from the back side of the in-axle adapter.

(b) Install the locking [16], washers [14], and screws [15] to hold the transducer [1] in place.

SUBTASK 32-42-11-420-009

(3) Connect the electrical connectors [12] to the transducer [1] and in-axle adapter [17] as follows:

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) Do this task: Inspection, Cleaning and Sealing of Electrical Connectors, TASK 20-60-07-910-802.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Wind the electrical wires so that the overall coil diameter is approximately 50% of the internal diameter of the landing gear axle diameter.

NOTE: If not coiled, the wire bundle section inside the axle is somewhat stiff and may result in hard contact with the inside of the axle.

(c) Connect the electrical connectors [12] to the back of the transducer [1] and in-axle adapter [17].

SUBTASK 32-42-11-420-008

(4) Connect the in-axle adapter [17] to the wheel assembly [2].

(a) Install the bolts [10] with the head to the inside of the axle.

(b) Install the washers [9] and nuts [8].

1) Check the self-locking nuts (TASK 20-50-11-910-801).

2) Tighten the nuts [8] to 65 ±2 in-lb (7.3 ±0.2 N·m).

SUBTASK 32-42-11-860-007

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-11-860-004

(6) Do this task: Transducer Operational Test, TASK 32-42-00-710-803.

SUBTASK 32-42-11-420-004

(7) Do this task: Main Landing Gear Hubcap Assembly Installation, TASK 32-49-03-400-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTISKID/AUTOBRAKE CONTROL UNIT - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the antiskid/autobrake control unit

(2) An installation of the antiskid/autobrake control unit.

B. The antiskid/autobrake control unit [1] is in the electronic equipment rack, E1-3, which is in the left side of the main equipment center.

TASK 32-42-21-000-801

2. Antiskid/Autobrake Control Unit Removal

(Figure 401)

A. References

Reference	Title
20-10-07-000-801	E/E Box Removal (P/B 401)
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)
20-40-12-400-803	Conductive Dust Cap and Connector Cover Installation (P/B 201)

B. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

C. Prepare for the Removal

SUBTASK 32-42-21-860-001

(1) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(2) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Antiskid/Autobrake Control Unit Removal

SUBTASK 32-42-21-840-001

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 32-42-21-020-001

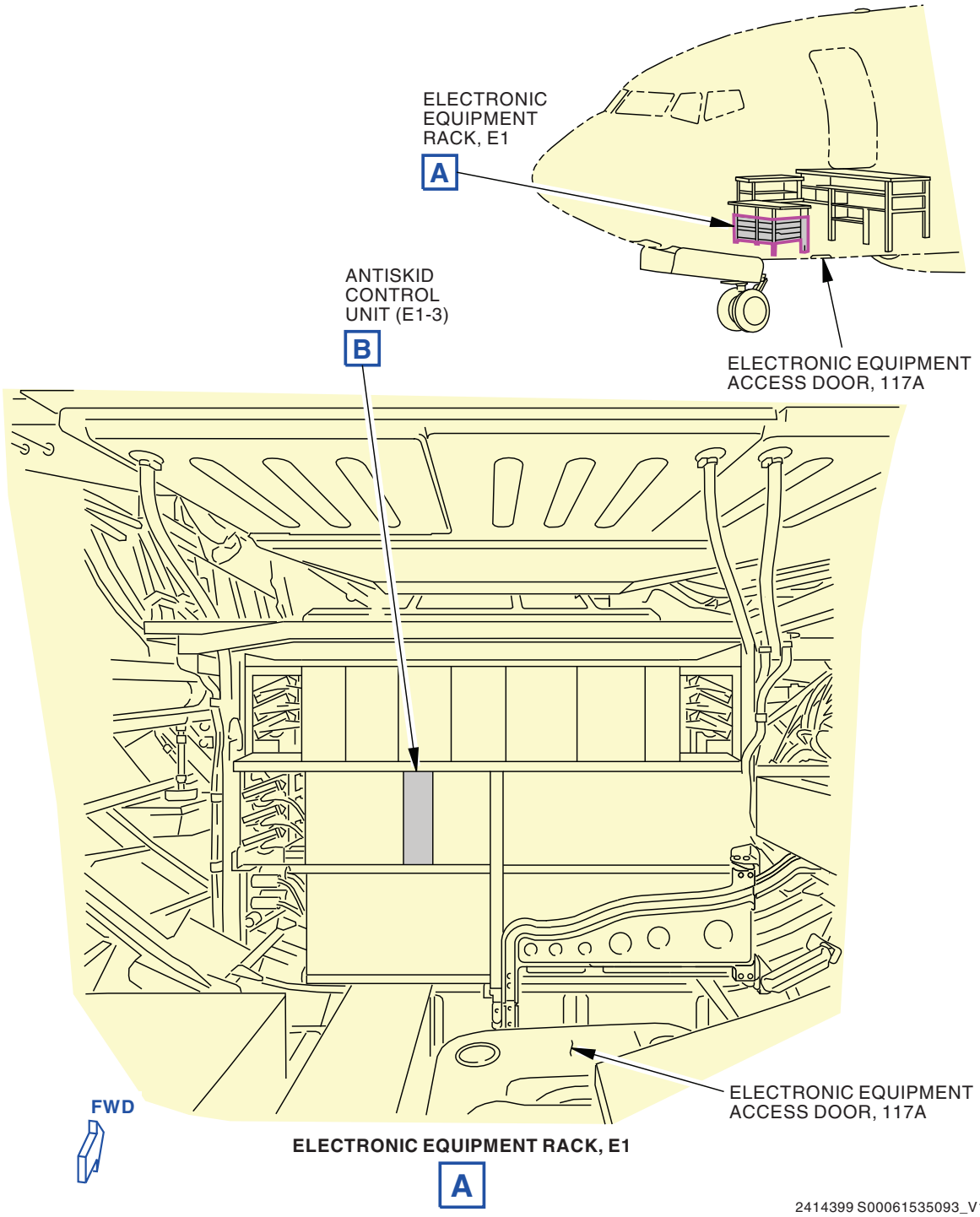
(2) To remove the antiskid/autobrake control unit [1] from the airplane, do this task: E/E Box Removal, TASK 20-10-07-000-801.

SUBTASK 32-42-21-840-002

(3) Do this task: Conductive Dust Cap and Connector Cover Installation, TASK 20-40-12-400-803.

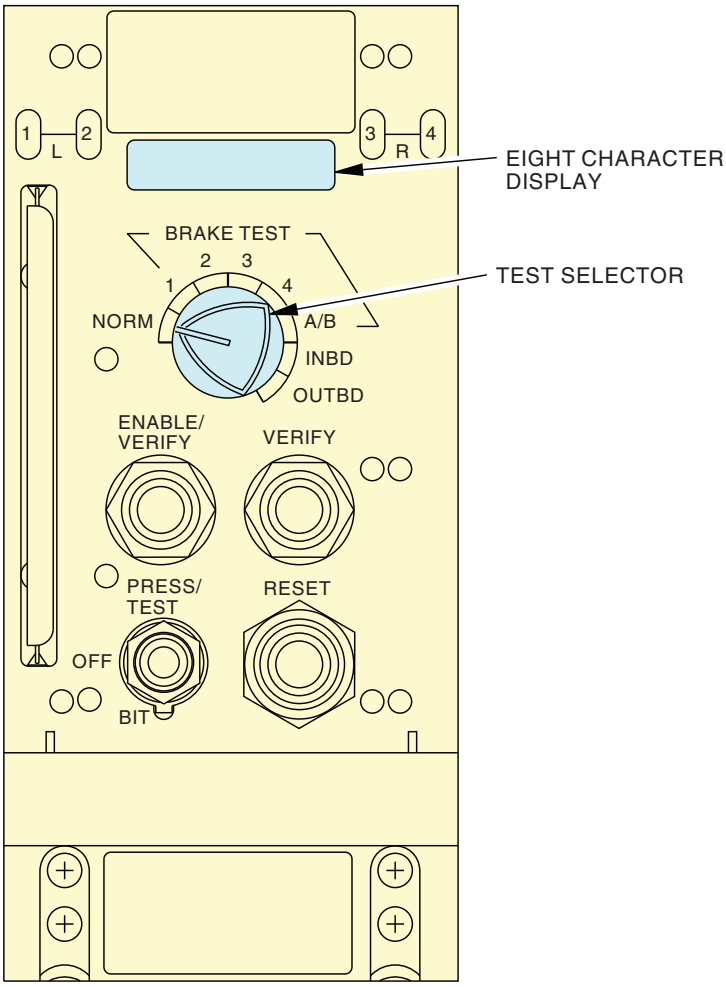
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Antiskid Control Unit Installation
Figure 401/32-42-21-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



[1] ANTISKID CONTROL UNIT

B

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Antiskid Control Unit Installation
Figure 401/32-42-21-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-21-400-801

3. Antiskid/Autobrake Control Unit Installation

(Figure 401)

A. References

Reference	Title
20-10-07-400-801	E/E Box Installation (P/B 401)
20-40-12-000-803	Conductive Dust Cap and Connector Cover Removal (P/B 201)
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
32-42-00-720-801	Antiskid/Autobrake Control Unit Operational Test (P/B 501)

B. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Unit	32-42-21-03-095	PTF ALL

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left

D. Antiskid/Autobrake Control Unit Installation

SUBTASK 32-42-21-840-003

CAUTION DO NOT TOUCH THE CONTROL UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE CONTROL UNIT.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 32-42-21-840-004

(2) Do this task: Conductive Dust Cap and Connector Cover Removal, TASK 20-40-12-000-803.

SUBTASK 32-42-21-420-001

(3) To install the antiskid/autobrake control unit [1], do this task: E/E Box Installation, TASK 20-10-07-400-801.

SUBTASK 32-42-21-860-002

(4) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	9	C00440	FLIGHT CONTROL AUTO SPEED BRAKE
B	11	C04040	GROUND SPOILER CONTROL

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING KEEP PERSONS AND EQUIPMENT CLEAR OF THE FLIGHT SPOILERS. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU OPEN/CLOSE THESE CIRCUIT BREAKERS. THIS CAN CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(5) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
C	10	C04032	SPOILER CONTROL ELEC PWR A
C	12	C04033	SPOILER CONTROL ELEC PWR B

E. Antiskid/Autobrake Control Unit Installation Test

SUBTASK 32-42-21-710-001

(1) Do this task: Antiskid/Autobrake Control Unit Operational Test, TASK 32-42-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ANTISKID VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the antiskid valve

(2) Installation of the antiskid valve.

B. There are two normal Unigain antiskid valves, located on the aft wall of the main landing gear wheel well.

C. There are two alternate antiskid valves, one for each main landing gear, located on the outboard walls of the main landing gear wheel well.

D. This procedure is applicable for all four antiskid valves.

E. In this procedure the antiskid valve will be referred to as the antiskid control valve [6] or antiskid control valve [49].

TASK 32-42-31-000-802

2. Antiskid Valve - Removal

(Figure 401, Figure 402)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for the Removal

SUBTASK 32-42-31-480-007

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-31-860-007

(2) Do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

SUBTASK 32-42-31-480-005

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-31-860-008

(4) Release the parking brake.

SUBTASK 32-42-31-840-002

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

SUBTASK 32-42-31-860-009

(6) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

E. Antiskid Valve Removal

SUBTASK 32-42-31-020-006

(1) Disconnect the electrical connectors [14] from the antiskid control valve [6] or antiskid control valve [49].

SUBTASK 32-42-31-020-007

(2) For the right normal antiskid control valve [49], do the these steps:

(a) Disconnect the brake hydraulic line [31] and brake hydraulic line [35] with elbows from the filters [45].

(b) Disconnect the return hydraulic line [32] and return hydraulic line [34] from the tee [56].

(c) Remove the tee [56].

(d) Disconnect the pressure hydraulic line [33] with elbow from the filter [45].

(e) Disconnect the pressure hydraulic line [30] from the filter [18].

(f) Install the caps/plugs, STD-14686, on the hydraulic lines.

(g) Remove the bolt [9], washer [55], washer [54], bonding jumper [53], and washer [52].

(h) Remove the bolts [47], washers [48] to remove the antiskid control valve [49] from the airplane.

SUBTASK 32-42-31-020-013

(3) For the left normal antiskid control valve [49], do the these steps:

(a) Disconnect the brake hydraulic line [36] and brake hydraulic line [39] with elbows from the filters [45].

(b) Disconnect the return hydraulic line [40] with elbow from the union [50].

(c) Disconnect the pressure hydraulic line [38] with elbow from the filter [45].

(d) Disconnect pressure hydraulic line [37] from filter [18].

(e) Install the caps/plugs, STD-14686, on the hydraulic lines.

(f) Remove the bolt [9], washer [55], washer [54], bonding jumper [53], and washer [52].

(g) Remove the bolts [47], washers [48] to remove the antiskid control valve [49] from the airplane.

SUBTASK 32-42-31-020-008

(4) For the alternate antiskid control valve [6], do these steps:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Disconnect the brake hydraulic line [41] from the union [12].

(b) Disconnect the return hydraulic line [42] from the check valve [13].

(c) Disconnect the pressure hydraulic line [43] and pressure hydraulic line [44] with the tee fitting from the filter [4].

(d) Install caps/plugs, STD-14686, on the hydraulic lines.

(e) Remove the bolts [8] and washers [7] to remove the antiskid control valve [6] from the airplane.

SUBTASK 32-42-31-020-011

(5) If the replacement of the normal antiskid control valve [49] does not have hydraulic fittings installed, do these steps:

(a) Remove the lockwires from the filters [45] and filter [18].

(b) Remove the union [50] and packing [51].

(c) Remove the filters [45] and packings [46].

(d) Remove the filter [18] and packing [19].

(e) Discard the packing [51], packings [46], and packing [19].

SUBTASK 32-42-31-020-015

(6) If the replacement of the alternate antiskid control valve [6] does not have hydraulic fittings installed, do these steps:

(a) Remove the lockwires from the filter [4], filter [11], and check valve [13].

(b) Remove the union [12] and packing [5].

(c) Remove the filter [11] and packing [10].

(d) Remove the check valve [13] and packing [5].

(e) Remove the filter [4] and packing [5].

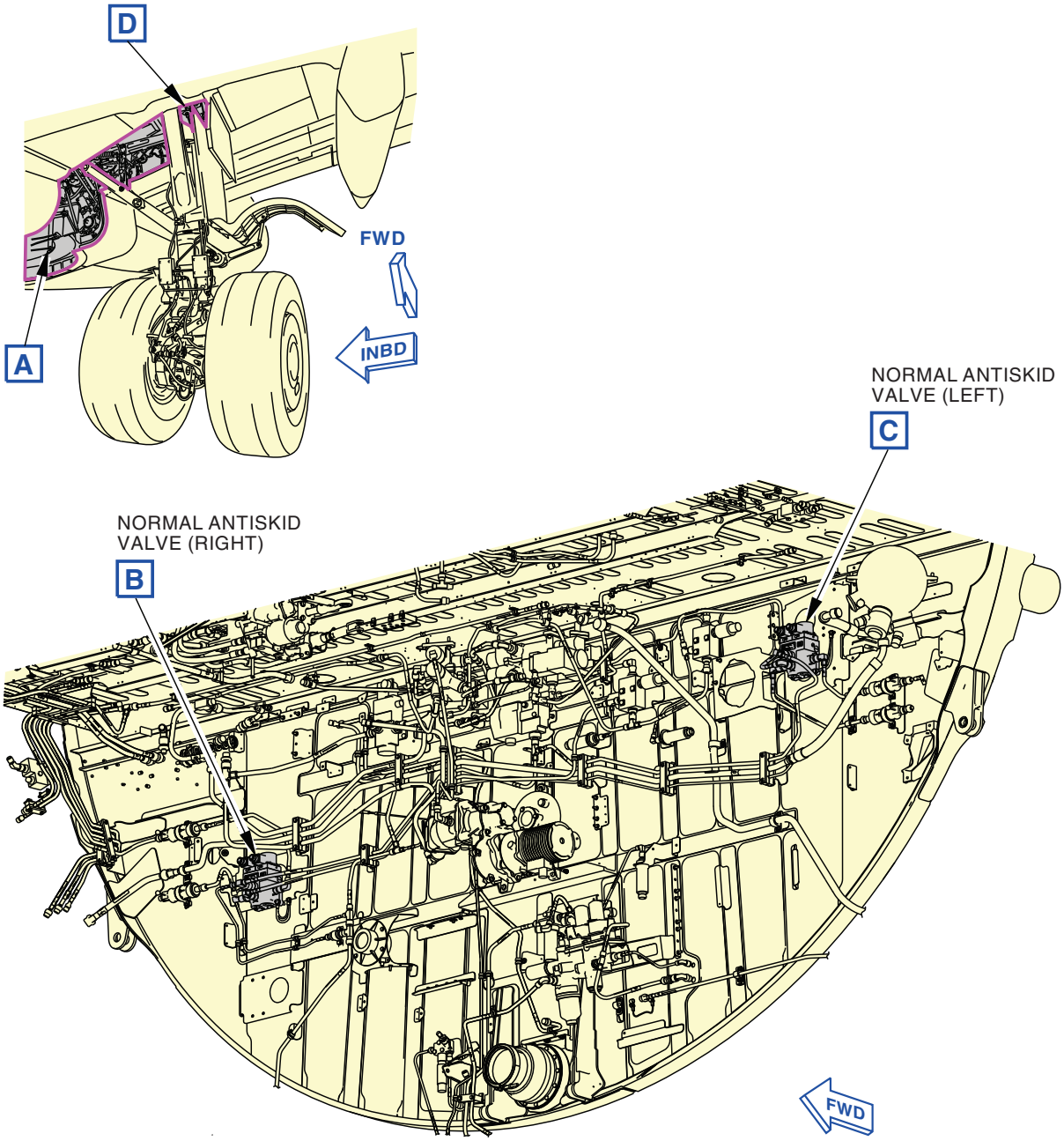
(f) Discard the packings [5] and packing [10].

SUBTASK 32-42-31-420-010

(7) Install the caps/plugs, STD-14686, in the ports of the antiskid control valve [6] or antiskid control valve [49].

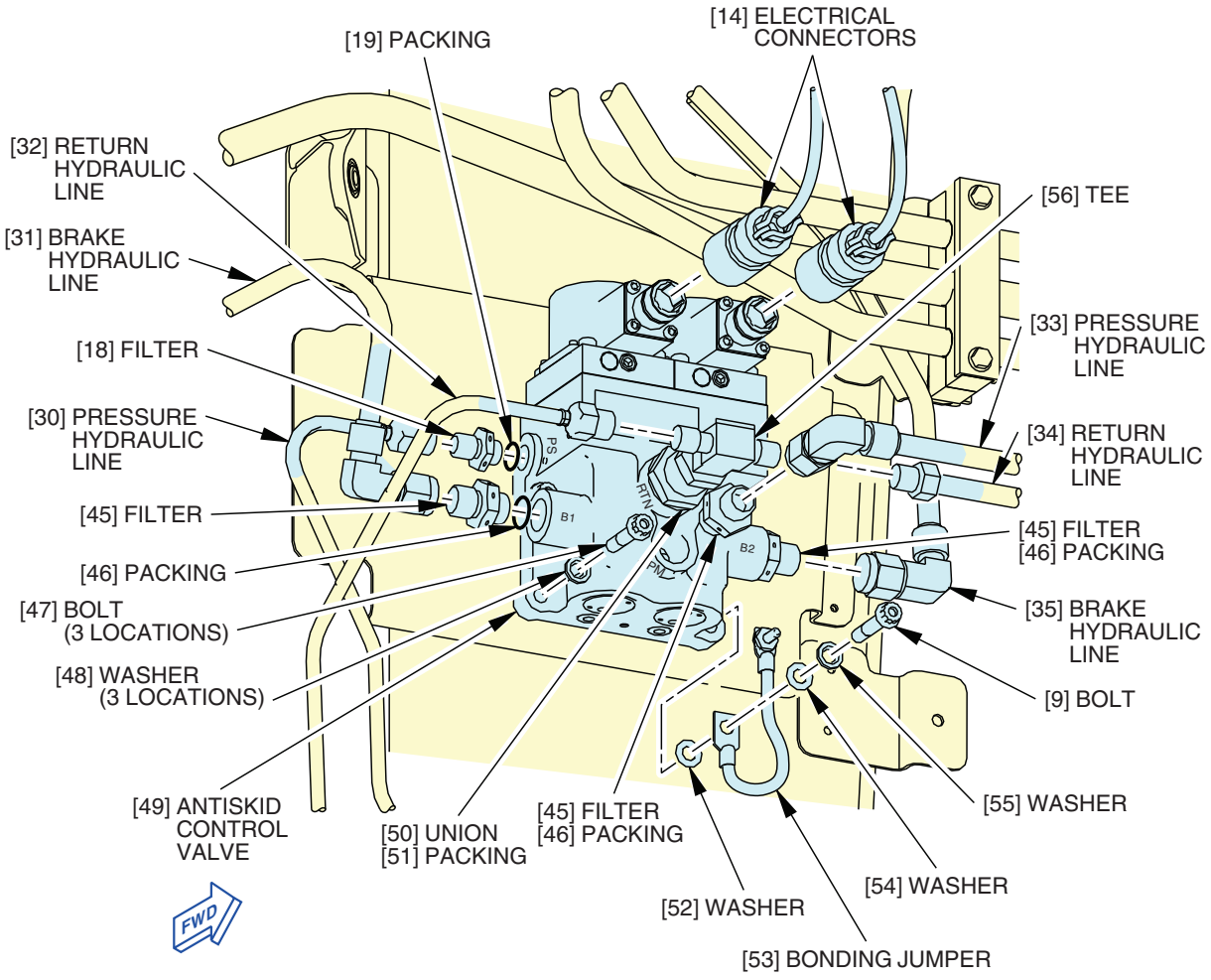
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Antiskid Valve Installation
Figure 401/32-42-31-990-803 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



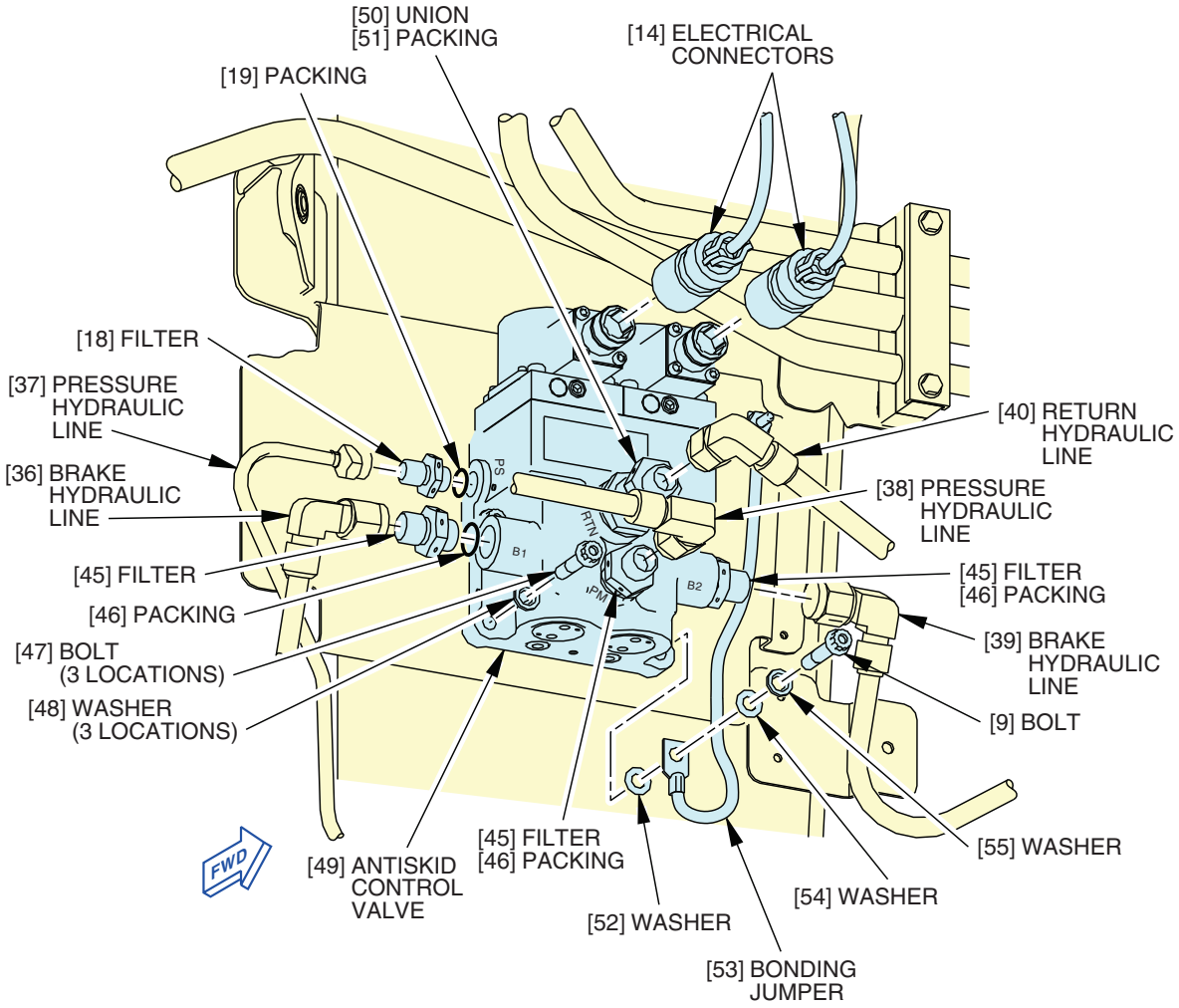
NORMAL ANTISKID VALVE (RIGHT)

B

2561715 S0000611436_V7

Antiskid Valve Installation
Figure 401/32-42-31-990-803 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



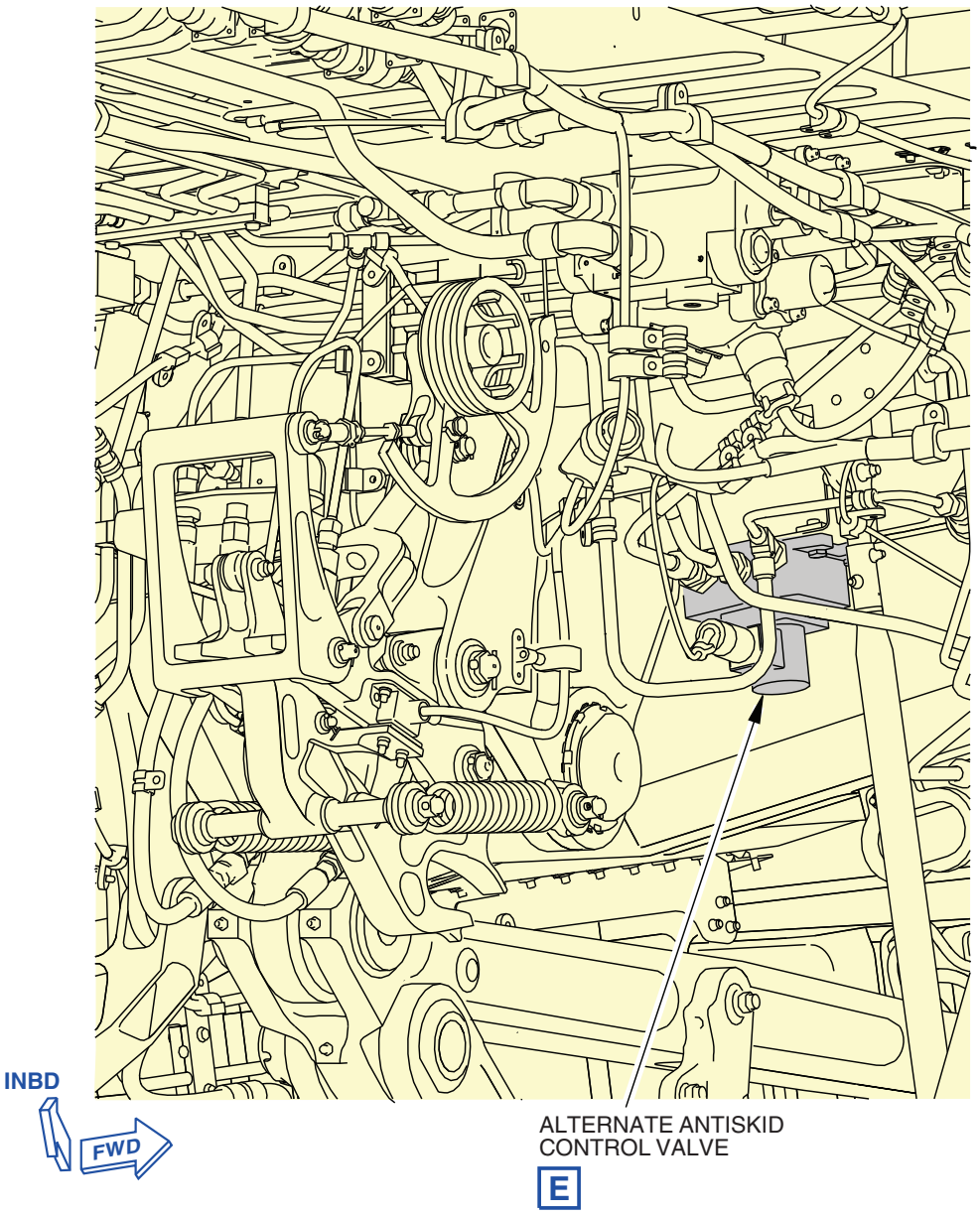
NORMAL ANTISKID VALVE (LEFT)

C

2578799 S0000620986_V5

Antiskid Valve Installation
Figure 401/32-42-31-990-803 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



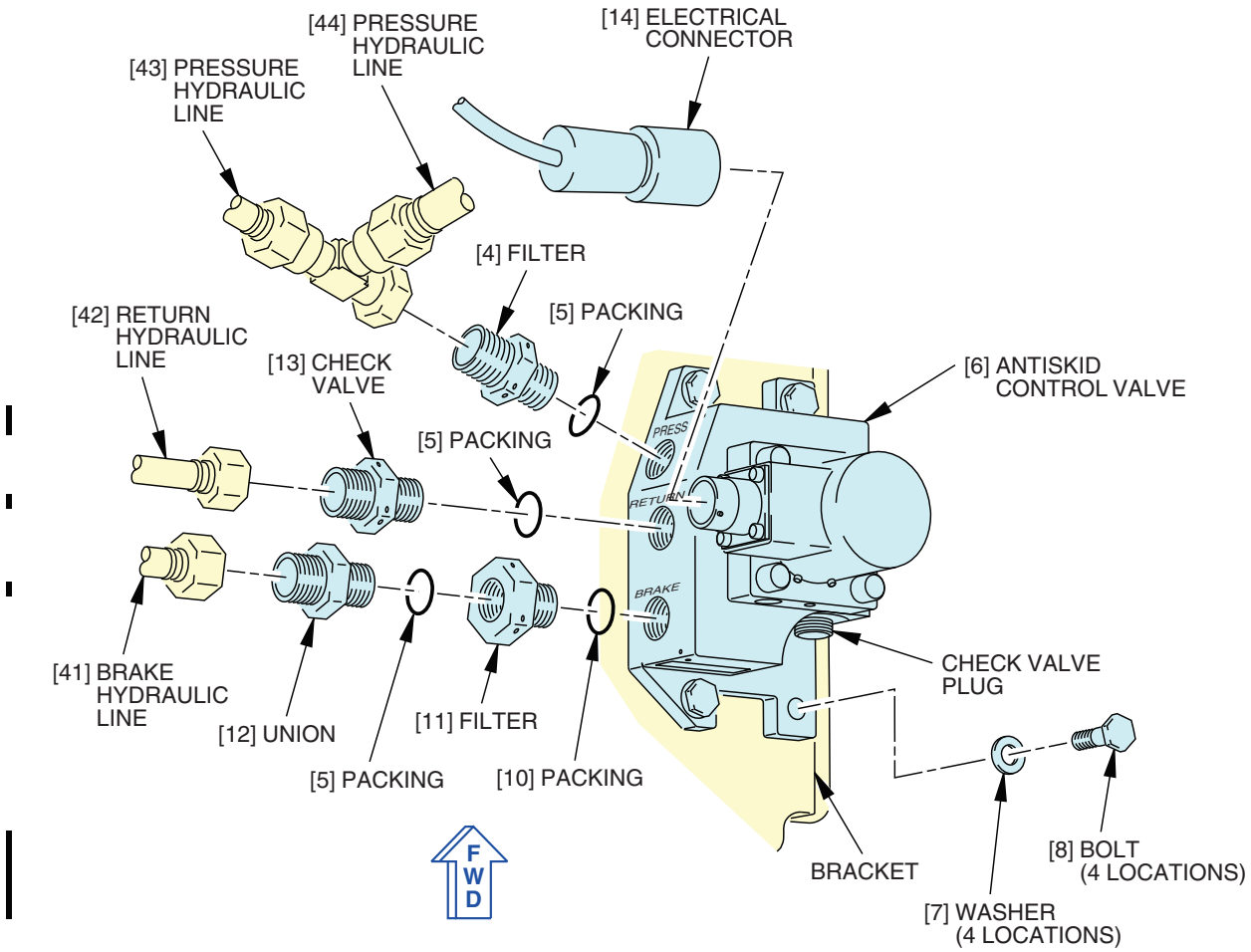
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE IS SHOWN, RIGHT SIDE IS EQUIVALENT)

D

2561713 S0000611435_V3

Antiskid Valve Installation
Figure 401/32-42-31-990-803 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



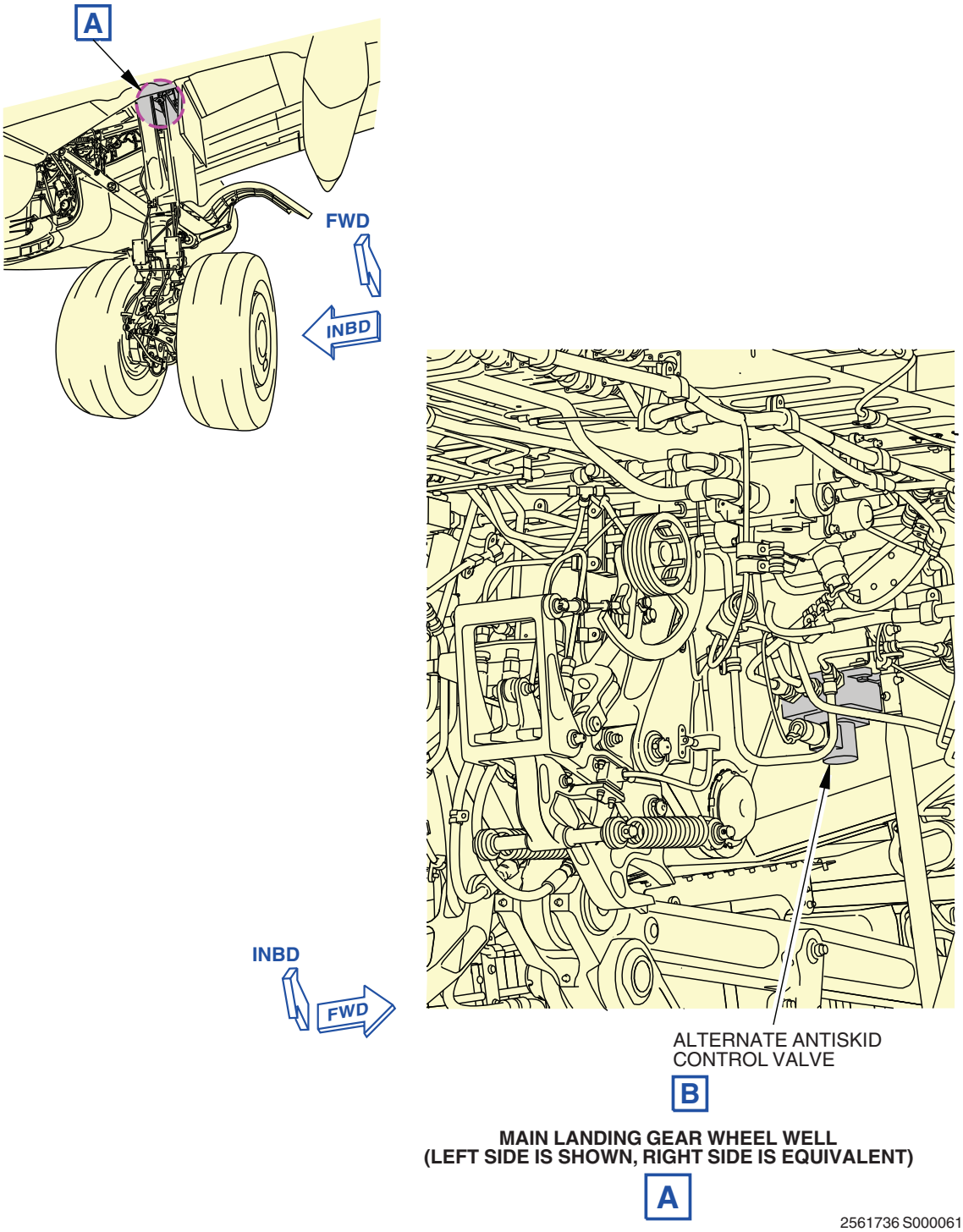
ALTERNATE ANTISKID VALVE

E

2561727 S0000611437_V4

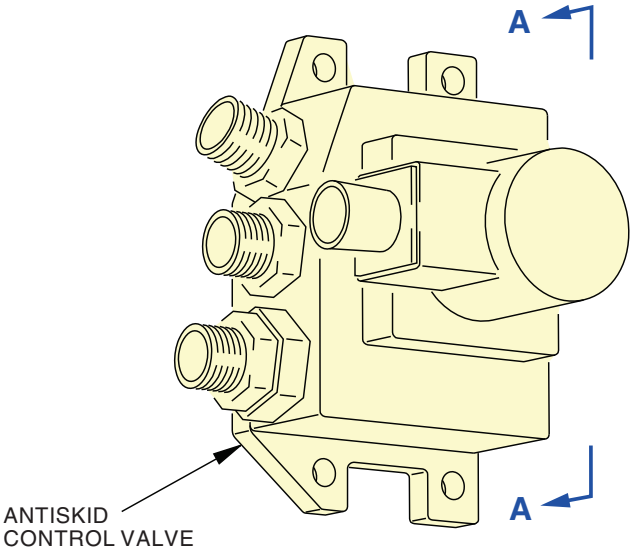
Antiskid Valve Installation
Figure 401/32-42-31-990-803 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



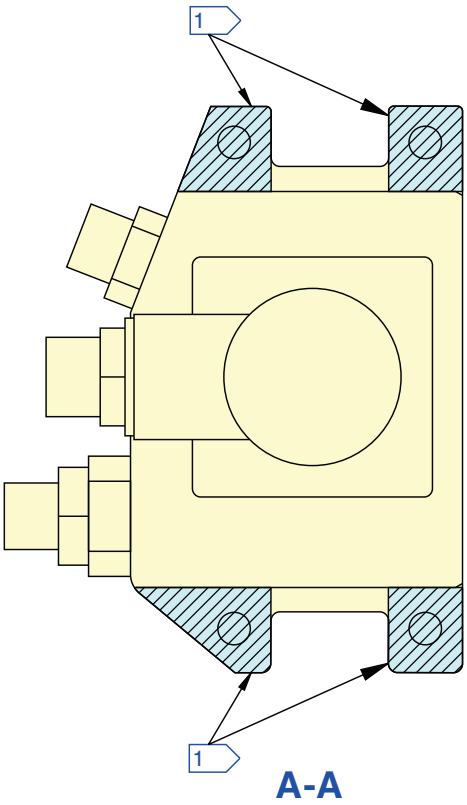
Cleaning of the Antiskid Valve
Figure 402/32-42-31-990-804 (Sheet 1 of 2)

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ANTISKID CONTROL VALVE (EXAMPLE)

B



1 CLEAN THIS AREA

2561762 S0000611439_V1

Cleaning of the Antiskid Valve
Figure 402/32-42-31-990-804 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-31-400-802

3. Antiskid Valve - Installation

(Figure 401, Figure 402)

A. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
32-41-00-870-802	Normal (System B) Hydraulic Brake System Bleeding (P/B 201)
32-41-00-870-803	Alternate (System A) Hydraulic Brake System Bleeding (P/B 201)
32-42-00-720-803	Antiskid Valve Operational Test (P/B 501)
51-22-00-390-806	Fillet Seal Application (P/B 201)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50170	Compound - Corrosion Inhibiting Compound, Soft Film, Exterior Use - Ardrox AV25	
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50225	Lockwire - MS20995C20, Corrosion Resistant Steel - 0.020 Inch (0.508 mm) Diameter	NASM20995

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(Continued)

Reference	Description	Specification
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38
G50264	Cable - Ferrule, End Fitting (Safety Kit)	BACC13AT2K

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Antiskid Valve Installation

SUBTASK 32-42-31-080-004

(1) Remove the caps/plugs, STD-14686, from the ports of the antiskid control valve [6] or antiskid control valve [49].

SUBTASK 32-42-31-420-011

(2) If the replacement of the normal antiskid control valve [49] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the union [50], filters [45], filter [18], new packing [19], new packings [46], and new packing [51] with hydraulic fluid, D00153.

(b) Install the packings [46] on the filters [45].

(c) Install the packing [51] on the union [50].

(d) Install the packing [19] on the filter [18].

(e) Install the filter [45] in the pressure port of the antiskid control valve [49].

1) Tighten the filter [45] to 257 in-lb (29.0 N·m) - 283 in-lb (32.0 N·m).

(f) Install the filters [45] in the brake ports of the antiskid control valve [49].

1) Tighten the filters [45] to 257 in-lb (29.0 N·m) - 283 in-lb (32.0 N·m).

(g) Install the union [50] in the return port of the antiskid control valve [49].

1) Tighten the union [50] to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

(h) Install the filter [18] in the pressure port of the antiskid control valve [49].

1) Tighten the filter [18] to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

(i) Install MS20995C20 lockwires, G50225, or cable, G50264, with the double twist method on the filters [45] and filter [18].

SUBTASK 32-42-31-420-020

(3) If the replacement of the alternate antiskid control valve [6] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the union [12], filter [4], filter [11], check valve [13], new packing [10], and new packings [5] for the antiskid control valve [6] with hydraulic fluid, D00153.

(b) Install the packings [5] on the filter [4], check valve [13], and union [12].

(c) Install the packing [10] on the filter [11].

(d) Install the filter [11] in the brake port of the antiskid control valve [6].

1) Tighten the filter [11] to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Install the union [12] in the filter [11]

1) Tighten the union [12] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(f) Install the check valve [13] in the return port of the antiskid control valve [6].

1) Tighten the check valve [13] to 161 in-lb (18.2 N·m) - 179 in-lb (20.2 N·m).

(g) Install the filter [4] in the pressure port of the alternate antiskid control valve [6].

1) Tighten the filter [4] to 257 in-lb (29.0 N·m) - 283 in-lb (32.0 N·m).

(h) Install MS20995C20 lockwires, G50225, or cable, G50264, with the double twist method on the filter [4], filter [11], and check valve [13].

SUBTASK 32-42-31-420-012

(4) For the normal antiskid control valve [49], do these steps:

(a) Do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

(b) For all cleaned areas, and if there is any damage to the finish on the antiskid control valve [49], manually apply coating, C00064.

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply Cor-Ban 27L Compound, G50237, to all of the areas of the fastener hole.

1) Make sure that you install the bolts [47] and washers [48] immediately.

(d) Apply sealant, A00247, to the mating surfaces of the bonding jumper [53] and antiskid control valve [49].

NOTE: Operators can defer the application of the sealant for the valve installation to avoid a flight delay. This step can be deferred for up to eight weeks but must be performed to put the airplane back to serviceable condition.

(e) Hold the antiskid control valve [49] in its position on the airplane.

(f) Install the bolts [47] and washers [48] that hold the antiskid control valve [49] to the airplane.

1) Tighten the bolts [47] to 63 in-lb (7.1 N·m) - 77 in-lb (8.7 N·m).

(g) Install the washer [52], bonding jumper [53], washer [54], washer [55], and bolt [9].

1) Tighten the bolt [9] to 63 in-lb (7.1 N·m) - 77 in-lb (8.7 N·m).

(h) Measure the electrical bonding resistance between the body of theantiskid control valve [49] and bonding jumper [53] (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.001 ohm (1.0 milliohms) or less.

SUBTASK 32-42-31-420-013

(5) For the alternate antiskid control valve [6], do these steps to help get a good electrical bond between the antiskid control valve [6] and bracket:

(a) Do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

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1) Clean the valve body and mating bracket surfaces at a minimum of two of the four lugs (Figure 402).

a) Do not clean more than the specified areas to prevent possible corrosion problems.

(b) For all cleaned areas, and if there is any damage to the finish on the antiskid control valve [6], manually apply coating, C00064.

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(c) Apply Cor-Ban 27L Compound, G50237, to all of the areas of the fastener hole.

1) Make sure that you install the bolts [8] and washers [7] immediately.

(d) Apply sealant, A00247, to the cleaned areas faying surfaces.

NOTE: Operators can defer the application of the sealant for the valve installation to avoid a flight delay. This step can be deferred for up to eight weeks but must be performed to put the airplane back to serviceable condition.

(e) Hold the antiskid control valve [6] in its position on the airplane.

(f) Install the bolts [8] and washers [7] that hold the antiskid control valve [6] to the airplane.

1) Tighten the bolts [8] to 63 in-lb (7.1 N·m) - 77 in-lb (8.7 N·m).

(g) Measure the electrical bonding resistance between the body of the antiskid control valve [6] and bracket (SWPM 20-20-00).

1) Use an intrinsically safe approved bonding meter, COM-1550.

2) Make sure that the electrical bonding resistance is 0.0025 ohm (2.5 milliohms) or less.

(h) Apply sealant, A00247, to the edge of the antiskid control valve [6] around the mating surface (TASK 51-22-00-390-806).

NOTE: Operators can defer the application of the sealant for the valve installation to avoid a flight delay. This step can be deferred for up to eight weeks but must be performed to put the airplane back to serviceable condition.

SUBTASK 32-42-31-080-005

(6) Remove the caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-42-31-420-014

(7) For the right normal antiskid control valve [49], do these steps:

(a) Connect the elbow of the pressure hydraulic line [33] to the filter [45].

1) Tighten the swivel nut of the elbow on the pressure hydraulic line [33] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(b) Install the tee [56] on the union [50].

(c) Connect the return hydraulic line [32] and return hydraulic line [34] to the tee [56].

1) Tighten the B-nut on return hydraulic line [32] to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

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2) Tighten the B-nut on return hydraulic line [34] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(d) Tighten the swivel nut of the tee [56] to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

(e) Connect the elbows on the pressure hydraulic line [31] and pressure hydraulic line [35] to the filters [45].

1) Tighten the swivel nuts of the elbows on pressure hydraulic line [31] and pressure hydraulic line [35] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(f) Connect the pressure hydraulic line [30] to the filter [18].

1) Tighten the B-nut on pressure hydraulic line [30] to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

SUBTASK 32-42-31-420-017

(8) For the left normal antiskid control valve [49], do the these steps:

(a) Connect the elbow of the brake hydraulic line [38] to the filter [45].

1) Tighten the swivel nut of the elbow on brake hydraulic line [38] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(b) Connect the elbow of the return hydraulic line [40] to the union [50].

1) Tighten the swivel nut of the elbow on return hydraulic line [40] to 475 in-lb (53.7 N·m) - 525 in-lb (59.3 N·m).

(c) Connect the elbow of the pressure hydraulic line [36] and pressure hydraulic line [39] to the filters [45].

1) Tighten the swivel nut of the elbow on brake hydraulic line [36] and pressure hydraulic line [39] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(d) Connect pressure hydraulic line [37] to the filter [18].

1) Tighten the B-nut on pressure hydraulic line [37] to 133 in-lb (15.0 N·m) - 147 in-lb (16.6 N·m).

SUBTASK 32-42-31-420-015

(9) For the alternate antiskid control valve [6], do these steps:

(a) Connect the brake hydraulic line [41] to the union [12].

1) Tighten the B-nut on brake hydraulic line [41] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(b) Connect the return hydraulic line [42] to the check valve [13].

1) Tighten the B-nut on return hydraulic line [42] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

(c) Connect the tee fitting with the pressure hydraulic line [43] and pressure hydraulic line [44] to the filter [4].

1) Tighten the swivel nut on the tee fitting of the pressure hydraulic line [43] and pressure hydraulic line [44] to 256 in-lb (28.9 N·m) - 284 in-lb (32.1 N·m).

SUBTASK 32-42-31-210-001

(10) Before you connect the electrical connectors [14] to the antiskid control valve [6] or antiskid control valves [49], examine the electrical connector [14] for corrosion.

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WARNING

DO THE STEPS BELOW WHEN YOU OPERATE THE AIRPLANE WHERE YOU USE DEICING FLUID (THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE). ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT. IF YOU DO NOT DO THESE STEPS, YOU CAN CAUSE DAMAGE TO EQUIPMENT OR INJURY TO PERSONNEL.

(a) If there was corrosion, correct the problem (SWPM 20-60-03).

(b) Apply D5026NS or ZC-026 compound, G50171, to the electrical connector [14] (SWPM 20-60-03).

SUBTASK 32-42-31-420-019

(11) Connect the electrical connectors [14] to the antiskid control valve [6] or antiskid control valves [49].

SUBTASK 32-42-31-860-010

(12) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

F. Antiskid Valve Installation Test

SUBTASK 32-42-31-790-002

(1) Do these steps to do a leak check of the antiskid control valve [6] or antiskid control valve [49]:

WARNING

MAKE SURE THAT EQUIPMENT AND PERSONNEL ARE CLEAR OF THE FLIGHT CONTROLS SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURY TO PERSONS AND DAMAGE TO EQUIPMENT.

(a) Do this task: Hydraulic System A or B Pressurization, TASK 29-11-00-860-801.

(b) Fully push the captain's brake pedals twelve times.

(c) Look for leaks around the antiskid control valve [6] or antiskid control valve [49].

SUBTASK 32-42-31-870-001

(2) Bleed the hydraulic system.

(a) For the normal antiskid control valve [49], do this task: Normal (System B) Hydraulic Brake System Bleeding, TASK 32-41-00-870-802.

(b) For the alternate antiskid control valve [6], do this task: Alternate (System A) Hydraulic Brake System Bleeding, TASK 32-41-00-870-803.

SUBTASK 32-42-31-710-003

(3) Do this task: Antiskid Valve Operational Test, TASK 32-42-00-720-803.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-31-860-011

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

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SUBTASK 32-42-31-390-004

(2) Apply Ardrox AV25 compound, G50170, to the external surface of the connector from the side of the connector (SWPM 20-60-03).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUTOBRAKE SELECTOR SWITCH - REMOVAL/INSTALLATION

1. General

A. This procedure contains these tasks:

(1) AUTO BRAKE Selector Switch Removal

(2) AUTO BRAKE Selector Switch Installation

TASK 32-42-61-000-801

2. Autobrake Selector Switch Removal

(Figure 401)

A. General

(1) This task is for the removal of the autobrake selector switch.

B. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
31-65-03-000-801	Multifunction Panel (P9-82) Removal (P/B 401)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Prepare for Removal

SUBTASK 32-42-61-490-001

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED IN ALL OF THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) Make sure that you install the landing gear downlock pins (TASK 32-00-01-480-801).

SUBTASK 32-42-61-490-002

(2) Make sure that you install the wheel chocks (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

SUBTASK 32-42-61-860-002

(3) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

E. Autobrake Selector Switch Removal

SUBTASK 32-42-61-020-001

(1) Remove the autobrake selector switch knob [2] from the autobrake selector switch shaft [7].

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SUBTASK 32-42-61-020-002

(2) Remove the knob [8] and knob [9].

SUBTASK 32-42-61-020-003

(3) Remove the lightplate [10].

SUBTASK 32-42-61-020-005

(4) Remove the multi-function control panel, P9-82 [6] from the P9 panel, do this task: TASK 31-65-03-000-801.

SUBTASK 32-42-61-020-007

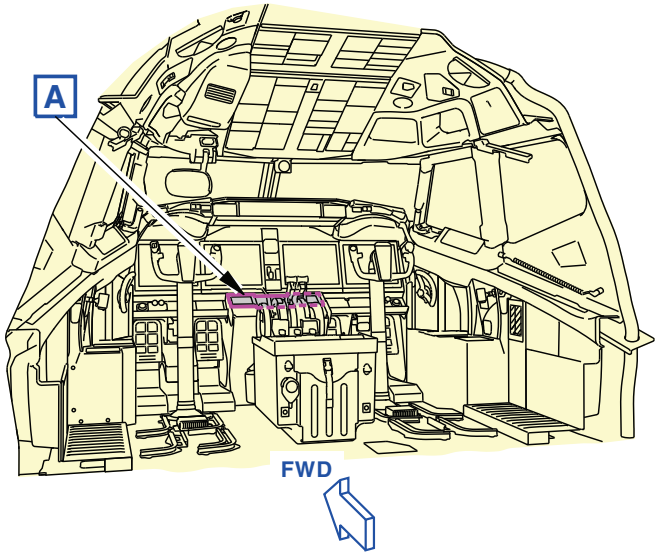
(5) Remove the nut [3] and washer [4] which hold the autobrake selector switch [1] in the control panel, P9-82 [6].

SUBTASK 32-42-61-020-008

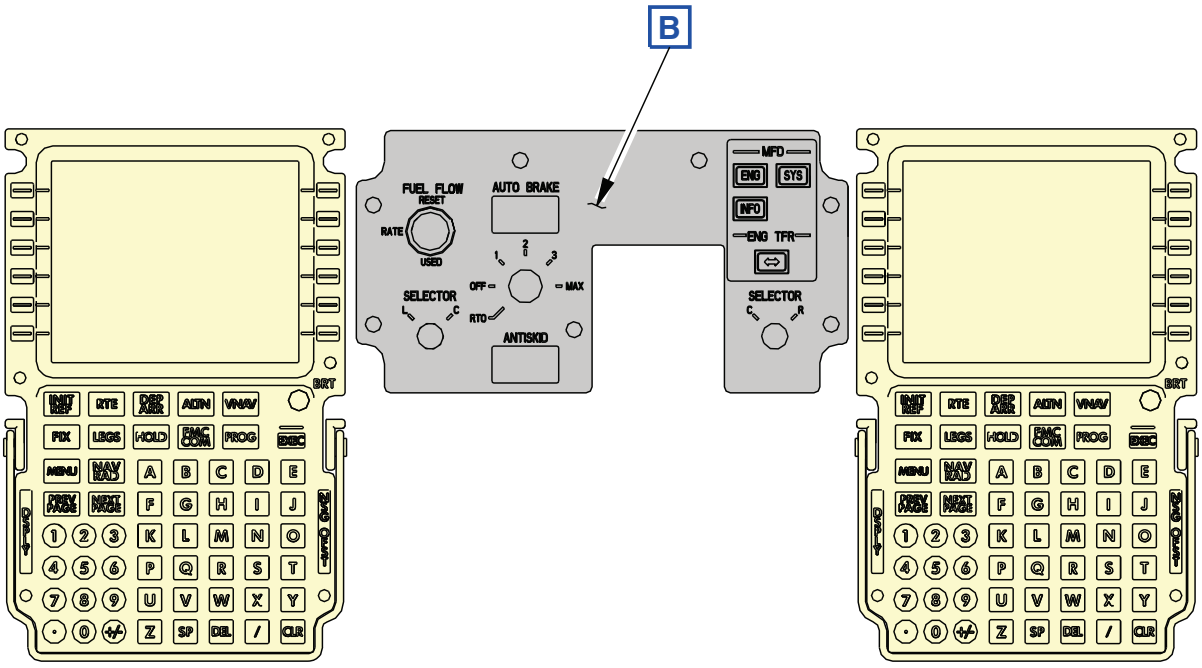
(6) Remove the locking ring [5] and autobrake selector switch [1] from the control panel, P9-82 [6].

——— END OF TASK ———

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FLIGHT COMPARTMENT



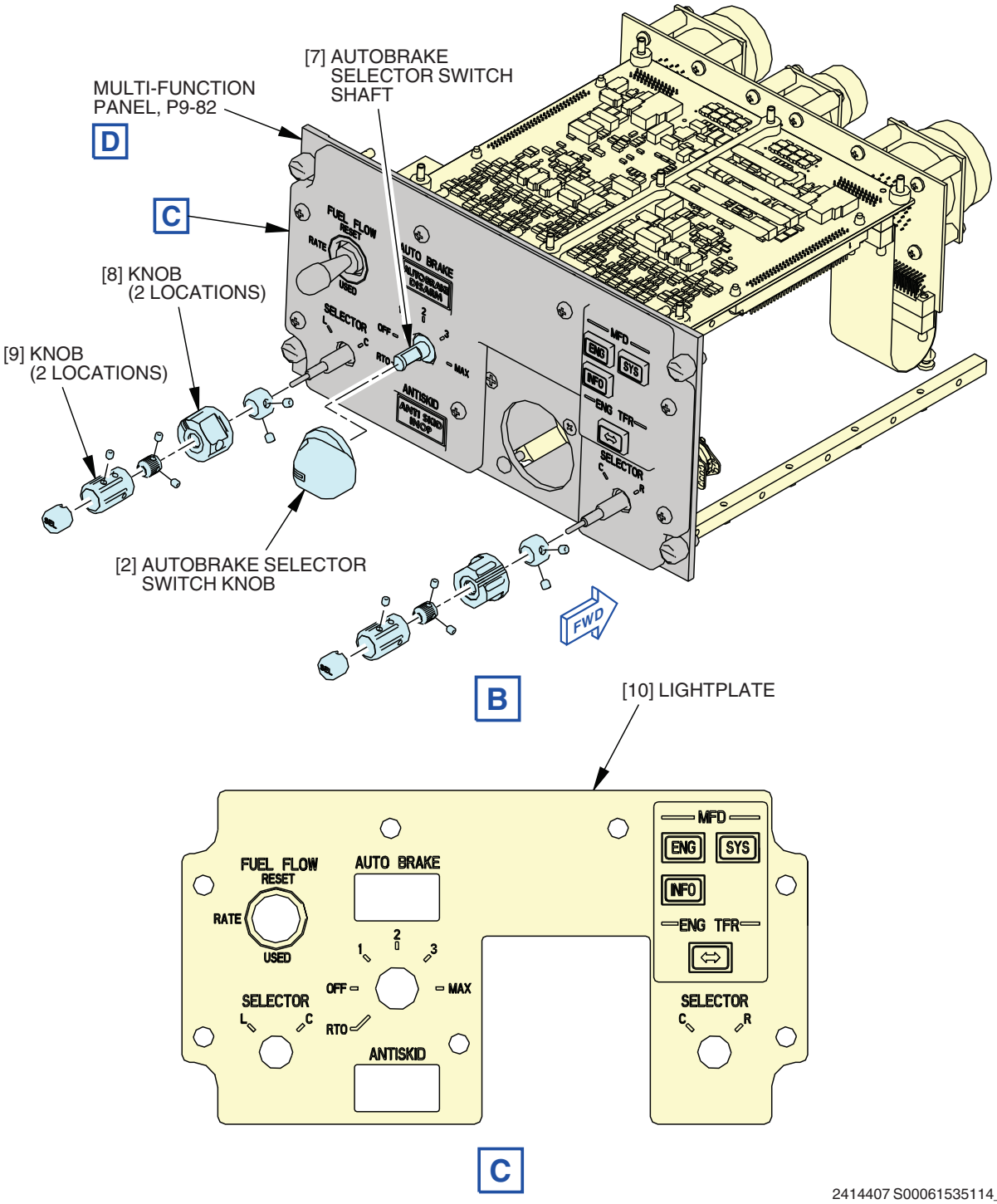
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A

2414406 S00061535113_V1

Autobrake Selector Switch Installation
Figure 401/32-42-61-990-802 (Sheet 1 of 3)

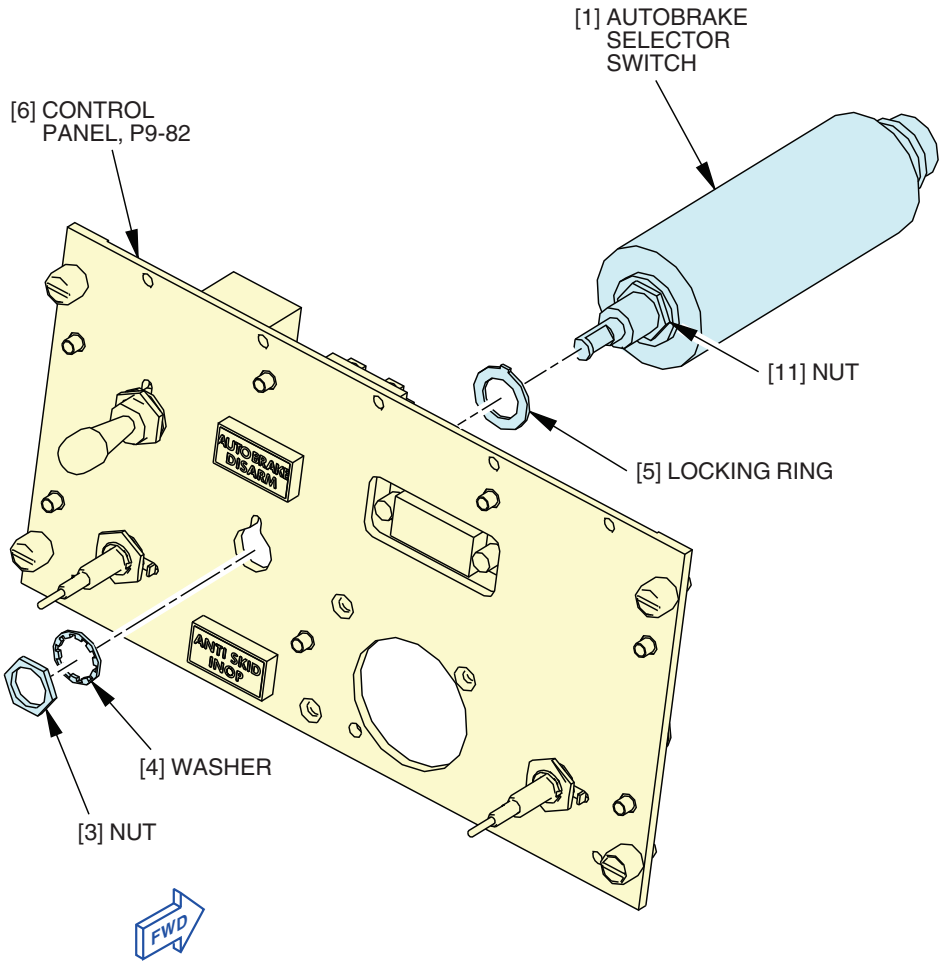
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2414407 S00061535114_V2

Autobrake Selector Switch Installation
Figure 401/32-42-61-990-802 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



MULTI-FUNCTION CONTROL PANEL, P9-82

D

2414408 S00061535115_V2

Autobrake Selector Switch Installation
Figure 401/32-42-61-990-802 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-61-400-801

3. Autobrake Selector Switch Installation

(Figure 401)

A. General

(1) This task is for the installation of the autobrake selector switch.

B. References

Reference	Title
10-11-05-500-801	Chock Installation Winds/Gusts Maximum 40 mph (35 Knots) (P/B 201)
10-11-05-500-802	Chock Installation in Winds More Than 35 Knots - Handling (P/B 201)
20-50-11-910-801	Standard Torque Values (P/B 201)
29-11-00-860-801	Hydraulic System A or B Pressurization (P/B 201)
29-11-00-860-805	Hydraulic System A or B Power Removal (P/B 201)
31-65-03-400-801	Multifunction Panel (P9-82) Installation (P/B 401)
34-21-00-820-801	Air Data Inertial Reference System - Alignment from the FMC CDU (P/B 201)
34-21-00-820-802	Air Data Inertial Reference System - Alignment from the ISDU (P/B 201)
34-21-00-820-803	Air Data Inertial Reference System - Automatic Alignment from the IRS MSU (P/B 201)

C. Location Zones

Zone	Area
117	Electrical and Electronics Compartment - Left
211	Flight Compartment - Left
212	Flight Compartment - Right

D. Autobrake Selector Switch Installation

SUBTASK 32-42-61-420-001

(1) Install the autobrake selector switch [1] in the control panel, P9-82 [6] as follows:

(a) Make sure that the nut [11] is in the correct location on the autobrake selector switch [1].

(b) Install the locking ring [5] to the autobrake selector switch [1].

(c) Install the autobrake selector switch [1] in the control panel, P9-82 [6].

(d) Install the washer [4] and nut [3] on the autobrake selector switch shaft [7] that holds the autobrake selector switch [1] to the control panel, P9-82 [6].

(e) Tighten the nut [3] (TASK 20-50-11-910-801).

(f) Install the lightplate [10].

(g) Install the autobrake selector switch knob [2] on the autobrake selector switch shaft [7].

(h) Install the knob [8] and knob [9].

SUBTASK 32-42-61-420-002

(2) Install the multi-function control panel, P9-82 [6] to the P9 panel, do this task: Multifunction Panel (P9-82) Installation, TASK 31-65-03-400-801.

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SUBTASK 32-42-61-865-001

(3) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1

E. Autobrake Selector Switch Installation Test

SUBTASK 32-42-61-860-003

(1) Make sure that the two thrust Levers are in the IDLE position.

SUBTASK 32-42-61-860-004

(2) Make sure that the landing gear control lever is in the DN position.

SUBTASK 32-42-61-860-005

(3) Make sure that the SPD BRK lever is in the DN position and the spoilers are stowed.

SUBTASK 32-42-61-860-006

(4) Make sure that the right and left Air Data Inertial Reference System (ADIRS)s are in the NAV mode as follows:

(a) Align the ADIRS (TASK 34-21-00-820-802 or TASK 34-21-00-820-801 or TASK 34-21-00-820-803).

SUBTASK 32-42-61-480-001

(5) Make sure that you install the wheel chocks (TASK 10-11-05-500-801 or TASK 10-11-05-500-802).

SUBTASK 32-42-61-860-007

(6) Release the PARKING BRAKE.

SUBTASK 32-42-61-860-008

WARNING

KEEP PERSONNEL AND EQUIPMENT AWAY FROM THE FLIGHT CONTROL SURFACES, THE THRUST REVERSERS, AND THE LANDING GEAR. THESE COMPONENTS CAN MOVE SUDDENLY WHEN YOU SUPPLY HYDRAULIC POWER. THIS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Pressurize the B hydraulic system (TASK 29-11-00-860-801).

SUBTASK 32-42-61-860-010

(8) Set the BRAKE TEST selector switch on the Antiskid/Autobrake Control Unit (AACU) to A/B.

NOTE: In next five steps, if the AUTO BRAKE DISARM light is ON when you push the ENABLE/VERIFY switches, the test will not successfully complete and the display on the front panel of the AACU will show BRK ####.

SUBTASK 32-42-61-860-011

(9) Do the steps that follow for AUTO BRAKE selector switch position 1.

(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF then to the 1 position.

(b) Push and then release the RESET switch on the AACU.

NOTE: The display on the front panel of the AACU will show MEM CLR.

(c) Push and hold the ENABLE/VERIFY switch on the AACU.

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(d) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(e) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B1 during the test.

(f) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(g) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-012

(10) Do the steps that follow for AUTO BRAKE selector switch position 2.

(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to 2 .

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B2 during the test.

(e) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(f) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-013

(11) Do the steps that follow for AUTO BRAKE selector switch position 3 .

(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to 3.

(b) Push an hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B3 during the test.

(e) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(f) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-014

(12) Do the steps that follow for the AUTO BRAKE selector switch position MAX.

(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] panel to MAX.

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK A/B4 during the test.

(e) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(f) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-015

(13) Do the steps that follow for AUTO BRAKE selector switch position RTO.

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(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to RTO.

(b) Push and hold the ENABLE/VERIFY switch on the AACU.

(c) While you hold the ENABLE/VERIFY switch, push and hold the VERIFY switch on the AACU.

(d) Release the ENABLE/VERIFY switch and the VERIFY switch at the same time.

NOTE: The AACU display will show BRK RTO during the test.

(e) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(f) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-016

(14) Do the steps that follow to test the brake metered pressure switches.

(a) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to 1.

(b) Push and hold the left brake pedal to the stop.

(c) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

(d) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF and then set it back to 1.

(e) Push and hold the right brake pedal to the stop.

(f) Make sure that the AUTO BRAKE DISARM light on the control panel, P9-82 [6] comes ON.

SUBTASK 32-42-61-860-017

(15) Set the autobrake selector switch knob [2] on the control panel, P9-82 [6] to OFF.

SUBTASK 32-42-61-860-018

(16) Set the BRAKE TEST selector switch on the AACU to NORM.

F. Put the Airplane Back to Its Usual Condition

SUBTASK 32-42-61-860-019

(1) Do this task: Hydraulic System A or B Power Removal, TASK 29-11-00-860-805.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUTOBRAKE SHUTTLE VALVE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Removal of the autobrake shuttle valve

(2) Installation of the autobrake shuttle valve.

B. The pressure switch removal and installation steps will be given as a part of the autobrake shuttle valve removal and installation task.

C. There are two autobrake shuttle valves, one for each landing gear, located on the aft wall of the main landing gear wheel well.

D. This procedure is applicable for the two shuttle valves.

TASK 32-42-71-000-801

2. Autobrake Shuttle Valve Removal

(Figure 401)

A. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

D. Prepare for the Removal

SUBTASK 32-42-71-480-005

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-71-860-001

(2) To remove pressure from the normal (system B) hydraulic system, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

(a) Make sure that the air charging valve stays open until the reservoir air pressure gauge shows 0 psi (0 kPa).

NOTE: This additional step makes complete depressurization of the reservoirs.

SUBTASK 32-42-71-480-002

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-71-860-002

(4) Release the PARKING BRAKE.

SUBTASK 32-42-71-840-001

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

(a) After you operate the brake pedals 12 times, open the air charging valve until the reservoir air pressure gauge shows 0 psi (0 kPa) (TASK 29-09-00-860-802).

(b) Close the air charging valve (TASK 29-09-00-860-802).

SUBTASK 32-42-71-860-003

(6) Set the PARKING BRAKE.

SUBTASK 32-42-71-860-004

(7) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

E. Autobrake Shuttle Valve Removal

SUBTASK 32-42-71-020-001

(1) Disconnect the electrical connector [1] from the pressure switch [2] on the autobrake shuttle valve [3].

SUBTASK 32-42-71-020-002

(2) Disconnect the hydraulic lines from the left autobrake shuttle valve [3]:

(a) Disconnect the hydraulic line [4] from the union [5].

(b) Disconnect the hydraulic line [9] from the union [5].

(c) Disconnect the hydraulic lines [7] and hydraulic line [13] from the tee coupling [18].

SUBTASK 32-42-71-020-006

(3) Disconnect the hydraulic lines from the right autobrake shuttle valve [3]:

(a) Disconnect the hydraulic line [17] from the union [5].

(b) Disconnect the hydraulic line [15] and hydraulic line [16] from the tee coupling [18].

(c) Disconnect the hydraulic line [14] and hydraulic line [9] from the tee coupling [18].

SUBTASK 32-42-71-480-003

(4) Install the caps/plugs, STD-14686, in the hydraulic lines.

SUBTASK 32-42-71-020-003

(5) Remove the bolts [11] and washers [10] that hold the autobrake shuttle valve [3] to the valve bracket [8].

SUBTASK 32-42-71-020-004

(6) If the replacement autobrake shuttle valve [3] does not have hydraulic fittings installed, do these steps:

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(a) Remove the unions [5] and packings [6].

(b) Discard the packings [6].

SUBTASK 32-42-71-020-005

(7) Do the steps that follow to remove the pressure switch [2] from the autobrake shuttle valve [3]:

(a) Remove the lockwire between the pressure switch [2] and autobrake shuttle valve [3].

(b) Remove the pressure switch [2] from the autobrake shuttle valve [3].

(c) Remove the packing [12] from the pressure switch [2].

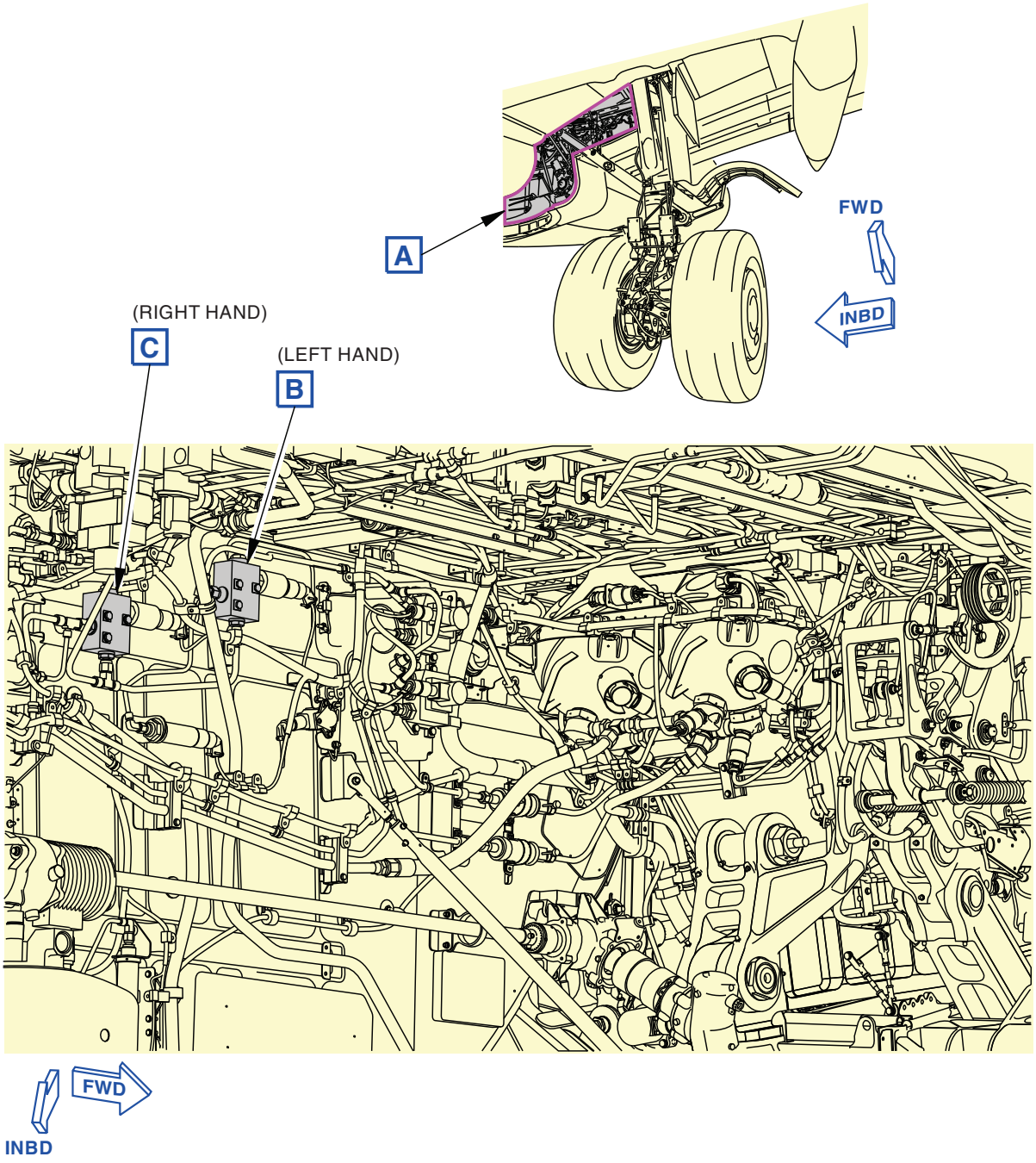
(d) Discard the packing [12].

SUBTASK 32-42-71-480-004

(8) Install the caps/plugs, STD-14686, in the ports of the autobrake shuttle valve [3].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



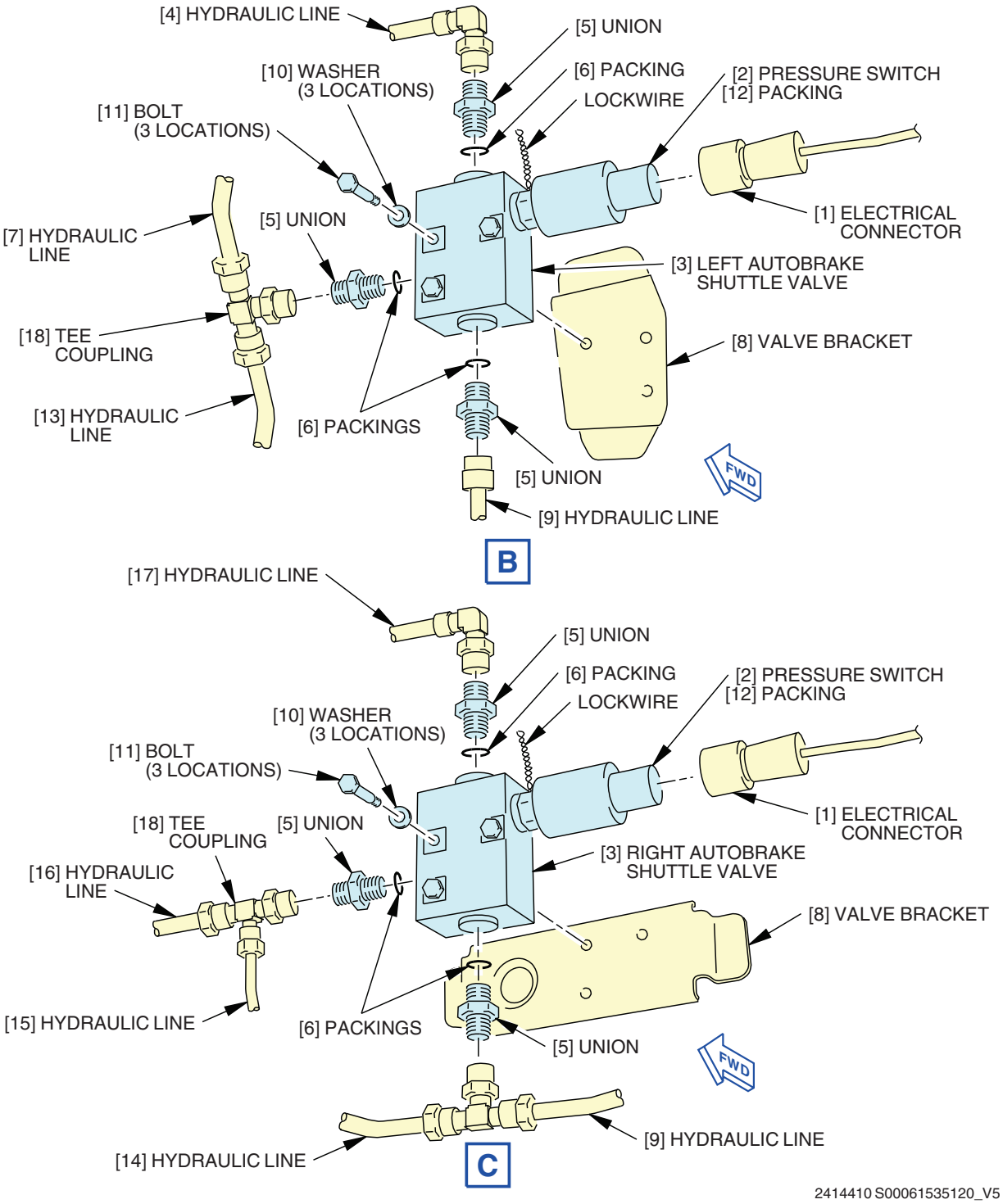
MAIN LANDING GEAR WHEEL WELL
(LEFT SIDE)

A

2414409 S00061535119_V2

Autobrake Shuttle Valve Installation
Figure 401/32-42-71-990-801 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Autobrake Shuttle Valve Installation
Figure 401/32-42-71-990-801 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-71-400-801

3. Autobrake Shuttle Valve Installation

(Figure 401)

A. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)
32-42-00-710-802	Autobrake Shuttle Valve Operational Test (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

B. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

C. Consumable Materials

Reference	Description	Specification
A00247	Sealant - Pressure And Environmental - Chromate Type	BMS5-95
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G01912	Lockwire - MS20995NC32, Monel - 0.032 Inch (0.8128 mm) Diameter	NASM20995
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

D. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Autobrake shuttle valve	32-42-71-01-010	PTF ALL
6	Packing	32-42-71-02-010	PTF ALL
12	Packing	32-42-71-02-060	PTF ALL

E. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

F. Autobrake Shuttle Valve Installation

SUBTASK 32-42-71-860-006

(1) Make sure that these circuit breakers are open and have safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-71-080-001

(2) Remove the caps/plugs, STD-14686, from the ports of the autobrake shuttle valve [3].

SUBTASK 32-42-71-420-001

(3) If the replacement autobrake shuttle valve [3] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the unions [5] and new packings [6] for the autobrake shuttle valve [3] with hydraulic fluid, D00153.

(b) Install the packings [6] on the unions [5].

(c) Install the unions [5] in the three hydraulic ports in the autobrake shuttle valve [3].

SUBTASK 32-42-71-420-002

(4) Do the steps that follow to install the pressure switch [2] on the autobrake shuttle valve [3]:

(a) Lubricate the threads of the pressure switch [2] and the new packing [12] with hydraulic fluid, D00153.

(b) Install the packing [12] on the pressure switch [2].

(c) Install the pressure switch [2] on the autobrake shuttle valve [3] and tighten to a torque of 170 in-lb (19.2 N·m).

(d) Install MS20995NC32 lockwire, G01912, between the pressure switch [2] and autobrake shuttle valve [3].

SUBTASK 32-42-71-100-001

(5) Do the steps below to get a good electrical bond between the autobrake shuttle valve [3] and valve bracket [8]:

(a) To fully clean the mating surface of the valve bracket [8] and the autobrake shuttle valve [3], do this task: Hand Clean Metal Surfaces with Abrasives, TASK 20-10-34-120-801.

(b) Manually apply coating, C00064, to all of the cleaned areas of the valve bracket [8].

(c) Fay surface seal the autobrake shuttle valve [3] to the valve bracket [8] with sealant, A00247.

SUBTASK 32-42-71-420-003

WARNING USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(6) Apply Cor-Ban 27L Compound, G50237, to the bolts [11] and washers [10] that hold the autobrake shuttle valve [3] to the valve bracket [8].

SUBTASK 32-42-71-420-004

(7) Install the bolts [11] and washers [10] to hold the autobrake shuttle valve [3] to the valve bracket [8].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-71-080-002

(8) Remove caps/plugs, STD-14686, from the hydraulic lines.

SUBTASK 32-42-71-420-005

(9) Connect the hydraulic lines to the left autobrake shuttle valve [3]:

(a) Connect the hydraulic line [4] to the union [5].

1) Tighten the B-nut on hydraulic line [4] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(b) Connect the hydraulic line [9] to the union [5].

1) Tighten the B-nut on hydraulic line [9] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(c) Connect the hydraulic lines [7] and hydraulic line [13] to the tee coupling [18].

1) Tighten the B-nut on hydraulic lines [7] to 270 ±14 in-lb (30.5 ±1.6 N·m).

2) Tighten the B-nut on hydraulic line [13] to 140 ±7 in-lb (15.82 ±0.79 N·m).

SUBTASK 32-42-71-420-007

(10) Connect the hydraulic lines to the right autobrake shuttle valve [3]:

(a) Connect the hydraulic line [17] to the union [5].

1) Tighten the B-nut on hydraulic line [17] to 270 ±14 in-lb (30.5 ±1.6 N·m).

(b) Connect the hydraulic line [15] and hydraulic line [16] to the tee coupling [18].

1) Tighten the B-nut on hydraulic line [16] to 270 ±14 in-lb (30.5 ±1.6 N·m).

2) Tighten the B-nut on hydraulic line [15] to 140 ±7 in-lb (15.8 ±0.8 N·m).

(c) Connect the hydraulic line [14] and hydraulic line [9] to the tee coupling [18].

1) Tighten the B-nuts on hydraulic line [14] and hydraulic line [9] to 270 ±14 in-lb (30.5 ±1.6 N·m).

SUBTASK 32-42-71-420-006

(11) Before you connect the electrical connector [1] to the autobrake shuttle valve [3], examine the electrical connector [1] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, refer to (SWPM 20-60-03) to correct the problem.

(b) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [1] (SWPM 20-60-03).

SUBTASK 32-42-71-420-008

(12) Connect the electrical connector [1] to the autobrake shuttle valve [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Autobrake Shuttle Valve Installation Test

SUBTASK 32-42-71-860-005

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-71-710-001

(2) For the normal brake system, do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

SUBTASK 32-42-71-710-002

(3) Do this task: Autobrake Shuttle Valve Operational Test, TASK 32-42-00-710-802.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

AUTOBRAKE PRESSURE CONTROL MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) A removal of the autobrake pressure control module

(2) An installation of the autobrake pressure control module

(3) A removal of the control module line replaceable units, (the servo valve, the solenoid valve, and the pressure switches).

(4) An installation of the control module line replaceable units, (the servo valve, the solenoid valve, and the pressure switches).

B. The autobrake pressure control module is on the ceiling of the main landing gear wheel well.

C. The autobrake pressure control module is referred to as the valve module [20] in this procedure.

TASK 32-42-81-000-801

2. Autobrake Pressure Control Module Valve Removal

(Figure 401)

A. General

(1) The autobrake pressure control module valve is referred to as the valve module [20] in this task.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Removal

SUBTASK 32-42-81-480-001

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-81-480-002

(2) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-860-011

(3) To remove pressure from the normal (system B) and the alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

(a) Make sure that the air charging valve stays open until the reservoir air pressure gauge shows 0 psi (0 kPa).

NOTE: This additional step makes complete depressurization of the reservoirs.

SUBTASK 32-42-81-860-001

(4) Release the PARKING BRAKE.

SUBTASK 32-42-81-840-001

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

(a) After you operate the brake pedals 12 times, open the air charging valve until the reservoir air pressure gauge shows 0 psi (0 kPa) (TASK 29-09-00-860-802).

(b) Close the air charging valve (TASK 29-09-00-860-802).

SUBTASK 32-42-81-860-003

(6) Set the PARKING BRAKE.

SUBTASK 32-42-81-860-004

(7) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

F. Autobrake Pressure Control Module Removal

SUBTASK 32-42-81-020-001

(1) Disconnect the hydraulic line [2], hydraulic line [9], and hydraulic line [11] from the valve module [20].

SUBTASK 32-42-81-480-003

(2) Install caps/plugs, STD-14686, in the hydraulic line [2], hydraulic line [9], and hydraulic line [11] and on the unions [13] and union [18].

SUBTASK 32-42-81-020-002

(3) Disconnect the electrical connector [1], electrical connector [6], electrical connector [7], and electrical connector [8] from the valve module [20].

SUBTASK 32-42-81-020-003

(4) Remove the bolts [3] and washer [4] which hold the valve module [20] to the airplane.

SUBTASK 32-42-81-020-004

(5) If the replacement valve module [20] does not have hydraulic fittings installed, do these steps:

(a) Remove the unions [13], union [18], packings [14], and packing [19].

(b) Discard the packings [14] and packing [19].

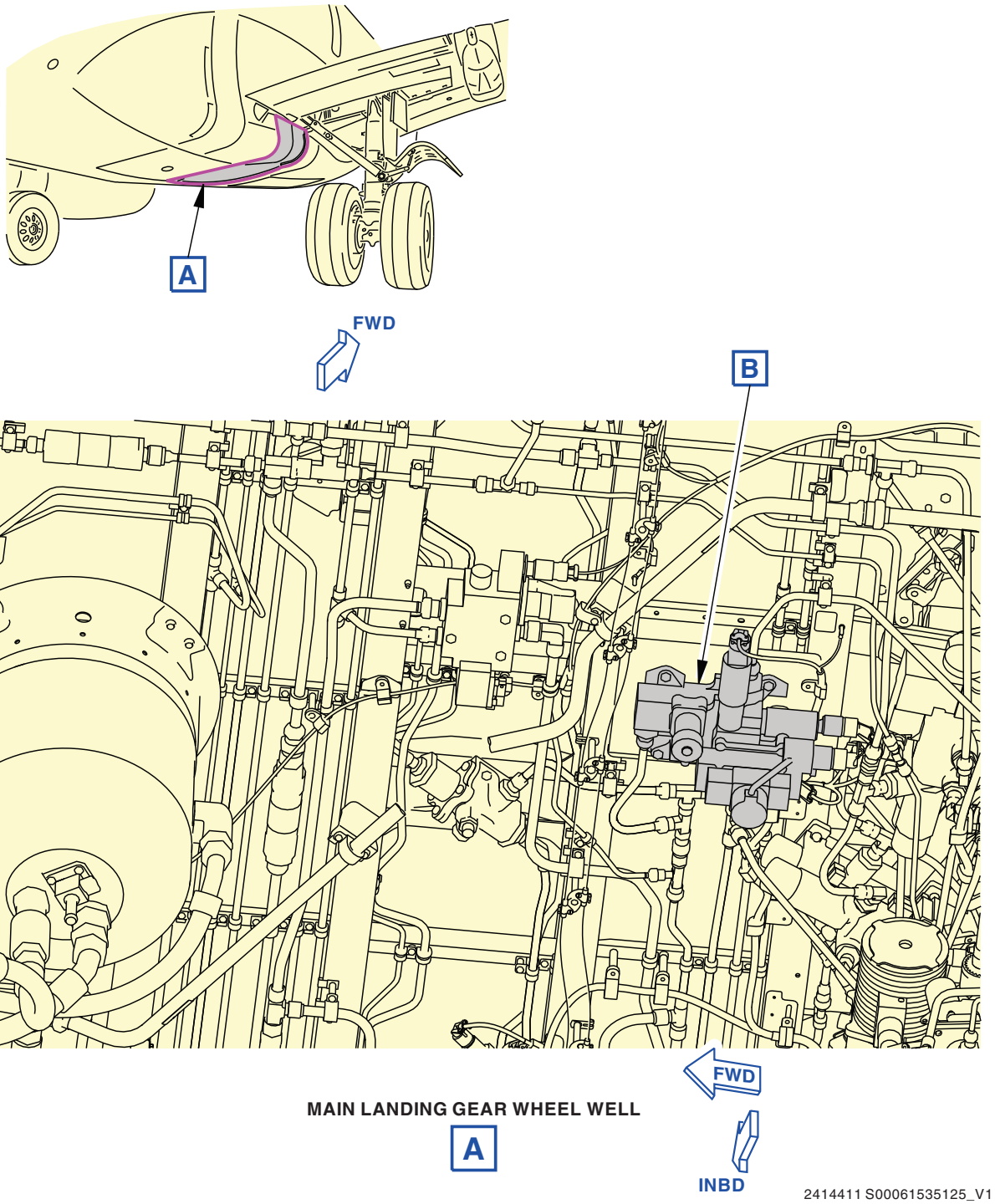
ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-480-004

(6) Install caps/plugs, STD-14686, in the hydraulic ports of the valve module [20].

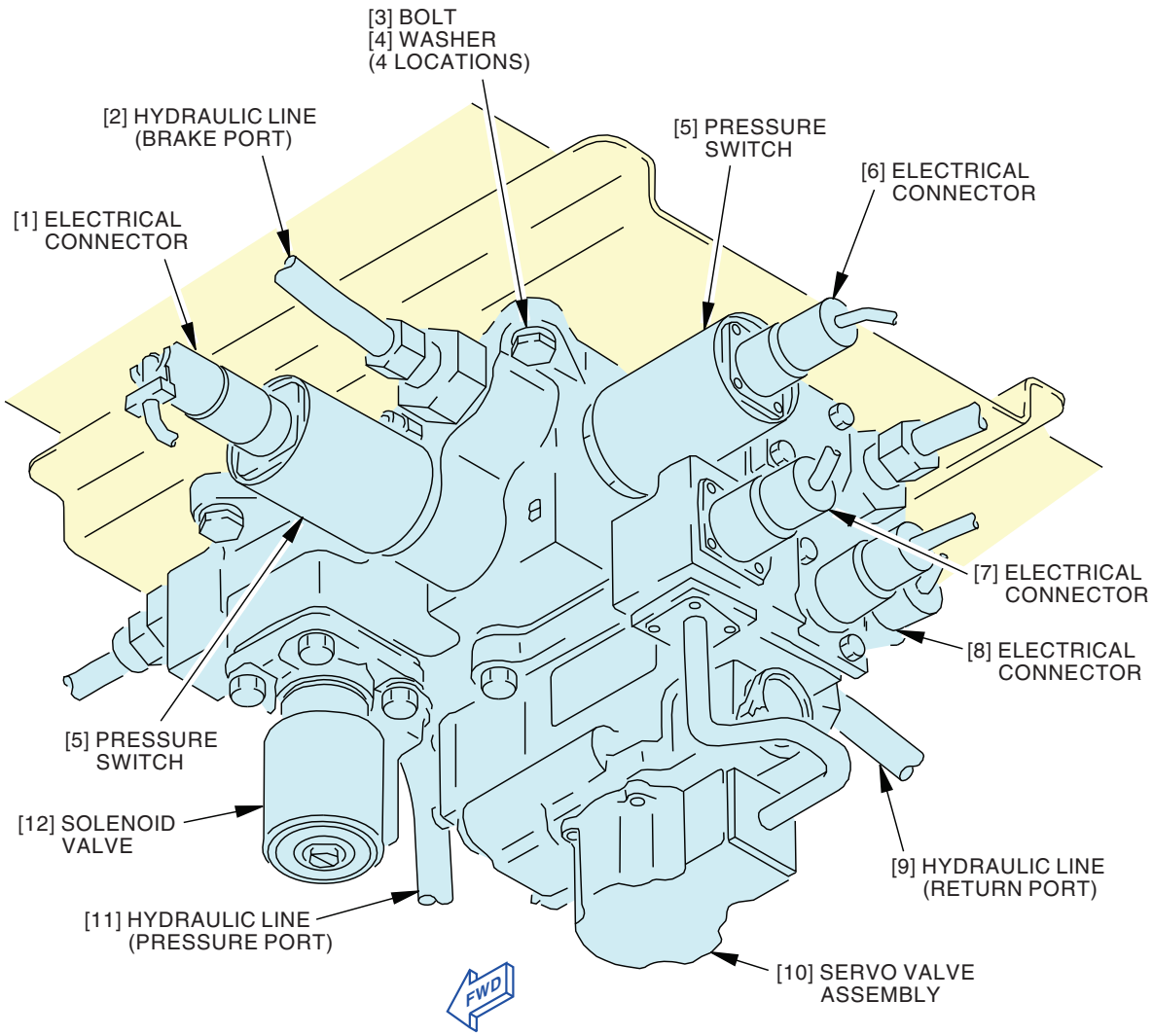
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Autobrake Pressure Control Module Installation
Figure 401/32-42-81-990-801 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

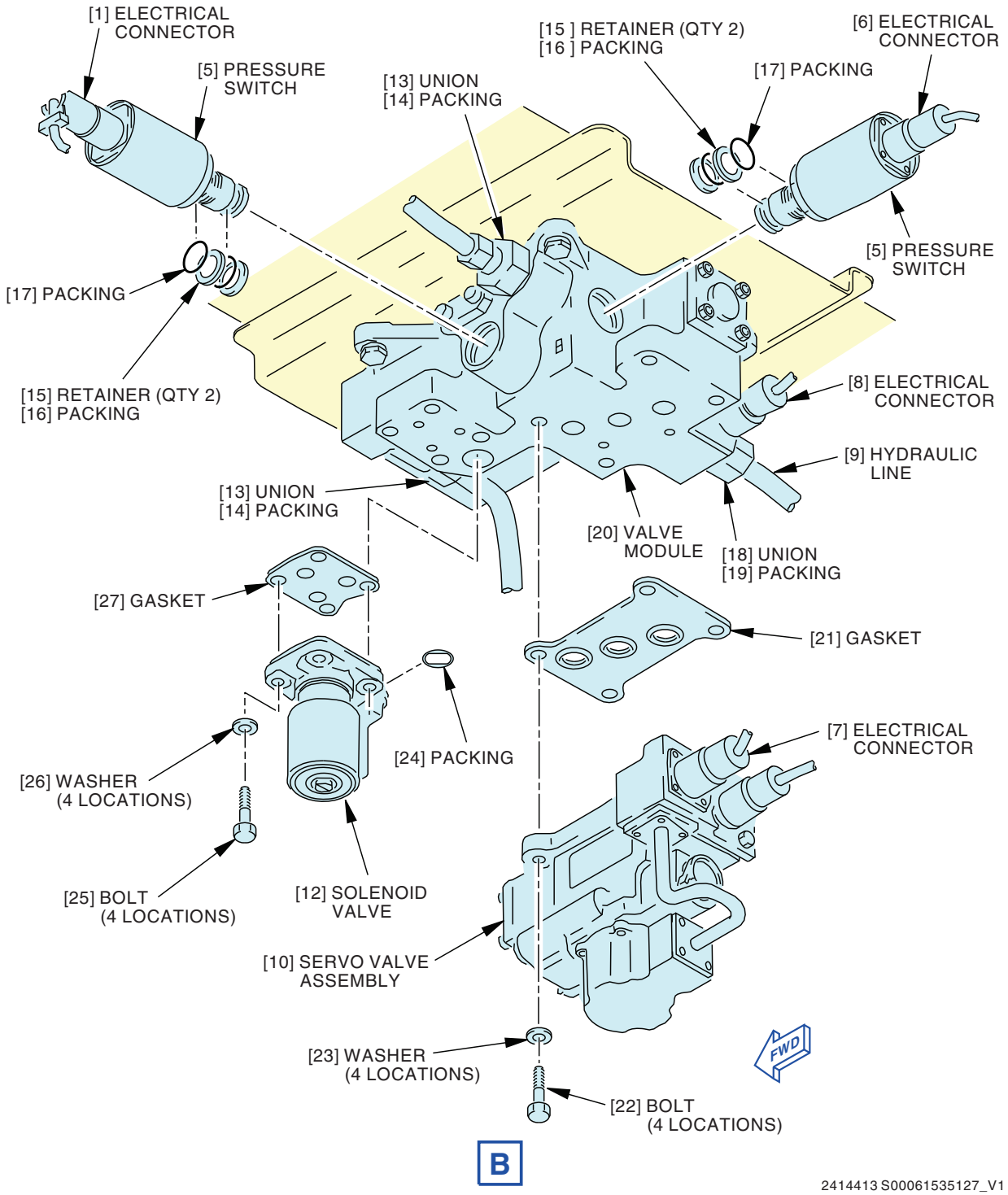


B

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Autobrake Pressure Control Module Installation
Figure 401/32-42-81-990-801 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Autobrake Pressure Control Module Installation
Figure 401/32-42-81-990-801 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-81-400-801

3. Autobrake Pressure Control Module Valve Installation

(Figure 401)

A. General

(1) The autobrake pressure control module valve is referred to as the valve module [20] in this task.

B. References

Reference	Title
20-10-34-120-801	Hand Clean Metal Surfaces with Abrasives (P/B 701)
32-41-00-870-801	Hydraulic Brake System Upstream of the Antiskid Valves Bleeding (P/B 201)
32-42-00-400-801	Autobrake Pressure Control Module Functional Test (P/B 501)
32-42-00-720-801	Antiskid/Autobrake Control Unit Operational Test (P/B 501)
SWPM 20-20-00	ELECTRICAL BONDING PROCESSES
SWPM 20-20-00, Section 14E	Standard Wiring Practices Manual
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
COM-1550	Bonding Meter - Approved, Intrinsically Safe (Approved for use in Class I, Divisions I & II hazardous (classified) locations. Outside these hazardous locations, COM-614 can be used in lieu of COM-1550). Part #: 620LK Supplier: 1CRL2 Part #: 720A-BK-KIT Supplier: 49374 Part #: M1 Supplier: 3AD17 Part #: M1B Supplier: 3AD17 Part #: T477W (C15292) Supplier: 06659
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Consumable Materials

Reference	Description	Specification
A02315	Sealant - Low Density, Synthetic Rubber. 2 Part	BMS5-142 Type II
C00064	Coating - Aluminum Chemical Conversion	BAC5719 Type II Class A (MIL-DTL-5541 Class 1A)
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	
G50237	Compound - Corrosion Inhibiting, Non-drying - Cor-Ban 27L	BMS3-38

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
14	Packing	32-42-81-01-030	PTF ALL
19	Packing	32-42-81-01-030	PTF ALL
20	Valve module	32-42-81-01-035	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

G. Autobrake Pressure Control Module Installation

SUBTASK 32-42-81-080-001

(1) Remove the caps/plugs, STD-14686, from the ports of the valve module [20].

SUBTASK 32-42-81-420-001

(2) If the replacement valve module [20] does not have hydraulic fittings installed, do these steps:

(a) Lubricate the unions [13], union [18], new packings [14], and new packing [19] for the valve module [20] with the hydraulic fluid, D00153.

(b) Install the packings [14] and packing [19] on the unions [13] and union [18].

(c) Install the unions [13] and union [18] in the valve module [20].

1) Tighten the union [13] and union [18] to 360 ±18 in-lb (41 ±2 N·m).

SUBTASK 32-42-81-100-001

(3) Do the steps below to help get a good electrical bond between the valve module [20] and the module bracket:

(a) Fully clean the mating surface of the bracket and the valve module [20] (TASK 20-10-34-120-801).

(b) Manually apply the coating, C00064, to all of the cleaned areas of the bracket.

SUBTASK 32-42-81-420-002

(4) Install the valve module [20]:

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION WHEN YOU USE COR-BAN 27L. IF IT GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER. GET MEDICAL AID. THIS MATERIAL CONTAINS FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(a) Apply Cor-Ban 27L Compound, G50237, to the bolts [3] and washers [4] that hold the valve module [20] to the bracket.

(b) Install the bolts [3] and washers [4] to hold the valve module [20] to the bracket.

(c) Do a check of the resistance between the valve module [20] and the bracket with an intrinsically safe approved bonding meter, COM-1550 (SWPM 20-20-00).

1) Make sure that the resistance is 0.0025 ohm (2.5 milliohms) or less.

(d) Remove caps/plugs, STD-14686, from the hydraulic line [2], hydraulic line [9], and hydraulic line [11].

(e) Connect the hydraulic line [2], hydraulic line [9], and hydraulic line [11] to the unions [13] and union [18] on the valve module [20].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(f) Tighten the B-nuts on hydraulic line [2], hydraulic line [9], and hydraulic line [11] to 170 ±9 in-lb (19.2 ±1.0 N·m).

(g) Before you connect the electrical connector [1], electrical connector [6], electrical connector [7], and electrical connector [8] to the valve module [20], examine the electrical connector [1], electrical connector [6], electrical connector [7], and electrical connector [8] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

1) If there is corrosion, refer to SWPM 20-60-03 to correct the problem.

2) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [1], electrical connector [6], electrical connector [7], and electrical connector [8] (SWPM 20-60-03).

3) Connect the electrical connector [1], electrical connector [6], electrical connector [7], and electrical connector [8] to the valve module [20].

SUBTASK 32-42-81-390-001

(5) Apply a fillet seal with sealant, A02315, around the mounting points between the valve module [20] and the bracket (SWPM 20-20-00, Section 14E).

H. Autobrake Pressure Control Module Installation Test

SUBTASK 32-42-81-860-005

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-81-710-001

(2) Do this task: Antiskid/Autobrake Control Unit Operational Test, TASK 32-42-00-720-801.

SUBTASK 32-42-81-710-003

(3) Do this task: Autobrake Pressure Control Module Functional Test, TASK 32-42-00-400-801.

SUBTASK 32-42-81-870-001

(4) Do this task: Hydraulic Brake System Upstream of the Antiskid Valves Bleeding, TASK 32-41-00-870-801.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 32-42-81-000-802

4. Autobrake Pressure Control Module Line Replaceable Units Removal

(Figure 401)

A. General

(1) The autobrake pressure control module is referred to as the valve module [20] in this task.

(2) This procedure has the removal of these components on the valve module [20]:

(a) solenoid valve [12]

(b) servo valve assembly [10]

(c) pressure switch [5]

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
29-09-00-860-802	Hydraulic Reservoirs Depressurization (P/B 201)
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

D. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

E. Prepare for the Removal

SUBTASK 32-42-81-480-005

WARNING MAKE SURE THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR COULD RETRACT AND CAUSE INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

SUBTASK 32-42-81-860-006

(2) To remove pressure from the normal (system B) and the alternate (system A) hydraulic systems, do this task: Hydraulic Reservoirs Depressurization, TASK 29-09-00-860-802.

(a) Make sure that the air charging valve stays open until the reservoir air pressure gauge shows 0 psi (0 kPa).

NOTE: This additional step makes complete depressurization of the reservoirs.

SUBTASK 32-42-81-480-006

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-42-81-860-007

(4) Release the PARKING BRAKE.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-840-002

(5) Operate the brake pedals until the gage at the brake accumulator shows no change in the pressure.

NOTE: You will have to operate the brake pedals approximately 12 times to fully release the pressure from the accumulator.

(a) After you operate the brake pedals 12 times, open the air charging valve until the reservoir air pressure gauge shows 0 psi (0 kPa) (TASK 29-09-00-860-802).

(b) Close the air charging valve (TASK 29-09-00-860-802).

SUBTASK 32-42-81-860-008

(6) Set the PARKING BRAKE.

SUBTASK 32-42-81-860-009

(7) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

F. Solenoid Valve Removal

SUBTASK 32-42-81-020-005

(1) Remove the bolts [25] and washers [26] that connect the solenoid valve [12] to the valve module [20].

SUBTASK 32-42-81-480-007

CAUTION

WHEN YOU REMOVE THE SOLENOID VALVE AND THE GASKET, MAKE SURE THAT CONTAMINATION CAN NOT GO INTO THE MODULE'S OPEN PORTS. DAMAGE TO THE VALVE CAN OCCUR IF LET CONTAMINATION GO INTO THE PORTS.

(2) Remove the solenoid valve [12] from the valve module [20].

SUBTASK 32-42-81-480-008

(3) Remove the packing [24] from the solenoid valve [12].

(a) Discard the packing [24].

SUBTASK 32-42-81-020-006

(4) Remove the gasket [27] from the valve module [20].

SUBTASK 32-42-81-480-009

(5) Install a cap/plug, STD-14686, in the port for the solenoid valve [12].

G. Servo Valve Removal

SUBTASK 32-42-81-020-007

(1) Disconnect the electrical connector [7] from the servo valve assembly [10].

SUBTASK 32-42-81-020-008

(2) Remove the bolts [22] and washers [23] that connect the servo valve assembly [10] to the valve module [20].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-480-010

CAUTION: WHEN YOU REMOVE THE SERVOVALVE AND THE GASKET, MAKE SURE THAT CONTAMINATION CAN NOT GO INTO THE MODULE'S OPEN PORTS. DAMAGE TO THE VALVE CAN OCCUR IF YOU LET CONTAMINATION GO INTO THE PORTS.

(3) Remove the servo valve assembly [10] from the valve module [20].

SUBTASK 32-42-81-020-009

(4) Remove the gasket [21].

SUBTASK 32-42-81-480-011

(5) Install caps/plugs, STD-14686, in the ports for the servo valve assembly [10].

H. Pressure Switch Removal

NOTE: These steps apply to the pressure switch [5] of the solenoid valve [12] and the pressure switch [5] of the servo valve assembly [10].

SUBTASK 32-42-81-020-010

(1) Disconnect the electrical connector [1] or electrical connector [6] from the pressure switch [5].

SUBTASK 32-42-81-480-012

(2) Remove the pressure switch [5] from the valve module [20].

SUBTASK 32-42-81-480-013

(3) Remove the packing [16], packing [17], and retainers [15] from the pressure switch [5].

(a) Discard the packing [16] and packing [17].

SUBTASK 32-42-81-480-014

(4) Install a cap/plug, STD-14686, in the port for the pressure switch [5].

——— END OF TASK ———

TASK 32-42-81-400-802

5. Autobrake Pressure Control Module Line Replaceable Units Installation

(Figure 401)

A. General

(1) The autobrake pressure control module is referred to as the valve module [20] in this task.

(2) The procedure that follows installs these components of the valve module [20]autobrake pressure control module:

(a) solenoid valve [12]

(b) servo valve assembly [10]

(c) pressure switch [5]

B. References

Reference	Title
32-42-00-720-801	Antiskid/Autobrake Control Unit Operational Test (P/B 501)
SWPM 20-60-03	Special Protection of Electrical Connectors

C. Tools/Equipment

Reference	Description
STD-14686	Fittings - Hydraulic, Caps/Plugs (ref. AMM 20-10-XX)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Consumable Materials

Reference	Description	Specification
D00153	Fluid - Hydraulic Fluid, Fire Resistant (Interchangeable And Intermixable With BMS 3-11 Type V)	BMS3-11 Type IV
G50136	Compound - Corrosion Inhibiting, Non-drying Paste	BMS3-38
G50171	Compound - Corrosion Inhibiting Compound, Interior Application - D5026NS or ZC-026	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Pressure switch	32-42-81-01-040	PTF ALL
10	Servo valve assembly	32-42-81-01-110	PTF ALL
12	Solenoid valve	32-42-81-01-080	PTF ALL
16	Packing	32-42-81-01-050	PTF ALL
17	Packing	32-42-81-01-045	PTF ALL
21	Gasket	32-42-81-01-095	PTF ALL
24	Packing	32-42-81-01-070	PTF ALL
27	Gasket	32-42-81-01-075	PTF ALL

F. Location Zones

Zone	Area
133	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Left
134	Main Landing Gear Wheel Well, Body Station 663.75 to Body Station 727.00 - Right

G. Solenoid Valve Installation

SUBTASK 32-42-81-420-003

(1) Install the new packing [24] on the solenoid valve [12].

SUBTASK 32-42-81-480-015

(2) Remove the cap/plug, STD-14686, from the port for the solenoid valve [12].

SUBTASK 32-42-81-480-016

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(3) Apply corrosion inhibiting material, G50136, to the bolts [25].

SUBTASK 32-42-81-020-011

(4) Install the gasket [27] on the valve module [20].

SUBTASK 32-42-81-020-012

(5) Put the solenoid valve [12] in its position on the valve module [20].

SUBTASK 32-42-81-420-004

(6) Install the bolts [25] and washers [26] to connect the solenoid valve [12] to the valve module [20].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-420-005

(7) Tighten the bolts [25] to 77.5 ±2.5 in-lb (8.8 ±0.3 N·m) in a crisscross sequence.

(a) Start with the bolt [25] closest to the nearest bolt [3] for the valve module [20]control module.

(b) Tighten all the bolts [25] to 7.5 ±2.5 in-lb (0.8 ±0.3 N·m), then 37.5 ±2.5 in-lb (4.2 ±0.3 N·m), and then to the final torque: 77.5 ±2.5 in-lb (8.8 ±0.3 N·m).

SUBTASK 32-42-81-420-006

(8) Use a 0.0015 in. (0.0381 mm) shim stock to examine the gasket [27] for gaps.

(a) Gaps more than 0.0015 in. (0.0381 mm) are not permitted.

NOTE: This applies to gaps on either side of the gasket [27]. There must be no gap more than 0.0015 in. (0.0381 mm) between the solenoid valve [12] and the gasket [27] and between the valve module [20] and the gasket [27].

H. Servo Valve Installation

SUBTASK 32-42-81-480-017

(1) Remove the caps/plugs, STD-14686, from the port for the servo valve assembly [10].

SUBTASK 32-42-81-480-018

WARNING

USE NITRILE GLOVES FOR SKIN PROTECTION AGAINST CORROSION INHIBITOR MATERIAL, G50136. IF CORROSION INHIBITOR MATERIAL, G50136 GETS ON YOUR SKIN, IMMEDIATELY REMOVE IT WITH WATER. IF THIS MATERIAL GETS IN YOUR EYES, IMMEDIATELY FLUSH YOUR EYES WITH WATER AND GET MEDICAL AID. THIS MATERIAL CONTAINS VERY POISONOUS AND FLAMMABLE AGENTS WHICH CAN CAUSE INJURIES TO PERSONNEL.

(2) Apply corrosion inhibiting material, G50136 to the bolts [22].

SUBTASK 32-42-81-020-013

(3) Install the gasket [21] on the valve module [20].

SUBTASK 32-42-81-020-014

(4) Put the servo valve assembly [10] in its position on the valve module [20].

SUBTASK 32-42-81-420-007

(5) Install the bolts [22] and washers [23] to connect the servo valve assembly [10] to the valve module [20].

SUBTASK 32-42-81-420-008

(6) Tighten the bolts [22] to 77.5 ±2.5 in-lb (8.8 ±0.3 N·m) in a crisscross sequence.

(a) Start with the bolt [22] closest to the solenoid valve [12].

(b) Tighten all the bolts [22] to 7.5 ±2.5 in-lb (0.8 ±0.3 N·m), then 37.5 ±2.5 in-lb (4.2 ±0.3 N·m), and then to the final torque: 77.5 ±2.5 in-lb (8.8 ±0.3 N·m).

SUBTASK 32-42-81-420-009

(7) Use a 0.0015 in. (0.0381 mm) shim stock to examine the gasket [21] for gaps.

(a) Gaps more than 0.0015 in. (0.0381 mm) are not permitted.

NOTE: This applies to gaps on either side of the gasket [21]. There must be no gap more than 0.0015 in. (0.0381 mm) between the servo valve assembly [10] and the gasket [21] and between the valve module [20] and the gasket [21].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 32-42-81-020-015

(8) Before you connect the electrical connector [7] to the servo valve assembly [10], examine the electrical connector [7] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, refer to SWPM 20-60-03 to correct the problem.

(b) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [7] (SWPM 20-60-03).

(9) Connect the electrical connector [7] to the servo valve assembly [10].

I. Pressure Switch Installation

NOTE: These steps apply to the pressure switch [5] of the solenoid valve [12] and the pressure switch [5] of the servo valve assembly [10] .

SUBTASK 32-42-81-420-010

(1) Lubricate the new packing [16], new packing [17], and retainers [15] with hydraulic fluid, D00153.

SUBTASK 32-42-81-420-011

(2) Install the packing [17] in the middle groove of the pressure switch [5].

SUBTASK 32-42-81-420-012

(3) Install a retainer [15], packing [16], and retainer [15] in the end groove of the pressure switch [5].

SUBTASK 32-42-81-480-019

(4) Install the pressure switch [5] on the valve module [20].

(a) For HR Textron (P/N 20102060 series) pressure switch [5], tighten the pressure switch [5] to a torque of 359.5 ±17.5 in-lb (40.6 ±2.0 N·m).

SUBTASK 32-42-81-420-013

(5) Before you connect the electrical connector [1] or electrical connector [6] to the pressure switch [5], examine the electrical connector [1] or electrical connector [6] for corrosion.

WARNING

DO THE STEPS BELOW IF THE AIRPLANE OPERATES WHERE DEICING FLUID THAT CONTAINS POTASSIUM FORMATE OR POTASSIUM ACETATE IS USED. ALSO, DO THE STEPS FOR ALL AIRPLANES THAT YOU FOUND CORROSION IN THE ELECTRICAL CONNECTORS IN THE MAIN WHEEL WELL. THE ELECTRICAL CONNECTORS ARE IN A SYSTEM THAT IS NECESSARY FOR SAFE FLIGHT.

(a) If there was corrosion, refer to SWPM 20-60-03 to correct the problem.

(b) Apply the D5026NS or ZC-026 compound, G50171, to the electrical connector [1] or electrical connector [6] (SWPM 20-60-03).

(c) Connect the electrical connector [1] or electrical connector [6] to the pressure switch [5].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

J. Put the Airplane Back to its Usual Condition

SUBTASK 32-42-81-860-010

(1) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
A	16	C01345	LANDING GEAR AUTOBRAKE BITE CONT 2
A	18	C00583	LANDING GEAR AUTOBRAKE BITE CONT 1
E	16	C00196	LANDING GEAR ANTISKID INBD
E	18	C00195	LANDING GEAR ANTISKID OUTBD

SUBTASK 32-42-81-710-002

(2) Do this task: Antiskid/Autobrake Control Unit Operational Test, TASK 32-42-00-720-801.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PARKING BRAKE SYSTEM - ADJUSTMENT/TEST

1. General

A. This procedure has these tasks:

(1) Adjustment of the parking brake system

(2) Operational test of the parking brake system

(3) Parking brake hold check.

TASK 32-44-00-820-801

2. Parking Brake System Adjustment

(Figure 501, Figure 502)

A. General

(1) This procedure has the steps to adjust the parking brake system for the captain or first officer.

B. References

Reference	Title
10-11-05 P/B 201	CHOCK INSTALLATION
32-00-01-480-801	Landing Gear Downlock Pins Installation (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-1585	Kit - Rigging Pin Part #: F70207-151 Supplier: 81205

D. Location Zones

Zone	Area
112	Area Forward of Nose Landing Gear Wheel Well
211	Flight Compartment - Left

E. Access Panels

Number	Name/Location
112A	Forward Access Door

F. Prepare for the Adjustment

SUBTASK 32-44-00-480-011

WARNING MAKE SURE THAT THE DOWNLOCK PINS ARE INSTALLED ON ALL THE LANDING GEAR. WITHOUT THE DOWNLOCK PINS, THE LANDING GEAR CAN RETRACT. THIS CAN CAUSE INJURIES TO PERSONS, AND DAMAGE TO EQUIPMENT.

(1) If the downlock pins are not installed in the nose and main landing gear, do this task: Landing Gear Downlock Pins Installation, TASK 32-00-01-480-801.

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SUBTASK 32-44-00-860-001

(2) Open this access panel:

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112A	Forward Access Door

to get access to the parking brake system.

SUBTASK 32-44-00-480-002

(3) Make sure that the tires have chocks installed around them (PAGEBLOCK 10-11-05/201).

SUBTASK 32-44-00-860-003

(4) Release the parking brake.

G. Parking Brake System Adjustment

SUBTASK 32-44-00-820-001

(1) Adjust the pawls [11] of the parking brake:

(a) Remove the cotter pins [10] from the pawls [11]].

(b) Loosen the nuts [9].

(c) Adjust the pawls [11] to make sure that there is a 0.13 in. (3.30 mm) - 0.17 in. (4.32 mm) gap between the support assembly [21] and bellcrank [13] when parking brake is engaged.

1) Make sure that both brake pedals are applied to get parking brake engaged.

(d) Make sure that both pawls [11] adjusted to the same height.

(e) Tighten the nuts [9].

(f) Install the cotter pins [10] on the pawls [11].

(g) Release the brake pedals.

SUBTASK 32-44-00-820-002

(2) Adjust the pawl stop:

(a) Install the rig pin LGB1, from rigging pin kit, SPL-1585, to the bellcrank rig pin holes [3] on captain side thought bellcranks [13] and structure.

(b) Rotate the pawls [11] until they touch the bellcrank bushings [12].

(c) Hold the pawls against the bushings [12].

(d) Adjust the pawl stop until the clearance between rubber bumper [4] and pawls [11] at the closest point is 0.34 in. (8.64 mm) - 0.46 in. (11.68 mm):

1) Remove the nut [7] from the pawl stop assembly.

2) Remove the pawl stop assembly (the bumper [4], screw [5], and washers [6]) from the pawl stop bracket [8].

3) Add or remove washers [6] until you get the correct clearance.

4) Install the pawl stop assembly (the bumper [4], screw [5], and washers [6]) on the pawl stop bracket [8].

5) Install the nut [7] on the pawl stop assembly.

(e) Release the bellcranks [13].

SUBTASK 32-44-00-820-003

(3) Adjust the parking brake lever [42] (Figure 501, Figure 502):

(a) Make sure that the rig pin LGB1, from rigging pin kit, SPL-1585, is installed.

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(b) Install a rig pin LGB2, from rigging pin kit, SPL-1585, through the jackshaft rig pin hole [15] and switch support bracket [18].

(c) Make sure that the pawl [11] is against rubber bumper [4].

(d) Adjust the length of the rod [1] and the rod [49] so that the rod end bearing holes align with the crank holes.

(e) Make sure that at least one half of the rod end is shown in the inspection holes.

(f) If necessary, remove the cotter pin [47], washer [46], and pin [45] from the parking brake lever [42] and the rod [44].

(g) Loosen the jamnut [48] on the bottom of the rod [44].

(h) Adjust the length of the rod [44] until the stop surface of the parking brake lever [42] is against the top of the control stand when the pin [45] is installed.

NOTE: If the rod [44] is adjusted with this method, the rod [44] itself must be turned in full 360 degree turns. The rod assembly is orientation sensitive.

(i) Tighten the jamnut [48] on the rod [44].

(j) If removed, install the pin [45], washer [46], and cotter pin [47] to hold the parking brake lever [42] to the rod [44].

(k) Make sure that the parking brake lever [42] moves smoothly.

(l) Remove the rig pin LGB2, from rigging pin kit, SPL-1585, from the jackshaft rig pin hole [15] and the switch support bracket [18].

(m) Remove the rig pin LGB1, from rigging pin kit, SPL-1585, from the bellcrank rig pin holes [3] on captain side thought bellcranks [13] and structure.

SUBTASK 32-44-00-820-004

(4) Adjust the switch [17] (Figure 501, Figure 502):

(a) Make sure that the rudder pedals are released.

(b) Pull the parking brake lever [42] on the control stand, until it stops, and hold it in this position.

(c) Make sure that the switch actuator roller [22] lightly touches the rod [1] end.

(d) Make sure that the parking brake light [43], next to the parking brake lever [42], does not come on.

(e) If the switch [17] does not operate in the correct range, do these steps:

1) Install the rig pin LGB1, from rigging pin kit, SPL-1585, to the bellcrank rig pin holes [3] on captain side thought bellcranks [13] and structure.

2) Loosen the screws [19] and nuts [20] that hold the switch actuator [16] and switch [17] to the airplane.

3) Rotate the pawls [11] until they touch the bellcrank bushings [12].

4) Adjust the switch actuator [16] until the switch actuator roller [22] touches the end of the rod [1].

5) Continue to rotate switch [17] until the switch clicks.

6) Tighten the screws [19] and nuts [20].

7) Remove the rig pin LGB1, rigging pin kit, SPL-1585 from the bellcrank rig pin holes [3] on captain side.

(f) Set the parking brake by depressing both brake pedals on the applicable side (captain's or first officer's) to the stops.

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(g) Pull the parking brake lever [42] and make sure that parking brake light [43] next to the parking brake lever [42] comes on.

NOTE: The parking brake light [43] should come on between 35 to 45 degrees of parking brake lever [42] travel from the released position.

(h) Continue to pull the parking brake lever [42] until it stops.

(i) With the parking brake lever [42] at the stops, release the brake pedals.

(j) Make sure that the brake pedals stay applied.

(k) Release the parking brake by depressing both pedal simultaneously.

(l) Make sure that the parking brake light [43] goes off.

(m) Make sure that the parking brake lever [42] returns to its usual position without assistance.

(n) Set the parking brake.

(o) Repeat steps (f) through (n) to adjust the first officer's rudder pedals.

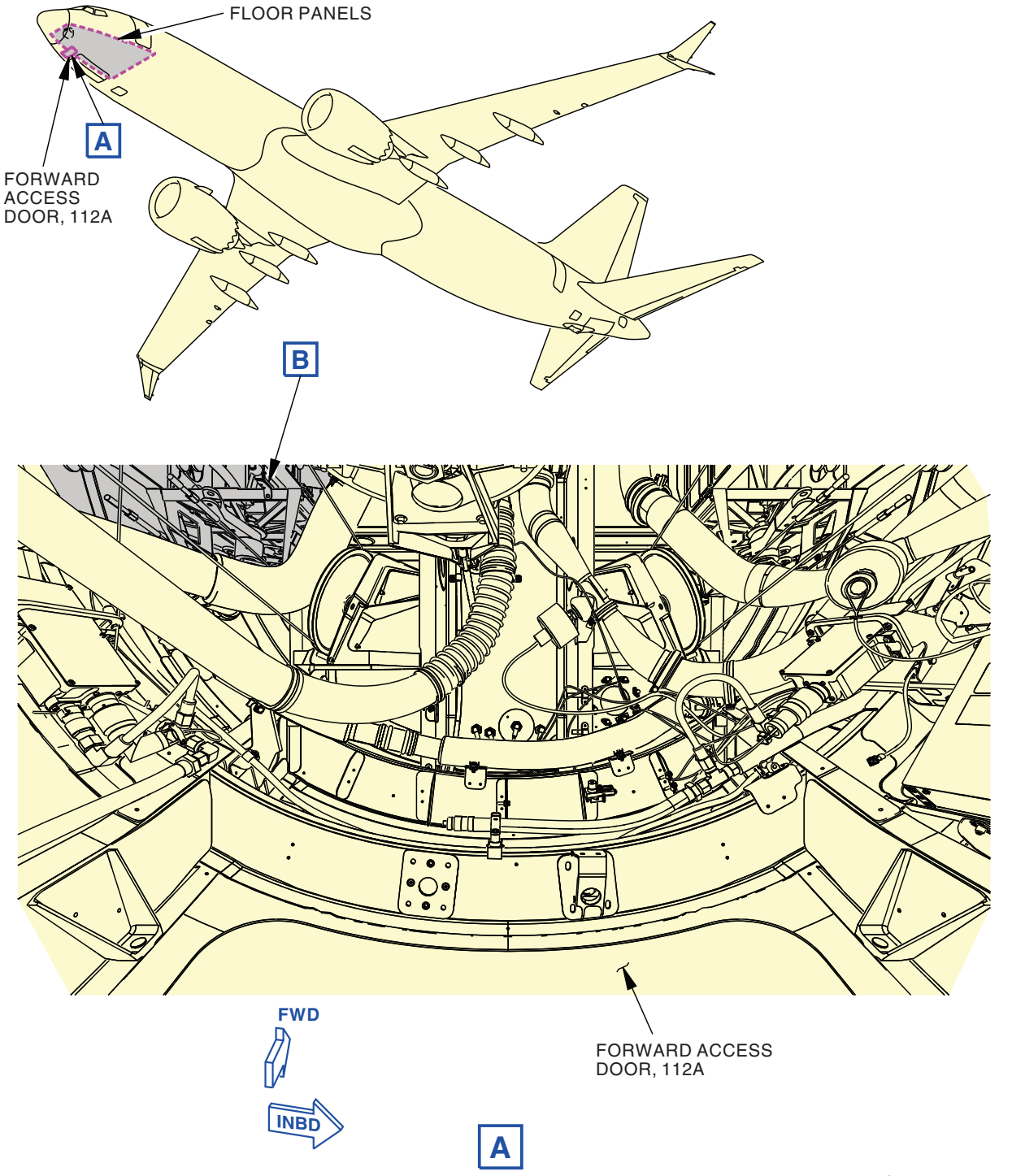
SUBTASK 32-44-00-860-004

(5) Close this access panel:

Number	Name/Location
112A	Forward Access Door

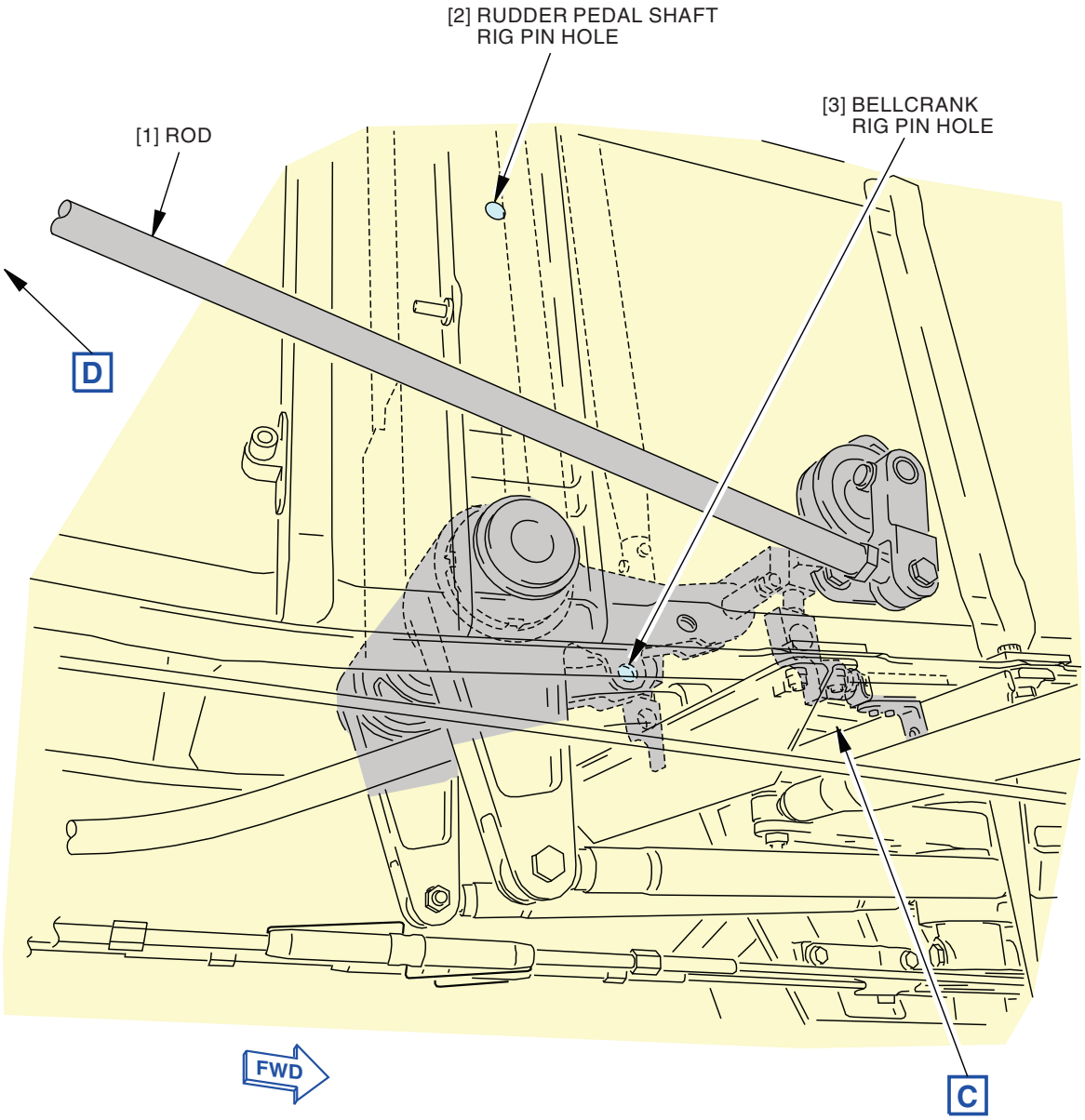
——— END OF TASK ———

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Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 1 of 6)

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Parking Brake System Adjustment
Figure 501/32-44-00-990-801 (Sheet 2 of 6)

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